STAC

GENERAL CATALOGUE
OF HARDWARE & ACCESSORIES

stac.es

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The products have specialized documentation so if you need more information do not hesitate to request it.

We have a qualified technical department that is at your complete disposal for any query. If you have not found what you are looking for, please contact us by email at ***info@stac.es*** *and will study a solution adapted to your needs.*

View or download the general catalogue of hardware and accessories in digital format from our website ***stac.es****.*

STAC

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Page diagram

NAME AND DESCRIPTION	A	MAIN CHARACTERISTICS OF THE PRODUCT	E
REFERENCE (S)	B	TYPES OF COMPATIBLE OPENING	F
UNITS / METRES PER BOX, TUBE OR REEL	C	TECHNICAL DIAGRAMS	G
FINISHES, MATERIALS AND OTHER SPECIFICATIONS	D	PRODUCT (S) WITH SIMILAR CHARACTERISTICS	H

A **POLARIS handle**

F

POLARIS HANDLE
Inward opening

B S5771 **

C 25

82 81 AL ZM AO PM 7

D

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- Supplied with transmission spindle of 7 x 29 mm.
- Available in different spindle lengths for outward opening: 7 x 64 mm (ref. S5775) or 7 x 75 mm (ref. T5775).
- Integrated rotating scutcheon for reduced installation time
- Audible indication of handle positions.

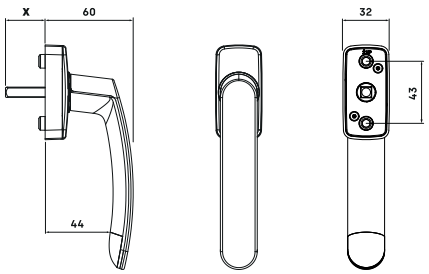
E

POLARIS HANDLE
Outward opening

H S5775 **

25

81 82 AL ZM AO PM 7



POLARIS HANDLE
Outward opening

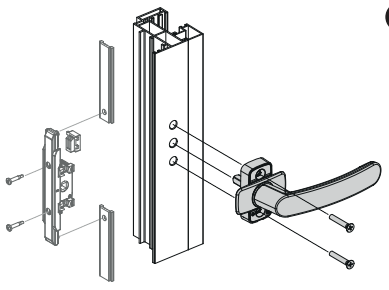
T5775 **

25

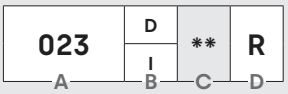
81 82 AL ZM AO PM 7

S5771 **	X = 27 mm	INWARD OPENING
S5775 **	X = 64 mm	OUTWARD OPENING
T5775 **	X = 75 mm	OUTWARD OPENING

G



REFERENCE BREAKDOWN



DESCRIPTION

023	PRODUCT REFERENCE ROOT	A	**	AVAILABLE IN DIFFERENT FINISHES	C
D	PRODUCT VALID FOR RIGH-HAND OPENING	B	R	PRODUCT WITH ADJUSTMENT	D
I	PRODUCT VALID FOR LEFT-HAND OPENING		S	THE PRODUCT INCLUDES SIRIUS CREMONE	

Other digits may appear larger to complete the product reference.

AVAILABLE FINISHES

IN STOCK

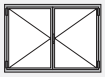
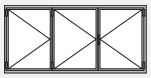
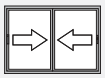
	00	PLAIN FINISH OR UNFINISH
	01	WHITE LACQUER FINISH
	02	BLACK LACQUER FINISH

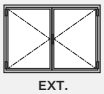
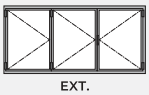
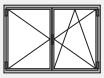
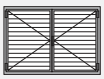
NOT AVAILABLE IN STOCK *

ANODIC	81	NATURAL ANODIZED FINISH	RAL	91	SPECIAL FINISH 1
	82	STAINLESS STEEL ANODIZED FINISH		92	SPECIAL FINISH 2
	83	ACIER GRATA ANODIZED FINISH		93	SPECIAL FINISH 3
			LIQUID PAINT	94	LIQUID PAINT FINISH

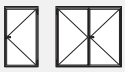
* *Please contact us for availability and delivery date.*

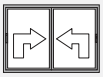
WINDOW TYPES

	SIDE-HUNG WINDOW
	SIDE-HUNG WINDOW WITH PASSIVE SASH
	SIDE-HUNG WINDOW WITH 3 OR MORE SASHES
	TILT & TURN WINDOW
	TOP-HUNG WINDOW
	SLIDING WINDOW
	SIDE-HUNG WINDOW WITH POLYGONAL SASH
	SINGLE SASH WINDOW SHUTTER

	OUTWARD OPENING WINDOW
	OUTWARD OPENING WINDOW WITH PASSIVE SASH
	OUTWARD OPENING WINDOW WITH 3 OR MORE SASHES
	TILT & TURN WINDOW WITH PASSIVE SASH
	CURTAIN WALL TOP-HUNG WINDOW
	BOTTOM-HUNG WINDOW
	OUTWARD OPENING WINDOW WITH POLYGONAL SASH
	DOUBLE SASH WINDOW SHUTTER

DOOR TYPES

	SINGLE / DOUBLE SASH DOOR

	LIFT & SLIDE

MATERIALS

AL	ALUMINIUM
ST	STEEL
PA	POLYAMIDE
POM	POLYOXYMETHYLENE (ACETAL)
PP	POLYPROPYLENE

ZM	ZAMAK
INX	STAINLESS STEEL
PE	POLYETHYLENE
PVC	POLYVINYL CHLORIDE
ABS	ACRYLONITRILE BUTADIENE STYRENE

Icon index

GROOVE TYPE

	EURO GROOVE 14 ~ 18 mm frame
	EURO GROOVE 10 ~ 14.5 mm frame
	EURO GROOVE 12 ~ 15 mm frame
	CENTRED GROOVE 18.5 ~ 25 mm frame

	16 GROOVE 14 ~ 18 mm frame
	16 GROOVE 10 ~ 14 mm frame
	40 20 GROOVE

OPENING TYPE

	TILT & TURN WINDOW WITH LOCKING TO FRAME
	TILT-BEFORE-TURN WINDOW
	SIDE-HUNG WINDOW WITH LOCKING TO FRAME

	TILT & TURN WINDOW WITH LOCKING TO INVERTER
	SIDE-HUNG WINDOW WITH LOCKING TO INVERTER

VALID OPENINGS

	SIDE-HUNG OPENING
	SIDE-HUNG OPENING INTERCHANGEABLE TO TILT & TURN OPENING
	TILT & TURN OPENING INTERCHANGEABLE TO BOTTOM-HUNG OPENING
	TILT & TURN AND SIDE-HUNG OPENING

	TILT & TURN OPENING
	TILT & TURN OPENING INTERCHANGEABLE TO SIDE-HUNG OPENING
	BOTTOM-HUNG OPENING INTERCHANGEABLE TO TILT & TURN OPENING

Icon index

PRODUCT HAND

	PRODUCT VALID FOR RIGHT-HAND OPENING
	PRE-MOUNTED FOR RIGHT-HAND AND INTERCHANGEABLE TO LEFT-HAND
	AMBIDEXTROUS PRODUCT

	PRODUCT VALID FOR LEFT-HAND OPENING
	PRE-MOUNTED FOR LEFT-HAND AND INTERCHANGEABLE TO RIGHT-HAND
	AMBIDEXTROUS PRODUCT BUT SUPPLIED UNMOUNTED

EVO HARDWARE

	PRODUCT INCLUDES EVO SOFT ACCESSORIES

	PRODUCT INCLUDES EVO SECURITY ACCESSORIES

LIFT & SLIDE SYSTEMS

	PRODUCT VALID FOR LS200 HARDWARE SYSTEM

	PRODUCT VALID FOR LS400 HARDWARE SYSTEM

TFS MACHINES

	PRODUCT VALID FOR TFS-01 MACHINE
	PRODUCT VALID FOR TFS-02 MACHINE
	PRODUCT VALID FOR TFS-04 MACHINE

	PRODUCT VALID FOR TFS-05 MACHINE
	PRODUCT VALID FOR TFS-06 MACHINE
	PRODUCT VALID FOR TFS-08 MACHINE

FAST FIXING SYSTEM

	You have the possibility to order the product with the built-in FAST fixing system. To order references with the FAST system, you must add an « F » before the finish (only for those products where this possibility is indicated).

A0790	**F**	**

01 SIDE-HUNG HINGES

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39 CARINA TAMIZ 4th sash hinges

40 NORMA hinges

41 4020 hinges

42 PEGASO hinge

BREAKDOWN

C	H	B	PM / PF	C	E

C	GROOVE TYPE
PM	FRAME PROFILE
PF	SASH PROFILE

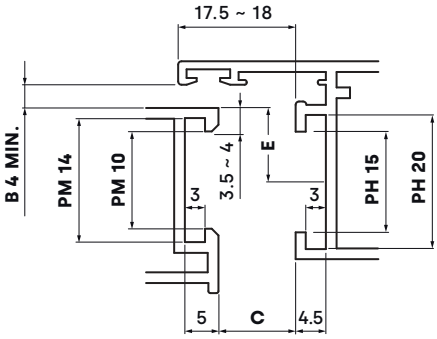
H	SASH
C	CHAMBER

B	HINGE PITCH
E	SHAFT

All units in the document are expressed in millimetres.

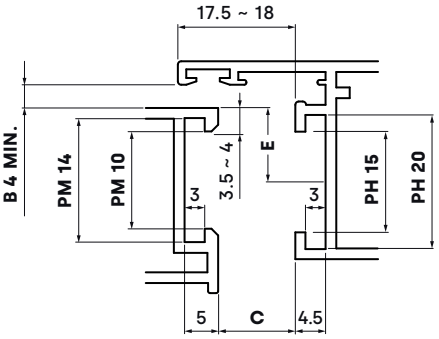
CR1

CE	-	4	10 / 14.5 15 / 20	11.5	10



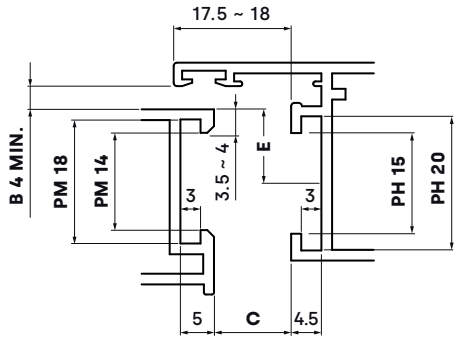
CR2

CE	-	4	12 / 15 15 / 20	11.5	10



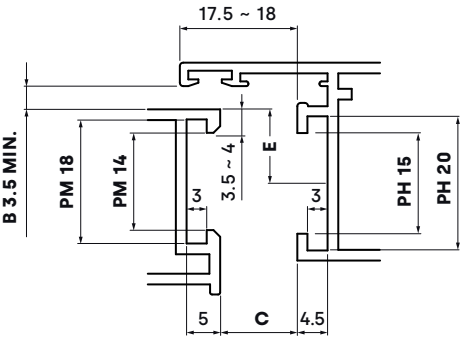
CR3

CE	-	4	14 / 18 15 / 20	11.5	10



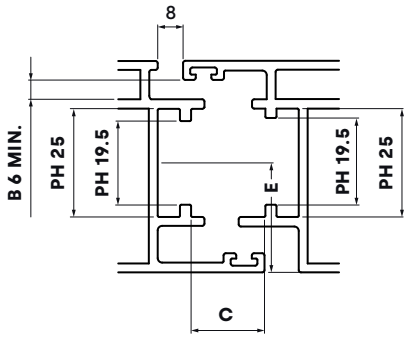
CR4

CE	-	3.5	14 / 18 15 / 20	11.5	10



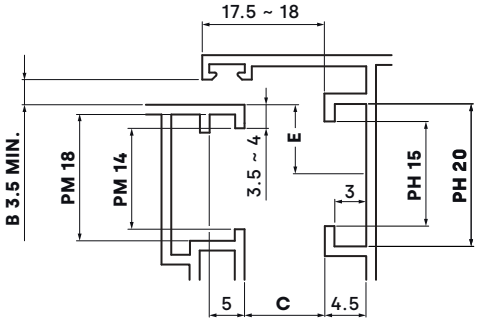
CR5

CC	-	6	19.5 / 25 19.5 / 25	15.6 / 17	25



CR6

CE	3[a]	3.5	14 / 18 15 / 20	11.5	10

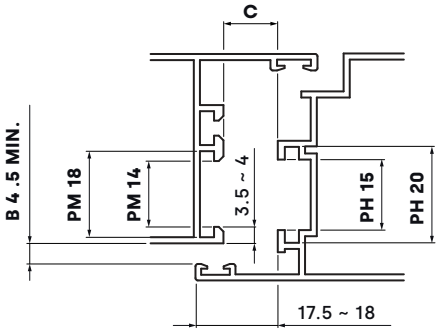


Reference groove for side-hung opening

C	GROOVE TYPE
CE	Euro Groove
CC	Centred Groove
4020	Groove 40 / 20
H	SASH
3ª	3rd sash
4ª	4th sash

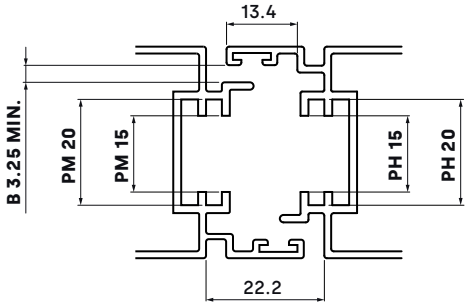
CR7

CE	4ª	4.5	14 / 18 15 / 20	11.5	10



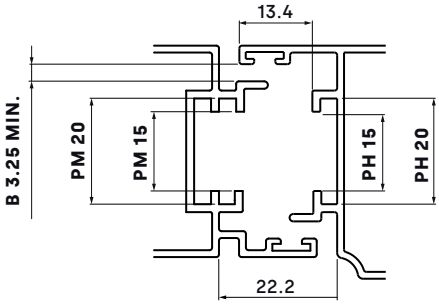
CR8

CC	3ª	3.25	15 / 20 15 / 20	-	-



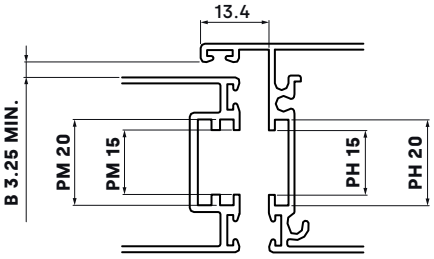
CR9

CC	4ª	3.25	15 / 20 15 / 20	-	-



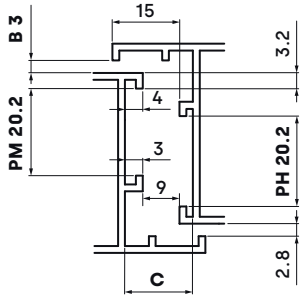
CR10

CC	4ª	3.25	15 / 20 15 / 20	-	-



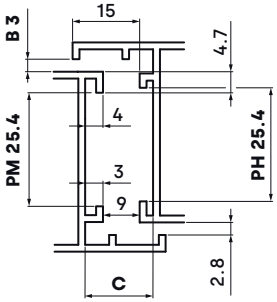
CR11

4020	-	3	20.2 20.2	15.2	-



CR12

4020	-	3	25.4 25.4	15.2	-



ACRUX

Hinge range

Wide range of profiles

Different opening types

Quick and easy assembly

High traffic public areas

ACRUX hinges

Small sized hinges for windows, doors and balcony doors with large sashes.

If there is one thing that makes the ACRUX range stand out, it is the versatility of its products and the capacity to adapt to the needs of the client that the hinges offer.

Our hinge range is the best option in the aluminium enclosure market for its working range in terms of size and performance.

Despite their small dimensions, the hinges support loads of up to 120 kg, reaching up to 150 kg with the three-leaf models.

» Wide range of profiles

The ACRUX range has different hinge models available to cover the widest range of profiles according to the customer's needs.

- **ACRUX CE hinges** for euro groove profiles with frame versions 10 / 14 mm or 14 / 18 mm.
- **ACRUX FR hinges** for profiles with centred groove (or french groove).
- **ACRUX 16 hinges** for profiles with a 16 mm groove with 13-millimetre shaft.

» Compatibility with different opening types

ACRUX hinges are compatible with different types of aluminium window, door and balcony door openings available on the market.

- Single or double-sash door with left or right-hand opening for both inward and outward access.
- Window or balcony door with heavy weight sashes and / or format with side-hung opening.
- Bottom-hung windows of large dimensions.

» Quick and easy installation

The hinges are installed on counter plates that are grooved, both on the frame profile and on the sash profile. The bolt has been specifically designed to facilitate fitting and insertion into the sash blade bushing.

» Public areas

The three-leaf models are specifically designed to be installed on public building doors with high traffic frequency, where people misuse them or where there is a high probability of accidents occurring.

Hinge for Euro Groove profiles with 14 / 18 mm frame.

• Adjustments: lateral adjustment of ± 1.2 millimetres and vertical adjustment of ± 3 millimetres by installing the kit with ref. STC-MC-0011.
• Antifriction washers made of acetal resin (POM) to guarantee optimum performance in terms of wear.

ACRUX 2-leaf hinge (ref. 010)

• Maximum load with two-leaf hinges: 100 kg with 2 units ~ 120 kg with 3 units.

ACRUX 3-leaf hinge (ref. P010D)

• Maximum load with three-leaf hinges: 120 kg with 2 units ~ 150 kg with 3 units.
• Designed for large doors, high weight and high frequency of use.

ACRUX HINGE CE 14.18

CE · CR3 · 81 82 · AL · IN

010	D / I	**

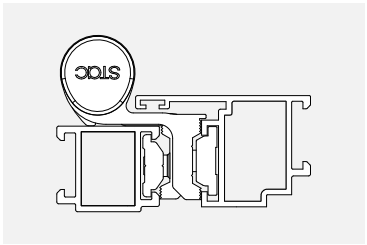
12

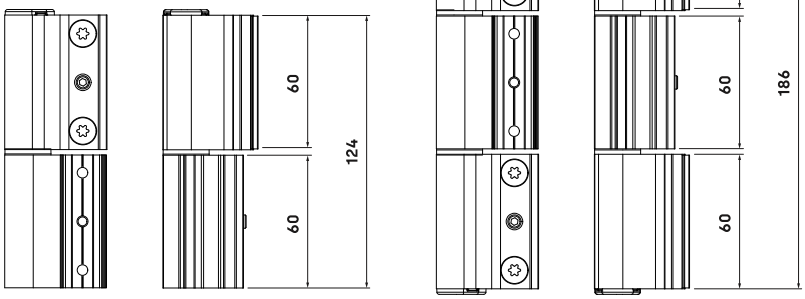
ACRUX HINGE CE 14.18

CE · CR3 · 81 82 · AL · IN

P010D	**

6





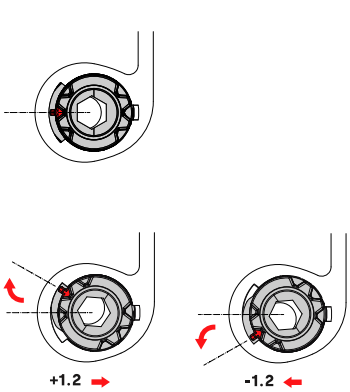
VERTICAL ADJUSTMENT KIT

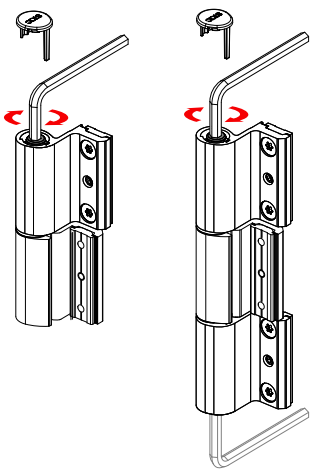
CE · CC · ZM

STC-MC-0011

50

LATERAL ADJUSTMENT ± 1.2 mm





ACRUX HINGE CE 10.14

010F	D / I	**

12

CE · CR1 · AL · IN

Hinge for Euro Groove profiles with 10 / 14 mm frame.

- Adjustments: lateral adjustment of ± 1.2 millimetres and vertical adjustment of ± 3 millimetres by installing the kit with ref. STC-MC-0011.
- Antifriction washers made of acetal resin (POM) to guarantee optimum performance in terms of wear.

ACRUX 2-leaf hinge (ref. 010F)

- Maximum load with two-leaf hinges: 100 kg with 2 units ~ 120 kg with 3 units.

ACRUX 3-leaf hinge (ref. P010FD)

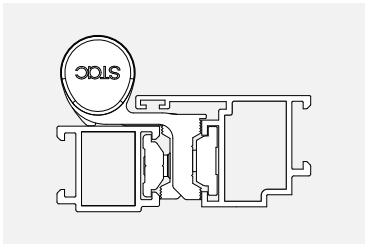
- Maximum load with three-leaf hinges: 120 kg with 2 units ~ 150 kg with 3 units.
- Designed for large doors, high weight and high frequency of use.

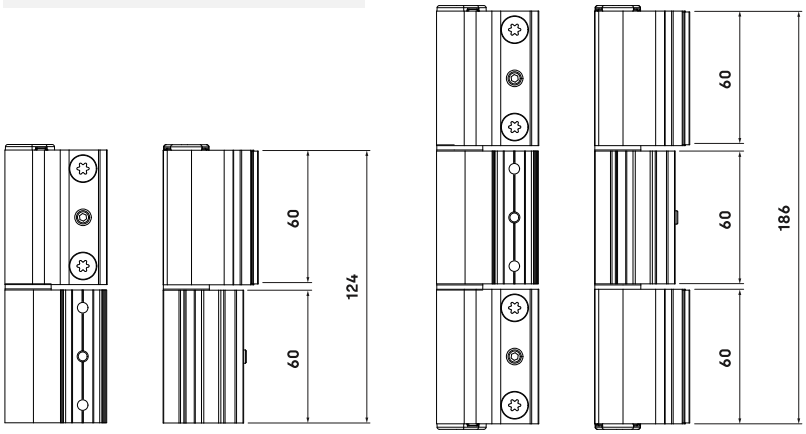
ACRUX HINGE CE 10.14

P010FD	**

6

CE · CR1 · AL · IN





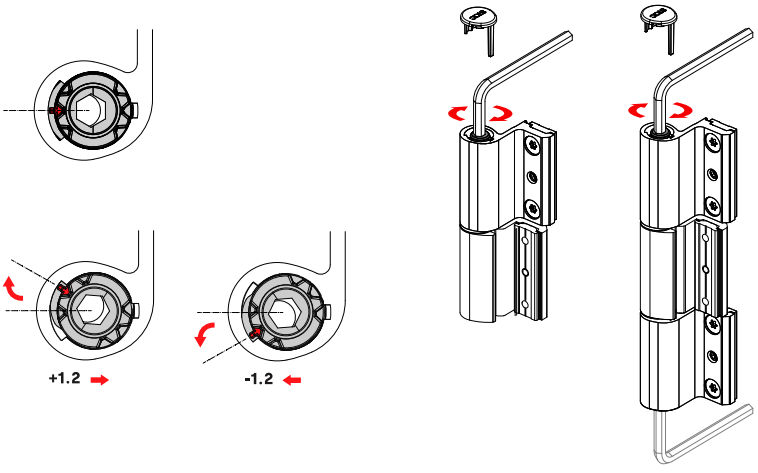
VERTICAL ADJUSTMENT KIT

STC-MC-0011

50

CE · CC · ZM

LATERAL ADJUSTMENT ± 1.2 mm



USAGE RECOMMENDATIONS

Considerations

Consult the technical data sheet to access the table of maximum weights supported by each hinge depending on the door dimensions. In any case, for public works, a weight reduction of 20% must be applied to the values shown in the table.

010 D/I ** / 010F D/I **	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	100 kg	120 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	80 kg	100 kg
WITH DOOR CLOSER WITH RESTRICTOR	-	80 kg
WITH DOOR CLOSER W/O RESTRICTOR	-	70 kg

P010D ** / P010FD **	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	120 kg	150 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	100 kg	130 kg
WITH DOOR CLOSER WITH RESTRICTOR	80 kg	110 kg
WITH DOOR CLOSER W/O RESTRICTOR	70 kg	100 kg

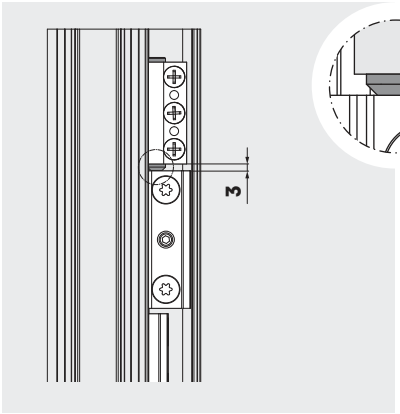
VERTICAL ADJUSTMENT ACCESSORY

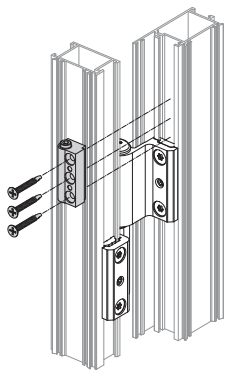
STC-MC-0011

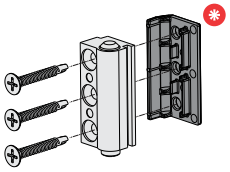
To obtain the adjustment range of 3 mm, install the adjustment device keeping a distance of 3 millimetres from the base of the device in respect to the contact surface with the hinge blade (1).

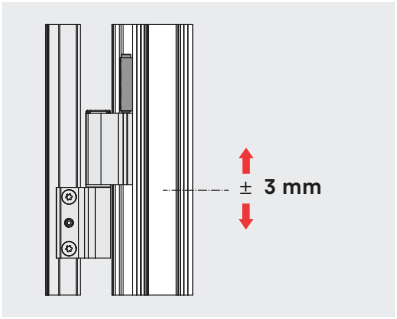
Before proceeding with the adjustment, it is necessary to loosen the hinge fastening screws (2).

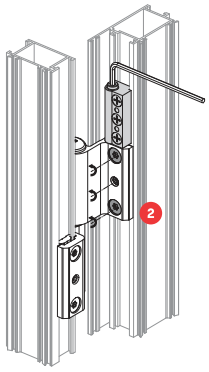
For Centred Groove profiles an adaptor shim is included. *











ACRUX FR HINGE

030	D / I	**

6

CC CR5 AL IN

Hinge for door profiles with Centred Groove.

• Adjustments: lateral adjustment of ± 1.2 millimetres and vertical adjustment of ± 3 millimetres by installing the kit with ref. STC-MC-0011.
• Antifriction washers made of acetal resin (POM) to guarantee optimum performance in terms of wear.

ACRUX 2-leaf hinge (ref. 030)

• Maximum load with two-leaf hinges: 100 kg with 2 units ~ 120 kg with 3 units.

ACRUX 3-leaf hinge (ref. P030D)

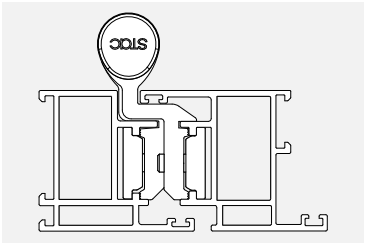
• Maximum load with three-leaf hinges: 120 kg with 2 units ~ 150 kg with 3 units.
• Designed for large doors, high weight and high frequency of use.

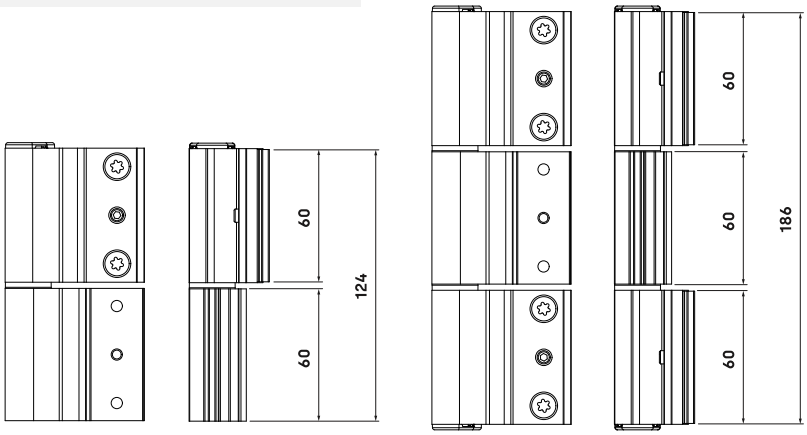
ACRUX FR HINGE

P030D	**

6

CC CR5 AL IN





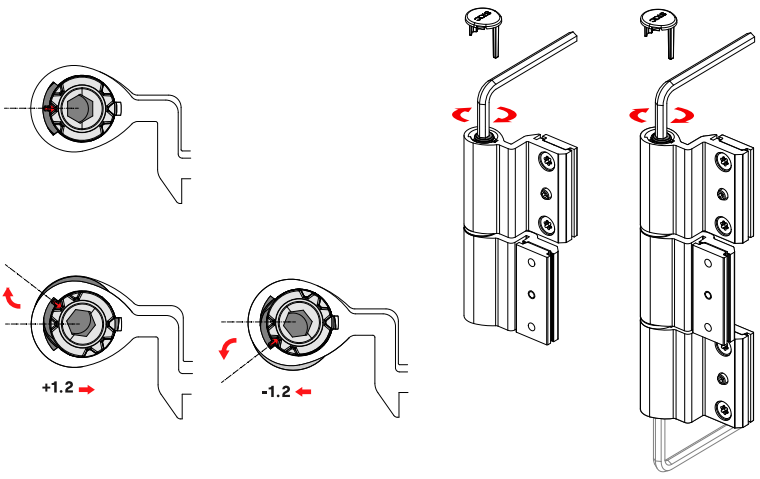
VERTICAL ADJUSTMENT KIT

STC-MC-0011

50

CE CC ZM

LATERAL ADJUSTMENT ± 1.2 mm



USAGE RECOMMENDATIONS

Considerations

Consult the technical data sheet to access the table of maximum weights supported by each hinge depending on the door dimensions. In any case, for public works, a weight reduction of 20% must be applied to the values shown in the table.

030 D / I **

	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	100 kg	120 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	80 kg	100 kg
WITH DOOR CLOSER WITH RESTRICTOR	-	80 kg
WITH DOOR CLOSER W/O RESTRICTOR	-	70 kg

P030D **

	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	120 kg	150 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	100 kg	130 kg
WITH DOOR CLOSER WITH RESTRICTOR	80 kg	110 kg
WITH DOOR CLOSER W/O RESTRICTOR	70 kg	100 kg

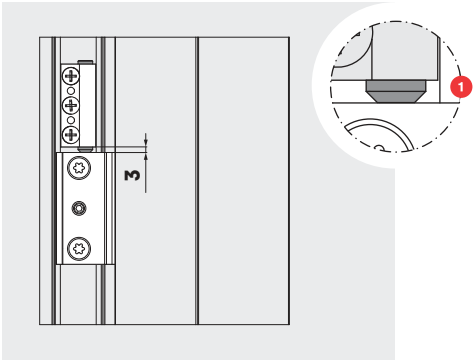
VERTICAL ADJUSTMENT ACCESSORY

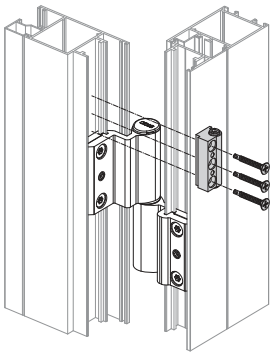
STC-MC-0011

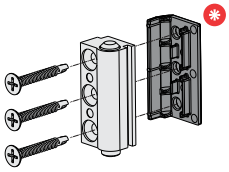
To obtain the adjustment range of 3 mm, install the adjustment device keeping a distance of 3 millimetres from the base of the device in respect to the contact surface with the hinge blade (1).

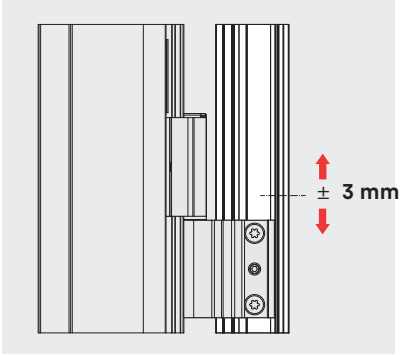
Before proceeding with the adjustment, it is necessary to loosen the hinge fastening screws (2).

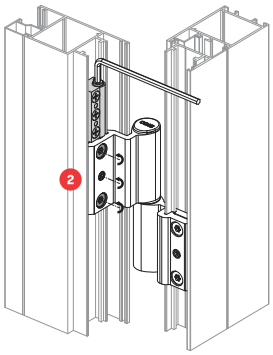
For Centred Groove profiles an adaptor shim is included. *











TAURUS HINGE

A011	D / I	**

12

AL IN PM

Adjustable hinges for door with Euro Groove profile.

- Horizontal adjustment of ± 1 mm, even with the door installed.
- Vertical adjustment up to ± 3 mm with reference STC-MC-0011.
- Direct installation in the profile without the need for any machining.

TAURUS hinge (ref. A011)

- Maximum load of 100 kg with 2 or more hinges.
- The hinge incorporates a set screw to prevent slippage, adjustment cap and acetal bushing with improved clip.
- Available hinge disassembled (ref. A011X).

TAURUS CLX hinge (ref. A011-160)

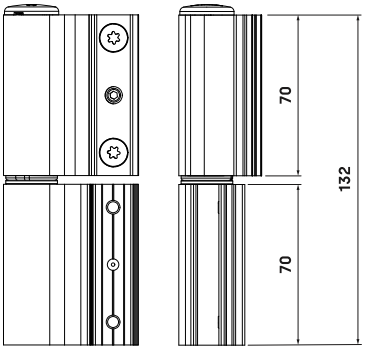
- Maximum load with 2 or more hinges is 160 kg.
- Incorporates a ball bearing for smooth and silent rotation.
- Supplied in sets with black lacquered body and coloured covers.

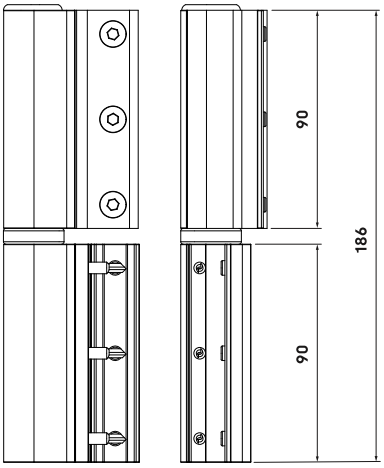
TAURUS CLX HINGE

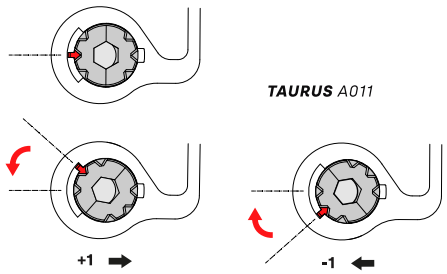
A011-160	D / I	**

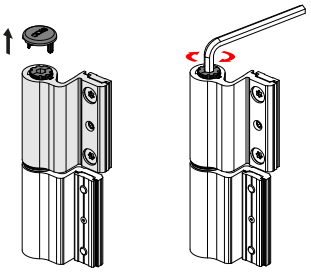
1

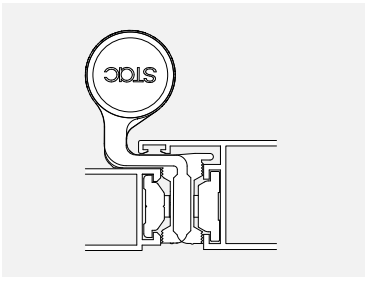
AL IN PM

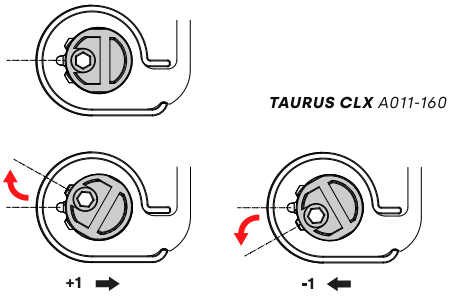


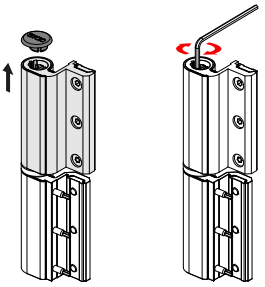


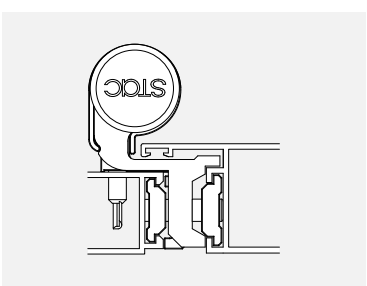












USAGE RECOMMENDATIONS

Considerations

Consult the technical data sheet to access the table of maximum weights supported by each hinge depending on the door dimensions. In any case, for public works, a weight reduction of 20% must be applied to the values shown in the table.

A011 D/I **	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	100 kg	100 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	80 kg	80 kg
WITH DOOR CLOSER WITH RESTRICTOR	-	-
WITH DOOR CLOSER W/O RESTRICTOR	-	-

A011 D/I ** -160	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	160 kg	160 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	140 kg	140 kg
WITH DOOR CLOSER WITH RESTRICTOR	120 kg	120 kg
WITH DOOR CLOSER W/O RESTRICTOR	110 kg	110 kg

ARIES

coplanar hinges
with 1 million cycles

Micrometric adjustment

Up to 180 kg per sash

Quick & easy assembly

Public buildings

ARIES hinges

During the life cycle of a door the hinges are subjected to an intensive use due to the high number of opening and closing cycles and type of usage to which they are continually subjected.

Our ARIES hinges line stands out thanks to the high quality and innovative design:

• Ideal for large and/or heavy doors and those with high usage levels. ARIES hinges can hang doors of up to 180 kg per sash.

• Micrometric, independent adjustment. Adjustment can be performed with the door leaf installed.

• Good behaviour against friction and wear of surfaces with relative movement.

» 1.000.000 cycles

The 3-leaf ARIES door hinge has been successfully tested in accordance with UNE-EN 1191 regulations to 1,000,000 cycles by the Applus+ testing laboratory. This is proof that STAC's coplanar hinges are the perfect choice for applications requiring a high performance level, as required in high-frequency- of-use doors and large sized doors in buildings with high levels of public transit, including cases of involuntary or intentional violent usage.

» Self-centering screws

• With a deeper hexagon trace, it is possible to apply more torque without over tightening the screw heads.

• Eliminates the use of the casing as a guide. This makes the installation of the hinges quicker and easier and also makes the set more long-lasting.

• The use of self-centering screws together with the ARIES T hinges significantly reduces stock by allowing these ARIES hinges both quick and traditional fixing.

» Public areas

The three-leaf models are specifically designed to be installed on public building doors with high traffic frequency, where people misuse them or where there is a high probability of accidents occurring.

Universal hinges with millimetre adjustment for door profiles.

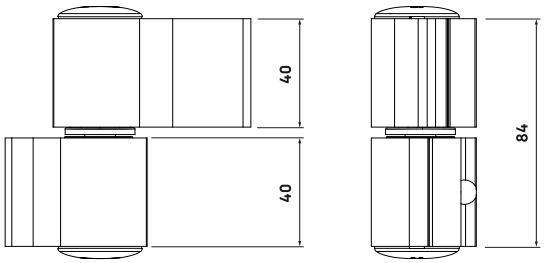
- Millimetric adjustment in three dimensions and independent of each other.
- Direct installation on the profile through screws.
- Compatible with standard square and rectangular profiles.
- Shims are available for profile series with projections (ref. 010TS, ref. 010CS).
- Adjustable drilling template to facilitate profile machining (ref. P-010-01).

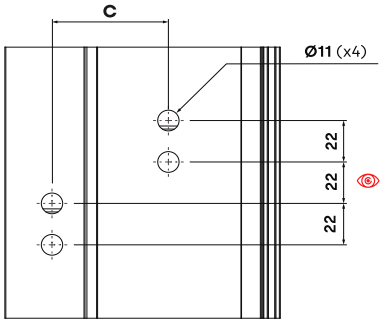
ARIES 2-LEAF HINGE

AL IN PA PM

010T2	40 60 66 80 93	**

2





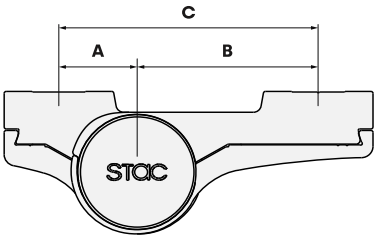
FOR REFERENCE 010T240 THE DISTANCE IS 20 mm

	A	B	C
010T240 **	20	20	**40**
010T260 **	20	40	**60**
010T266 **	20	46.5	**66.5**
010T280 **	40	40	**90**
010T293 **	46.5	46.5	**93**

» COMPATIBLE FIXING SYSTEMS 010T2

A05KT2	TRADITIONAL FIXING WITH ECO COUNTERPLATES
A05KT2-L40	TRADITIONAL FIXING WITH ECO L40 COUNTERPLATES
P05KT2	TRADITIONAL FIXING WITH RPT COUNTERPLATES
A0505041	FIXING DIRECTLY TO THE PROFILE (Ø12 x 50 mm)
A0505042	FIXING DIRECTLY TO THE PROFILE (Ø12 x 60 mm)

CONSULT "HINGE ACCESSORIES"



ARIES 3-LEAF HINGE

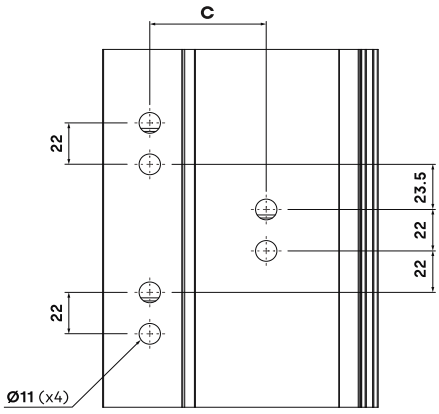
010T3	60	**
	66	
	80	
	93	

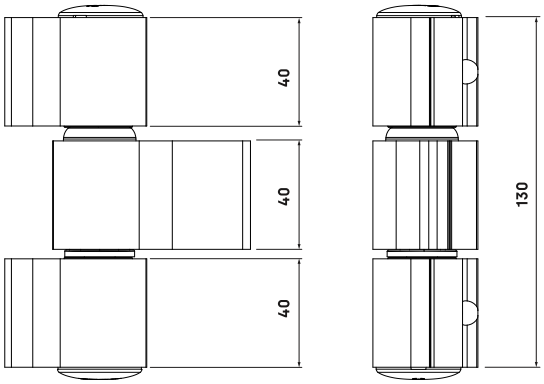
2

AL IN
PA PM

Universal hinges with millimetre adjustment especially designed for large and heavy doors and high frequency of use.

- Millimetric adjustment in three dimensions and independent of each other.
- Direct installation on the profile through screws.
- Shims are available for profile series with projections (ref. 010TS, ref. 010CS).
- Adjustable drilling template to facilitate profile machining (ref. P-010-01).
- Successfully tested at 1 million cycles with 100 kg of weight (inform nº 16~12559-1551).



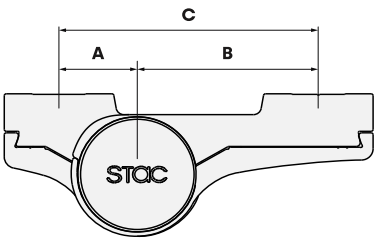


» **COMPATIBLE FIXING SYSTEMS 010T3**

A05KT3	TRADITIONAL FIXING WITH ECO COUNTERPLATES
A05KT3-L40	TRADITIONAL FIXING WITH ECO L40 COUNTERPLATES
P05KT3	TRADITIONAL FIXING WITH RPT COUNTERPLATES
A0505041	FIXING DIRECTLY TO THE PROFILE (Ø12 x 50 mm)
A0505042	FIXING DIRECTLY TO THE PROFILE (Ø12 x 60 mm)

CONSULT "HINGE ACCESSORIES"

	A	B	C
010T360 **	20	40	**60**
010T366 **	20	46.5	**66.5**
010T380 **	40	40	**80**
010T393 **	46.5	46.5	**93**



USAGE RECOMMENDATIONS

Considerations

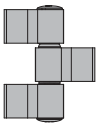
Consult the technical data sheet to access the table of maximum weights supported by each hinge depending on the door dimensions. In any case, for public works, a weight reduction of 20% must be applied to the values shown in the table.

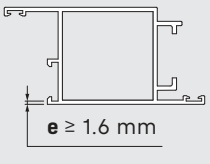
010T2 **

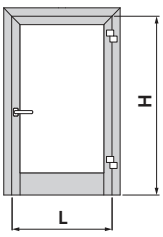
	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	160 kg	180 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	140 kg	160 kg
WITH DOOR CLOSER WITH RESTRICTOR	100 kg	120 kg
WITH DOOR CLOSER W/O RESTRICTOR	125 kg	150 kg

010T3 **

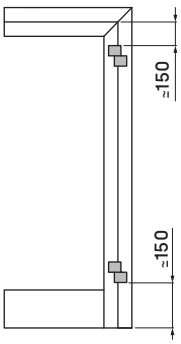
	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	160 kg	180 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	150 kg	170 kg
WITH DOOR CLOSER WITH RESTRICTOR	120 kg	140 kg
WITH DOOR CLOSER W/O RESTRICTOR	140 kg	160 kg



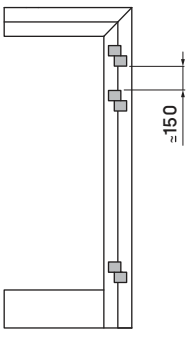




The recommendations for the use of ARIES hinges apply for a profile thickness ≥ 1.6 millimetres. If the thickness is less, please consult STAC's technical department.

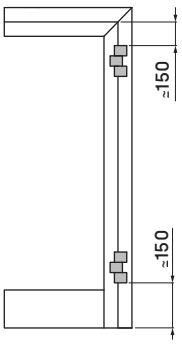


H / L ≥ 2

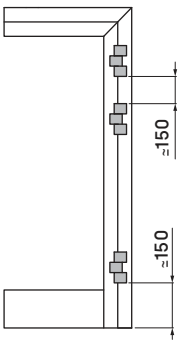


H / L < 2

H > 2200, L > 1200 (mm)



H / L ≥ 2



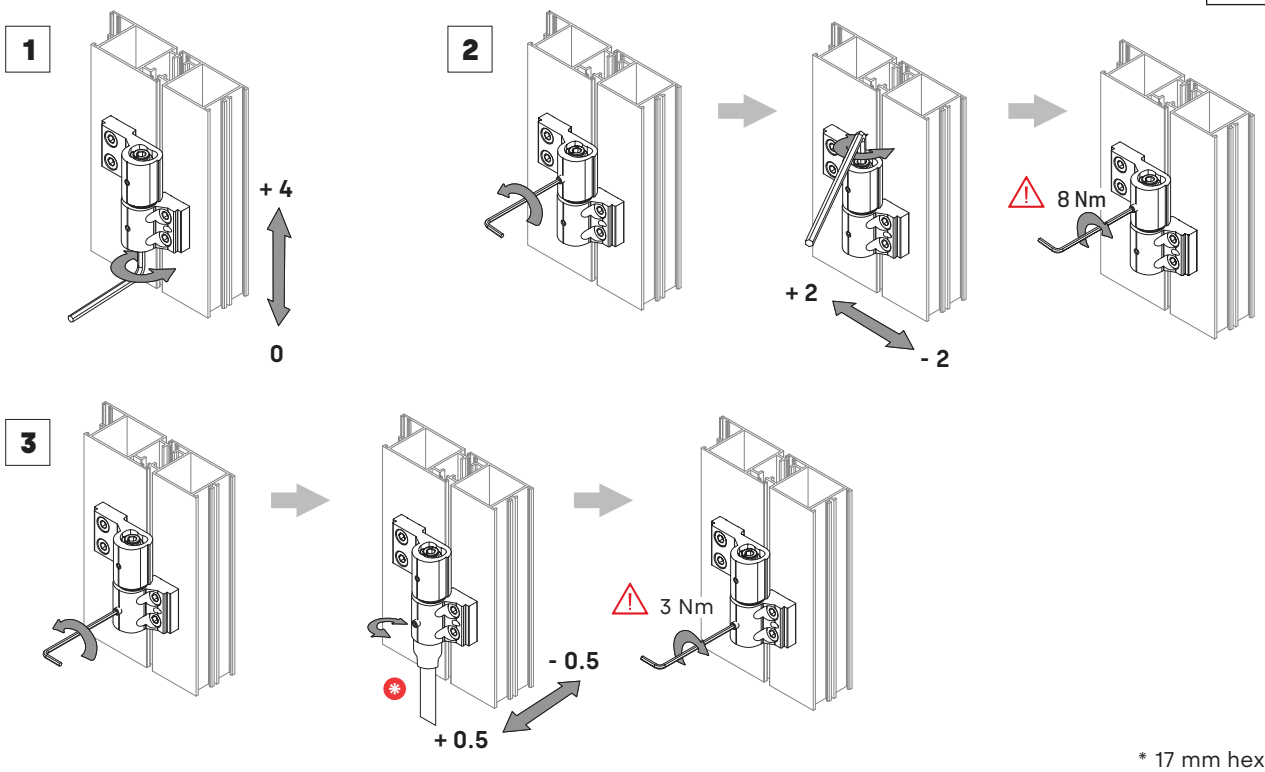
H / L < 2

H > 2200, L > 1200 (mm)

ADJUSTMENTS

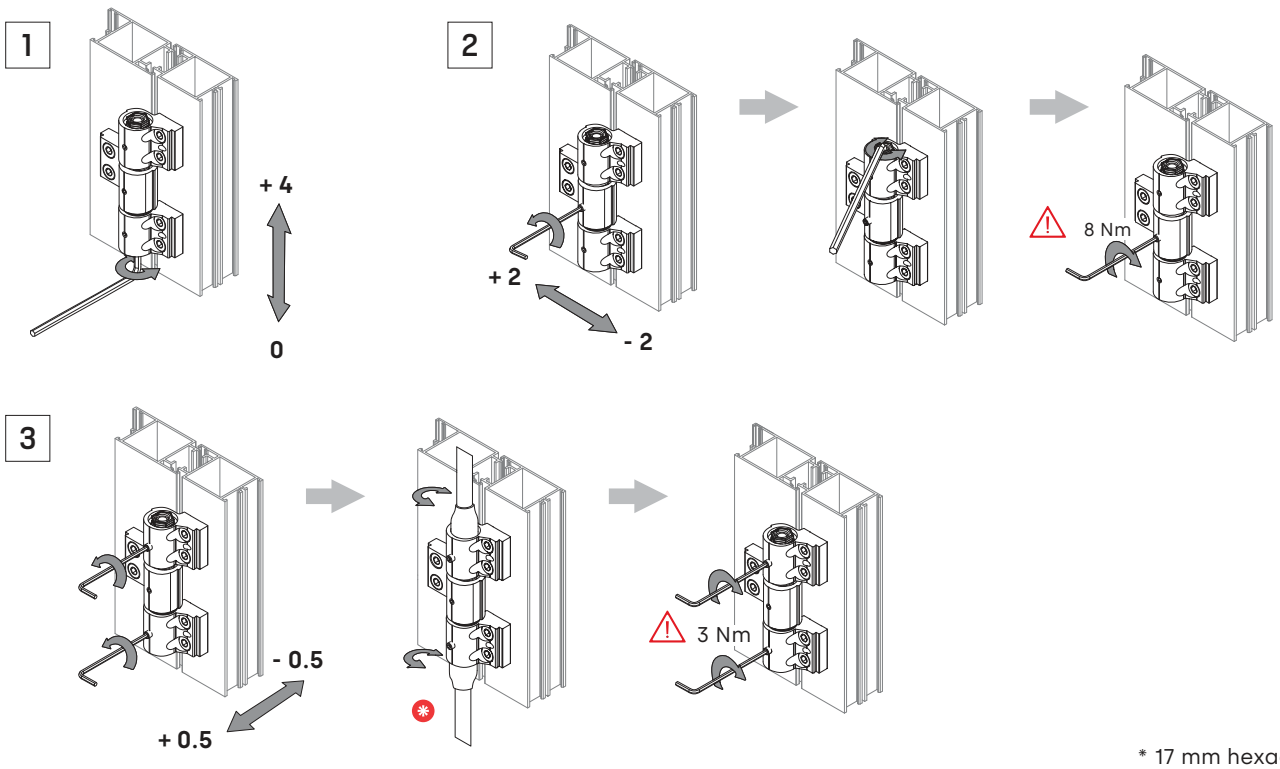
Considerations ARIES hinges adjustments are always made with the door closed.

010T2 **



* 17 mm hexagonal key.

010T3 **



* 17 mm hexagonal key.

Shims for fixing ARIES hinges on profile series with projections.

Shims for self-tapping screws (ref. 010TS).

• Aluminium shims for fixing the hinges to the profile by means of steel screws.
• Self-tapping screws are not included (ref. A0505041, ref. A0505042).

Shims with bushes (ref. 010CS).

• Aluminium shims with bushes for fixing the hinges to the profile by means of ECO L40 counterplates.
• The installation of the shims is carried out by means of screws (ref. 0505090) or by means of SLIM L40 fixing kits (ref. A05KT2-L40, ref. A05KT3-L40).

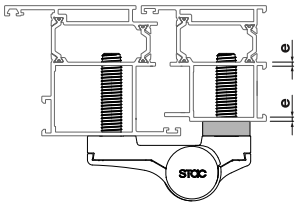
SHIMS FOR SELF-TAPPING SCREWS

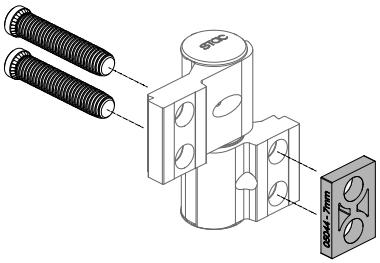
010TS	07 08 10	**

AL

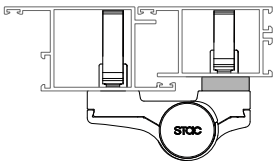
2

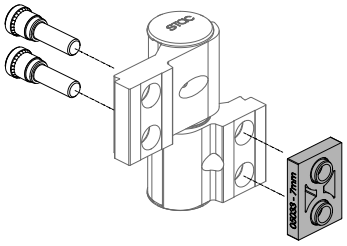
QUICK FIXING
e ≥ 1.6 millimetres





TRADICIONAL FIXING





SHIMS WITH BUSHES

010CS	07 08 10	**

AL

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COMPATIBLE FIXING SYSTEMS 010TS

A0505041	FIXING DIRECTLY TO THE PROFILE (Ø12 x 50 mm)
A0505042	FIXING DIRECTLY TO THE PROFILE (Ø12 x 60 mm)

COMPATIBLE FIXING SYSTEMS 010CS

A05KT2-L40	TRADITIONAL FIXING WITH ECO L40 COUNTERPLATES
A05KT3-L40	TRADITIONAL FIXING WITH ECO L40 COUNTERPLATES
0505090	SELF-CENTERING SCREWS (M8 x 38)

SLIM FIXING KIT (2P)
Screws with 28 mm bushing

A05KT2

10

AL

RPT FIXING KIT (2P)
Screws with 28 mm bushing

P05KT2

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Fixing kits with counterplates for ARIES hinges.

• Specific fixing kits are available for 2-leaf o 3-leaf hinges.

SLIM fixing kits (ref. A05KT2, ref. A05KT3)

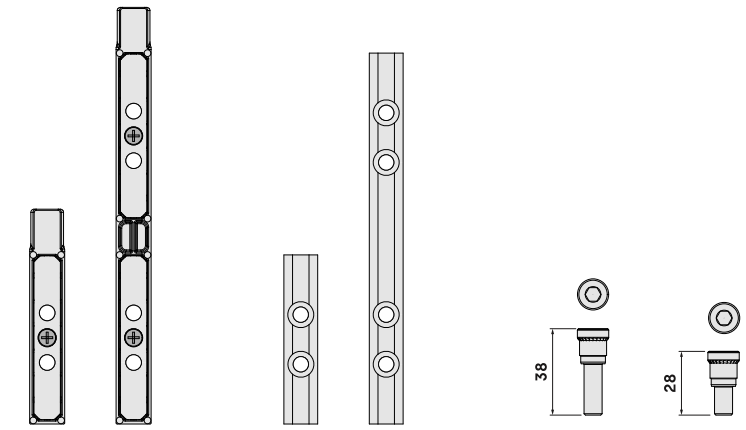
• Supplied with ECO threaded backplates, centring bushing screws and positioning springs.
• The holes are centred longitudinally on the plate; therefore, they do not have a position.

RPT fixing kits (ref. P05KT2, ref. P05KT3)

• Supplied with RPT threaded backplates, centring bushing screws and positioning springs.
• The RPT backplates are made of polyamide, providing greater insulation and they are recommended for use in profiles with thermal break.

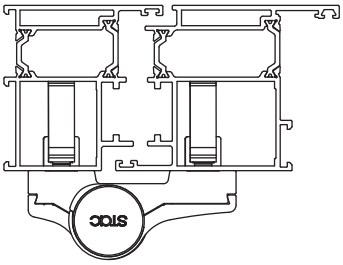
Screws with centring bushing

• The M8 screws are supplied in lengths of 28 or 38 mm.
• The greater depth of the hexagon socket allows a higher torque to be applied without over tightening the screw heads.



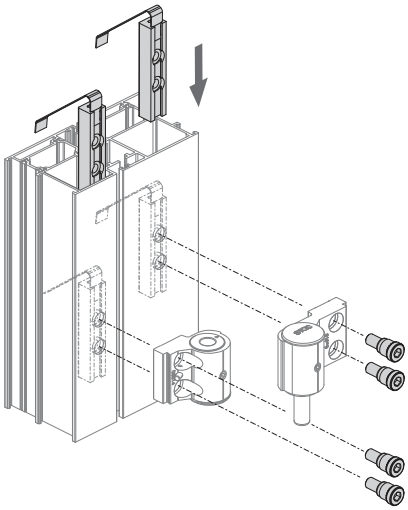
RPT BACKPLATES *ECO BACKPLATES* *SCREW WITH BUSHING*

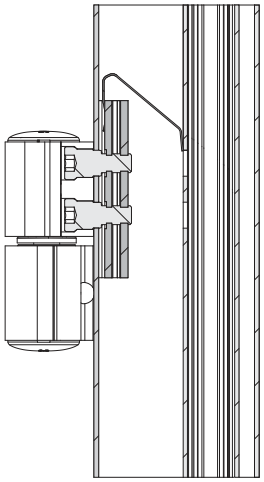
SLIM FIXING SYSTEM A05KT2		*ARIES*
A05KT2	**ECO BACKPLATES** AND SCREW WITH BUSHING L = 28 mm	*010T2*
A05KT2-L40	**ECO BACKPLATES** AND SCREW WITH BUSHING L = 38 mm	
A05KT3	**ECO BACKPLATES** AND SCREW WITH BUSHING L = 28 mm	*010T3*
A05KT3-L40	**ECO BACKPLATES** AND SCREW WITH BUSHING L = 38 mm	
RPT FIXING SYSTEM P05KT2		*ARIES*
P05KT2	**RPT BACKPLATES** AND SCREW WITH BUSHING L = 28 mm	*010T2*
P05KT2-L40	**RPT BACKPLATES** AND SCREW WITH BUSHING L = 38 mm	
P05KT3	**RPT BACKPLATES** AND SCREW WITH BUSHING L = 28 mm	*010T3*
P05KT3-L40	**RPT BACKPLATES** AND SCREW WITH BUSHING L = 38 mm	

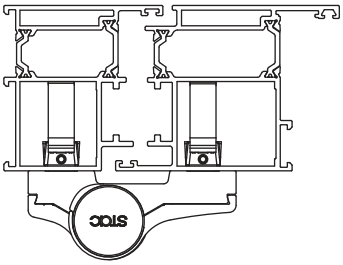


Recommended tightening torque:
20 N.m

A05KT	BACKPLATES	SCREWS	COMPATIBILITY
A05KT2	ECO BACKPLATES	M8, L = 28 mm	ARIES 2-LEAVES
A05KT2-L40	ECO BACKPLATES	M8, L = 38 mm	
A05KT3	ECO BACKPLATES	M8, L = 28 mm	ARIES 3-LEAVES
A05KT3-L40	ECO BACKPLATES	M8, L = 38 mm	

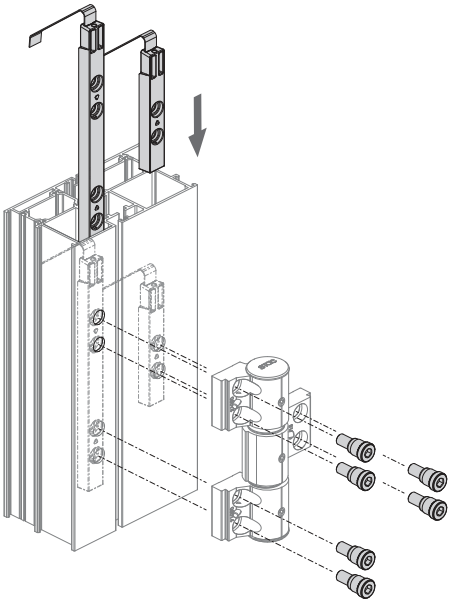


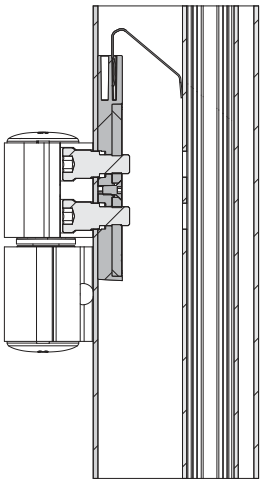




Recommended tightening torque:
20 N.m

P05KT	BACKPLATES	SCREWS	COMPATIBILITY
P05KT2	RPT BACKPLATES	M8, L = 28 mm	ARIES 2-LEAVES
P05KT3	RPT BACKPLATES	M8, L = 28 mm	ARIES 3-LEAVES





SELF-TAPPING SCREW
12 x 50 millimetres

0505041

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Screws for fast and simple fixing of ARIES hinges to the profile.

Self-tapping screw (ref. 0505041)

- The quick fixing screws are valid for series with double chamber and with walls ≥ 1.6 mm.
- Supplied in zinc-plated steel with a diameter of 12 mm and a length of 50 or 60 mm (ref. 0505042).

Screw with centring bushing (ref. 0505089)

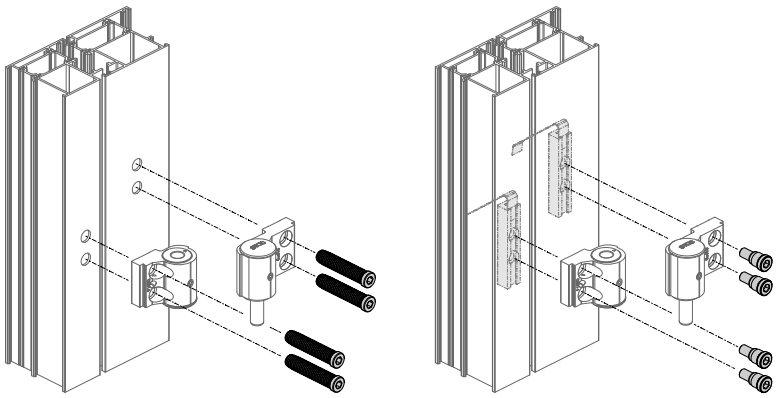
- Screw with centring bushing which allow the installation of the hinges to the profile through ECO or RPT counterplates.
- The M8 screw is supplied in length of 28 mm or 38mm (ref. 0505090).
- The screw with centring bushing have a greater hexagonal depth. This allows a higher torque to be applied without passing the screw head.

SCREW WITH CENTRING BUSHING
M8 x 28 millimetres

0505089

100

AC

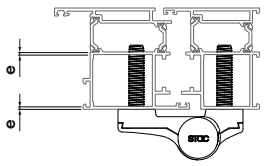


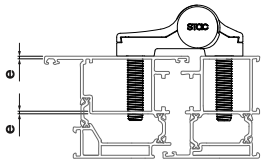
SELF-TAPPING SCREWS		ARIES
0505041	SELF-TAPPING SCREW **Ø12 millimetres, L = 50 millimetres**	010T2 010T3
0505042	SELF-TAPPING SCREW **Ø12 millimetres, L = 60 millimetres**	

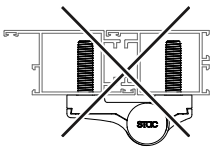
SCREW WITH BUSHING		ARIES
0505089	SCREW WITH BUSHING **M8 millimetres, L = 28 millimetres**	010T2 010T3
0505090	SCREW WITH BUSHING **M8 millimetres, L = 38 millimetres**	

For series with double chamber and with walls over than 1.6 mm

e ≥ 1.6 millimetres







Adjustable template that facilitates and optimises the drilling of the profiles.

- The drilling template adapts to all available ARIES hinge sizes.
- The drilling template is supplied with bolts to fit 4, 5, 6 and 8 mm coplanar profile traces.

DRILLING TEMPLATE

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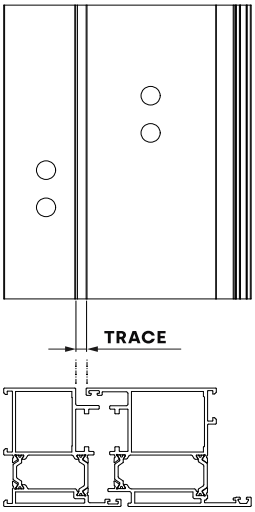

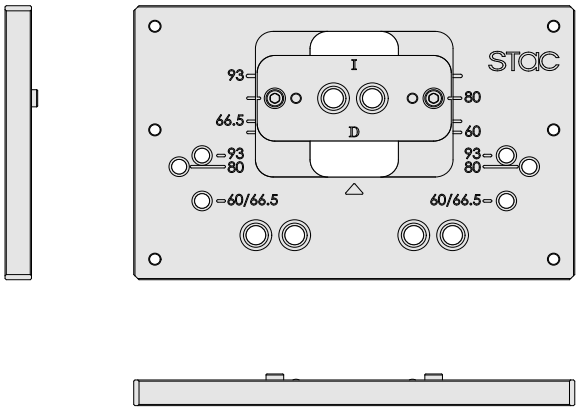

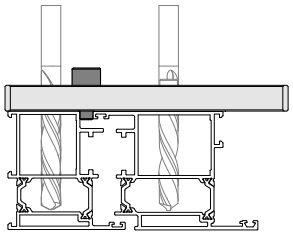

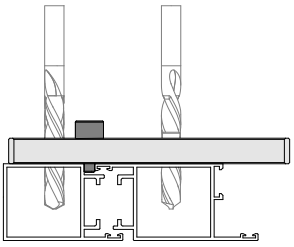

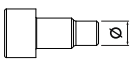

BOLTS WITH COPLANAR PROFILE TRACES

	Ø4 millimetres
	Ø5 millimetres
	Ø6 millimetres
	Ø8 millimetres

BOLTS ARE INCLUDED IN THE REFERENCE

HERCULES HINGE

C0170 | **

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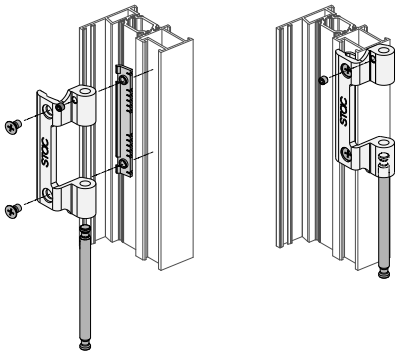
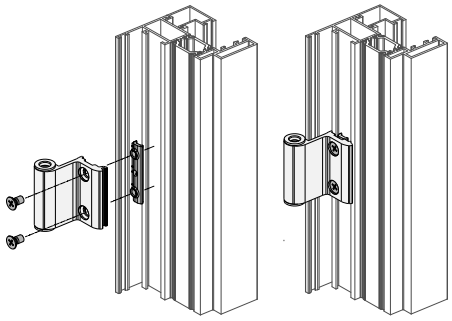
Hinge for Euro Groove windows profiles.

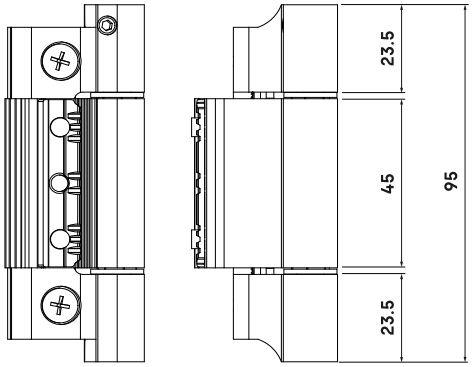
- Maximum load with two or more hinges is 90 kgs.
- The hinges do not keep opening hand, they are ambidextrous.
- Self-assembly hinge: the geometry of the plates and backplates allows direct mounting without grooves.
- The back plates incorporate a special ribbing in order to avoid looseness.
- On-site assembly is made easier by the set screw which allows the bolt to be removed.
- The installation is recommended when there are transversal stresses: sashes with width > 1200 mm height / width ratio

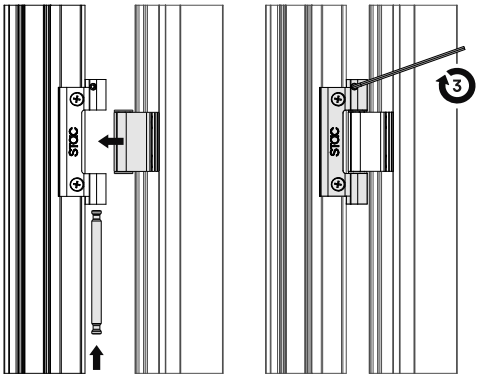


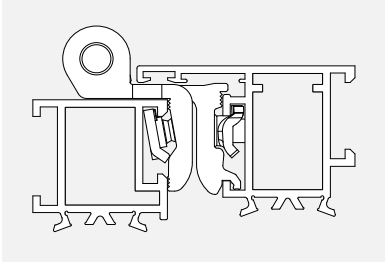
LEAF
FRAME PROFILE



LEAF
SASH PROFILE







Window hinges for Euro Groove profiles.

- Maximum load 60 kg with 2 hinges and 75 kg with 3 hinges.
- Supplied with zamak counterplates incorporating a set screw to prevent slippage.
- The bushing covers the pin so that it is insulated from the weather.
- Two-part reversible hinges that are supplied assembled for right-hand or left-hand installation.
- Disassembled hinges available (ref. B019X, ref. B019XR).

CARINA hinge with adjustment (ref. 019R)

- The eccentric pin adjustment allows horizontal adjustment of ± 0.4 mm and tightening of 0.2 mm.

CARINA HINGE
Zamak counterplates

CR4

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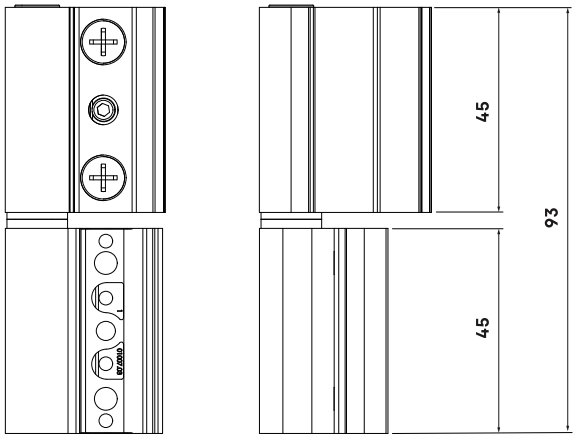

CARINA HINGE
Adjustment by means of eccentric pin

CR4

AL ZM IN PA

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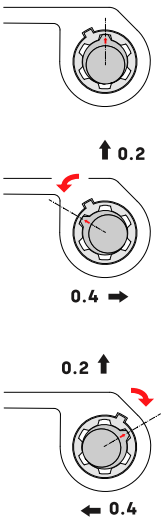

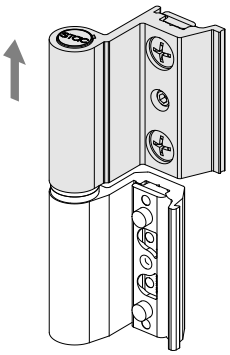

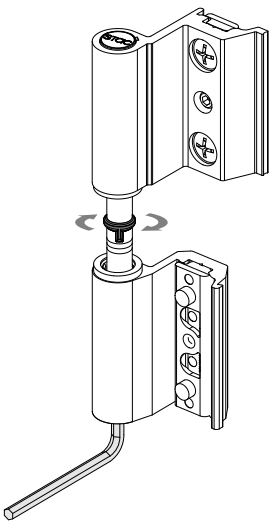

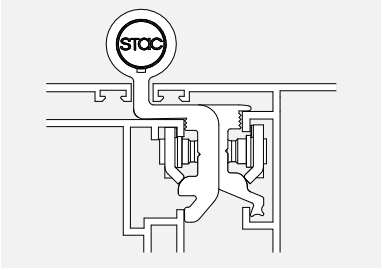

CARINA PLUS HINGE
Stainless steel counterplates

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CR4

AL IN PA

Window hinges for Euro Groove profiles.

- Maximum load 60 kg with 2 hinges and 75 kg with 3 hinges.
- Supplied with stainless steel anti-slip counterplates to double the gripping capacity and improve the pre-assembly of the hinges.
- The bushing covers the pin so that it is insulated from the weather.
- Two-part reversible hinges that are supplied assembled for right-hand or left-hand installation.
- Disassembled hinges available (ref. B023X, ref. B023XR).

CARINA PLUS hinge with adjustment (ref. 023R)

- The eccentric pin adjustment allows horizontal adjustment of ± 0.4 mm and tightening of 0.2 mm.

CARINA PLUS HINGE
Adjustment by means of eccentric pin

023	D / I	**	R

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CR4

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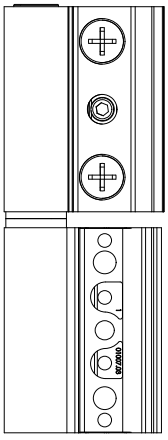

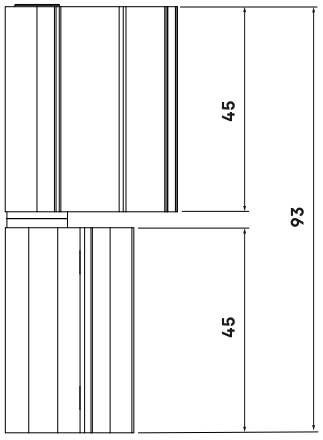

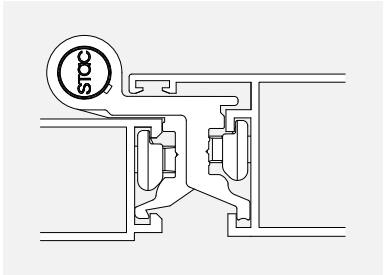

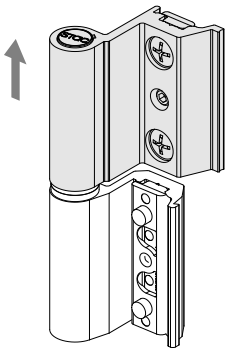

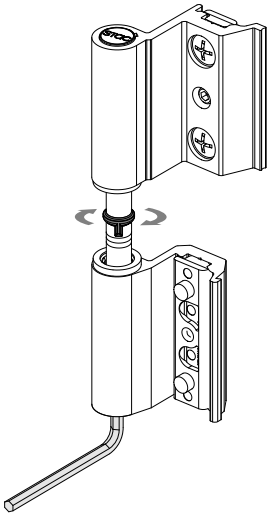

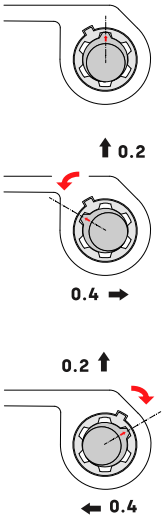

3rd sash window hinges for Euro Groove profiles.

- Maximum load 50 kg with 2 hinges and 60 kg with 3 hinges.
- Two-part reversible hinges that are supplied partially assembled for right-hand or left-hand installation.
- Self-assembly hinge: the geometry of the backplates and counterplates allows direct mounting without grooving.
- Compatibility with Euro Groove coplanar profiles.
- The counterplates have special serrations to prevent slipping.
- The bushing covers the pin so that it is insulated from the weather.

CARINA 3RD hinge with adjustment (ref. 014DR)

- The eccentric pin adjustment allows horizontal adjustment of ± 0.4 mm and tightening of 0.2 mm.

CARINA 3RD SASH HINGE
Stainless steel backplates

CE
CR6

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CARINA 3RD SASH HINGE
Adjustment by means of eccentric pin

CE
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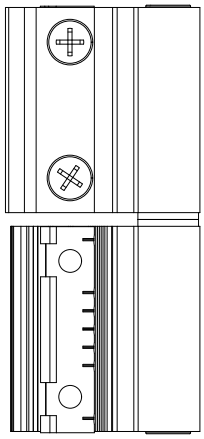

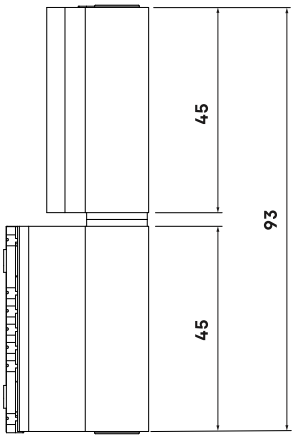

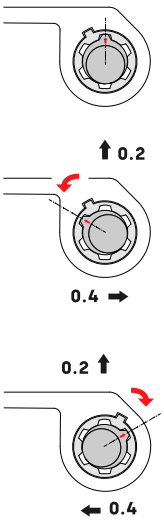

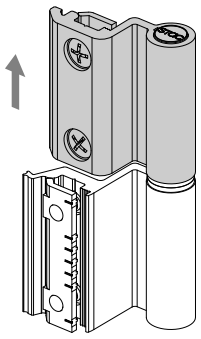

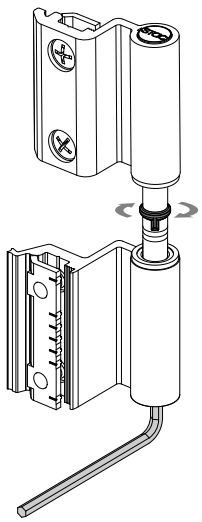

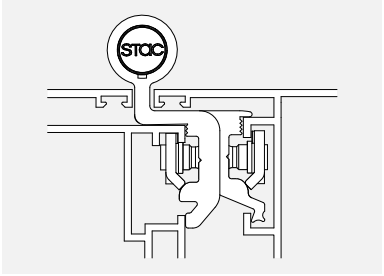

CARINA 3RD SASH HINGE
Centred Groove

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CC

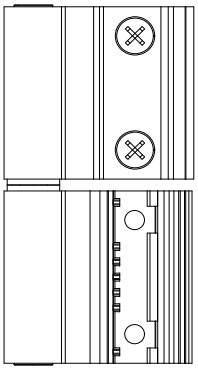
CR8

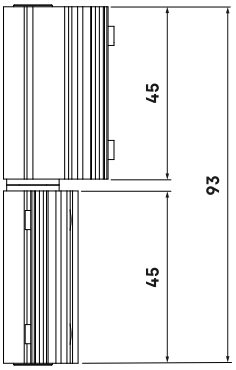
AL IN

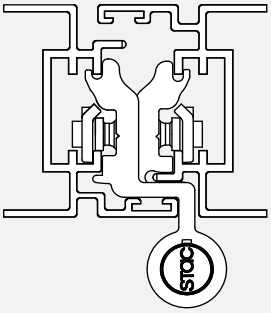
PA

3rd sash window hinge for Centred Groove profiles.

- Maximum load 50 kg with 2 hinges and 60 kg with 3 hinges.
- Self-assembly hinge: the geometry of the leaves and counterplates allows direct mounting without grooving.
- The counterplates have special serrations to prevent sagging.
- Supplied semi-assembled for installation in right-hand or left-hand openings.







4th sash hinges for windows with Euro Groove or Centred Groove profiles.

• Self-assembly hinges: the geometry of the leaves and counterplates allows direct assembly without grooving.

CARINA 2-leaf hinge (ref. 018)

• Valid for Euro Groove profiles.
• Adjustment: horizontal adjustment of ± 1 millimetre.
• Direct installation on the profile without the need for machining.
• Bushing made of thermoplastic resin (acetal) to avoid noise and wear.

CARINA 3-leaf hinge (ref. A0108)

• Valid for Centred Groove profiles.
• The counterplates have special serrations to avoid slippage.
• The bushing covers the bolt so that it is insulated from the weather.
• Three-body reversible hinge supplied semi-assembled for installation in right or left-hand openings.

CARINA 4TH SASH HINGE
Euro Groove

CE
CR7
AL IN
PA

018	D / I	**

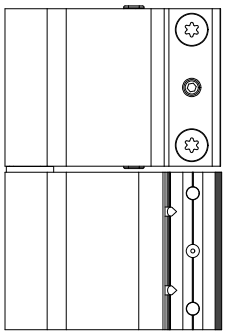
20

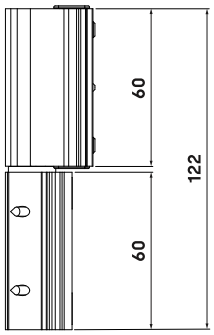
CARINA 4TH SASH HINGE
Centred Groove

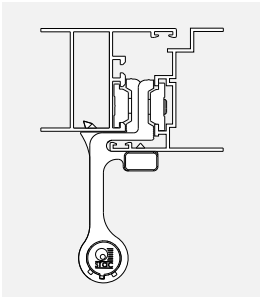
CC
CR9
AL IN
PA

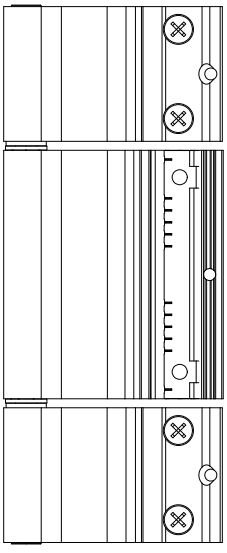
A0108	D / I	**

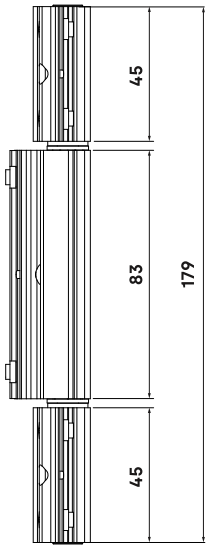
10

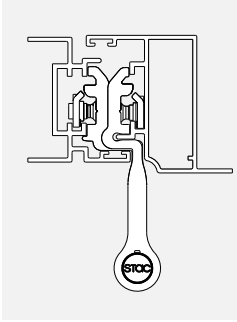


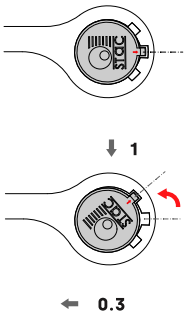


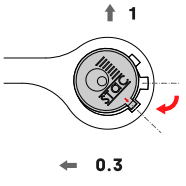


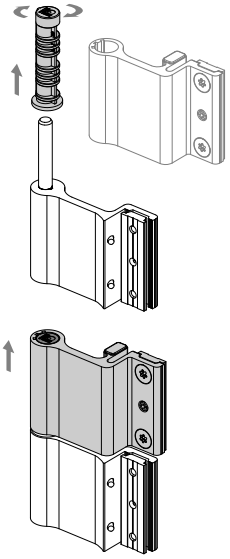












CARINA TAMIZ 3RD SASH HINGE
Centred Groove

A0133	D / I	**

40

CC

CR10

AL IN PA

3rd sash window hinges for Centred Groove profiles.

- Self-assembly hinges: the geometry of the backplates and counterplates allows direct assembly without grooving.
- The counterplates have special serrations to prevent the hinges from slipping.
- The hinges incorporate a third hole for a threaded screw to secure them to the profile.

CARINA TAMIZ 2-leaf hinge (ref. A0133)

- Maximum load 50 kg with 2 hinges and 60 kg with 3 hinges.

CARINA TAMIZ 3-leaf hinge (ref. A0135D)

- Maximum load 60 kg with 2 hinges and 80 kg with 3 hinges.

CARINA TAMIZ 3RD SASH HINGE
Centred Groove

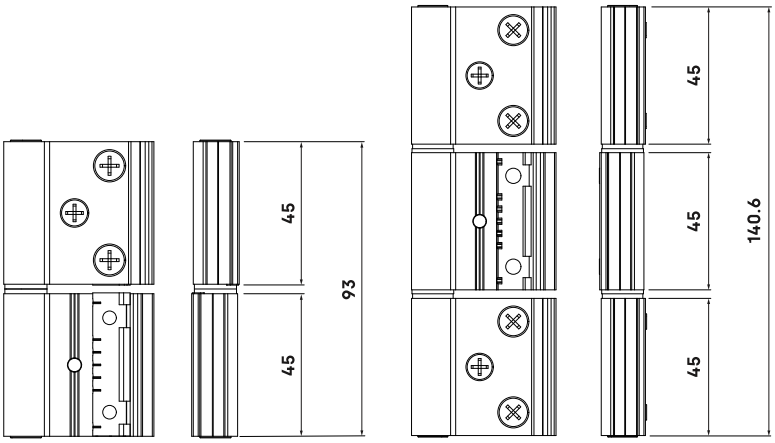
A0135D	**

-

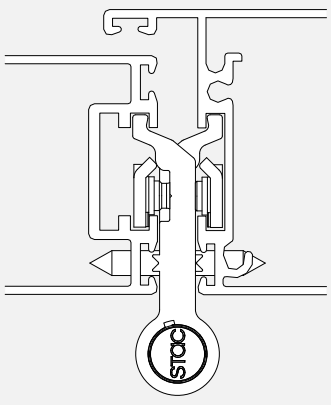
CC

CR10

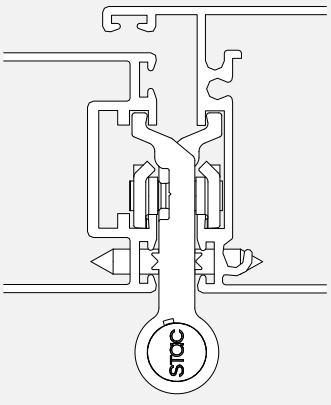
AL IN PA



A0133



A0135D



4th sash window hinge for Centred Groove profiles.

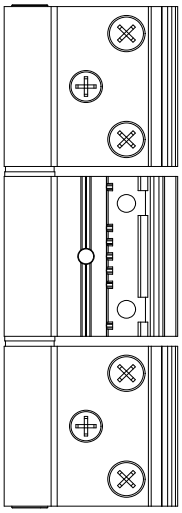
- Maximum loading: consult technical data sheet.
- Self-assembly hinge: the geometry of the backplates and counterplates allows direct mounting without grooving.
- The counterplates have special serrations to prevent slipping.
- Incorporate a third hole for a self-tapping screw to secure the hinge to the profile.

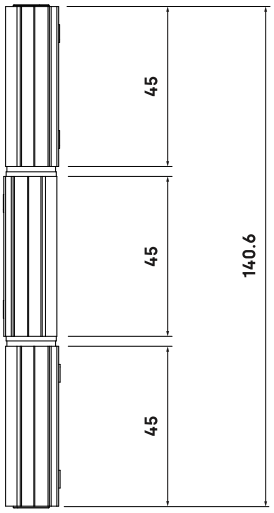
CARINA TAMIZ 4TH SASH HINGE
Centred Groove

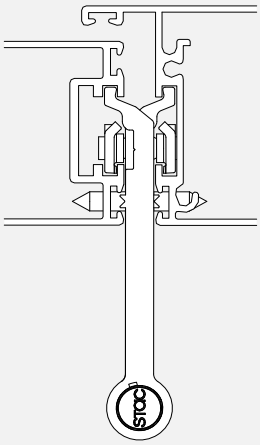
CC
CR10
AL IN
PA

A0134	D / I	**

30







NORMA ECO HINGE
Zamak counterplates

024	D / I	**

50

CR4

AL ZM IN PA

NORMA FAST HINGE
Stainless steel counterplates

025	D / I	**

50

CR4

AL IN PA

NORMA FAST HINGE
Stainless steel counterplates

P025D	**

6

CR4

AL IN PA

Window hinges for Euro Groove profiles.

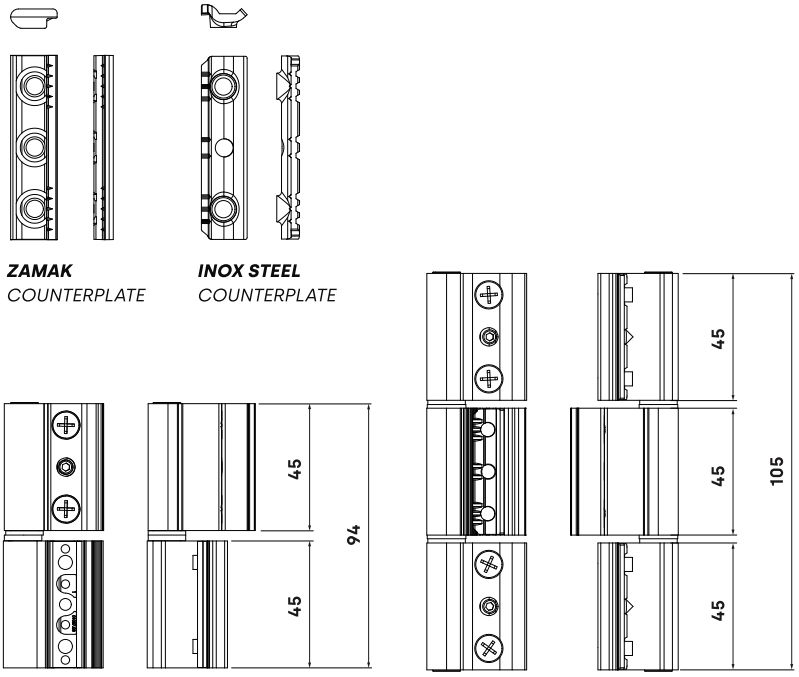
- Maximum load with two-leaf hinges: 60 kg with 2 units ~ 75 kg with 3 units.
- Maximum load with three-leaf hinges: 75 kg with 2 units ~ 90 kg with 3 units.
- The hinge has an eccentric pin that allows a ± 0.4 mm horizontal adjustment and 0.2mm tightening adjustment.
- The hinge frame leaf is universal to fit most of the frame Euro Groove window profiles.
- Self-assembly hinge: the geometry of the leaves and counterplates allows direct mounting without grooving.

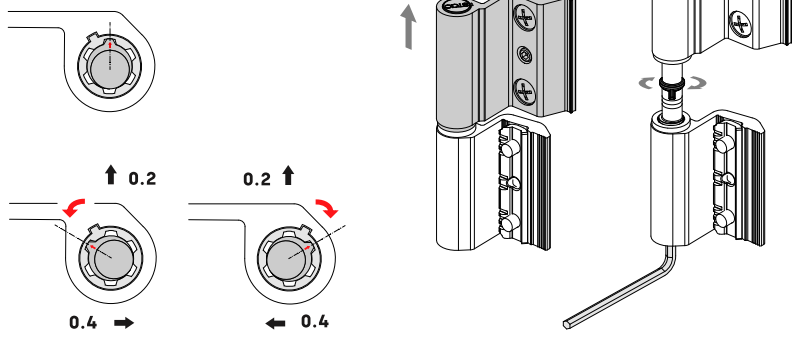
NORMA ECO hinge (ref. 024)

- Supplied with zamak counterplates.

NORMA FAST hinges (ref. 025, ref. P025D)

- Supplied with stainless steel counterplates.
- Stainless steel anti-slip counterplates doubles the gripping capacity and improves the pre-assembly of the hinges.





Hinges for profiles with conventional 4000 and 6000 series.

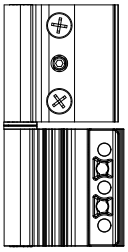
- Maximum load: 60 kg with 2 hinges and 75 kg with 3 hinges.
- The hinge has an eccentric pin that allows a ± 0.5 mm lateral adjustment.
- Supplied with self-assembly zamak counterplates with fixing to the profile by means of grub screws.

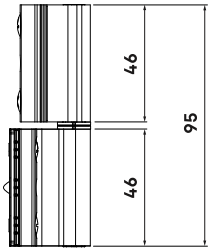
4020 HINGE
Zamak counterplates

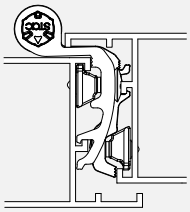
4020
CR11
AL ZM
IN PA

026	D / I	**

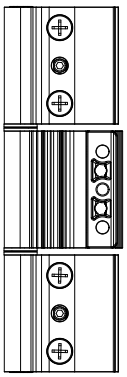
40

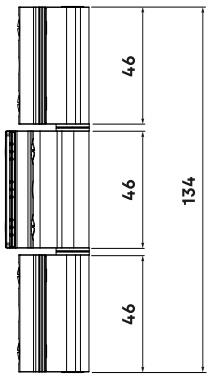


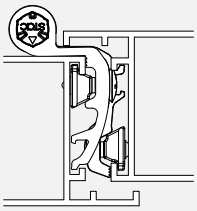




SERIE 4000







SERIE 6000

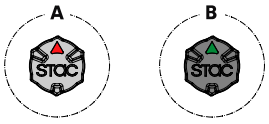
4020 HINGE
Zamak counterplates

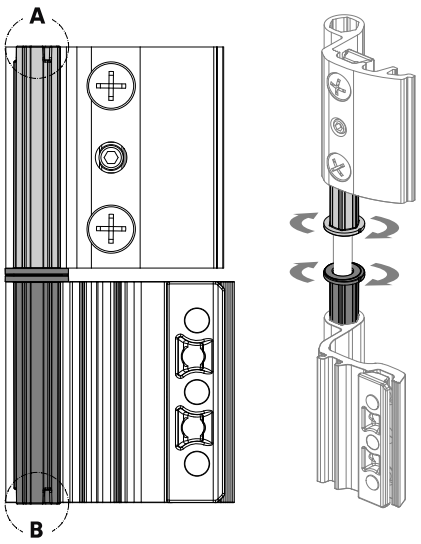
4020
CR12
AL ZM
IN PA

029	D / I	**

25

BUSHING A	BUSHING B	HORIZONTAL	TIGHTENING	CHAMBER
		+0.5 mm	+0.5 mm	15.7 mm
		-	+0.5 mm	15.2 mm
		-0.5 mm	-	14.7 mm
		+1 mm	+0.5 mm	16.2 mm
		-1 mm	+0.5 mm	14.2 mm





PEGASO HINGE

100001 **

50

ZM

IN

Hardware specifically developed for counter windows.

PEGASO hinge (ref. 100001)

• Ambidextrous hinge with maximum load per sash of 50 kg with two hinges.

PEGASO fastener (ref. 100002)

• Pressure lock with multiple positioning options in order to adapt to counter windows by means of symmetrical spacers that allow a perfect fit in its installation.

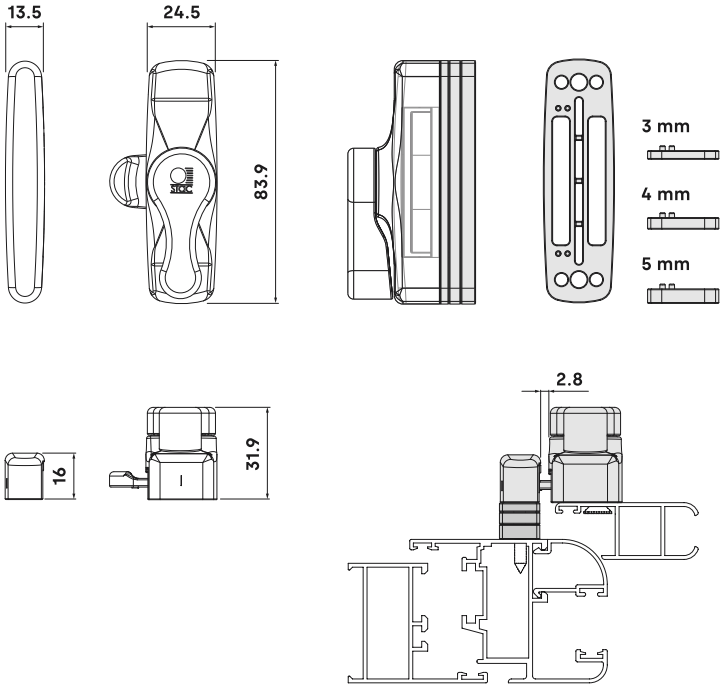
PEGASO LOCK

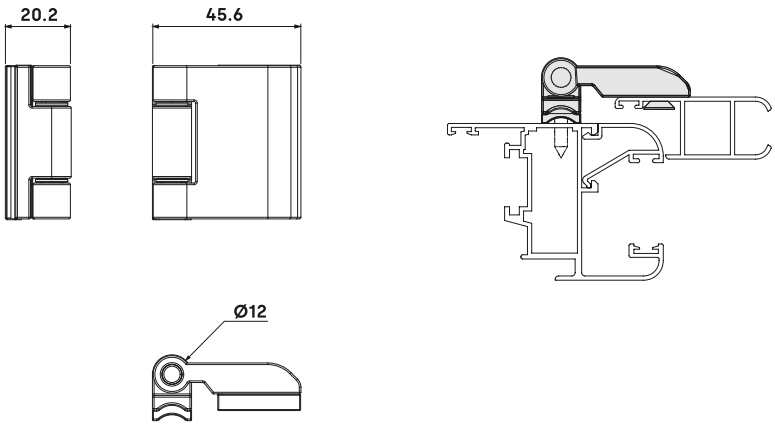
100002 **

25

ZM

IN





02 WINDOW HANDLES

44 DELTA handle

50 MINIMALIST handle

53 PVC MINIMALIST handle

54 SIRIUS handle

60 POLARIS handle

64 POLARIS detachable handle

66 LYRA handle

67 ALTAIR handle

70 KAIROX handle

76 Handle accessories

Quick installation

No visible screws

Corrosion resistance

RAL colour chart

DELTA range

The design must give priority to the usefulness of the product, so we have focused on developing a mechanism that is as integrated as possible in the current aluminium window and door frames, avoiding those features that would take us away from this objective.

» Compatibility with current systems

- Windows with side-hung or tilt & turn openings.
- Sliding windows with multipoint lock systems.
- Windows with Euro Groove or Groove 16 profiles.
- Single and double-sash doors with both inward and outward opening.

» Personalisation through colour

Projects find a unique look through new colours or finishes.

The range of handles and cremones is fully customisable and is available in white or black lacquer as well as in the colours that make up the RAL colour chart, a colour matching system that defines colours for paints, plastics and coatings.

» Product aesthetic

The aesthetic of a product is an integral part of its usefulness, as products used on a daily basis tend to have an indirect effect on people and their well-being.

We have placed special emphasis on the ergonomics of use by eliminating any obtrusive edges for the user.

» Corrosion resistance

The parts that make up the handles have been treated to guarantee resistance to corrosion, exceeding 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

According to the parameters described in the standard, a grade 4 classification has been obtained, which allows the possibility of using the handles in environments with severe climatic conditions.

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- Supplied with transmission spindle of 7 x 34 mm (for outward opening ref. 7014).
- The integrated cover shield conceals the transmission mechanism and makes it easier to install the handle by means of a simple "clip".

DELTA PRO HANDLE
Inward opening (7 x 26)

7015 **

25

AL ZM IN PM

7

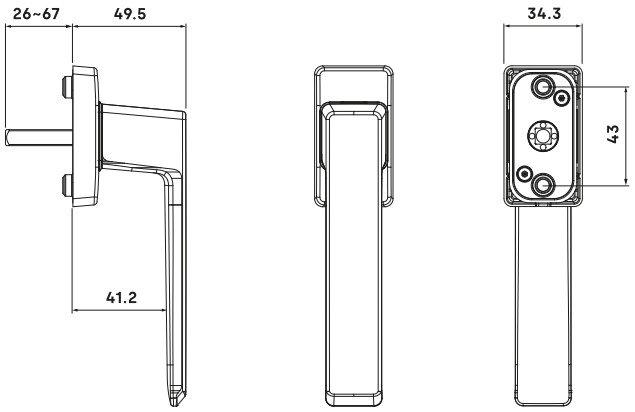
DELTA PRO HANDLE
Outward opening (7 x 67)

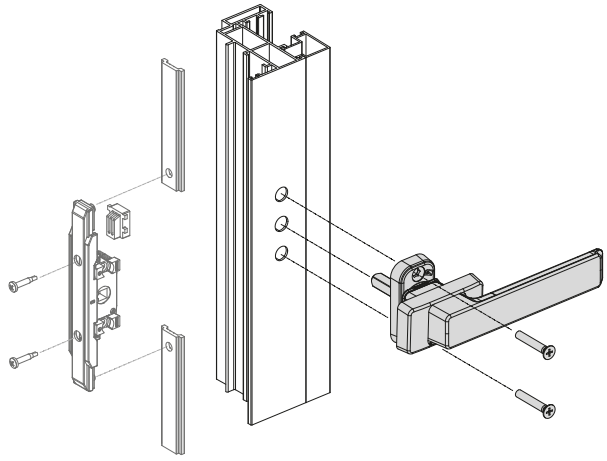
7014 **

25

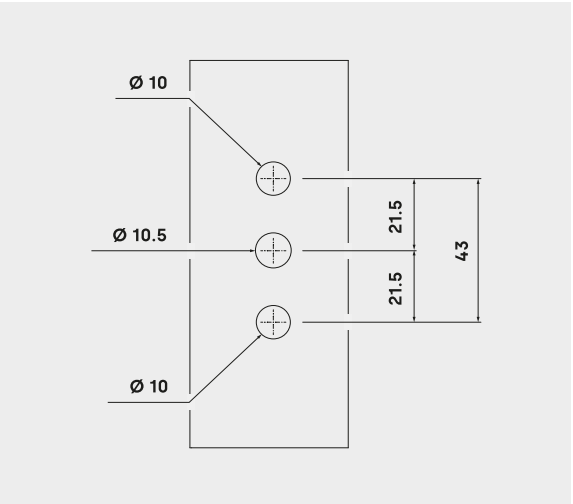
AL ZM IN PM

7





PROFILE MACHINING



DELTA HANDLE
Inward opening (7 x 26)

7007 | **

25

AL ZM
AC PM

7

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- Supplied with transmission spindle of 7 x 26 mm (for outward opening ref. 7008).
- The handles incorporate a caged transmission for the handle scutcheon and reinforced with a zamak support that makes it more resistant.
- Fixing kits for double handle installation and outward opening.

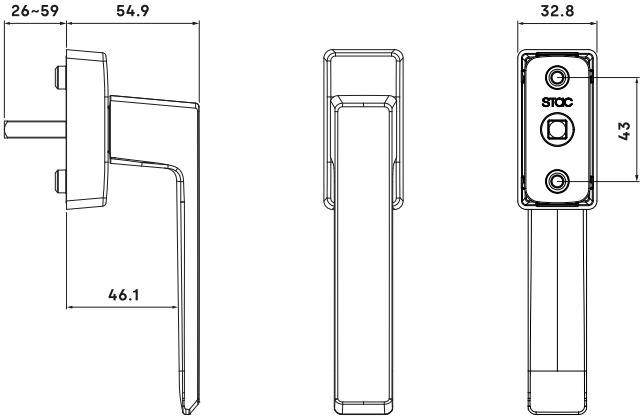
DELTA HANDLE
Outward opening (7 x 59)

7008 | **

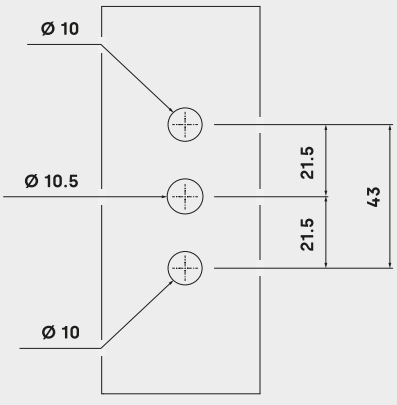
25

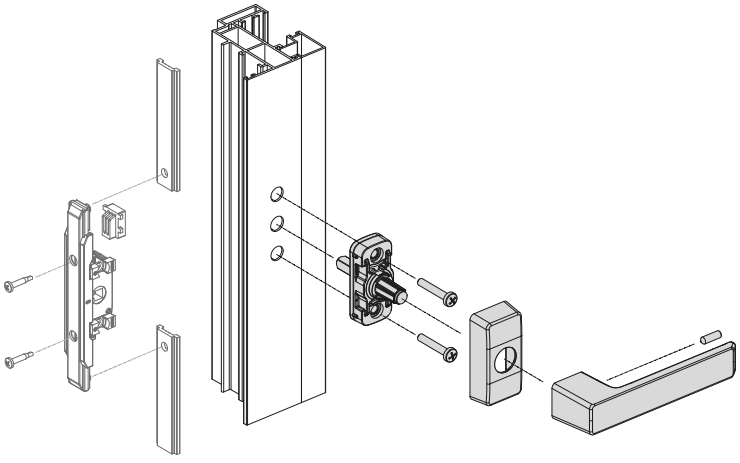
AL ZM
AC PA

7



PROFILE MACHINING





Offset handles for sliding window with 7 mm transmission spindle for multipoint lock.

- Valid for right-hand opening (left-hand opening ref. 7025).
- Allows different fixing options to the profile.

DELTA OFFSET HANDLE
Right-hand opening

7024 **

25

AL ZM AC PM

7

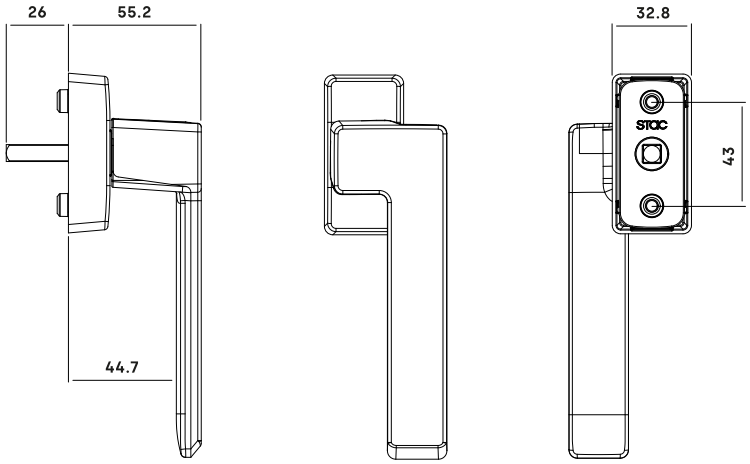
DELTA OFFSET HANDLE
Left-hand opening

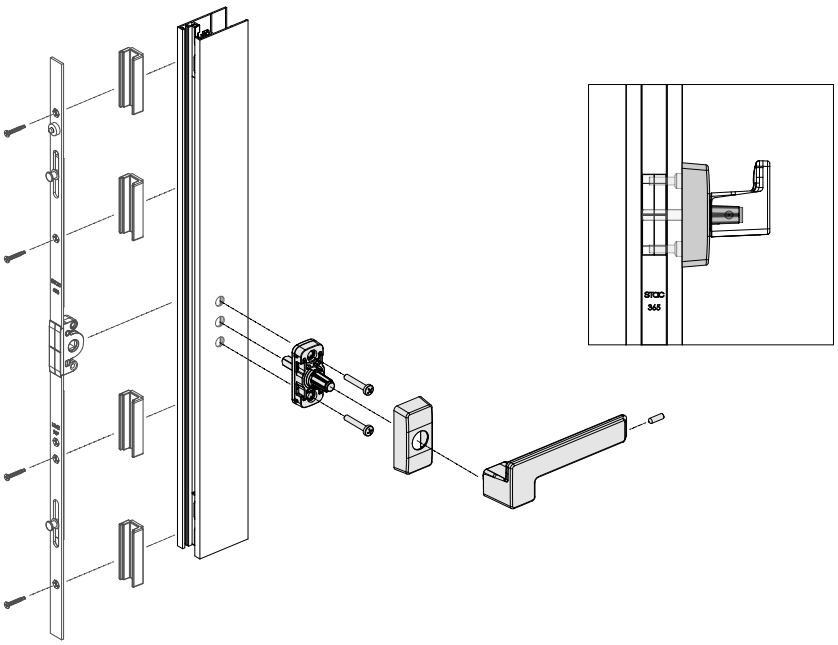
7025 **

25

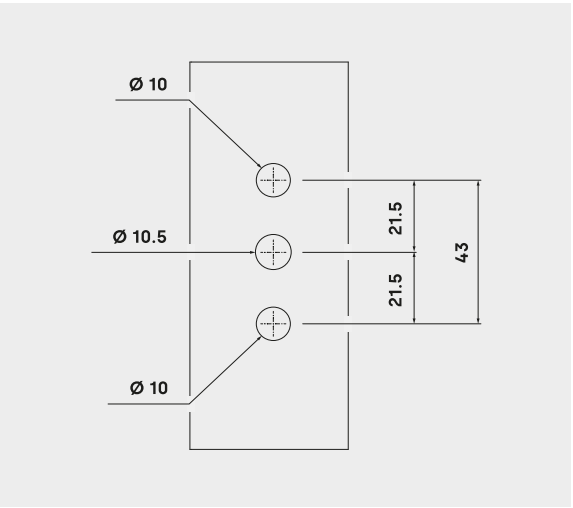
AL ZM AC PM

7





PROFILE MACHINING



DELTA REINFORCED HANDLE

7009 | **

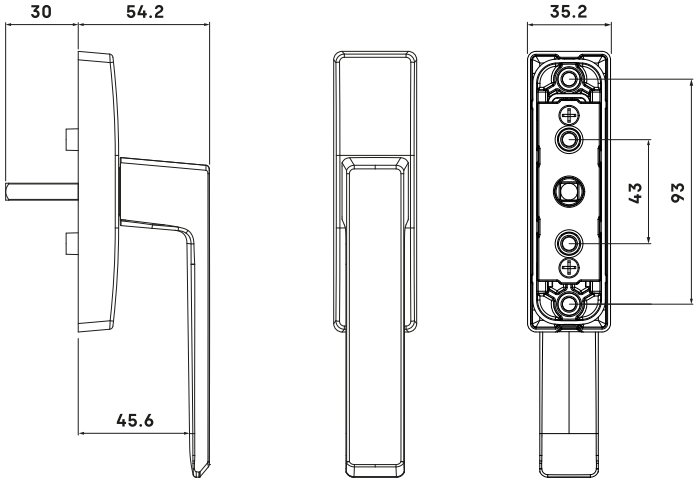
15

AL ZM AC

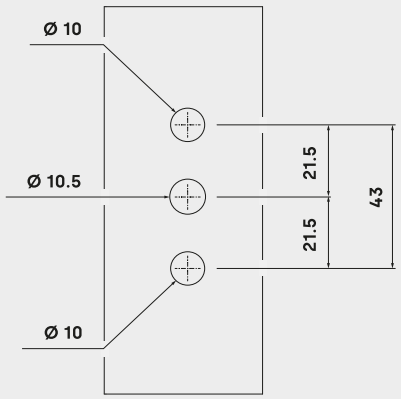
7

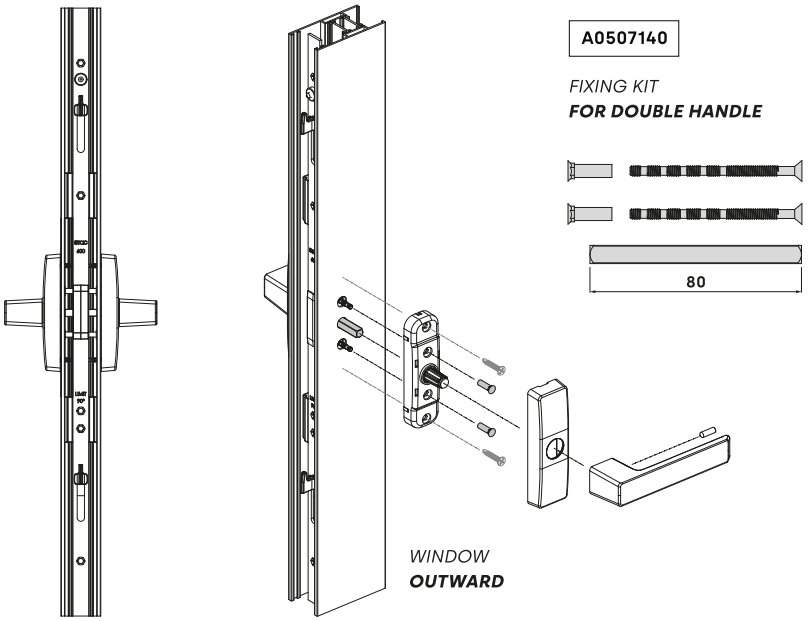
Reinforced handle for side-hung or tilt & turn hinged system and sliding windows.

- The handle is compatible with transmission boxes and multipoint lock.
- Double fixing the handle to both the frame and the hardware prevents distortion, giving the locking system as a whole great stability and reinforcement.
- Fixing to the profile is carried out by means of self-tapping screws without the need for additional machining.
- A security screw kit is available for double handle configurations that prevent the hardware from being manipulated from the outside (ref. A0507140).



PROFILE MACHINING





Reinforced offset handles for sliding windows with multipoint lock.

• Double fixing the handle to both the frame and the hardware prevents distortion, giving the locking system as a whole great stability and reinforcement.
• Fixing to the profile is carried out by means of self-tapping screws without the need for additional machining.
• A security screw kit is available for double handle configurations that prevent the hardware from being manipulated from the outside (ref. A0507140).

DELTA REINFORCED OFFSET HANDLE
Right-hand opening

7016 **

15

AL ZM AC

7

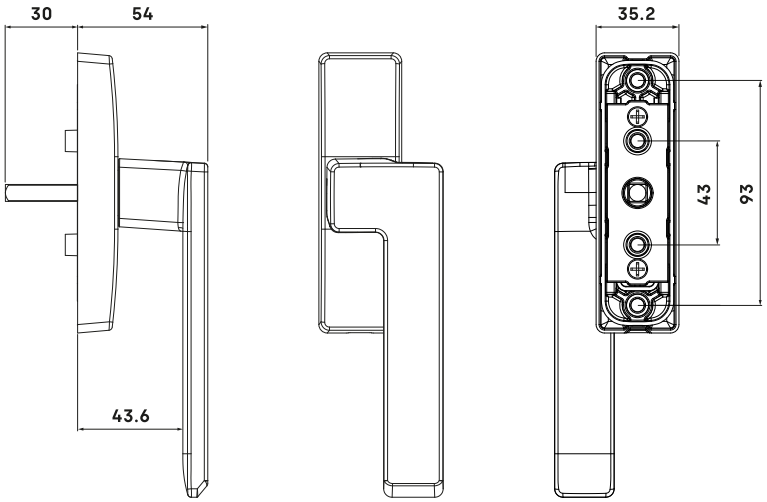
DELTA REINFORCED OFFSET HANDLE
Left-hand opening

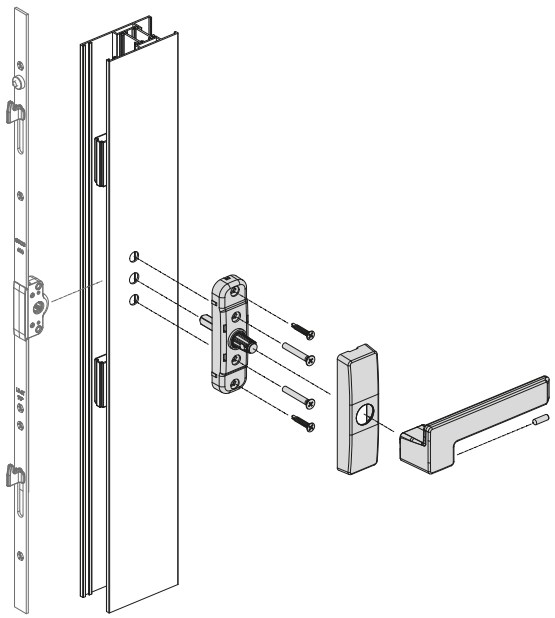
7017 **

15

AL ZM AC

7





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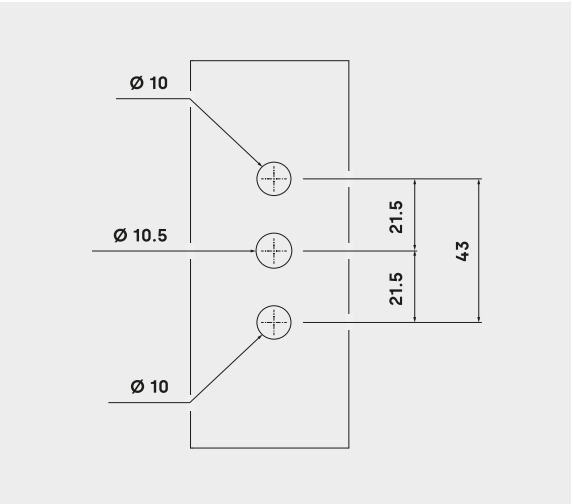


minimalist
delta & **gamma**

clic !

Direct installation

Highest durability

RAL colours

Tested robustness

MINIMALIST range

For people who crave a sense of creativity, quality and style in their spaces, STAC introduces the minimalist line of door and window handles to offer a complete range of technical solutions of the highest quality.

» Minimalist

During the design process of the minimalist handles we have done away with all unnecessary elements, developing a hidden mechanism that only exposes the handle.

In this way, we offer technical solutions of great robustness and reliability throughout their service life in accordance with the strictest quality standards.

» Complete range

Our range is compatible with multiple opening systems, with particular emphasis on:

- Windows with side-hung or tilt & turn opening.
- Sliding windows with multipoint lock systems.
- Windows with Euro Groove or Groove 16 profiles.
- Single and double-sash doors with both inward and outward opening.

» Direct installation

The handle is supplied with a positioning and retention mechanism that allows it to be installed directly with a **"click"**, and is just as easy and practical to remove.

» Full range of finishes

There are ever more projects which try to stand out through their used of colour and finishes; the hardware can be painted in the same colour as the aluminium carpentry. Minimalist range is available in both black and white lacquer and the colours that make up the RAL chart.

» Corrosion resistance

The parts that make up the handles have been treated to guarantee resistance to corrosion, exceeding 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

According to the parameters described in the standard, a grade 4 classification has been obtained, which allows the possibility of using the handles in environments with severe climatic conditions.

Minimalist handles for side-hung or tilt & turn hinged system.

- Handle with concealed mechanism without cover.
- They are supplied with 8 millimetres transmission spindle.
- The guide system minimises the clearances.
- Curved aesthetics available: GAMMA minimalist handle (ref. 7703).
- Available in individual kits (ref. KA7701, ref. K7703) including minimalist handle with 8 millimetres backset gearbox (ref. A770001).

DELTA MINIMALIST HANDLE

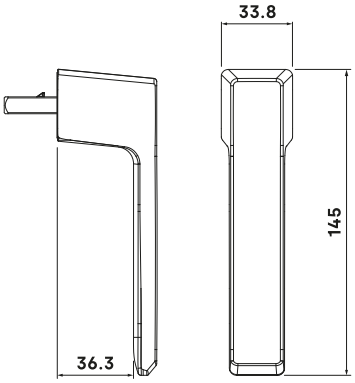
8 mm transmission spindle

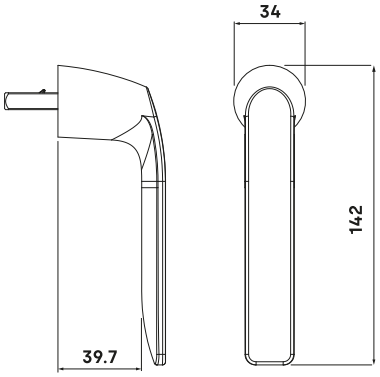
AL AC PM

8

A7701	**

25





GAMMA MINIMALIST HANDLE

8 mm transmission spindle

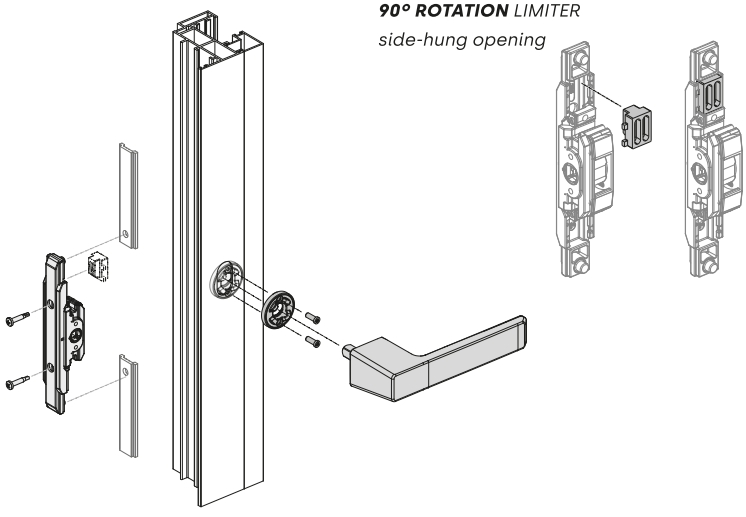
AL AC PM

8

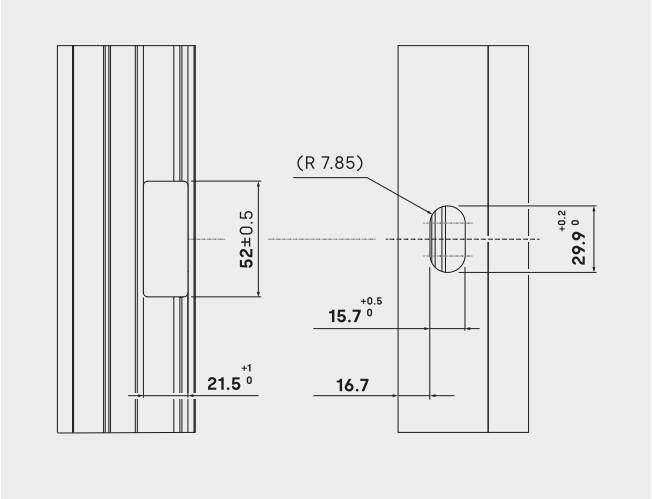
7703	**

25

90° ROTATION *LIMITER*
side-hung opening



PROFILE MACHINING



DELTA MINIMALIST HANDLE

Kit with minimalist handle and gearbox

KA7701 **

1

AL IN PM

8

Kits with minimalist handle and gearbox ref. A770001.

- Available with DELTA (ref. KA7701) or GAMMA (ref. K7703) minimalist handle.
- Handle without decorative escutcheon and concealed transmission mechanism.
- The reference is supplied in an individual box with 8 millimetre transmission spinlde.
- The guiding system minimises play.

GAMMA MINIMALIST HANDLE

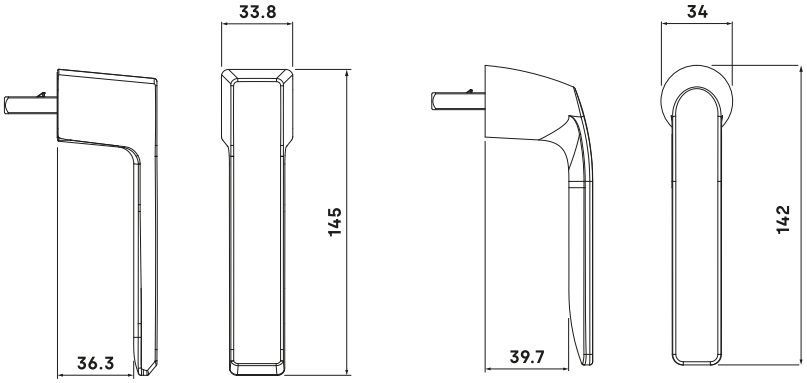
Kit with minimalist handle and gearbox

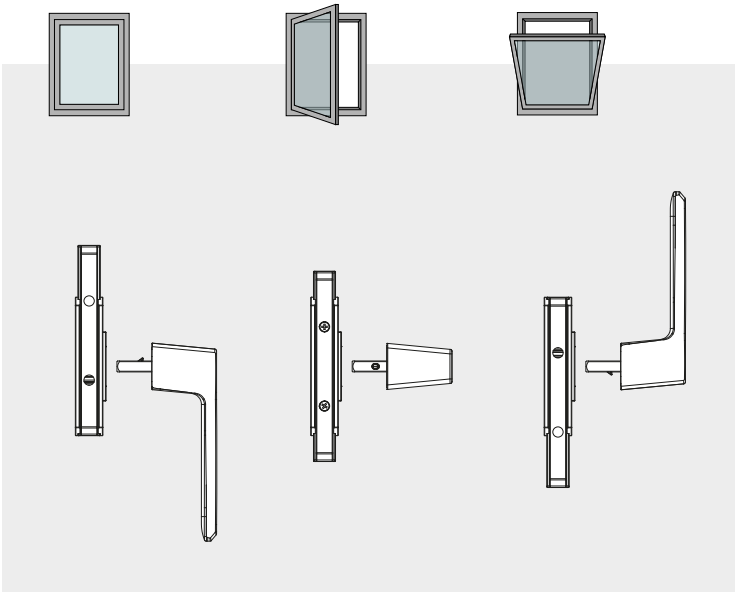
K7703 **

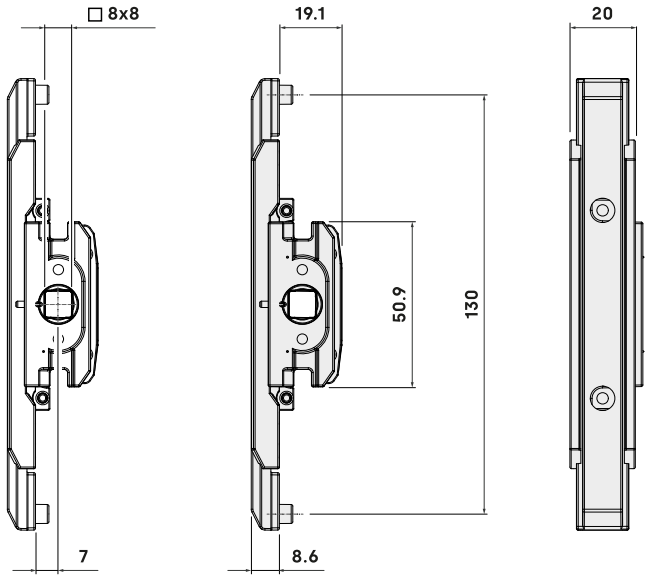
1

AL AC PM

8







Minimalist handles without embellishing cover and concealed drive mechanism, suitable for PVC systems with flanges.

- Available: DELTA (ref. A7731) and GAMMA (ref. 7733) minimalist handles.
- Compatible with hinged systems, side-hung, tilt & turn and bottom-hung openings in PVC.
- Installation on PVC groove profiles with flanges (minimum 23 mm).
- Minimalist handles are adaptable to any backset size of multipoint locks.
- The indexing mechanism allows the handle positions to be marked every 90º and the guiding system minimises possible play.

DELTA MINIMALIST HANDLE
PVC systems

AL ZM AC IN

A7731	**

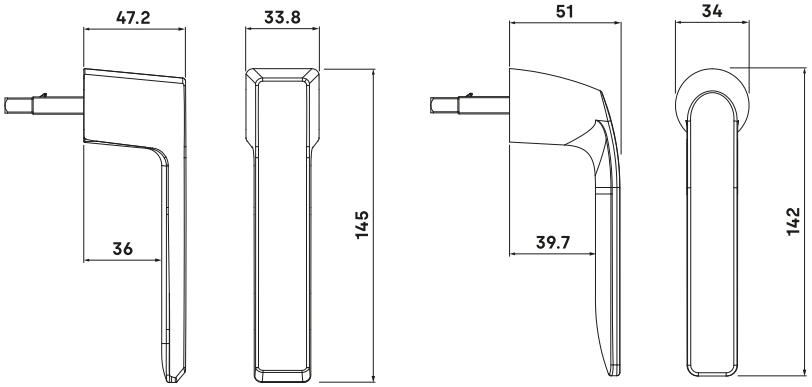
25

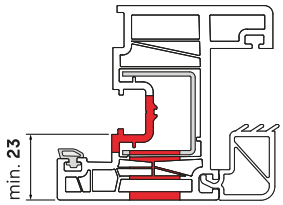
GAMMA MINIMALIST HANDLE
PVC systems

AL ZM AC IN

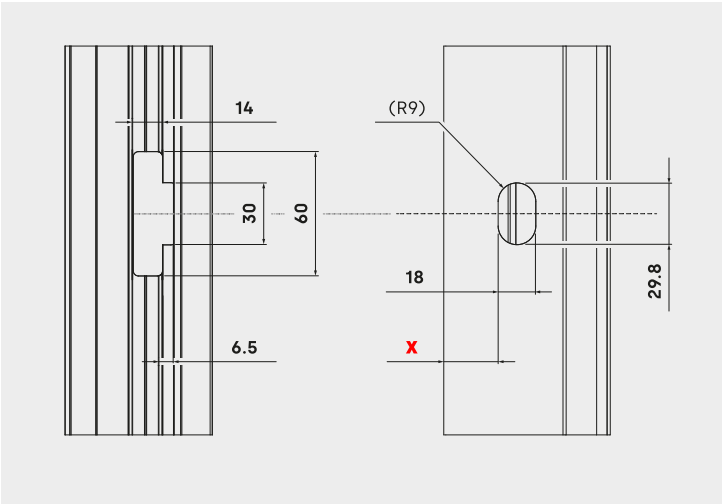
7733	**

25





PROFILE MACHINING



MACHINING DIMENSION

MULTIPOINT LOCK WITH 8 BACKSET	X = 26 mm
MULTIPOINT LOCK WITH 15 BACKSET	X = 19 mm

SIRIUS PRO HANDLE
Inward opening

V5771 | **

25

AL ZM IN PM

7

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- They are supplied with spindle of 7 x 34 mm (for outward opening ref. V5775).
- The integrated cover shield conceals the transmission mechanism and makes it easier to install the handle by means of a simple "clip".

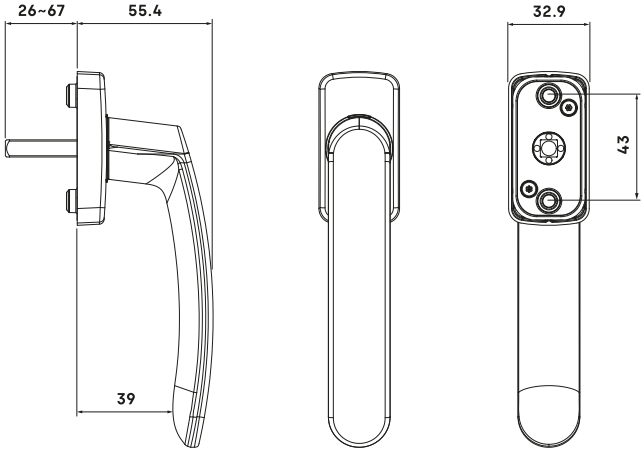
SIRIUS PRO HANDLE
Outward opening

V5775 | **

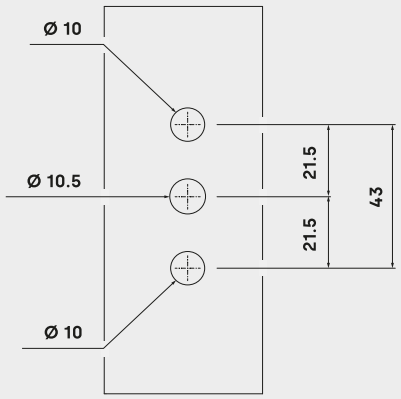
25

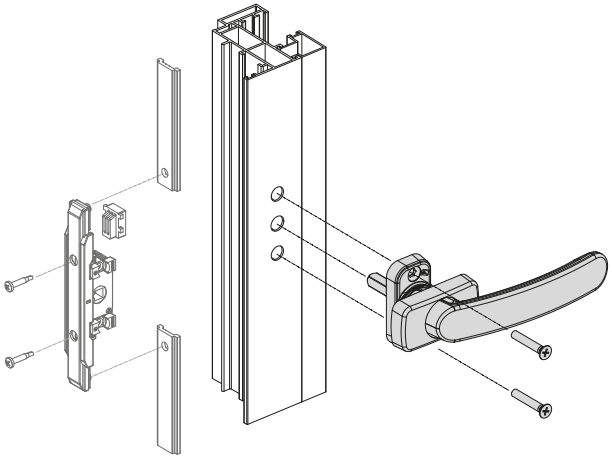
AL ZM IN PM

7



PROFILE MACHINIG





Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- Supplied with transmission spindle of 7 x 26 mm (for inward opening ref. A0775).
- The handles incorporate a caged transmission for the handle scutcheon and reinforced with a zamak support that makes it more resistant.

SIRIUS HANDLE
Inward opening

A0771	**

25

AL ZM AC PM

7

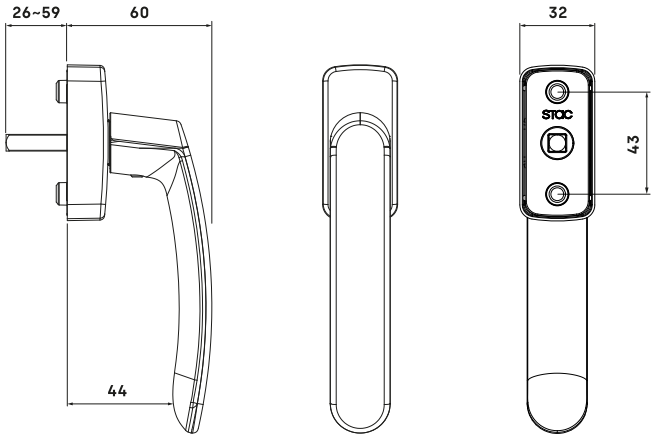
SIRIUS HANDLE
Outward opening

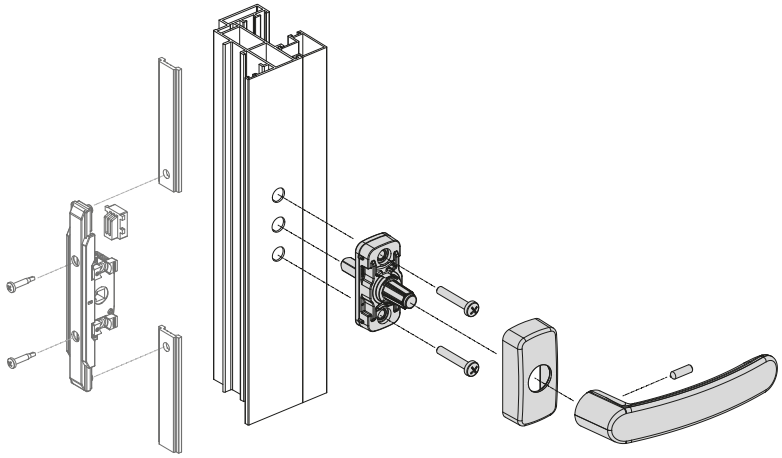
A0775	**

25

AL ZM AC PM

7





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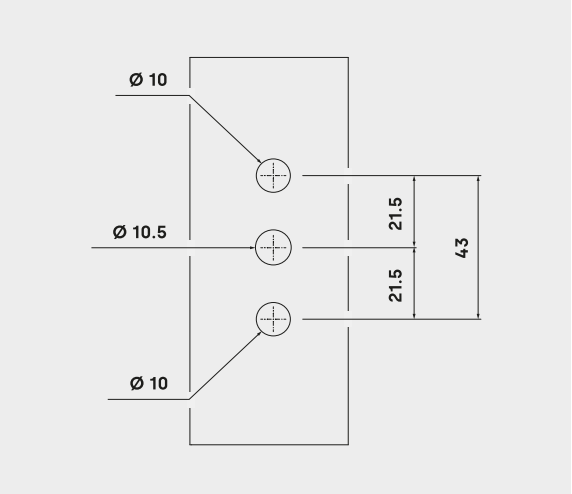


SIRIUS HANDLE
Fixing with self-tapping screws

R0771 **

25

AL ZM AC PM

7

Handles for side-hung or tilt & turn hinged system with gearbox and sliding window with multipoint lock.

• Fixing system using self-tapping screws or metric nuts that allow the installation of a double handle (inside and outside window area).
• In sliding windows, the inside handle can be fixed to the multipoint lock while the outside handle can be fixed by means of self-tapping screws or metric nuts.
• The scutcheon incorporates a caged drive mechanism reinforced with a zamak bracket to make the device stronger.

Fixing with self-tapping screws (ref. R0771)

• Supplied with fixing base cover (self-tapping screws not included)

Fixing with metric nuts (ref. S0771)

• Supplied with base cover and fixing kit (DIN 7985 M5 x 20 mm screws and M5 metric nuts).

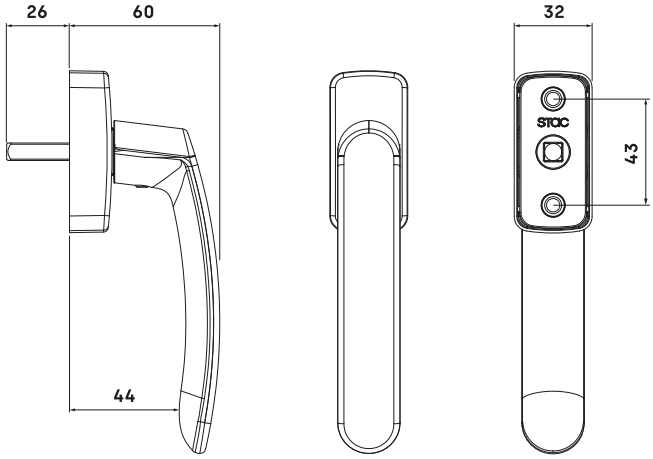
SIRIUS HANDLE
Fixing with metric nuts

S0771 **

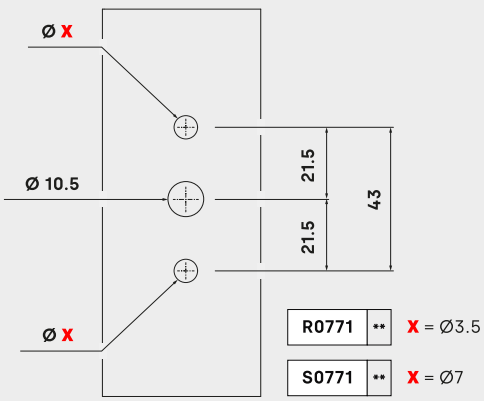
25

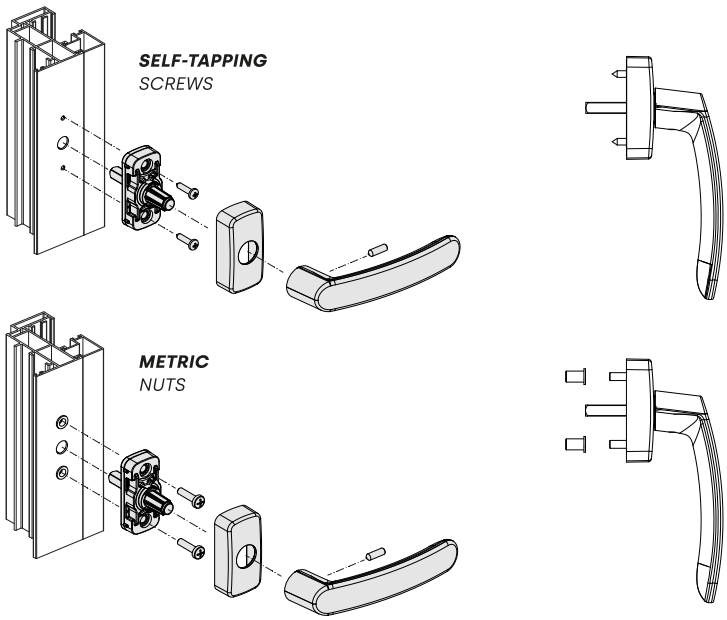
AL ZM AC PM

7



PROFILE MACHINING





Offset handles for sliding windows with multipoint lock.

- Valid for right-hand opening (left-hand opening ref. B0778).
- Compatible with both inward and outward configurations.
- The handles are suitable for the insertion of threaded rivets for fixing to the profile by means of millimetre-threaded screws.

SIRIUS OFFSET HANDLE
Right-hand opening

B0777 **

25

AL ZM AC PM

7

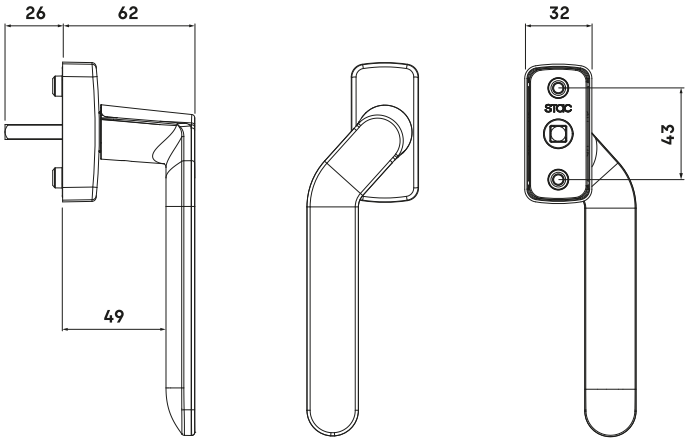
SIRIUS OFFSET HANDLE
Left-hand opening

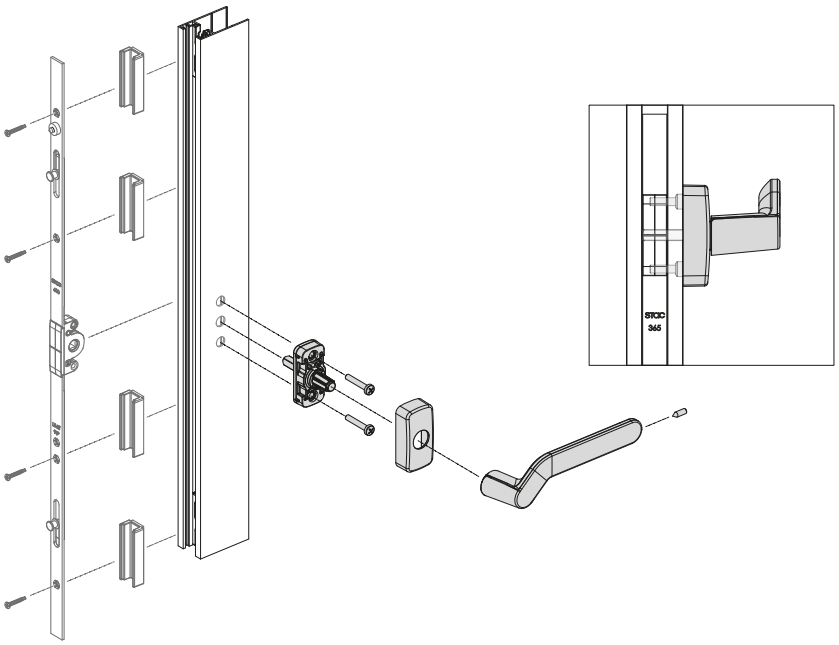
B0778 **

25

AL ZM AC PM

7





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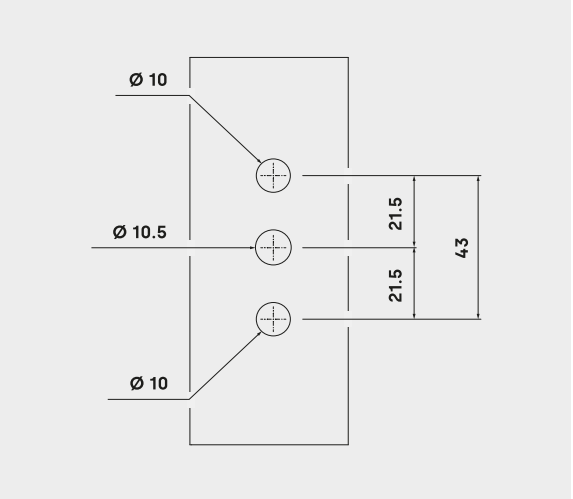


SIRIUS OFFSET HANDLE
Fixing with self-tapping screws

R0777 | **

25

AL ZN AC PN

7

Offset handles for sliding windows with multipoint lock.

- Valid for right-hand opening (left-hand opening ref. R0778, ref. S0778).
- The inward handle can be fixed to the multipoint lock while the outward handle can be fixed by means of self-tapping screws or metric nuts.
- The scutcheon incorporates a caged drive mechanism reinforced with a zamak bracket to make the device stronger.

Fixing with self-tapping screws (ref. R0777)

- Supplied with fixing base cover (DIN 7981 self-tapping screws not included)

Fixing with metric nuts (ref. S0777)

- Supplied with base cover and fixing kit (DIN 7985 M5 x 20 mm screws and M5 metric nuts).

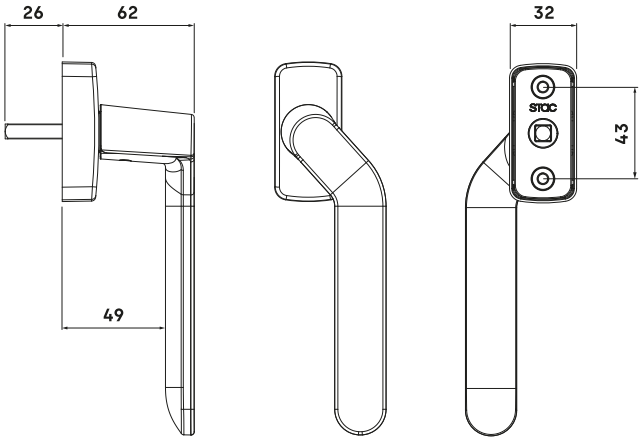
SIRIUS OFFSET HANDLE
Fixing with metric nuts

S0777 | **

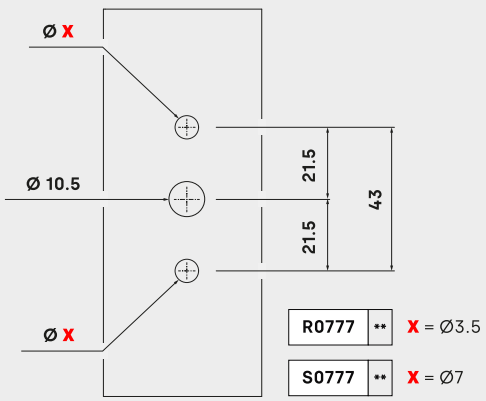
25

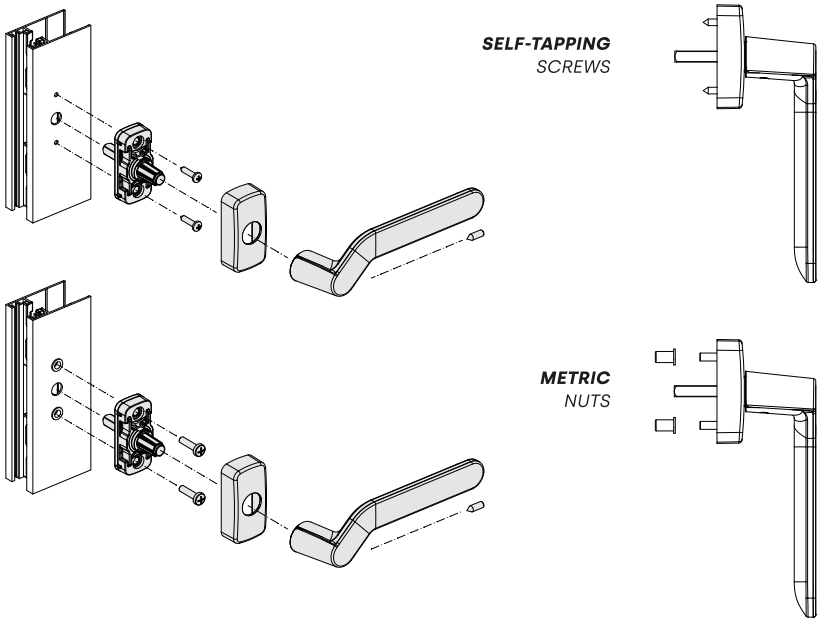
AL ZN AC PN

7



PROFILE MACHINING





Offset handles for sliding windows with multipoint lock.

• Valid for left-hand opening (right-hand opening refs. R0777, ref. S0777).
• The inward handle can be fixed to the multipoint lock while the outward handle can be fixed by means of self-tapping screws or metric nuts.
• The scutcheon incorporates a caged drive mechanism reinforced with a zamak bracket to make the device stronger.

Fixing with self-tapping screws (ref. R0778)

• Supplied with fixing base cover (DIN 7981 self-tapping screws not included)

Fixing with metric nuts (ref. S0778)

• Supplied with base cover and fixing kit (DIN 7985 M5 x 20 mm screws and M5 metric nuts).

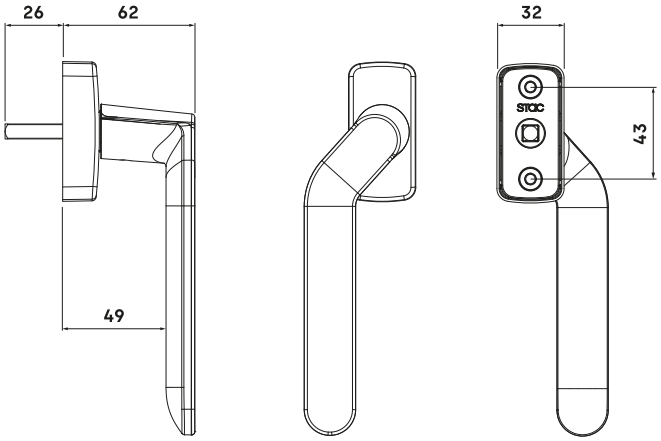
SIRIUS OFFSET HANDLE
Fixing with self-tapping screws

R0778 **

25

AL ZM AC PM

7



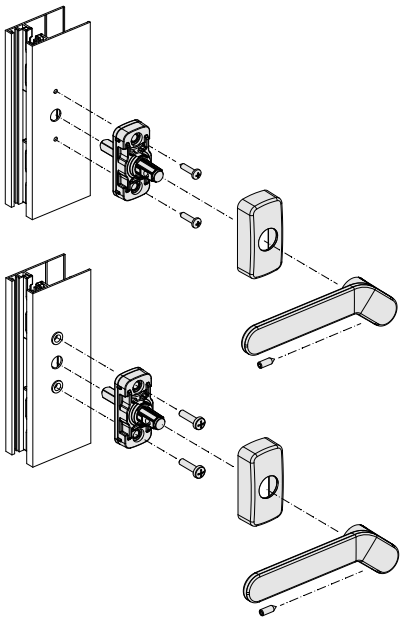
SIRIUS OFFSET HANDLE
Fixing with metric nuts

S0778 **

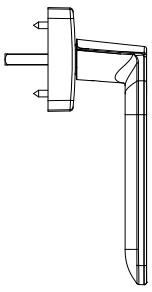
25

AL ZM AC PM

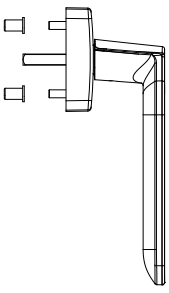
7



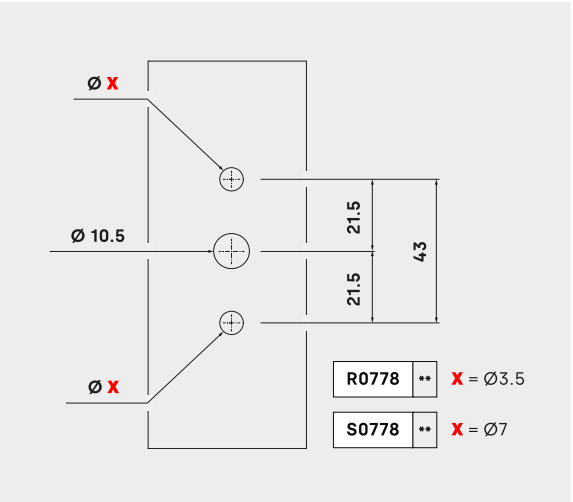
SELF-TAPPING
SCREWS



METRIC
NUTS



PROFILE MACHINING



POLARIS HANDLE
Inward opening

S5771 **

25

82 81

AL ZM
AC PN

7

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- Supplied with transmission spindle of 7 x 29 mm.
- Available in different spindle lengths for outward opening: 7 x 64 mm (ref. S5775) or 7 x 75 mm (ref. T5775).
- Integrated rotating scutcheon for reduced installation time
- Audible indication of handle positions.

POLARIS HANDLE
Outward opening

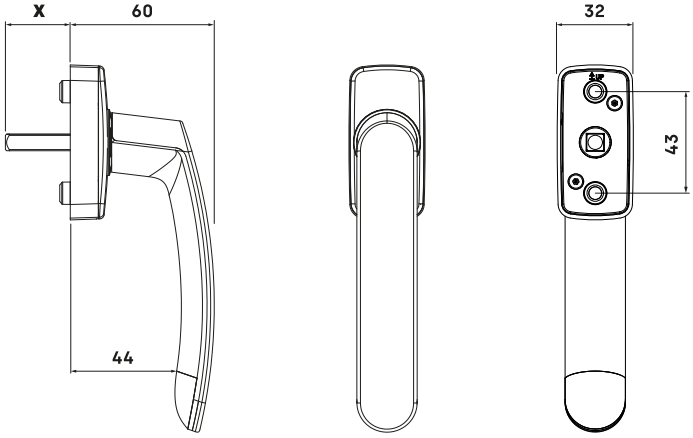
S5775 **

25

81 82

AL ZM
AC PN

7



POLARIS HANDLE
Outward opening

T5775 **

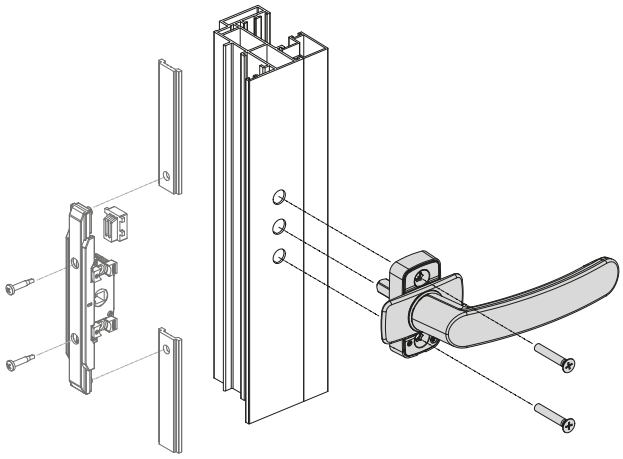
25

81 82

AL ZM
AC PN

7

S5771 **	X = 27 mm	INWARD OPENING
S5775 **	X = 64 mm	OUTWARD OPENING
T5775 **	X = 75 mm	OUTWARD OPENING



Handles with reduced handle for side-hung or tilt & turn hinged system and sliding windows.

- The ideal solution for installations of windows, balcony doors or sliding doors where the shutters do not leave the space necessary for the mounting of a traditional handle.
- The handles are valid for gearboxes and multipoint locks.
- Supplied with 7 x 38 mm (ref. S5771S) or 7 x 64 mm (ref. S5775S) spindle.
- Rotary cover ring that allows for a noticeable reduction in handle installation times.

POLARIS SLIM HANDLE
Inward opening

S5771S **

5

AL ZM AC PM

7

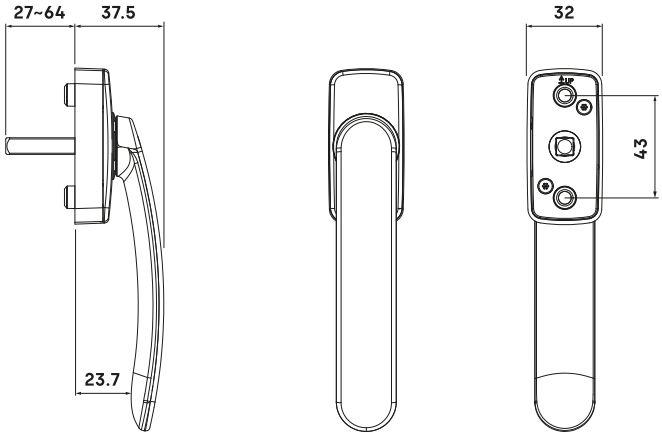
POLARIS SLIM HANDLE
Outward opening

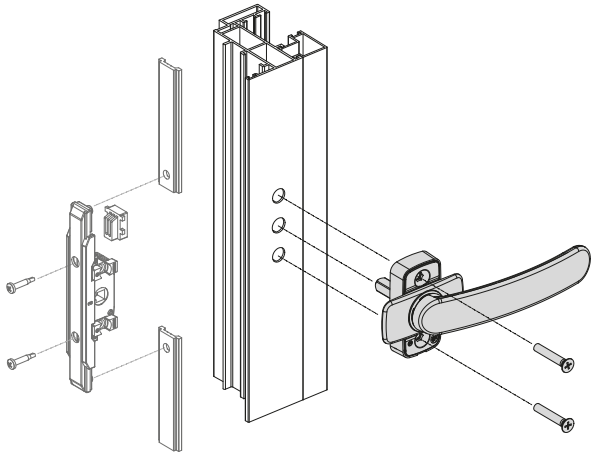
S5775S **

5

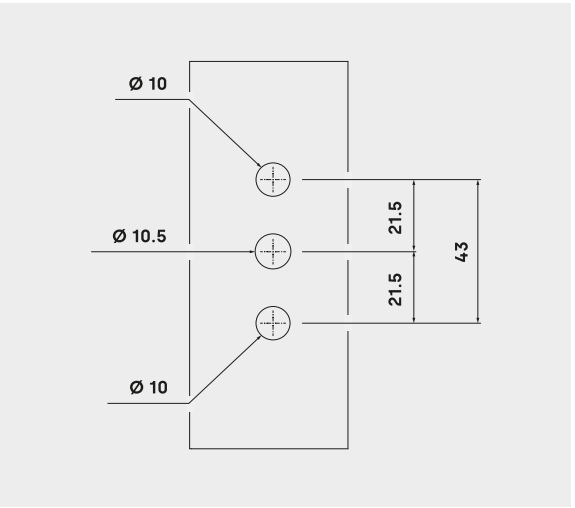
AL ZM AC PM

7





PROFILE MACHINING



POLARIS BLOCK HANDLE
Inward opening

S5773 **

20

AL ZM AC PM

7

POLARIS BLOCK HANDLE
Outward opening

S5779 **

20

AL ZM AC PM

7

POLARIS BLOCK PLUS HANDLE

S5778 **

20

AL ZM AC PM

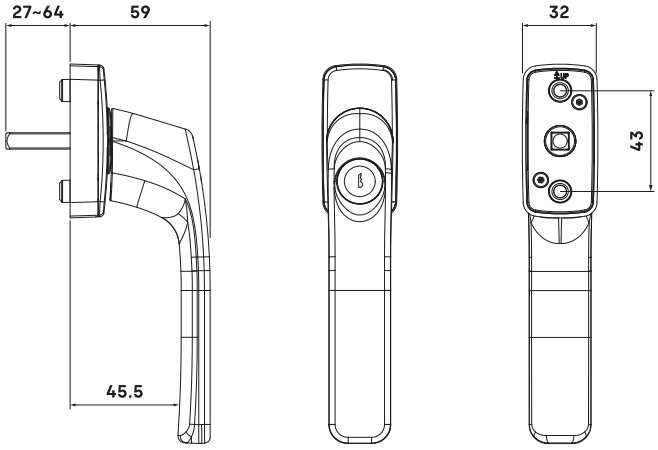
7

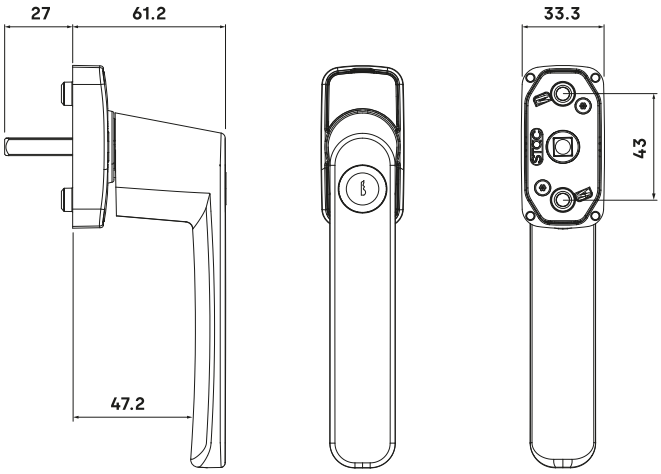
Window handles with locking system for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearboxes and multipoint locks.
- The handle is locked with a button and unlocked with a key.
- Supplied with transmission spindle of 7 x 27 mm(ref. S5773) or 7 x 64 mm (ref. S5779).
- Rotary cover ring that allows for a noticeable reduction in handle installation times.

POLARIS BLOCK PLUS handle (ref. S5778)

- Handle with locking and limitation of handle rotation by a key.
- Designed to provide greater security on windows and / or to make some kind of limitation in the opening sequence. Such as in windows with tilt-before-turn locking or with ventilation friction stay.





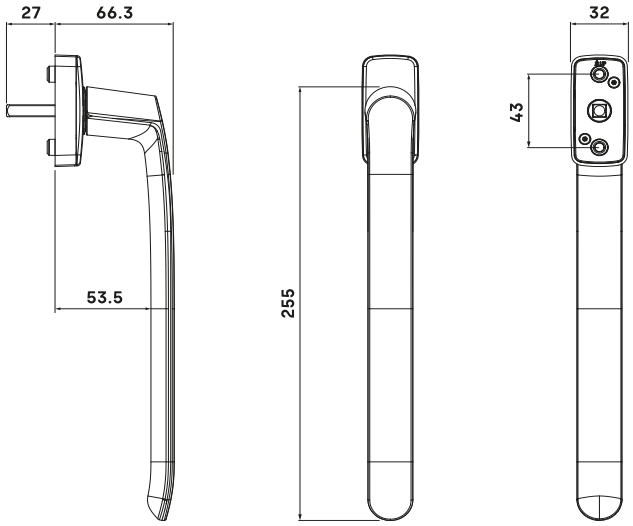
Window handle with longer handle length for side-hung or tilt & turn hinged system and sliding windows.

- The handle is valid for gearboxes and multipoint locks.
- Special length of the handle that facilitates the operation of the hardware.
- Reduces the force required to move the hardware by up to 50%.
- Rotary scutcheon that allows for a noticeable reduction in handle installation times.

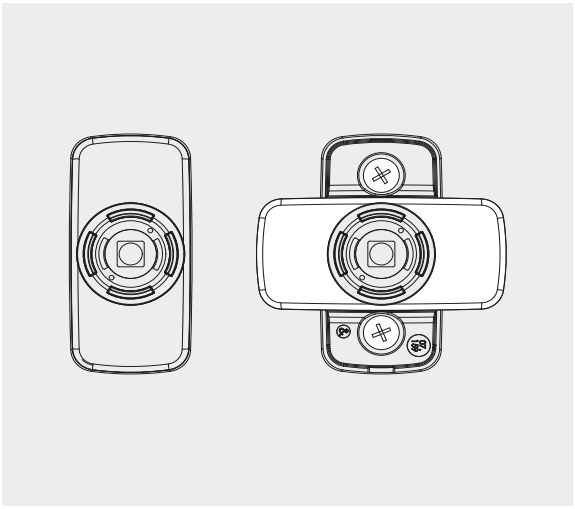
POLARIS XXL HANDLE

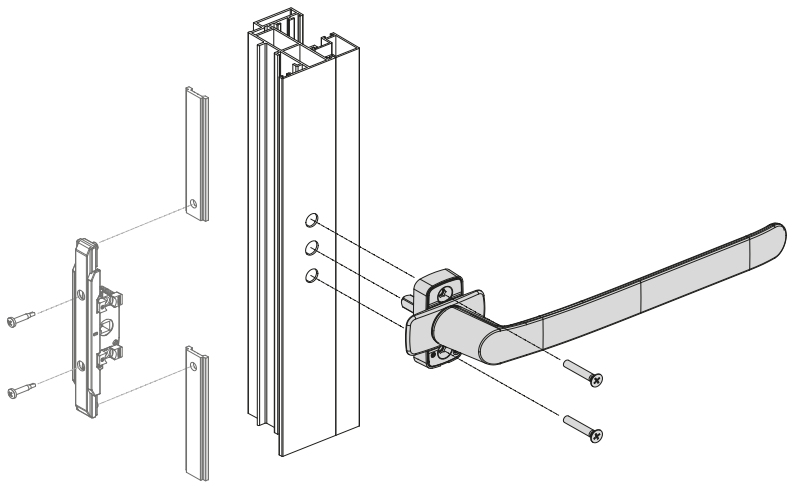
S5777 **

20



SCUTCHEON WITH ROTATION





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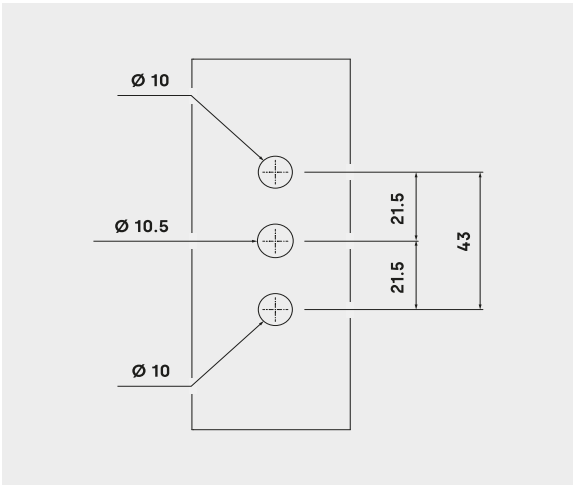


POLARIS HANDLE

detachable handle

Direct clip

Audible indication

RAL colours

Proven robustness

POLARIS base for detachable handle

Nowadays, safety has become an indispensable requirement when selecting hardware or accessories.

In a market that has evolved towards systems that incorporate mechanisms that allow limiting the operation of the window through removable handles, we have managed to position our SIRIUS cremone as a reference within the options of cremone with removable handle, standing out for its standard of quality and innovation.

In response to the growing demand for a smaller accessory for square handles, we have updated the POLARIS hardware range with a detachable handle option.

» Direct and grooved handle installation

The lever is securely mounted on the escutcheon. The escutcheon is equipped with a zamak mushroom-shaped support that allows direct and grooved clipping (*). We have succeeded in evolving the cavity of the bracket used for clipping, which is generally square in shape, by a double T-shaped slot.

The channel groove is hidden in the transmission shaft and is imperceptible from the front and is only 4 mm high with respect to the base. (*)

» Retention system

The polyamide retaining clip prevents the handle from slipping out of the groove of the zamak support. In this way, the knob is fully functional and operational for as long as required. To release the knob, simply press lightly on the retaining clip and release the knob.

» Operational drive

The double T-shaped transmission parts allow the handle to be completely restricted on the base, making it possible to pull the handle itself in both the casement and tilt and turn positions of the window.

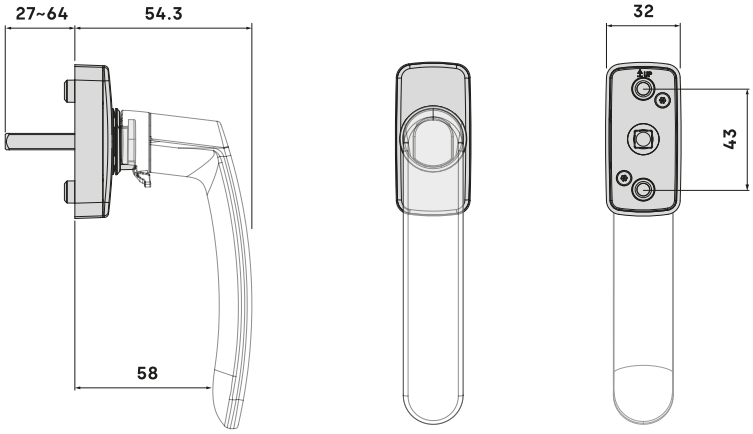
» Airing position

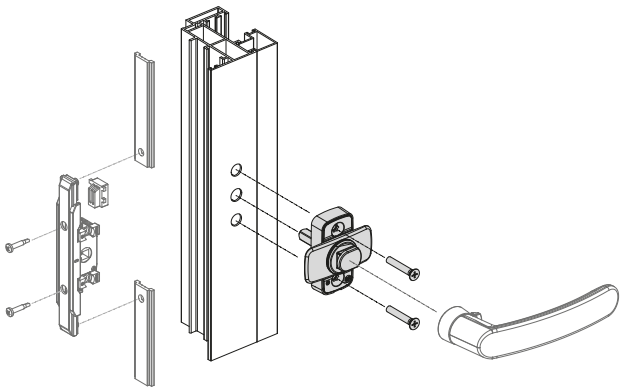
The micro-ventilation system favours circulation of air and allows for savings without losing the comfort of a fresh environment and without brusque temperature fluctuations.

The Polaris detachable handle allows controlled air flow into the internal space in the micro-ventilation position in compliance with the minimum ventilation levels set by the Spanish Building Code (Código Técnico de la Edificación).

POLARIS handle compatible with SIRIUS detachable handle for hinged side-hung, tilt & turn systems or sliding openings.

- Supplied with 7 x 27 mm spindle (ref. T5774 for outward opening with 7 x 64 mm spindle).
- The detachable handle is clipped directly onto the scutcheon of the handles.
- The user can unlock and release the detachable handle by slightly pressing on the locking clip.
- For windows with multipoint locks, it is possible to set a micro-ventilation position.
- The references do not include the SIRIUS detachable handle (ref. 0507167).





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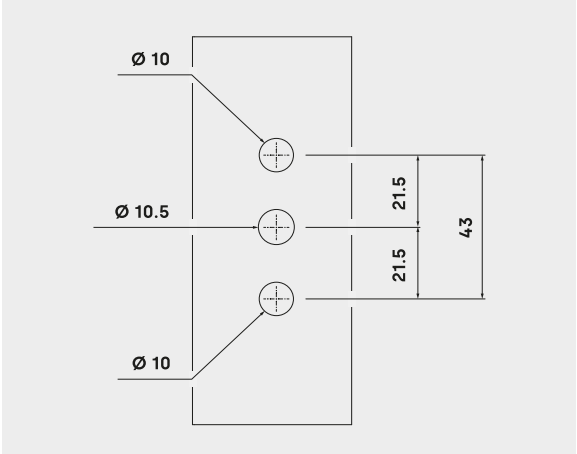


POLARIS DETACHABLE BASE
Inward opening

S5774 **

5

ZM
AC
7

POLARIS DETACHABLE HANDLE BASE
Outward opening

T5774 **

5

ZM
AC
7

SIRIUS DETACHABLE HANDLE

0507167 **

5

AL ZM
IN PA

LYRA BOX HANDLE
Inward opening

0770 | **

25

AL ZM
IN PA

7

Handles for side-hung or tilt & turn hinged system and sliding windows.

- The handles are valid for gearbox and multipoint locks.
- Supplied with 7 x 27 mm spindle (ref. P0770 for outward opening with 7 x 60 mm spindle).
- Integrated rotating escutcheon reduces handle installation time.

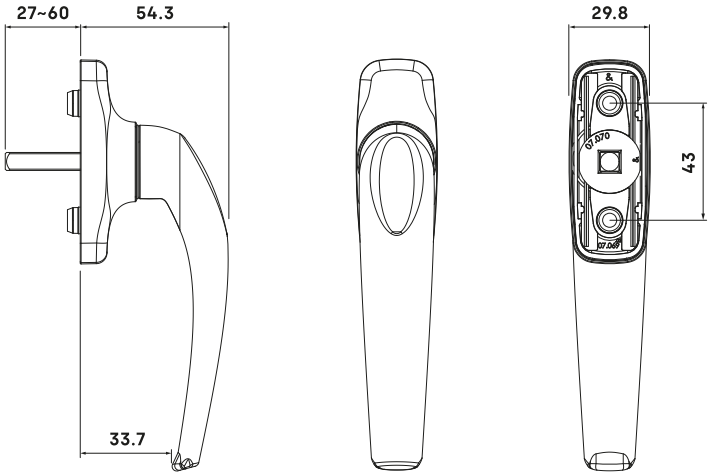
LYRA BOX HANDLE
Outward opening

P0770 | **

25

AL ZM
IN PA

7



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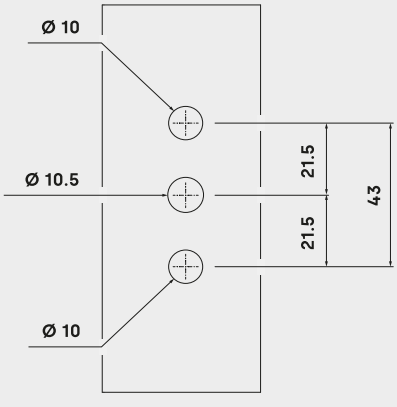
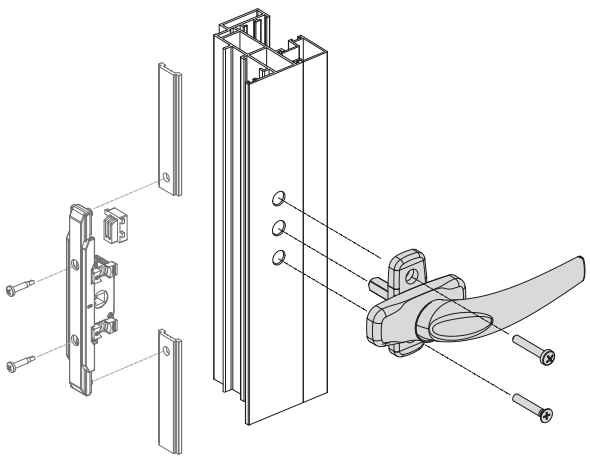




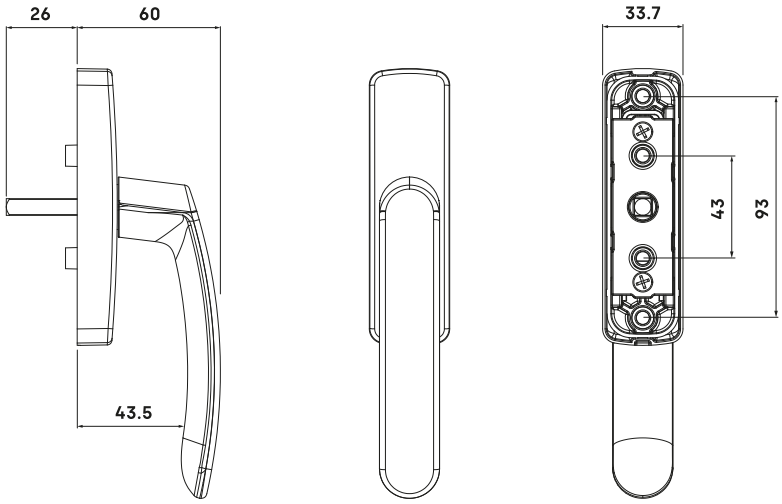
Reinforced handle side-hung, tilt & turn or sliding windows.

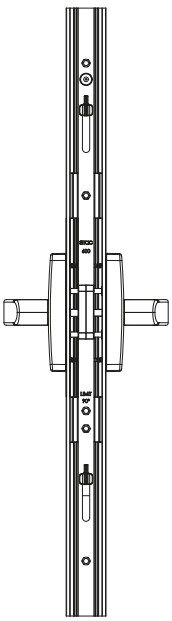
- The handles are valid for gearbox and multipoint locks.
- The double fixing of the handle to the profile and to the hardware prevents deformation and gives great stability and reinforcement to the locking system.
- The fixation to the profile is made by means of self-tapping screws without the need of additional mechanizations.
- Security screw kit available for double handle that prevent manipulation of the hardware from the outside (ref. A0507140).

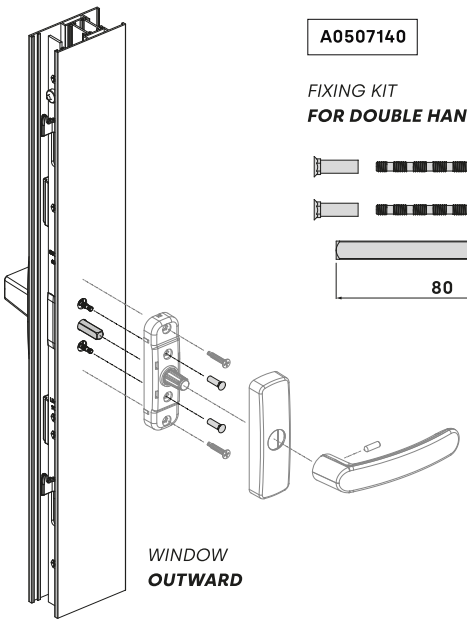
ALTAIR HANDLE

0771	**	R

15

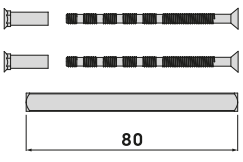






A0507140

FIXING KIT
FOR DOUBLE HANDLE



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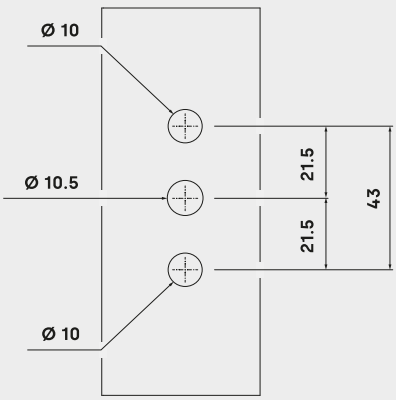


ALTAIR OFFSET HANDLE
Right-hand opening

B0777	**	R

15

AL ZM AC PM

7

Reinforced offset handles for sliding windows with multipoint lock.

- The handles are valid for gearbox and multipoint locks.
- Double fixing the handle to both the frame and the hardware prevents distortion, giving the locking system as a whole great stability and reinforcement.
- Fixing to the profile is carried out by means of self-tapping screws without the need for additional machining.
- A security screw kit is available for double handle configurations that prevent the hardware from being manipulated from the outside (ref. A0507140).

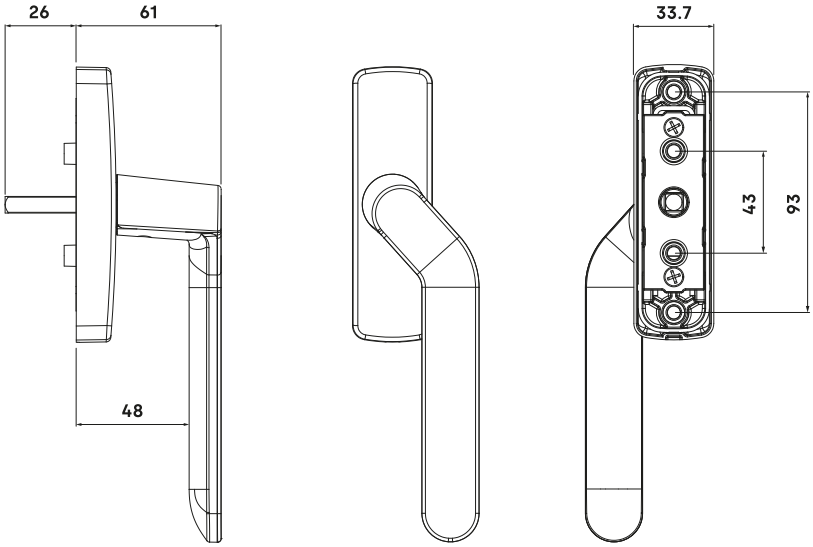
ALTAIR OFFSET HANDLE
Left-hand opening

B0778	**	R

15

AL ZM AC PM

7



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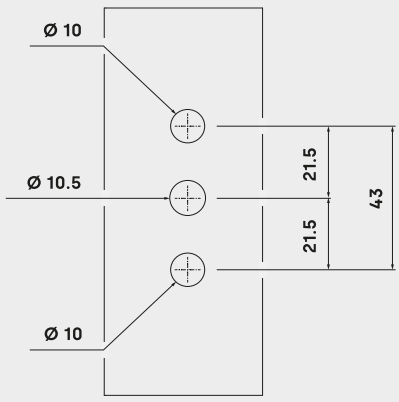
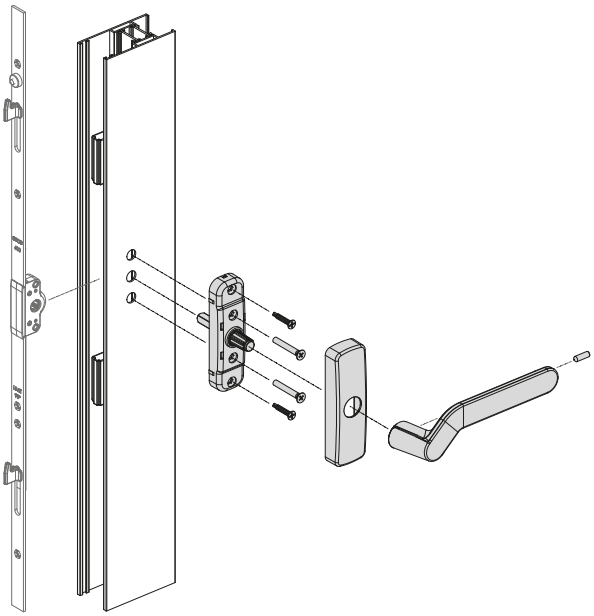




Reinforced offset handles for sliding windows with multipoint lock.

- The handles are valid for gearbox and multipoint locks.
- The handle is higher (58 millimetres) to avoid knocks against the profile of the central knot.
- Double fixing the handle to both the frame and the hardware prevents distortion, giving the locking system as a whole great stability and reinforcement.
- Fixing to the profile is carried out by means of self-tapping screws without the need for additional machining.
- A security screw kit is available for double handle configurations that prevent the hardware from being manipulated from the outside (ref. A0507140).

ALTAIR PLUS OFFSET HANDLE
Right-hand opening

P0777	**	R

15

AL ZM AC PM

7

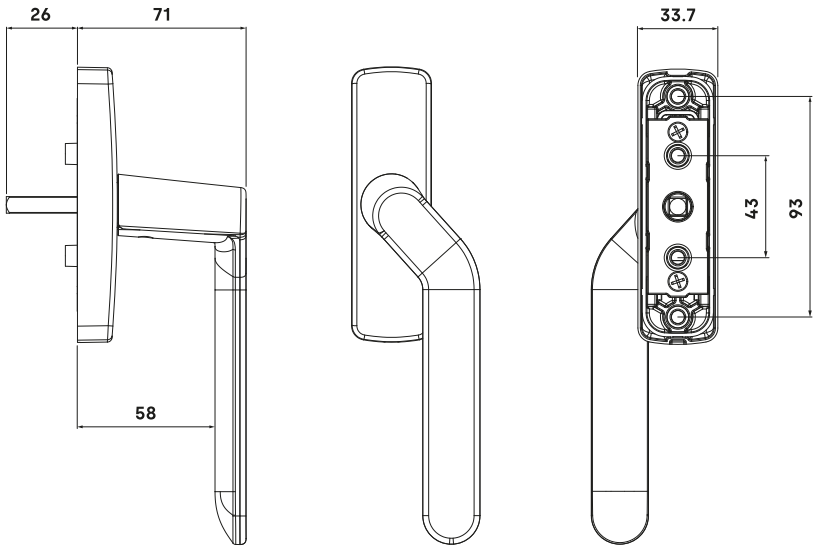
ALTAIR PLUS OFFSET HANDLE
Left-hand opening

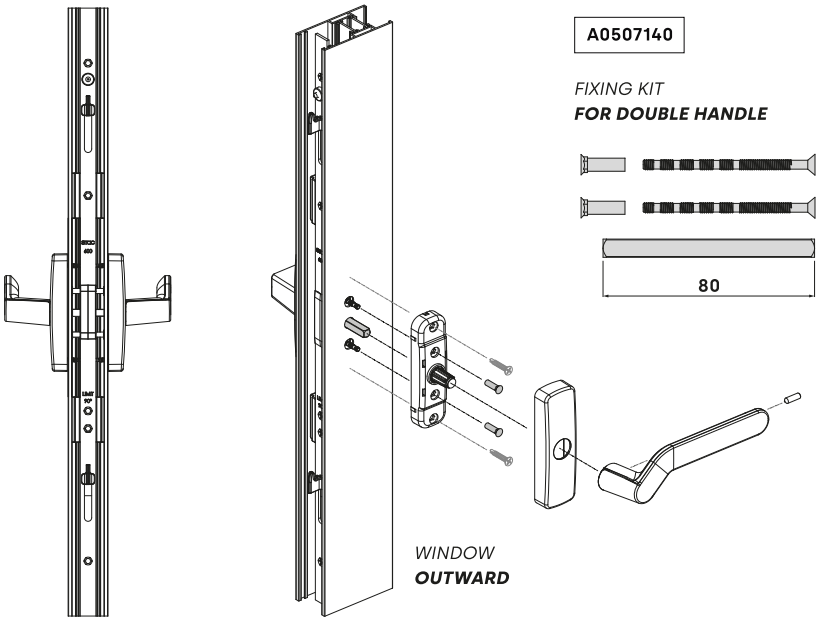
P0778	**	R

15

AL ZM AC PM

7





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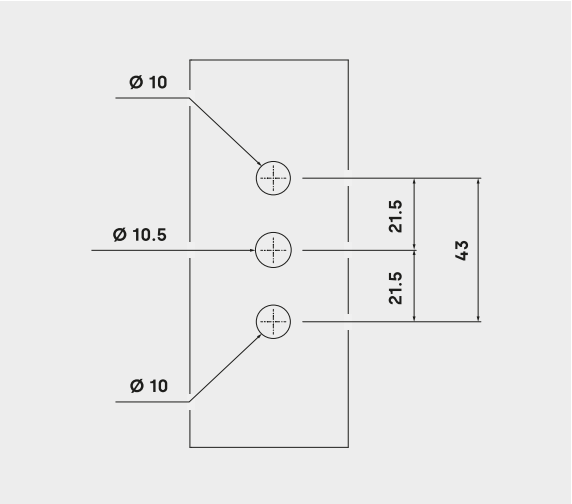


KAIROX

AISI 304 stainless steel

AISI 304

Micro-ventilation

Maximum durability

Multiple designs

KAIROX handles

More than just a product needed to open windows or doors, handles are elements that add value and style.

Windows help to brighten and enlarge rooms and, combined with the right handle detail, can make an impression from the entrance, whether it is a single-family, residential or commercial property.

Our KAIROX line of handles, made up of complete solutions in AISI 304 stainless steel, with a clean and careful design where we add the latest developments to guarantee its durability.

» AISI 304 stainless steel

Steel allows its original characteristics to maintain unaltered over time, even with intensive use or in severe climatic conditions, preserving its aesthetic appearance, resistance and hygiene.

As it is a corrosion resistant material, it does not require any additional surface protection to improve its appearance and durability, which is why it is recommended for use in public areas as it allows maintenance and cleaning tasks to be carried out in a simple and easy way.

» Double locking aluminium system

The KAIROX BLOCK handle can be blocked using a key in the closed position with the handle at 0º (traditional tilt & turn system) and in the ventilation position with the handle at 180º (traditional tilt & turn system).

Locking in the tilt position is particularly advisable in public areas: common areas of shopping centres, health centres, schools, airports, passenger stations, ...

» Complete range

The range is suitable for different types of openings:

- Windows with side-hung or tilt & turn opening.
- Sliding windows with multipoint lock systems.
- Single and double-sash doors with inward opening.

Handles for hinged systems, side-hung or tilt & turn windows with inward or outward opening.

- The handles are compatible with gearbox and multipoint locks.
- The handle and the cover shield are made of AISI 304 stainless steel.
- They are supplied with screws and zinc-plated steel spindle of 7 x 84 millimetres that allows both window openings: inward and outward.
- Position for micro-ventilation to allow ventilation of the room.

KAIROX RR BLOCK handle (ref. 260214)

- Model with handle locking by pressing the button and unlocking by means of a key.

KAIROX CC HANDLE

260110

1

ZM

IN

7

KAIROX RR BLOCK HANDLE

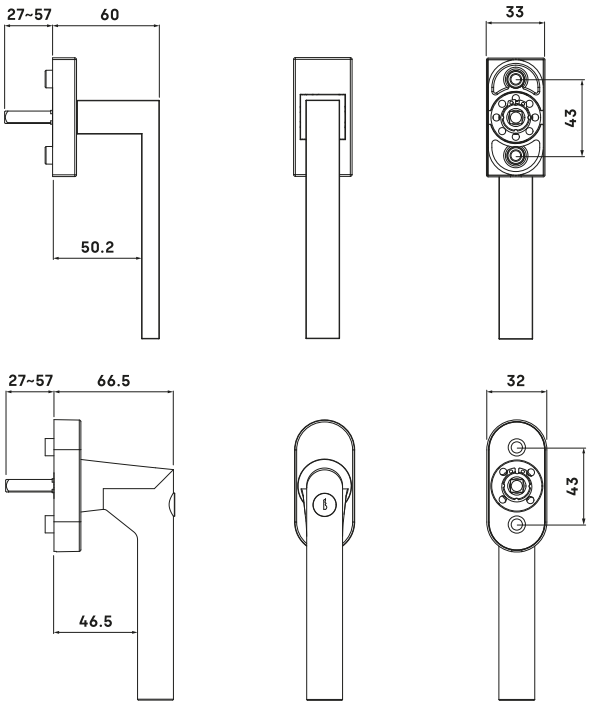
260214

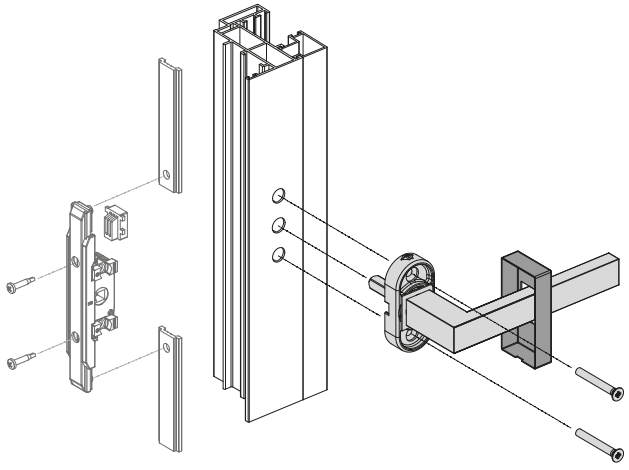
1

AL

IN

7





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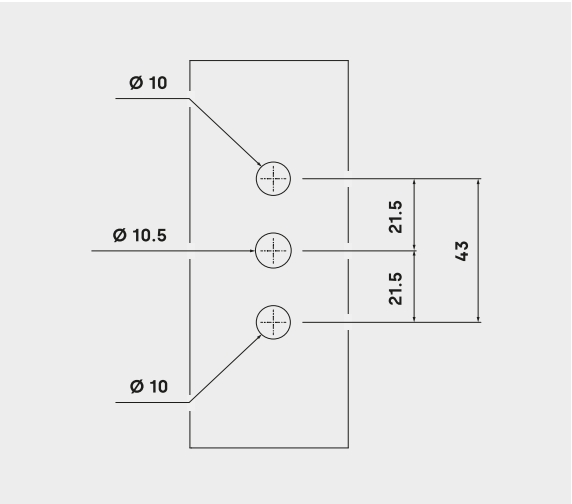


KAIROX CR HANDLE

260310

1

ZM

IN

7

Handles for hinged systems, side-hung or tilt & turn windows with inward or outward opening.

- The handles are compatible with gearbox and multipoint locks.
- The handle and the cover shield are made of AISI 304 stainless steel.
- They are supplied with screws and zinc-plated steel spindle of 7 x 84 millimetres that allows both window openings: inward and outward.
- Position for micro-ventilation to allow ventilation of the room.

KAIROX RR HANDLE

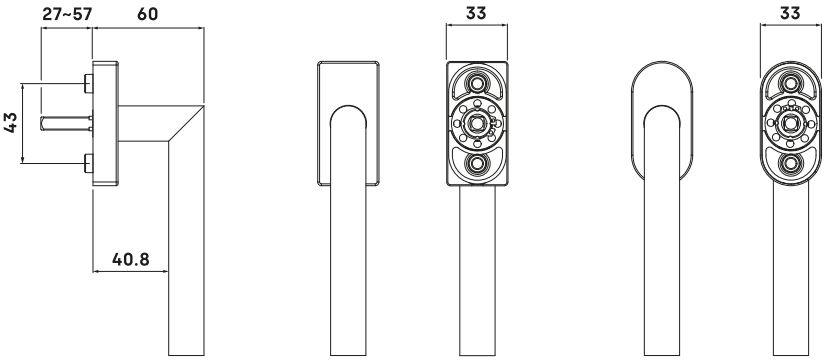
260210

1

ZM

IN

7



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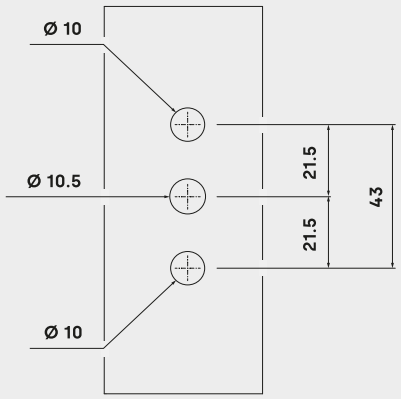
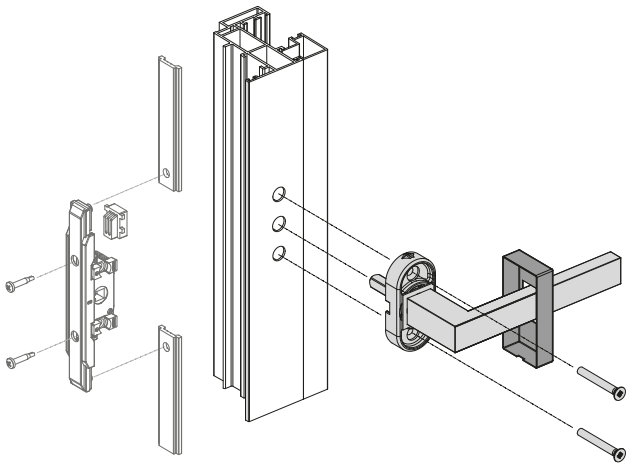




Offset handles for sliding windows with multipoint lock.

- The handles are compatible with gearbox and multipoint locks.
- The handle and the cover shield are made of AISI 304 stainless steel.
- They are supplied with screws and zinc-plated steel spindle of 7 x 84 millimetres that allows both window openings: inward and outward.
- Position for micro-ventilation to allow ventilation of the room.

KAIROX CR HANDLE
Right-hand opening

260311

1

ZM

IN

7

KAIROX RR HANDLE
Right-hand opening

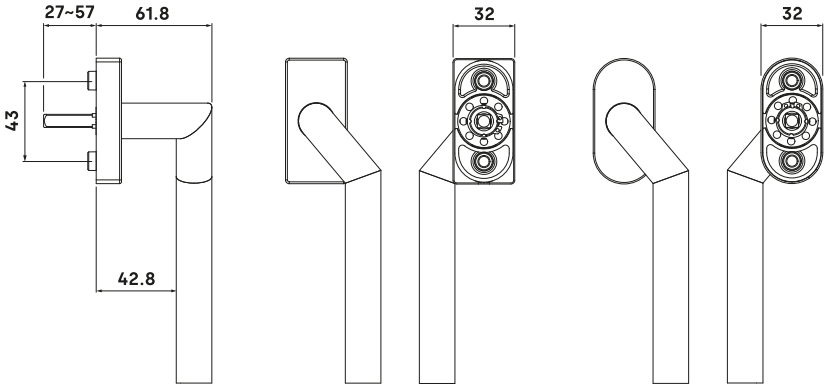
260211

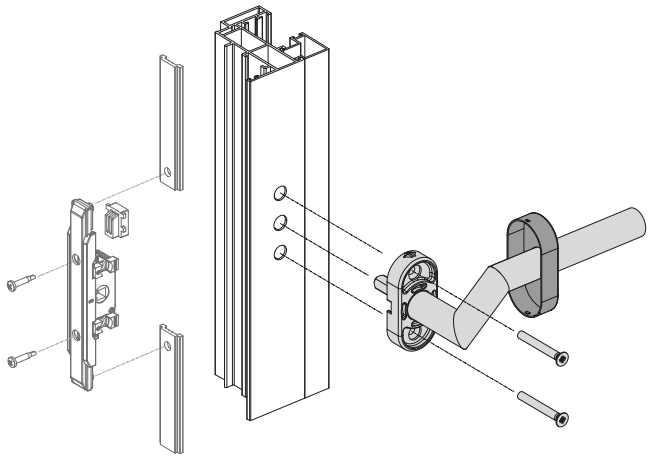
1

ZM

IN

7





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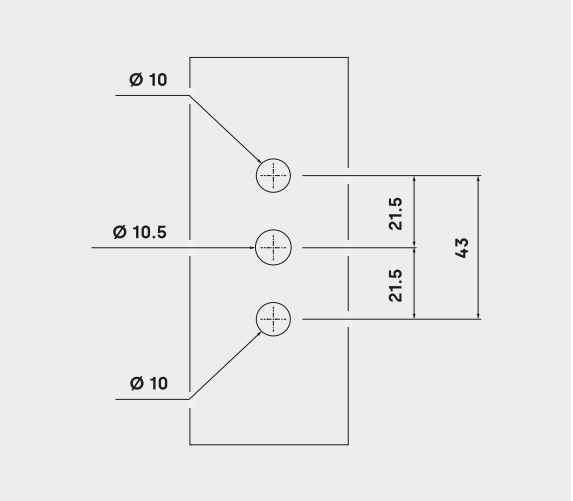


KAIROX CR HANDLE
Left-hand opening

260312

1

ZM

IN

7

Offset handles for sliding windows with multipoint lock.

- The handles are compatible with gearbox and multipoint locks.
- The handle and the cover shield are made of AISI 304 stainless steel.
- They are supplied with screws and zinc-plated steel spindle of 7 x 84 millimetres that allows both window openings: inward and outward.
- Position for micro-ventilation to allow ventilation of the room.

KAIROX RR HANDLE
Left-hand opening

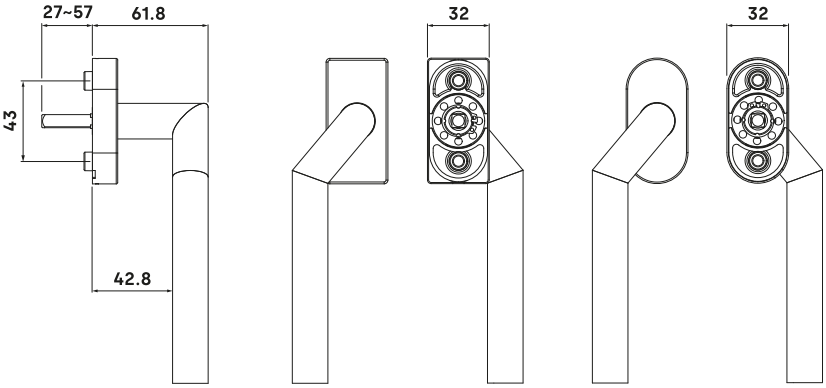
260212

1

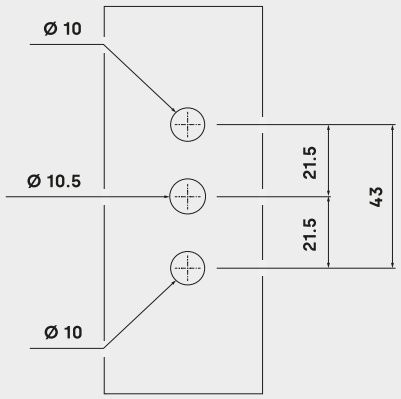
ZM

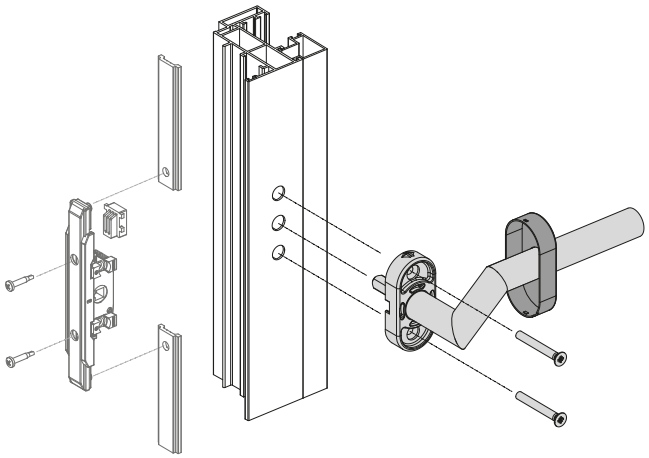
IN

7



PROFILE MACHINING





Handles for hinged systems, side-hung or tilt & turn windows with inward or outward opening.

- The handles are compatible with gearbox and multipoint locks.
- The handle and the cover shield are made of AISI 304 stainless steel.
- They are supplied with screws and zinc-plated steel spindle of 7 x 84 millimetres that allows both window openings: inward and outward.
- Position for micro-ventilation to allow ventilation of the room.

KAIROX CRC HANDLE

ZM
IN
7

260313

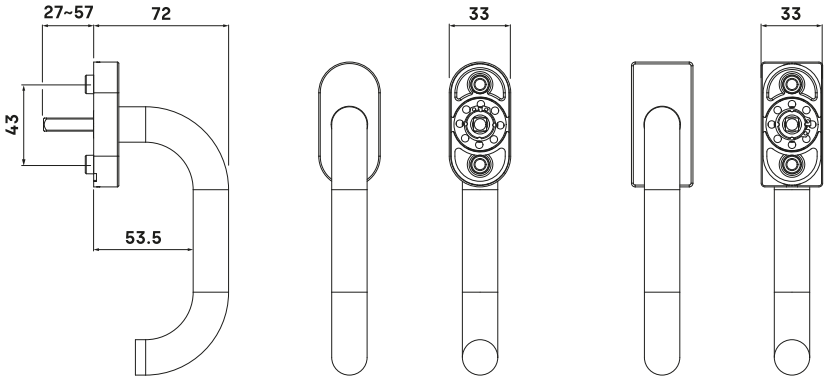
1

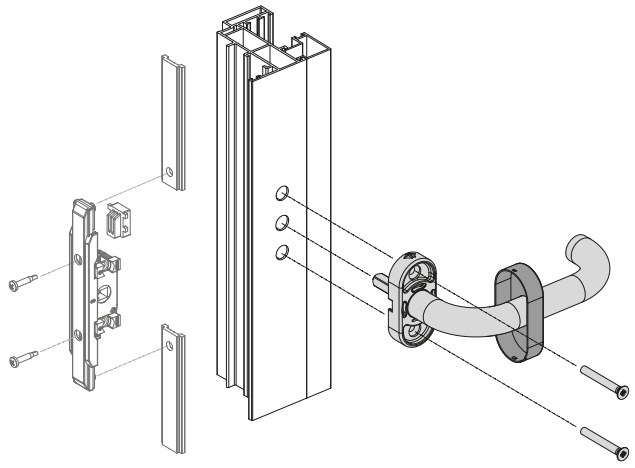
KAIROX RRC HANDLE

ZM
IN
7

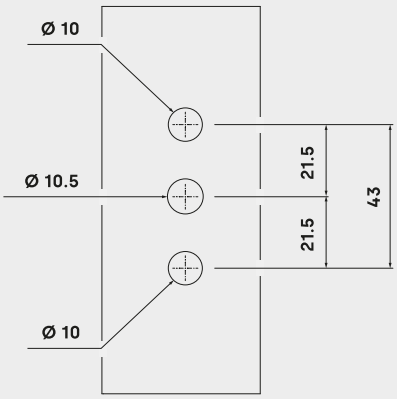
260213

1





PROFILE MACHINING



OUTWARD OPENING KIT

K0775-01

100

AC

IN

Fixing kit for outward opening on hinged windows.

Kit for outward opening K0775-01

• The kit is supplied with 7 x 7 x 130 mm transmission spindle and DIN 965 M5 x 100 mm screws.

Kit for outward opening K0775-02

• Kit specifically designed for SIRIUS or DELTA mechanisms.
• The kit is supplied with 7 x 7 x 100 mm transmission spindle and DIN 7985 M5 x 90 mm screws.

Kit for outward opening K0775-03

• The kit is supplied with 7 x 7 x 90 mm transmission spindle and DIN 965 M5 x 90 mm screws.

OUTWARD OPENING KIT

K0775-02

100

AC

IN

	PROFILE WIDTH	SPINDLE	SCREWS
K0775-01	J < 110	7 x 7 x 130	DIN 965 M5 x 100
K0775-02	J < 90	7 x 7 x 100	DIN 7985 M5 x 90
K0775-03	J < 90	7 x 7 x 90	DIN 965 M5 x 90

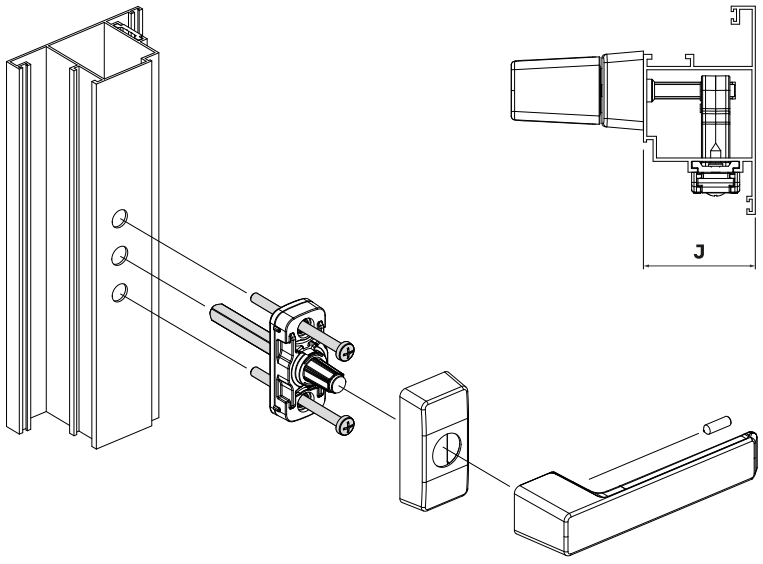
OUTWARD OPENING KIT

K0775-03

100

AC

IN



» FIXING KITS FOR DOUBLE HANDLE CONFIGURATIONS

ACCESSORIES	K0771-01	K0771-02	K5771-09	A0507140
POLYAMIDE SHIMS *with fibreglass*	✔	✔	-	-
DIN 965 M4 SCREWS *length (mm)*	L = 80	L = 100	L = 80	L = 65
7 x 7 SPINDLE *length (mm)*	L = 100	L = 130	L = 75	L = 80
M4 CAP NUT *thread depth (mm)*	P = 13.6	P = 13.6	P = 5.25	P = 13.6

COMPATIBLE HANDLES		K0771-01	K0771-02	K5771-09	A0507140
SIRIUS / DELTA	*Inward op.*	**J min ~ max** = 34 - 62	**J min ~ max** = 34 - 88	-	-
	Outward op.	**J min ~ max** = 35 - 62	**J min ~ max** = 35 - 88	-	-
Fixing screws		**C** = J + 12	**C** = J + 12	-	-
Spindle		**C** = J + 32	**C** = J + 32	-	-
POLARIS	*Inward op.*	**J min ~ max** = 34 - 72	**J min ~ max** = 34 - 92	**J min ~ max** = 34 - 55	**J min ~ max** = 34 - 57
	Outward op.	**J min ~ max** = 35 - 72	**J min ~ max** = 35 - 92	**J min ~ max** = 35 - 55	**J min ~ max** = 35 - 57
Fixing screws		**C** = J + 8	**C** = J + 8	**C** = J + 18	**C** = J + 8
Spindle		**C** = J + 20	**C** = J + 20	**C** = J + 20	**C** = J + 20
SIRIUS PRO / DELTA PRO	*Inward op.*	**J min ~ max** = 34 - 75	**J min ~ max** = 34 - 95	**J min ~ max** = 34 - 60	**J min ~ max** = 34 - 60
	Outward op.	-	-	**J min ~ max** = 35 - 60	-
Fixing screws		**C** = J + 4	**C** = J + 4	**C** = J + 14	**C** = J + 4
Spindle		**C** = J + 13	**C** = J + 13	**C** = J + 13	**C** = J + 13
ALTAIR	*Inward op.*	**J min ~ max** = 34 - 75	**J min ~ max** = 34 - 95	**J min ~ max** = 34 - 60	**J min ~ max** = 34 - 60
	Outward op.	**J min ~ max** = 35 - 75	**J min ~ max** = 35 - 95	**J min ~ max** = 35 - 60	**J min ~ max** = 35 - 60
Fixing screws		**C** = J + 11	**C** = J + 11	**C** = J + 21	**C** = J + 11
Spindle		**C** = J + 26	**C** = J + 26	**C** = J + 26	**C** = J + 26

(**J**) Profile width (mm), (**C**) Cutting length (mm)

FIXING KIT FOR DOUBLE HANDLE

K0771-01

100

IN

PA

Fixing kits that allow double handle installations and prevent access from the outside of the window.

• The polyamide shims allow the countersunk head screws to be fitted to SIRIUS or DELTA mechanisms.
• The cap nuts prevent manipulation and disassembly of the hardware and so must be installed on the outside face of the window.

Fixing kit for double handle K0771-01

• The fixing kit is supplied with polyamide shims, 7 x 7 x 100 mm transmission spindle, M4 x 80 mm DIN 965 screws and M4 cap nuts.

Fixing kit for double handle K0771-02

• The fixing kit is supplied with polyamide shims, 7 x 7 x 130 mm transmission spindle and DIN 965 M4 x 100 mm screws and M4 cap nuts.

Polyamide shims are not required for PRO or POLARIS mechanisms.

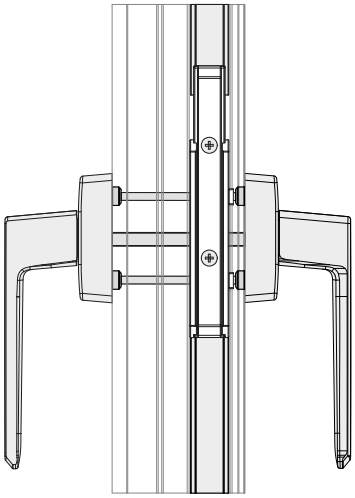
FIXING KIT FOR DOUBLE HANDLE

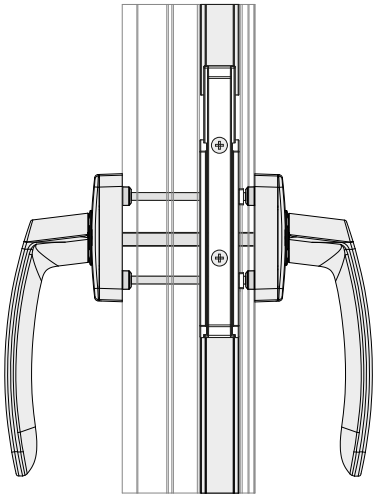
K0771-02

50

IN

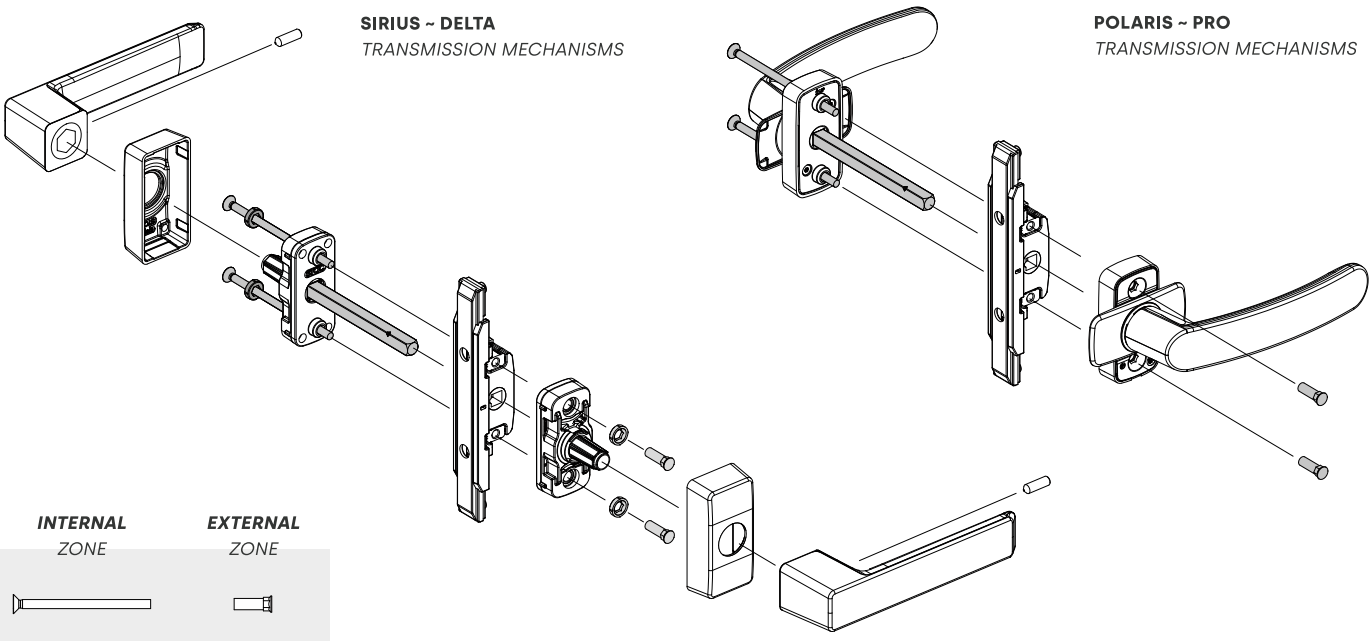
PA





SIRIUS ~ DELTA
TRANSMISSION MECHANISMS

POLARIS ~ PRO
TRANSMISSION MECHANISMS



Fixing kits that allow double handle installations and prevent access from the outside of the window.

• The cap nuts prevent manipulation and disassembly of the hardware and so must be installed on the outside face of the window.

Fixing kit for double handle K5771-09

• The fixing kit is supplied with polyamide shims, 7 x 7 x 75 mm transmission spindle, M4 x 80 mm DIN 965 screws and M4 cap nuts.
• Compatible with DELTA FLAP pull handle with cam.

Kit with security screws A0507140

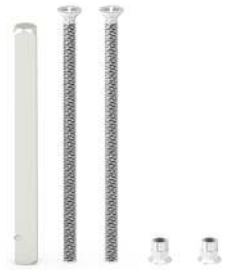
• The fixing kit is supplied with polyamide shims, 7 x 7 x 80 mm transmission spindle and M4 x 65 mm DIN 965 screws and M4 cap nuts.

FIXING KIT FOR DOUBLE HANDLE

AC
IN

K5771-09

50

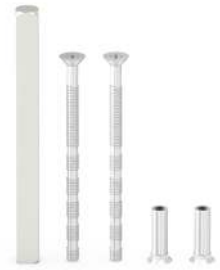


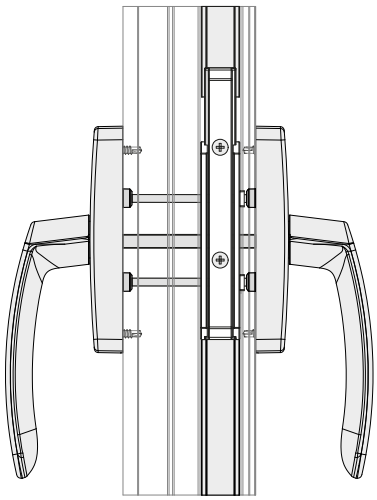
FIXING KIT WITH SECURITY SCREWS

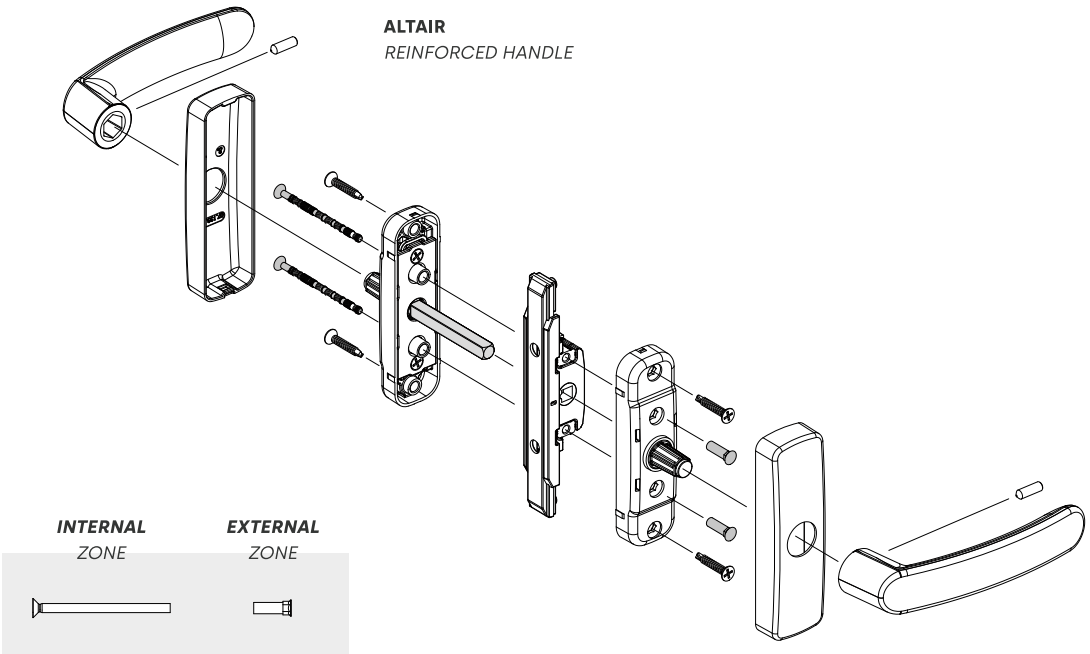
AC

A0507140

50







FIXING KIT WITH METRIC RIVET NUTS

K0771-03

50

AC

IN

Alternative fixing systems for window handles with SIRIUS mechanisms.

• Provide a more robust fixing by attaching to the profile in cases such as multi-point when there is no handle on the outside.

Kit with metric rivet nuts (ref. K0771-03)

• Fixing to the profile by means of metric threaded screws and rivets.
• Supplied with base plate, M5 x 20 mm DIN 7985 screws and M5 nuts.

Kit with self-tapping screws (ref. K0771-04)

• Fixing to the profile by means of self-tapping screws.
• Supplied with base plate and 4.2 x 19 mm DIN 7981 self-tapping screws.

FIXING KIT WITH SELF-TAPPING SCREWS

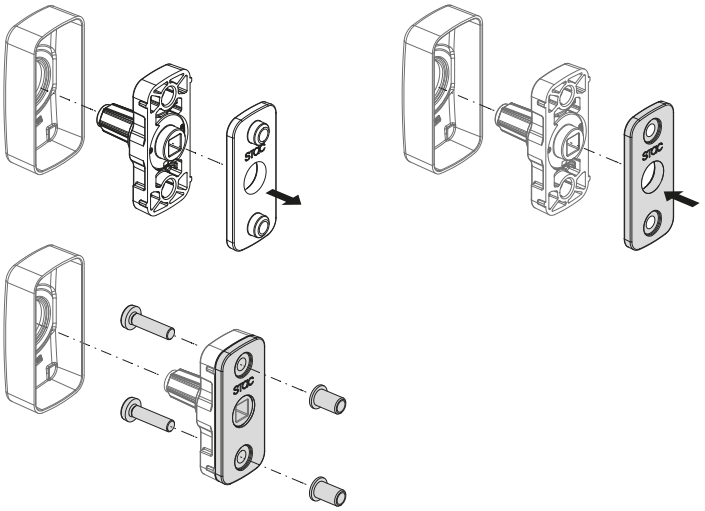
K0771-04

200

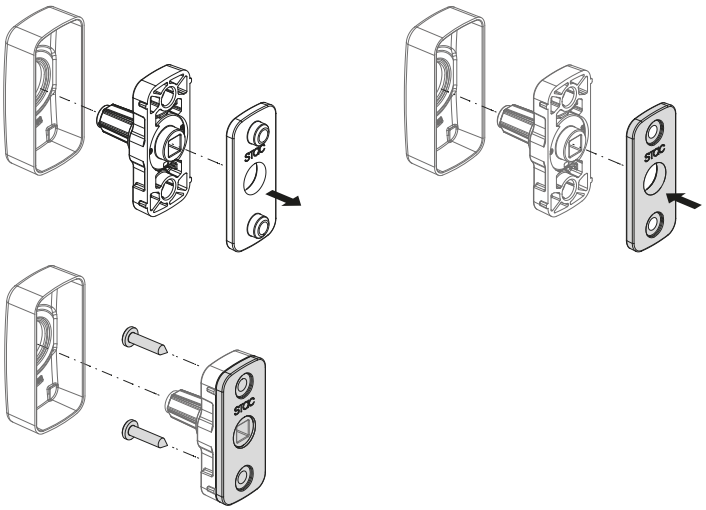
AC

IN

K0771-03



K0771-04



Alternative fixing systems for window handles with SIRIUS mechanisms.

• Provide a more robust fixing by attaching to the profile in cases such as multi-point when there is no handle on the outside.

Kit with metric rivet nuts (ref. K5771-03)

• Fixing to the profile by means of metric threaded screws and rivets.
• Supplied with base plate, M5 x 20 mm DIN 965 screws and M5 nuts.

Kit with self-tapping screws (ref. K5771-04)

• Fixing to the profile by means of self-tapping screws.
• Supplied with base plate and 4.8 x 19 mm DIN 7982 self-tapping screws .

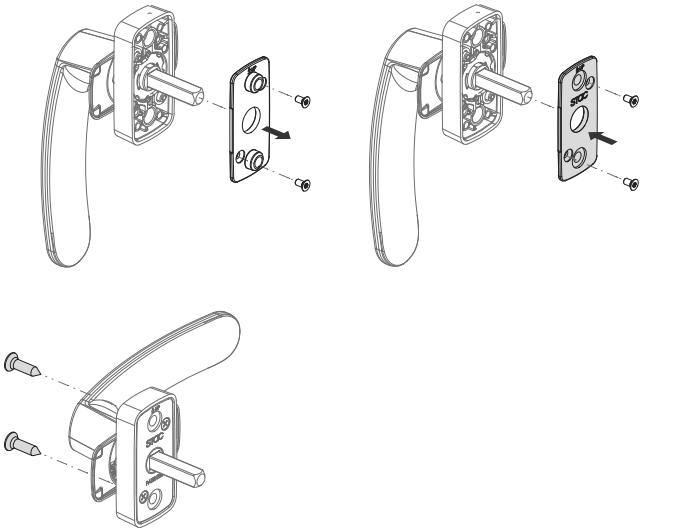
FIXING KIT WITH METRIC RIVET NUT

ZM
IN

K5771-03

5

K5771-03



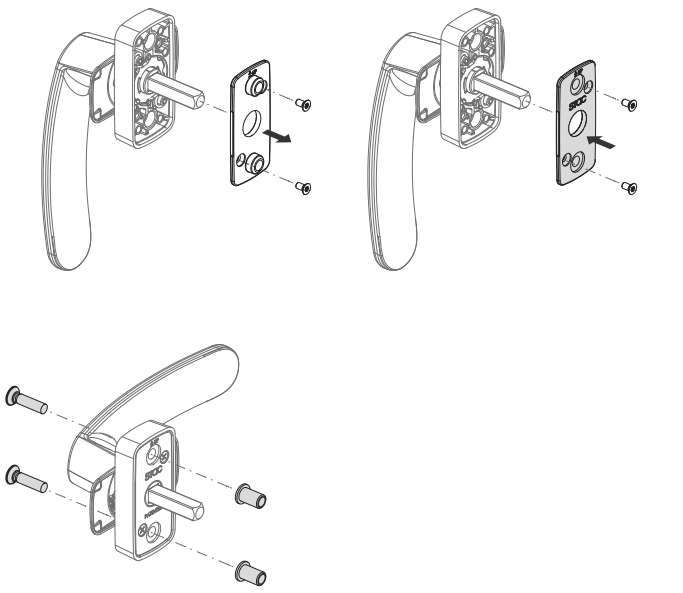
FIXING KIT WITH SELF-TAPPING SCREWS

ZM
IN

K5771-04

5

K5771-04



03 CREMONE HANDLES

84 **DELTA cremone**

87 **SIRIUS cremone**

94 **SIRIUS detachable handle**

97 **LYRA cremone**

102 **LEO cremone**

106 **HYDRA cremone**

DELTA

minimalist aesthetic range

Quick installation

No visible screws

Corrosion resistance

RAL colour chart

DELTA range

The design must give priority to the usefulness of the product, so we have focused on developing a mechanism that is as integrated as possible in the current aluminium window and door frames, avoiding those features that would take us away from this objective.

» Compatibility with current systems

- Windows with side-hung or tilt & turn openings.
- Sliding windows with multipoint lock systems.
- Windows with Euro Groove or Groove 16 profiles.
- Single and double-sash doors with both inward and outward opening.

» Personalisation through colour

Projects find a unique look through new colours or finishes.

The range of handles and cremones is fully customisable and is available in white or black lacquer as well as in the colours that make up the RAL colour chart, a colour matching system that defines colours for paints, plastics and coatings.

» Product aesthetic

The aesthetic of a product is an integral part of its usefulness, as products used on a daily basis tend to have an indirect effect on people and their well-being.

We have placed special emphasis on the ergonomics of use by eliminating any obtrusive edges for the user.

» Corrosion resistance

The parts that make up the handles have been treated to guarantee resistance to corrosion, exceeding 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

According to the parameters described in the standard, a grade 4 classification has been obtained, which allows the possibility of using the handles in environments with severe climatic conditions.

Cremones for side-hung or tilt & turn hinged systems.

- The cremones are compatible with the FAST fixing system.
- The levers and the pinion are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

DELTA CREMONE
With transmission cover

7001 **

20

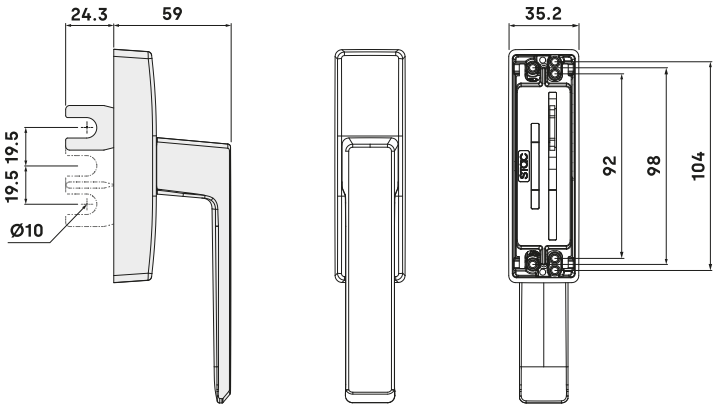
AL ZM IN PA

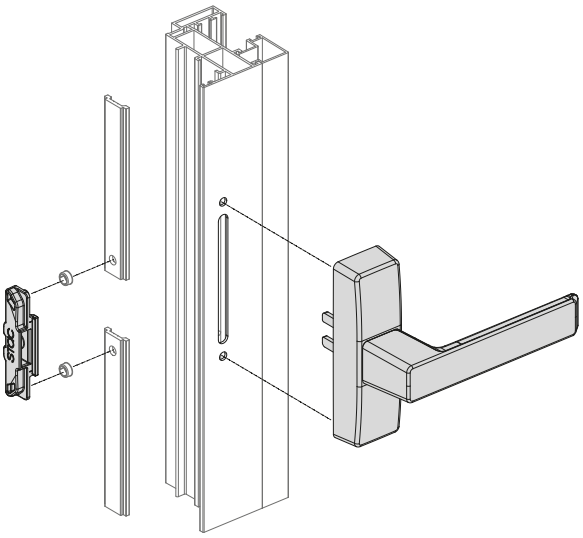
DELTA CREMONE
Without transmission cover

7002 **

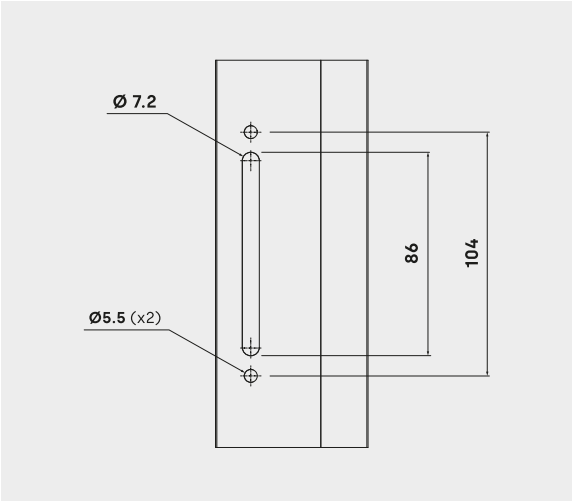
20

AL ZM IN PA





PROFILE MACHINING



DELTA CREMONE
Side-hung opening

7004 **

20

AL ZM
AC PA

Cremones for side-hung hinged systems.

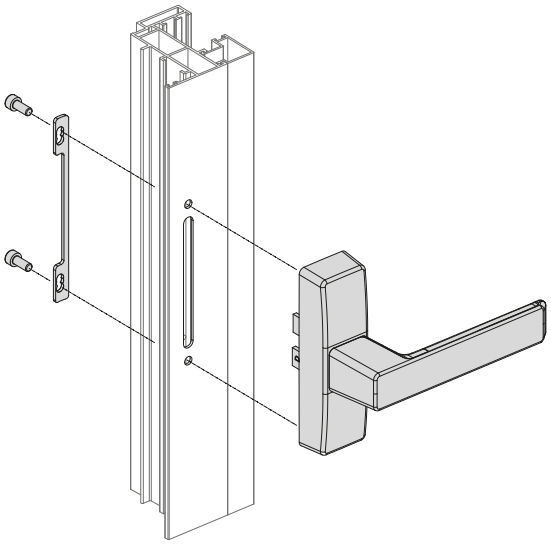
- Cremone available for unidirectional opening (ref. 7004).
- The cremone is compatible with the FAST fastening system.
- The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

DELTA CREMONE
French side-hung opening

7003 **

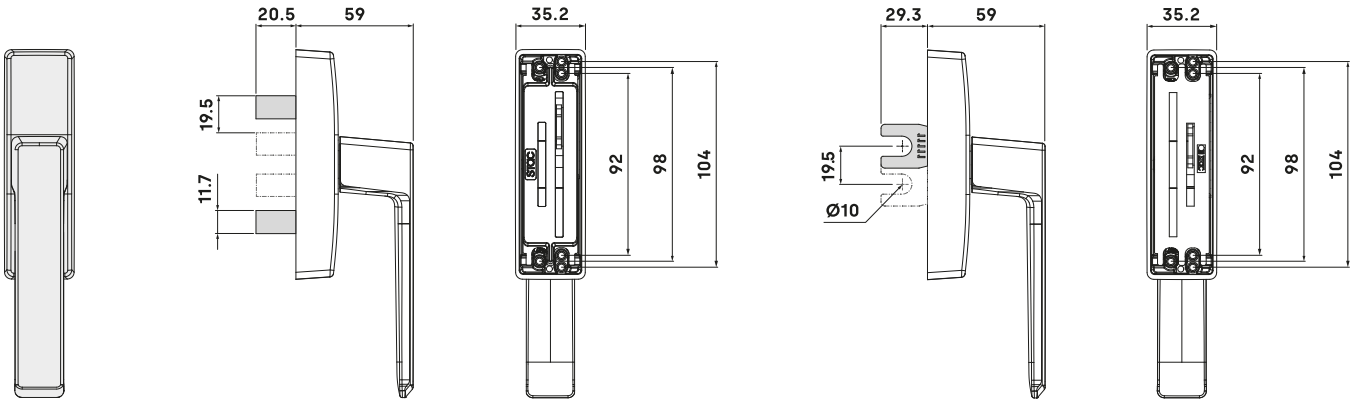
20

AL ZM
IN PA



7004 **

SIDE-HUNG OPENING



7003 **

FRENCH STYLE OPENING

Cremones for side-hung hinged systems.

- The cremones are compatible with the FAST fastening system.
- The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.
- For each box of right-hand or left-hand opening cremones, five kits of left-hand (ref. A0790) or right-hand (ref. A0791) opening levers are supplied.

SIRIUS CREMONE
Right-hand opening

A0790	**

20

82 81

AL ZM

IN PA

SIRIUS CREMONE
Left-hand opening

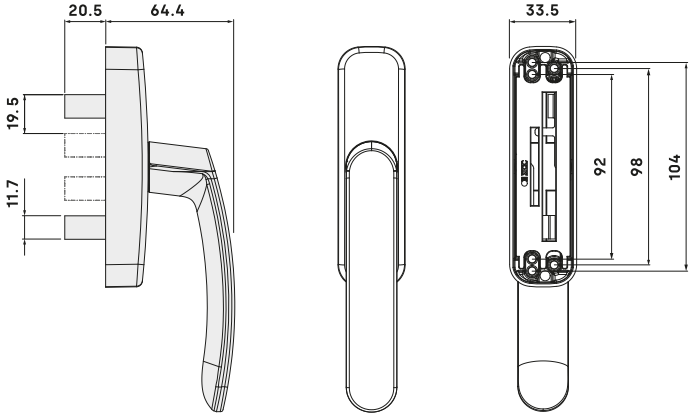
A0791	**

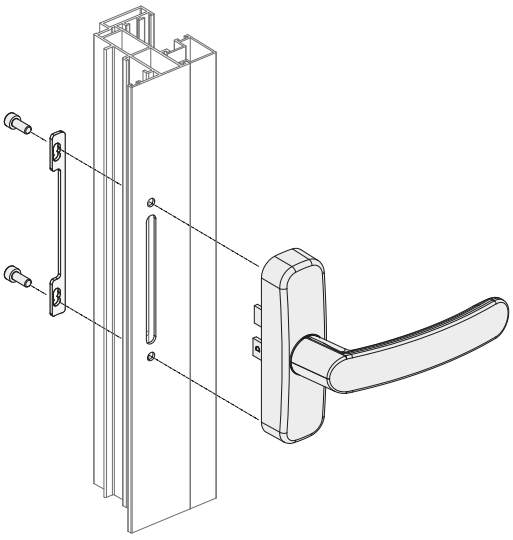
20

82 81

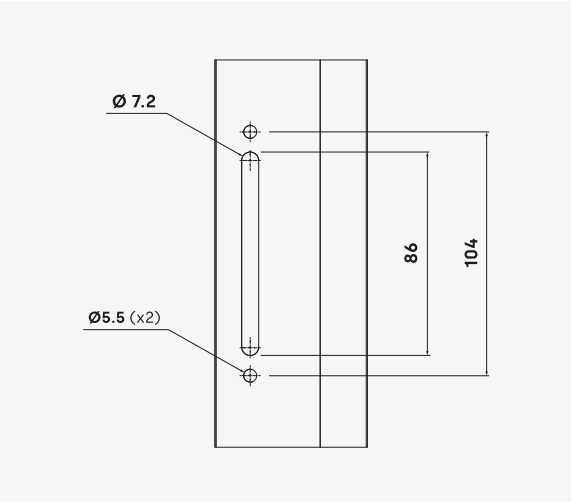
AL ZM

IN PA





PROFILE MACHINING



SIRIUS CREMONE (24 SHORT LEVER)

French side-hung opening

A0794 **

20

82 81

AL ZM IN PA

Handles for side-hung windows.

- Incorporates 24 millimetres transmission lever (ref. P0794 for 29 millimetres lever).
- The cremones are compatible with the FAST fastening system.
- The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

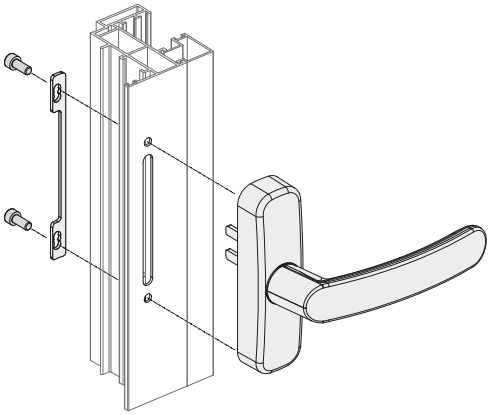
SIRIUS CREMONE (29 LONG LEVER)

French side-hung opening

P0794 **

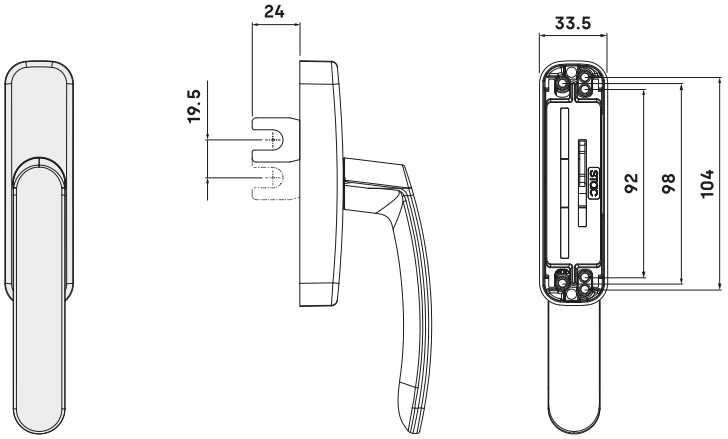
20

AL ZM IN PA



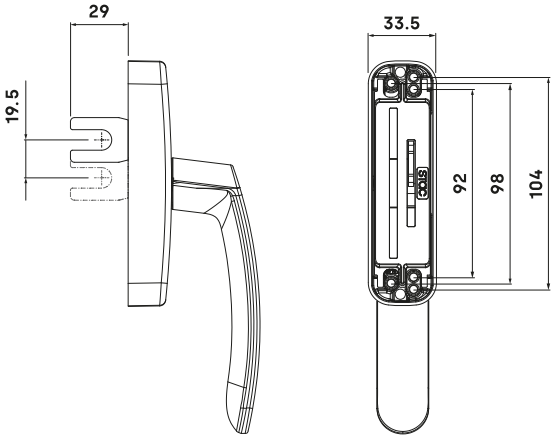
A0794 **

SHORT LEVER 24



P0794 **

LONG LEVER 29



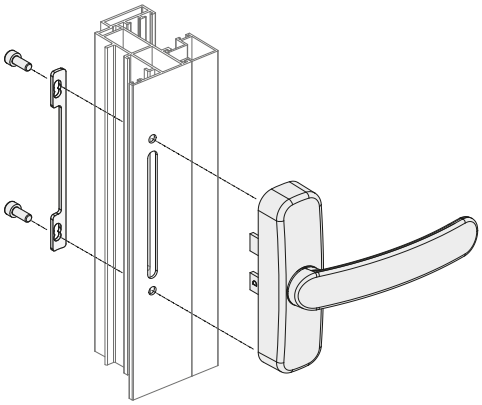
Cremone with reduced handle for side-hung windows.

• The ideal solution for installations of windows, balcony doors or sliding doors where the shutters do not leave the space necessary for the mounting of a traditional cremone.
• The cremone is compatible with the FAST fastening system.
• The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.
• Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

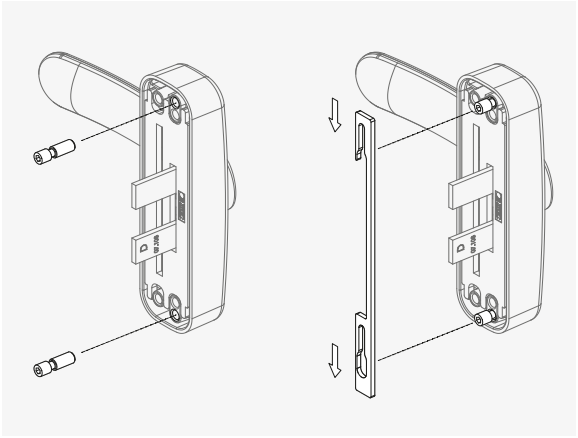
SIRIUS SLIM CREMONE
Side-hung opening

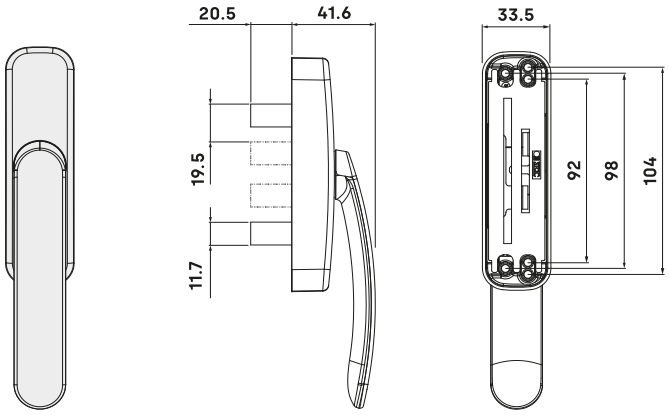
A0790S	**

5

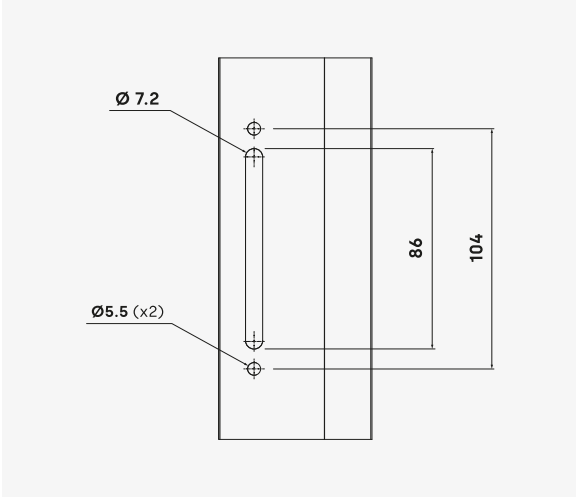


FAST FIXING SYSTEMS COUNTERPLATE + SCREWS





PROFILE MACHINING



SIRIUS BLOCK CREMONE

Side-hung opening

A0793	**

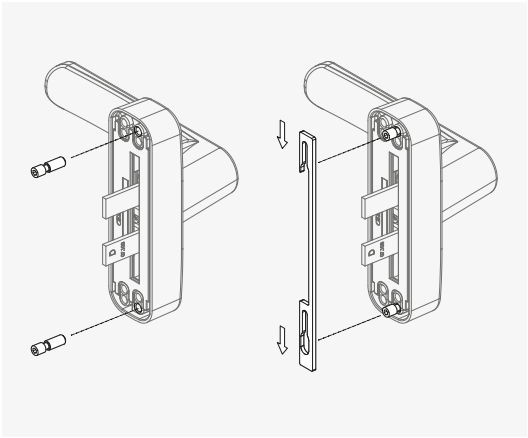
20

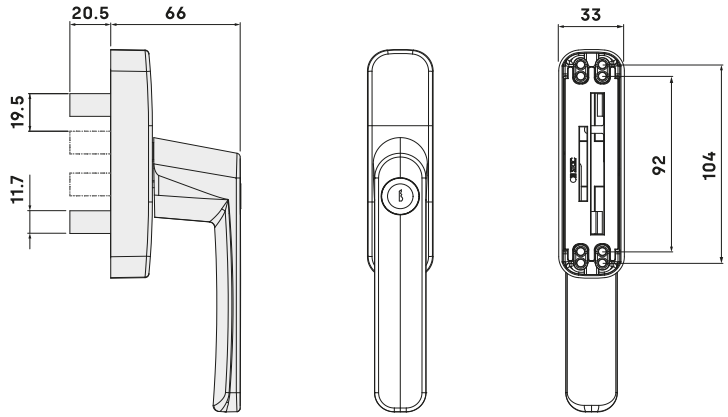
AL ZM
IN PA

Cremone with key lock compatible with side-hung windows.

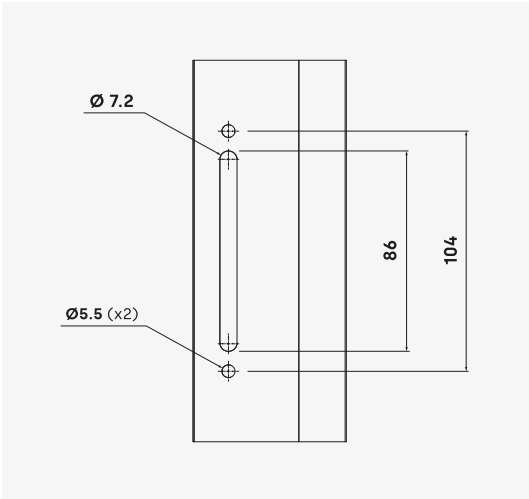
- The lock of the cremone handle is done by means of a key.
- The cremone is compatible with the FAST quick fastening system.
- The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.

FAST FIXING SYSTEM COUNTERPLATE + SCREWS

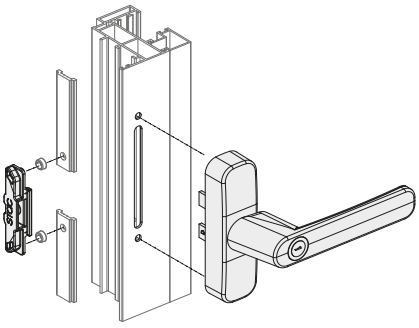




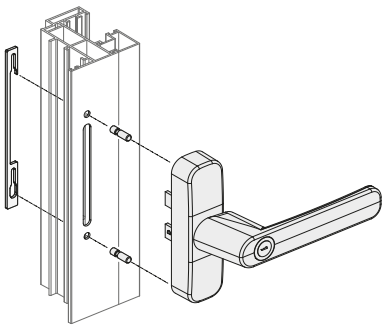
PROFILE MACHINING



TRANSMISSION COVER



*FIXING WITH **FAST** COUNTERPLATE*



Cremones for side-hung or tilt & turn hinged systems.

- The cremones are compatible with the FAST fixing system.
- The levers and the pinion are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

SIRIUS CREMONE
With transmission cover

A1154	**

20

82 81

AL ZM

IN PA

SIRIUS CREMONE
Without transmission cover

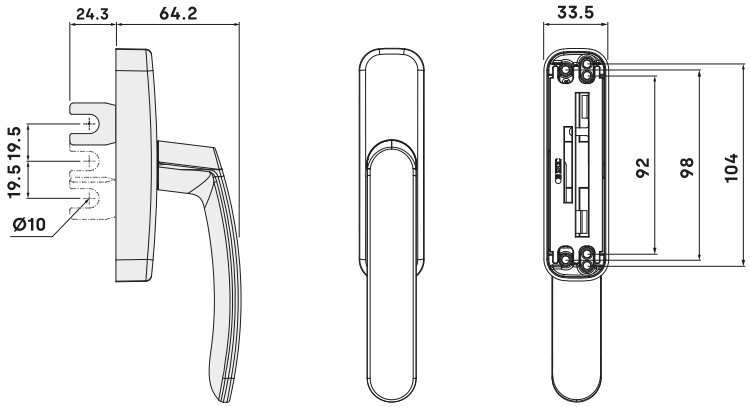
A1157	**

20

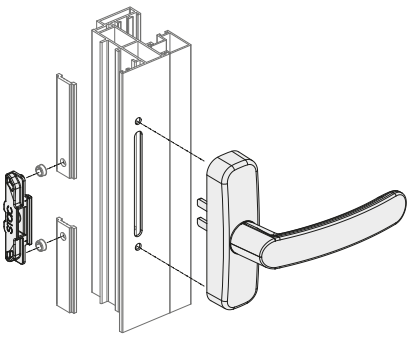
82 81

AL ZM

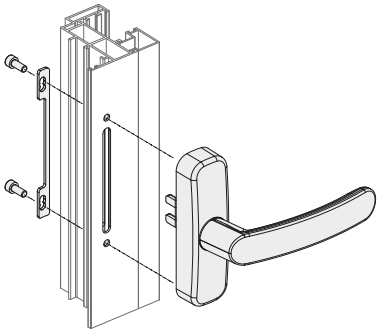
IN PA



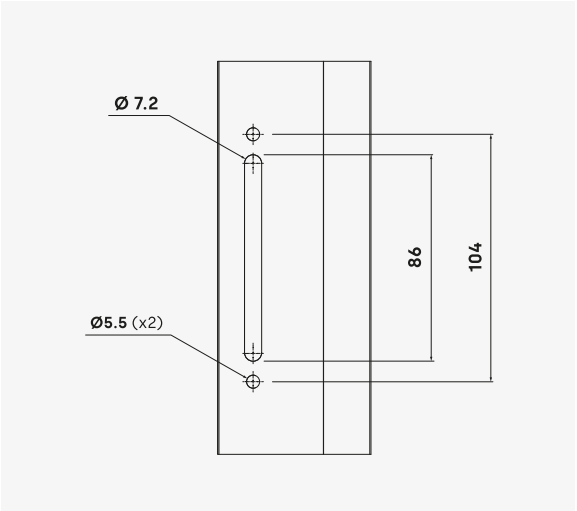
TRANSMISSION COVER



FIXING WITH COUNTERPLATE



PROFILE MACHINING



SIRIUS SLIM CREMONE

With transmission cover

A1154S	**

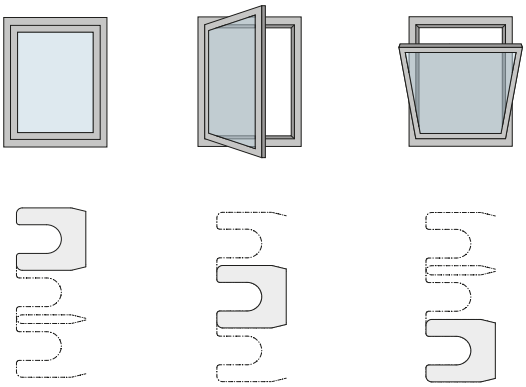
5

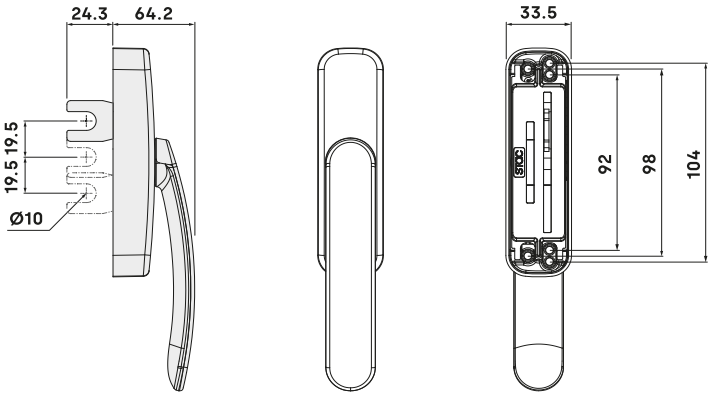
AL ZM
IN PA

Cremone with reduced handle for side-hung or til & turn windows.

- The ideal solution for installations of windows, balcony doors or sliding doors where the shutters do not leave the space necessary for the mounting of a traditional cremone.
- The cremone is compatible with the FAST fastening system.
- The levers and the pinion of the cremone are made of steel and they provide high resistance and durability to cycles.
- Internal mechanisms adjusted to eliminate clearances and ensure perfect operation.

TRANSMISSION LEVER POSITIONS





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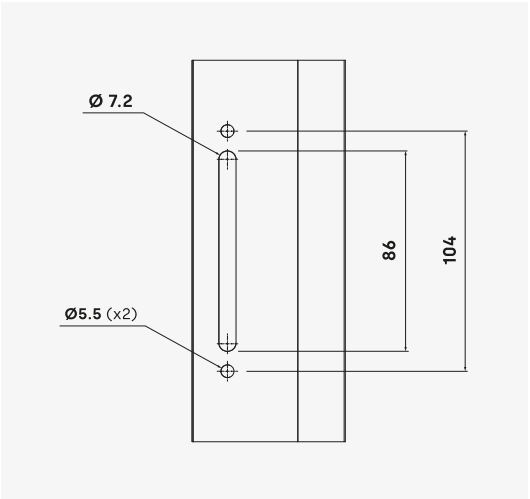
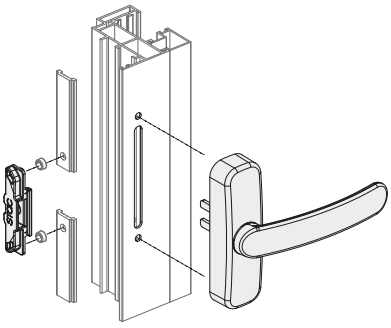
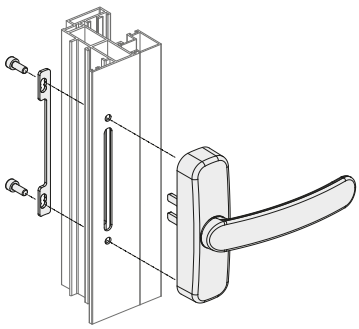


TRANSMISSION COVER



FIXING WITH COUNTERPLATE



Cremones with a security lock for hinge side-hung or tilt & turn systems.

• The lock of the cremone handle is done by means of a key.
• In windows with tilt-before-turn operation, the cremone allows two locking positions to be achieved:
– Turning key 90º, the cremone allows to tilt but not to open the window.
– Turning key 180º, the cremone is blocked, preventing both opening and tilting the window.

SIRIUS BLOCK CREMONE
With transmission cover

A1160 **

20

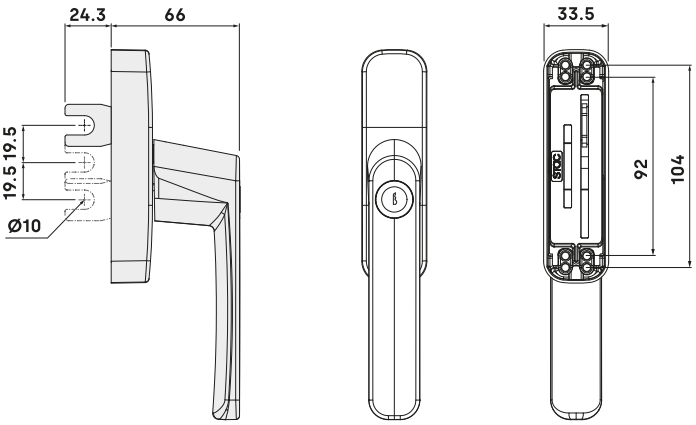
AL ZM IN PA

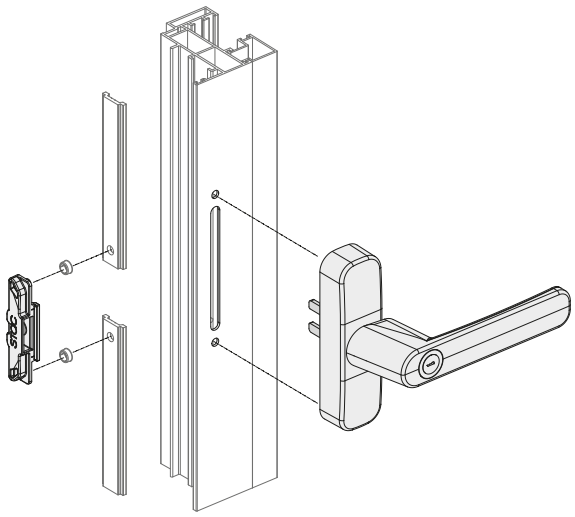
SIRIUS BLOCK CREMONE
Without transmission cover

A1163 **

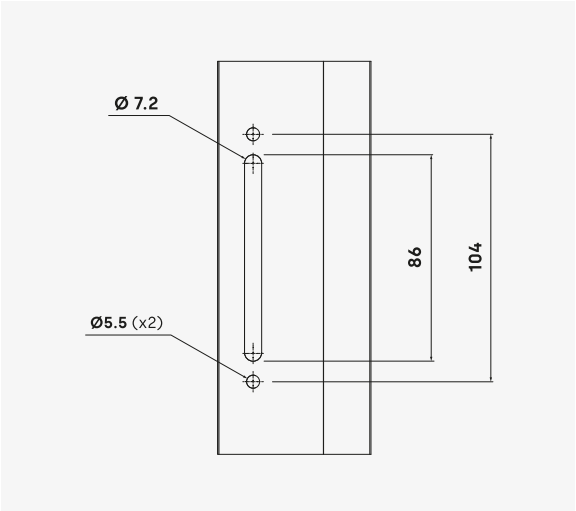
20

AL ZM IN PA





PROFILE MACHINING



SIRIUS CREMONE

detachable handle

Direct clip

Audible indication

RAL colour chart

Proven robustness

SIRIUS base for detachable handle

Nowadays, safety has become an indispensable requirement when selecting hardware or accessories.

In a market that has evolved towards systems that incorporate mechanisms that allow limiting the operation of the window through removable handles, we have managed to position our SIRIUS cremone as a reference within the options of cremone with removable handle, standing out for its standard of quality and innovation.

» Direct and grooved handle installation

The handle is securely mounted on the escutcheon. The escutcheon is equipped with a zamak mushroom-shaped support that allows direct and grooved clipping (*). We have succeeded in evolving the cavity of the bracket used for clipping, which is generally square in shape, by a double T-shaped slot.

The channel groove is hidden in the transmission shaft and is imperceptible from the front and is only 4 mm high with respect to the base. (*)

» Retention system

The polyamide retaining clip prevents the handle from slipping out of the groove of the zamak support. In this way, the handle is fully functional and operational for as long as required. To release the handle, simply press lightly on the retaining clip and release it.

» Operational drive

The double T-shaped transmission parts allow the handle to be completely restricted on the base, making it possible to pull the handle itself in both the side-hung and tilt & turn positions of the window.

» Maximum cycling durability

The double-slotted T-shaped bracket is designed to maximise the contact faces between the handle and escutcheon drive parts, while moving the contact point away from the axis of rotation to reduce torque.

SIRIUS cremone base compatible with SIRIUS detachable handle.

- Mechanism valid for hinged side-hung or tilt & turn systems.
- The detachable handle is clipped directly onto the scutcheon.
- The user can unlock and release the detachable handle by slightly pressing on the locking clip.
- Reference do not include SIRIUS handle (ref. 0507167).

SIRIUS DETACHABLE CREMONE
With transmission cover

A11541	**

5

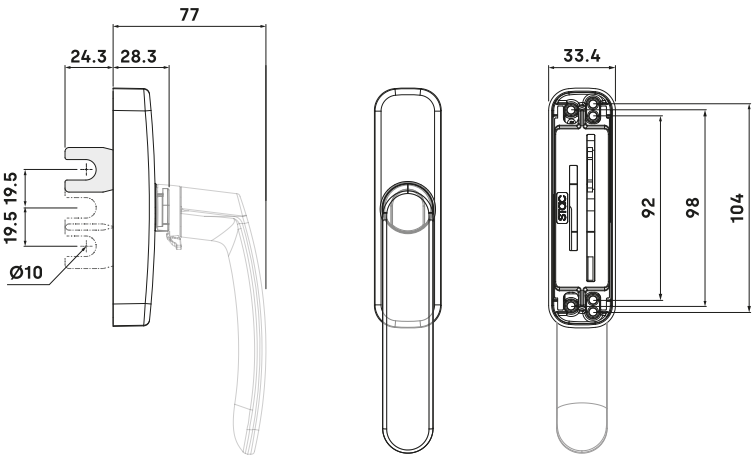
ZM IN PA PM

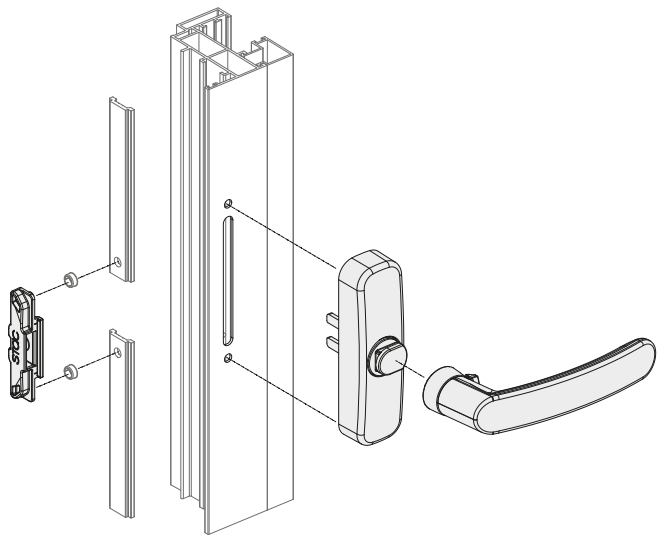
SIRIUS DETACHABLE CREMONE
Without transmission cover

A11571	**

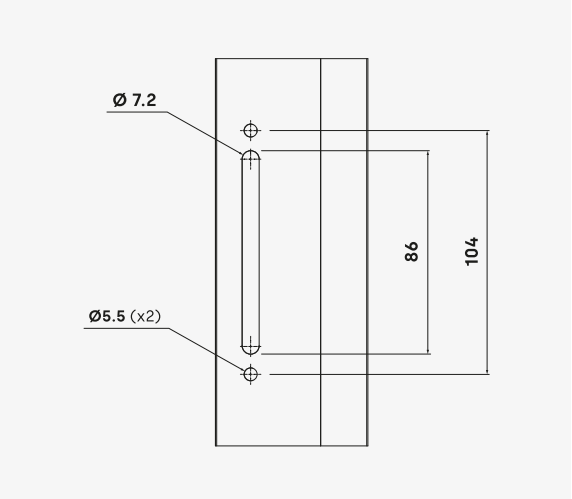
5

ZM IN PA





PROFILE MACHINING



SIRIUS DETACHABLE CREMONE
Side-hung opening

A07901 **

5

ZM IN PA

SIRIUS cremone base compatible with SIRIUS detachable handle.

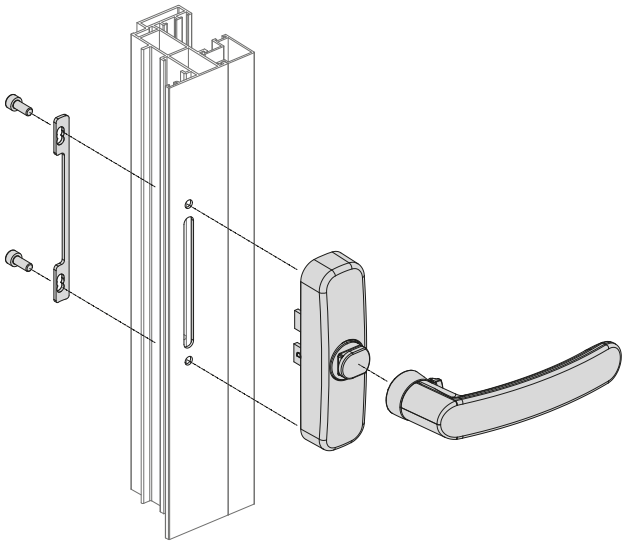
- Mechanism valid for hinged side-hung systems.
- Cremone available for unidirectional opening (ref. A07941).
- The detachable handle is clipped directly onto the scutcheon.
- The user can unlock and release the detachable handle by slightly pressing on the locking clip.
- Reference do not include SIRIUS handle (ref. 0507167).

SIRIUS DETACHABLE CREMONE
French side-hung opening

A07941 **

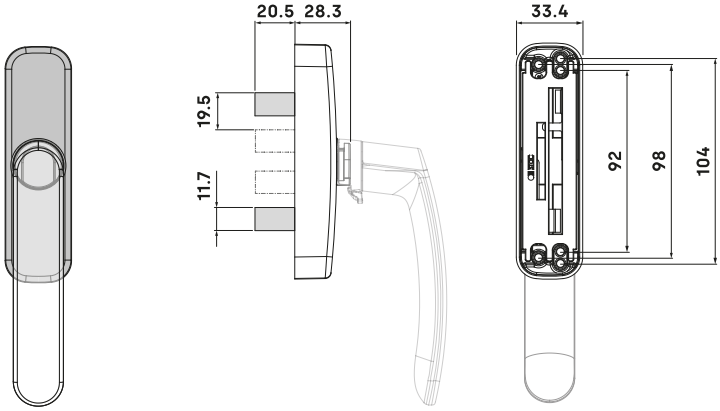
5

ZM IN PA



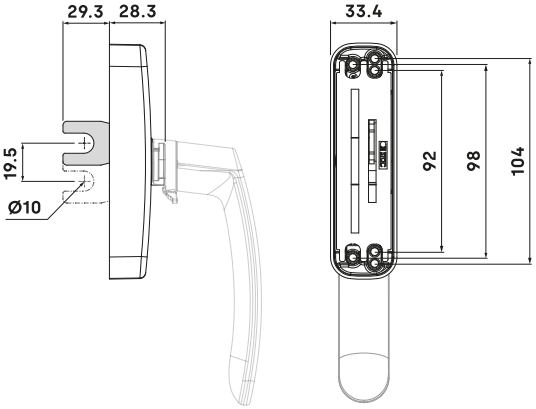
A07901 **

SIDE-HUNG OPENING



A07941 **

FRENCH STYLE OPENING



Cremones for hinge side-hung or tilt & turn systems.

- The transmission of the movement to the rods is made by the transmission cover or by rivets on the rod itself.
- The cremones can be used on side-hung windows by limiting the turn of the cremone to 90º.
- Available cremone with rectangular shield (ref. 1124).

LYRA CREMONE
With transmission cover

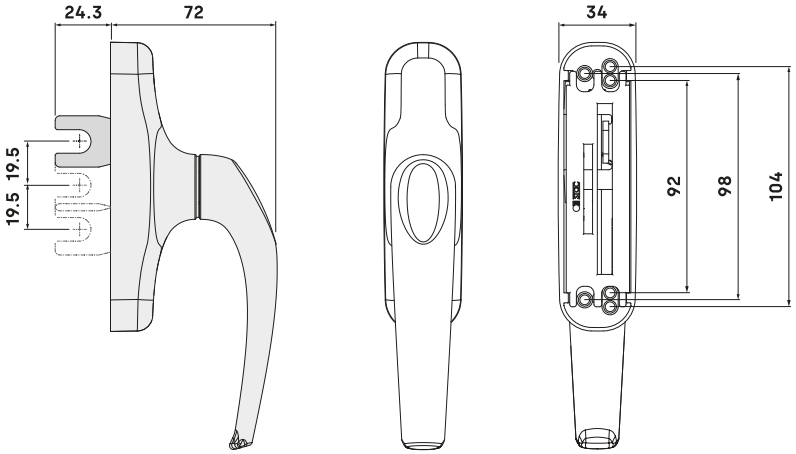
1104 **

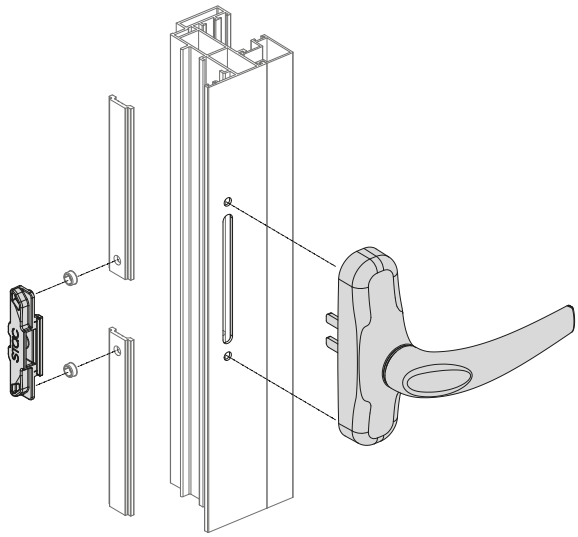
25

CREMONE LYRA
Without transmission cover

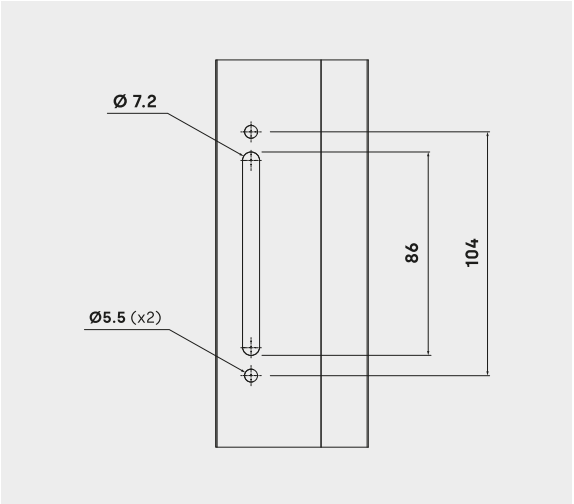
1135 **

25





PROFILE MACHINING



LYRA SQUARED CREMONE
With transmission cover

1124 | **

25

AL ZM
IN PA

Cremones for hinge side-hung or tilt & turn systems.

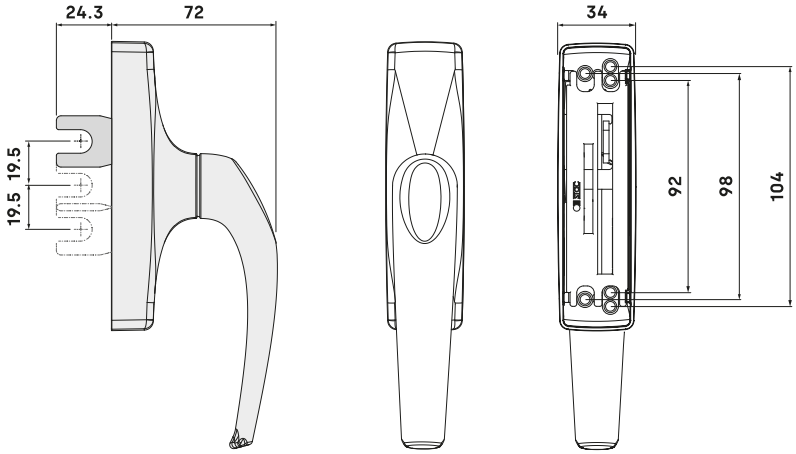
- The transmission of the movement to the rods is made by the transmission cover or by rivets on the rod itself.
- The cremones can be used on side-hung windows by limiting the turn of the cremone to 90°.
- Available cremone with round shield (ref. 1104).

LYRA SQUARED CREMONE
Without transmission cover

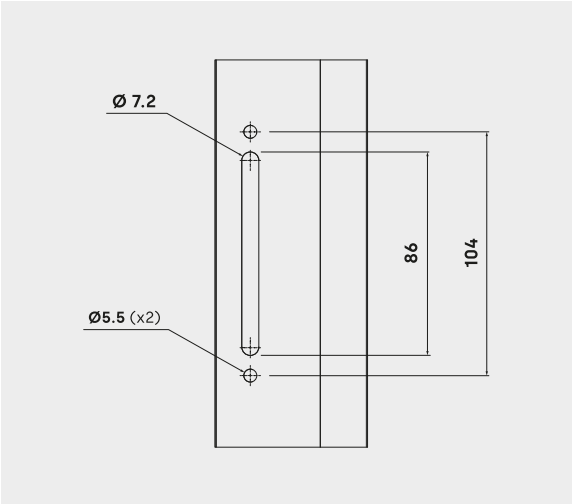
1135C | **

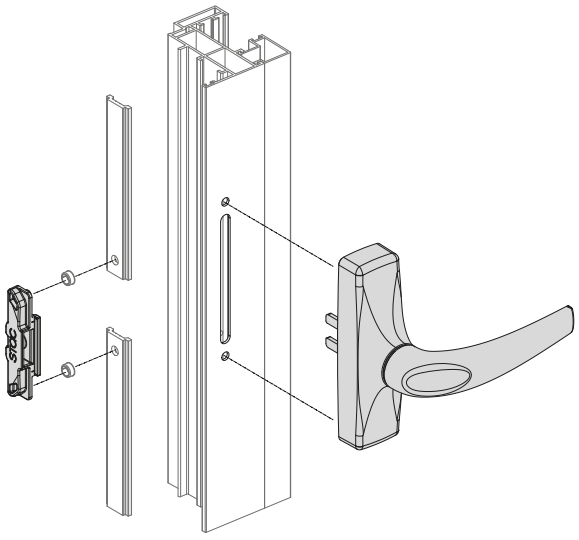
25

AL ZM
IN PA



PROFILE MACHINING





Cremone with handle locking device for hinged side-hung systems.

- The lock of the cremone handle is done by means of a key.
- The cremones can be used on side-hung windows by limiting the turn of the cremone to 90º.
- There are two locking positions for tilt-before-turn windows:
  - Turning key 90º, the cremone allows to tilt but not to open the window.
  - Turning key 180º, the cremone is blocked, preventing both opening and tilting the window.

LYRA BLOCK CREMONE
With transmission cover

AL ZM
IN PA

1129 **

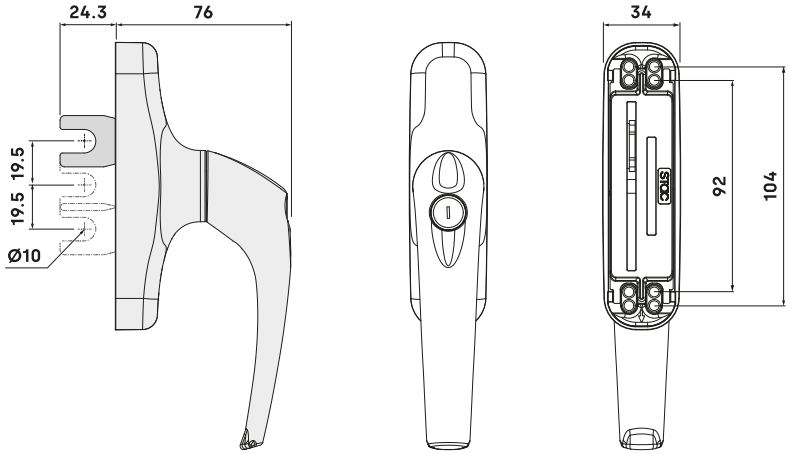
20

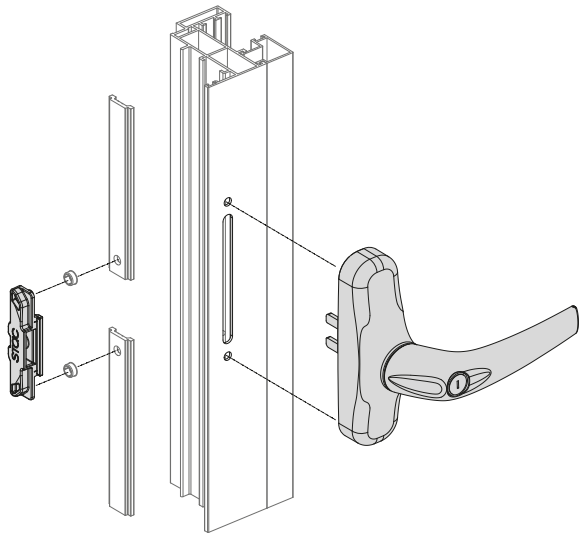
LYRA BLOCK CREMONE
Without transmission cover

AL ZM
IN PA

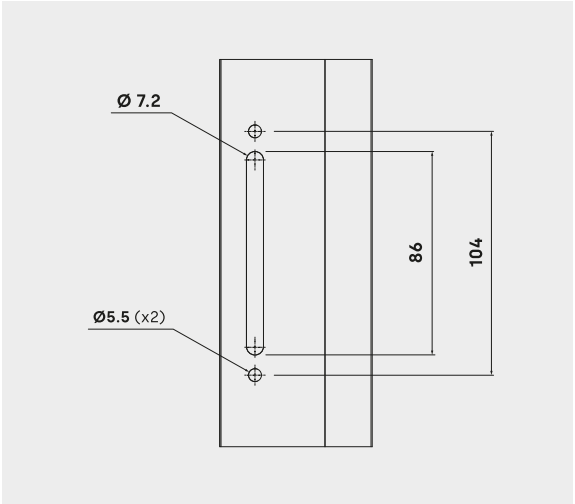
1145 **

20





PROFILE MACHINING



LYRA DETACHABLE CREMONE
With transmission cover

11041 **

5

AL ZM IN PA

LYRA cremone base compatible with LEO detachable handle.

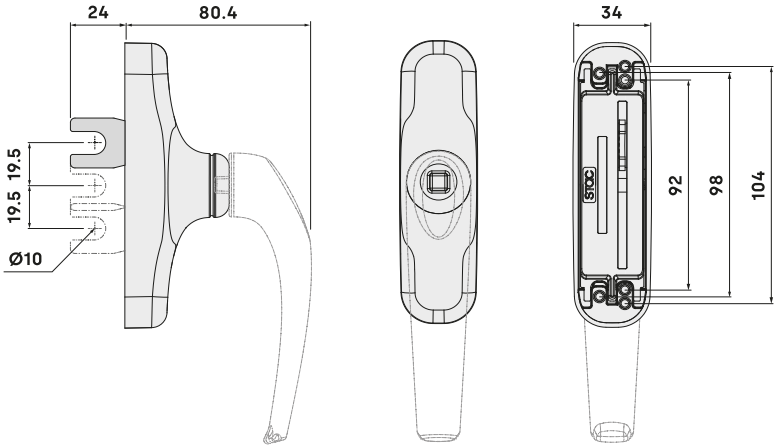
- Mechanism valid for hinged side-hung systems or tilt & turn systems.
- Cremone designed for installation in public places where opening to the outside world must be controlled: medical premises, educational centres, shopping malls,...
- When the handle is removed, the movement of the cremone is disabled, preventing the window from being opened.
- The cremone can be used on side-hung windows by limiting the turn of the cremone to 90º.
- Available cremone with rectangular shield (ref. 11041C).

LYRA DETACHABLE CREMONE
Without transmission cover

11351 **

5

AL ZM IN PA

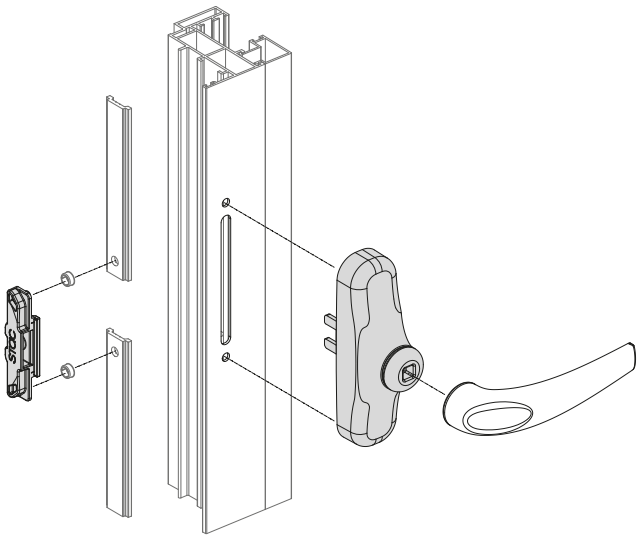


LEO REMOVABLE HANDLE

0507078 **

5

AL



LYRA cremone base compatible with LEO detachable handle.

- Mechanism valid for hinged side-hung systems or tilt & turn systems.
- Cremone designed for installation in public places where opening to the outside world must be controlled: medical premises, educational centres, shopping malls,...
- When the handle is removed, the movement of the cremone is disabled, preventing the window from being opened.
- The cremone can be used on side-hung windows by limiting the turn of the cremone to 90º.
- Available cremone with rounded shield (ref. 11041).

LYRA REMOVABLE SQUARE CREMONE

With transmission cover

11041C	**

5

AL ZM IN PA

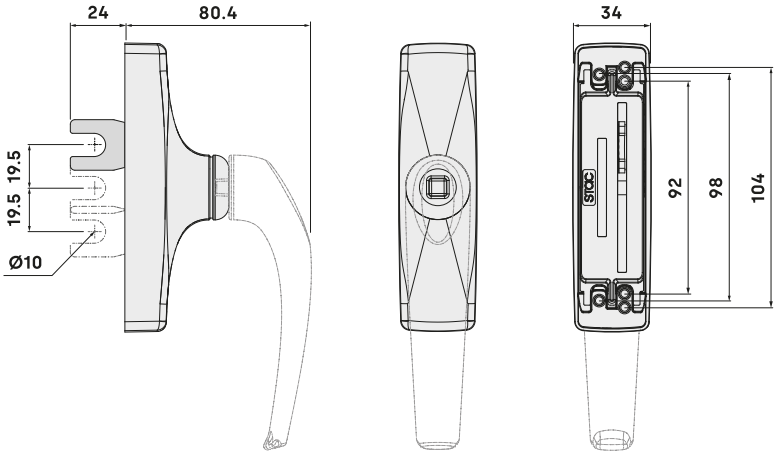
LYRA REMOVABLE SQUARE CREMONE

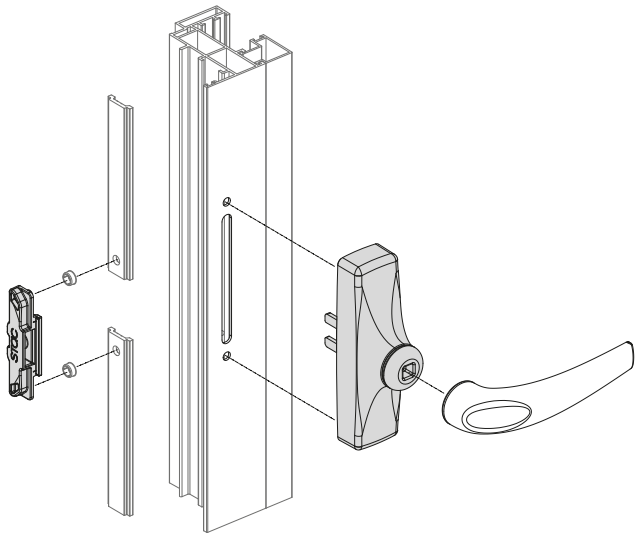
Without transmission cover

11351C	**

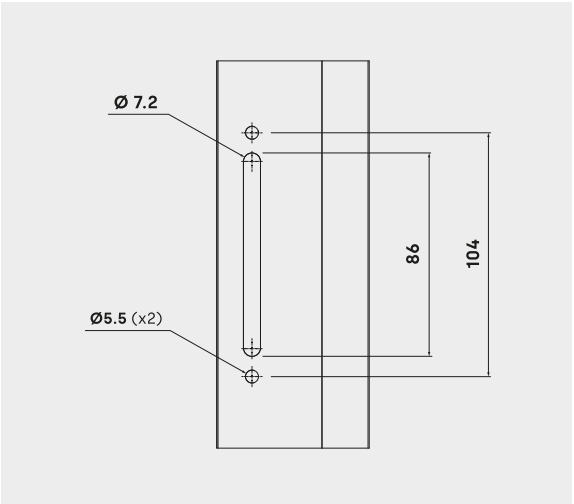
5

AL ZM IN PA





PROFILE MACHINING



LEO CREMONE
Side-hung opening

0710 **

25

AL ZM
IN PA

Cremones for hinged side-hung systems.

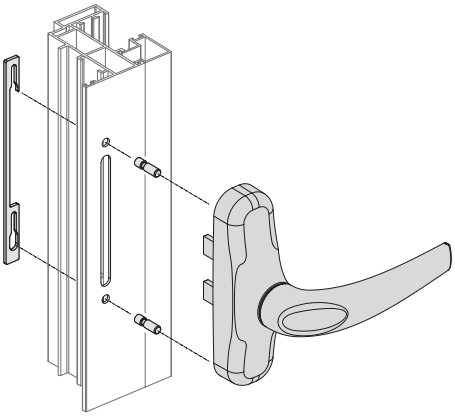
- The references are supplied for right-handed opening but can be easily moved to the left-handed by changing the position of the transmission levers.
- The cremones are compatible with the FAST fastening system.
- Available cremone with rectangular shield (ref. 0720).

LEO SQUARE CREMONE
Side-hung opening

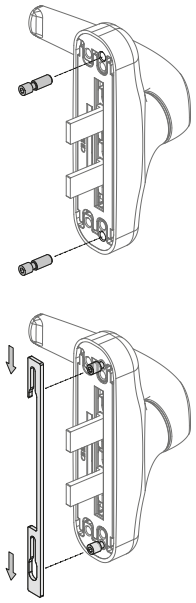
0720 **

25

AL ZM
IN PA

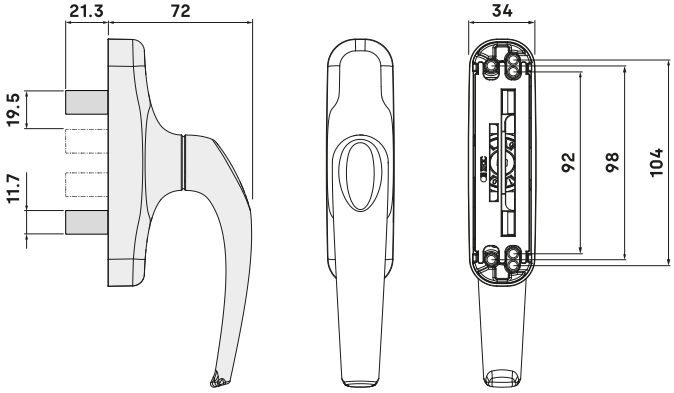


FAST FIXING SYSTEM



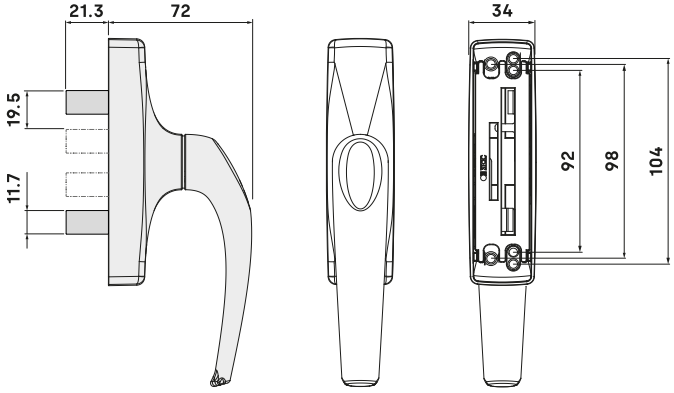
7010 **

ROUNDED SCUTCHEON



7020 **

RECTANGULAR SCUTCHEON



Handles for side-hung windows.

• Incorporates 24 millimetres transmission lever (ref. P0760 for 29 millimetres lever).
• The references are supplied for right-handed opening but can be easily moved to the left-handed by changing the position of the transmission levers.
• The cremones are compatible with the FAST fastening system.

LEO CREMONE (24 SHORT LEVER)

French side-hung opening

0760	**

25

AL ZM IN PA

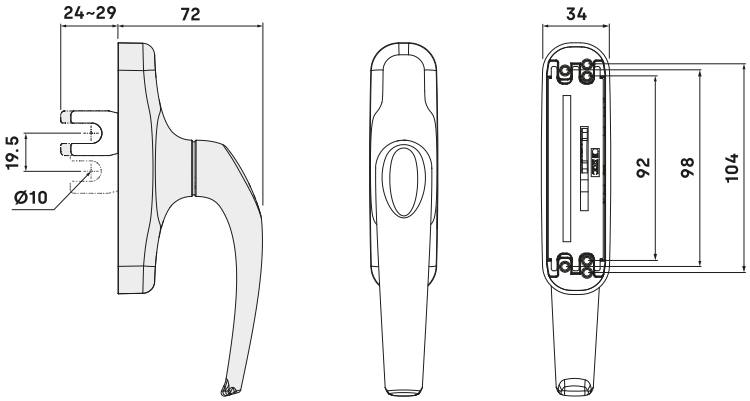
LEO CREMONE (29 LONG LEVER)

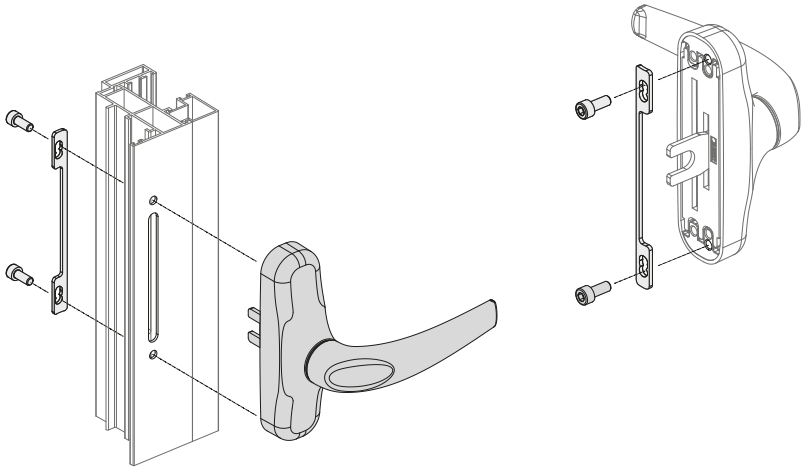
French side-hung opening

P0760	**

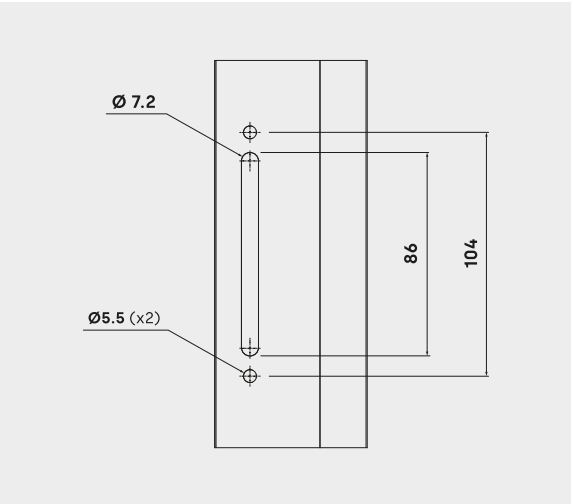
25

AL ZM IN PA





PROFILE MACHINING



LEO BLOCK CREMONE

Side-hung opening

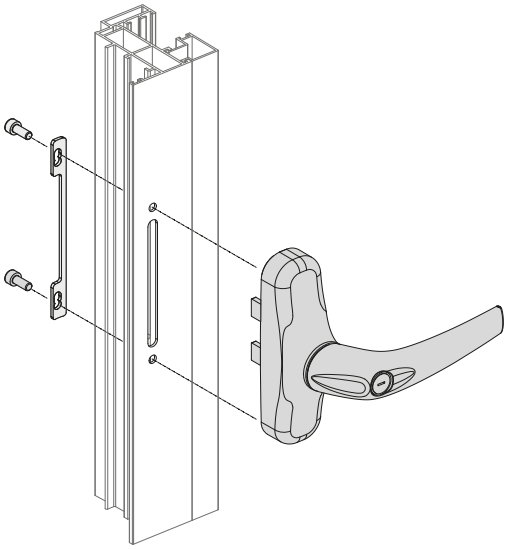
0730 **

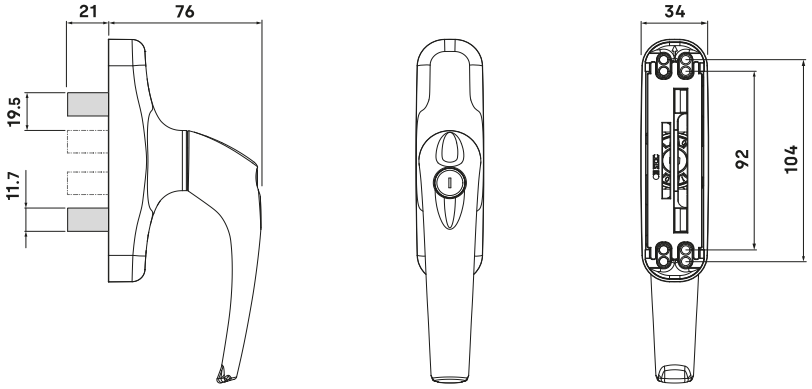
20

AL ZM IN PA

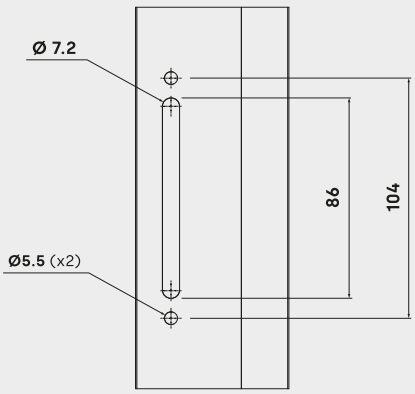
Cremone with key lock compatible with side-hung windows.

- The lock of the cremone handle is done by means of a key.
- The references are supplied for right-handed opening but can be easily moved to the left-handed by changing the position of the transmission levers.
- The cremone is compatible with the FAST fastening system.

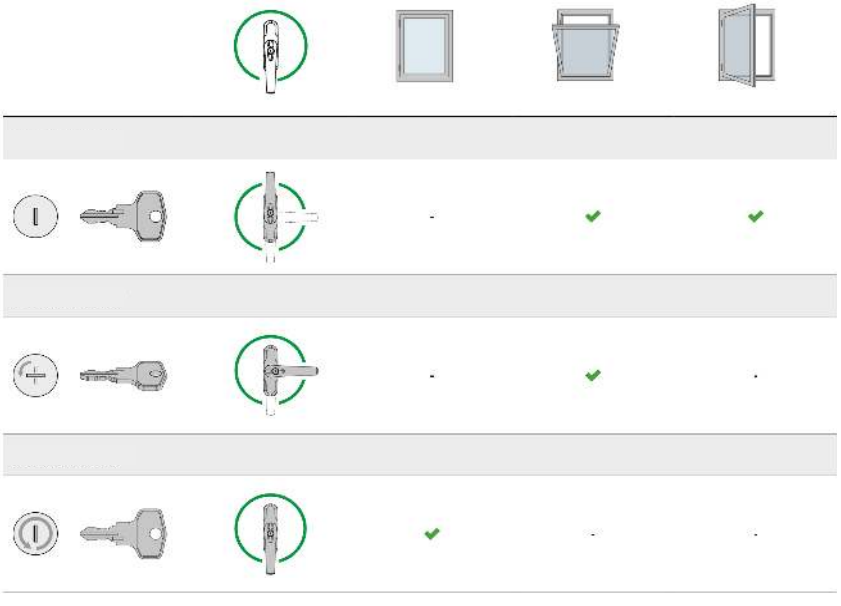




PROFILE MACHINING

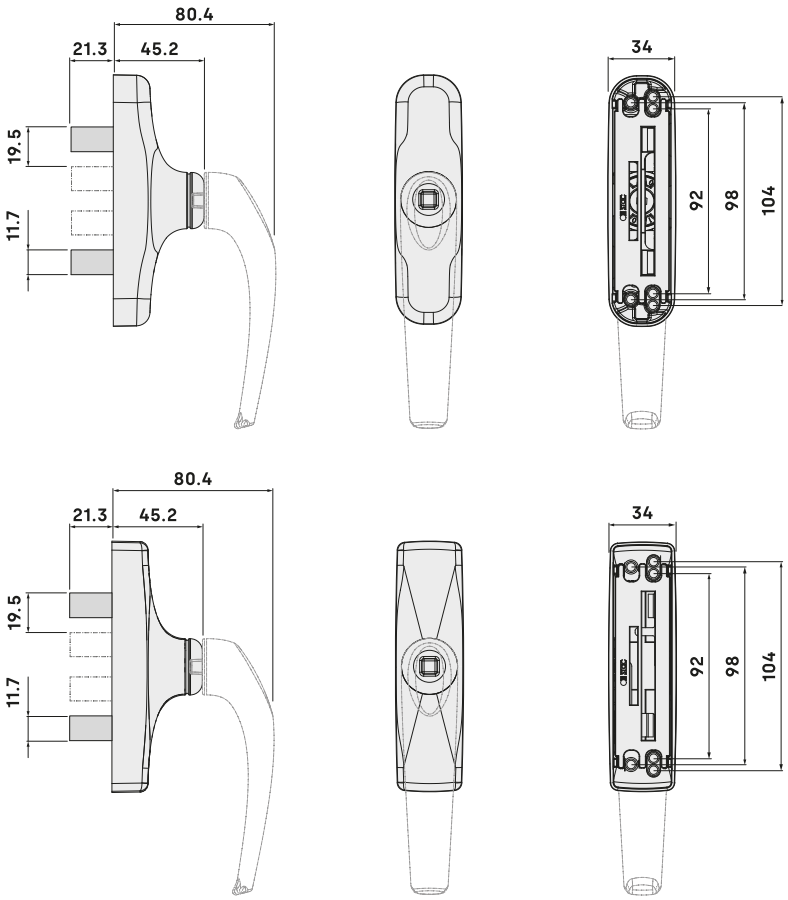


TILT-BEFORE-TURN POSITIONS

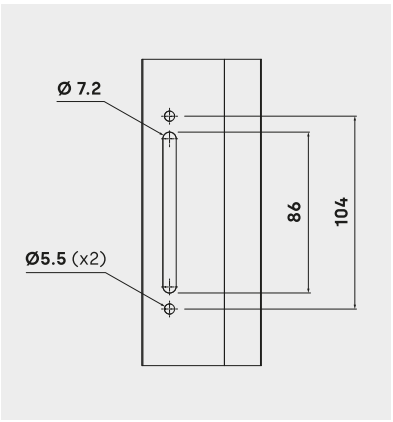


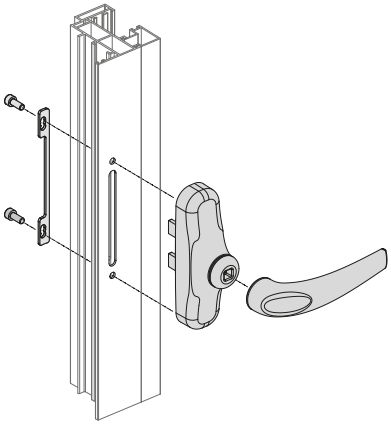
LEO cremone base compatible with LEO detachable handle.

- Mechanism valid for side-hung systems windows.
- Cremone designed for installation in public places where opening to the outside world must be controlled: medical premises, educational centres, shopping malls,...
- When the handle is removed, the movement of the cremone is disabled, preventing the window from being opened.
- Available cremone with rectangular shield (ref. 0721).
- The reference does not include the detachable LEO handle (ref. 0507078).



PROFILE MACHINING





LEO REMOVABLE CREMONE

Side-hung opening

0711	**

5

LEO REMOVABLE SQUARE CREMONE

Side-hung opening

0721	**

5

LEO REMOVABLE HANDLE

0507078	**

5

HYDRA CREMONE
Short lever (36.8 mm)

A0740 **

15

AL ZM
IN PA

Reversible window cremone for hinged side-hung windows with outward opening.

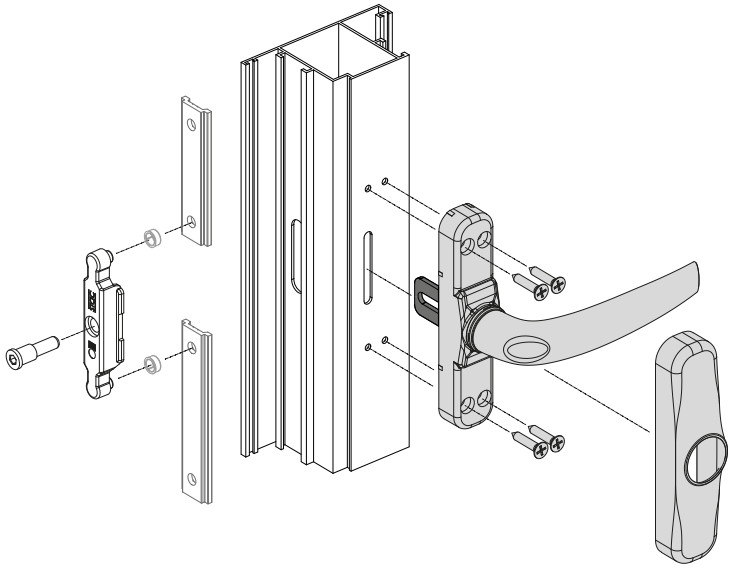
- The cremone is supplied with a short 36.8 mm transmission lever.
- Available with long lever of 47.7 mm (ref. A0750) and extra long lever of 49.8 mm (ref. A0750L).
- It has a closing system composed of the cremone itself, a bolt and a piece that acts as a support and transmission (not included).
- The base of the cremone is fixed to the profile by means of four self-tapping screws or metric thread.
- The shield allows the fixing screws to be concealed.

HYDRA CREMONE
Long lever (49.8 mm)

A0750 **

15

AL ZM
IN PA



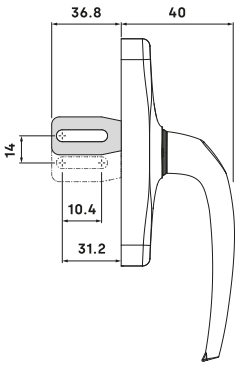
HYDRA CREMONE
Long lever (49.8 mm)

A0750L **

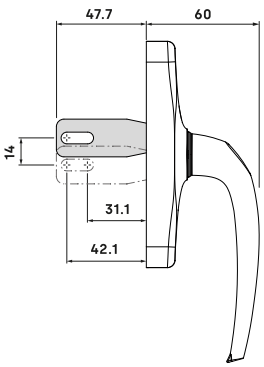
15

AL ZM
IN PA

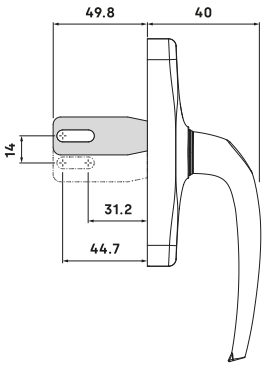
A0740 ** SHORT LEVER 36.8



A0750 ** LONG LEVER 47.7



A0750L ** EXTRA LONG LEVER 49.8



04 LIFT & SLIDE PULL HANDLES

108 **DELTA pull handle**

112 **MINIMALIST pull handle**

114 **ATRIA pull handle**

116 **DELTA finger pull**

116 **ATRIA finger pull**

117 **Pull handle accessories**

lift & slide

DELTA

Keyhole cover plate

Simple installation

Pull handle kits

RAL colour chart

DELTA range for lift & slide systems

The design stands out mainly for its minimalist, useful and functional lines. The priority of all design should be the usability of the product. That's why we focused on developing a mechanism that is integrated into the aluminium windows as much as possible, removing the superfluous characteristics that took us away from this objective.

Clean lines never go out of fashion. The design of the handle concentrates the style and character of the system, allowing us greater creative freedom.

» Pull handle and finger pull

The lift & slide pull handles allow the operation (opening and closing manoeuvres) and lifting of sashes of large weights and dimensions. The finger pull also makes it easy to slide the sash intuitively from the exterior.

The pull handles are available with and without keyhole cover plate, allowing the user to guarantee the requirements of security and resistance against attempted theft, break-ins, or malicious use.

The finger pull has a footprint depth of 9.5 millimetres, which allows it to be installed on most of the series on the market.

» Personalisation through colour

Projects strive for uniqueness through the use of new colours or finishes.

The DELTA range can be personalised and is available in both black and white lacquer and the colours that make up the RAL chart, a colour matching system that defines colours for paints, plastics and coatings.

The handles can be disassembled, allowing them to be offered in bare metal so that customers can lacquer and assemble them easily.

» Corrosion resistance

The components that make up the handles have been treated to ensure corrosion resistance exceeding 240 hours in saline mist chamber in compliance with the UNE-EN 1670 standard.

In accordance with the parameters described in the standard, a grade 4 rating has been obtained that allows the handles to be used in environments with severe climatic conditions.

Pull handle without keyhole cover plate, which allows the operation and lifting of the sashes in sliding windows.

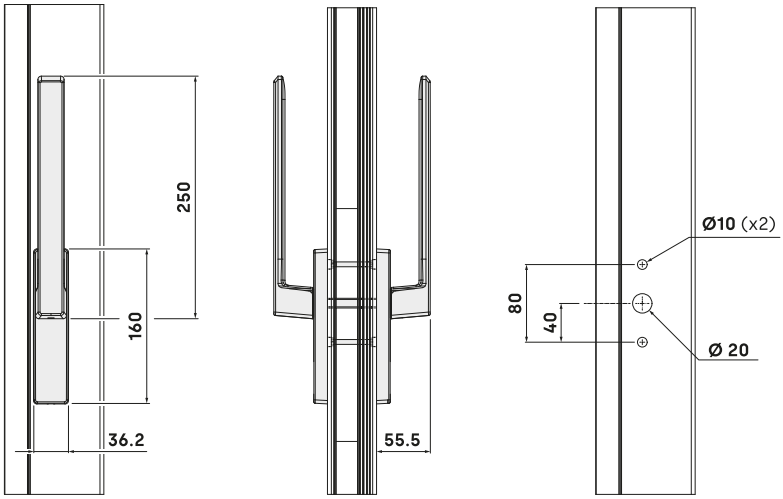
- Available in double handle set (ref. 7020) or with ATRIA 9.5 finger pull (ref. 7022).
- The handle lever can be disassembled, allowing the handle to be offered in bare metal so that customers can lacquer and assemble them easily.

DELTA PULL HANDLE
Without keyhole cover plate

7018	**

1

AL ZM AC IN

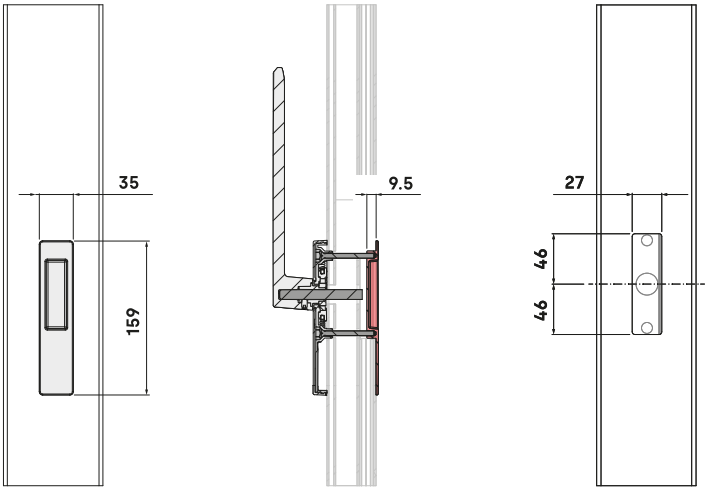


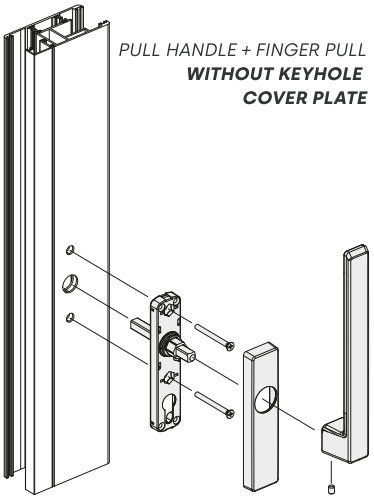
DELTA PULL HANDLE (SET)
Without keyhole cover plate

7020	**

1

AL ZM AC IN





DELTA PULL HANDLE
With keyhole cover plate

7019 **

1

AL ZM
AC IN

Pull handle with keyhole cover plate, to guarantee safety requirements, which allows the operation and lifting of the sashes in sliding windows.

• Available in double handle set (ref. 7021) or with ATRIA 9.5 mm finger pull (ref. 7023).
• The handle lever can be disassembled, allowing the handle to be offered in bare metal so that customers can lacquer and assemble them easily.

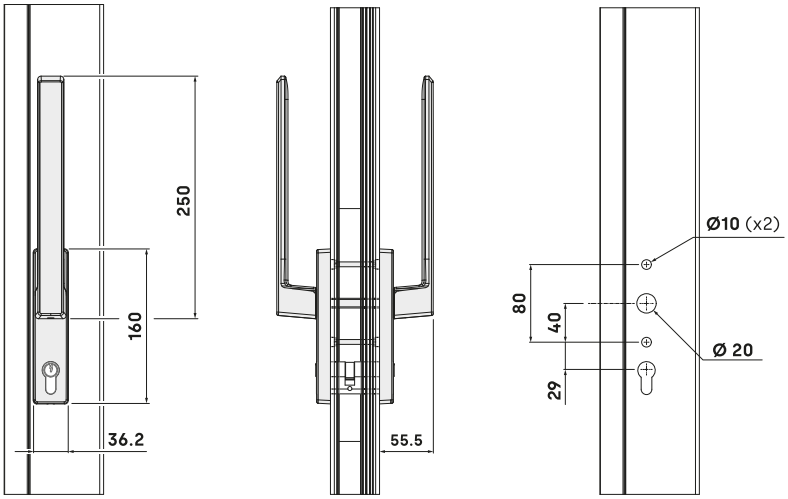
The pull handle is valid for European cylinder with 15 mm cam (not included).

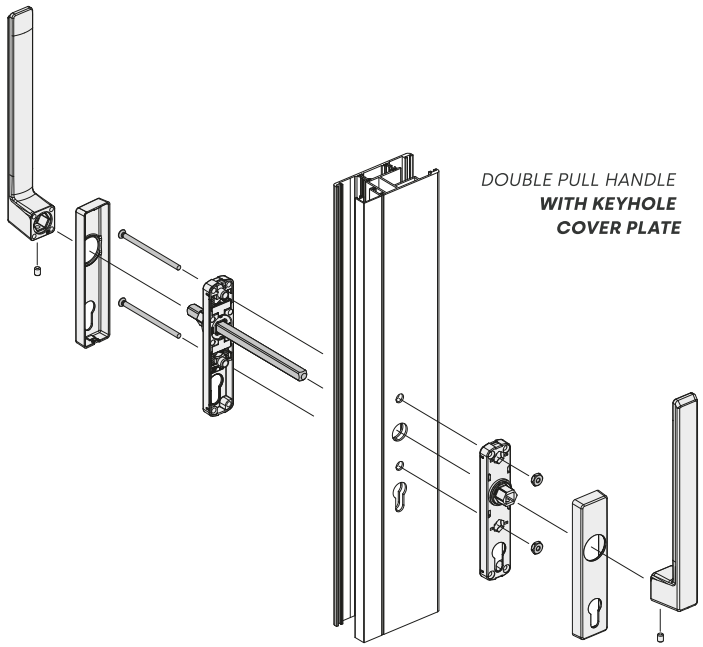
DELTA PULL HANDLE (SET)
With keyhole cover plate

7021 **

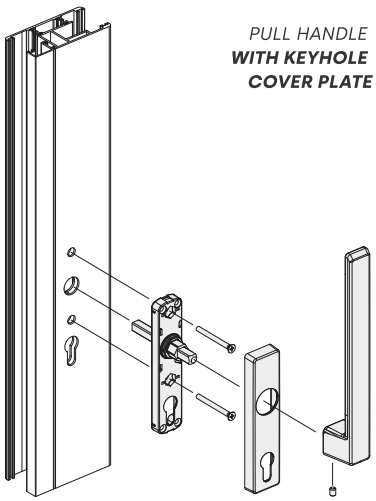
1

AL ZM
AC IN





DOUBLE PULL HANDLE
WITH KEYHOLE COVER PLATE



PULL HANDLE
WITH KEYHOLE COVER PLATE

Accessory kit which consists of a pull handle and a finger pull which allows the operation, lifting and movement of the sashes in sliding windows from the outside.

• The handle lever can be disassembled, allowing the handle to be offered in bare metal so that customers can lacquer and assemble them easily.
• The finger pull, with a 9.5 footprint, allows easy and intuitive movement of the door sash from the outside of the sliding window.

Without keyhole cover plate (ref. 7022)

• Supplied with pull handle without keyhole cover plate and a 9.5 mm footprint depth DELTA finger pull (ref. 2320).

With keyhole cover plate (ref. 7023)

• Supplied with pull handle with keyhole cover plate and a 9.5 mm footprint depth DELTA finger pull (ref. 2320).

DELTA PULL HANDLE AND 9.5 FINGER PULL
Without keyhole cover plate

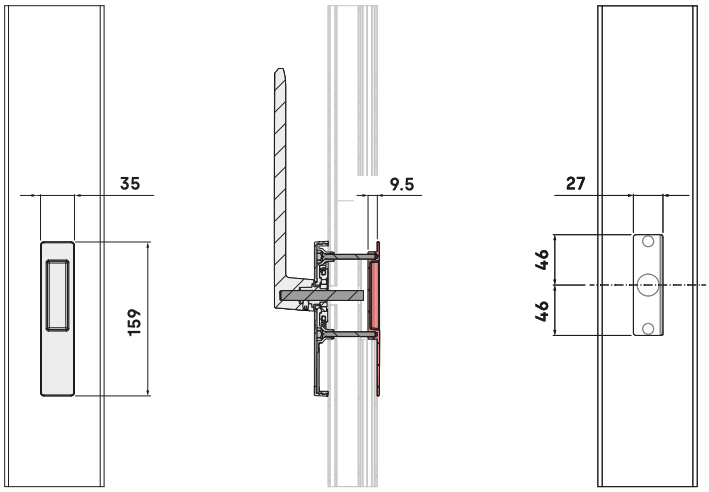
7022	**

1

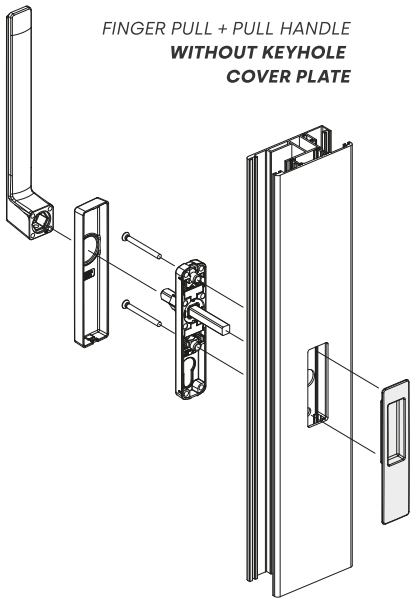
DELTA PULL HANDLE AND 9.5 FINGER PULL
With keyhole cover plate

7023	**

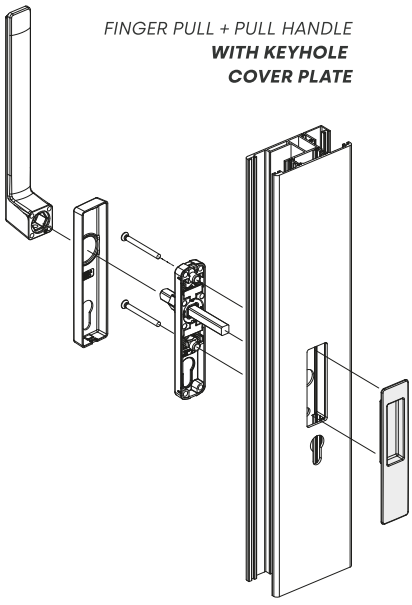
1

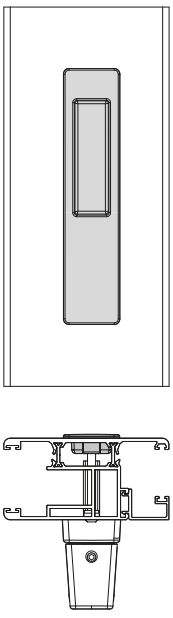


FINGER PULL + PULL HANDLE
WITHOUT KEYHOLE COVER PLATE



FINGER PULL + PULL HANDLE
WITH KEYHOLE COVER PLATE





minimalist pull handle

with concealed device

clic !

Direct clip

No visible trim

Quick installation

RAL colour chart

MINIMALIST pull handle

The trend toward minimalism in the world of architecture has spread internationally in recent years. In line with this trend, we provide innovative solutions to generate unique experiences.

We present the first, and totally new, minimalist, architectural pull handle.

» Pull handle without trim

The pull handle stands out for its simplicity and clean style, where the lines, forms and materials all come together in harmony. Despite its visual simplicity, the design is not casual and everything is planned with exhaustive precision.

The minimalist handles for lift & slide systems have no visible trim, the drive mechanism is completely hidden inside the body. In this way, both the fixing system and the indexing mechanism, which allows the opening and closing positions to be marked at 180° of rotation, are hidden from view.

» High performance

Both the strength and durability of the mechanism are important factors in the design stage, ensuring the same performance as a conventional handle.

The perception of quality is complete, highlighting the refined finish, the careful design and the smooth rotation and movement.

The practicality and simplicity mean that the installer can significantly reduce installation times and better optimize their time.

Minimalist pull handle without trim that allows the operation and elevation of the sliding window.

- Transmission and fixing system hidden in the body itself.
- Indexing system allowing the opening and closing positions to be marked.
- Supplied with 10 x 10 x 180 millimetres transmission spindle.

ATRIA finger pull (ref. 2324)

- The reference is supplied with ATRIA 9.5 mm finger pull, anti-fraction piece for drive gear and set screws, is only valid for configurations with minimalist pull handle.

MINIMALIST PULL HANDLE

2327 **

1

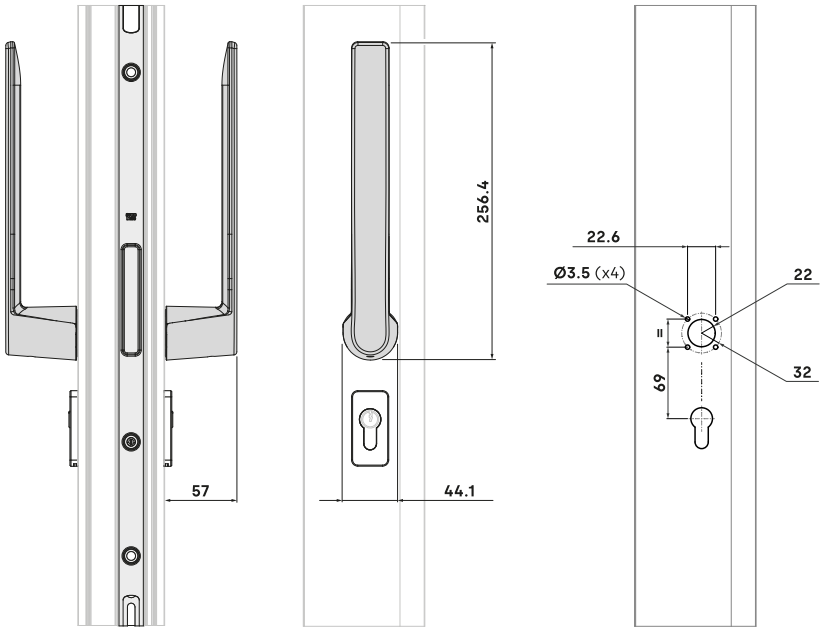
AL

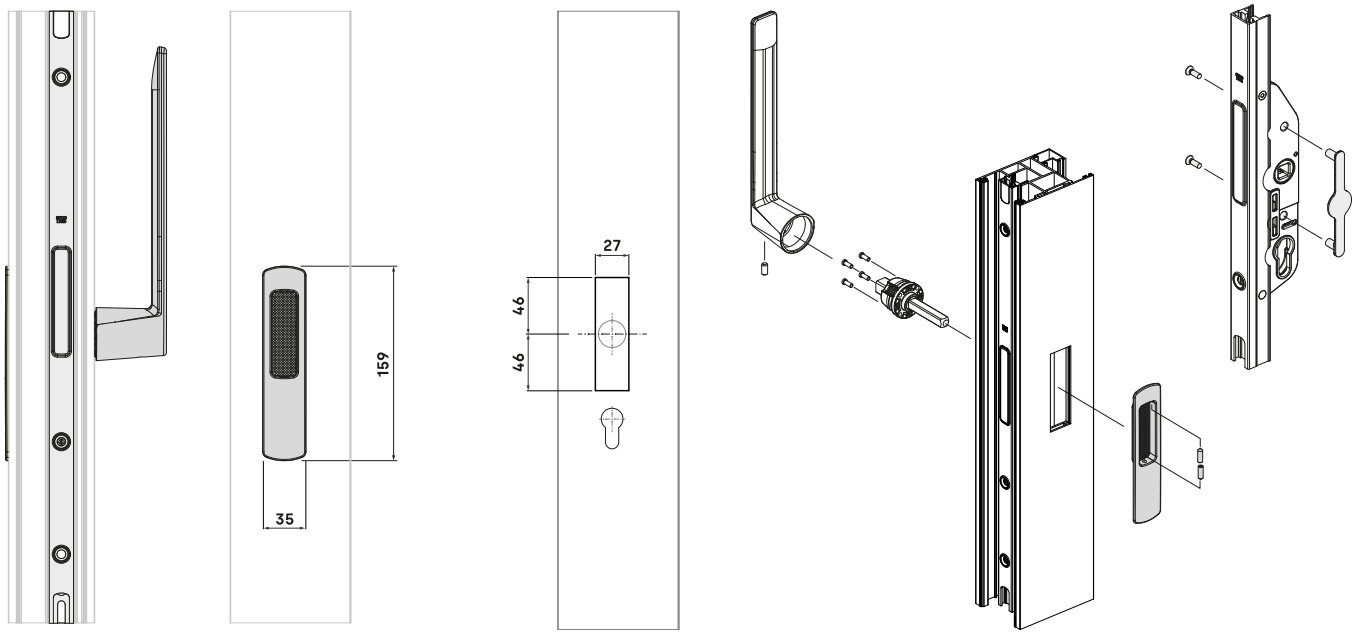
ATRIA 9.5 FINGER PULL
With set screws

2324 **

10

ZM





ATRIA PULL HANDLE
Without keyhole cover plate

2312 **

1

82 81

AL ZM

AC IN

Pull handle without keyhole cover plate, which allows the operation and lifting of the sashes in sliding windows.

- Available in double handle set (ref. 2314), with ATRIA 9.5 finger pull (ref. 2316) or ATRIA PLUS 18.5 finger pull (ref. 2326).
- The handle lever can be disassembled, allowing the handle to be offered in bare metal so that customers can lacquer and assemble them easily.
- Available with anodised finish: stainless steel or natural.

ATRIA PULL HANDLE AND 9.5 FINGER PULL SET
Without keyhole cover plate

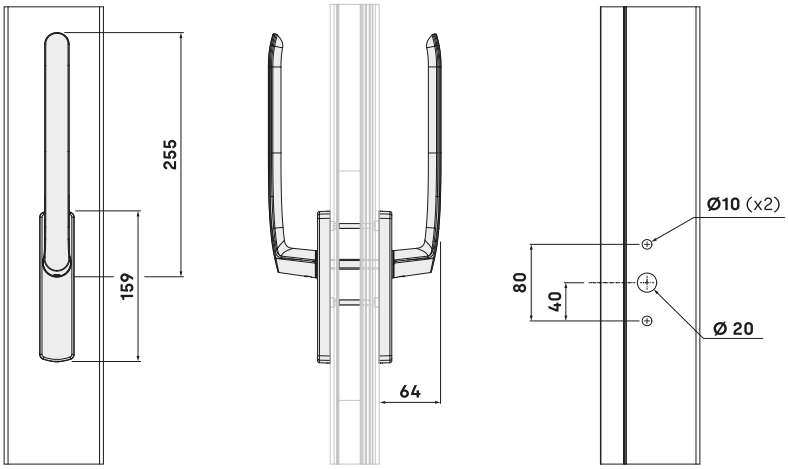
2316 **

1

82 81

AL ZM

AC IN



ATRIA PULL HANDLE AND 18.5 FINGER PULL SET
Without keyhole cover plate

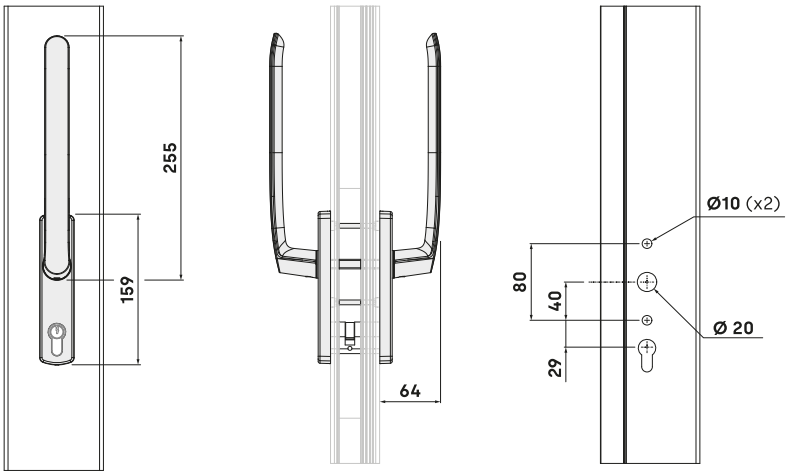
2326 **

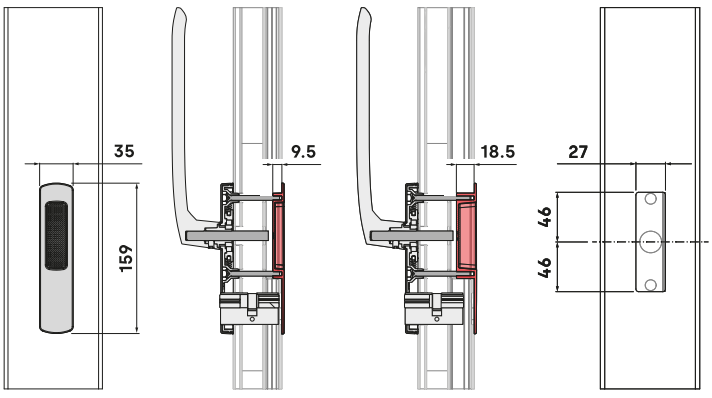
1

82 81

AL ZM

AC IN

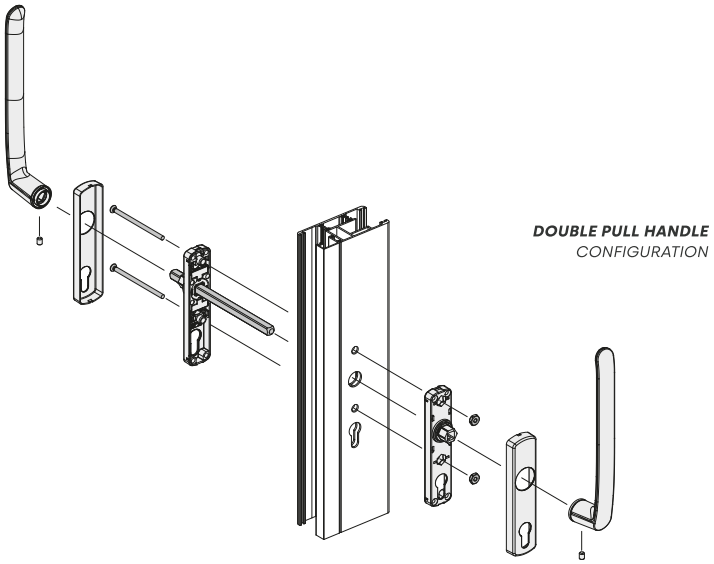




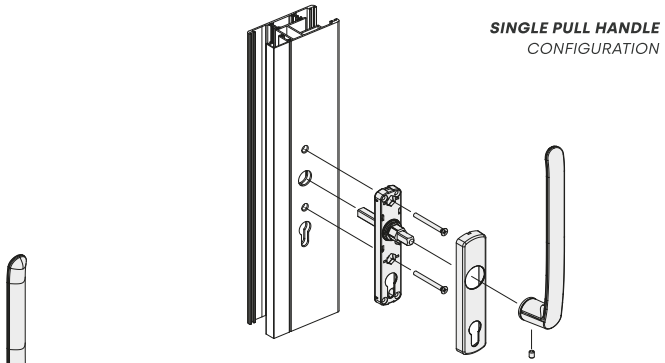
Pull handle with keyhole cover plate, to guarantee safety requirements, which allows the operation and lifting of the sashes in sliding windows.

- Available in double handle set (ref. 2313), with ATRIA 9.5 finger pull (ref. 2315) or ATRIA PLUS 18.5 finger pull (ref. 2325).
- The handle lever can be disassembled, allowing the handle to be offered in bare metal so that customers can lacquer and assemble them easily.
- Available with anodised finish: stainless steel or natural.

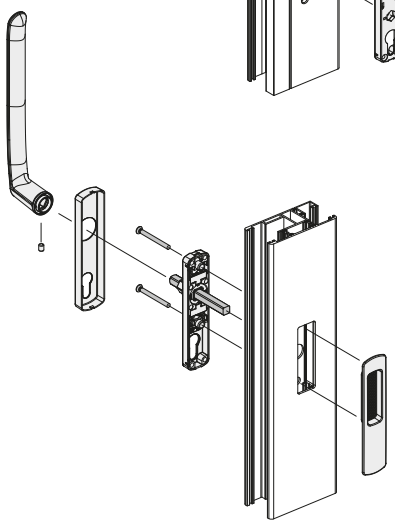
The handle is valid for European cylinder with 15 mm cam (not included).



DOUBLE PULL HANDLE
CONFIGURATION



SINGLE PULL HANDLE
CONFIGURATION



FINGER PULL
CONFIGURATION

ATRIA PULL HANDLE
With keyhole cover plate

2311	**

1

81 82

AL ZM
AC IN

ATRIA PULL HANDLE AND 9.5 FINGER PULL SET
With keyhole cover plate

2315	**

1

81 82

AL ZM
AC IN

ATRIA PULL HANDLE AND 18.5 FINGER PULL SET
With keyhole cover plate

2325	**

1

81 82

AL ZM
AC IN

ATRIA 9.5 FINGER PULL
9.5 mm depth

2320 **

20

81 82

ZM

Finger pulls that allows the movement of the sash easily and intuitively from the exterior of the lift & slide system.

• Aesthetics adapted to the ATRIA and DELTA pull handle range.

ATRIA 9.5 finger pull (ref. 2320)

• The finger pull is available with 9.5 millimetre footprint depth.
• Available with anodised finish: stainless steel or natural.

ATRIA 18.5 finger pull (ref. 2323)

• The finger pull is available with 18.5 millimetre footprint depth.

DELTA 9.5 finger pull (ref. 2323)

• The finger pull is available with 9.5 millimetre footprint depth.

ATRIA PLUS 18.5 FINGER PULL
18.5 mm depth

2322 **

20

81 82

ZM

DELTA FINGER PULL
9.5 mm depth

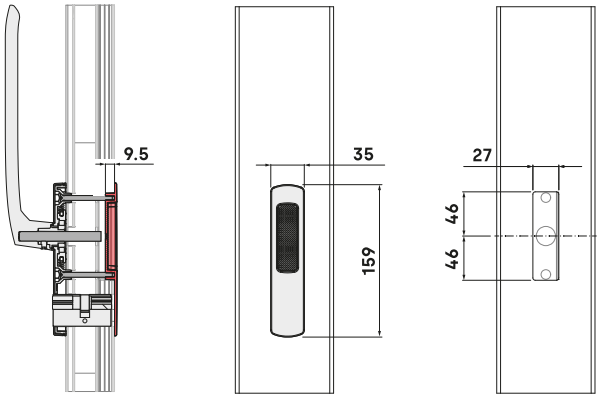
2323 **

20

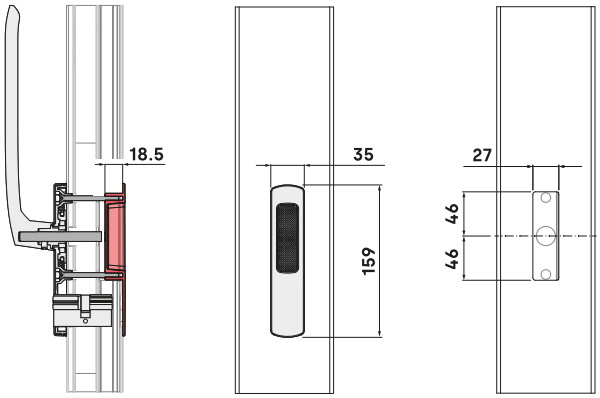
81 82

ZM

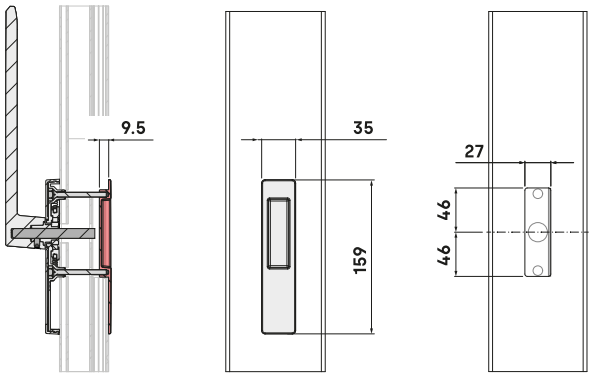
2320 ** ATRIA 9.5



2322 ** ATRIA 18.5



2323 ** DELTA 9.5



Fixing kit for single pull-handles with threaded inserts.

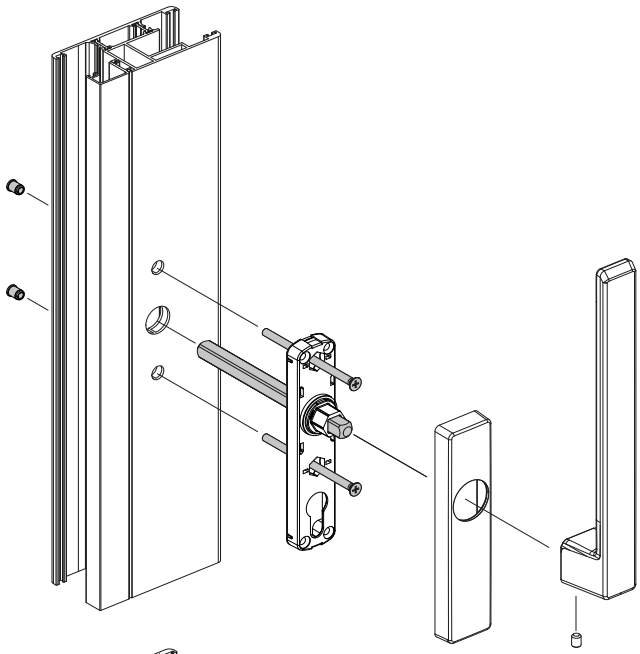
• For sash thicknesses of • Metric stainless steel screws with PH head and brass inserts.
• Zinc plated spindle with a corrosion resistance of >240 hours in a saline mist chamber.

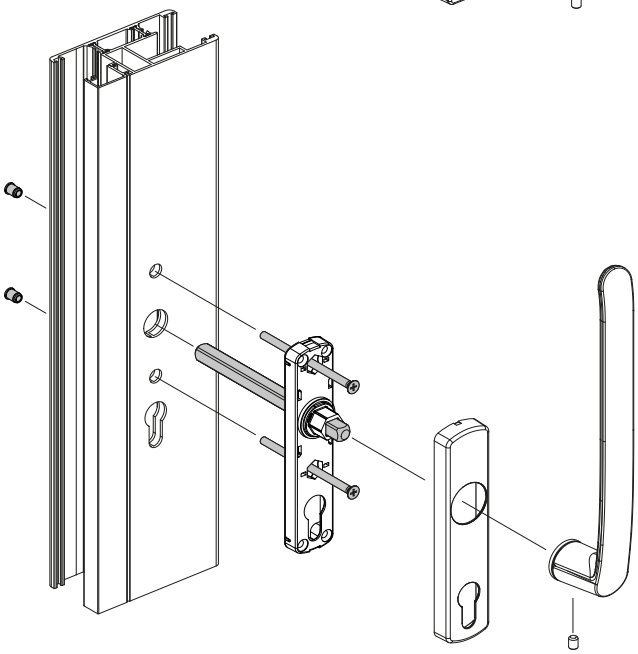
FIXING KIT
Single pull-handles configurations

AC
IN

236003

100





05 DOOR HANDLES

120 DELTA handle

122 POLARIS handle

126 MINIMALIST handle

128 SIRIUS handle

129 GEMINIS handle

130 LIBRA handle

131 INDO handle

132 KAIROX handle

134 Cylinder cover plates

DELTA

minimalist aesthetic range

Quick installation

No visible screws

Corrosion resistance

RAL colour chart

DELTA handle

For all those who long for a sense of creativity, quality and innovation in their spaces, we bring you a complete range of technical solutions. The DELTA range with geometric handle complies with the quality standards described in the UNE-EN 13126 standard.

» Compatibility with current systems

- Single and double-sash doors with both inward and outward opening.
- Sliding windows with multipoint lock systems.
- Windows with Euro Groove or Groove 16 profiles.

» Personalisation through colour

Projects find a unique look through new colours or finishes.

The range of handles and cremones is fully customisable and is available in white or black lacquer as well as in the colours that make up the RAL colour chart, a colour matching system that defines colours for paints, plastics and coatings.

» Minimalist, useful and functional design

The design must give priority to the usefulness of the product, so we have focused on developing a mechanism that is an integrated as possible in the current enclosures, avoiding those features that would take us away from this objective.

The simple design is the one that stands the test of time. The handle design focuses on the style and character of the system, which allows us great creative freedom.

» Product aesthetic

The aesthetic of a product is an integral part of its usefulness, as products used on a daily basis tend to have an indirect effect on people and their well-being. We have placed special emphasis on the ergonomics of use by eliminating any obtrusive edges for the user.

» Corrosion resistance

The components have been treated to guarantee a corrosion resistance of more than 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

Door handle set with reinforced transmission mechanism.

- Maximum recovery spring actuation system.
- The handle does not keep opening hand, it is ambidextrous.
- The reference is supplied with a zinc-plated steel transmission spindle 8 x 129 mm.
- The base of the transmission mechanism allows the fixing of up to 4 screws with double positioning each.
- The base of the transmission mechanism has been reinforced through its manufacture in zamak, giving greater hardness and resistance to the base.

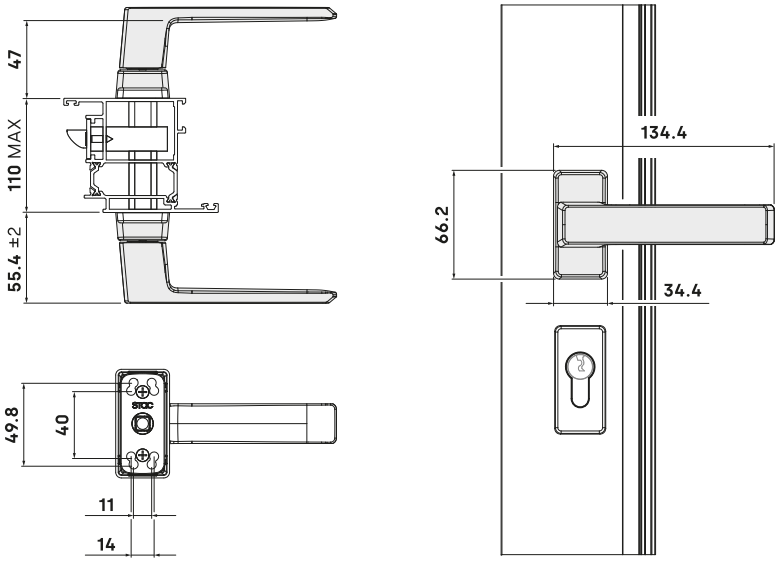
DELTA HANDLE
Handle set

7005 **

20

AL ZM
AC IN

8



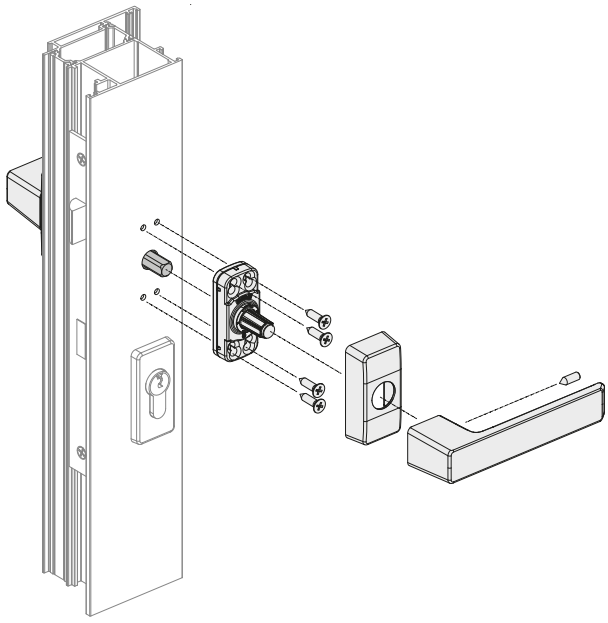
DELTA HANDLE
Single handle

7006 **

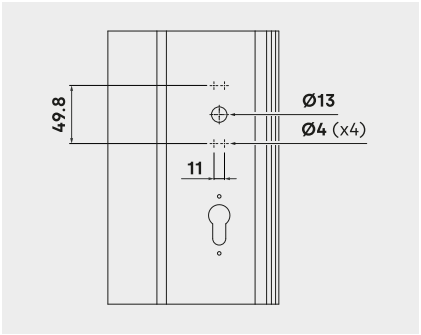
25

AL ZM
AC IN

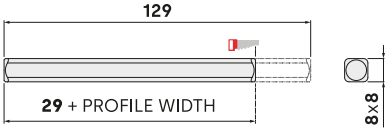
8



PROFILE MACHINING



SPINDLE CUTTING



Quick installation

No visible screws

Corrosion resistance

RAL colour chart

POLARIS handle

High performance handle range which was conceived taking in to consideration intensive use in buildings with a high frequency of passage or of large dimensions such as public use doors with door closers where the handle has to constantly withstand all types of knocks and handling. The advanced design of the handles allows for very simple assembly on to all types of profiles with very different aluminium thicknesses.

» Transmission mechanism

Base transmission mechanism that stands out primarily for its metal components which provide high cycle strength and durability, certified through multiple internal and external tests. Its features include an integrated rotary cover ring that allows for a noticeable reduction in handle installation times.

» SLIM handle

The SLIM handle is envisaged as the ideal solution for installations of balcony doors or sliding doors where the shutters do not leave the space necessary for the mounting of a traditional cremone or handle. Further, its versatility allows for conventional use as a standard handle in any type of opening existing in the door and window market.

» Proven robustness

The handles have been tested in accordance with UNE-EN 1906, "Building hardware. Lever handles and knob furniture", obtaining a grade 2 classification in category of use: suitable for use on doors with an average frequency of use by attentive persons and where there are some risks of misuse, such as interior office doors.

» Corrosion resistance

The components have been treated to guarantee a corrosion resistance of more than 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

In accordance with the parameters described in the standard, a grade 4 classification has been obtained, allowing the handles to be used in environments with severe climatic conditions.

Door handle set with recovery spring, specially designed for use on doors with high frequency of use.

- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with a zinc-plated steel transmission spindle 8 x 130 mm.
- Different fixing systems to the profile are available: by means of self-tapping screws, rivet nuts or security screws.

POLARIS HANDLE
Handle set

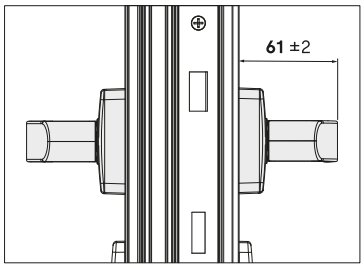
05182 **

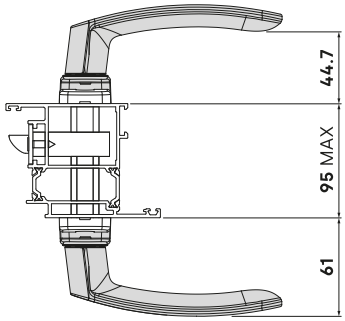
12

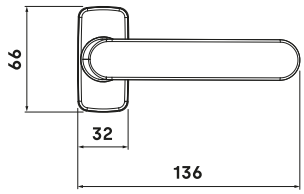
AL ZM AC IN

8

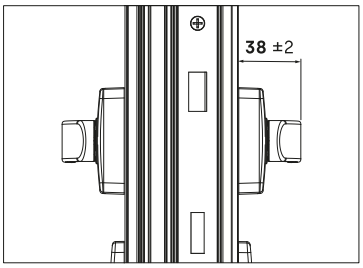
05182 **

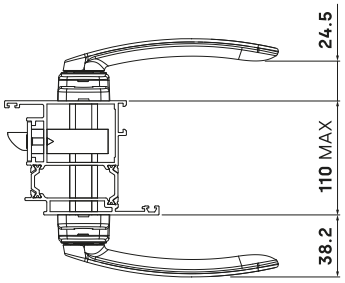


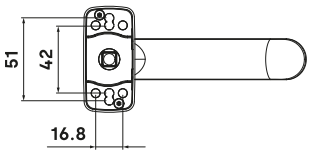




05182S **







POLARIS HANDLE
Single handle

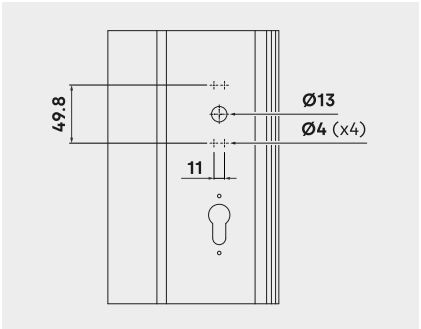
05183 **

20

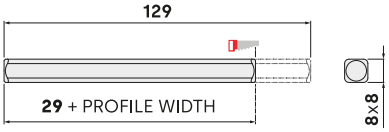
AL ZM AC IN

8

PROFILE MACHINING



SPINDLE CUTTING



FIXING KITS FOR POLARIS HANDLES		MIN PROFILE WIDTH	MAX PROFILE WIDTH
KT0518-01	WITH RIVET NUTS	-	-
KT0518-02	WITH SECURITY SCREWS	-	D = 60
KT0518-03	WITH SECURITY SCREWS	D = 60	D = 80

POLARIS SLIM HANDLE
Handle set

05182S **

5

AL ZM AC IN

8

Door handle set with recovery spring, specially designed for use on doors with high frequency of use.

- The handle is envisaged as the ideal solution for installations of doors where the shutters do not leave the space necessary for the mounting of a traditional handle.
- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with a zinc-plated steel transmission spindle 8 x 130 mm.
- Different fixing systems to the profile are available: by means of self-tapping screws, rivet nuts or security screws.

POLARIS SLIM HANDLE
Single handle

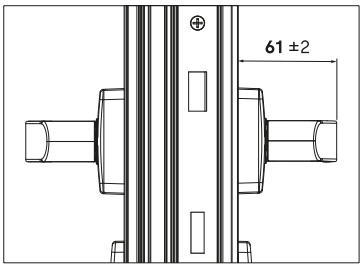
05183S **

5

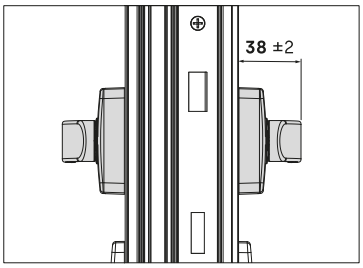
AL ZM AC IN

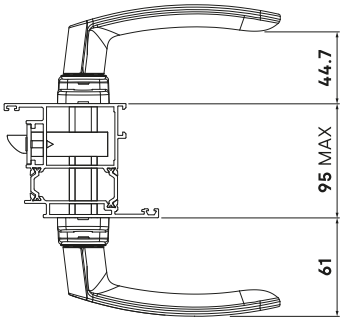
8

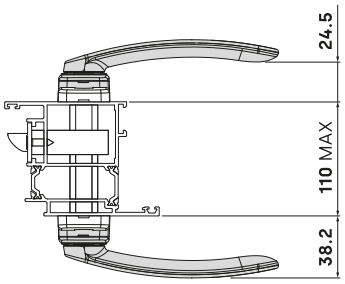
05182 **

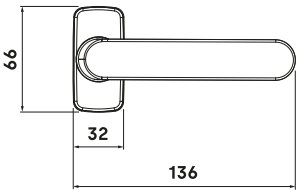


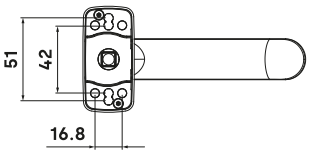
05182S **



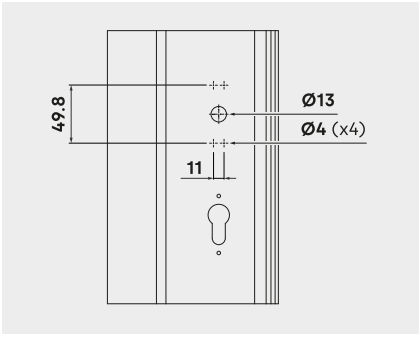




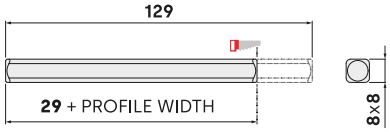




PROFILE MACHINING



SPINDLE CUTTING



FIXING KITS FOR POLARIS HANDLES		MIN PROFILE WIDTH	MAX PROFILE WIDTH
KT0518-01	WITH RIVET NUTS	-	-
KT0518-02	WITH SECURITY SCREWS	-	D = 60
KT0518-03	WITH SECURITY SCREWS	D = 60	D = 80

Fixing kits for POLARIS door handles

FIXING KIT

KT0518

Rivet nuts (ref. KT0518-01)

• Supplied with DIN 965 x 20 mm M5 metric screws and M5 rivet nuts.

Security screws (ref. KT0518-02)

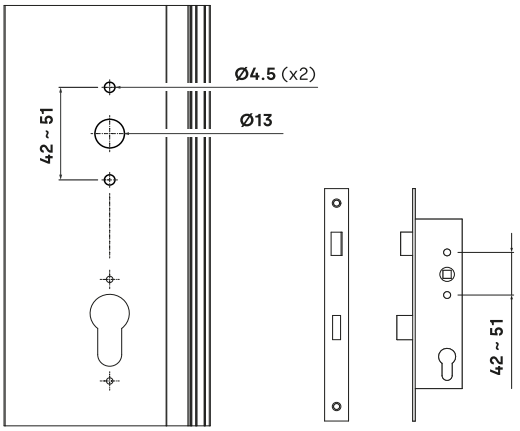
• Supplied with metric DIN 965 x 65 mm M4 screws and M4 cap nuts.

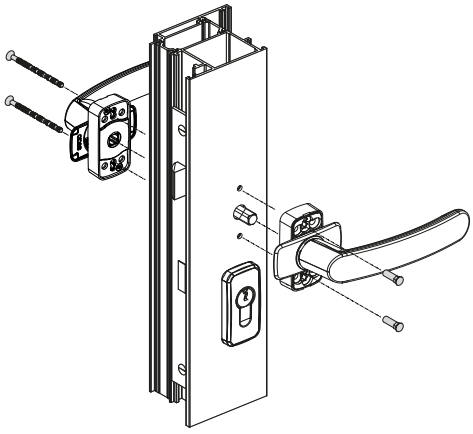
Security screws (ref. KT0518-03)

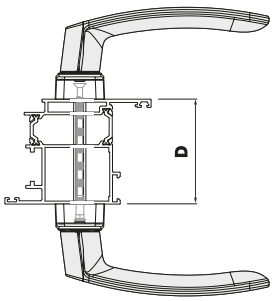
• Supplied with metric DIN 965 M4 x 80 mm screws and M4 cap nuts.

	MIN PROFILE WIDTH	**MAX** PROFILE WIDTH	**SCREWS**	**NUTS**	
KT0518-01	-	-	DIN 965 M5 x 20	M5	*100*
KT0518-02	-	D = 60	DIN 965 M4 x 65	M4	*50*
KT0518-03	D = 60	D = 80	DIN 965 M4 x 80	M4	*50*

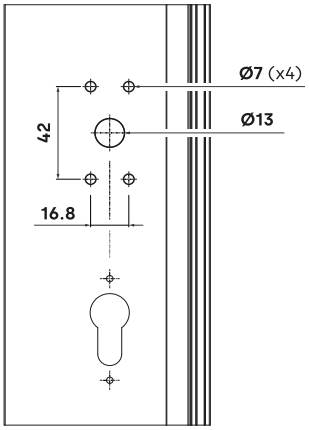
FIXING WITH
SECURITY SCREWS

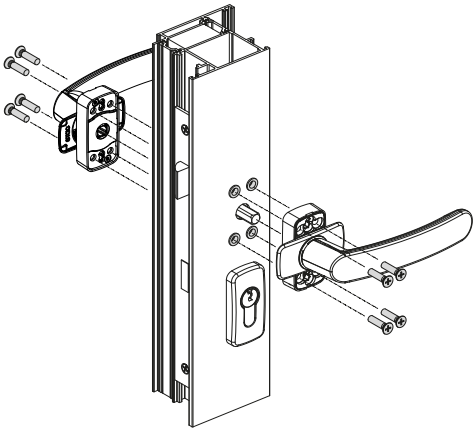


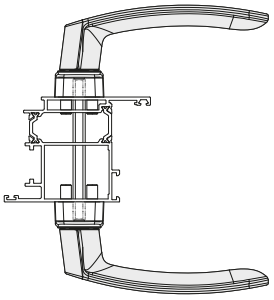




FIXING WITH
RIVET NUTS







DELTA MINIMALIST HANDLE
Handle set

A7771	**

1

AL ZM
AC IN

8

Minimalist door handle set with recovery spring.

- Handle without escutcheon and with concealed drive mechanism
- The reference is supplied with a zinc-plated steel transmission spindle 8 x 100 mm.
- Available in simple configuration with right-hand opening (ref. A7781) or left-hand opening (ref. A7791).
- Fixing systems to the profile: by means of TORX screws or rivet nuts.

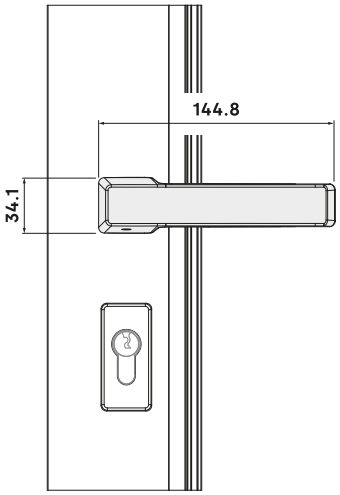
DELTA MINIMALIST HANDLE
Right-hand opening

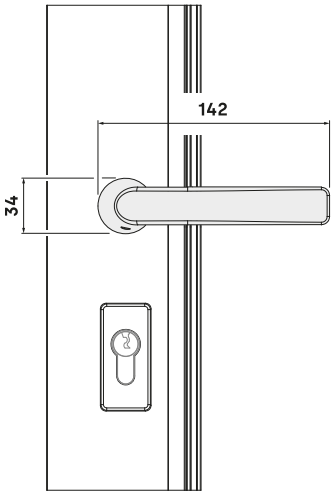
A7781	**

1

AL ZM
AC IN

8





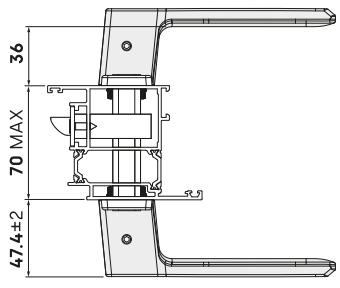
DELTA MINIMALIST HANDLE
Left-hand opening

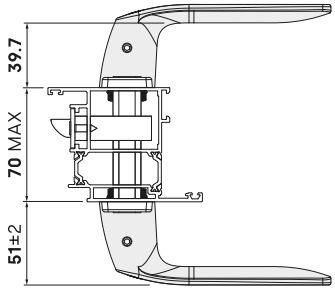
A7791	**

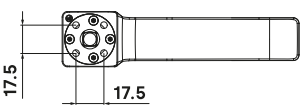
1

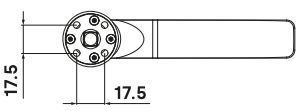
AL ZM
AC IN

8









Minimalist door handle set with recovery spring.

- Handle without escutcheon and with concealed drive mechanism
- The reference is supplied with a zinc-plated steel transmission spindle 8 x 100 mm.
- Available in simple configuration with right-hand opening (ref. 7783) or left-hand opening (ref. 7793).
- Fixing systems to the profile: by means of TORX screws or rivet nuts.

GAMMA MINIMALIST HANDLE
Handle set

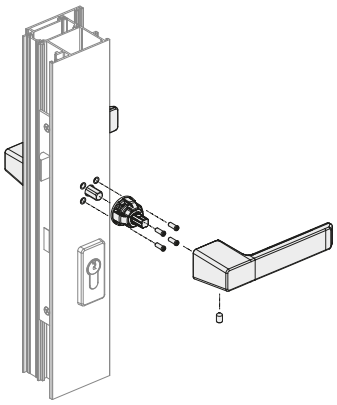
7773 **

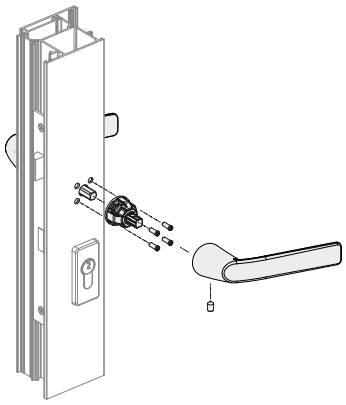
1

AL ZM AC IN

8

MINIMALIST FIXING	MIN PROFILE WIDTH	
TORX SCREWS	1.6 mm	Ø3 (x4)
RIVET NUTS	-	Ø6 (x4)





GAMMA MINIMALIST HANDLE
Right-hand opening

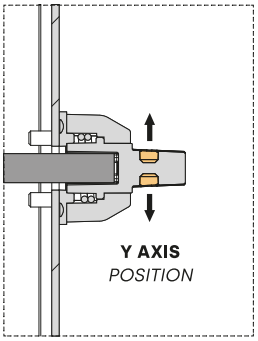
7783 **

1

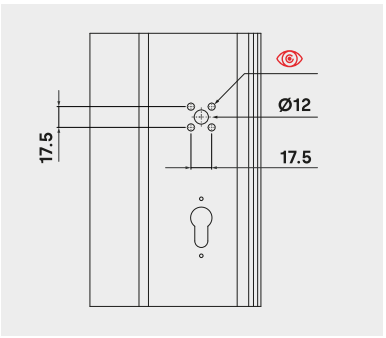
AL ZM AC IN

8

BASE FIXING



PROFILE MACHINING



GAMMA MINIMALIST HANDLE
Left-hand opening

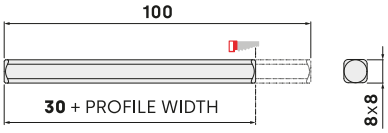
7793 **

1

AL ZM AC IN

8

SPINDLE CUTTING



SIRIUS HANDLE
Handle set

A05082 **

20

AL ZM AC IN

8

Reinforced door handle set with recovery spring.

- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with zinc-plated steel transmission spindle 8 x 129 mm.
- The base of the transmission mechanism has been reinforced through its manufacture in zamak, giving greater hardness and resistance to the base.
- The base of the transmission mechanism allows the fixing of up to 4 screws with double positioning each.

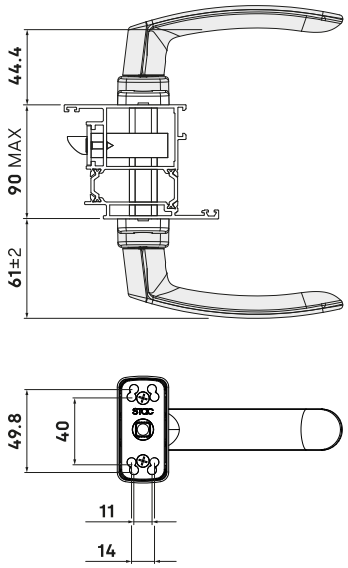
SIRIUS HANDLE
Single handle

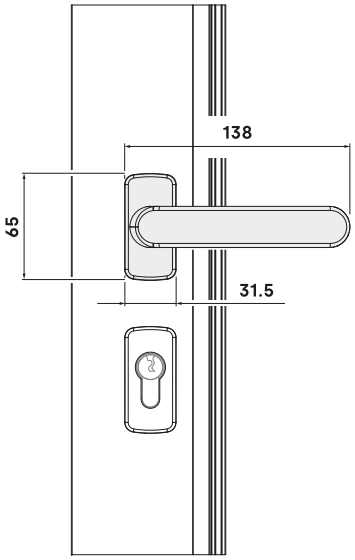
A05083 **

25

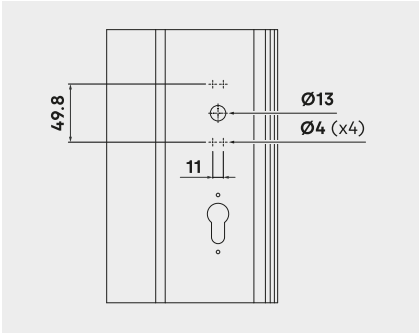
AL ZM AC IN

8

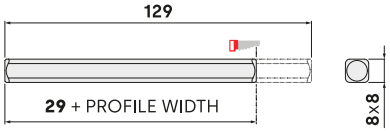


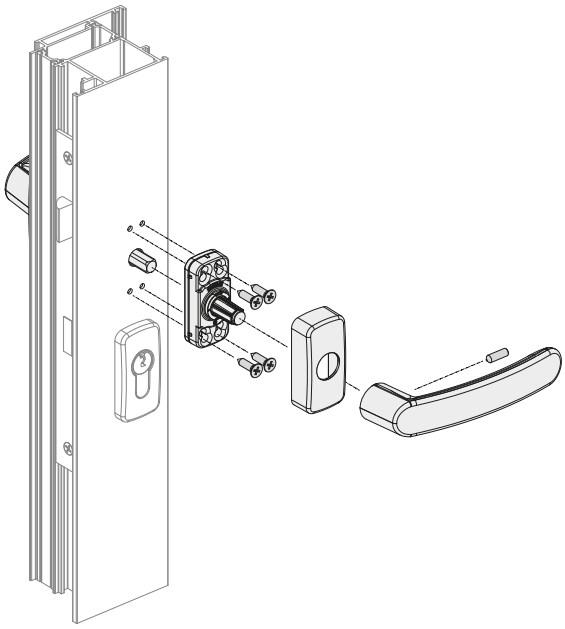


PROFILE MACHINING



SPINDLE CUTTING





Door handle set with recovery spring.

- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with zinc-plated steel transmission spindle 8 x 129 mm.
- The base of the transmission mechanism allows the fixing of up to 4 screws with double positioning each.
- The screw covers hide all the fixing mechanism and increases the style and contributes to a practical and rapid installation.

GEMINIS HANDLE
Handle set

05080 **

12

AL ZM
AC IN

8

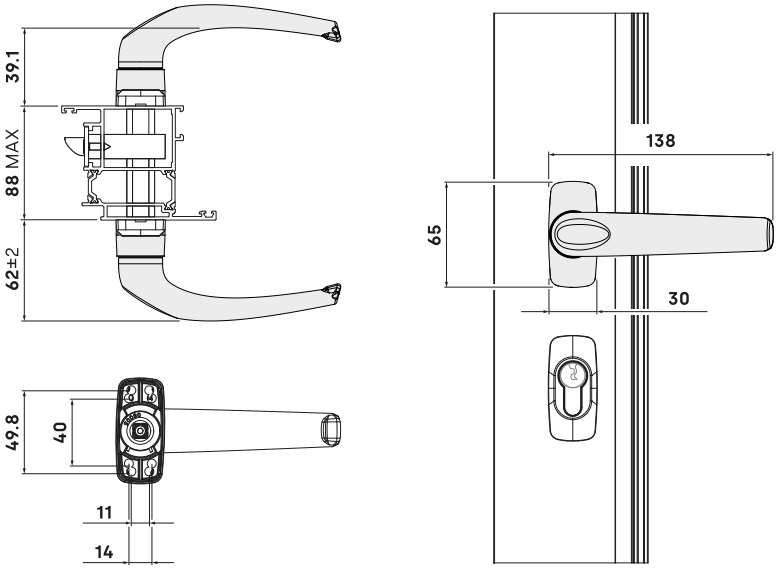
GEMINIS HANDLE
Single handle

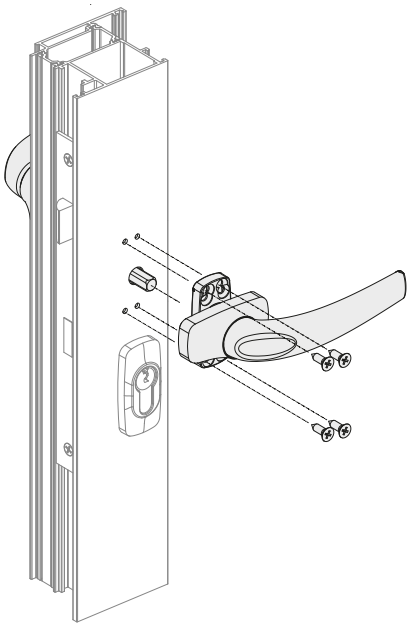
081 **

24

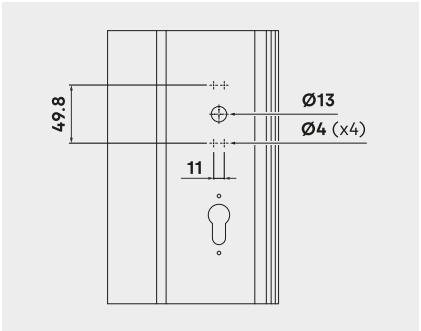
AL ZM
AC IN

8

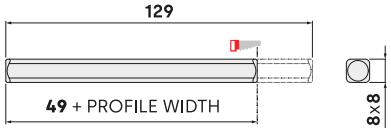




PROFILE MACHINING



SPINDLE CUTTING



LIBRA HANDLE
Handle set

082 **

20

AL ZM
AC IN

8

Slim door handle set with recovery spring.

- The handle is envisaged as the ideal solution for installations of doors where the shutters do not leave the space necessary for the mounting of a traditional handle.
- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with zinc-plated steel transmission spindle 8 x 129 mm.
- The screw covers hide all the fixing mechanism and increases the style and contributes to a practical and rapid installation.
- The base of the transmission mechanism allows the fixing of up to 4 screws with double positioning each.

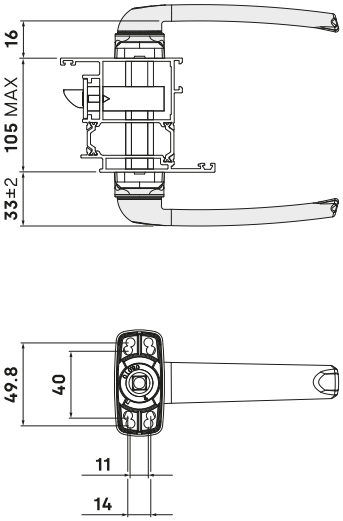
LIBRA HANDLE
Single handle

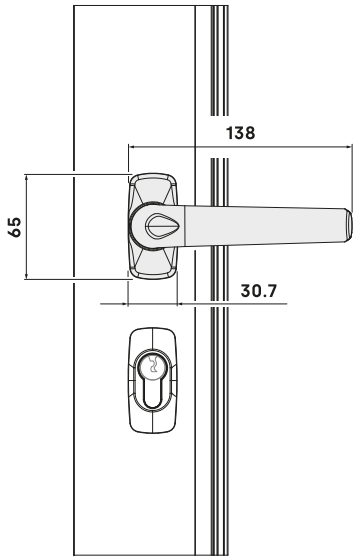
083 **

25

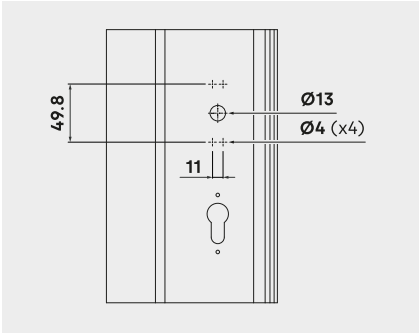
AL ZM
AC IN

8

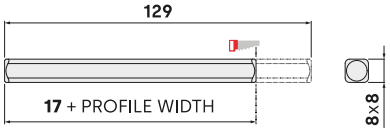


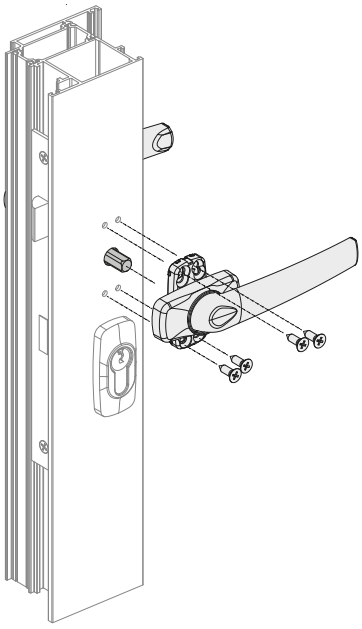


PROFILE MACHINING



SPINDLE CUTTING





Door handle set with recovery spring.

- The handles are ambidextrous, they can be installed in both right and left-handed openings.
- The reference is supplied with zinc-plated steel transmission spindle 8 x 129 mm.
- The screw covers hide all the fixing mechanism and increases the style and contributes to a practical and rapid installation.

INDO HANDLE
Handle set

A05086	**

12

AL ZM AC IN

8

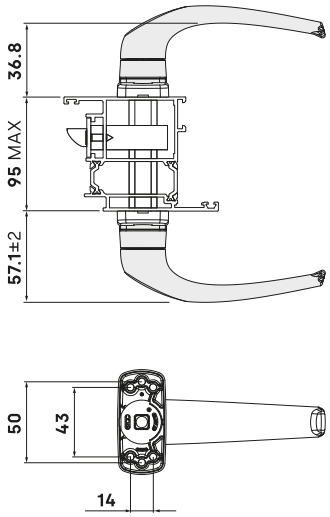
INDO HANDLE
Single handle

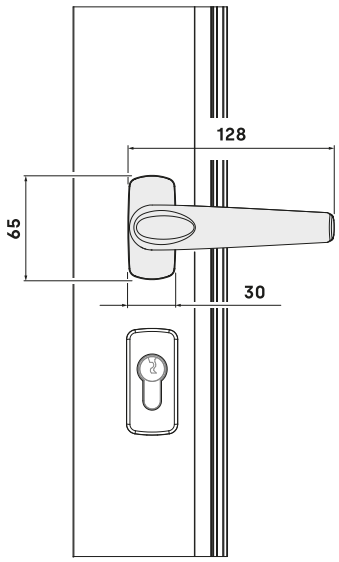
A05087	**

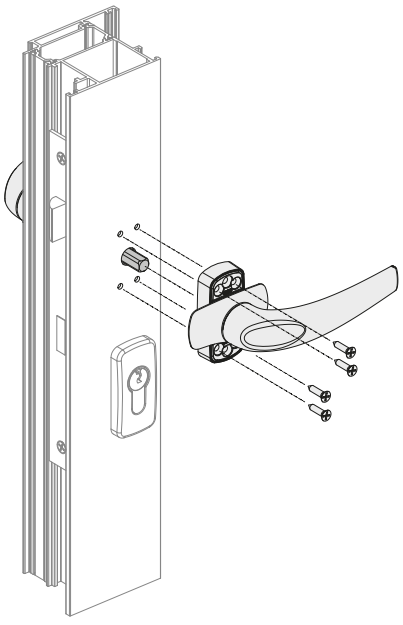
24

AL ZM AC IN

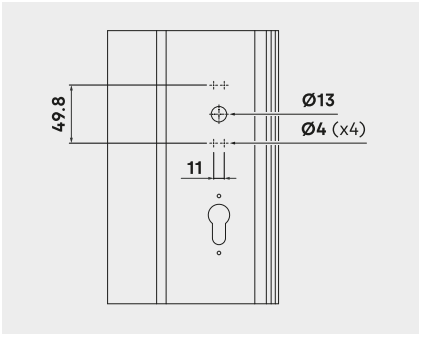
8



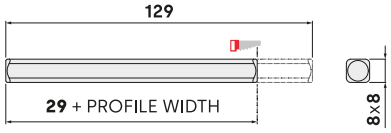




PROFILE MACHINING



SPINDLE CUTTING



KAIROX CC HANDLE
Handle set

260120

1

ZM

IN

8

Reinforced door handle set with recovery spring.

- Handle, rosette and lock cover made of INOX AISI 304.
- The reference is supplied with screws and zinc-plated transmission spindle 8 x 110 mm.
- The handles are fixed directly to the profile by means of self-tapping screws.

KAIROX RR HANDLE
Handle set

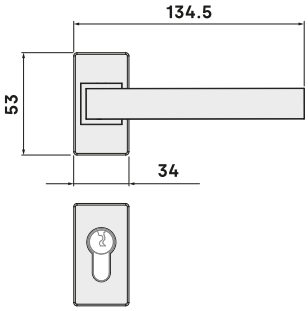
260220

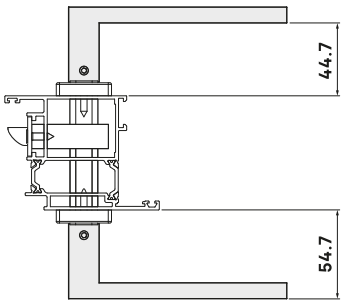
1

ZM

IN

8





KAIROX CR HANDLE
Handle set

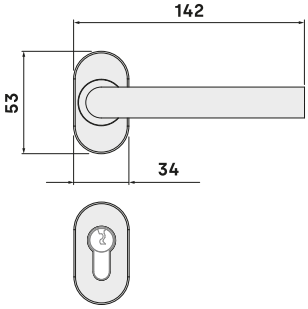
260320

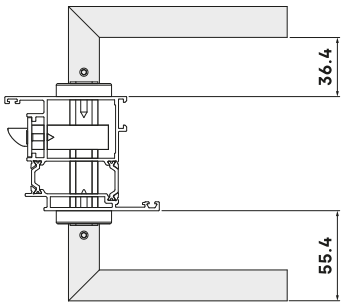
1

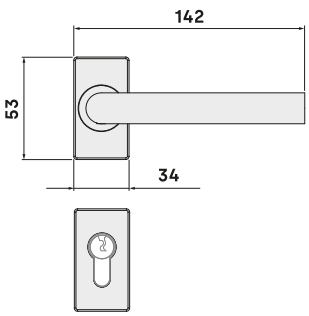
ZM

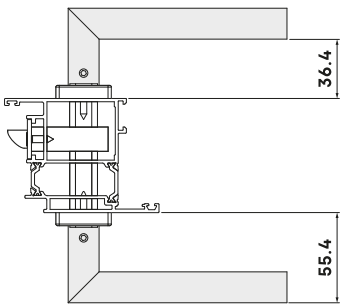
IN

8









Reinforced door handle set with recovery spring.

- Handle, rosette and lock cover made of INOX AISI 304.
- The reference is supplied with screws and zinc-plated transmission spindle 8 x 110 mm.
- The handles are fixed directly to the profile by means of self-tapping screws.

KAIROX RRC HANDLE

Handle set

260221

1

AL

IN

8

KAIROX CRC HANDLE

Handle set

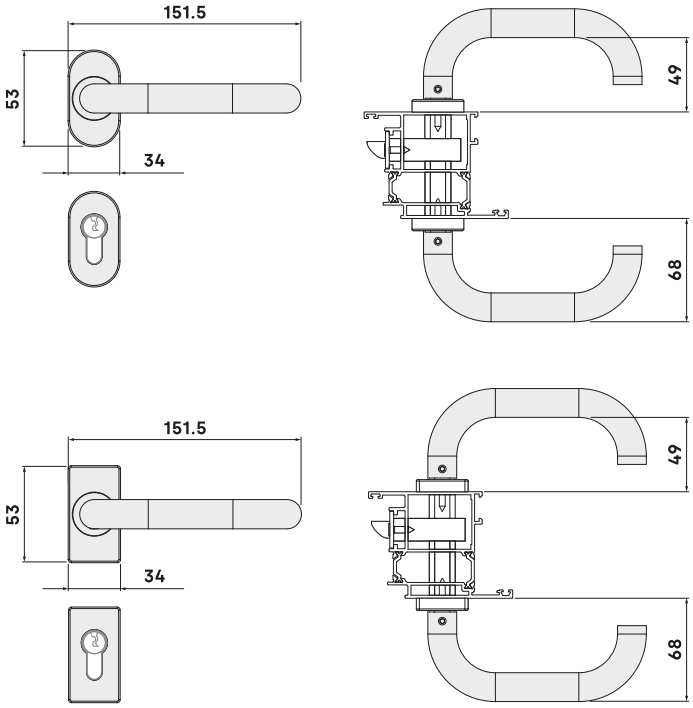
260321

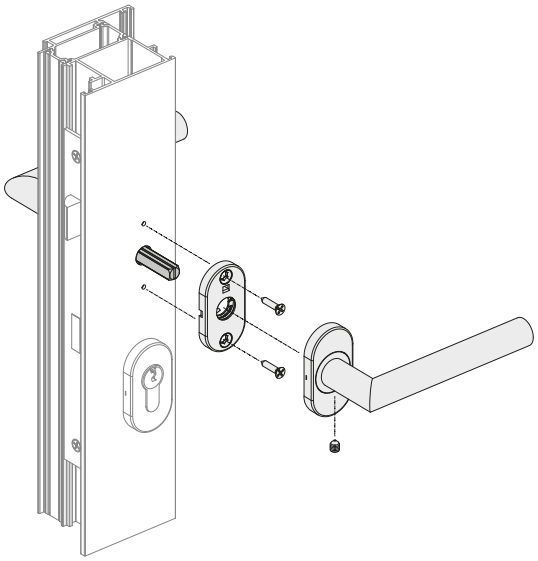
1

ZM

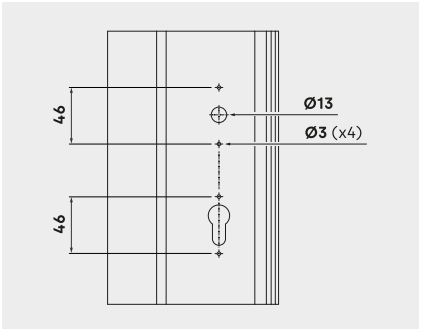
IN

8

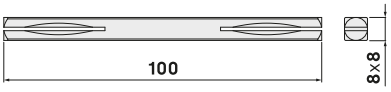




PROFILE MACHINING



TRANSMISSION SPINDLE



DELTA COVER PLATE

0508077 **

20

ZM

Cover plates compatible with standard European cylinders locks.

- The design of the screw-covering shield means it obscures the attachment system.
- Fixing is carried out directly to the profile by means of self-tapping screws.

DELTA cover plate (ref. 0508077)

- Shield depth: 6.3 mm.
- The shield slides over the grooved zamak base.

SLIM cover plate (ref. 0508053)

- Shield depth: 6.2 mm.
- Available in anodized finish: natural or stainless steel.

The references do not include the fixing screws and the European cylinder.

SLIM COVER PLATE

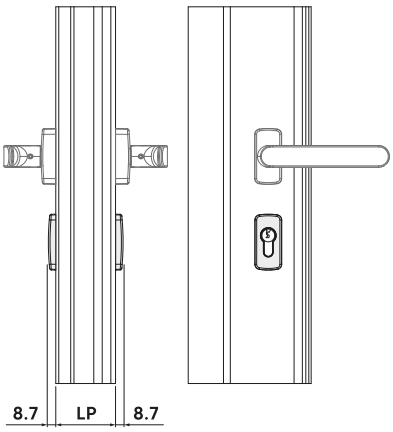
0508053

50

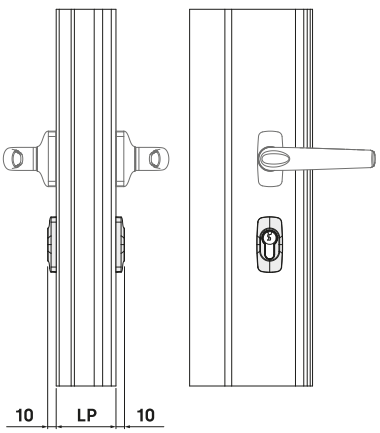
81 82

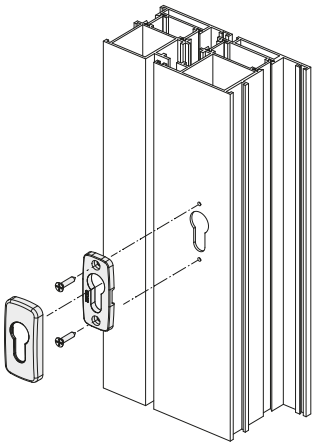
AL

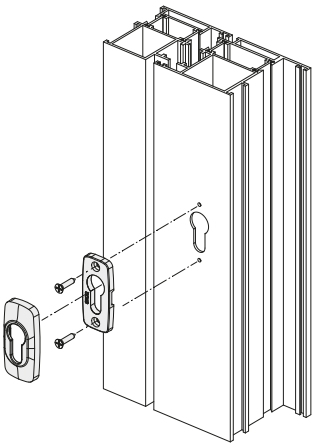
0508029 **



0508028 **







TO SELECT THE CYLINDER SHOULD BE CONSIDERED:

PROFILE LENGTH (**LP**) + COVER PLATE DEPTH (**PE**)

COVER PLATES DEPTH

0508077 **	DELTA COVER PLATE	6.3 mm
0508053 **	SLIM COVER PLATE	6.2 mm
0508029 **	SIRIUS COVER PLATE	8.7 mm
0508028 **	GEMINIS COVER PLATE	10 mm
260100 **	KAIROX COVER PLATE	8 mm
260200 **	KAIROX COVER PLATE	8 mm

Cover plates compatible with standard European cylinders locks.

- The design of the screw-covering shield means it obscures the attachment system.
- Fixing is carried out directly to the profile by means of self-tapping screws.

SIRIUS cover plate (ref. 0508029)

- Shield depth: 8.7 mm.

GEMINIS cover plate (ref. 0508028)

- Shield depth: 10 mm.

The references do not include the fixing screws and the European cylinder.

SIRIUS COVER PLATE

0508029 **

75

ZM

PE

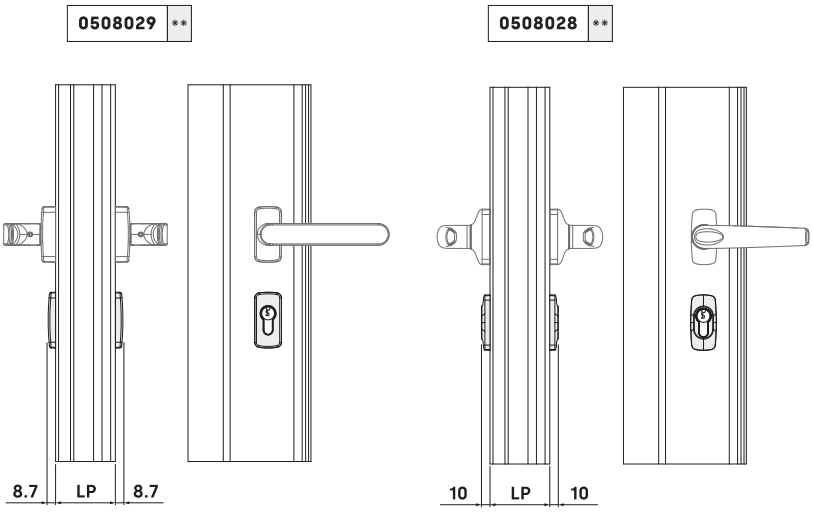
GEMINIS COVER PLATE

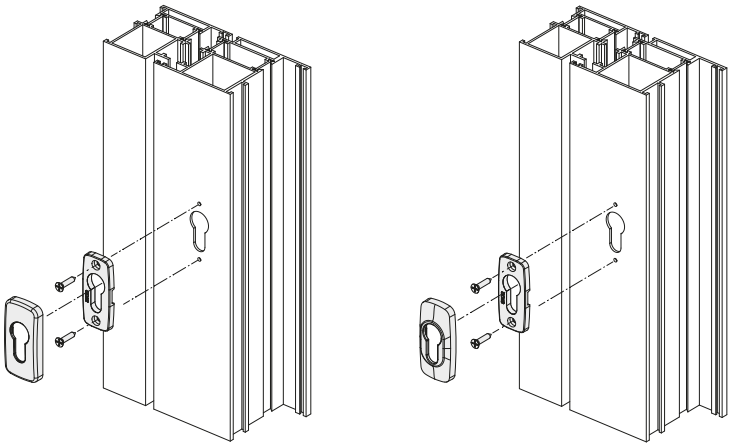
0508028 **

75

ZM

PE





TO SELECT THE CYLINDER SHOULD BE CONSIDERED:

PROFILE LENGTH (**LP**) + COVER PLATE DEPTH (**PE**)

COVER PLATES DEPTH

0508077 **	DELTA COVER PLATE	6.3 mm
0508053 **	SLIM COVER PLATE	6.2 mm
0508029 **	SIRIUS COVER PLATE	8.7 mm
0508028 **	GEMINIS COVER PLATE	10 mm
260100 **	KAIROX COVER PLATE	8 mm
260200 **	KAIROX COVER PLATE	8 mm

KAIROX COVER PLATE
Straight aesthetic

260100 **

IN

20

AISI 304 stainless steel cover plates compatible with standard European cylinders locks.

- The design of the screw-covering shield means it obscures the attachment system.
- Fixing is carried out directly to the profile by means of self-tapping screws.

KAIROX straight cover plate (ref. 260100)

- Cover plate with straight aesthetic.
- Shield depth: 8 mm.

KAIROX round cover plate (ref. 260200)

- Cover plate with round aesthetic.
- Shield depth: 8 mm.

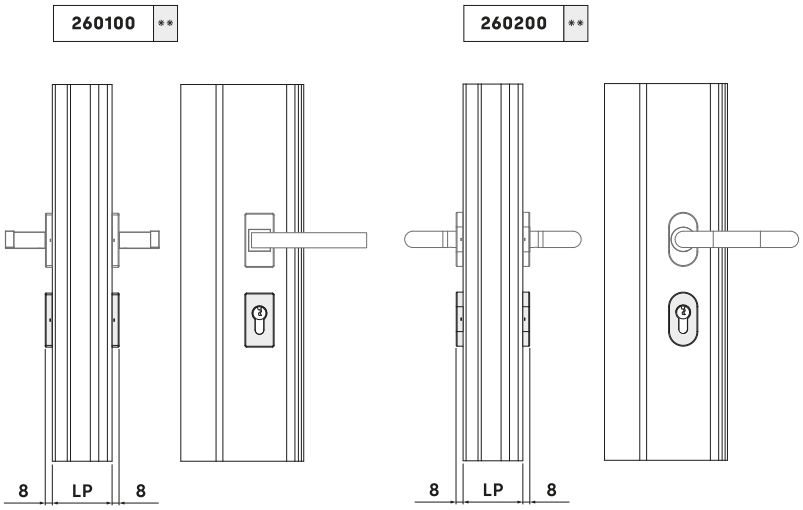
The references do not include European cylinder.

KAIROX COVER PLATE
Round aesthetic

260200 **

IN

20

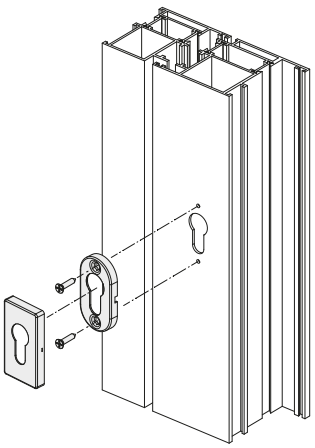


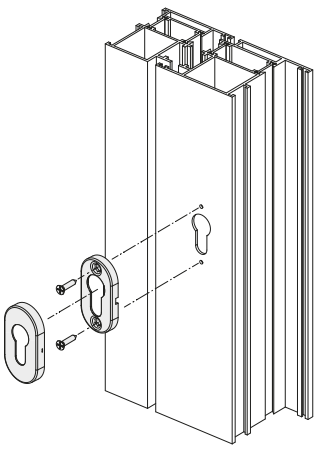
TO SELECT THE CYLINDER SHOULD BE CONSIDERED:

PROFILE LENGTH (**LP**) + COVER PLATE DEPTH (**PE**)

COVER PLATES DEPTH

0508077 **	DELTA COVER PLATE	6.3 mm
0508053 **	SLIM COVER PLATE	6.2 mm
0508029 **	SIRIUS COVER PLATE	8.7 mm
0508028 **	GEMINIS COVER PLATE	10 mm
260100 **	KAIROX COVER PLATE	8 mm
260200 **	KAIROX COVER PLATE	8 mm





06 PULL HANDLES

138 **TAU door pull handle**

143 **DELTA EASY pull handle**

144 **NAOS door pull handle**

145 **FENIX door pull handle**

TAU

minimalist pull handles

High degree of customisation

Anodized and lacquered finishes

Areas of public concurrence

Robustness and durability

TAU pull handles

TAU pull handles offer a special appeal to architects and decorators through elegant, discreet and timeless designs. The pull handles have been designed with elementary geometric composition, proportions and simplicity of form in mind.

» Multiple customisation options

Available with straight or offset aesthetics and with simple or double configurations with independent fixing systems, they can be adapted to all types of solutions, both avant-garde and more conventional.

• Straight aesthetics in single or double configuration.

• Offset aesthetics for right-hand or left-hand openings in single or double configuration.

» Architecture is colour

More and more projects are looking for their uniqueness through colours or finishes and where the accessories are lacquered in the same colours as the carpentry. Handles allow different finishes to be combined between the handle and the trims.

» Anodised: natural and acier grata

Through the anodising process we artificially increase the thickness of the natural aluminium oxide layer, achieving a high level of protection against corrosion and considerably increasing the strength and durability characteristics of this material.

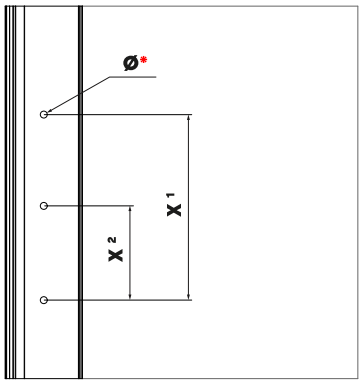
This makes it particularly suitable for projects with severe climatic conditions, as well as projects in coastal areas.

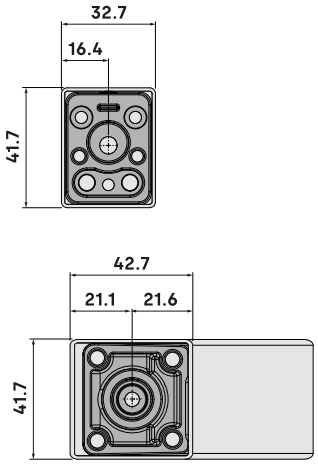
» High frequency use

The handles have been specifically designed for installation on doors of public buildings or in public areas where there is a high frequency of use by users or other persons who may be careless in their use or who may even deliberately abuse them: common areas of shopping centres, health facilities, educational establishments, office buildings, airports, bus / train stations,...

TAU pull handle with straight or offset aesthetics.

- Designed for entrances to public buildings or public areas with high footfall.
- Handle length: 450 millimetres.
- Offset aesthetics: left-hand (ref. 052412) and right-hand (ref. 052413).
- Fixing kits are available for single or double configurations.





* (Ø) THE DIAMETER DEPENDS ON THE FIXING KIT

REF	LENGTH	X¹	X²	FIXING KITS **05244**~					
				~001	~002	~003	~004	~005	~006
052411	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052451	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052461	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9

REF.	LENGTH	X¹	X²	FIXING KITS **05244**~					
				~001	~002	~003	~004	~005	~006
052412	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052452	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052462	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9
052413	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052453	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052463	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9

05244001	L = 450, 800	CONCEALED FIXING **FOR SINGLE PULL HANDLE**
05244002	L = 450, 800	FIXING **FOR SINGLE PULL HANDLE**
05244003	L = 450, 800	FIXING **FOR SINGLE PULL HANDLE**

TAU PULL HANDLE 450
Ambidextrous opening

052411 **

1

83 81

AL

ZM

TAU 450 OFFSET PULL HANDLE
Left-hand opening

052412 **

1

83 81

AL

ZM

TAU 450 OFFSET HANDLE
Right-hand opening

052413 **

1

83 81

AL

ZM

TAU PULL HANDLE 800
Ambidextrous opening

052451 **

1

83 81

AL

ZM

TAU pull handle with straight or offset aesthetics.

- Designed for entrances to public buildings or public areas with high footfall.
- Handle length: 800 millimetres.
- Offset aesthetics: left-hand (ref. 052452) and right-hand (ref. 052453).
- Fixing kits are available for single or double configurations.

TAU 800 OFFSET PULL HANDLE
Left-hand opening

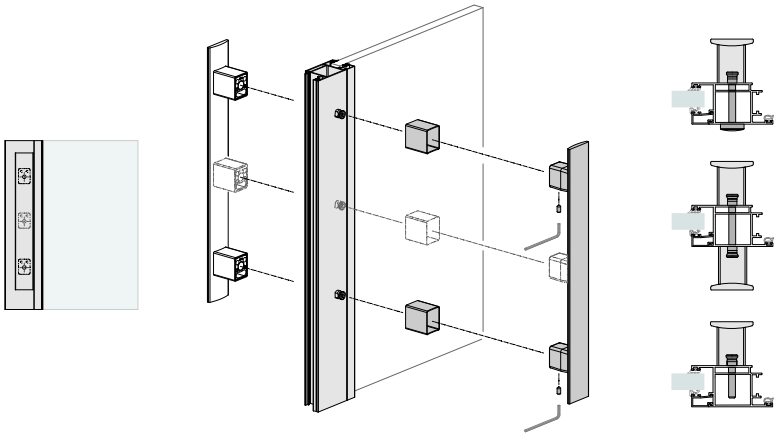
052452 **

1

83 81

AL

ZM



TAU 800 OFFSET HANDLE
Right-hand opening

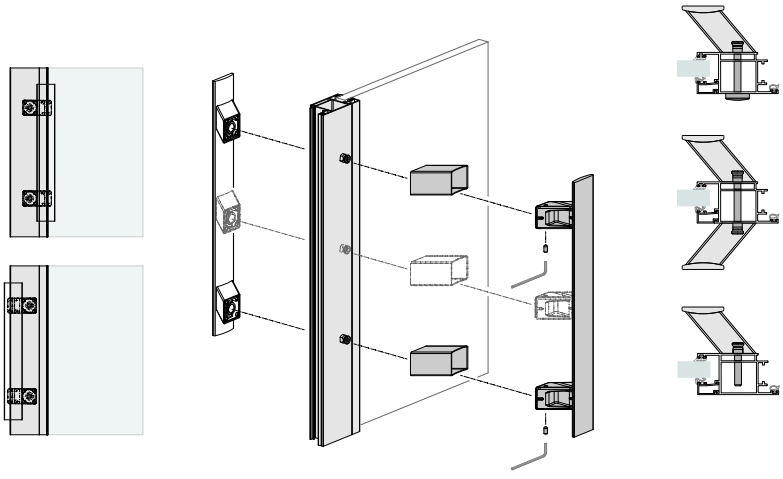
052453 **

1

83 81

AL

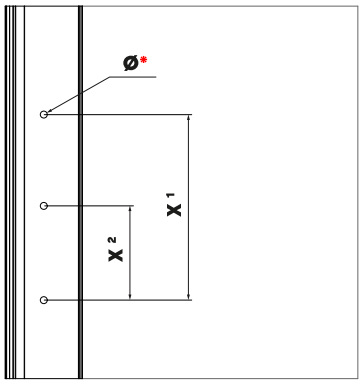
ZM

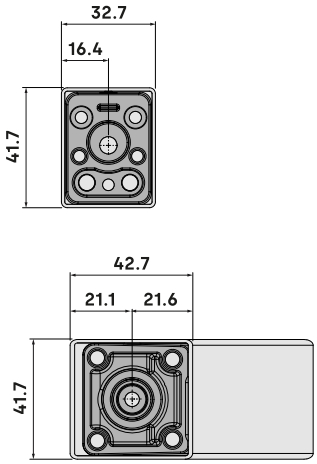


05244001	L = 450, 800	CONCEALED FIXING **FOR SINGLE PULL HANDLE**
05244002	L = 450, 800	FIXING **FOR DOUBLE PULL HANDLE**
05244003	L = 450, 800	FIXING **FOR SINGLE PULL HANDLE**

TAU pull handle with straight or offset aesthetics.

- Designed for entrances to public buildings or public areas with high footfall.
- Handle length: 1200 millimetres.
- Offset aesthetics: left-hand (ref. 052462) and right-hand (ref. 052463).
- Fixing kits are available for single or double configurations.





* (Ø) THE DIAMETER DEPENDS ON THE FIXING KIT

REF	LENGTH	X¹	X²	FIXING KITS **05244**~					
				~001	~002	~003	~004	~005	~006
052411	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052451	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052461	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9

REF.	LENGTH	X¹	X²	FIXING KITS **05244**~					
				~001	~002	~003	~004	~005	~006
052412	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052452	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052462	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9
052413	450	320	-	Ø7.5	Ø9	Ø9	-	-	-
052453	800	630	-	Ø7.5	Ø9	Ø9	-	-	-
052463	1200	1030	515	-	-	-	Ø7.5	Ø9	Ø9

05244004	L = 1200	CONCEALED FIXING **FOR SINGLE PULL HANDLE**
05244005	L = 1200	FIXING **FOR DOUBLE PULL HANDLE**
05244006	L = 1200	FIXING **FOR DOUBLE PULL HANDLE**

TAU PULL HANDLE 1200

Ambidextrous opening

83 81

AL

ZM

052461 **

1

TAU 1200 OFFSET PULL HANDLE

Left-hand opening

83 81

AL

ZM

052462 **

1

TAU 1200 OFFSET HANDLE

Right-hand opening

83 81

AL

ZM

052463 **

1

CONCEALED FASTENING SYSTEM

05244001

10

AL

IN

Fixing kits for TAU pull handles that allow single or double configurations.

Concealed fixing kit (ref. 05244001, ref. 05244004)

• Profiles with thermal break or with internal partition walls by means of a diabolo installation and where the bolt is screwed directly to the profile.

Double fixing kit (ref. 05244002, ref. 05244005)

• Installation of a double handle by means of a diabolo on each side of the profile.

Single fixing kit (ref. 05244003, ref. 05244006)

• Profiles with a single tubular where fixing is required on both sides of the profile.

Different fixing kits are available depending on the length of the handle. See reference table.

DOUBLE FASTENING SYSTEM

05244002

10

AL

IN

SINGLE FIXING SYSTEM

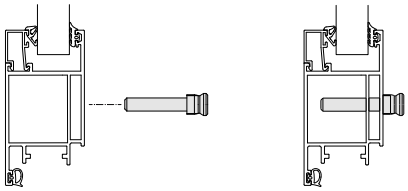
05244003

10

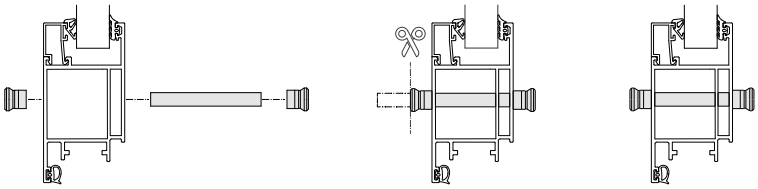
AL

ZM

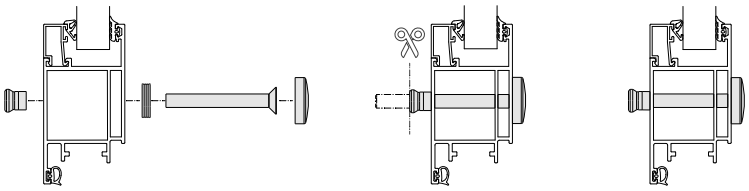
05244001 05244004



05244002 05244005



05244003 05244006



	PULL HANDLE	
05244001	L = 450, 800	CONCEALED FIXING FOR SINGLE PULL HANDLE
05244004	L = 1200	
05244002	L = 450, 800	FIXING FOR SINGLE PULL HANDLE
05244005	L = 1200	
05244003	L = 450, 800	FIXING FOR DOUBLE PULL HANDLE
05244006	L = 1200	

DELTA EASY minimalist pull handles for hinged or sliding balcony door systems.

• The handle allows its installation in can be installed on sliding windows, both indoors and outdoors.

DELTA EASY pull handle (ref. 0508084)

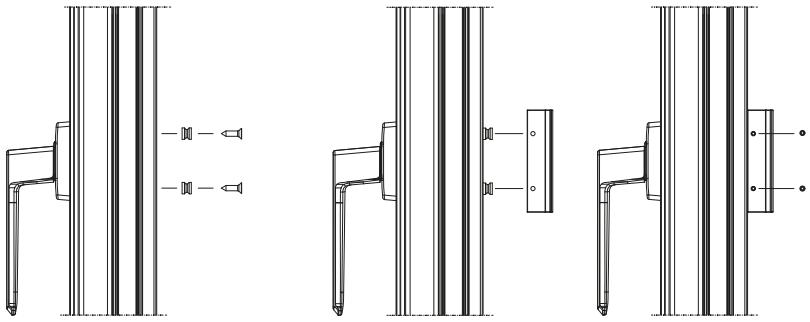
• Frontal fixation with diabolo screws: allows posterior mounting on an already existing frame.

DELTA EASY pull handle (ref. 0508086)

• Side mounting with screws to the profile: for profiles which allow installation by screwing from the inner side of the profile.
• Side mounting with screws from the handle for installations with double configuration handle ~ pull handle (outside / inside).
• Fixing kit available for double pulling configuration with standard handles (ref. K5771-10).

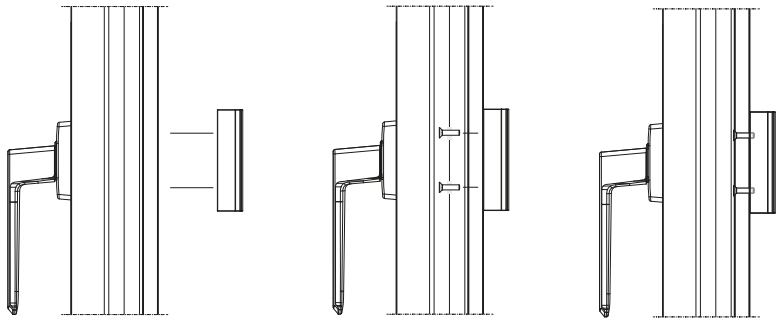
FRONT INSTALLATION
WITH DIABOLO SCREWS

0508084



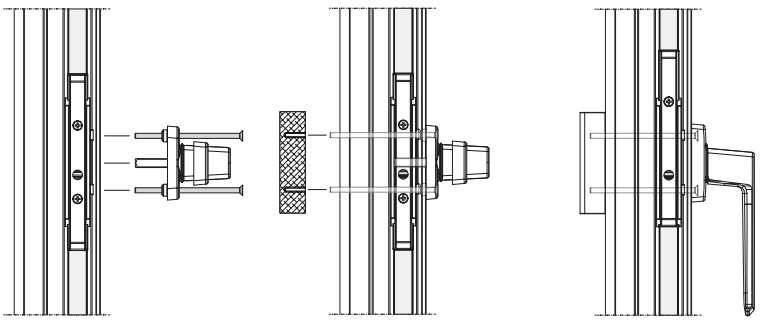
SIDE INSTALLATION TO THE PROFILE
WITH SCREWS

0508086



SIDE INSTALLATION FROM THE HANDLE
WITH SCREWS

K5771-10 0508086



DELTA EASY PULL HANDLE

0508084

10

81 83

AL

DELTA EASY PULL HANDLE

0508086

10

81 83

AL

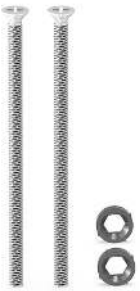
DELTA EASY DOUBLE FIXING KIT

K5771-10

50

AC

IN



NAOS PULL HANDLE
Pull handle set

0508030 **

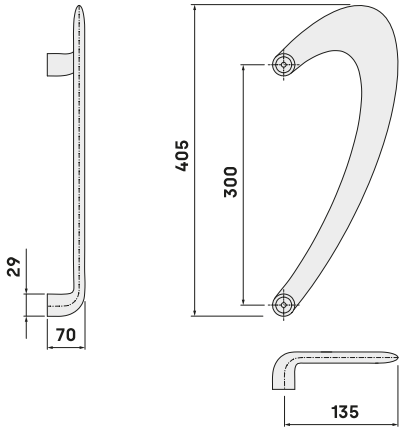
1

AL

AC

Pull handles set for entrances to public buildings and commercial establishments.

- Hidden fixing system which impedes dismantling from the outside thus avoiding attempted forcing of the handle and theft.
- Rapid installation and simple mechanism.
- Available for right-hand opening doors (Ref. 0508031) and left-hand opening doors (Ref. 0508032).



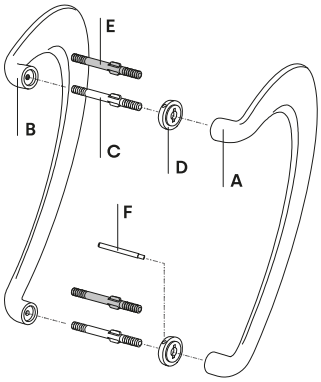
NAOS PULL HANDLE
Right-hand opening

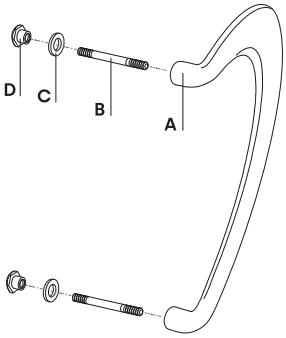
0508031 **

1

AL

AC





NAOS PULL HANDLE
Left-hand opening

0508032 **

1

AL

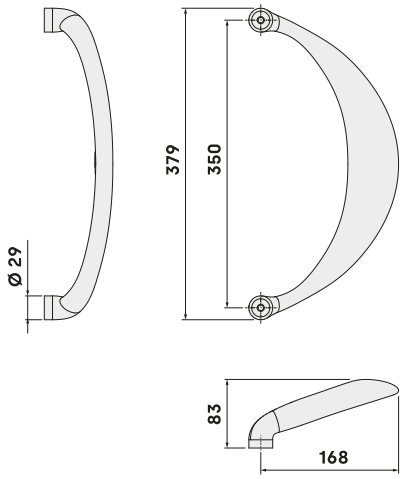
AC

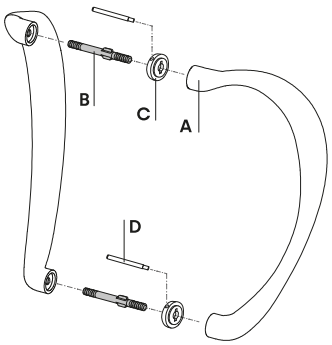
DESCRIPTION	
A RIGHT PULL HANDLE	1
B LEFT PULL HANDLE	1
C RIGHT SCREWS (M8 x 126)	2
D FIXING PLATES (Ø29x 10)	2
E LEFT SCREWS (M8 x 126)	2
F FIXING KEY	1

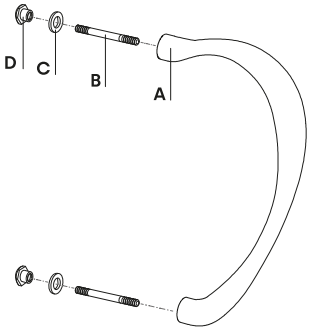
DESCRIPTION	
A RIGHT PULL HANDLE	1
B RIGHT SCREWS (M8 x 114)	2
C STAINLESS STEEL WASHERS	2
D NUTS	2

Pull handles set for entrances to public buildings and commercial establishments.

- Rapid installation and simple profile mechanism.
- The pull handles do not keep opening hand, they are ambidextrous.
- Hidden fixing system which impedes dismantling from the outside thus avoiding attempted forcing of the handle and theft.







DESCRIPTION	
A PULL HANDLE	2
B RIGHT SCREWS (M8 x 126)	2
C FIXING PLATES (Ø29x 10)	2
D FIXING KEY	2

DESCRIPTION	
A PULL HANDLE	1
B RIGHT SCREWS (M8 x 114)	2
C STAINLESS STEEL WASHERS	2
D NUTS	2

FENIX PULL HANDLE
Pull handle set

0508033 **

1

AL

AC

FENIX PULL HANDLE
Single pull handle

0508034 **

1

AL

AC

07 SIDE-HUNG OPENING

148 One-sash kit

148 Double-sash kit

153 Passive sash kit

157 Outward opening

162 CH180 PLUS hinges

163 Side-hung accessories

Ø6 SIDE-HUNG KIT
Single sash configurations

07K1H

50

Kits of bolts and keepers for passive sash side-hung windows with single or double configuration.

- Universal frame keepers made of zamak.
- Available with polyamide keepers (ref. 07K1HPA, ref. 07K2HPA).
- 6 mm polyamide pointers reinforced with glass-fibre and 8 mm magnifying washers.
- Available zamak pointers kits (ref. R07K1H, ref. 07K2H).
- The double-leaf keeper (ref. 110501) has an anti-sliding groove that allows a progressive tightening adjustment, increasing the adjustment of the locking system.

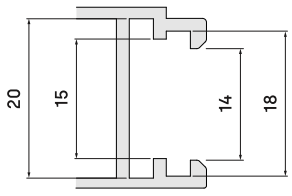
Ø6 SIDE-HUNG KIT
Double sash configurations

07K2H

		DIAMETER	POINTERS	KEEPER
07K1H		Ø6, Ø8	07085 PA	0507105 ZM
» 07K1HD6		Ø6, Ø8	07085 PA	0507105 ZM
07K1HD8		Ø8	07085 PA	0507105 ZM
07K1HPA		Ø8	07002 PA	0507059 PA
R07K1H		Ø6, Ø8	0407205 ZM	0507105 ZM
07K2H		Ø6, Ø8	07085 PA	110501 ZM
» 07K2HD6		Ø6, Ø8	07085 PA	110501 ZM
07K2HD8		Ø8	07085 PA	110501 ZM
07K2HPA		Ø8	07002 PA	0507059 PA
R07K2H		Ø6, Ø8	0407205 ZM	110501 ZM

(») *INCREASE WASHERS (Ø8) ARE NOT SERVED BAGGED*

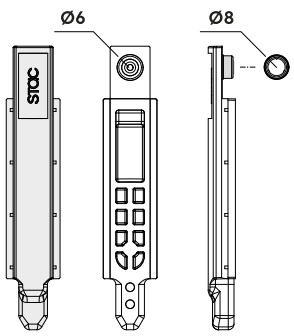
***SASH~ FRAME** INVERTER PROFILE*



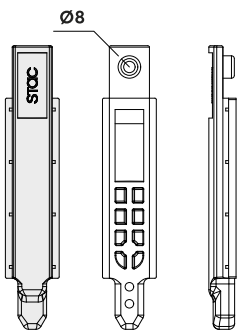
EURO GROOVE

FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

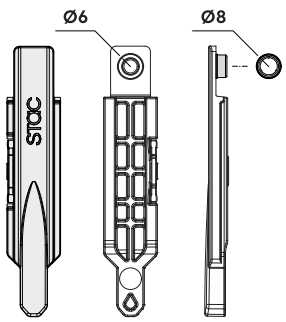
*POINTER **07085***



*POINTER **07002***



*POINTER **0407205***



Kits of bolts with adjustment and keepers for passive sash side-hung windows with single or double configuration.

- Pointers with metal bushing and metal end for 1 mm adjustment.
- Supplied with pointers, bolts and keepers made of zamak with 8 mm magnifying washers.
- Available kits with double sash keepers (ref. R07K1HRD6, ref. R07K2HRD6).
- The double-leaf keeper (ref. 110501) has an anti-sliding groove that allows a progressive tightening adjustment, increasing the adjustment of the locking system.

Ø6 SIDE-HUNG KIT WITH ADJUSTMENT
Single sash configurations

ce ce ZM PA

07K1HRD6

50

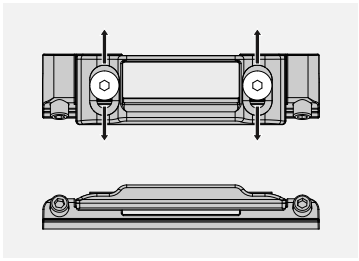
		DIAMETER	POINTERS	KEEPER
07K1HRD6		Ø6, Ø8	0507082 ZM	0507105 ZM
R07K1HRD6		Ø6, Ø8	0507082 ZM	110501 ZM
07K2HRD6		Ø6, Ø8	0507082 ZM	0507105 ZM
R07K2HRD6		Ø6, Ø8	0507082 ZM	110501 ZM

Ø6 SIDE-HUNG KIT WITH ADJUSTMENT
Double sash configurations

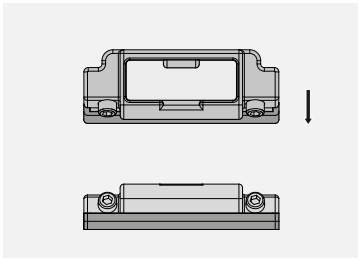
ce ce ZM PA

07K2HRD6

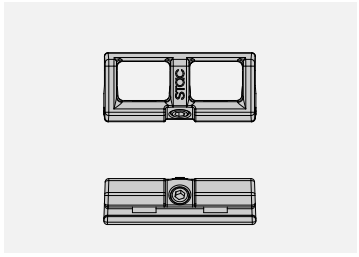
110501 ADJUSTMENT ±2.5



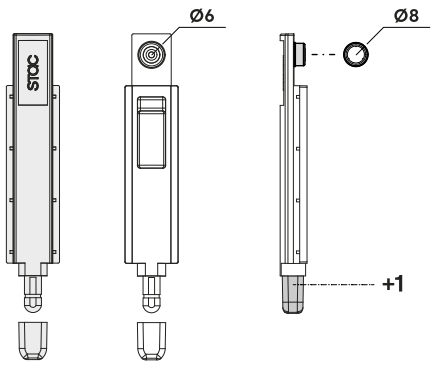
0507059 ADJUSTMENT +1



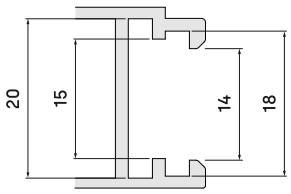
0507105



*POINTER **0507082** ADJUSTMENT +1*



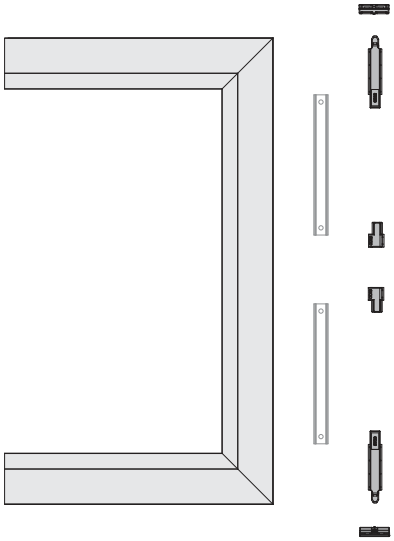
***SASH~ FRAME** INVERTER PROFILE*



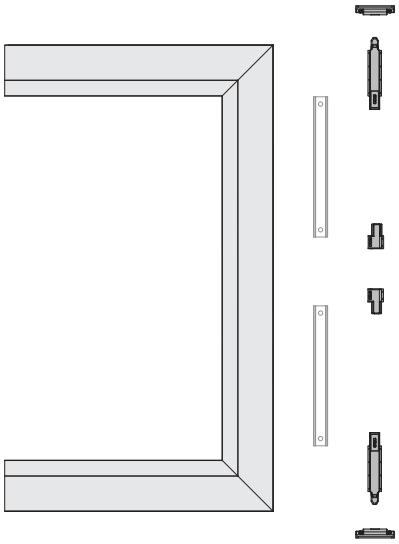
EURO GROOVE

FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

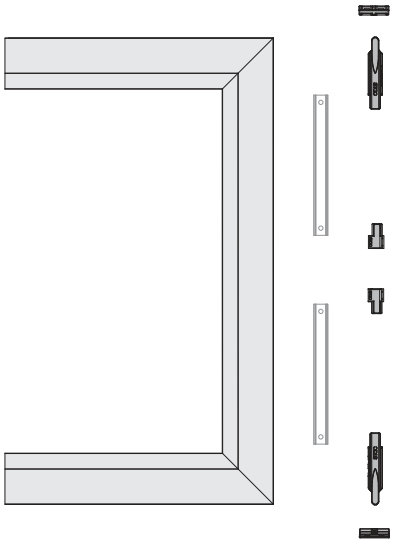
» SINGLE SASH CONFIGURATION



	POINTER	KEEPER
07K1H	**07085**	0507105
07K1HD6	**07085**	0507105
07K1HD8	**07085**	0507105

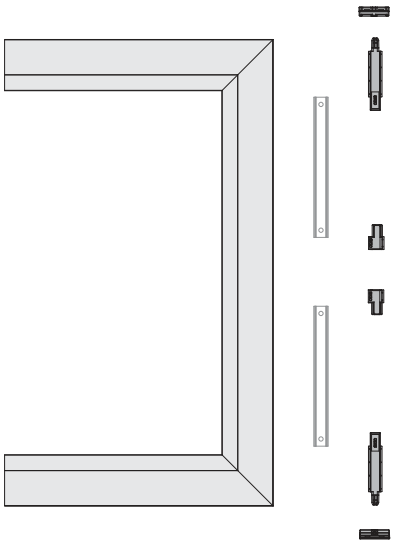


	POINTER	KEEPER
07K1HPA	**07002**	0507059

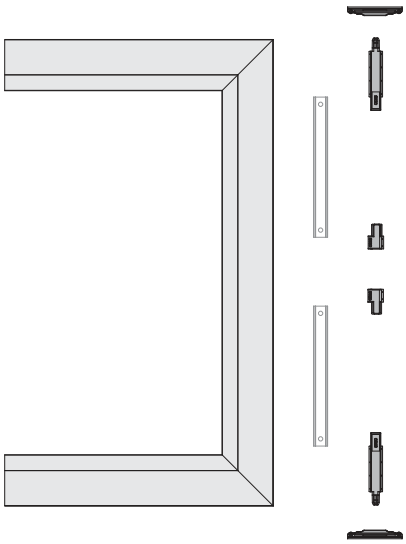


	POINTER	KEEPER
R07K1H	**0407205**	0507105

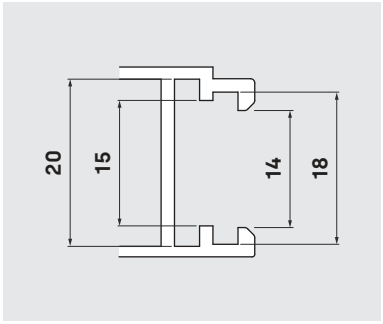
» SINGLE SASH CONFIGURATION WITH ADJUSTMENT



	POINTER	KEEPER
07K1HRD6	**0507082**	0507105



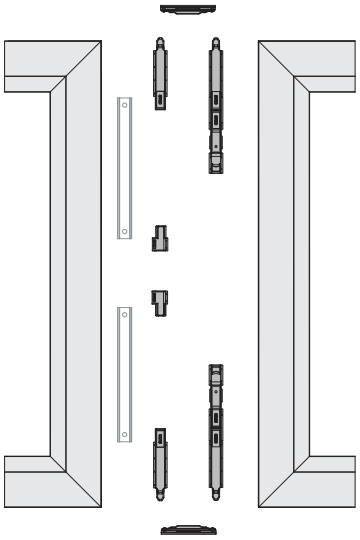
	POINTER	KEEPER
R07K1HRD6	**0507082**	110501



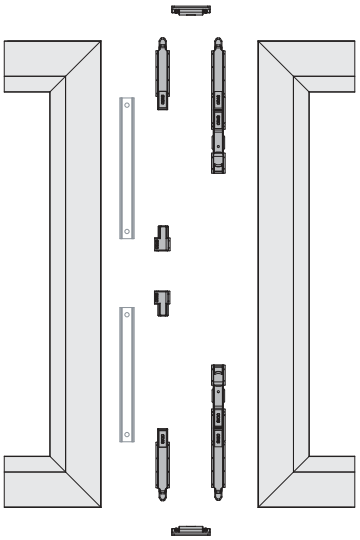
SASH ~ FRAME INVERTER PROFILE

EURO GROOVE	
FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

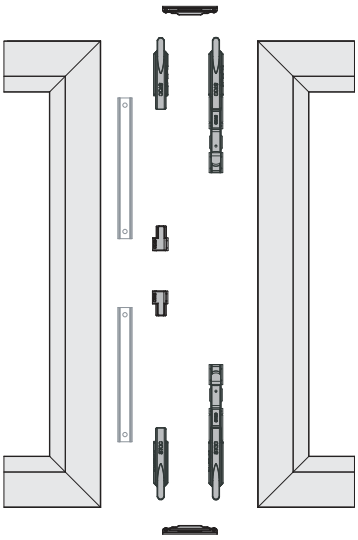
» DOUBLE SASH CONFIGURATION



	POINTER	KEEPER
07K2H	**07085**	110501
07K2HD6	**07085**	110501
07K2HD8	**07085**	110501

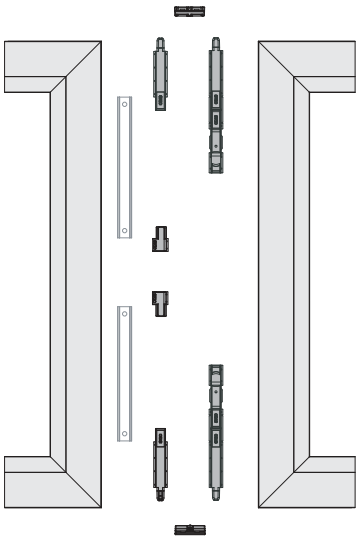


	POINTER	KEEPER
07K2HPA	**07002**	0507059

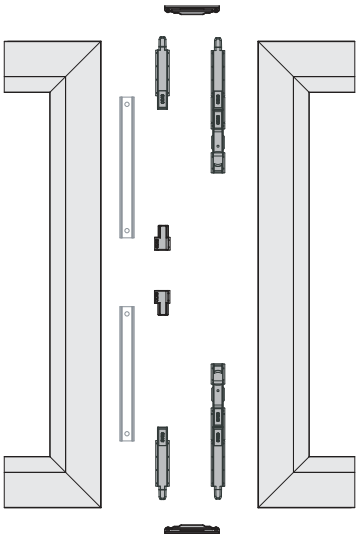


	POINTER	KEEPER
R07K2H	**0407205**	110501

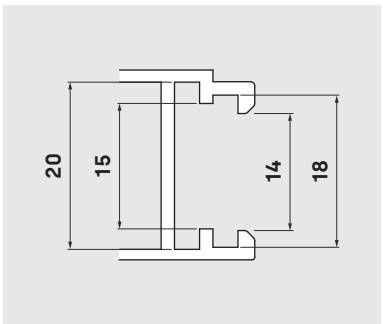
» DOUBLE SASH CONFIGURATION WITH ADJUSTMENT



	POINTER	KEEPER
07K2HRD6	**0507082**	0507105



	POINTER	KEEPER
R07K2HRD6	**0507082**	110501



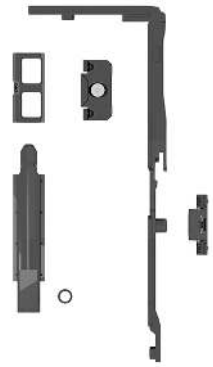
SASH ~ FRAME INVERTER PROFILE

EURO GROOVE	
FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

FRENCH-STYLE OPENING KIT Ø6

113601

10

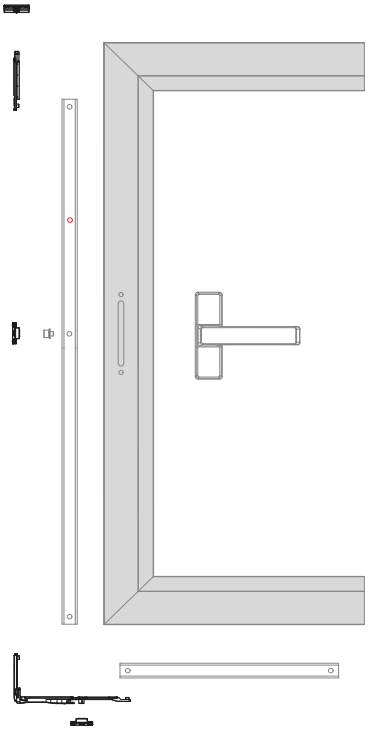


ZM AC PA

Kit for french-style sinde-hung window.

- Supplied with an upper pointer with Ø6 mm connection, a lower corner drive and keepers for frame profile.
- The movement is transmitted by means of a unidirectional drive cremone (not included).
- Compatible with french-style (unidirectional) or tilt & turn cremones.
- Additional locking points can be added to the lower part of the sash.
- Transmission cover not included (ref. 0511193).

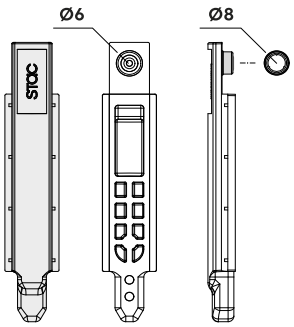
Translated with www.DeepL.com/Translator (free version)



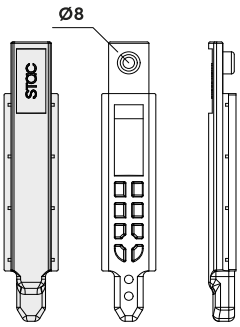
		DIAMETER	POINTER	KEEPER
»	113601	Ø6, Ø8	07085 PA	0507105 ZM
	113600	Ø6	07002 PA	0507105 ZM

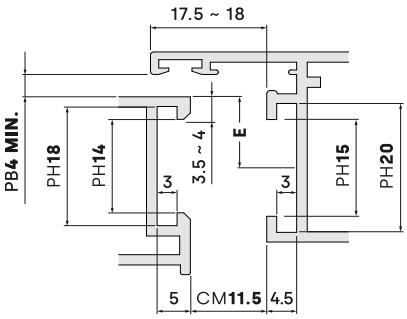
(») *THE REFERENCE INCLUDES INCREASE WASHER FOR Ø8*

*POINTER **07085***



*POINTER **07002***





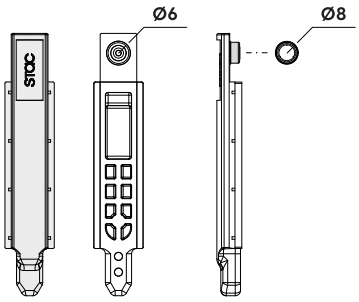
EURO GROOVE	
FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

Kit of bolts for passive sash in side-hung windows.

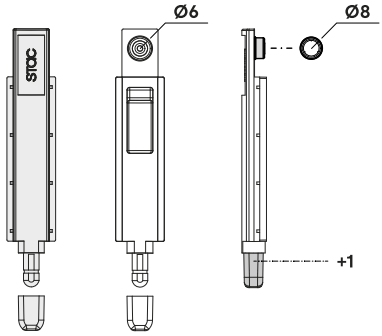
• Supplied with bolts and pointers made of 6 millimetre glass-fibre reinforced polyamide and washers to increase to 8 millimetres.
• Available with pointers with bushing and metal points for 1 mm adjustment (ref. 07K3HRD6, ref. ref. 07K4HRD6).

		DIAMETER	POINTER	KEEPER
P07K2H		Ø6, Ø8	07085 PA	-
07K3HRD6		Ø6, Ø8	0507082 ZM	-
07K4HRD6		Ø6, Ø8	0507082 ZM	-

*POINTER **07085***

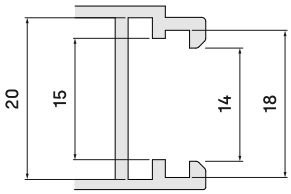


*POINTER **0507082** ADJUSTMENT +1*



P07K2H, 07K3HRD6

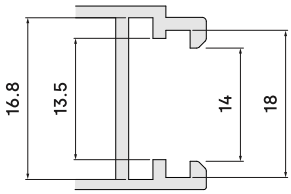
***SASH ~ FRAME** INVERTER PROFILE*



EURO GROOVE	
FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

07K4HRD6

***SASH ~ FRAME** INVERTER PROFILE*



EURO GROOVE	
FRAME PROFILE	16.8 - 13.5 mm
SASH PROFILE	14 - 18 mm

Ø6 PASSIVE SASH KIT
Passive sash configurations

CE ZM PA

P07K2H

50

PASSIVE SASH SIDE-HUNG KIT 15 ~ 20 WITH ADJUSTMENT
15 - 20 mm frame profile

CE CE ZM PA

07K3HRD6

50

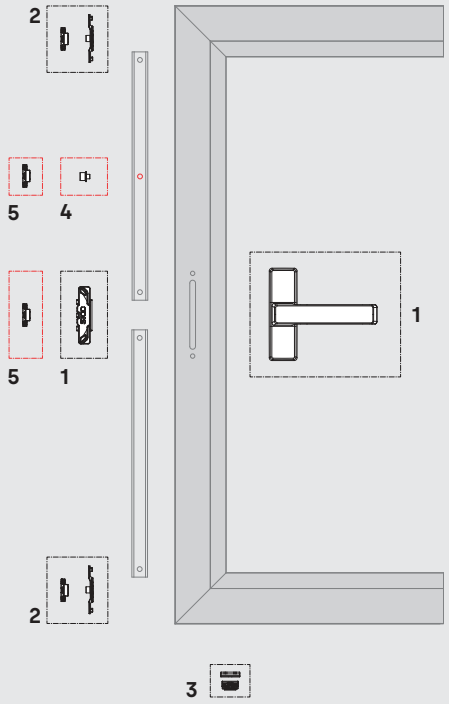
PASSIVE SASH SIDE-HUNG KIT 13.5 ~ 16. WITH ADJUSTMENT
13.5 - 16.8 mm frame profile

CE CE ZM PA

07K4HRD6

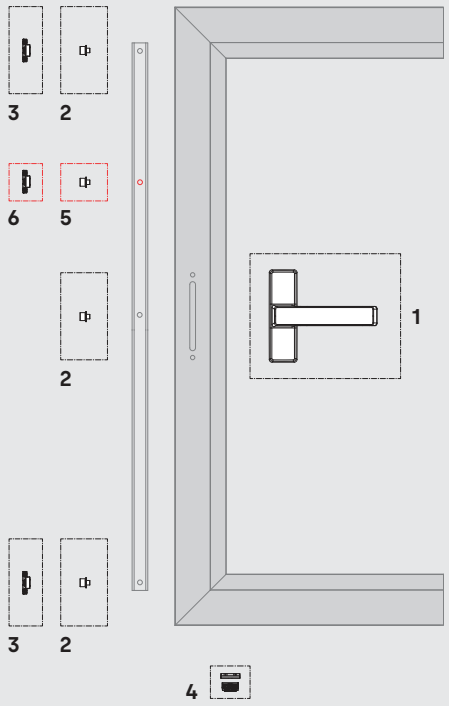
50

Accessories for sash height greater than 1200 millimetres



LOCKING TO THE INVERTER
WITH TRANSMISSION COVER

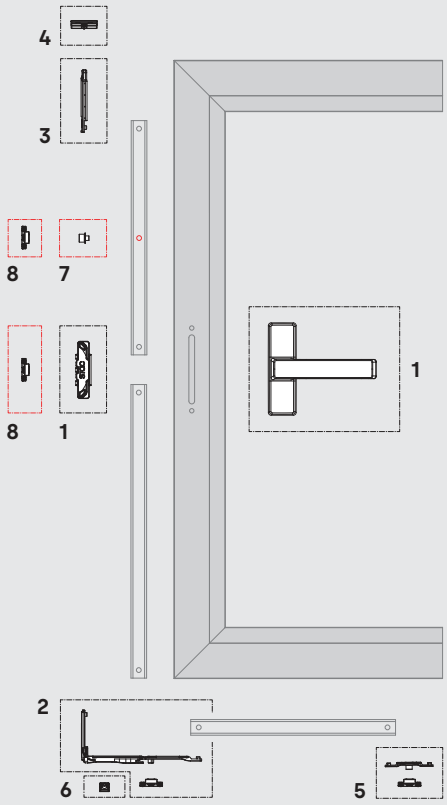
1	**CREMONE**	*Consult cremone section*	
2	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
3	**0710621**	**POSITIONING LATCH KIT**	100
4	**0517002**	**Ø6 TRANSMISSION RIVET**	50
5	**110200**	**ADJUSTABLE LOCKING POINT**	100



LOCKING TO THE INVERTER
WITH TRANSMISSION RIVET

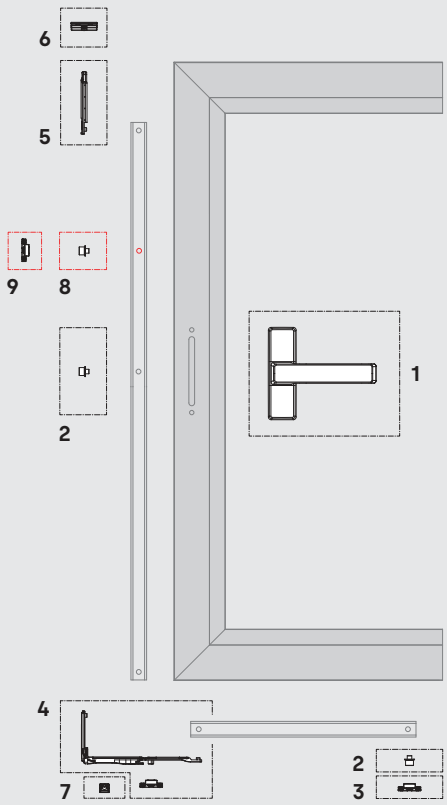
1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 TRANSMISSION RIVET**	50
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**0710621**	**POSITIONING LATCH KIT**	100
5	**0517002**	**Ø6 TRANSMISSION RIVET**	50
6	**110200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height greater than 1200 millimetres



LOCKING TO FRAME WITH TRANSMISSION COVER
CORNER DRIVE AND POINTER

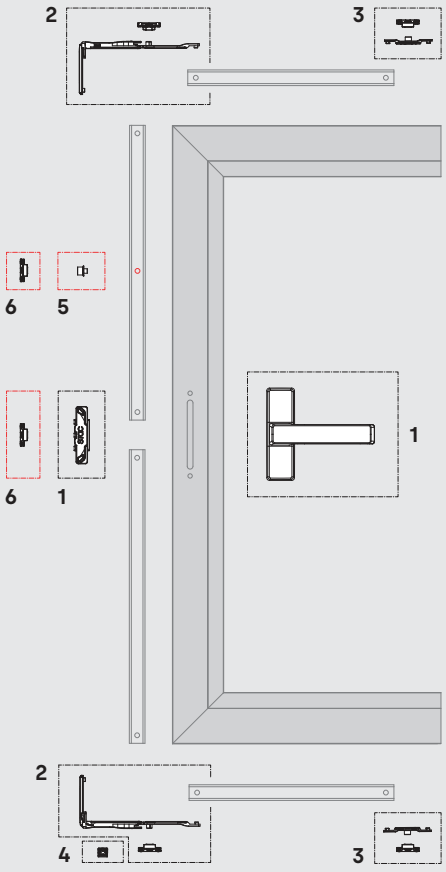
1	**CREMONE**	*Consult cremone section*	
2	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
3	**07085**	**Ø6 SIDE-HUNG OPENING POINTER**	1000
4	**0507105**	**UNIVERSAL LOCKING POINT**	200
5	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
6	**0517193**	**POSITIONING LATCH**	100
7	**0517002**	**Ø6 TRANSMISSION RIVET**	50
8	**110200**	**ADJUSTABLE LOCKING POINT**	100

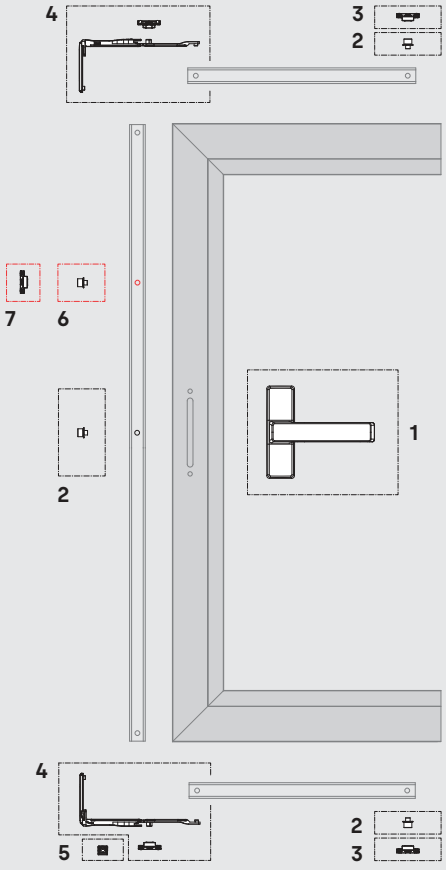


LOCKING TO FRAME WITH TRANSMISSION RIVET
CORNER DRIVE AND POINTER

1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 TRANSMISSION RIVET**	50
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**07085**	**Ø6 SIDE-HUNG OPENING POINTER**	1000
6	**0507105**	**UNIVERSAL LOCKING POINT**	200
7	**0517193**	**POSITIONING LATCH**	100
8	**0517002**	**Ø6 TRANSMISSION RIVET**	50
9	**110200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height greater than 1200 millimetres





LOCKING TO FRAME WITH TRANSMISSION COVER
DOUBLE CORNER DRIVE

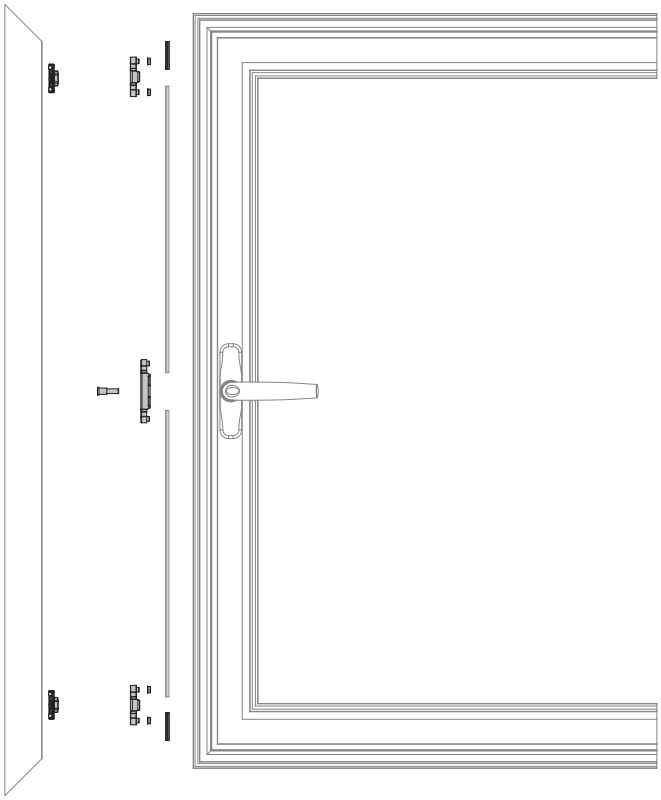
1	**CREMONE**	*Consult cremone section*	
2	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**0517193**	**POSITIONING LATCH**	100
5	**0517002**	**Ø6 TRANSMISSION RIVET**	50
6	**110200**	**ADJUSTABLE LOCKING POINT**	100

LOCKING TO FRAME WITH TRANSMISSION RIVET
DOUBLE CORNER DRIVE

1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 TRANSMISSION RIVET**	50
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**0517002**	**Ø6 TRANSMISSION RIVET**	50
7	**110200**	**ADJUSTABLE LOCKING POINT**	100

Accessory kits that allow transmission on outward opening windows.

• Kit made up of transmission parts, sliders, adjustable locking points and washers to increase to Ø8.
• Available with different bolt lengths: 19 mm (ref. 07044), 26 mm (ref. 07060) or 36 mm (ref. 07129).
• Available transmission kits consisting of transmission part with short bolt (ref. 074K01) or with long bolt (ref. 075K01).



OUTWARD OPENING KIT (07044)
With short bolt

CE ZM PA

074K01C

20

OUTWARD OPENING KIT (07060)
With long bolt

CE ZM PA

074K01L

20

OUTWARD OPENING KIT (07129)
With special bolt

CE ZM PA

075K01CE1

20

OUTWARD OPENING KIT		BOLT
074K01C	SHORT BOLT (19 mm) + TRANSMISION KIT	07044
074K01L	LONG BOLT (26 mm) + TRANSMISION KIT	07060
075K01CE1	SPECIAL BOLT (36 mm) + TRANSMISION KIT	07129

TRANSMISSION KIT WITH SHORT BOLT

074K01

20

ZM

IN

TRANSMISSION KIT WITH LONG BOLT

075K01

20

ZM

IN

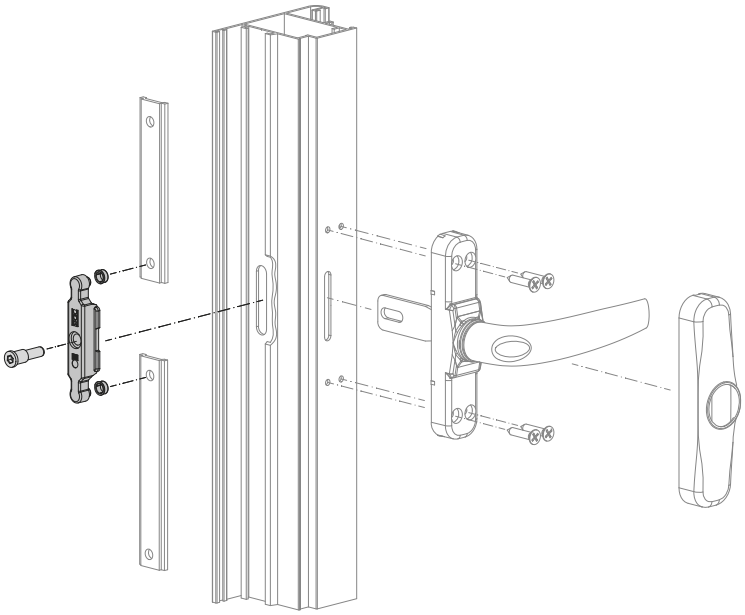
BOLTS

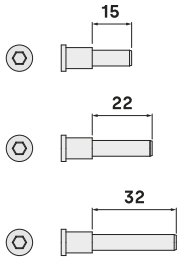
BULONES

-

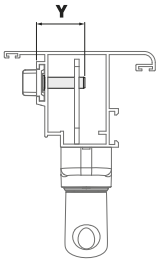
Accessory kits that allow transmission on outward opening windows.

- Supplied with short bolt (ref. 07044).
- Kit made up of transmission parts, sliders, adjustable locking points and washers to increase to Ø8.
- Available transmission kits consisting of transmission part with large bolt (ref. 075K01).

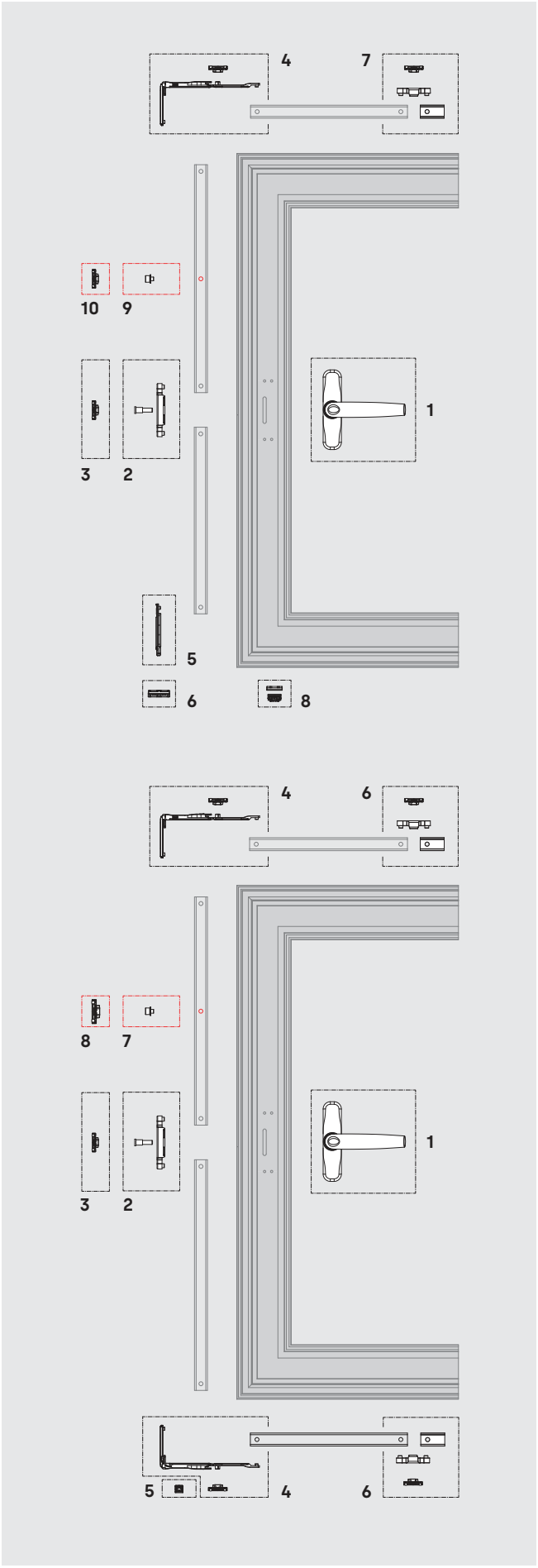




BOLTS		DIMENSION (Y)
07044	SHORT BOLT (Ø10 mm)	19 mm
07060	LONG BOLT (Ø10 mm)	26 mm
07129	SPECIAL BOLT (Ø10 mm)	36 mm



Accessories for sash height greater than 1200 millimetres



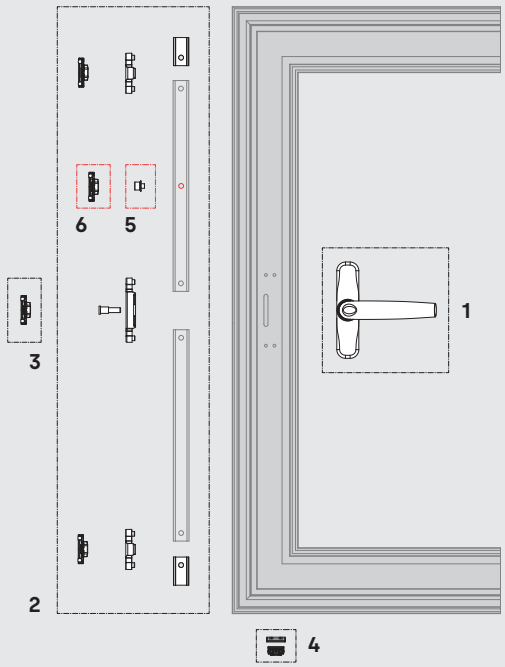
LOCKING TO FRAME WITH TRANSMISSION KIT
CORNER DRIVE AND POINTER

1	**HYDRA CREMONE**	*Consult cremone section*	
2	**074K01**	**TRANSMISSION KIT** (SHORT BOLT)	20
	075K01	**TRANSMISSION KIT** (LONG BOLT)	
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**07085**	**Ø6 SIDE-HUNG OPENING POINTER**	1000
6	**0507105**	**UNIVERSAL LOCKING POINT**	200
7	**113000**	**SUPPLEMENTARY LOCKING POINT KIT**	50
8	**0710621**	**POSITIONING LATCH KIT**	100
9	**0517002**	**Ø6 TRANSMISSION RIVET**	50
10	**110200**	**ADJUSTABLE LOCKING POINT**	100

LOCKING TO FRAME WITH TRANSMISSION KIT
DOUBLE CORNER DRIVE

1	**HYDRA CREMONE**	*Consult cremone section*	
2	**074K01**	**TRANSMISSION KIT** (SHORT BOLT)	20
	075K01	**TRANSMISSION KIT** (LONG BOLT)	
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**113000**	**SUPPLEMENTARY LOCKING POINT KIT**	50
7	**0517002**	**Ø6 TRANSMISSION RIVET**	50
8	**110200**	**ADJUSTABLE LOCKING POINT**	100

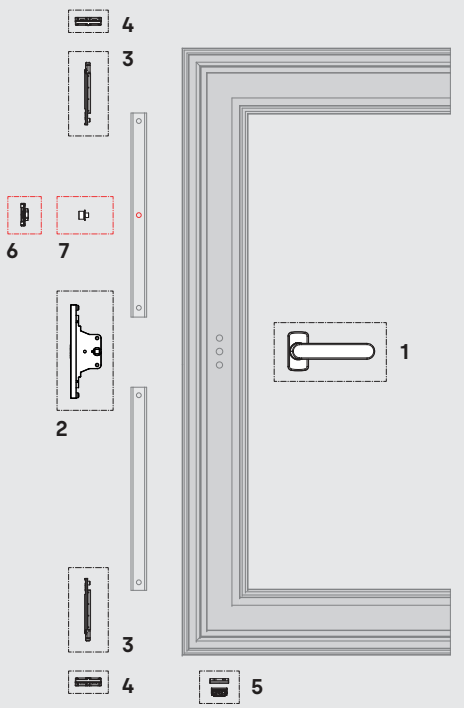
Accessories for sash height greater than 1200 millimetres



LOCKING TO INVERTER
WITH OUTWARD OPENING KIT

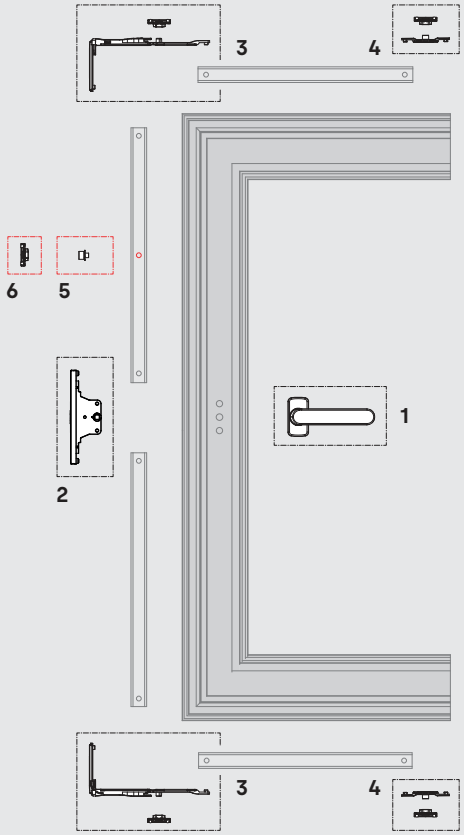
1	**HYDRA CREMONE**	*Consult cremone section*	
2	**074K01C**	**OUTWARD OPENING KIT** (SHORT BOLT)	20
	074K01L	**OUTWARD OPENING KIT** (LONG BOLT)	
3	**110200**	**ADJUSTABLE LOCKING POINT**	100
4	**0710621**	**POSITIONING LATCH KIT**	100
5	**0517002**	**Ø6 TRANSMISSION RIVET**	50
6	**110200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height greater than 1200 millimetres



LOCKING TO INVERTER WITH GEARBOX
DOUBLE POINTER

1	**HANDLE**	*Consult the section of compatible handles with outward opening*	
2	**0507248**	**TWO-WAY GEARBOX**	25
3	**07085**	**Ø6 SIDE-HUNG OPENING POINTER**	1000
4	**0507105**	**UNIVERSAL KEEPER**	200
5	**0710621**	**POSITIONING LATCH KIT**	100
6	**0517002**	**Ø6 TRANSMISSION RIVET**	50
7	**110200**	**ADJUSTABLE LOCKING POINT**	100



LOCKING TO FRAME WITH GEARBOX
WITH DOUBLE CORNER DRIVE

1	**HANDLE**	*Consult the section of compatible handles with outward opening*	
2	**0507248**	**TWO-WAY GEARBOX**	25
3	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
4	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
5	**0517002**	**Ø6 TRANSMISSION RIVET**	50
6	**110200**	**ADJUSTABLE LOCKING POINT**	100

CH180 PLUS HINGE KIT

Side-hung opening

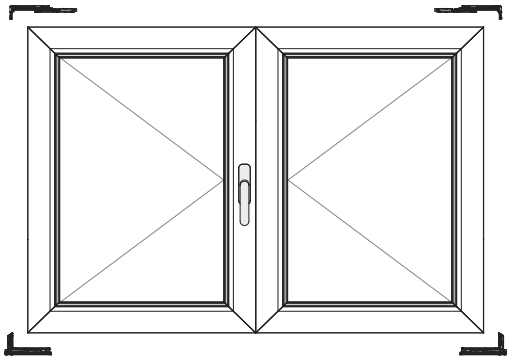
H1214	R
	L

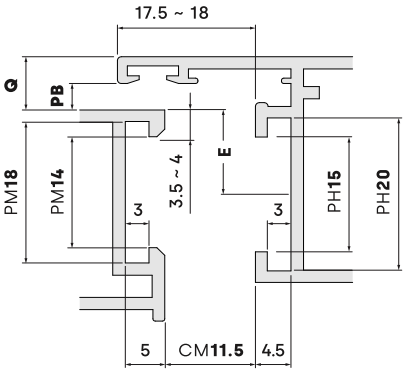
1

AL ZM IN PE

CH180 PLUS hinges for side-hung windows.

- Maximum load per sash of 100 kg and opening up to 180°.
- Supplied with opening hand: right-hand (ref. H1214R) or left-hand (ref. H1214L).
- Upper hinge adjustments: tightening (±1.5 mm) and slipping (±3 mm).
- Lower hinge adjustments: height (-0.5, 2.5 mm), tightening (0.5 mm) and lateral (±1.5 mm).





EURO GROOVE

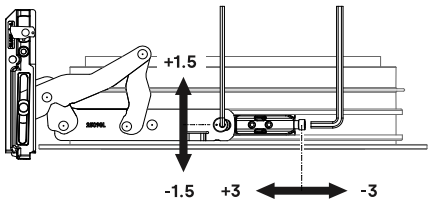
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
SHAFT **E**	10 mm

MAXIMUM ANGLE (PB)

3	175°
4.5	180°
5.5	180°

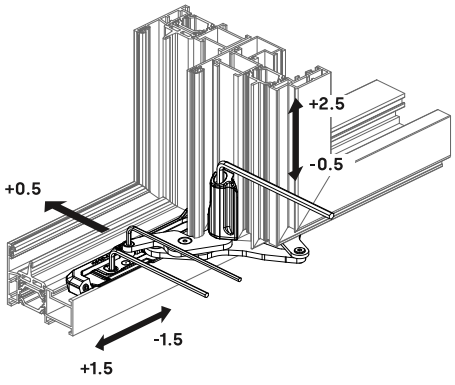
MAXIMUM ANGLE (Q)

3	130°
4.5	130°
5.5	90°



UPPER HINGE ADJUSTMENTS

PRESSURE	±1.5 mm
DROP	±3 mm



LOWER HINGE ADJUSTMENTS

PRESSURE	+0.5 mm
HEIGHT	-0.5, +2.5 mm
LATERAL	±1.5 mm

Accessories for frame profile that allows the locking of side-hung windows by means of levers.

Double-sash keeper (ref. 110501)

- Double-sash configurations with ± 2.5 mm tightening adjustment.
- Anti-slip groove allowing progressive adjustment by increasing the adjustment of the locking system.

Polyamide keeper (ref. 0507059)

- Supplied in polyamide PA6.
- Single sash configurations with 1 mm tightening adjustment.

Zamak keeper (ref. 0507105)

- Supplied in zamak.
- Single sash configurations.

DOUBLE-SASH KEEPER

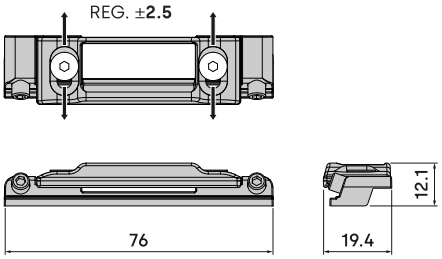
110501

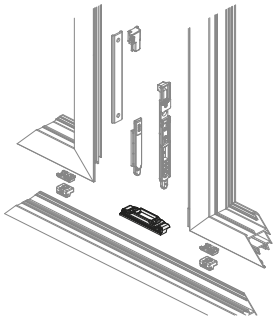
200

ZM

IN

110501





KEEPER (PA)

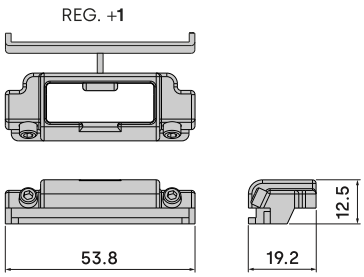
IN

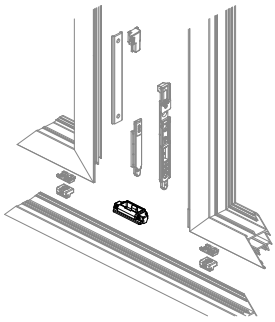
PA

0507059

100

0507059





KEEPER (ZM)

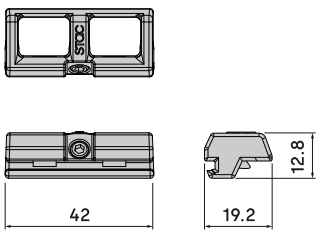
ZM

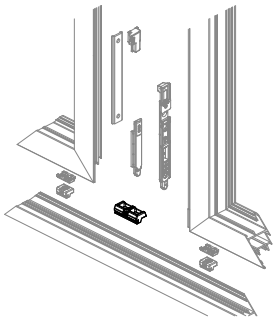
IN

0507105

200

0507105





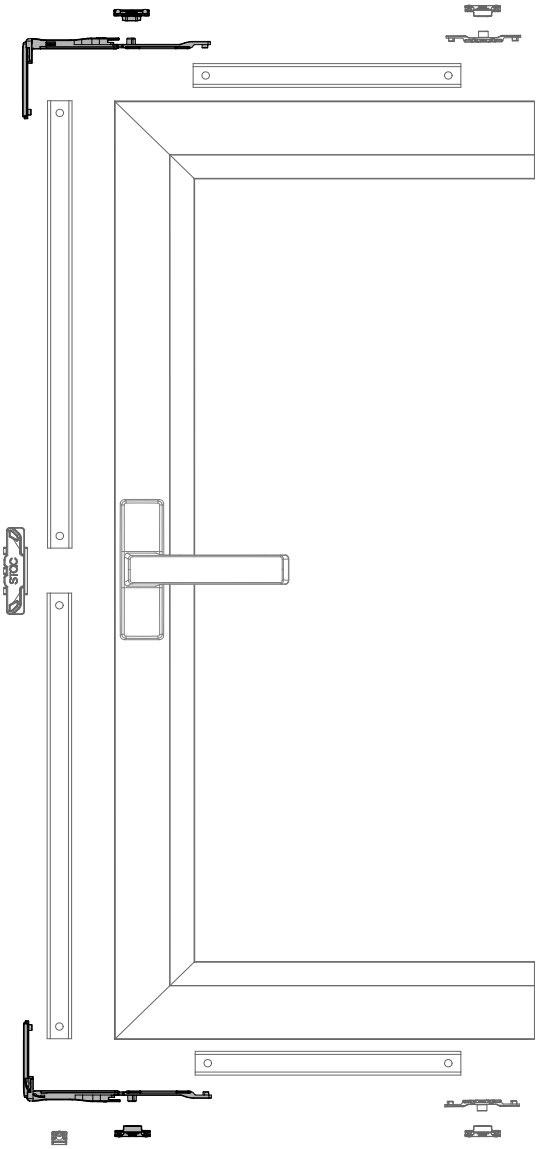
CORNER DRIVE

113420

20

Corner drive for side-hung, top-hung and bottom-hung windows.

- The corner drive is supplied with a keeper point with adjustment.
- Valid for locking to the frame profile configurations.



08 TILT & TURN OPENING

166 CLX 160 hinges

168 CLX 160 tilt & turn

172 CH180 PLUS hinges

174 CH180 PLUS tilt & turn

176 CLASSIC hinges

178 CLASSIC tilt & turn

182 SMART hinges

184 SMART tilt & turn

188 Tilt & turn friction stays

190 Double tilt & turn systems

192 Corner drives

193 Tilt & turn accessories

195 Bulk packaging

CLX160

Euro Groove

Load up 160 kg

Ambidextrous hinges

3-dimensional adjustment

CLX160

CLX160 hinges are designed for high loads on tilt & turn windows and offer a wide adjustment range.

They allow a load capacity of up to 160 kg per sash and incorporate safety measures to prevent the sash from falling in the event of breakage due to malicious blows or breakage of the frame groove.

With this hardware, which is compatible with all traditional STAC tilt & turn elements, customers have a competitive and high-performance solution for everyday windows.

» Safe system

Sash profile leaf

Security pin in the lower sash hinge leaf counterplate that prevents the sash falling where the counterplate has broken due to strong blows or displacements that affect how the hinges work.

Frame profile leaf

The stabilising screw provides an additional fixing point, giving the blade a better load balance, reducing the forces it must support and avoiding possible breakages in the groove.

» Ambidextrous hinges

The evolution of the sash blade has given the hinge greater versatility during installation in windows. Whilst also reducing the client's stock management load.

The simplified leaf design allows both the right and left leaf to be included in the tilt & turn kits, allowing the kits to become ambidextrous.

» Simple and effective adjustments

With highly loaded windows it is essential that the adjustments are simple, effective and direct but above all that they are reliable.

All adjustment points are located in easy-to-reach areas and can be regulated with the leaf in place.

- Height adjustment +3, -0.5 mm with a 5 mm allen key.
- Lateral adjustment ±1.3 mm with a 2.5 mm allen key.
- Tightening adjustment +0.3 mm using a bushing.

CLX tilt & turn hinges specially designed for high loads in side-hung and tilt & turn windows.

- The hinges allow a maximum load per sash of 160 kg.
- Specific kits for active sash (ref. T1113) and passive sash (ref. T1114).
- Ambidextrous hinges: the hinges are supplied pre-assembled for right-handed openings, but they include both the left and the right-leaf.
- Specific tilt & turn friction stays for traditional manoeuvre (ref. T110301) or tilt-before-turn (ref. T110302) not included.

Available in an anodised finish with a maximum load per sash of 120 kg, the opening hand of the sash must be indicated.

CLX 160 HINGES
Active sash

AL ZM
IN PA

T1113	**

25

CLX 160 HINGES
Passive sash

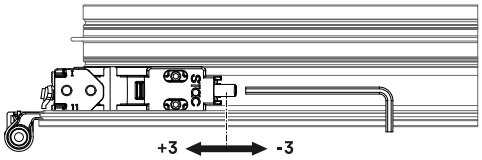
AL ZM
AC PA

T1114	**

20

UPPER HINGES ADJUSTMENTS PASSIVE SASH

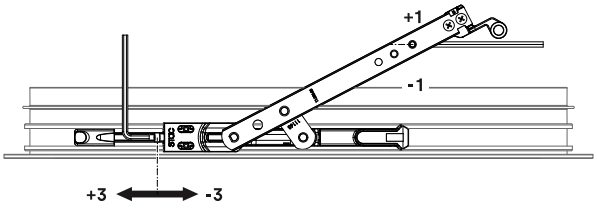
DROP ±3 mm



UPPER HINGES ADJUSTMENTS ACTIVE SASH

PRESSURE ±1 mm

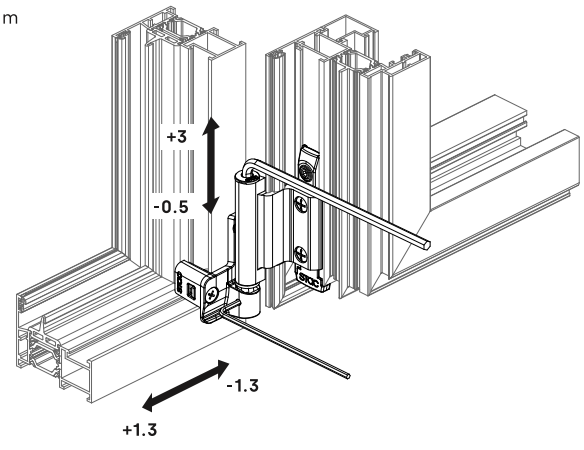
DROP ±3 mm



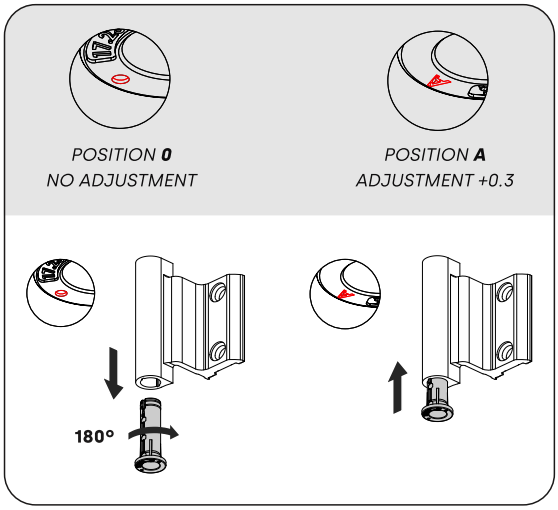
LOWER HINGES ADJUSTMENTS

HEIGHT −3, +0.5 mm

LATERAL ±1.3 mm



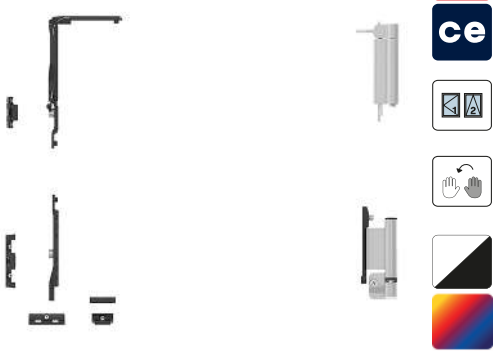
PRESSURE ADJUSTMENT



CLX 160 KIT FOR LOCKING TO THE INVERTER
Without cremone

T1133 | **

24



CLX 160 KIT FOR LOCKING TO THE INVERTER
With LYRA cremone

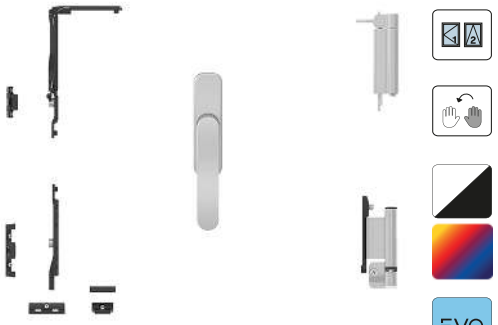
T1118 | **

16

CLX 160 KIT FOR LOCKING TO THE INVERTER
With SIRIUS cremone

T1118 | ** | S

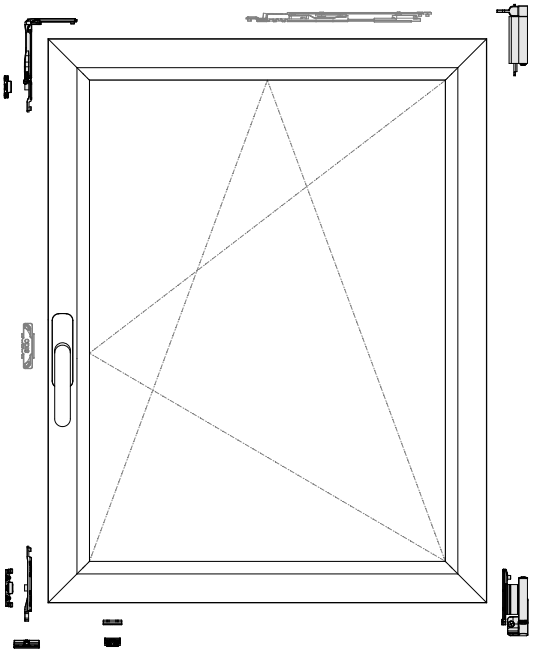
16



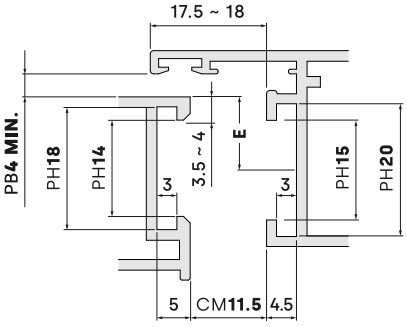
Base kit with CLX hinges for tilt & turn openings with locking to window inverter.

- Maximum load per sash: 160 kg.
- The reference is supplied without cremone.
- Available with LYRA (ref. T1118) cremone or SIRIUS cremone (ref. T1118S).
- Specific tilt & turn friction stay for traditional manoeuvre (locking to inverter or locking to frame) not included in the kit (ref. T110301).
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.

The kits do not include tilt & turn friction stays. See compatible friction stays.



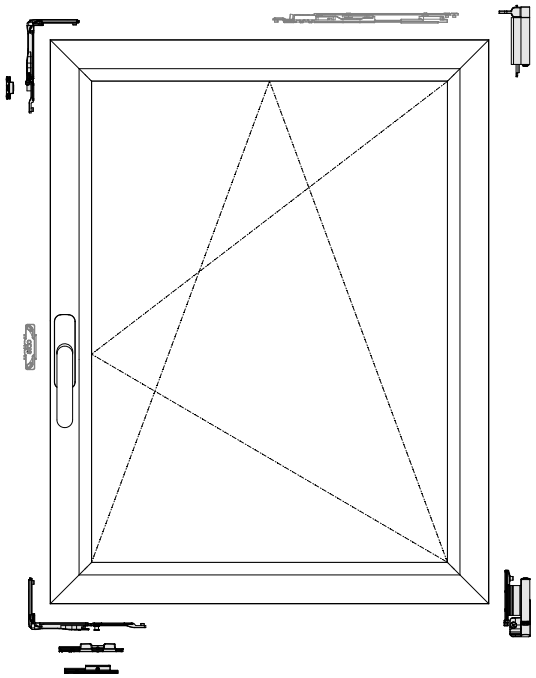
EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

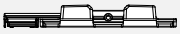


CLX 160 hinge kit for tilt & turn windows with locking to the window frame.

- Maximum load per sash: 160 kg.
- The reference is supplied without cremone.
- Available with SIRIUS cremone (ref. T1132S).
- Specific tilt & turn friction stay for traditional manoeuvre (locking to inverter or locking to frame) not included in the kit (ref. T110301).
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.

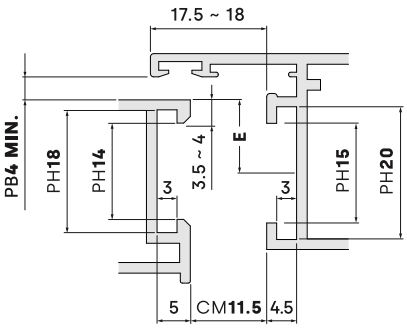
The kits do not include a tilt & turn friction stays. See compatible friction stays.





LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER



EURO GROOVE

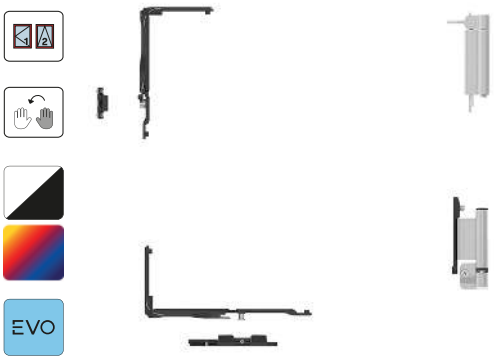
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

CLX 160 KIT FOR LOCKING TO THE FRAME

Without cremone

T1139	**

24

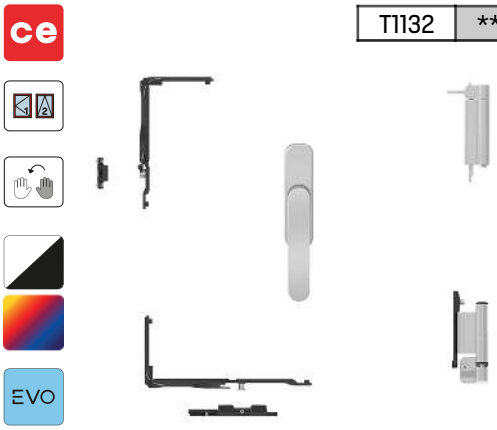


CLX 160 KIT FOR LOCKING TO THE FRAME

With SIRIUS cremone

T1132	**	S

16



CLX 160 KIT FOR TILT-BEFORE-TURN

Without cremone

T1123 **

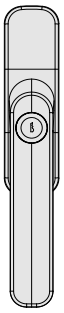
24

Base kit with CLX hinges for windows with tilt-before-turn manoeuvre

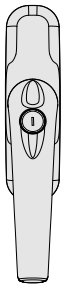
- Maximum load per sash: 160 kg.
- The reference is supplied without cremone.
- Specific tilt & turn friction stay for tilt-before-turn manoeuvre not included in the kit (ref. T110302).
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.
- The tilt-before-turn kit is suitable for use with key-operated cremone locks provides two locking positions that allow the window to tilt but not to open in the side-hung position.

The kit does not include a tilt & turn friction stay. See compatible friction stays.

		TRANSMISSION COVER 0511193
A1160 **	SIRIUS BLOCK	✔
A1163 **		-
1129 **	LYRA BLOCK	✔
1145 **		-

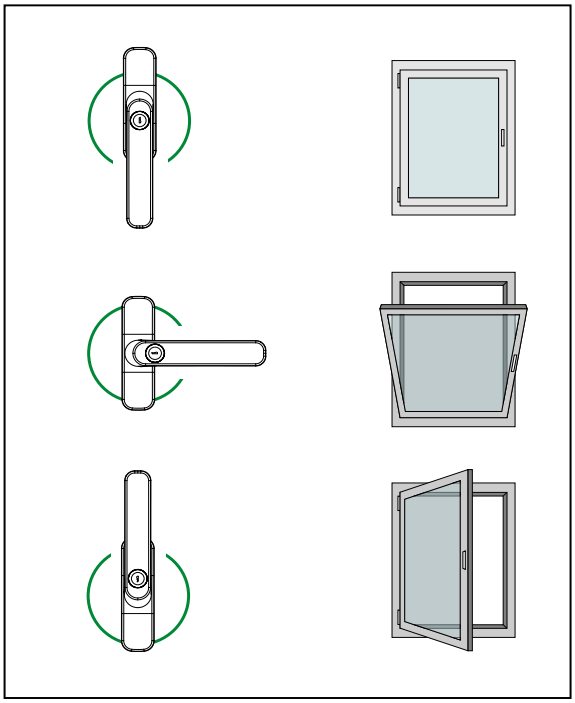


SIRIUS BLOCK
A1160, A1163

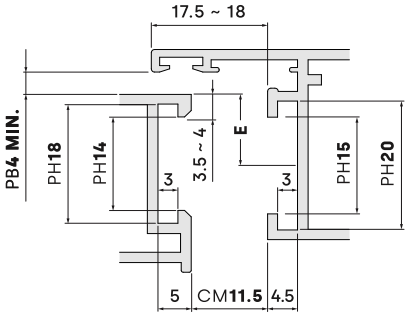


LYRA BLOCK
1129, 1145

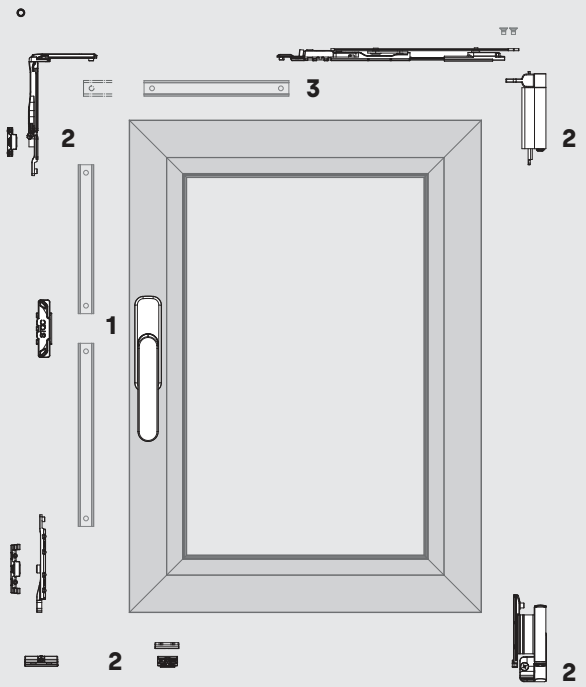
CREMONE WITH KEY + TILT-BEFORE-TURN WINDOW



EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

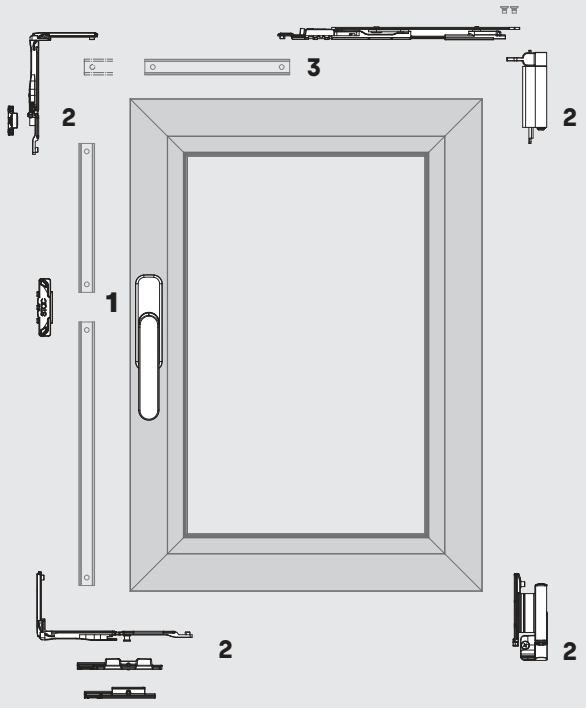


The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes, please refer to the technical catalog.



CLX TILT & TURN KIT
LOCKING TO INVERTER

1	**CREMONE** Refer to « cremones »			
2	**T1133**	**	**LOCKING TO INVERTER WITHOUT CREMONE KIT**	24
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10



CLX TILT & TURN KIT
LOCKING TO FRAME / TILT-BEFORE-TURN

1	**CREMONE** Refer to « cremones »			
2	**T1139**	**	**LOCKING TO FRAME WITHOUT CREMONE KIT**	24
	T1123	**	**TILT-BEFORE-TURN WITHOUT CREMONE KIT**	
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10

Euro Groove

180°

Full opening

Quick and simple installation

3-dimensional adjustment

CH180 PLUS

The trend in architecture and design for pure, simple spaces and objects has meant that closure systems have had to evolve in this direction, offering systems with ever larger fenestrations and doors / windows to make the light the leading element. This has meant that accessories have been reduced to the bare essentials.

Different concealed hinge fittings adapted to Euro Groove systems are available on the market. However, customers and system designers demand a fully opening concealed hinge. The CH180 PLUS hinge was created in response to this need.

» Multiple opening systems

The concealed hinge is suitable for a wide range of aluminium opening systems:

- Bottom-hung windows.
- Side-hung windows.
- Tilt & turn windows (traditional manoeuvre as a tilt-before-turn).

» Full 180° opening

It allows smooth 180° openings with almost no wasteful movements through adjustment and simplification of the hinge articulation kinematics.

» Three dimensional adjustment

All adjustment points are located in easily accessible areas and can be carried out with the door leaf loaded. The hinges incorporate three-dimensional adjustment to facilitate on-site adjustments.

- Horizontal adjustment: ±1.5 millimetres
- Closure adjustment: + 0.5 millimetres
- Vertical adjustment: + 2.5, – 0.5 millimetres

» Rapid and simple installation

The hinges have been designed with the fitter in mind. To this end, all parts are supplied pre-assembled and no type of machining or special operations on the profiles is required.

In the lower hinge, there are two well differentiated parts which are pre-mounted to the frame and sasg and they are then attached together during the assembly of the sash into the frame.

Tilt & turn concealed hinges valid for side-hung or tilt & turn windows.

- Maximum load per sash is 100 kg and maximum opening is 180º.
- The hinges keep the opening hand: right-hand opening (R) or left-hand opening (L).
- Supplied with integrated tilt & turn friction stay (small or large).
- Available for traditional manoeuvre (ref. R1, R3) or tilt-before-turn (ref. R2, R4).
- Compatible supplementary corner drives (ref. H122100, ref. H122100S).

CH180 PLUS HINGES (LARGE FRICTION STAY)

Traditional manoeuvre

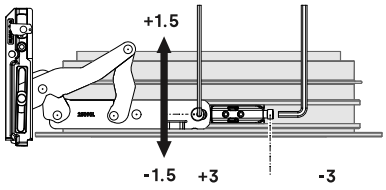
H1213	R1 / L1

1

AL ZM IN PA

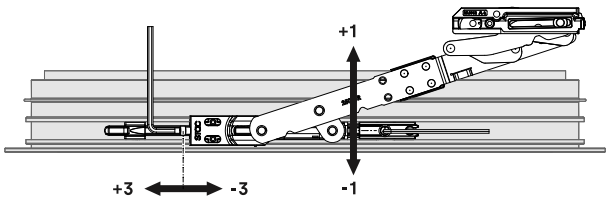
UPPER HINGES ADJUSTMENTS PASSIVE SASH

PRESSURE	±1.5 mm
DROP	±3 mm



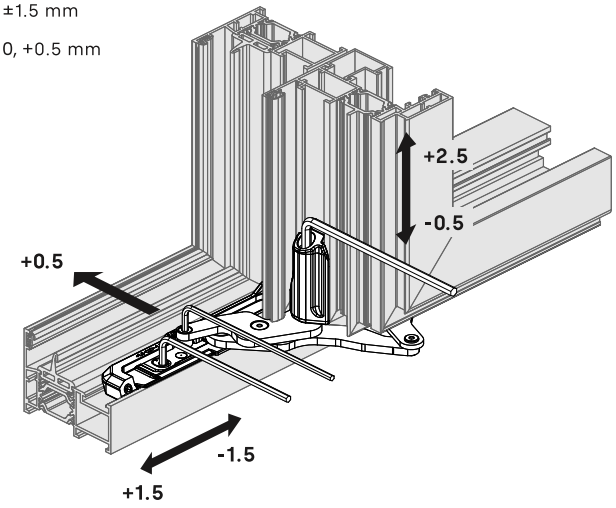
UPPER HINGES ADJUSTMENTS ACTIVE SASH

PRESSURE	±1 mm
DROP	±3 mm



LOWER HINGES ADJUSTMENTS

HEIGHT	+2.5, -0.5 mm
LATERAL	±1.5 mm
PRESSURE	0, +0.5 mm



CH180 PLUS HINGES (SMALL FRICTION STAY)

Traditional manoeuvre

H1213	R3 / L3

1

AL ZM IN PA

	OPENING TYPE	FRICTION STAY
H1213 R1 / L1	LOCKING TO INVERTER / FRAME	LARGE
H1213 R2 / L2	TILT-BEFORE-TURN	LARGE
H1213 R3 / L3	LOCKING TO INVERTER / FRAME	SMALL
H1213 R4 / L4	TILT-BEFORE-TURN	SMALL

Tilt & turn kit with CH180 PLUS concealed hinges.

- Maximum load per sash is 100 kg and maximum opening is 180º.
- Hinges keep the opening hand of the window: right-hand opening (R) or left-hand opening (L).
- The top hinge is supplied with an integrated large tilt & turn friction stay.
- Compatible supplementary corner drives (ref. H122100, ref. H122100S).
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.

CH180 PLUS KIT FOR LOCKING TO THE INVERTER

Large friction stay

H1233	R1
	L1

1

CH180 PLUS KIT FOR LOCKING TO THE FRAME

Large friction stay

H1239	R1
	L1

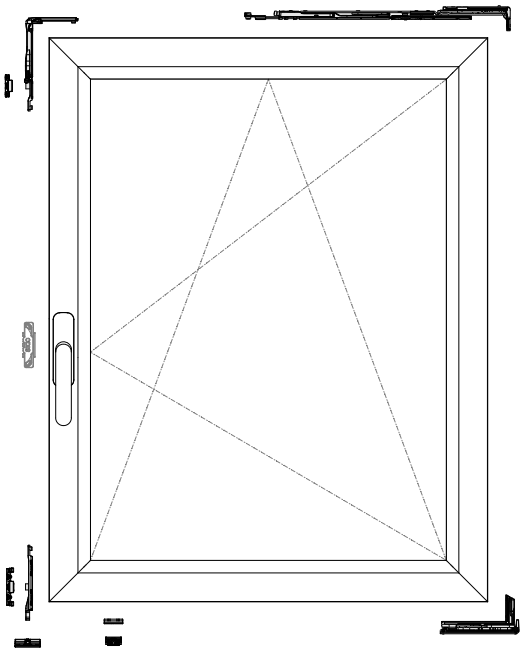
1

CH180 PLUS KIT FOR TILT-BEFORE-TURN

Large friction stay

H1223	R2
	L2

1

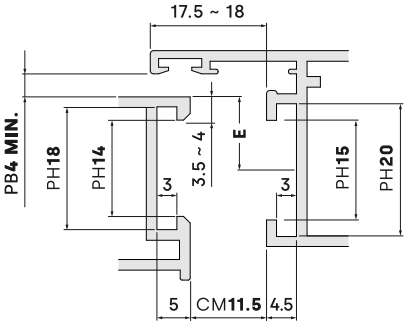


LARGE FRICTION STAY TRADICIONAL MANEUVER

SMALL FRICTION STAY TILT-BEFORE-TURN

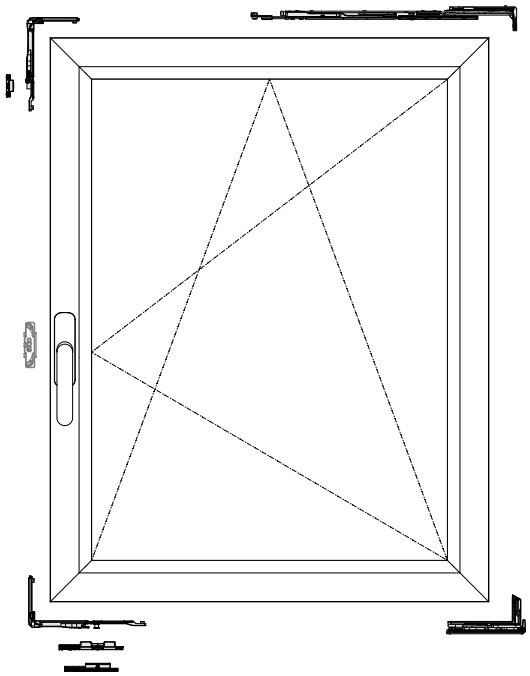
EURO GROOVE

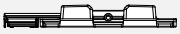
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm



Tilt & turn kit with CH180 PLUS concealed hinges.

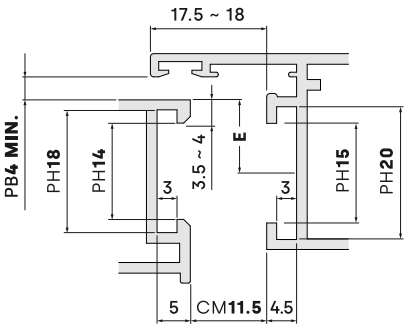
- Maximum load per sash is 100 kg and maximum opening is 180º.
- Hinges keep the opening hand of the window: right-hand opening (R) or left-hand opening (L).
- The top hinge is supplied with an integrated small tilt & turn friction stay.
- Compatible supplementary corner drives (ref. H122100, ref. H122100S).
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.





LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER



EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

CH180 PLUS KIT FOR LOCKING TO THE INVERTER

Small friction stay

H1233	R3 / L3

1

CH180 PLUS KIT FOR LOCKING TO THE FRAME

Small friction stay

H1239	R3 / L3

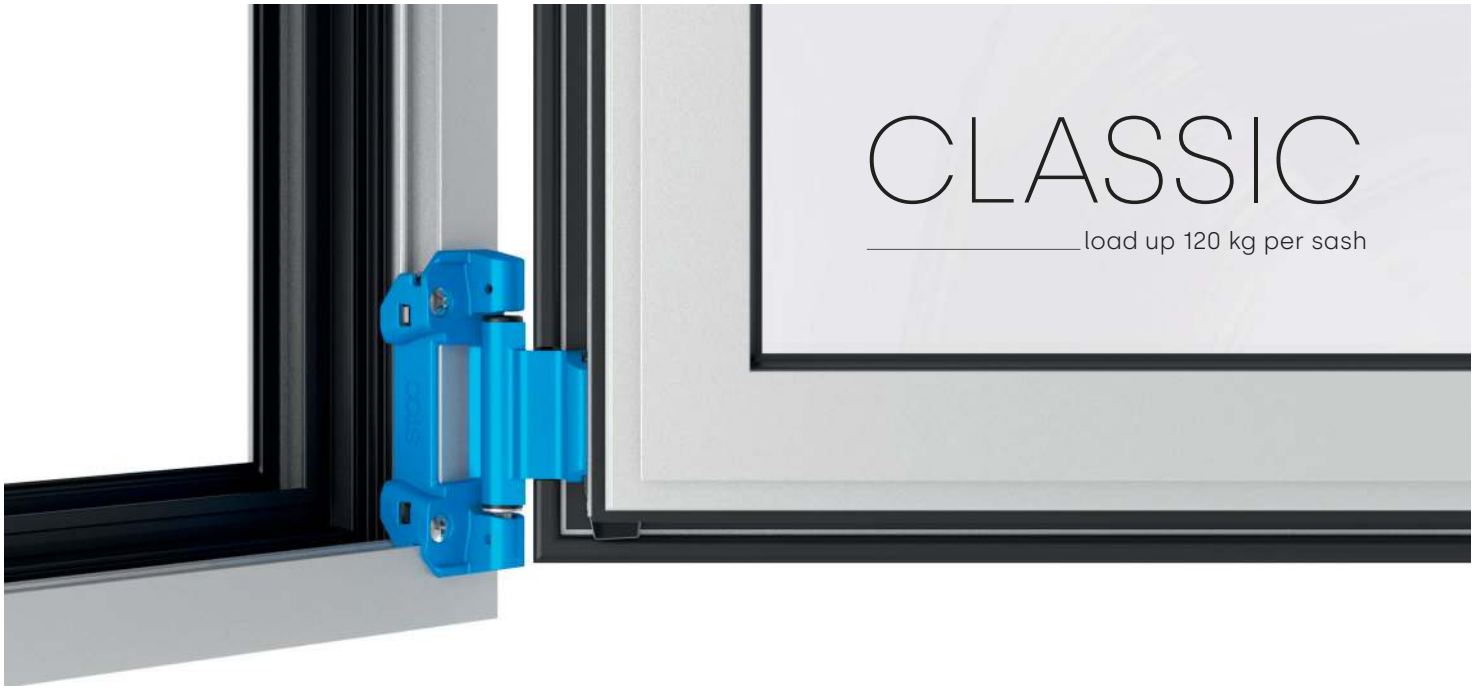
1

CH180 PLUS KIT FOR TILT-BEFORE-TURN

Small friction stay

H1223	R4 / L4

1



Euro Groove

Maximum load of 120 kg

Ambidextrous hinges

Corrosion resistance

CLASSIC

The CLASSIC hinges for Euro Groove profiles have been designed and developed to fit the most common types of windows in the aluminium enclosure market and are compatible with all traditional STAC tilt & turn elements.

The hinges have a frame fixing screw that considerably reduces fatigue stresses and the sash leaf is compatible with hinge pitches from 3 millimetres.

» Maximum load of 120 kg

CLASSIC tilt & turn hinges allow window configuration up to 120 kg per sash. In addition, the location of the window or the height of the cremone may vary the recommended configurations. From 1000 millimetres in height or width and 80 kg in weight, it is recommended that the profile used is a balcony door profile.

» Ambidextrous hinges

Hinges are ambidextrous, i.e. they do not keep the opening hand of the window. The sash hinges have the lower hinge axis already inserted into the sash leaf at the factory. By default, the hinges are pre-assembled on the right-hand side and are easily interchangeable on the left-hand side.

» Corrosion resistance

The zamak parts have been treated to ensure a corrosion resistance of more than 240 hours in saline mist chamber in compliance with UNE-EN 1670 obtain a grade 4 rating that gives high corrosion resistance and usability in environments with severe weather conditions.

Tilt & turn hinges designed for side-hung and tilt & turn windows.

- Maximum load per sash 120 kg.
- Specific kits for active sash (ref. B1113) and passive sash (ref. B1114).
- They are supplied pre-assembled for right-hand opening but are interchangeable.
- The active leaf hinges are compatible with the false friction stay (ref. T110100) which provides horizontal adjustment of ± 3 mm and tightening of ± 1.5 mm.

CLASSIC HINGES
Active sash

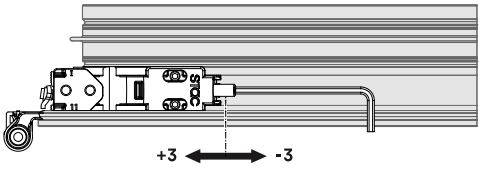
B1113 | **

25

AL ZM AC IN

UPPER HINGES ADJUSTMENTS PASSIVE SASH

DROP ±3 mm



CLASSIC HINGES
Passive sash

B1114 | **

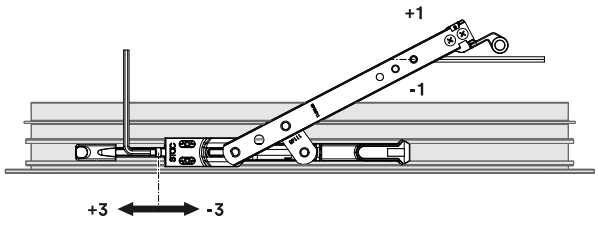
20

AL ZM AC IN

UPPER HINGES ADJUSTMENTS ACTIVE SASH

PRESSURE ±1 mm

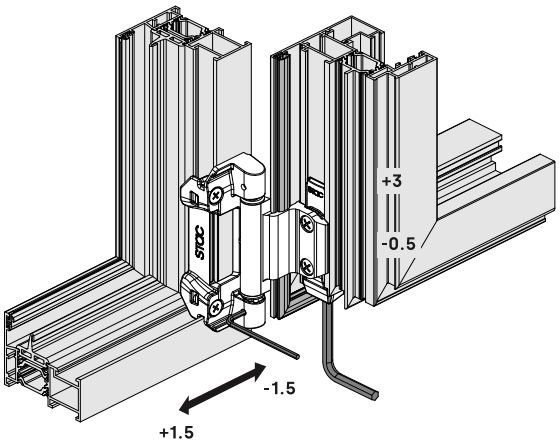
DROP ±3 mm



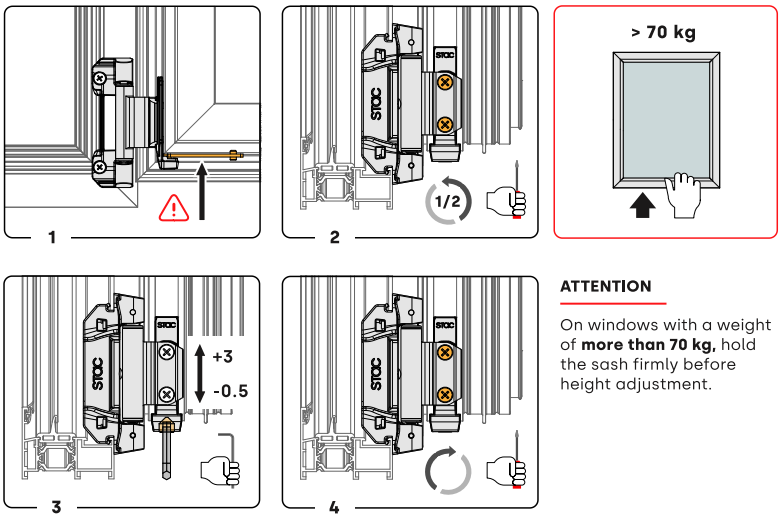
LOWER HINGES ADJUSTMENTS

HEIGHT −3, +0.5 mm

LATERAL ±1.5 mm



HEIGHT ADJUSTMENTS



ATTENTION

On windows with a weight of **more than 70 kg,** hold the sash firmly before height adjustment.

CLASSIC KIT FOR LOCKING TO THE INVERTER
Without cremone

1133 | **

24

AL ZM IN PA

CLASSIC KIT FOR LOCKING TO THE INVERTER
With LYRA cremone

1118 | **

16

AL ZM IN PA

CLASSIC KIT FOR LOCKING TO THE INVERTER
With SIRIUS cremone

1118 | ** | S

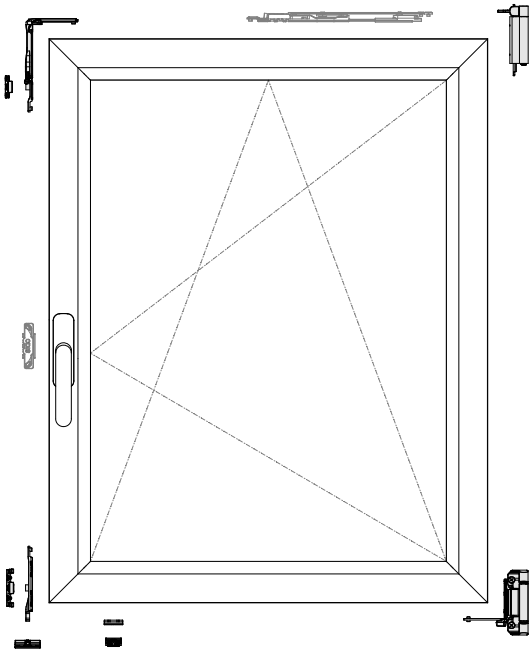
16

AL ZM IN PA

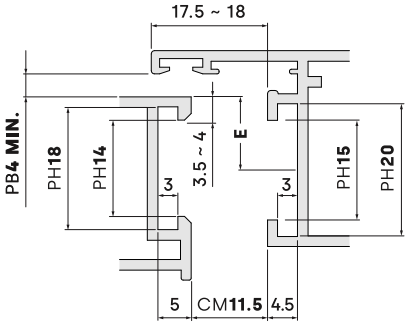
Base kit with CLASSIC hinges for tilt & turn openings with locking to the window inverter.

- Maximum load per sash 120 kg.
- The reference is supplied without cremone.
- Available with LYRA (ref. 1118) cremone or SIRIUS cremone (ref. 1118S).
- The accessories integrate adjustable locking points.

The kits do not include tilt & turn friction stays. See compatible friction stays.



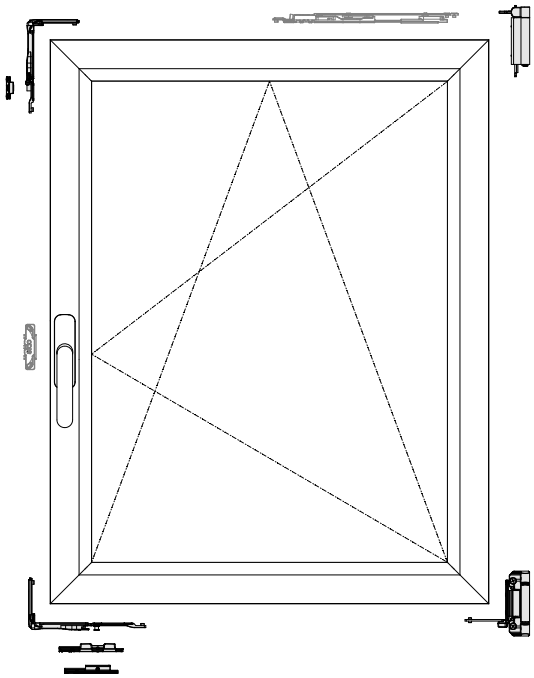
EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

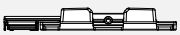


Base kit with CLASSIC hinges for tilt & turn openings with locking to the window frame.

- Maximum load per sash 120 kg.
- The reference is supplied without cremone.
- Available with LYRA (ref. 1132) cremone or SIRIUS cremone (ref. 1132S).
- The accessories integrate adjustable locking points.

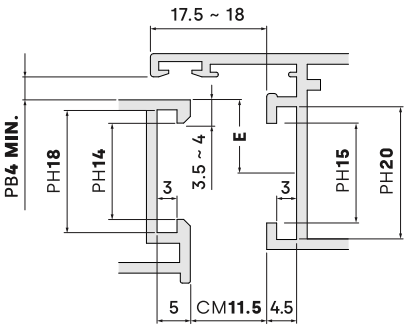
The kits do not include tilt & turn friction stays. See compatible friction stays.





LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER



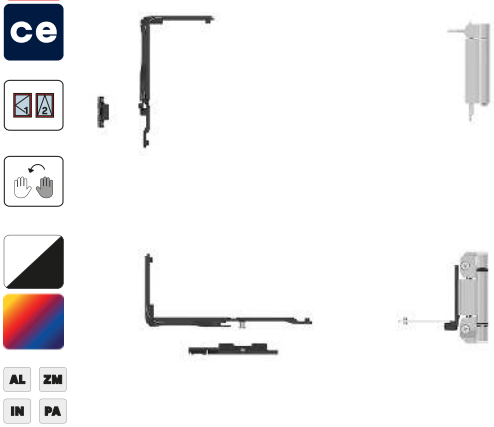
EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

CLASSIC KIT FOR LOCKING TO THE FRAME
Without cremone

1139 **

24



AL ZM IN PA

CLASSIC KIT FOR LOCKING TO THE FRAME
With LYRA cremone

1132 **

16

AL ZM IN PA

CLASSIC KIT FOR LOCKING TO THE FRAME
With SIRIUS cremone

1132 ** S

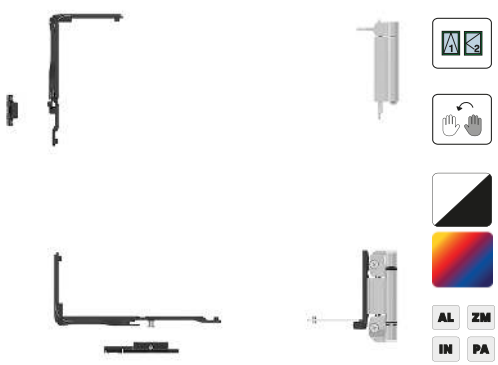
16

AL ZM IN PA

CLASSIC KIT FOR TILT-BEFORE-TURN
Without cremone

1123 **

24

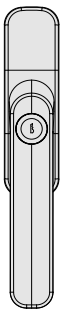


Base kit with CLASSIC hinges for windows with tilt-before-turn manoeuvre

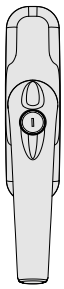
- Maximum load per sash: 120 kg.
- The reference is supplied without cremone.
- The accessories integrate adjustable locking points.
- Tilt-before-turn kit **is suitable for use with keyed cremone** as they provide two locking positions that allow the window to tilt but not to open in the turn position.

The kit does not include a tilt & turn friction stay. See compatible friction stays.

		TRANSMISION COVER 0511193
A1160 **	SIRIUS BLOCK	✔
A1163 **		-
1129 **	LYRA BLOCK	✔
1145 **		-

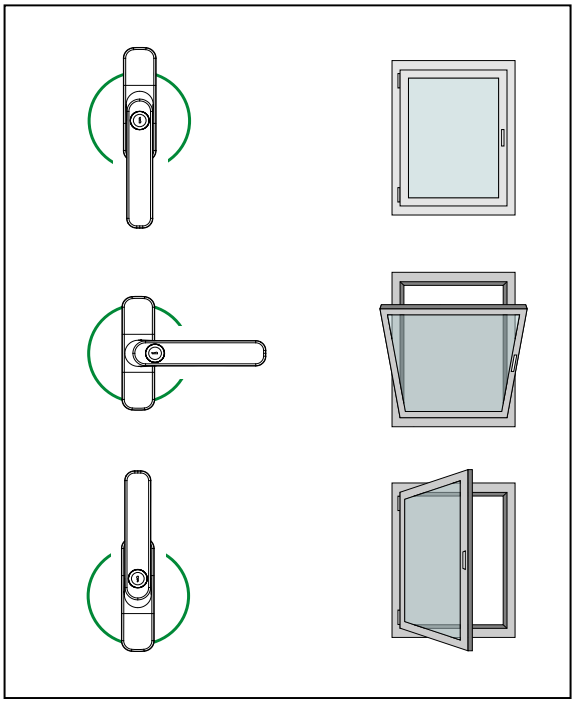


SIRIUS BLOCK
A1160, A1163

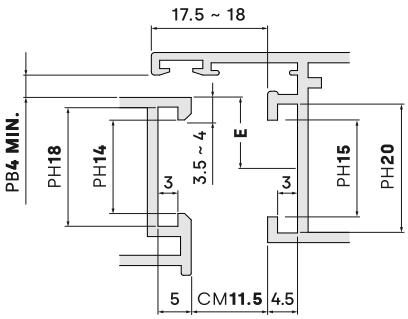


LYRA BLOCK
1129, 1145

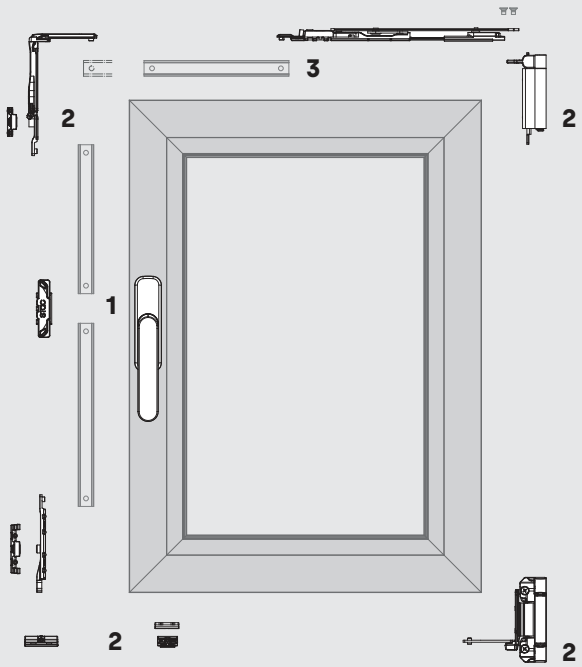
CREMONE WITH KEY + TILT-BEFORE-TURN WINDOW



EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

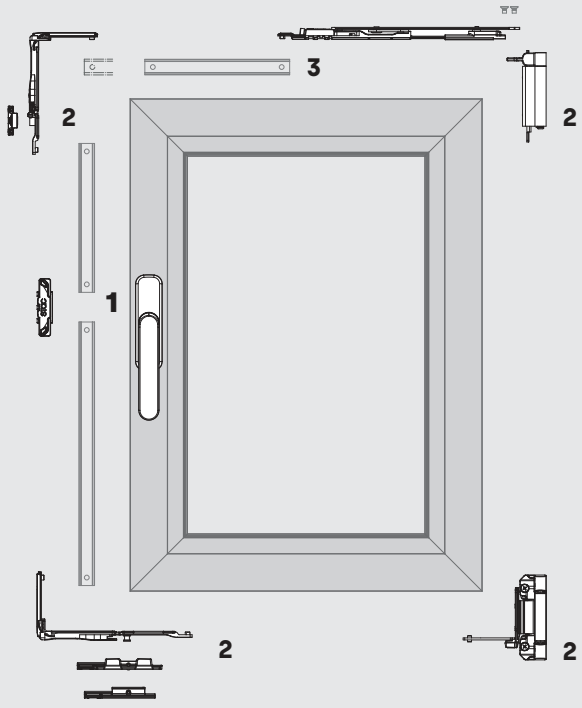


The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes, please refer to the technical catalog.



CLASSIC TILT & TURN KIT
LOCKING TO INVERTER

1	**CREMONE** Refer to « cremones »			
2	**1133**	**	**LOCKING TO INVERTER WITHOUT CREMONE KIT**	24
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10



CLASSIC TILT & TURN KIT
LOCKING TO FRAME / TILT-BEFORE-TURN

1	**CREMONE** Refer to« cremones »			
2	**1139**	**	**LOCKING TO FRAME WITHOUT CREMONE KIT**	24
	1123	**	**TILT-BEFORE-TURN WITHOUT CREMONE KIT**	
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10

Euro Groove

Encased hardware

Ambidextrous hinges

3-dimensional adjustment

SMART

SMART tilt & turn hinges for Euro Groove profiles have been designed and developed to fit the most common window types in the aluminium window market.

With this hardware that is fully compatible with all the traditional tilt and turn elements developed by STAC, customers have a competitive and high-performance solution for everyday window.

» Smooth closing

All SMART tilt & turn systems include EVO-SOFT encased locking points which is why the hardware with bushings in the locking points prolongs the durability of the keepers and reduces wear and tear due to friction, while facilitating smoother closing of the windows.

» Ambidextrous hinges

SMART hinges are ambidextrous, i.e. they do not keep the window opening hand, this simplifying order management and considerably reducing stock.

» Three-dimensional adjustment

In windows, simple, effective and straightforward adjustments are necessary, but above all they must be reliable. The SMART bottom hinge allows three-dimensional adjustment (height, lateral and pressure adjustment).

» UNE-EN 13126-8 standard

The SMART hinges have passed with satisfactory results all the tests carried out by the TECNALIA test laboratory as described in the UNE-EN 13126-8:2018 standard, "Requirements and test methods for tilt & turn, tilt-first and turn-only hardware".

Tilt & turn hinges designed for side-hung and tilt & turn windows.

- Maximum load per sash 90 kg.
- Specific kits for active sash (ref. R1113) and passive sash (ref. R1114).
- Reversible hinges: they are valid for both the active and passive sash.
- They are supplied with zamak back plates.
- The active sash hinges are compatible with the false friction stay (ref. T110100) which provides horizontal adjustment of ± 3 mm and tightening of ± 1.5 mm.

SMART HINGES
Active sash

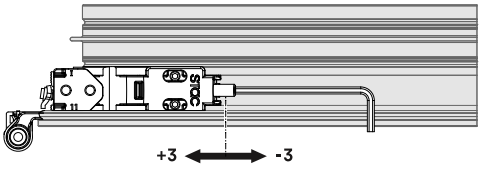
R1113 **

25

AL ZM AC IN

UPPER HINGES ADJUSTMENTS PASSIVE SASH

DROP ±3 mm



SMART HINGES
Passive sash

R1114 **

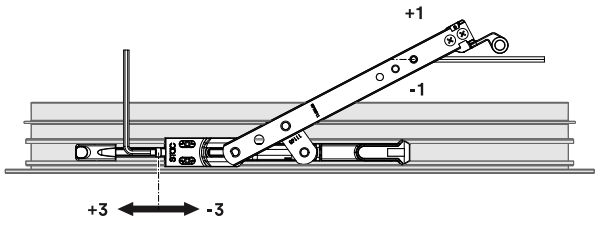
20

AL ZM AC IN

UPPER HINGES ADJUSTMENTS ACTIVE SASH

PRESSURE ±1 mm

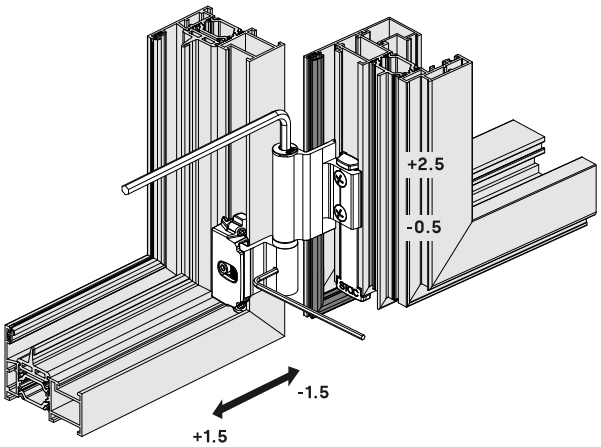
DROP ±3 mm



LOWER HINGES ADJUSTMENTS

HEIGHT +2.5, -0.5 mm

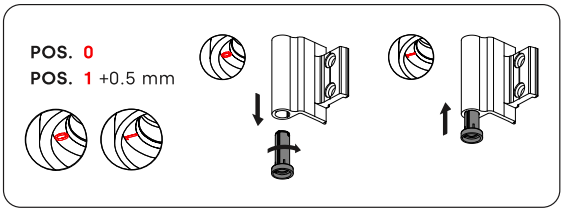
LATERAL ±1.5 mm



PRESSURE ADJUSTMENT 0, +0.5

PRESSURE	SASH WEIGHT	BUSHING
0	< 60 kg	**0**
+0.5	60 - 90 kg	**1** (+180°)

BY DEFAULT, THE HINGES ARE SUPPLIED WITH PRESSURE SETTING 0.



SMART KIT FOR LOCKING TO THE INVERTER
Without cremone

R1133 | **

24

SMART KIT FOR LOCKING TO THE INVERTER
With LYRA cremone

R1118 | **

16

SMART KIT FOR LOCKING TO THE INVERTER
With SIRIUS cremone

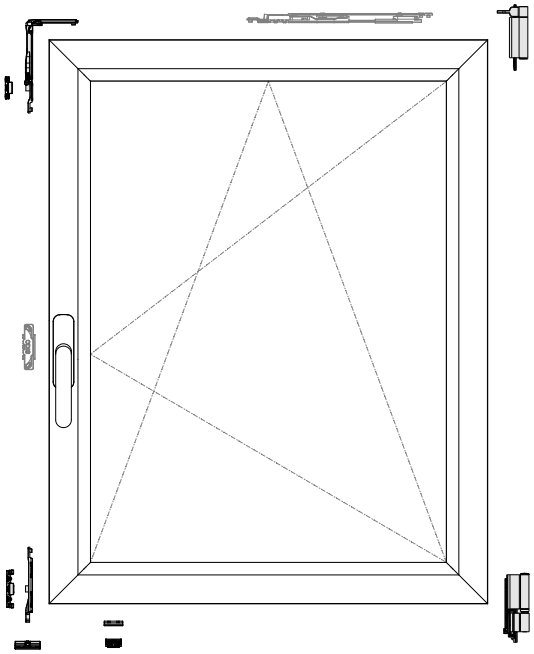
R1118 | ** | S

16

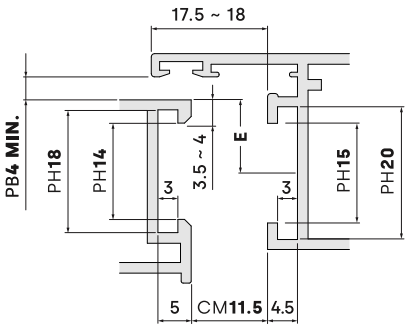
Base kit with SMART hinges for tilt & turn openings with locking to the window inverter.

- Maximum load per sash 90 kg.
- The reference is supplied without cremone.
- Available with LYRA cremone (ref. R1118) or SIRIUS cremone (ref. R1118S).
- They are supplied pre-assembled for right-hand opening.
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.

The kits do not include tilt & turn friction stays. See compatible friction stays.



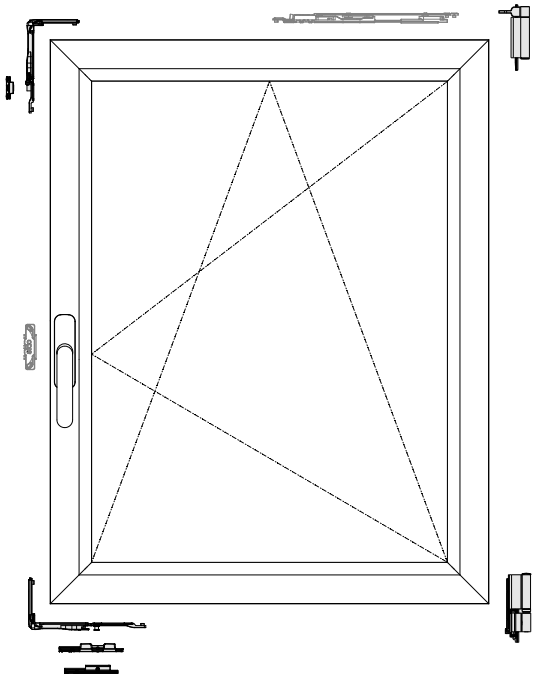
EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm



Base kit with SMART hinges for tilt & turn openings with locking to the window frame.

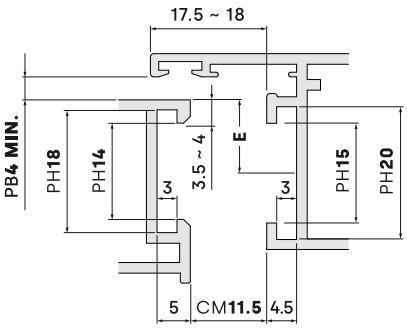
- Maximum load per sash 90 kg.
- The reference is supplied without cremone.
- Available with LYRA cremone (ref. R1132) or SIRIUS cremone (ref. R1132S).
- They are supplied pre-assembled for right-hand opening.
- The accessories are equipped with EVO-SOFT locking hardware, which prolongs the durability of the locks and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.

The kits do not include a tilt & turn friction stays. See compatible friction stays.



LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER



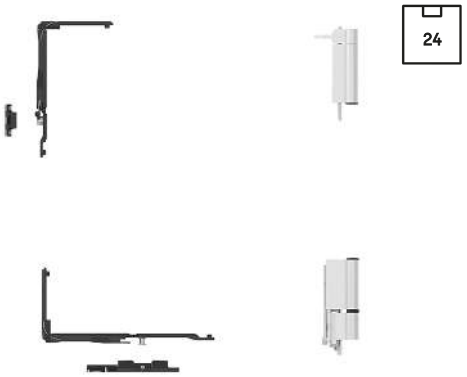
EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

SMART KIT FOR LOCKING TO THE FRAME
Without cremone

R1139 | **

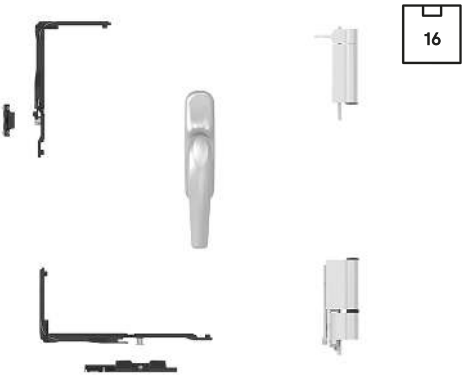
24



SMART KIT FOR LOCKING TO THE FRAME
With LYRA cremone

R1132 | **

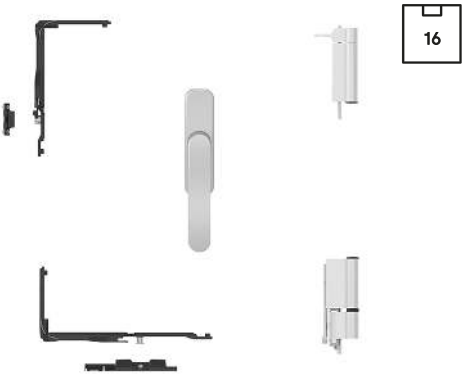
16



SMART KIT FOR LOCKING TO THE FRAME
With SIRIUS cremone

R1132 | ** | S

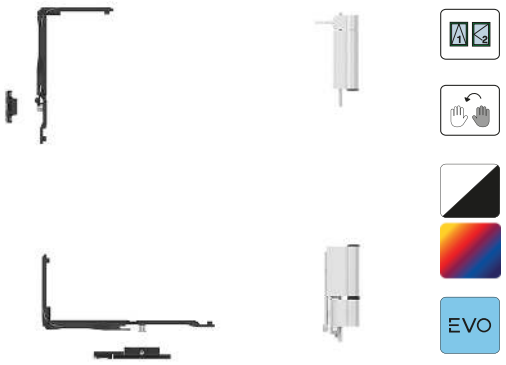
16



SMART KIT FOR TILT-BEFORE-TURN
Without cremone

R1123 **

24

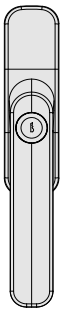


Base kit with SMART hinges for windows with tilt-before-turn manoeuvre.

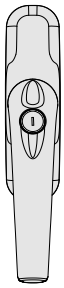
- Maximum load per sash 90 kg.
- The reference is supplied without cremone.
- The accessories are equipped with EVO-SOFT encased hardware, which prolongs the durability of the keepers and reduces wear due to friction, while facilitating a smoother window closing manoeuvre.
- Tilt-before-turn kit is suitable for use with keyed cremone as they provide two locking positions that allow the window to tilt but not to open in the turn position.

The kit do not include tilt & turn friction stay. See compatible friction stays.

		TRANSMISSION COVER 0511193
A1160 **	SIRIUS BLOCK	✔
A1163 **		-
1129 **	LYRA BLOCK	✔
1145 **		-

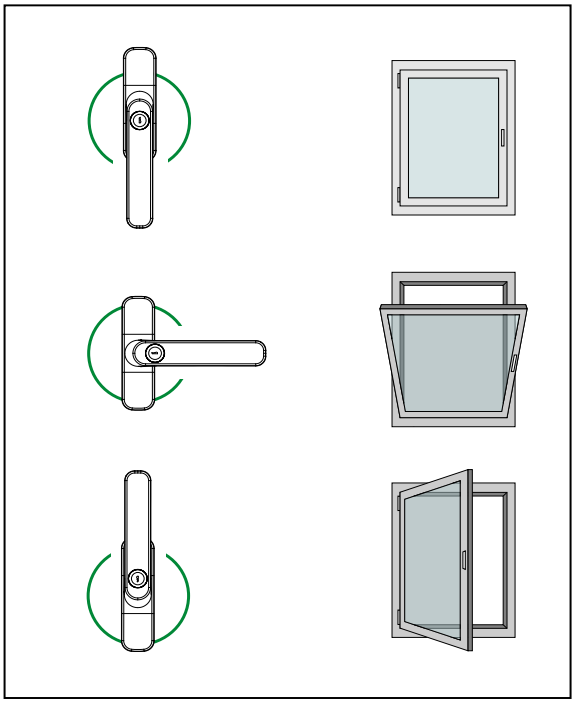


SIRIUS BLOCK
A1160, A1163

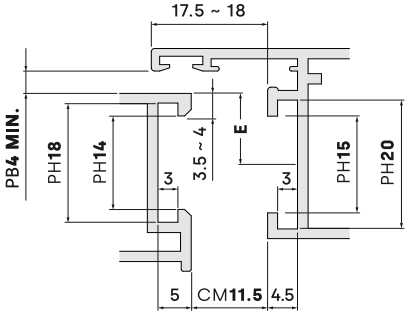


LYRA BLOCK
1129, 1145

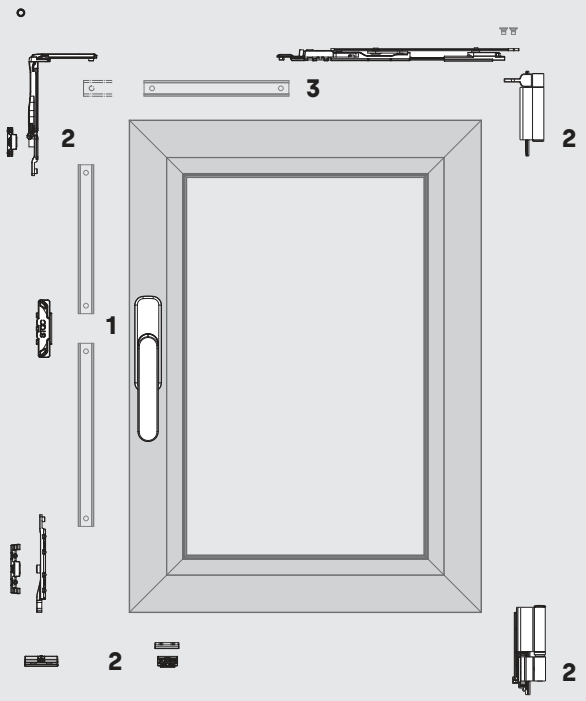
CREMONE WITH KEY + TILT-BEFORE-TURN WINDOW



EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

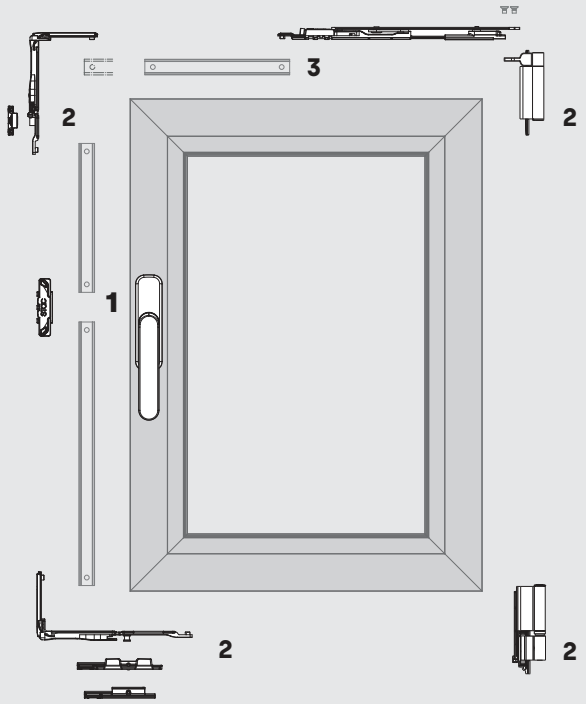


The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes, please refer to the technical catalog.



SMART TILT & TURN KIT
LOCKING TO INVERTER

1	**CREMONE** Refer to « cremones »			
2	**R1133**	**	**LOCKING TO INVERTER WITHOUT CREMONE KIT**	24
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10



SMART TILT & TURN KIT
LOCKING TO FRAME / TILT-BEFORE-TURN

1	**CREMONE** Refer to « cremones »			
2	**R1139**	**	**LOCKING TO FRAME WITHOUT CREMONE KIT**	24
	R1123	**	**TILT-BEFORE-TURN WITHOUT CREMONE KIT**	
3	**FRICTION STAY** Refer to « tilt & turn friction stay »			10

TILT & TURN FRICTION STAY
Traditional manoeuvre

COMPAS MT

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ZM

IN

Friction stays for Euro Groove profiles that allow tilt & turn openings in combination with different accessories.

- Specific references for tilt & turn configurations with traditional manoeuvre (locking to the inverter and locking to the frame) or tilt-before-turn.
- Encased eccentric that allows adjustment of the locking pressure (± 1 mm).
- The sliding elements are fitted with an anti-looseness clip.
- The MICRO friction stay (ref. 110309) allows the addition of a supplementary corner drive for residential rehabilitation.
- Available supplementary friction stay (ref. A130301) compatible with the large friction stays (ref. B110301, ref. T110301) that reduces the stresses and vibrations of the main friction stay when the window is in the tilt position.

TILT & TURN FRICTION STAY
Tilt-before-turn

COMPAS ML

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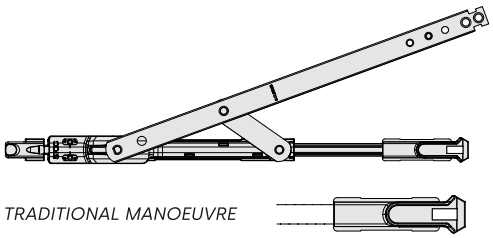
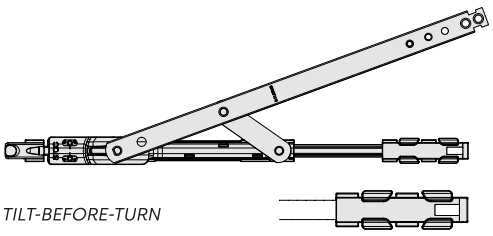
IN

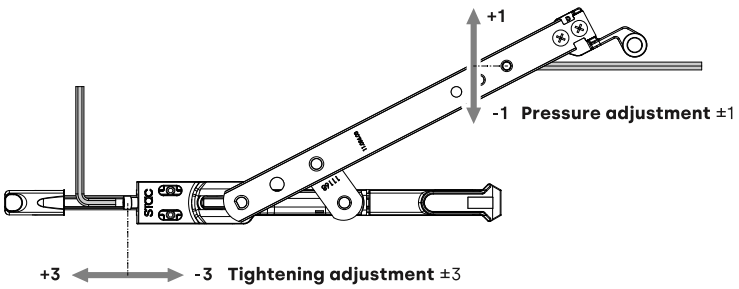
TRADITIONAL MANOEUVRE		WEIGHT (kg)	SUPPLEMENTARY (A130301)
110309	MICRO FRICTION STAY	120	-
B110302	SMALL FRICTION STAY	120	-
B110301	LARGE FRICTION STAY	120	✔
T110301	LARGE CLX FRICTION STAY	160	✔
TILT-BEFORE-TURN MANOEUVRE			
A110305	SMALL FRICTION STAY	120	-
A110304	LARGE FRICTION STAY	120	✔
T110304	LARGE CLX FRICTION STAY	160	✔

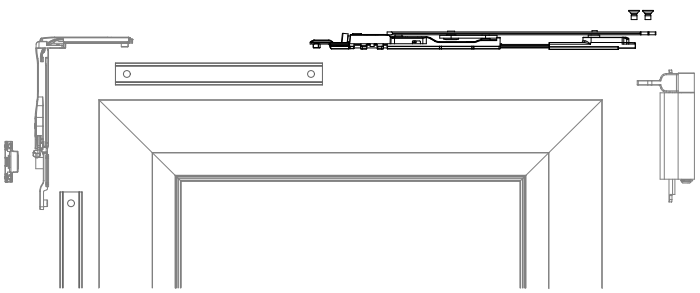


TRADITIONAL MANOEUVRE



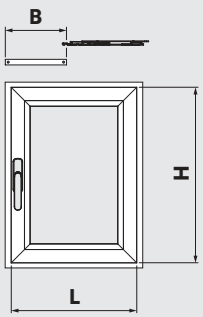
TILT-BEFORE-TURN





		L min	L max	H min	H max	X max	P max	B rod	
CLASSIC, SMART, CLX 120									
110309	**MICRO FRICTION STAY**	345	700	450	2600	110	120 kg	L-309	-
B110302 / A110305	**SMALL FRICTION STAY**	442	750	500	2600	138	120 kg	L-373	-
B110301 / A110304	**LARGE FRICTION STAY**	570	1200	700	2600	183	120 kg	L-495	-
B110301 + **A130301**	**LARGE FRICTION STAY** WITH **SUPPLEMENTARY**	910	1500	700	2600	183	120 kg	L-595	-
A110304 + **A130301**		1200						L-965	325
CLX 160									
T110301 / T110304	**CLX LARGE FRICTION STAY**	570	1200	500	2800	135	160 kg	L-495	-
T110301 + **A130301**	**CLX LARGE FRICTION STAY** WITH **SUPPLEMENTARY**	910	1500	500	2800	135	160 kg	L-595	-
T110304 + **A130301**		1200						L-965	325
CH180 PLUS									
H1213R3 or H1213L3 / H1213R4 or H1213L4	**CH180 PLUS** WITH **SMALL FRICTION STAY**	470	1000	550	2600	90	100 kg	L-432	-
H1213R1 or H1213L1 / H1213R2 or H1213L2	**CH180 PLUS** WITH **LARGE FRICTION STAY**	590	1200	650	2600	130	100 kg	L-516	-
H1213R1 + **A130301** / H1213L1 + **A130301**	**CH180 PLUS** WITH **SUPPLEMENTARY**	930	2500	650	2600	130	100 kg	L-616	-
H1213R2 + **A130301** / H1213L2 + **A130301**	**CH180 PLUS** WITH **SUPPLEMENTARY**	1220	2500	650	2600	130	100 kg	L-986	325

Consult the weight table of each hinge according to the dimensions of the window in the technical data sheet. CLX160 hinges can be used with standard friction stays (max. 120 kg).



L = window width.

H = window height.

X = window opening in tilt position.

P = weight per sash supported by each hinge.

B = transmission rod length.

DOUBLE
TILT & TURN

Maximise ventilation capacity

Euro Groove

Refurbishments

Maximum ventilation

Encased hardware

Double tilt & turn system

Aesthetic trends in new constructions are focused on the need to install large windows without obstructing the view and the light. But what happens when building conditions require the use of double-sash openings of reduced dimensions, or in the case of refurbishment of old buildings where it is not possible to alter the aesthetics of the building due to updating or modernisation?

STAC offers a totally new and practical solution that maximises the ventilation capacity of these openings without the need to open them completely, thus improving the safety and practicality of the building.

» Maximize ventilation

Our aim is to maximise the ventilation capacity of small-format double sash windows and balconies.

- **Increased ventilation** compared to double-sash systems where only the active sash is allowed to tilt.
- **Ventilation equal** to that provided by traditional single-sash tilt & turn systems.

» Renovations and refurbishments

The Euro Groove double sash tilt & turn system is specifically designed for refurbishment where windows and/or balcony doors are renovated to improve the insulation of the house or its aesthetics, but in which it is not desired or not possible to increase its dimensions, causing the sash width to be very reduced.

» Single or double configuration

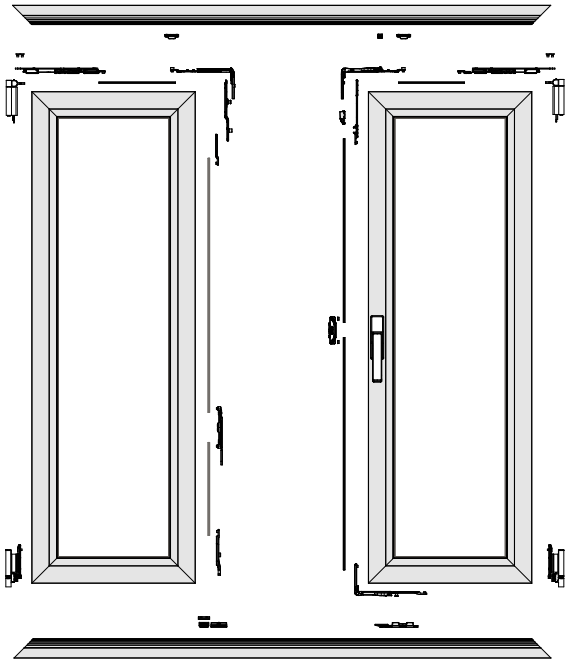
Allows the fitting of both sashes to be manoeuvred at the same time from the handle. The system, by simply changing the position of the connection bushing on the passive sash, gives you the choice between two operating modes:

- **Single setup:** independent opening of both the active and passive sash in the turn position and opening of the active sash in the tilt position.
- **Double setup:** independent opening of both the active and passive sash in the turn position and the opening of both sashes at the same time in the tilt position.

Hardware and accessories kit for opening double-sash tilt & turn windows and balcony doors with Euro Groove profiles with a maximum weight of 60 kg per sash.

- Supplied without tilt & turn hinges.
- Available with CLX tilt & turn hinges (ref. T110234).
- Developed for refurbishment where windows and/or balcony doors are renovated in order to improve the insulation of the house but where it is not possible to increase the size of the window.
- The system allows the hardware to be operated on both window sashes at the same time from the handle or cremone.
- Single or double configuration: allows you to tilt the active sash or both sashes at the same time.
- Does not keep opening hand: compatible with right or left hand openings.

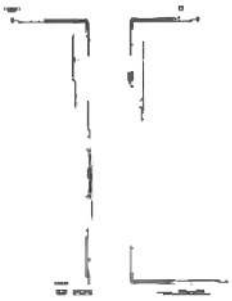
It is important to note that in the tilt position the user must move the combined weight of both sashes.



DOUBLE TILT & TURN SYSTEM
Without hinges

110240

1



DOUBLE TILT & TURN SYSTEM (CLX 160)
With CLX 160 hinges

T110243 **

1

HINGES AND FRICTIONS STAYS		SASH CONFIGURATIONS			
	FRICTION STAY	WIDTH (min · max)	HEIGHT (min · max)	TILT OPENING	WEIGHT
CLX160	MICRO ref. 110309	L min · max = **435 · 700**	H min · max = **675 · 2600**	110 millimetres	60 kg max.
	SMALL ref. B110302	L min · max = **500 · 700**	H min · max = **675 · 2600**	138 millimetres	
CH180+	SMALL ref. H1213 R3 o L3	L min · max = **560 · 750**	H min · max = **675 · 2600**	90 millimetres	60 kg max.

SUPPLEMENTARY CORNER DRIVE

B112100

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Supplementary corner drives for tilt & turn opening windows.

• The supplementary corner drives provide additional locking points by coupling them to the terminal of the active sash or to the tail of the tilt & turn friction stay.
• It is recommended that it be installed at sash heights of over 1200 mm vertically and 1200 mm horizontally.

CORNER DRIVE (ref. B112100)

• Simple installation as the coupling with the friction stay is done without having to dismantle the hardware.
• Compact support casing with the aim of reducing contact with the upper hinges.

CH180 PLUS CORNER DRIVE (ref. H122100)

• Specifically designed for use with the CH180 concealed hinges.
• Supplied with EVO SOFT encased hardware (ref. H122100) or with EVO SECURITY locking points (ref. H122100S).

SUPPLEMENTARY CORNER DRIVE (CH180 PLUS)

H122100

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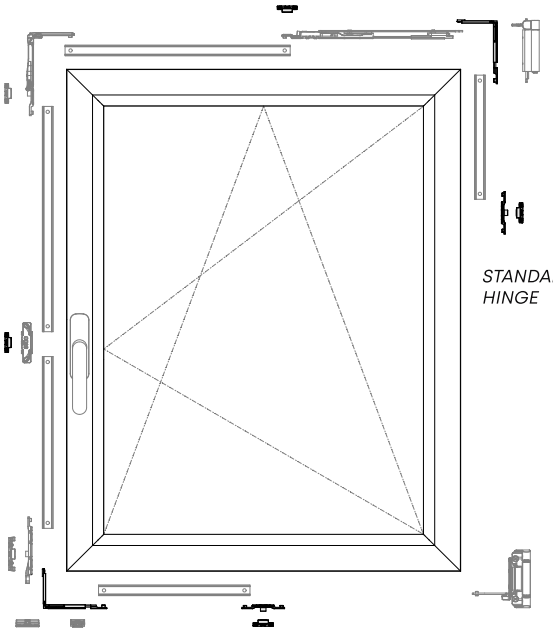
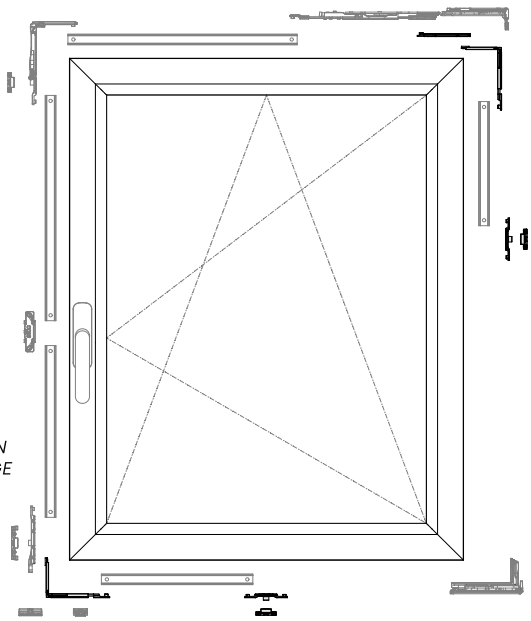
	COMPATIBLES HINGES	HARDWARE
A112100	STANDARD TILT & TURN	STANDARD
B112100	STANDARD TILT & TURN	STANDARD
H122100	CH180 PLUS TILT & TURN	**EVO SOFT**
H122100S	CH180 PLUS TILT & TURN	**EVO SECURITY**

It is recommended to be intalled in windows from 1200 x 1200 mm



STANDARD TILT & TURN HINGE



CH180 PLUS TILT & TURN HINGE

It can be used for hinged openings of up to 160 kg with the same adjustments as the tilt & turn openings.

• Compatible with CLX 160 (ref. T1113), CLASSIC (ref. B1113) and SMART (ref. R1113) tilt & turn hinges with active sash.
• Suitable for window profiles with a hinge pitch between 3 and 3.5 mm.
• Horizontal adjustment: ± 3 millimetres and tightening adjustment: ± 1.5 millimetres.

Supplementary frictions stay (ref. A130301)

• It acts as a reinforcement for the main friction stay, reducing stresses and vibrations when the window is in the tilt position.
• Its use is recommended in windows with sash lengths equal to or greater than 1200 millimetres.
• It is supplied pre-assembled for rapid installation.
• The friction stay is compatible with bottom-hung openings.

TOP HINGE ADAPTER WITH ADJUSTMENT

CE

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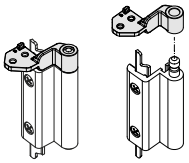
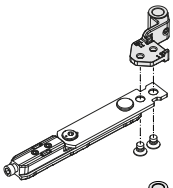
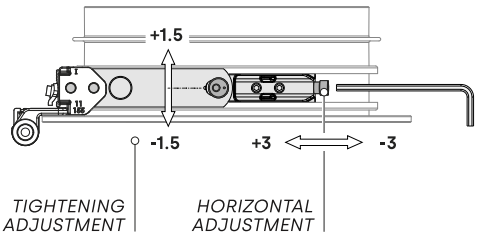
T110100

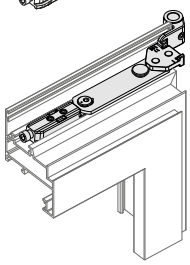
100

T110100	COMPATIBLE HINGES
T1113 **	CLX 160 HINGE
B1113 **	CLASSIC 120 HINGE
R1113 **	SMART 90 HINGE









SUPPLEMENTARY FRICTION STAY

CE

AL

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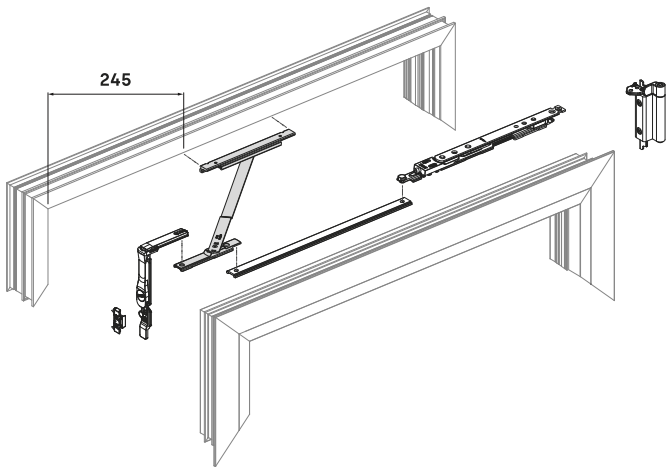
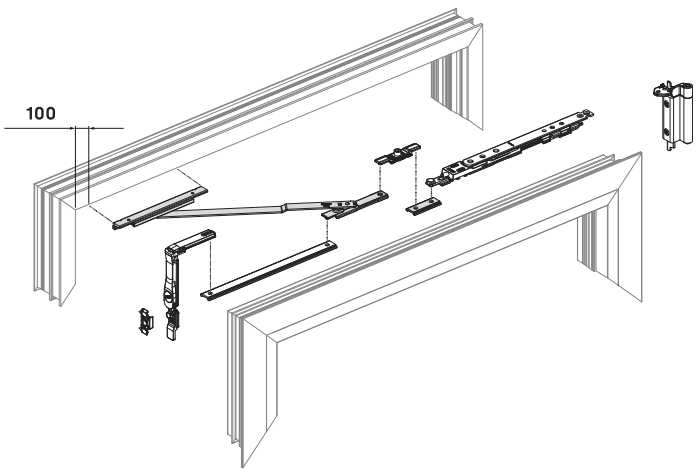
A130301

15

A130301 CONFIGURATION **TRADICIONAL MANEUVER**



A130301 CONFIGURATION **TILT-BEFORE-TURN**



TEMPLATE FOR HARDWARE INSTALLATION

STC-P-01

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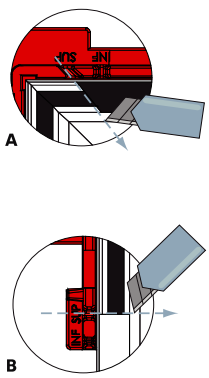
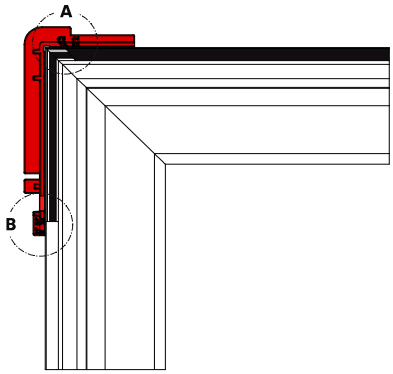
PP

Assembly template to facilitate the installation of the hardware for tilt & turn openings.

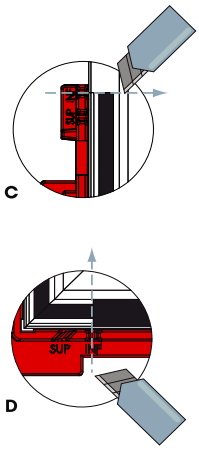
- Correctly position the hardware on the window: tilt & turn friction stay, pointer, keeper, positioning pieces...
- Cutting of the rubbers in the areas where the tilt & turn hinges work.

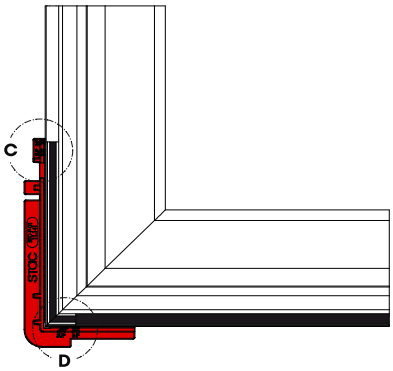
UPPER HINGE
RUBBER CUTTING



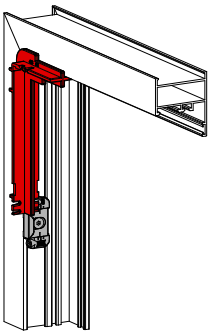


LOWER HINGE
RUBBER CUTTING

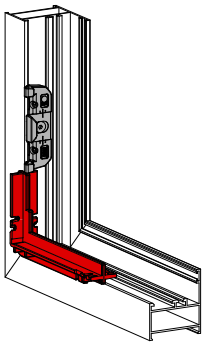




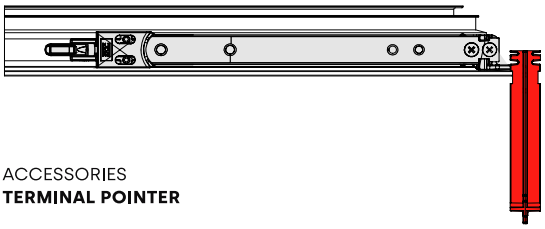
ACCESSORIES
UPPER KEEPER



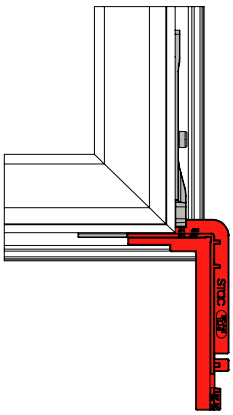
ACCESSORIES
LOWER KEEPER



ACCESSORIES
TILT & TURN FRICTION STAY



ACCESSORIES
TERMINAL POINTER



Packaging designed to facilitate the assembly of large productions of tilt & turn kits.

- The advantage of industrial packaging is the reduction of installation times as the bags in which the hinges are usually served are eliminated.
- The distribution of the pieces has been designed to save time in those carpentries where there are two installation lines: one for the frame profiles and one for the sash profiles.
- Blister 1 contains all the frame leaves with the plates and screws are pre-assembled.
- Blister 2 contains all the sash leaves with the shaft already assembled from the factory.
- Blister 3 contains the upper hinges completely preassembled to the right hand side.

CLASSIC HINGES
Active sash

CE

AL ZM
AC IN

1141	**

20

CLASSIC HINGES
Passive sash

CE

AL ZM
AC IN

1142	**

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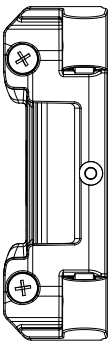
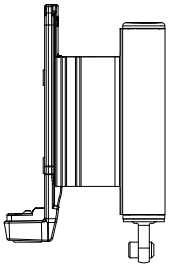
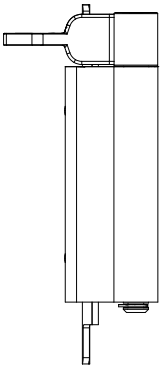


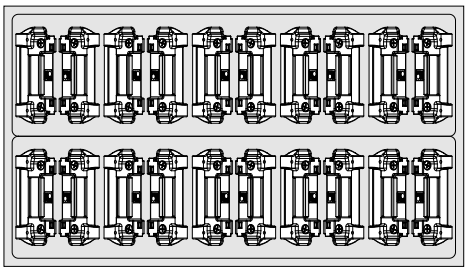
FRAME LEAF



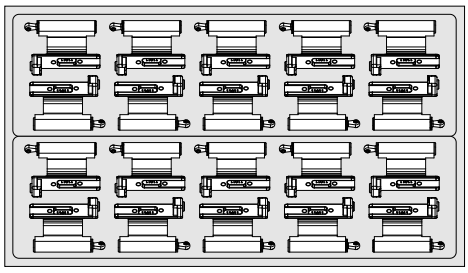
SASH LEAF



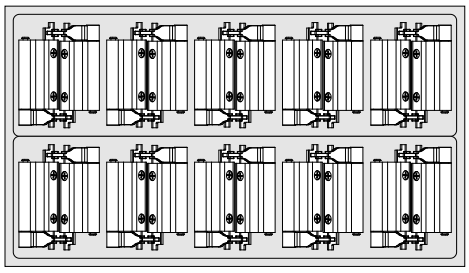
UPPER HINGE



BLISTER 1 **FRAME LEAF**



BLISTER 2 **SASH LEAF**



BLISTER 3 **UPPER HINGES**

BULK PACKING (LOCKING TO THE INVERTER)
Polyamide keeper

114000

20

Kit for tilt & turn opening with locking to the window inverter in industrial packaging for the assembly of large productions.

- The industrial packaging includes all sash and frame parts that are not dependent on the colour of the hardware.
- Supplied with PA6 polyamide plastic keeper (ref. 0507059).
- Available kit (ref. 114000Z) with zamak keeper (ref. 0507105).
- The accessories are supplied in bulk pre-assembled and separated into two groups: sash and frame elements eliminating the traditional packaging bags.

Cremone transmission cover (ref. 0511110) and terminal pointer (ref. 0511020) are not included.

BULK PACKING (LOCKING TO THE INVERTER)
Zamak keeper

114000Z

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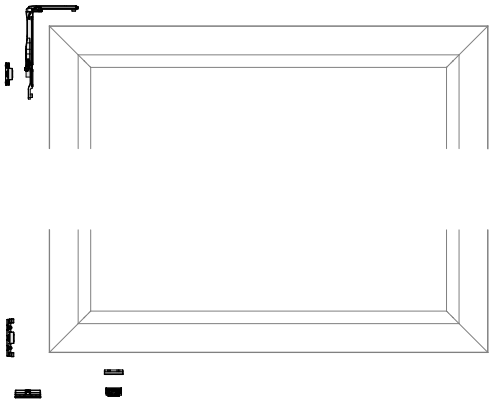
LOCKING TO INVERTER TILT & TURN KIT		KEEPER
114000	WITH POLYAMIDE KEEPERS	0507059
114000Z	WITH ZAMAK KEEPERS	0507105

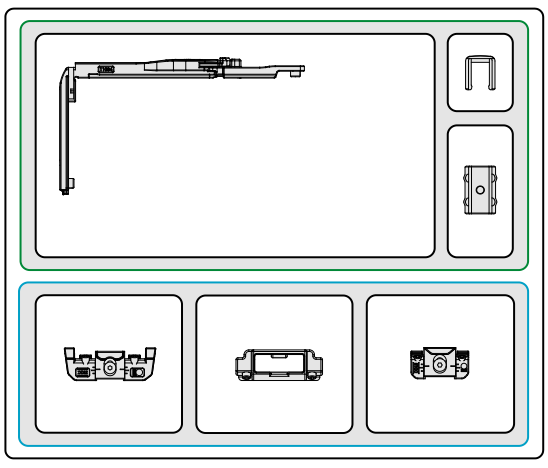
Terminal pointer (0511020) and transmision cover (0511110) not included



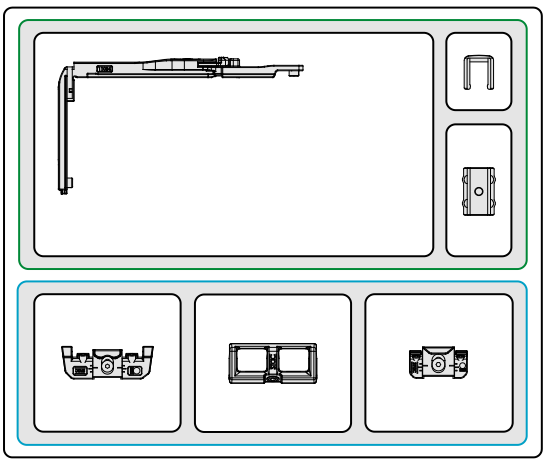
ACCESSORIES
SASH PROFILE

ACCESSORIES
FRAME PROFILE

114000



114000Z



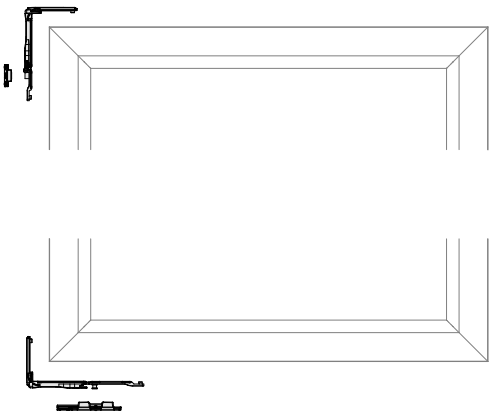
Kit for tilt & turn opening with locking to the window frame.

• This reference includes all sash and frame parts that are not dependent on the colour of the hardware.
• All bags are eliminated from traditional packaging.
• The accessories are supplied in bulk pre-assembled and separated into two groups: sash and frame elements.

The reference does not include the cremone transmission cover (ref. 0511110).

LOCKING TO FRAME TILT & TURN KIT		KEEPER
A114010	LOCKING TO FRAME	0511145
A114020	LOCKING TO FRAME WITH TILT-BEFORE-TURN	0511146

Cremone transmision cover (0511110) not included.



BULK PACKAGING (LOCKING TO THE FRAME)
Locking to frame

A114010

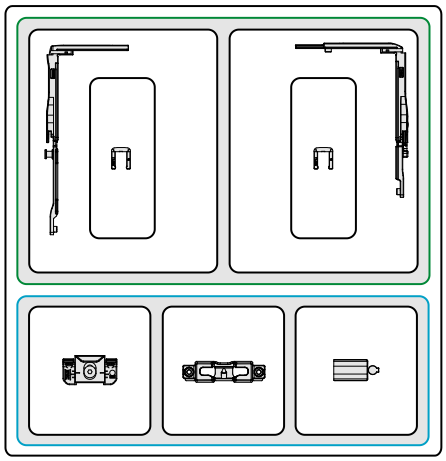
20

BULK PACKAGING (LOCKING TO THE FRAME)
Tilt-before-turn

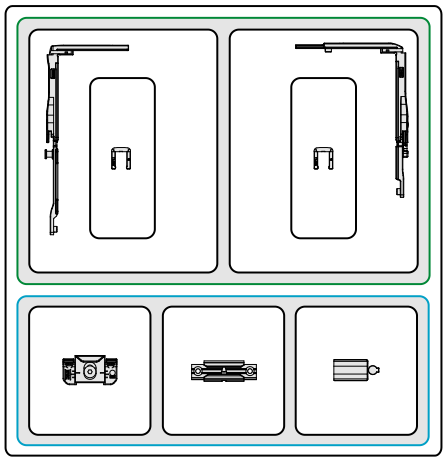
A114020

20

A114010



A114020



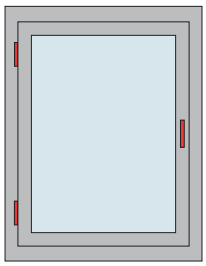
ACCESSORIES
SASH PROFILE

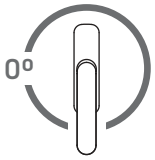
ACCESSORIES
FRAME PROFILE

TILT & TURN WINDOW
POSITIONS

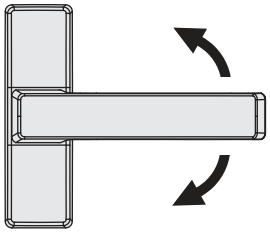
TILT & TURN OPENING

0°	CLOSED WINDOW



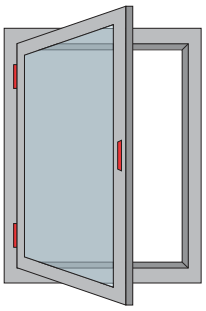


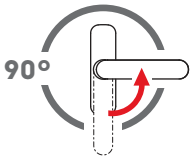
IN CREMONES OR HANDLES THE MAXIMUM FORCE APPLICABLE IS **10 kilograms**.

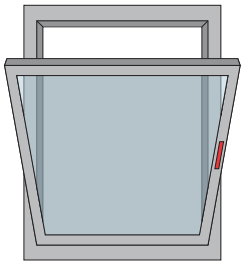


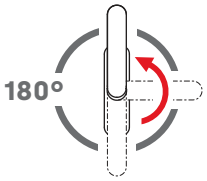
LOCKING TO INVERTER ~ LOCKING TO FRAME

90°	TURN WINDOW POSITION
180°	TILT WINDOW POSITION



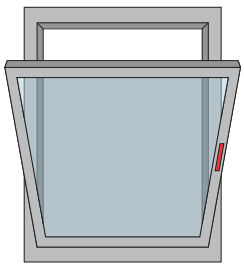


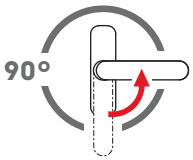


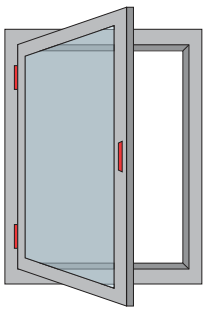


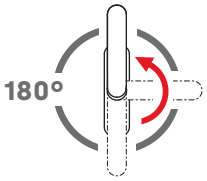
TILT-BEFORE-TURN

90°	TILT WINDOW POSITION
180°	TURN WINDOW POSITION

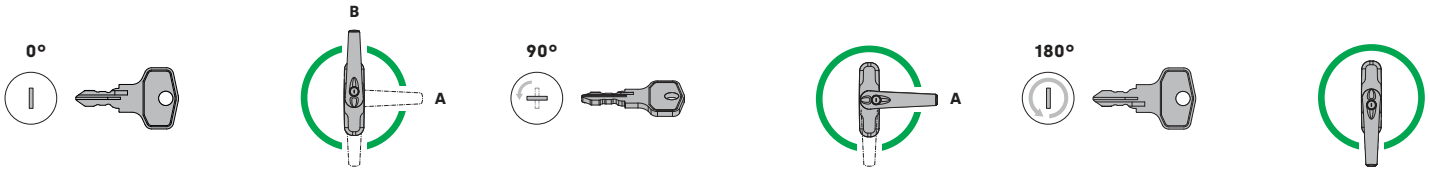




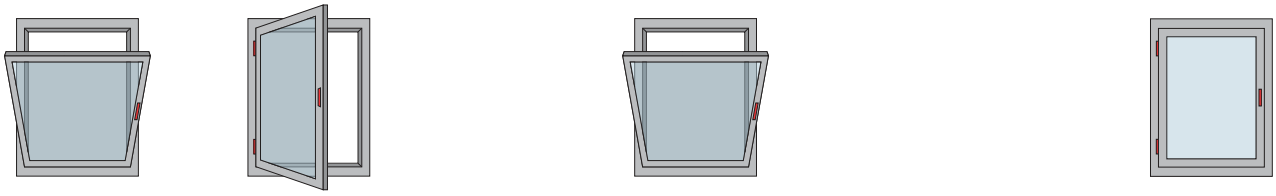




LOCKING OF THE POSITION IN TILT-BEFORE-TURN
WITH LYRA BLOCK CREMONE



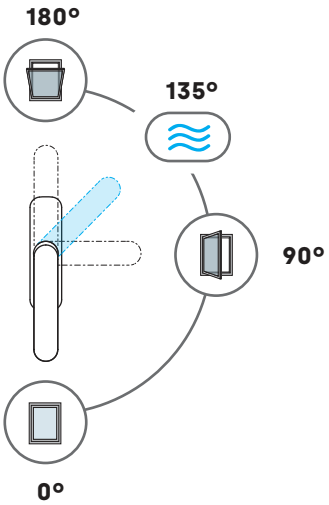
0° KEY	**90°** KEY	**180°** KEY
It allows tilt (**A**) and side-hung opening (**B**)	It only allows a tilt opening (**A**)	It does not allow any opening



MICRO VENTILATION POSITIONS
CREMONES

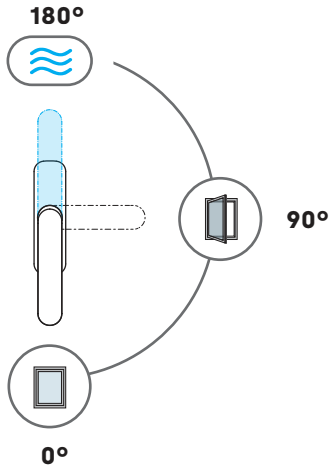
TILT & TURN OPENINGS
POSITIONS

0°	CLOSED WINDOW
90°	OPEN WINDOW
135°	MICRO VENTILATION
180°	TILT WINDOW



SIDE-HUNG OPENINGS
POSITIONS

0°	CLOSED WINDOW
90°	OPEN WINDOW
180°	MICRO VENTILATION



09

EVO HARDWARE

202 CLASSIC EVO SOFT tilt & turn

206 CLASSIC EVO SECURITY tilt & turn

210 CLX 160 EVO SECURITY tilt & turn

211 CH180 PLUS EVO SECURITY tilt & turn

212 EVO SOFT accessories

214 EVO SECURITY accessories

Encased hardware

Reduced effort

Time optimisation

Variety of systems

EVO SOFT hardware

The EVO SOFT system adapts the main advantages offered by the Groove 16 hardware to the Euro Groove systems with both side-hung and tilt & turn opening.

» Reduced actuating force

The recessed hardware considerably reduce frictional wear and tear and extend the service life of the locks.

At the same time, the recessed points allow for a smoother closing of the window, with up to 50% less effort required compared to traditional tilt & turn hardware.

» Compatibility with different types of aperture

The accessories with EVO SOFT hardware are compatible with various hinge systems:

- Tilt & turn windows.
- Side-hung windows with inward and outward opening.
- Bottom-hung or top-hung windows.
- Balcony doors.

» Optimisation of installation time in the workshop

A very important part of the development of accessories is the optimisation of installation time in the workshop.

We have a recessed rivet for transmission rods compatible with our TFS range of automatic and semi-automatic die-cutting machines: TFS-04, TFS-05 and TFS06.

The recessed rivet can be used to complement or replace the intermediate locking points on any type of hinged configuration with side-hung or tilt & turn opening.

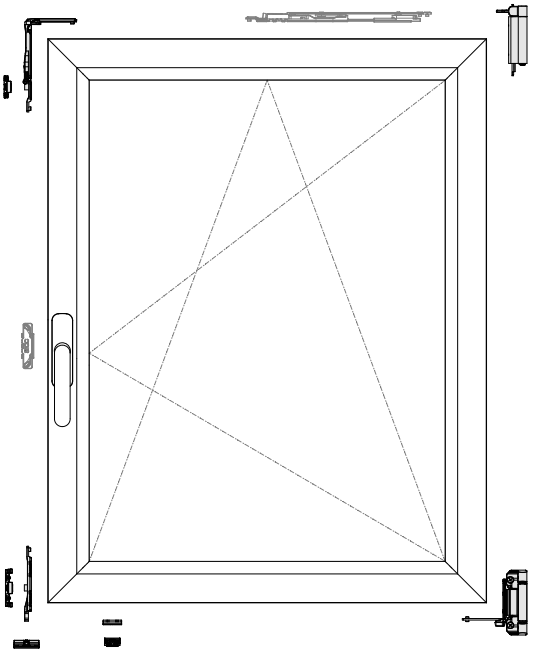
» Tilt & turn kits

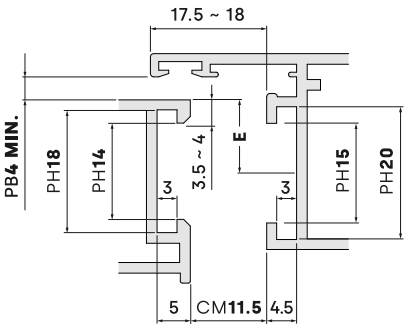
By default, accessories with EVO SOFT hardware are included in all CLX, CH180 PLUS and SMART hinged tilt & turn kits, for both locking to the inverter and locking to the frame configurations.

Base kits with CLASSIC hinges and EVO SOFT hardware for tilt & turn openings with locking to the window inverter.

- Maximum load per sash is 120 kg.
- The reference is supplied without cremone.
- Available with LYRA cremone (ref. C1118) or SIRIUS cremone (ref. C1118S).
- All sliding elements are fitted with an anti-looseness clip and the encased locking points allow for adjustment.
- The accessories include encased hardware that prolongs the durability of the keepers and reduces wear and tear due to friction, while facilitating smoother closing of the windows.

The tilt & turn friction stays are not supplied in the kits. See friction stays.



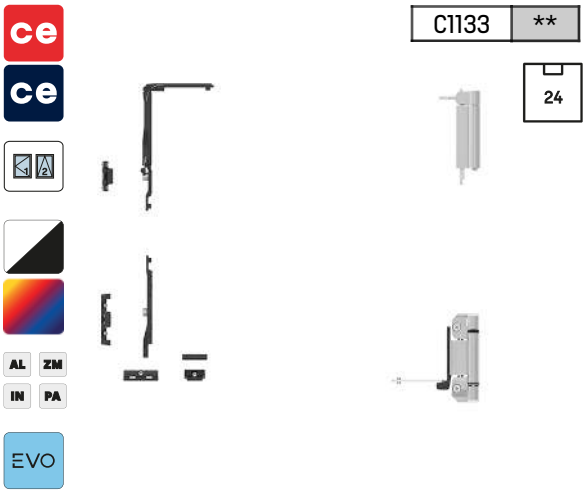


EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

EVO SOFT CLASSIC KIT (LOCKING TO THE INVERTER)
Without cremone

C1133 **

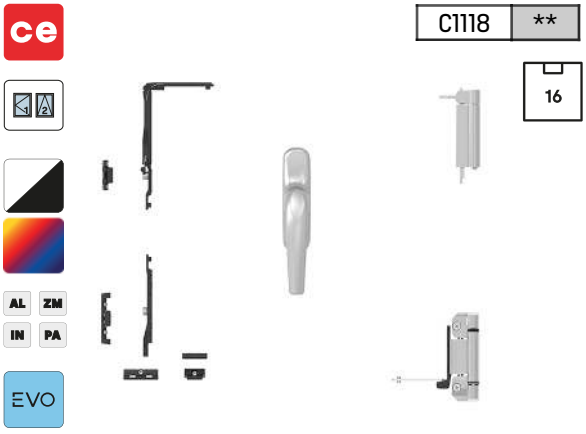
24



EVO SOFT CLASSIC KIT (LOCKING TO THE INVERTER)
With LYRA cremone

C1118 **

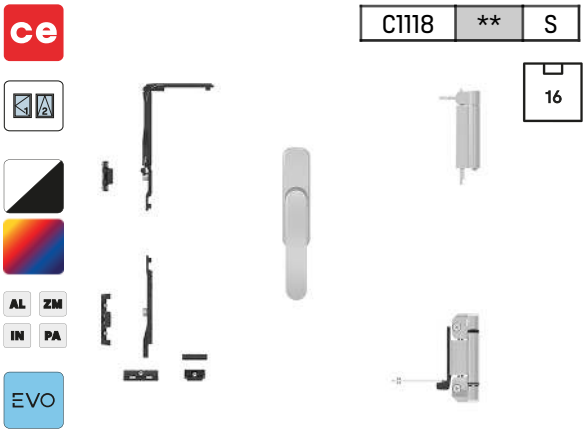
16



EVO SOFT CLASSIC KIT (LOCKING TO THE INVERTER)
With SIRIUS cremone

C1118 ** S

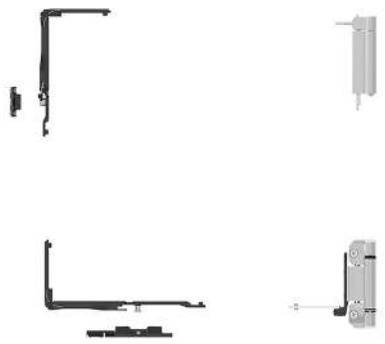
16



EVO SOFT CLASSIC KIT (LOCKING TO THE FRAME)
Without cremone

C1139 | **

24



AL ZM IN PA

EVO

EVO SOFT CLASSIC KIT (LOCKING TO THE FRAME)
With LYRA cremone

C1132 | **

16

AL ZM IN PA

EVO

EVO SOFT CLASSIC KIT (LOCKING TO THE FRAME)
With SIRIUS cremone

C1132 | ** | S

16

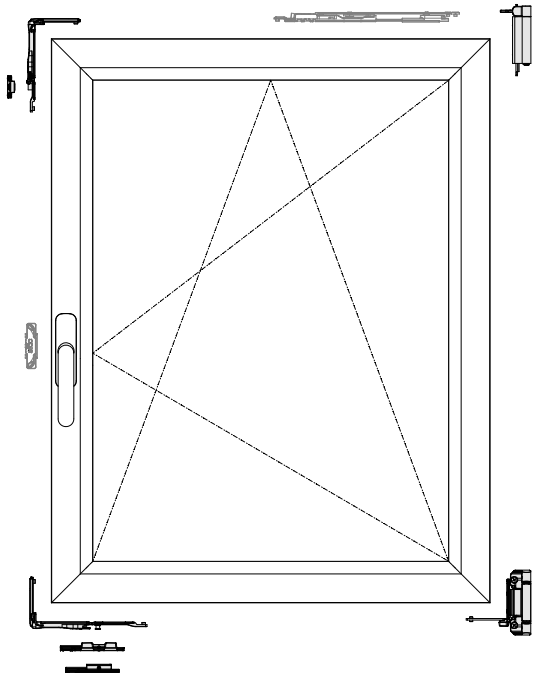
AL ZM IN PA

EVO

Base kits with CLASSIC hinges and EVO SOFT hardware for tilt & turn openings with locking to the window frame.

- Maximum load per sash is 120 kg.
- The reference is supplied without cremone.
- Available with LYRA cremone (ref. C1132) or SIRIUS cremone (ref. C1132S).
- All sliding elements are fitted with an anti-looseness clip and the encased locking points allow for adjustment.
- The lower corner drive is supplied with eccentric locking point and the tilting keeper with the integrated sash positioning latch.
- The accessories include encased hardware that prolongs the durability of the keepers and reduces wear and tear due to friction, while facilitating smoother closing of the windows.

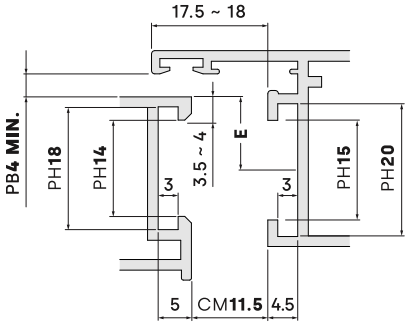
The tilt & turn friction stays are not supplied in the kits. See friction stays.



LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER

EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm



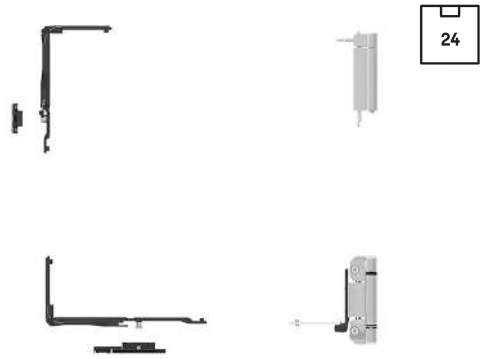
Base kit with CLASSIC hinges and EVO SOFT hardware pour fenêtres avec manœuvre logique.

- Maximum load per sash is 120 kg.
- The reference is supplied without cremone.
- All sliding elements are fitted with an anti-looseness clip and the encased locking points allow for adjustment.
- The lower corner drive is supplied with eccentric locking point and the tilting keeper with the integrated sash positioning latch.
- The accessories include encased hardware that prolongs the durability of the keepers and reduces wear and tear due to friction, while facilitating smoother closing of the windows.
- Tilt-before-turn kit is suitable for use with cremones with key as it provides two locking positions that allow the window to tilt but not to open in the side-hung position.

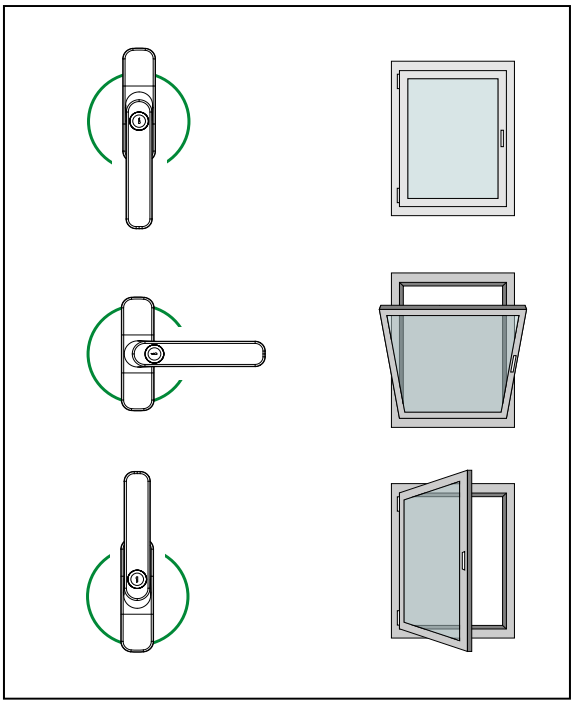
The tilt & turn friction stays are not supplied in the kits. See friction stays.

EVO SOFT CLASSIC KIT (TILT-BEFORE-TURN)
Without cremone

AL ZM IN PA EVO

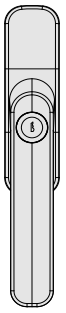


CREMONE WITH KEY + TILT-BEFORE-TURN WINDOWS

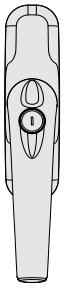


TRANSMISSION COVER 0511193

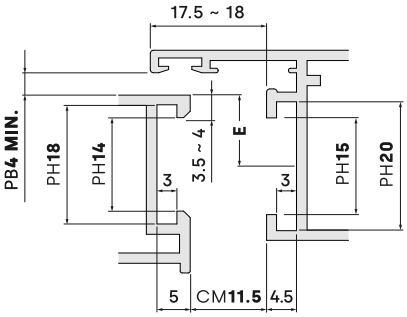
A1160 **	SIRIUS BLOCK	✔
A1163 **		-
1129 **	LYRA BLOCK	✔
1145 **		-



SIRIUS BLOCK
A1160, A1163



LYRA BLOCK
1129, 1145



EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SAHFT **E**	10 mm

Eccentric locking points

RC2 certification

Tightening adjustment

Simple installation

EVO SECURITY hardware

Security is a determining factor when choosing the accessories that allow us to configure our locking system. The EVO SECURITY hardware is characterised by its eccentric locking points and anti-burglary strikers that allow us to efficiently improve our protection against forced entry, burglary and breakage, increasing the security of the windows to certified RC2 levels according to European standard UNE-EN 1627.

» Eccentric locking points

The eccentric locking points allow adjustment of the clamping pressure up to 1 millimetre, offering better manoeuvrability and allowing adaptation to the different series on the market.

» Customisable hardware « EVO HYBRID »

The tilt & turn systems with hybrid hardware combine the protection provided by the hardware with security locking points and the smooth operation of the reccesed hardware, being fully configurable according to the level of security and/or smoothness that you want to give to the enclosure by adding the additional locking points that best suit the needs of each home.

From the basic kit (ref. S1123) it is possible to configure windows combining EVO SOFT recessed hardware and EVO SECURITY security hardware according to the level of security and softness that you want to give to the enclosure.

» RC2 Anti-intrusion security

Security windows offer protection against unauthorised entry of persons and must therefore be designed and constructed to be particularly resistant to mechanical influences.

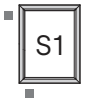
The RC2 standard is a certificate that determines the level of security of windows and doors against break-in. The certificate is awarded on the basis of stringent tests and a joint verification of the hardware, the glass and the aluminium system series.

In order to achieve perfect security, two types of break-in tests are carried out:

Static tests in which the windows are subjected to high loads at their most critical points.

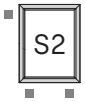
Real intrusion tests in which a real intrusion attempt is made through the windows with specific tools and measured times.

SECURITY LEVELS
FOR SINGLE-FAMILY HOUSES



S1 CONFIGURATION

Protection from an inexperienced and opportunist thief and from simple acts of vandalism using physical force and non-use of tools.



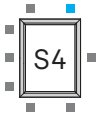
S2 CONFIGURATION

Efficient protection from chance delinquents equipped with lever tools who are not sure what they will achieve. Generally they will not risk more than 3 minutes of trying to force the windows.

SECURITY LEVELS
FOR COMMERCIAL AREAS AND PUBLIC BUILDINGS

S3 CONFIGURATION *(up to 6 security points per sash)*

Advanced protection from experienced delinquents that are equipped with large screwdrivers and wedges and know exactly what they want to get. Planned attack moments and increased attack times of up to 15 minutes



S4 CONFIGURATION *(up to 7 security points per sash)*

Advanced protection with the same characteristics as S3 but in large size windows and balconies.

RC2 CONFIGURATION *(8 to 10 security points per sash)*

The RC2 standard is a certificate that determines the level of security of the enclosures against burglary. In order to achieve perfect security, two types of intrusion attempt tests are carried out:

- Static tests in which windows are subjected to high loads at their most critical points.
- Real intrusion tests in which a real intrusion attempt is made through the windows with specific tools and measured times.

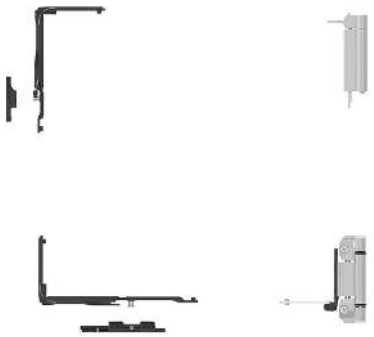
The RC2 configuration is certified in conjunction with the glass and aluminium system series.

For a higher level of protection, we recommend installing the EVO SECURITY hardware on aluminium transmission rods and single-sash groove.

EVO SECURITY CLASSIC KIT (LOCKING TO THE FRAME)
Without cremone

S1139 | **

24



AL ZM AC PA

EVO

EVO SECURITY CLASSIC KIT (LOCKING TO THE FRAME)
With LYRA cremone

S1132 | **

16

AL ZM IN PA

EVO

EVO SECURITY CLASSIC KIT (LOCKING TO THE FRAME)
With SIRIUS cremone

S1132 | ** | S

16

AL ZM IN PA

EVO

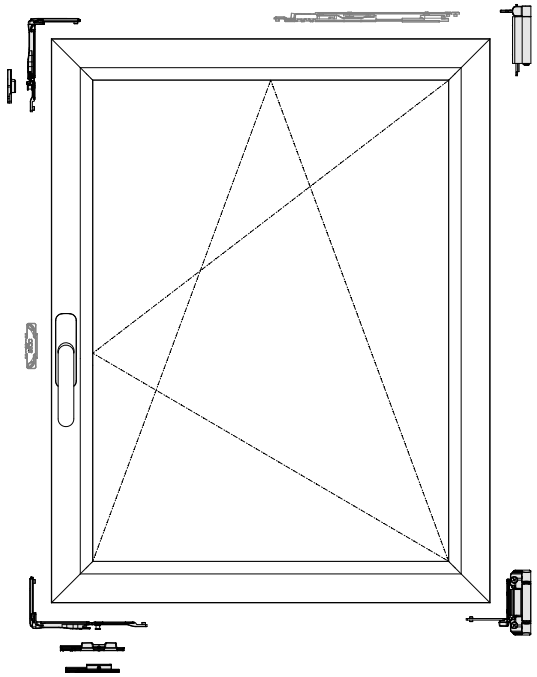
Base kit with CLASSIC hinges and EVO SECURITY hardware for tilt & turn openings with locking to the window frame.

- Maximum load per sash is 120 kg.
- Available with LYRA cremone (ref. S1132) or SIRIUS cremone (ref. S1132S).
- The accessories integrate hardware with eccentric locking points and anti-picking keepers that improve protection against forced entry, burglary or breakage.

The kits do not include tilt & turn friction stays. See compatible friction stays.

EVO HYBRID configuration

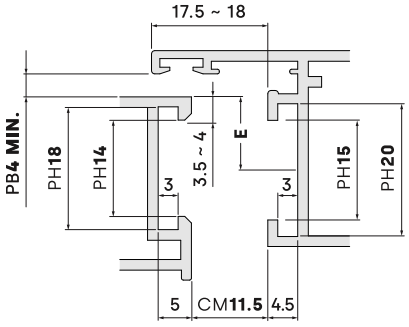
Starting from the basic kit (ref. S1139), it is possible to configure windows combining EVO SOFT encased hardware and EVO SECURITY hardware according to the level of security and softness that you wish to give to the window. For more information, all EVO HYBRID configurations are listed in the technical catalogue.



LOCKING TO FRAME KEEPER

TILT-BEFORE-TURN KEEPER

EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm



EVO SECURITY tilt & turn kit with CLASSIC hinges for windows with tilt-before-turn manoeuvre

- Maximum load per sash is 120 kg.
- The reference is supplied without cremone.
- The accessories integrate hardware with eccentric locking points and anti-fraction locks that improve protection against forced entry, theft or breakage.
- Tilt-before-turn kit is suitable for use with keyed cremone as they provide two locking positions that allow the window to tilt but not to open in the turn position.

The kit does not include tilt & turn friction stay. See compatible friction stays.

EVO HYBRID configuration

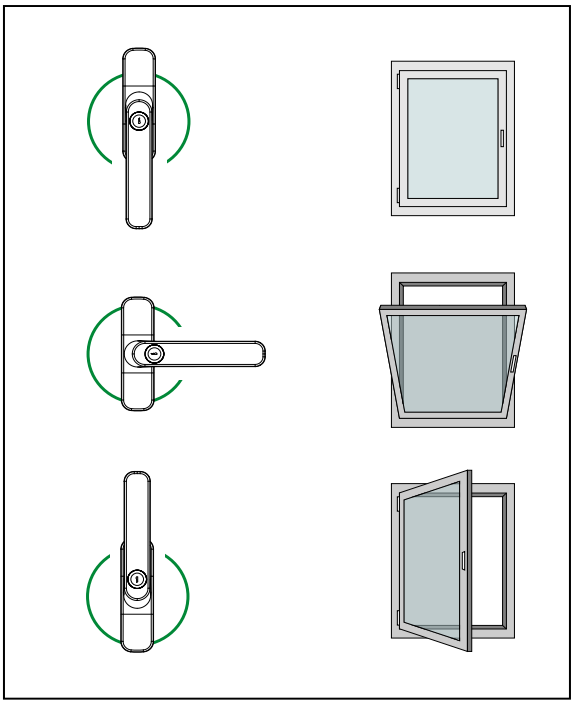
Starting from the basic kit (ref. S1123), it is possible to configure windows combining EVO SOFT encased hardware and EVO SECURITY hardware according to the level of security and softness that you wish to give to the window. For more information, all EVO HYBRID configurations are listed in the technical catalogue.

EVO SECURITY CLASSIC KIT (TILT-BEFORE-TURN)
Without cremone

S1123	**

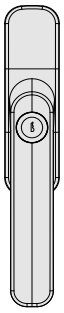
24

CREMONE WITH KEY + TILT-BEFORE-TURN WINDOW

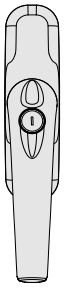


TRANSMISSION COVER 0511193

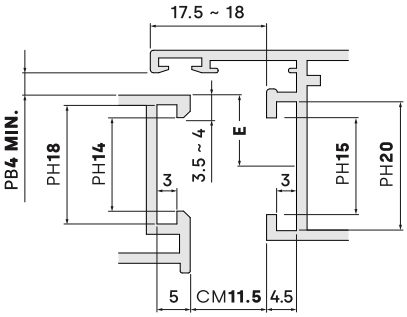
A1160 **	SIRIUS BLOCK	✔
A1163 **		-
1129 **	LYRA BLOCK	✔
1145 **		-



SIRIUS BLOCK
A1160, A1163



LYRA BLOCK
1129, 1145



EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

EVO SECURITY CLX 160 KIT (LOCKING TO THE FRAME)

Without cremone

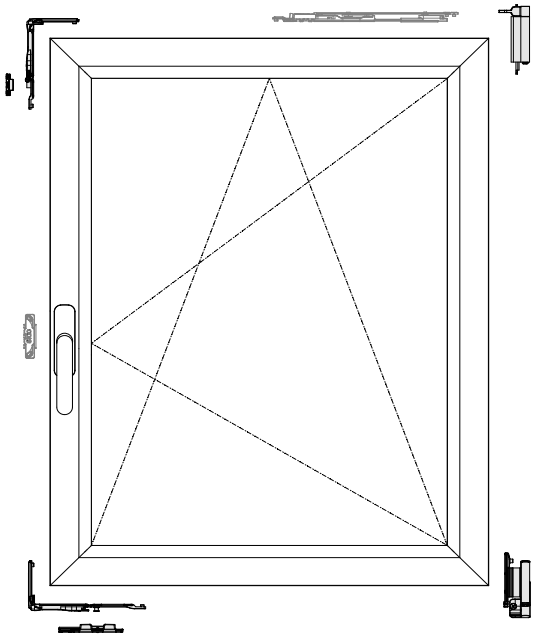
TS1139	**

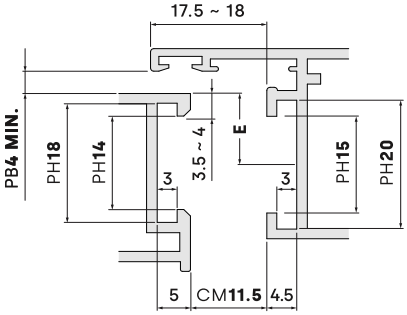
24

CLX 160 hinge kit for tilt & turn windows with locking to the window frame.

- Maximum load per sash: 160 kg.
- The reference is supplied without cremone.
- Specific tilt & turn friction stay (ref. T110301) for traditional manoeuvre not included in the kit.
- The accessories integrate hardware with eccentric locking points and anti-fraction locks that improve protection against forced entry, theft or breakage.

The kit does not include a tilt & turn friction stay. See compatible friction stays.

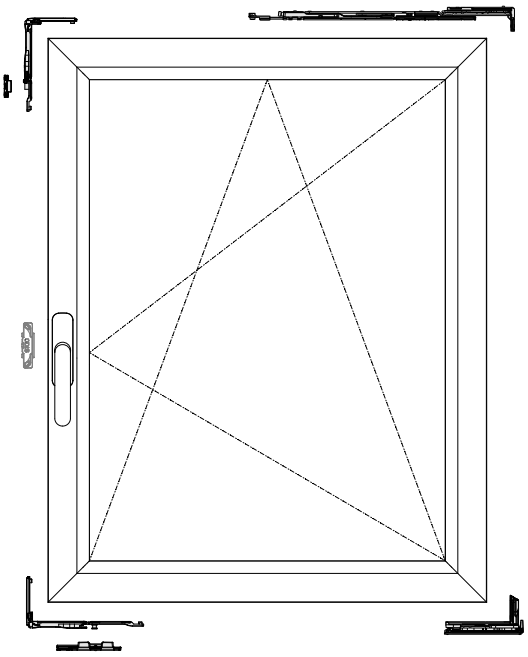




EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

Base kit with CH180 PLUS concealed hinges and EVO SECURITY hardware for tilt & turn openings with locking to the window frame.

- Maximum load per sash is 100 kg and maximum opening is 180º.
- Hinges keep the opening hand of the window: right-hand opening (R) or left-hand opening (L).
- The top hinge is supplied with an integrated large tilt & turn friction stay.
- Compatible supplementary corner drives (ref. H122100, ref. H122100S).
- The accessories integrate hardware with eccentric locking points and anti-fraction locks that improve protection against forced entry, theft or breakage.



EVO SECURITY CH180 PLUS HINGE KIT (LOCKING TO THE FRAME)

Large friction stay

SH1239	R1 / L1

1

AL ZM IN PA EVO

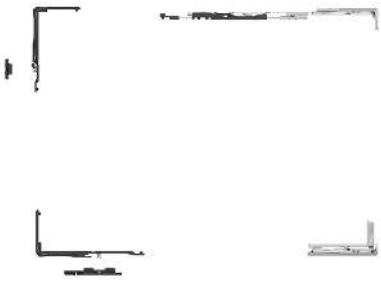
EVO SECURITY CH180 PLUS HINGE KIT (LOCKING TO THE FRAME)

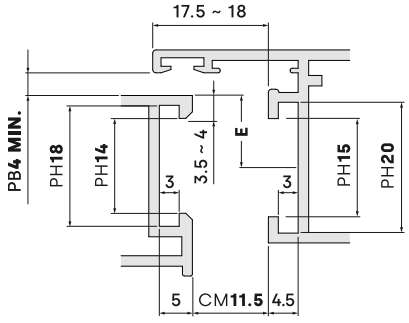
Small friction stay

SH1239	R3 / L3

1

AL ZM IN PA EVO





EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm

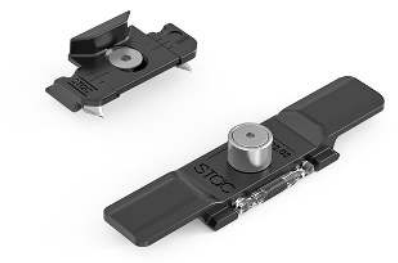
LARGE FRICTION STAY

SMALL FRICTION STAY

EVO SOFT INTERMEDIATE LOCKING POINT KIT

113003C

50



ce ce ZM IN EVO

EVO SOFT TERMINAL LOCKING POINT KIT

113004C

50

ce ce ZM IN EVO

EVO SOFT Ø6 RIVET

0511149

50

ce C16 ZM IN EVO

Accessories that integrate encased hardware that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.

Intermediate locking point kit (ref. 113003C)

- In large window configurations, it is possible to insert as many locking points as required.
- The reference is supplied with anti-looseness clip.

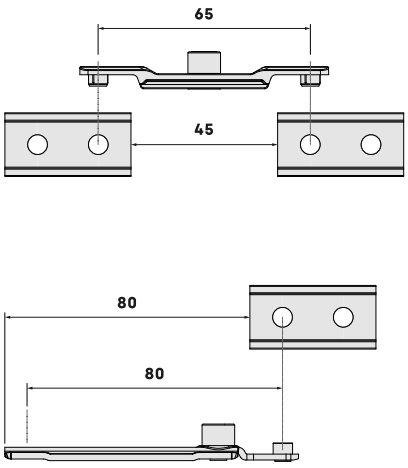
Terminal locking point kit (ref. 113004C)

- Accessory developed to be installed after a sash hardware: tilt & turn friction stay, corner drive or transmission cover.

Ø6 rivet (ref. 0511149)

- Encased locking point for windows with Euro Groove or 16 Groove profile.
- It allows to obtain a locking point in whatever point where the operating rod passes through without having to be machined.
- The cremone transmission lever acts directly on the rivet, allowing the transmission cover to be replaced.
- Valid for rod die-cutting machines: TFS-04, TFS-05 and TFS-06.

DISCOUNT



Accessories that integrate encased hardware that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.

Supplementary corner drive (ref. A112103C)

- Supplementary corner drive for tilt & turn windows.
- It is fitted to the terminal of the active sash or at the end of the friction stay to provide additional locking points.
- It is recommended for use with sashes above 1200 mm vertically or 1200 mm horizontally.

Supplementary corner drive (ref. B112103C)

- Supplementary corner drive for tilt & turn windows.
- It is fitted to the terminal of the active sash or at the end of the friction stay to provide additional locking points.
- It is recommended for use with sashes above 1200 mm vertically or 1200 mm horizontally.

Horizontal rcorner drive (ref. 113420C)

- Horizontal corner drive kit for locking to frame side-hung, bottom-hung or top-hung openings.

EVO SOFT SUPPLEMENTARY CORNER DRIVE KIT

ce ZM IN EVO

A112103C

50

EVO SOFT INTERMEDIATE LOCKING POINT KIT

ce ce ZM IN EVO

B112103C

50

EVO SOFT HORIZONTAL CORNER DRIVE KIT

ce ce ZM IN EVO

113420C

20

EVO SECURITY INTERMEDIATE LOCKING POINT KIT

113005S

50

ce ZM IN EVO

Accessories that integrate hardware with eccentric locking points and anti-fraction locks that improve protection against forced entry, theft or breakage, increasing the security of the window to RC2 certified levels.

Intermediate locking point kit (ref. 113005S)

• In large window configurations, it is possible to insert as many locking points as required.

Terminal locking point kit (ref. 113004S)

• Accessory developed to be installed after a sash hardware: tilt & turn friction stay, corner drive or transmission cover.

Keeper (ref. 0511135)

• Anti-fraction keeper supplied with set screws that pre-assembled over the profile without the need for machining.

EVO SECURITY TERMINAL LOCKING POINT KIT

113004S

150

ce ZM IN EVO

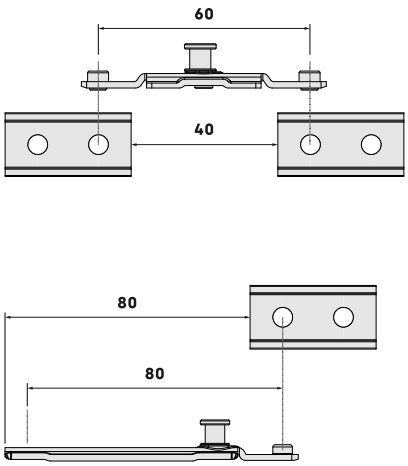
EVO SECURITY KEEPER

0511135

50

ce ZM IN EVO

DISCOUNT



Accessories that integrate hardware with eccentric locking points and anti-fraction locks that improve protection against forced entry, theft or breakage, increasing the security of the window to RC2 certified levels.

Supplementary corner drive (ref. A112103S)

• Supplementary corner drive for tilt & turn windows.
• It is fitted to the terminal of the active sash or at the end of the friction stay to provide additional locking points.
• It is recommended for use with sashes above 1200 mm vertically or 1200 mm horizontally.

Supplementary corner drive (ref. B112103S)

• Supplementary corner drive for tilt & turn windows.
• It is fitted to the terminal of the active sash or at the end of the friction stay to provide additional locking points.
• It is recommended for use with sashes above 1200 mm vertically or 1200 mm horizontally.

Horizontal corner drive (ref. 113420S)

• Horizontal corner drive kit for locking to frame side-hung, bottom-hung or top-hung openings.

EVO SECURITY SUPPLEMENTARY CORNER DRIVE

ce
ZM
IN
EVO

A112103S

50

EVO SECURITY SUPPLEMENTARY CORNER DRIVE

ce
ZM
IN
EVO

B112103S

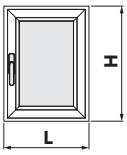
50

EVO SECURITY HORIZONTAL CORNER DRIVE

ce
ZM
IN
EVO

113420S

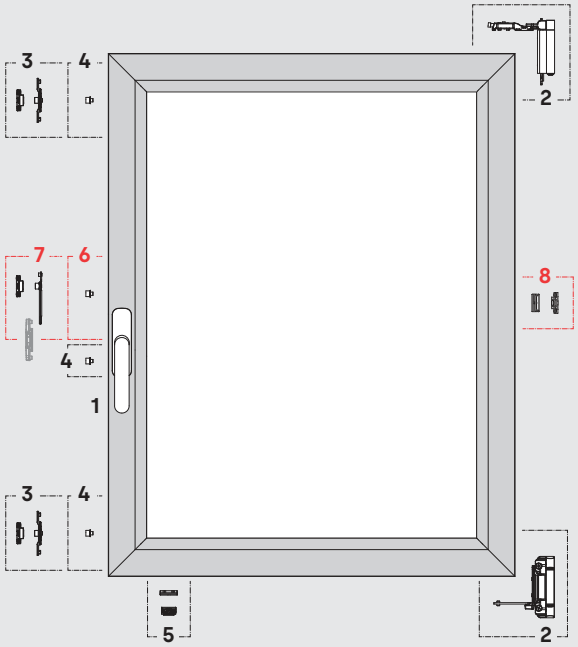
20



The hardware specified in these configurations is for side-hung windows with a maximum of 1200 x 1200 mm. For other window sizes please refer to the technical catalogue.

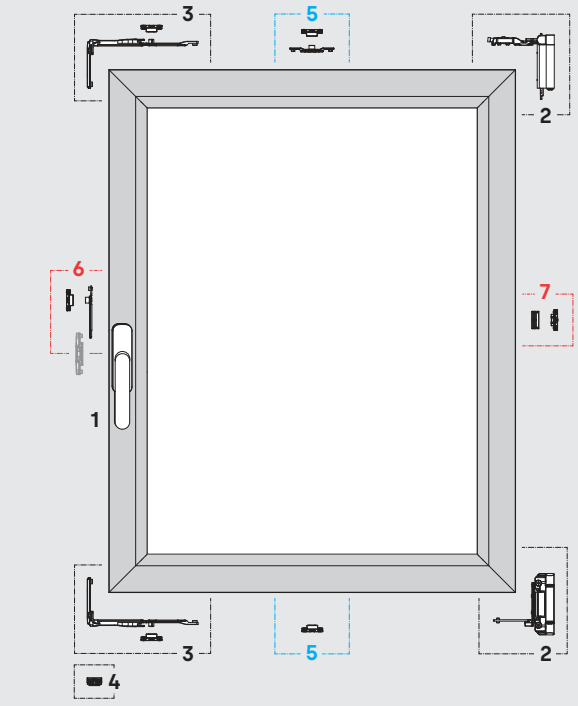
L > 1200 milimetres

H > 1200 milimetres



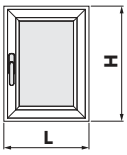
CLASSIC SIDE-HUNG KIT WITH EVO SOFT HARDWARE
LOCKING TO INVERTER

1	**CREMONE**		*Consult cremone section*	
2	**B1114**	**	**CLASSIC HINGE KIT** *PASSIVE SASH*	20
3	**113003C**		**INTERMEDIATE LOCKING POINT KIT** *EVO SOFT*	50
4	**0511149**		**Ø6 TRANSMISSION RIVET** *EVO SOFT*	50
5	**0710621**		**POSITIONING LATCH KIT**	100
6	**0511149**		**Ø6 TRANSMISSION RIVET** *EVO SOFT*	50
7	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50
8	**112500**		**CONCEALED LOCKING POINT**	50



CLASSIC SIDE-HUNG KIT WITH EVO SOFT HARDWARE
LOCKING TO FRAME

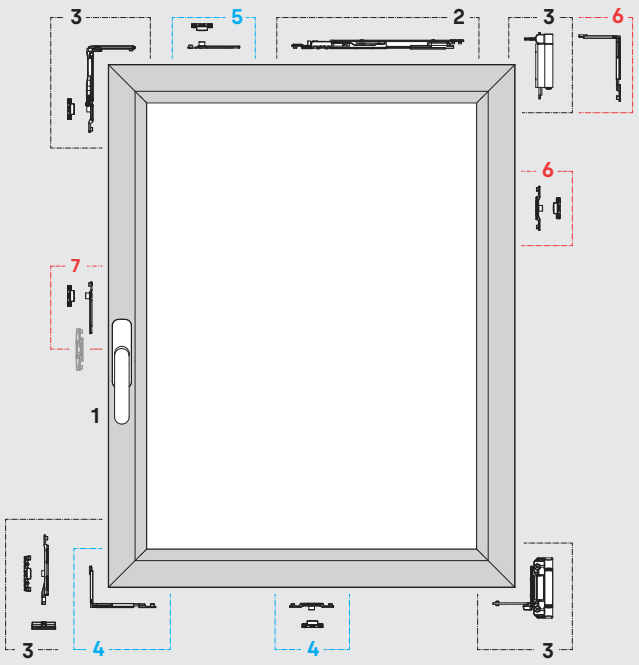
1	**CREMONE**		*Consult cremone section*	
2	**B1114**	**	**CLASSIC HINGE KIT** *PASSIVE SASH*	20
3	**113420C**		**HORIZONTAL CORNER DRIVE KIT** *EVO SOFT*	20
4	**0517193**		**POSITIONING LATCH**	100
5	**113003C**		**INTERMEDIATE LOCKING POINT KIT** *EVO SOFT*	50
6	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50
7	**112500**		**CONCEALED LOCKING POINT**	50



The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes please refer to the technical catalogue.

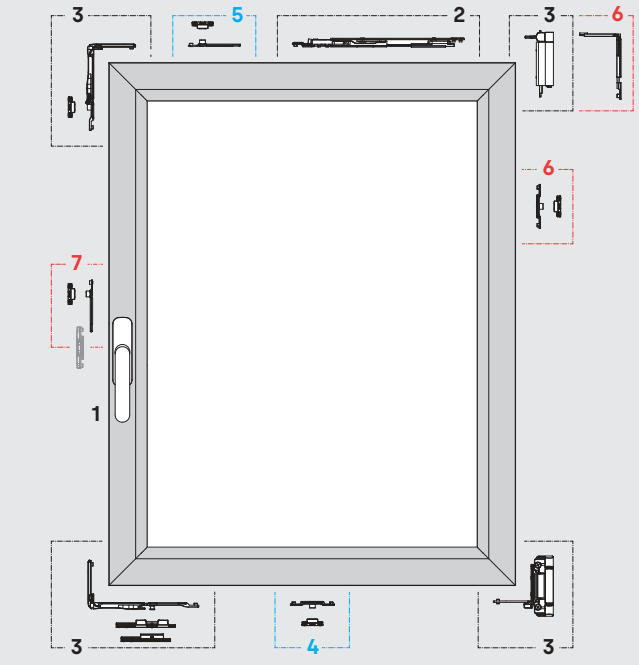
L > 1200 milimetres

H > 1200 milimetres



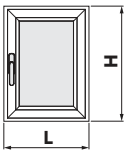
CLASSIC TILT & TURN KIT WITH EVO SOFT HARDWARE
LOCKING TO INVERTER

1	**CREMONE**		*Consult cremone section*	
2	**FRICTION STAY**		*Consult tilt & turn opening section*	
3	**C1133**	**	**EVO SOFT LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
4	**B112103C**		**SUPPLEMENTARY CORNER DRIVE KIT** *EVO SOFT*	50
5	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50
6	**B112103C**		**SUPPLEMENTARY CORNER DRIVE KIT** *EVO SOFT*	50
7	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50



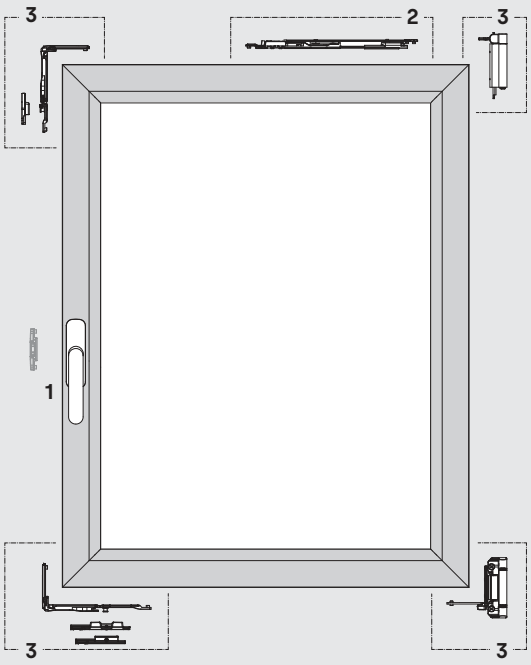
CLASSIC TILT & TURN KIT WITH EVO SOFT HARDWARE
LOCKING TO FRAME OR TILT-BEFORE-TURN

1	**CREMONE**		*Consult cremone section*	
2	**FRICTION STAY**		*Consult tilt & turn opening section*	
3	**C1139**	**	**EVO SOFT LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
	C1123	**	**EVO SOFT TILT-BEFORE-TURN KIT** *WITHOUT CREMONE*	
4	**113003C**		**INTERMEDIATE LOCKING POINT KIT** *EVO SOFT*	50
5	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50
6	**B112103C**		**SUPPLEMENTARY CORNER DRIVE KIT** *EVO SOFT*	50
7	**113004C**		**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50



The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes please refer to the technical catalogue.

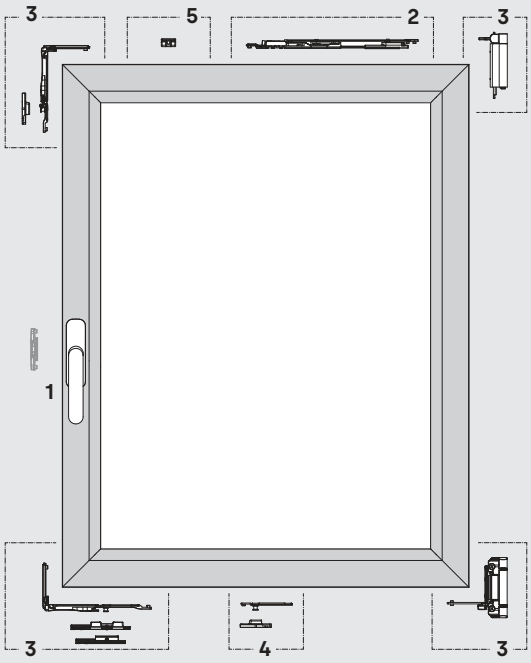
L > 1200 milimetres H > 1200 milimetres



S1

TILT & TURN KIT WITH EVO SECURITY HARDWARE
LOCKING TO FRAME

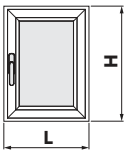
1	**CREMONE**		*Consult cremone section*	
2	**FRICTION STAY**		*Consult tilt & turn opening section*	
3	**S1139**	**	**EVO SECURITY LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
	S1123	**	**EVO SECURITY TILT-BEFORE-TURN KIT** *WITHOUT CREMONE*	



S2

TILT & TURN KIT WITH EVO SECURITY HARDWARE
LOCKING TO FRAME

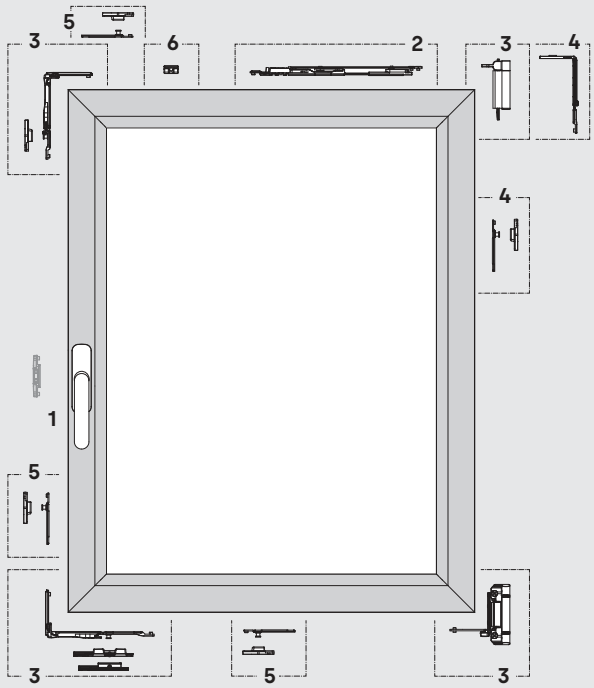
1	**CREMONE**		*Consult cremone section*	
2	**FRICTION STAY**		*Consult tilt & turn opening section*	
3	**S1139**	**	**EVO SECURITY LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
	S1123	**	**EVO SECURITY TILT-BEFORE-TURN KIT** *WITHOUT CREMONE*	
4	**113004S**		**TERMINAL LOCKING POINT KIT** *EVO SECURITY*	50
5	**0507106**		**POSITIONING LATCH**	250

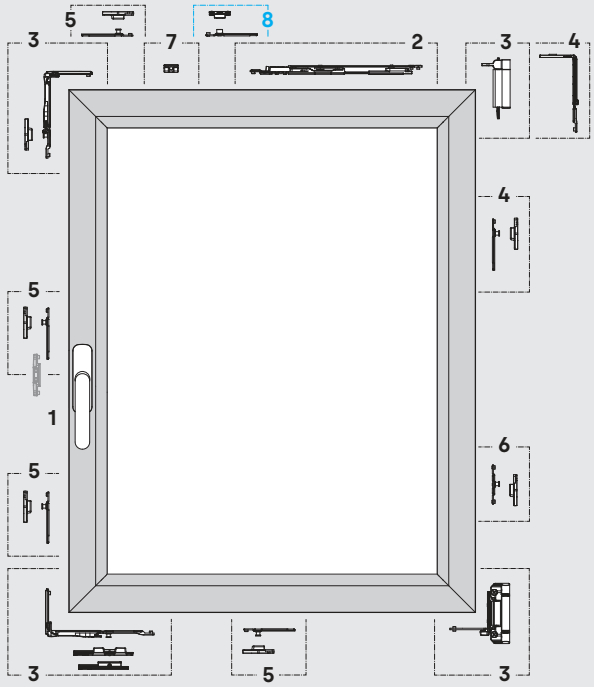


The hardware specified in these configurations is for tilt and turn windows with a maximum of 1200 x 1200 mm. For other window sizes please refer to the technical catalogue.

L > 1200 milimetres

H > 1200 milimetres





S3

TILT & TURN KIT WITH EVO SECURITY HARDWARE
LOCKING TO FRAME

1	**CREMONE**	*Consult cremone section*	
2	**FRICTION STAY**	*Consult tilt & turn opening section*	
3	**S1139** **	**EVO SECURITY LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
	S1123 **	**EVO SECURITY TILT-BEFORE-TURN KIT** *WITHOUT CREMONE*	
4	**A112103S**	**SUPPLEMENTARY CORNER DRIVE KIT** *EVO SECURITY*	50
5	**113004S**	**TERMINAL LOCKING POINT KIT** *EVO SECURITY*	50
6	**0507106**	**POSITIONING LATCH**	250

S4

TILT & TURN KIT WITH EVO SECURITY HARDWARE
LOCKING TO FRAME

1	**CREMONE**	*Consult cremone section*	
2	**FRICTION STAY**	*Consult tilt & turn opening section*	
3	**S1139** **	**EVO SECURITY LOCKING TO FRAME KIT** *WITHOUT CREMONE*	24
	S1123 **	**EVO SECURITY TILT-BEFORE-TURN KIT** *WITHOUT CREMONE*	
4	**A112103S**	**SUPPLEMENTARY CORNER DRIVE KIT** *EVO SECURITY*	50
5	**113004S**	**TERMINAL LOCKING POINT KIT** *EVO SECURITY*	50
6	**113005S**	**INTERMEDIATE LOCKING POINT KIT** *EVO SECURITY*	50
7	**0507106**	**POSITIONING LATCH**	250
8	**113004C**	**TERMINAL LOCKING POINT KIT** *EVO SOFT*	50

10 BOTTOM-HUNG OPENING

222 **CH180 PLUS hinges**

223 **Catch**

224 **Limiting friction stays**

CH180 PLUS HINGES
Bottom-hun window

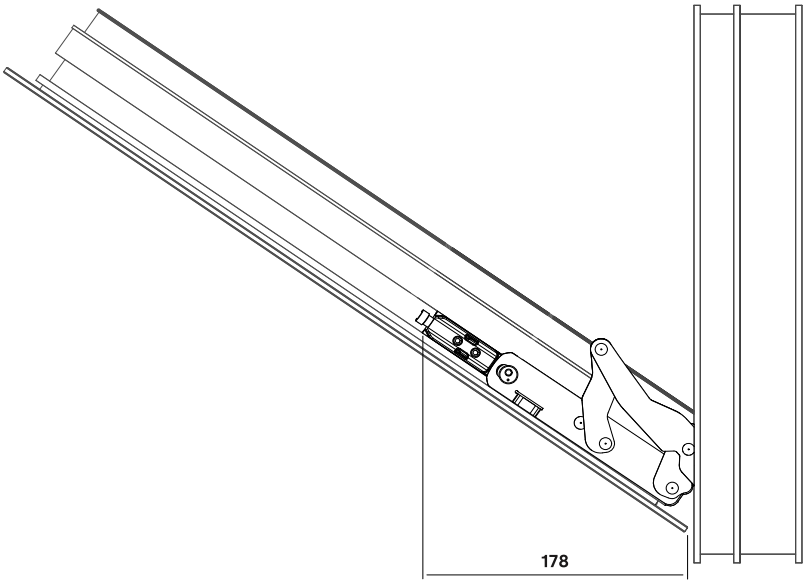
H1215

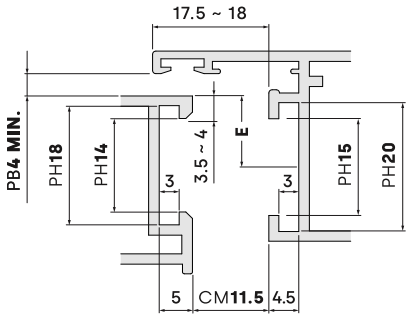
1

AL ZM
IN PA

CH180 PLUS hinges for bottom-hung windows with Euro Groove profile.

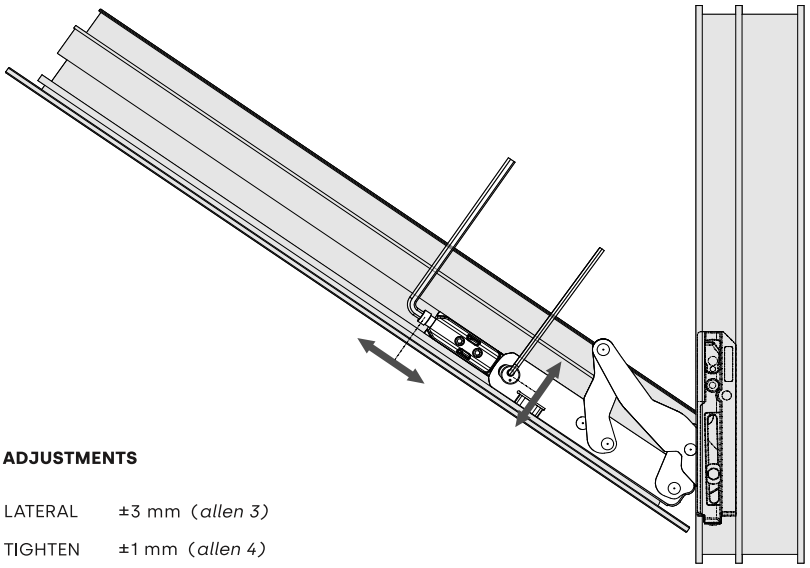
- Maximum load per sash: 100 kg.
- Lateral adjustment: ±3 mm.
- Tightening adjustment: ± 1 mm.





EURO GROOVE

FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	4 mm
SHAFT **E**	10 mm



ADJUSTMENTS

LATERAL	±3 mm (*allen 3*)
TIGHTEN	±1 mm (*allen 4*)

Adjustable lock for bottom-hung opening windows with Euro Groove profiles.

- ± 1.5 mm tightening adjustment by means of a set screw.
- Bolts for a perfect adaptation to the profile.
- The lock installation does not require any type of machining on the profile.

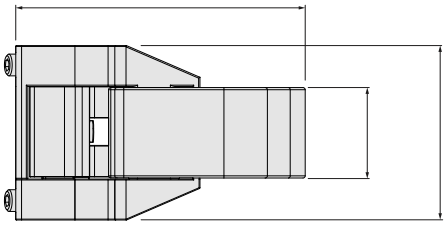
STRIKE LOCK

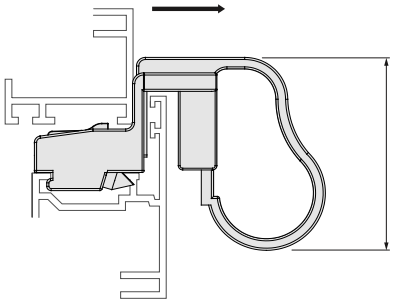
1560	**

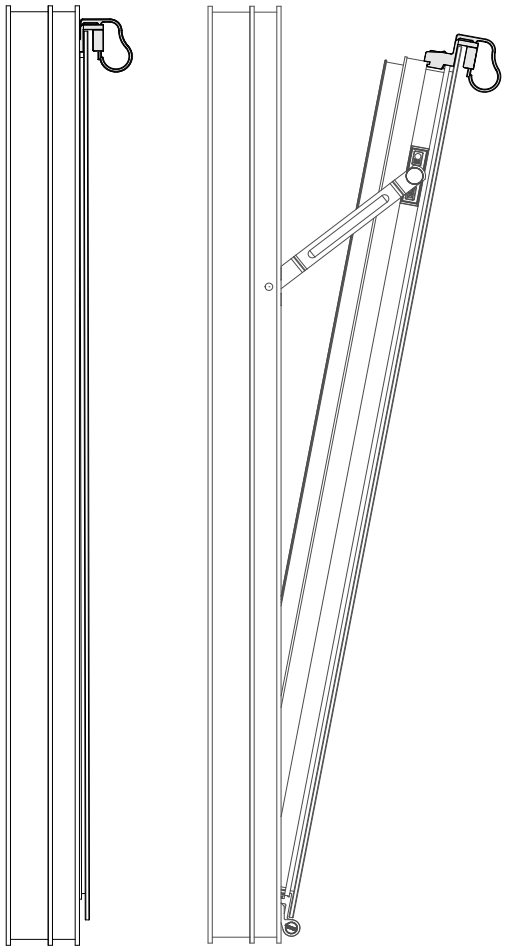
50

AL ZM

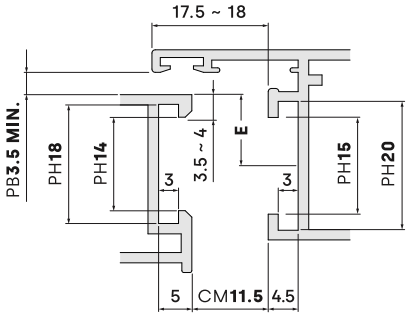
IN







EURO GROOVE	
FRAME PROFILE **PM**	14 - 18 mm
SASH PROFILE **PH**	15 - 20 mm
CHAMBER **CM**	11.5 mm
HINGE PITCH **PB**	3.5 mm
SHAFT **E**	10 mm



SHORT FRICTION STAY

A130100

40

ZM IN PA

A friction stay that restricts openings to 90° in side-hung, bottom-hung and turn-tilt windows.

- Available friction stay for 130° openings (ref. A130101).
- Brake system integrated into the adjustable slider.
- The polyamide slider allows silent operation.
- The position of the limit stop allows the sash to open more or less
- The function of the friction stays is cancelled by releasing the screw cap.

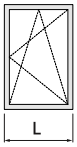
LONG FRICTION STAY

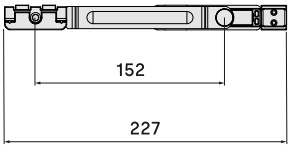
A130101

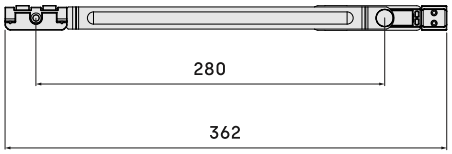
40

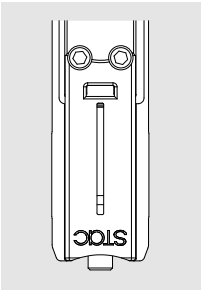
AL IN PA

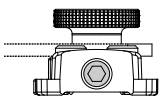
		L min
A130100	SHORT **LIMITING FRICTION STAY**	270 mm
A130101	LONG **LIMITING FRICTION STAY**	400 mm





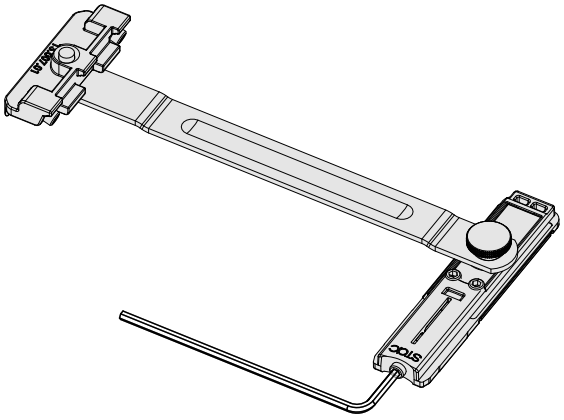






LIMITING FRICTION STAY
WITH ADJUSTABLE BRAKE

ALLEN 2.5 millimetres

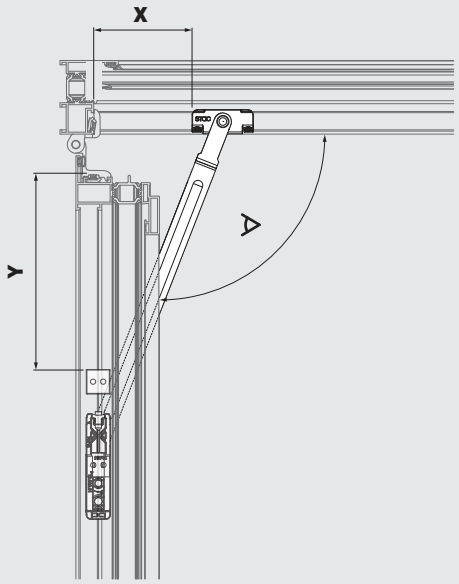


CONFIGURATIONS
FOR SIDE-HUNG WINDOWS

	▷ (max)	**X** (mm)	**Y** (mm)
A130100	90º	34	48
A130101	90º	80	166
A130101	130º	80	106

Previous configurations

It is not possible to limit side-hung sash or tilt & turn sash with an opening less than 30º. The LR limiting friction stay (ref. 130110) is available for this purpose.

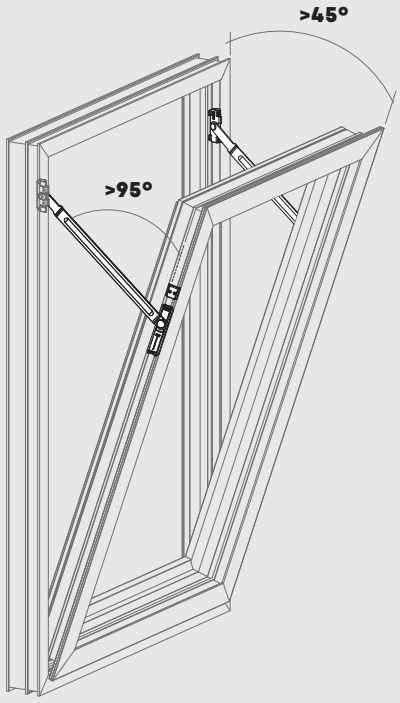


CONFIGURATIONS
FOR BOTTOM-HUNG WINDOWS

	ALTURA MÍNIMA DE HOJA (mm)
A130100	270
A130101	385

Previous configurations

- Sash fold angle may not exceed 45º.
- The angle formed by the limiting friction platen and the sash must always be greater than 95º.
- The block must be installed in a position that allows the conditions described in the previous points.
- The fixed part of the limiting friction stay shall be installed as high as the frame profile will allow.



Installation notes

- In public areas, from 40 kg of sash weight, two additional arms must be installed (if necessary, replace the side locking points).
- For sash weights over 50 kg, the friction stay brake must be adjusted to reduce the inertia of the sash when it hits the stop.
- The brake adjustment is made on all windows. For sash weights > 50 kg, it must be tightened so that it does not hit against the stop.
- For profiles with anodised finishes, the maximum hardness setting is significantly reduced.

SHORT RS LIMITING FRICTION STAY

130120

30

ZM IN PA

A friction stay that restricts openings to 90° in side-hung, tilt & turn or bottom-hung windows.

- Available friction stay for 130° openings (ref. 130121).
- The retention position is carried out in the window's limiting position.
- Integrated security system that allows access to the unlocking by means of a TORX PIN key.
- Brake system integrated into the adjustable slider.

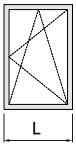
LONG RS LIMITING FRICTION STAY

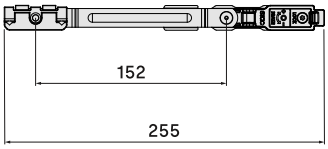
130121

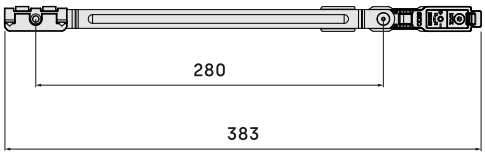
10

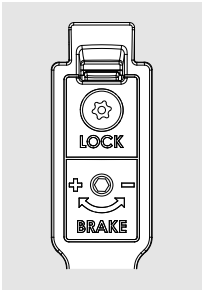
ZM IN PA

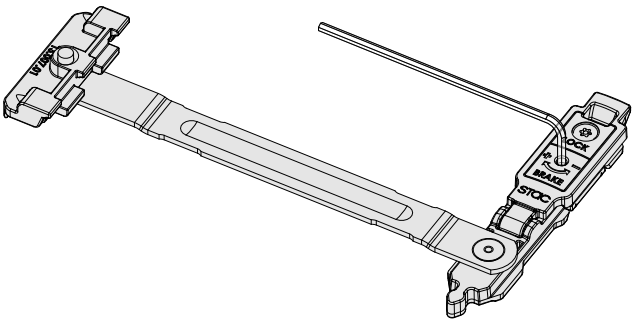
		L min
130120	SHORT **RS LIMITING FRICTION STAY**	290 mm
130121	LONG **RS LIMITING FRICTION STAY**	420 mm











LIMITING FRICTION STAY
WITH SECURITY LOCKING

TORX

LIMITING FRICTION STAY
WITH ADJUSTABLE BRAKE

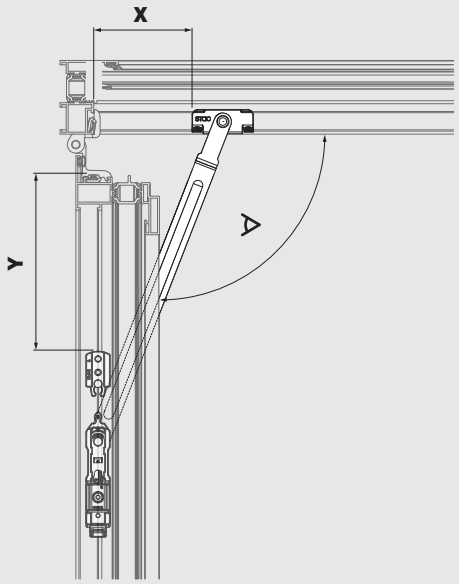
ALLEN 2.5 millimetres

CONFIGURATIONS
FOR SIDE-HUNG WINDOWS

	▷ (max)	**X** (mm)	**Y** (mm)
130120	90º	34	45
130121	90º	34	163
130121	130º	80	103

Previous considerations

It is not possible to limit side-hung sash or tilt & turn sash with an opening less than 30º. The LR limiting friction stay (ref. 130110) is available for this purpose.

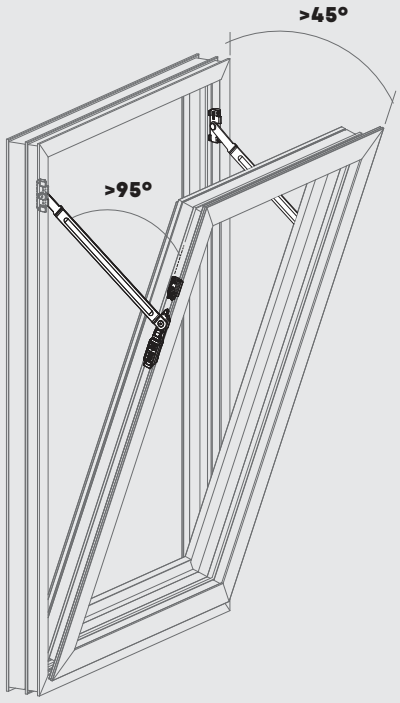


CONFIGURATIONS
FOR BOTTOM-HUNG WINDOWS

	SASH MINIMUM HEIGHT (mm)
130120	290
130121	405

Previous considerations

- Sash fold angle may not exceed 45º.
- The angle formed by the limiting friction platen and the sash must always be greater than 95º.
- The block must be installed in a position that allows the conditions described in the previous points.
- The fixed part of the limiting friction stay shall be installed as high as the frame profile will allow.



Installation notes

- In public areas, from 40 kg of sash weight, two additional arms must be installed (if necessary, replace the side locking points).
- For sash weights over 50 kg, the friction stay brake must be adjusted to reduce the inertia of the sash when it hits the stop.
- The brake adjustment is made on all windows. For sash weights > 50 kg, it must be tightened so that it does not hit against the stop.
- For profiles with anodised finishes, the maximum hardness setting is significantly reduced.

11 TOP-HUNG OPENING

230 PLEXUS handle

232 SIRIUS cremone

235 PIVOT handle

236 SIRIUS pressure fasteners

238 LINCE pressure fasteners

240 ORION pressure fasteners

242 Limiting friction stays

245 Top-hung opening friction stays

247 Top-hung opening accessories

Euro Groove

Quick and simple installation

Corrosion resistance

Customisation

PLEXUS handle

Our design line considers one of its main goals to be the simplification of the visual impact of its hardware without sacrificing any functionality. To do so, we have focused on several approaches: minimalist, integrated and even invisible designs.

This has led us to the launch of the PLEXUS handle, specifically designed for top-hung windows, perfectly embodying the spirit of the current design objectives.

» Top-hung windows

This handle fits the most innovative Euro Groove top-hung opening systems, can be installed on any profile with a minimum of 13 mm to the groove, and perfectly integrates with our Euro Groove hardware.

- Developed for perimeter profiles.
- Ambidextrous: has no opening hand.
- Single or double handle configurations.

» Minimalist and functional design

Simple and sophisticated lines remain unchanged with time. The handle is especially attractive to architects and decorators thanks to its elegant, discreet and timeless design. Designed with a focus on elementary geometric composition, the simplicity of the shapes and the precision of the finishes.

» Customisation

More and more projects are looking for their uniqueness through colour and where the accessories are lacquered in the same colours as the carpentry

The PLEXUS handle is fully customisable and is available in white or black lacquer as well as in the colours that make up the RAL chart, a system of colour combinations that defines colours for paints, plastics and coatings.

» Corrosion resistance

The components have been treated to guarantee over 240 hours of corrosion resistance in a saline mist chamber in accordance with European standard UNE-EN 1670.

According to the parameters described in the standard, a grade 4 classification has been obtained, meaning the product may be used in environments with extreme climates.

Ambidextrous handle specifically designed for facades (curtain wall).

- Installation on profiles with a maximum of 19 mm in length up to the groove.
- Compatible with Euro Groove perimeter window profiles.
- Single or double handle configurations
- Simple installation system and profile machining.

PLEXUS HANDLE

ce

ZM

IN

1551 **

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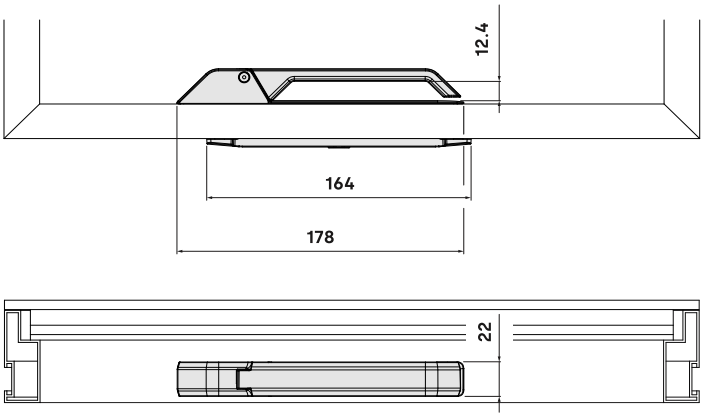
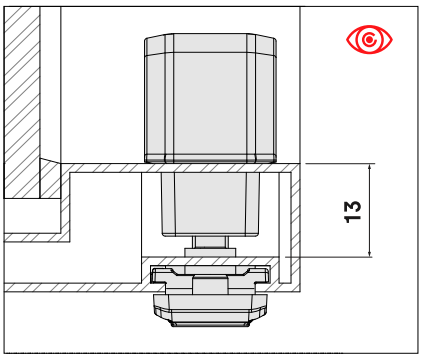
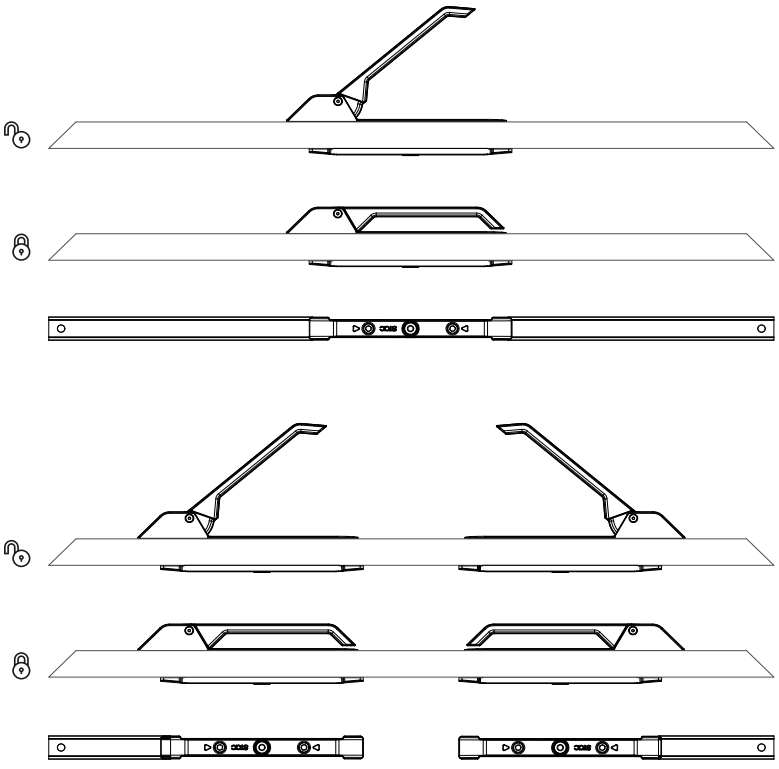
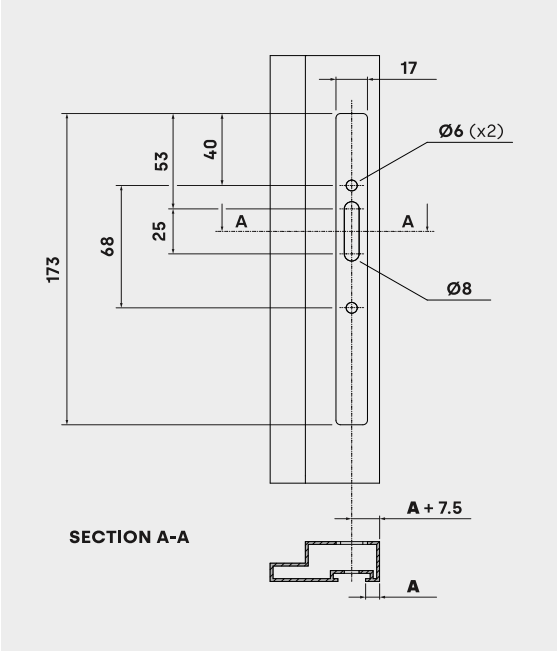




MINIMUM PROFILE DEPTH
13 millimetres





SIRIUS CREMONE
Top-hung opening

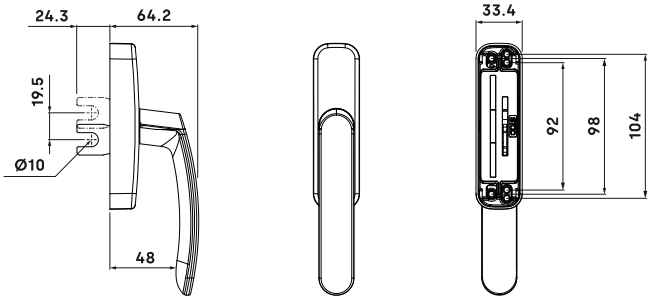
1166 | **

20

AL ZM IN PA

Cremone with the special displaced lever for top-hung windows.

- The displaced levers and pinion are supplied in steel for high cycle strength and durability.
- Internal mechanisms adjusted to the millimetre to eliminate any type of play and ensure perfect operation.
- Available a cremone with handle lock (ref. A1167) which, by means of the push button, allows the movement to be locked in different positions.
- By default, all cremone are supplied without the top-hung transmission kit (ref. 180100).

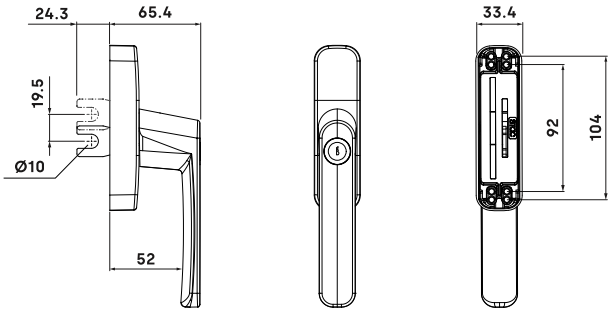


SIRIUS BLOCK CREMONE
Top-hung opening

A1167 | **

20

AL ZM IN PA

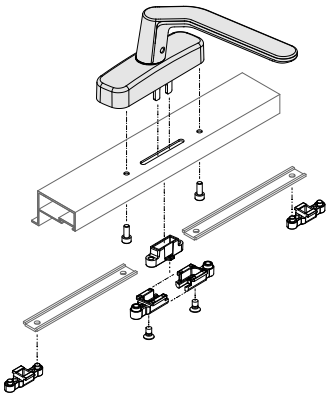


TOP-HUNG WINDOW TRANSMISSION

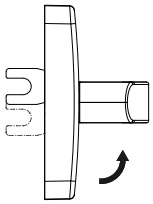
180100

50

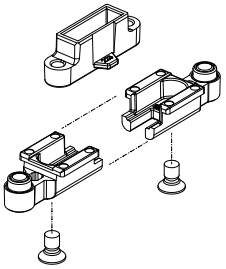
ZM PA

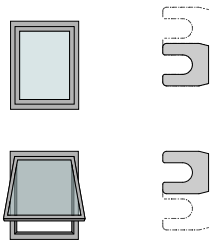


INSTALLATION
POSITION HANDLE AT 90°



POSITION OF
TRANSMISSION LEVERS





Offset cremone with the special displaced lever for top-hung windows.

- Valid for right-hand opening (left-hand opening ref. 1162).
- The displaced levers and pinion are supplied in steel for high cycle strength and durability.
- Internal mechanisms adjusted to the millimetre to eliminate any type of play and ensure perfect operation.
- By default, all cremone are supplied without the top-hung transmission kit (ref. 180100).

SIRIUS OFFSET CREMONE
Right-hand opening

1161	**

20

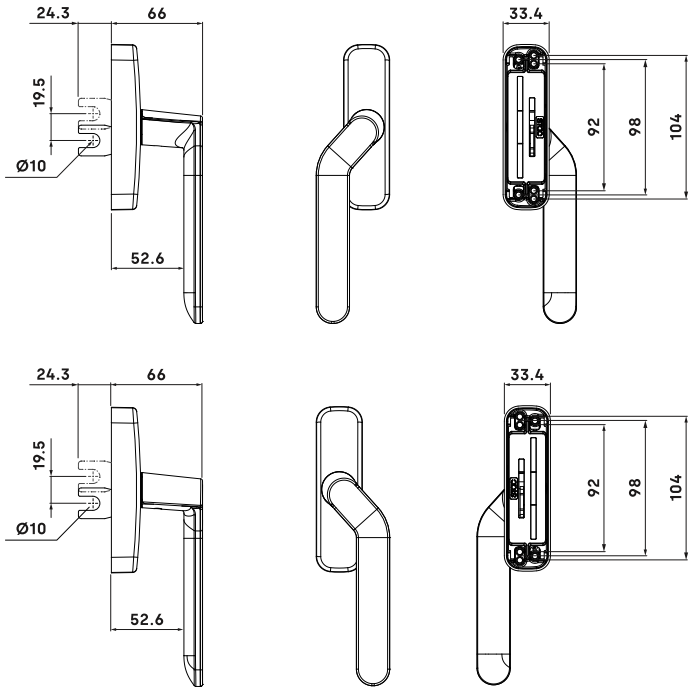
AL ZM IN PA

SIRIUS OFFSET CREMONE
Left-hand opening

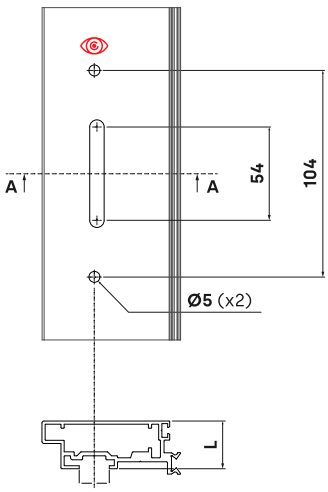
1162	**

20

AL ZM IN PA

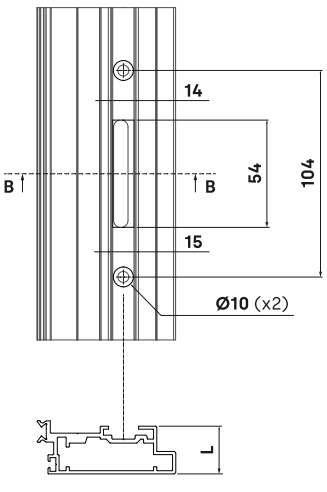


TOP-HUNG CREMONE
SIRIUS, SIRIUS BLOCK

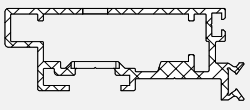


L min. *15 mm,* ***L max.*** *29 mm*

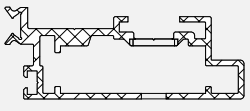
TRANSMISSION KIT
ref. 180100



L min. *15 mm,* ***L max.*** *29 mm*



SECTION A-A



SECTION B-B

SIRIUS OFFSET CREMONE
Right-hand opening

R1161 **

20

AL ZM IN PA

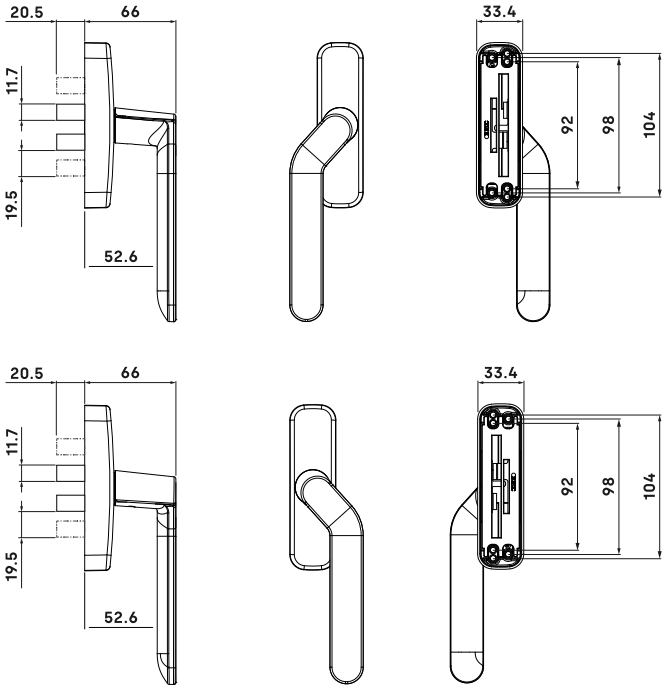
Offset cremone with the bidirectional displaced lever for top-hung windows.

- Valid for right-hand opening (left-hand opening ref. R1162).
- The displaced levers and pinion are supplied in steel for high cycle strength and durability.
- Internal mechanisms adjusted to the millimetre to eliminate any type of play and ensure perfect operation.
- By default, all cremone are supplied without the top-hung transmission kit (ref. 180101).

SIRIUS OFFSET CREMONE
Left-hand opening

R1162 **

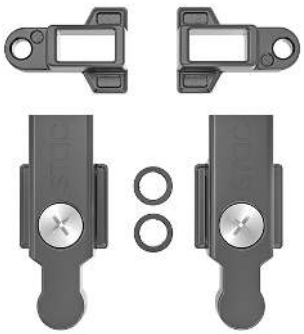
AL ZM IN PA



TWO-WAY TOP-HUNG TRANSMISSION KIT

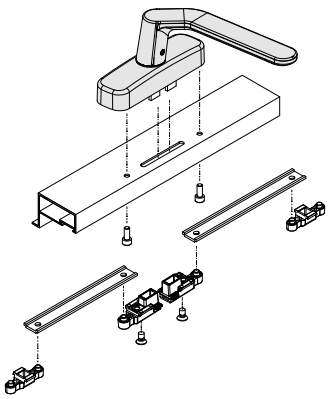
180101

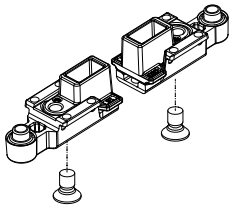
50



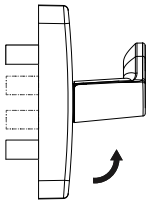
ce

ZM

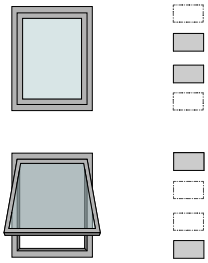




INSTALLATION
POSITION HANDLE AT 90°



POSITION OF
TRANSMISSION LEVERS



Facade handles designed for top-hung windows that allow pressure closing.

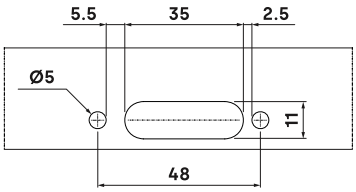
- Valid for right-hand opening (left-hand opening ref. A1512).
- The handles incorporate two keepers and three pivots to adjust to the different systems.
- Includes 10.5 mm short keeper and pivot with a height of 28.5 mm.
- References with short or long keepers and 25 or 32 millimetre high pivots are available (see table of references).

FACADE HANDLE
Right-hand opening

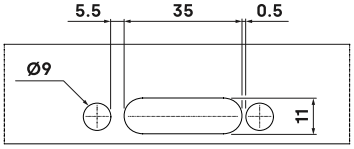
	PIVOT	KEEPER	OPENING
A1511 **	28.5 mm	10.5 mm	RIGHT
A1512 **			LEFT
A1521 **	32 mm	13.5 mm	RIGHT
A1522 **			LEFT
A1531 **	25 mm	10.5 mm	RIGHT
A1532 **			LEFT

FACADE HANDLE
Left-hand opening

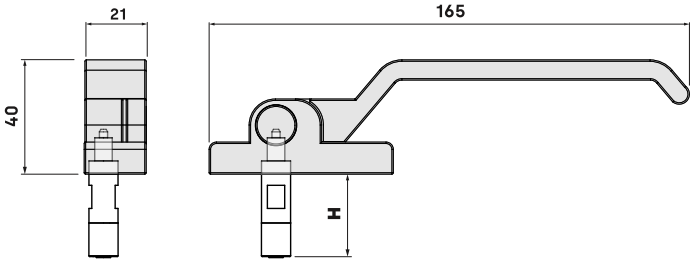
***UPPER ZONE** HANDLE*



***LOWER ZONE** GROOVE*

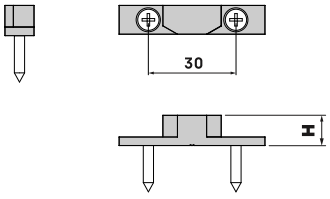


PIVOT HANDLE



		H (mm)
153100	SHORT KEEPER	10.5
153200	LONG KEEPER	13.5
154000	MINI PIVOT	25
154100	SHORT PIVOT	28.5
154200	LONG PIVOT	32

***KEEPER** 153100, 153200*



Pressure fastener

SIRIUS

Adaptation shims

Covers to hide screws

Rotating scutcheon

Corrosion resistance

SIRIUS pressure fastener

We have renewed our line of pressure fasteners for top-hung openings with an updated, cohesive and functional design.

The design line of the handle favours integration with the rest of the SIRIUS family of products, both cremone and handles.

» Adaptability

It includes shims made of ABS in order to adapt to different profile series and provide greater pressure in the closing manoeuvre of the system.

» Clean aesthetics

Through a system of springs, the screw trims allow to hide the fixing mechanism, favouring the aesthetics of the product.

» Quick and simple installation

The lock is supplied with a rotating scutcheon for simple and quick installation. This allows the installer to significantly reduce installation times and optimise the time spent.

» Robustness and durability

The SIRIUS pressure fastener has been subjected to multiple tests in accordance with the UNE-EN 13126-2 standard, *"Building hardware. Hardware for windows and door height windows. Requirements and test methods. Part 2: Window fastener handles"* obtaining results that exceed the established criteria.

» Corrosion resistance

The parts that make up the fastener have been treated to guarantee resistance to corrosion, exceeding 240 hours in a salline mist chamber in compliance with the UNE-EN 1670 standard.

According to the parameters described in the standard, a grade 4 classification has been obtained, meaning the product may be used in environments with extreme climates.

Pressure lock with SIRIUS aesthetic valid for outward and inward top-hung windows.

- Valid for right-hand opening (left-hand opening ref. 1509).
- Supplied with 1, 3 and 4 mm shims that allow height adjustment up to 12 millimetres to suit different profile series and provide greater pressure to the closure.
- Cover to conceal the handle fixings and the back lock.
- The integrated rotating escutcheon allows a reduction of the installation time of the lock.

SIRIUS PRESSURE LOCK
Right-hand opening

1508 **

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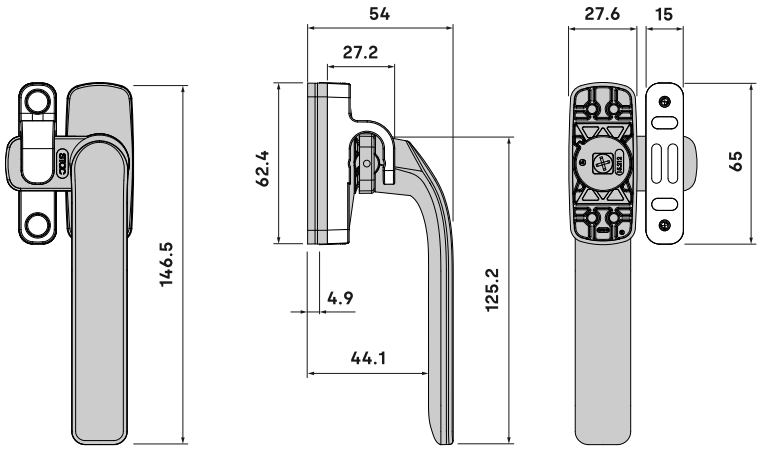
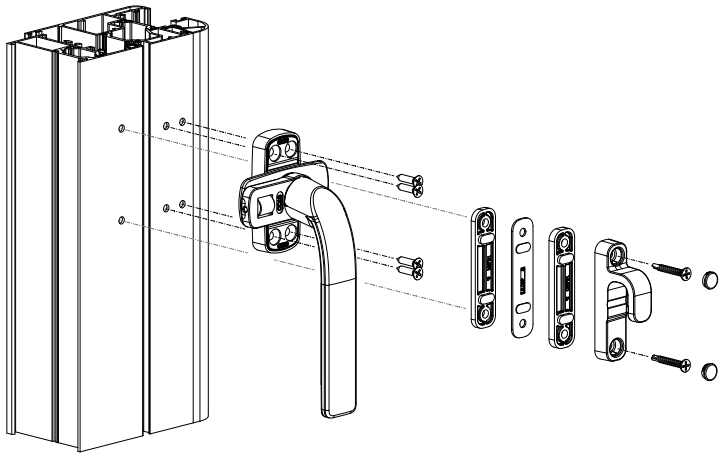
AL ZM IN PA

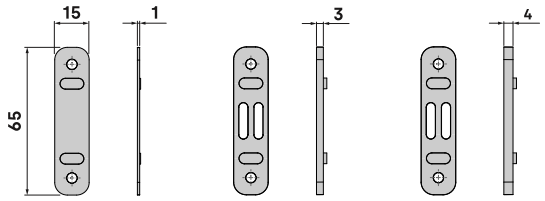
SIRIUS PRESSURE LOCK
Left-hand opening

1509 **

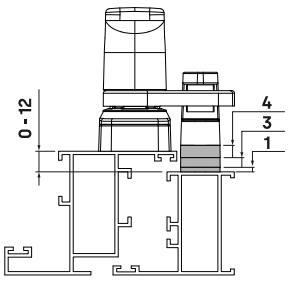
AL ZM IN PA



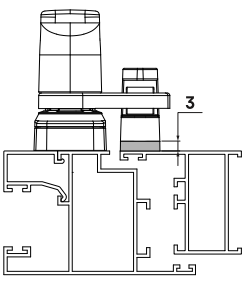




Adaptation shims (0 - 12 mm)



***INWARD** OPENING*



***OUTWARD** OPENING*

LINCE PRESSURE LOCK
Right-hand opening

A1506 **

30

AL ZM
IN PA

Pressure lock with LINCE aesthetic valid for outward and inward top-hung windows.

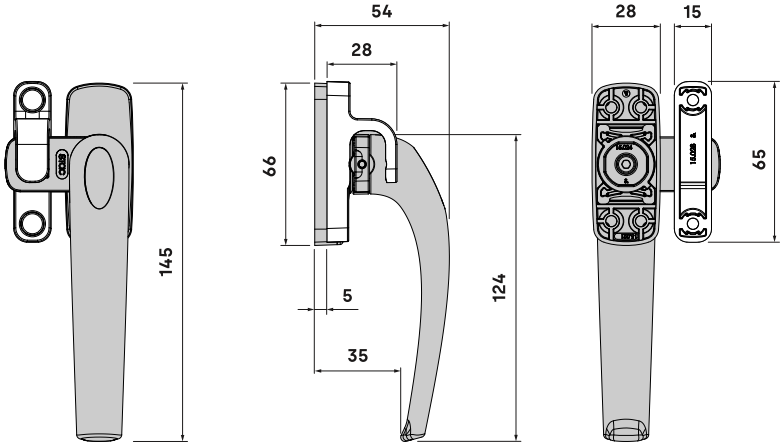
- Valid for right-hand opening (left-hand opening ref. A1507).
- Supplied with 1, 3 and 4 mm shims that allow height adjustment up to 12 millimetres to suit different profile series and provide greater pressure to the closure.
- Cover to conceal the handle fixings and the back lock.
- The integrated rotating escutcheon allows a reduction of the installation time of the lock.

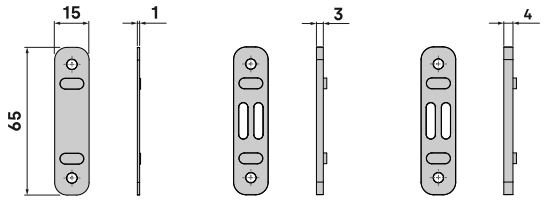
LINCE PRESSURE LOCK
Left-hand opening

A1507 **

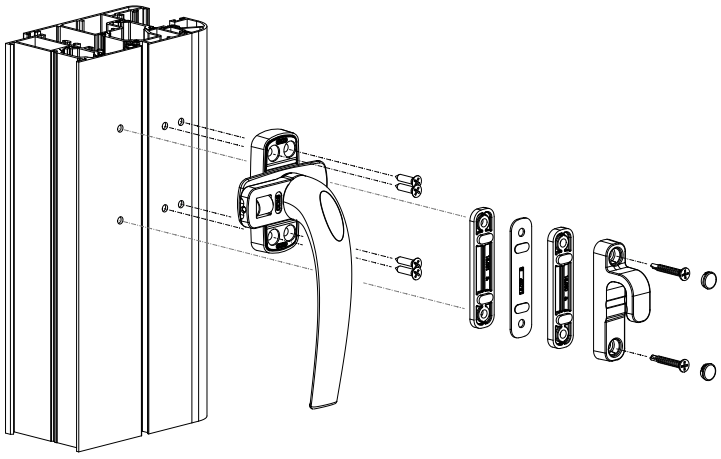
30

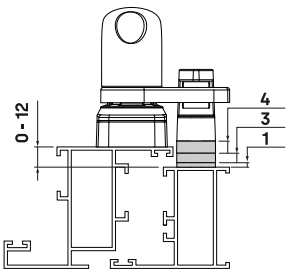
AL ZM
IN PA



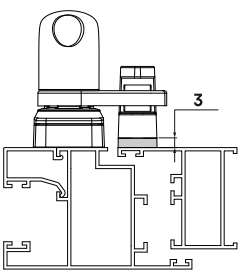


Adaptation shims (0 - 12 mm)





***INWARD** OPENING*



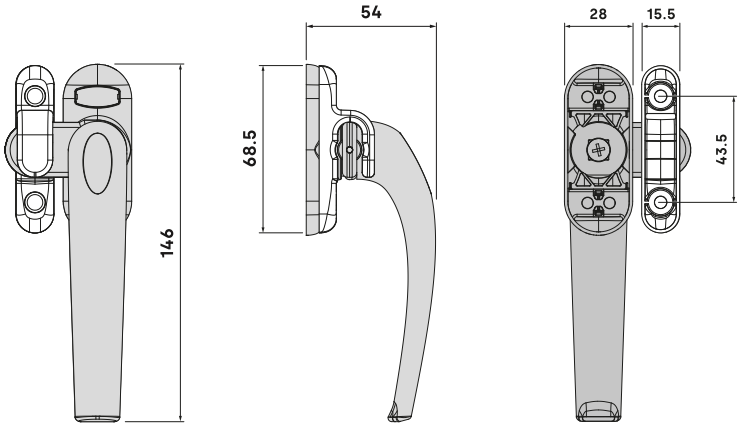
***OUTWARD** OPENING*

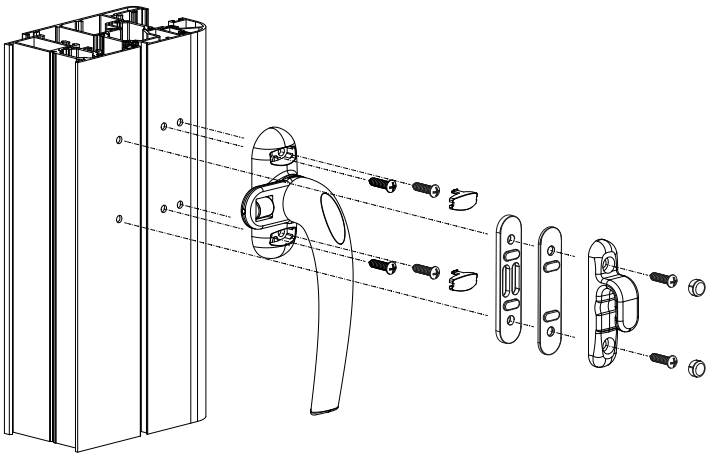
Pressure lock with LINCE aesthetic valid for outward and inward top-hung windows.

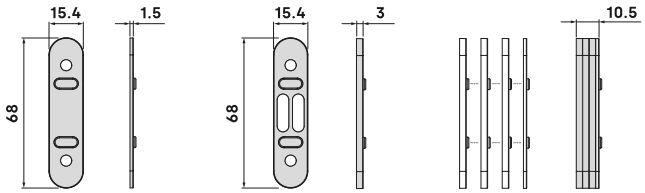
- Valid for right-hand opening (left-hand opening ref. 1513).
- Supplied with 1.5 and 3 mm shims that allow height adjustment up to 10.5 millimetres to suit different profile series and provide greater pressure to the closure.
- Cover caps to hide the fixing screws of the handle and the counter lock.

LINCE ECO PRESSURE LOCK
Right-hand opening

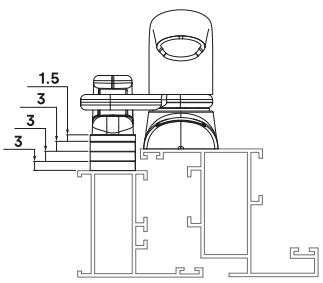
LINCE ECO PRESSURE LOCK
Left-hand opening



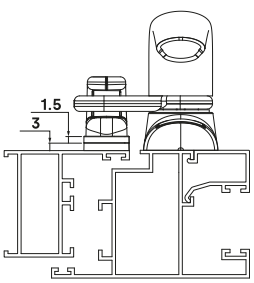




Adaptation shims (0 - 10.5 mm)



***INWARD** OPENING*



***OUTWARD** OPENING*

ORION PRESSURE LOCK
Right-hand opening

1501 | **

30

AL ZM
IN PA

Pressure lock designed for inward and outward bottom-hung windows.

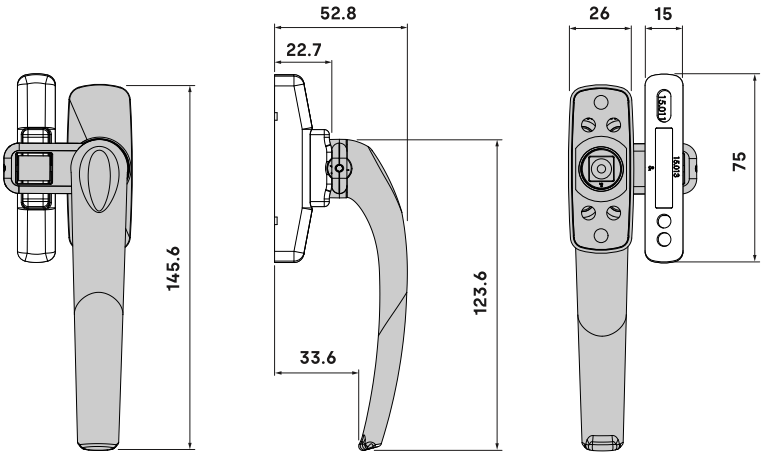
- Valid for right-hand opening (left-hand opening ref. 1502).
- The base supports different fixing options via the screws.
- Cover to conceal the handle fixings and the back lock.
- Polyamide supplements that provide greater closing pressure (1, 3 and 5 mm).
- It is supplied with polyethylene shims for the counter-closure devices, which can be adapted to different profile series (1, 2 and 3 mm).

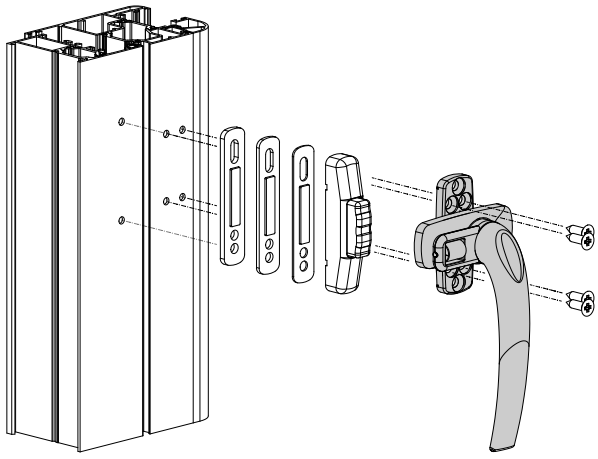
ORION PRESSURE LOCK
Left-hand opening

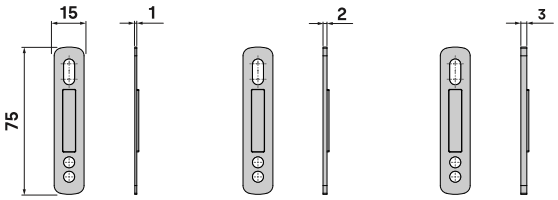
1502 | **

30

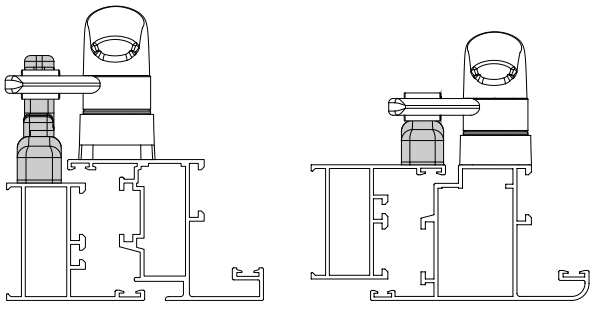
AL ZM
IN PA







Adaptation shims (0 - 6 mm)



***INWARD** OPENING*

***OUTWARD** OPENING*

Counter-locks developed for the ORION pressure lock.

- Valid for inward opening (ref. 1503) or outward opening (ref. 1504).
- They are equipped with a slotted locking point for easy installation on the profile.
- Supplied with covers to conceal the fixing screws.
- Compatible with polyethylene shims (not included) to fit different profile series.

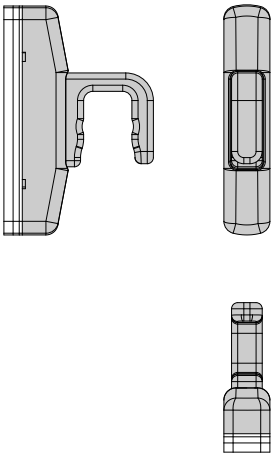
ORION COUNTER-LOCK
Inward opening

1503 **

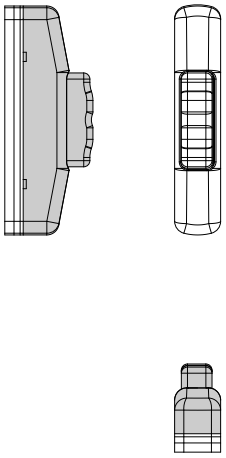
30

AL ZM PA

COUNTER-LOCK
INWARD OPENING



COUNTER-LOCK
OUTWARD OPENING



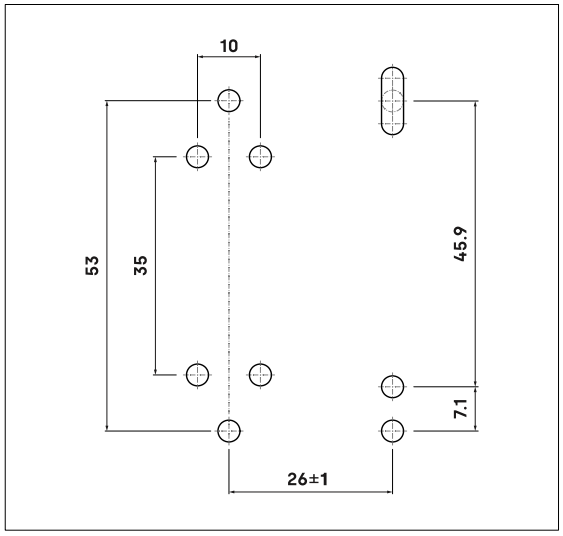
ORION COUNTER-LOCK
Outward opening

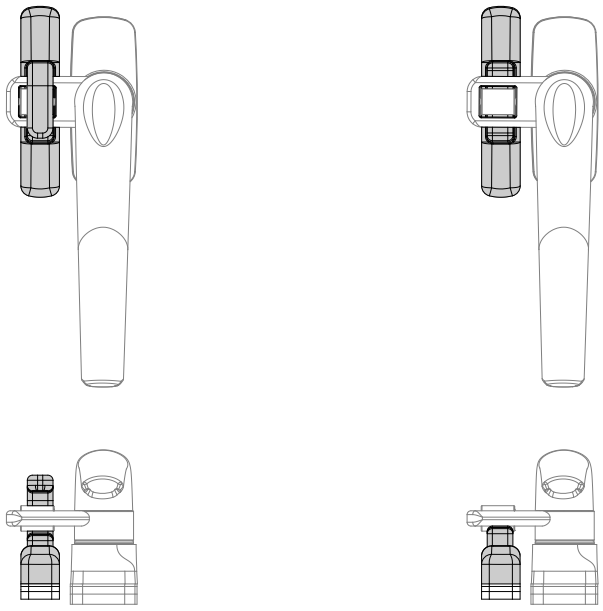
1504 **

30

AL ZM PA

MACHINNING
THE COUNTER-LOCK HAS A SLOTTED
LOCKING POINT FOR AN EASIER INSTALLATION





LIMITING RETAINING FRICTION STAY

130801

20

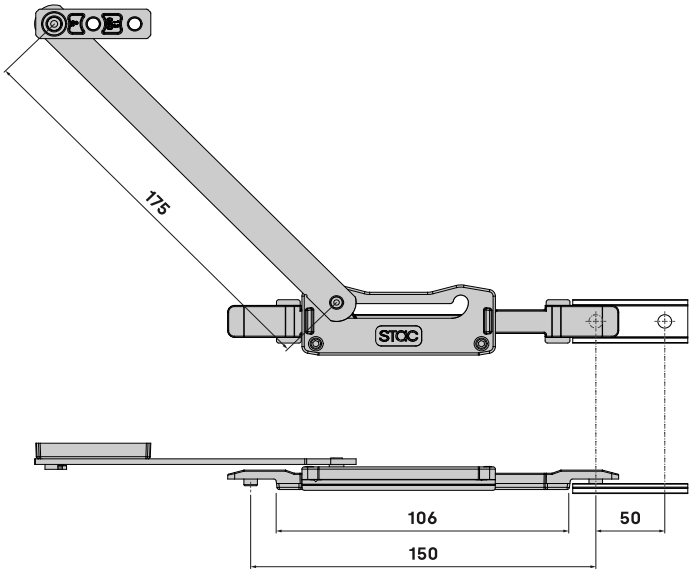
ZM

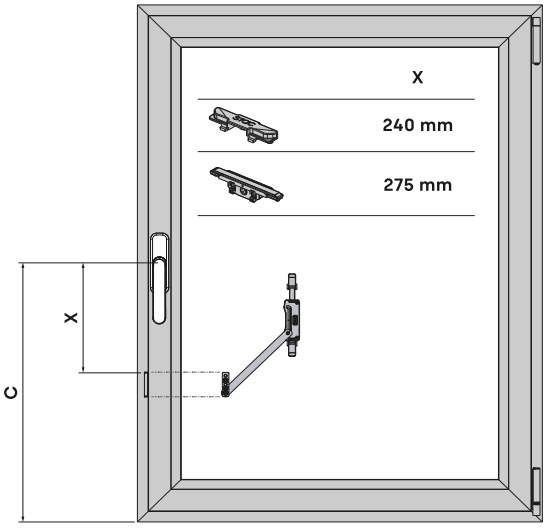
IN

Limiting retaining friction stay for top-hung windows or exterior openings with Euro Groove profile.

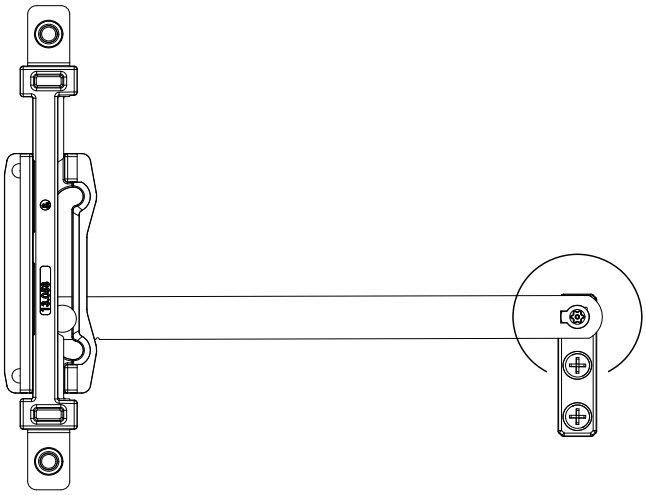
- The use of bi-directional cremone or double handle PLEXUS is required.
- The lock is operated by the cremone in the limit position.
- Disconnection of the limiter in the frame is by means of a TORX PIN key.
- It has different degrees of opening according to its position on the sash.

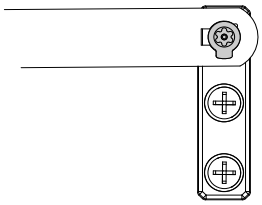
Not compatible with the anti-false manoeuvring corner drive (ref. 110900).



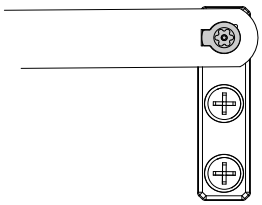


C *CREMONE HEIGHT*





LIMITER CONNECTED
TORX PIN TR25



LIMITER DISCONNECTED
TORX PIN TR25

Limiting friction stay set for top-hung opening windows.

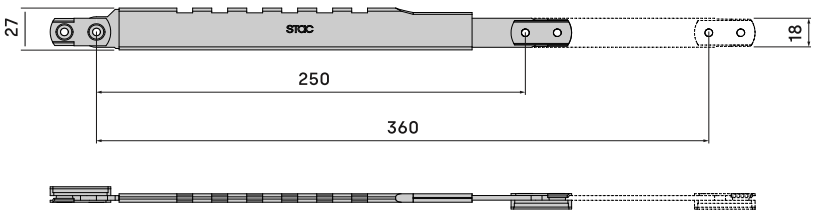
- Valid for Euro Groove profiles or 40 ~ 20 Groove (4000/6000 series).
- Allows the opening in 5 defined positions
- The window opening is adjustable with positioning.
- The sash is closed by placing the friction stay in the maximum open position.

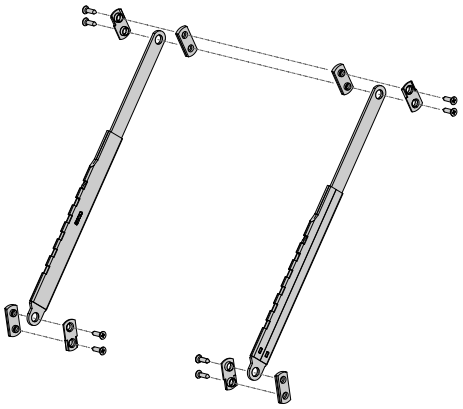
EXTENDABLE LIMITING FRICTION STAY

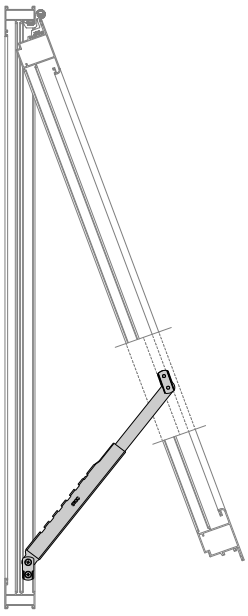
IN

18009

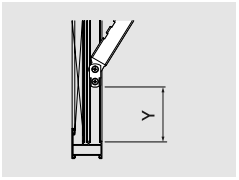
15





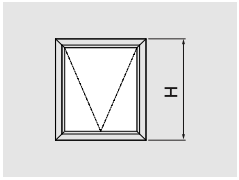


Y = 0



H = 1200	X maximum	X unlocking
800	380	430
1000	360	400
1200	340	385
1400	330	390
1600	320	360

H = 1200



Y = 0	X maximum	X unlocking
100	380	430
200	430	480
300	565	490
400	575	640
500	700	770

STAINLESS STEEL LIMITING FRICTION STAY

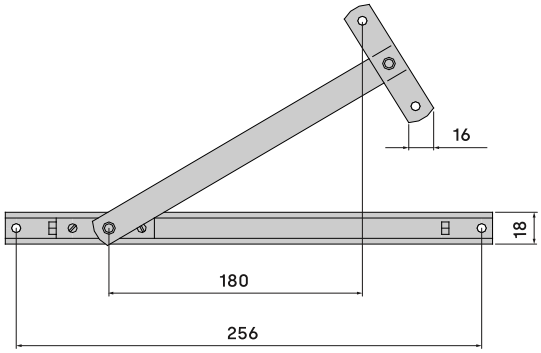
18010

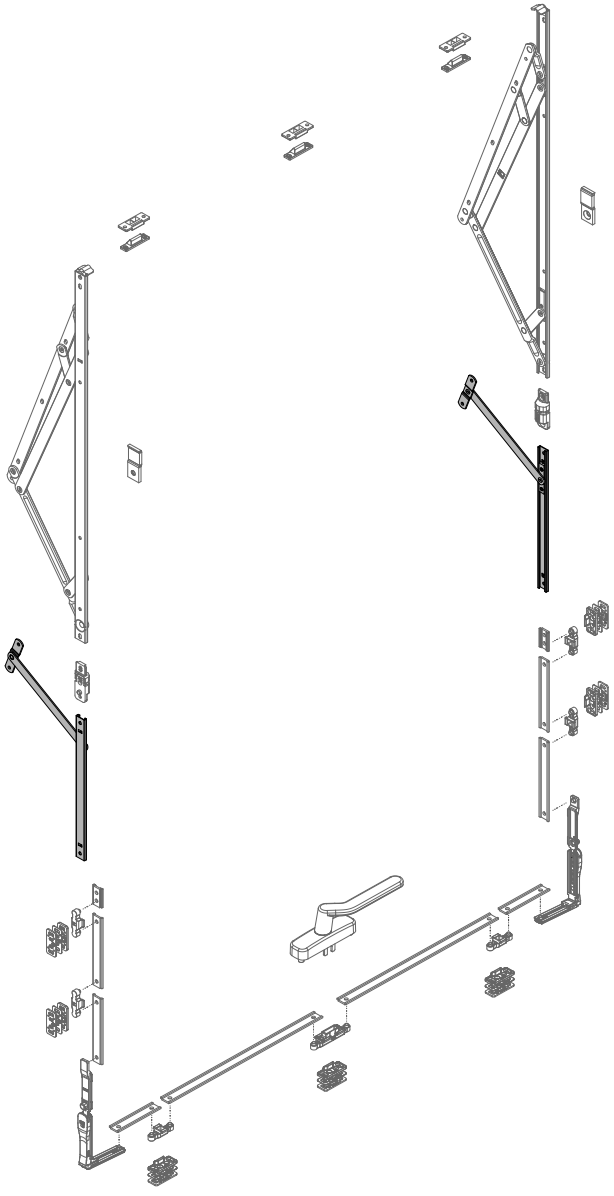
10

IN

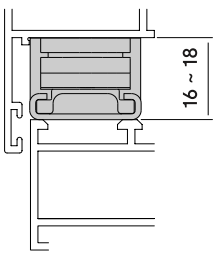
Limiting friction stay set for top-hung windows.

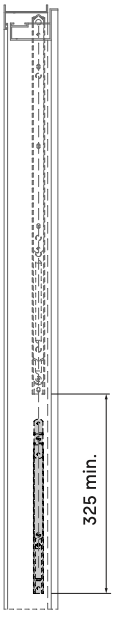
- Valid for Euro Groove profiles or flat frame.
- The use of standard limiting friction stays is recommended as additional elements to the top-hung friction stays from 1800 mm sash height
- The use of limiting friction stays is recommended in areas exposed to strong winds or with restrictions on the limitation of opening.

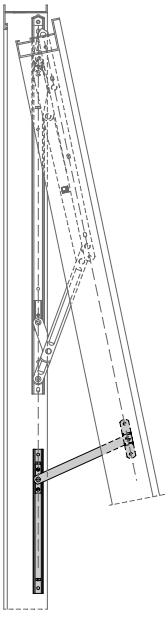




*CHAMBER **16 ~ 18***







Friction stay with limitation set designed for top-hung opening systems with 16 ~ 18 chamber.

- They are supplied in AISI 304 stainless steel in sets of 2.
- Regulating skid that allows pressure adjustment.
- Reinforced projecting frictions stays available (ref. 18002 and ref. 18003).

Height adjuster (ref. 18200)

- Accessory for adjusting the height of the projecting friction stay.
- Supplied in sets of 2 units.
- It is recommended for installation on sashes with a height greater than 1000 mm.
- The regulator is not compatible with the standard projecting friction stays (ref. 18006, 18007 and 18008).

**Check availability.*

	FRICTION STAY **LENGTH**	MAX. **WIDTH**	MAX. **HEIGHT**	MAX. **WEIGHT**	OPENING **ANGLE**	FRICTION STAY **MATERIAL**	
18002	400	1200	900	30 kg	87°	AISI 304	25
18003	550	1200	1300	40 kg	87°	AISI 304	20
18006	400	1200	750	24 kg	87°	AISI 304	20
18007	500	1200	1000	28 kg	40°	AISI 304	25
18008	600	1200	1100	35 kg	40°	AISI 304	20

TOP-HUNG FRICTION STAY

IN

180

25

ADJUSTER FOR TOP-HUNG FRICTION STAY

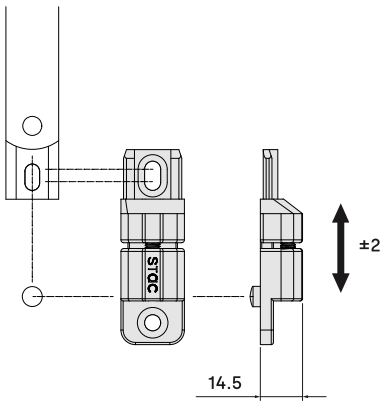
ZM

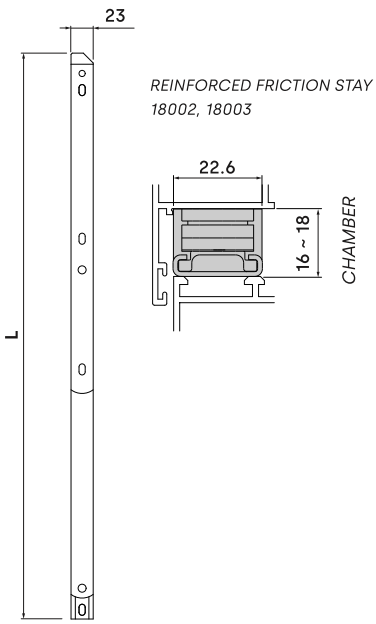
IN

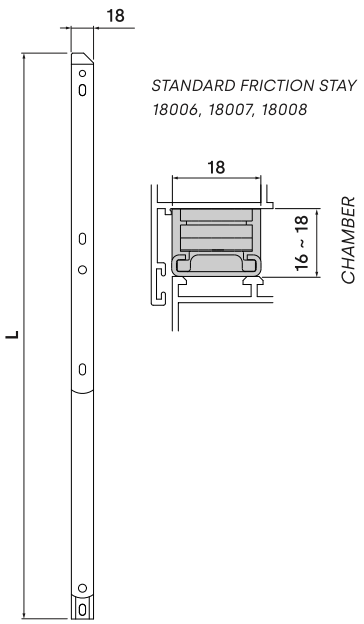
18200

100

HEIGHT ADJUSTMENT
18002, 18003







CURTAIN WALL FRICTION STAY

A180

20

IN

Friction stays sets, with limiting function, developed for curtain wall systems with chamber 16 ~ 18.

- They are supplied in AISI 304 stainless steel (available in AISI 316).
- Regulating skid that allows pressure adjustment.

Height adjuster (ref. 18200)

- Accessory for adjusting the height of the top-hung friction stay.
- Its installation is recommended in sashes with a height greater than 1000 mm.
- Supplied in sets of 2 units.

Limiting plate (ref. A0518014)

- Accessory to limit the opening of the curtain wall friction stay.
- Compatible with top-hung friction stays ref. A18011 and ref. A18012.
- Supplied with screw DIN 7982 4.8 x 16 millimetres.

ADJUSTER FOR TOP-HUNG FRICTION STAY

18200

100

ZM

IN

	LENGTH	WIDTH	HEIGHT	MAX. WEIGHT	OPENING ANGLE	MATERIAL
A18011	455	500-1500	500-1200	80 kg	30° / 25°*	AISI 304
B18011	455	500-1500	500-1200	80 kg	30° / 25°*	AISI 316
A18012	705	500-2200	1200-2200	130 kg	20° / 15°* / 10°*	AISI 304
B18012	160	500-2200	1200-2200	130 kg	20° / 15°* / 10°*	AISI 316
A18017	555	500-1800	600-1800	110 kg	20° / 25°*	AISI 304
A18018	600	500-1800	650-1800	120 kg	15° / 20°*	AISI 304
A18032	410	500-1200	500-1200	80 kg	21° / 15°*	AISI 304

(*) ANGLE WITH OPENING LIMITATION

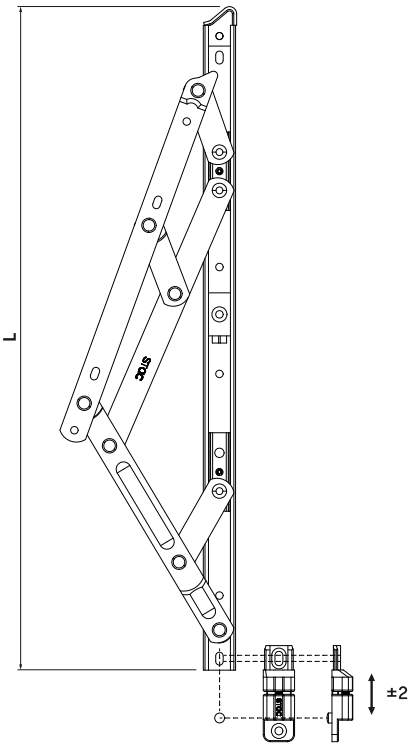
LIMITING PLATE

A0518014

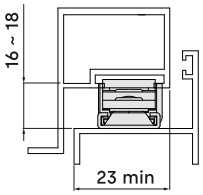
200

ZM

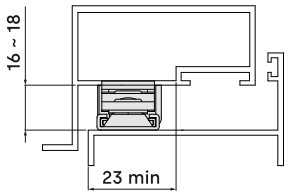
IN



*CHAMBER **16 ~ 18***



*CHAMBER **16 ~ 18***



Locking points for top-hung opening system.

Supplementary locking point kit (ref. 113000).

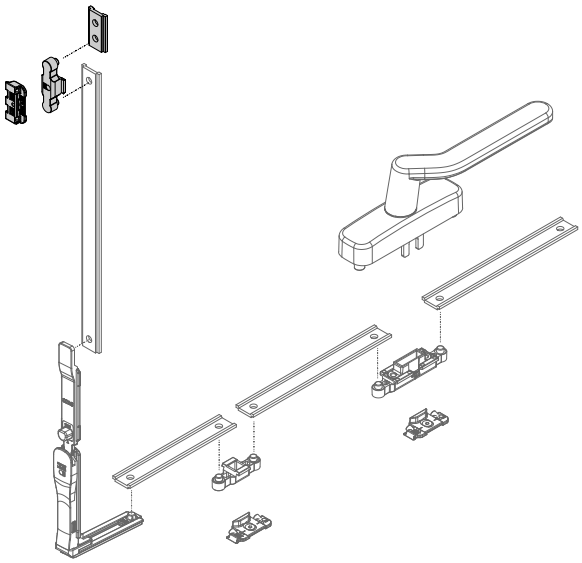
• Valid for systems with Euro Groove profile.
• Supplied with a sliding locking point that acts as a supplementary locking point, an adjustable keeper (ref. 110200), connecting rod and Ø8 mm magnifying washers.

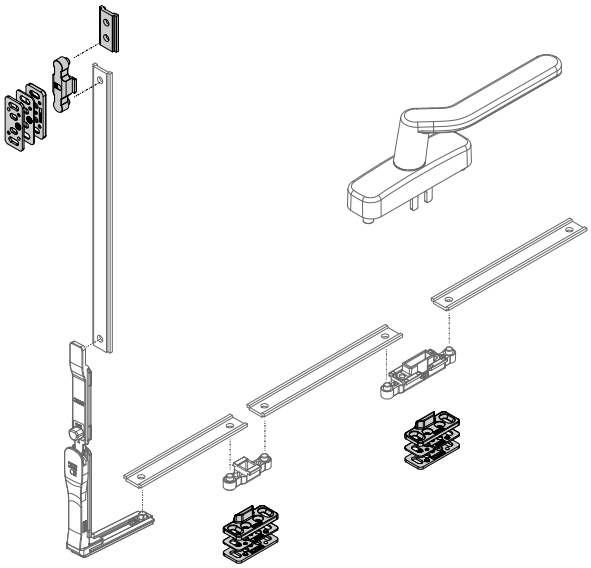
Adjustable locking point kit (ref. 183000).

• Valid for flat frame systems.
• Supplied with frame element, zamak shims, connecting rod and Ø8 mm magnifying washers.
• The shims, of 1 and 2 millimetres, allow the height to be adjusted in relation to the profile.

Adjustable locking point (ref. 180200).

• Valid for flat frame systems.
• Supplied with frame elements and zamak shims that allow the height to be adjusted in relation to the profile.





SUPPLEMENTARY LOCKING POINT KIT

ce

ZM

PA

113000

50

TOP-HUNG LOCKING POINT KIT

ZM

PA

183000

50

ADJUSTABLE LOCKING POINT

ZM

PA

180200

100

CURTAIN WALL HIDDEN LOCKING POINT

0518015

100

ZM

IN

Accessories for top-hung opening window systems or curtain wall.

Concealed locking point (ref. 0518015).

- Valid for curtain wall systems with Euro Groove or flat frame.
- The elements act as a supplementary locking point.
- Recommended for installation from 1200 mm sash height upwards.

Drive gear (ref. 18016).

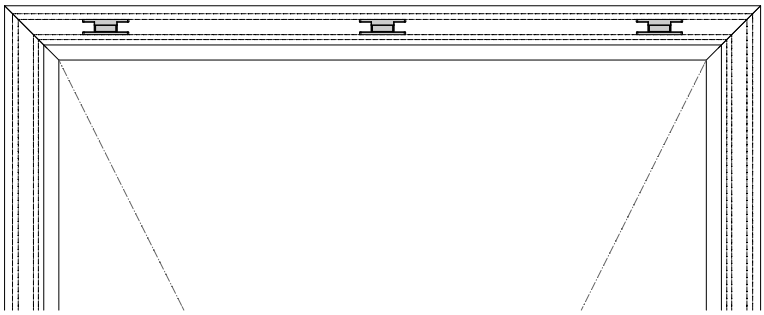
- Valid for Euro Groove systems
- Aluminium grooved accessory for connecting the cremone transmission sash to the rod plates.

TRANSMISSION ROD

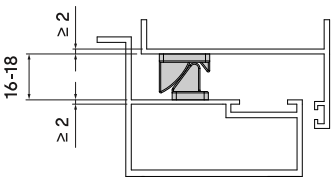
0518016

100

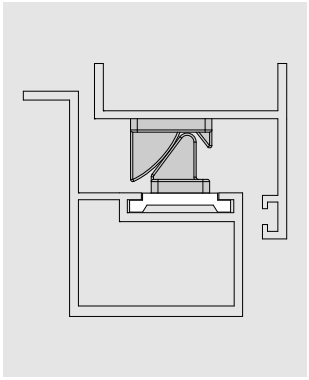
AL



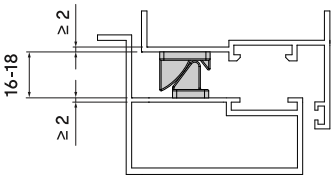
***FLAT** FRAME*

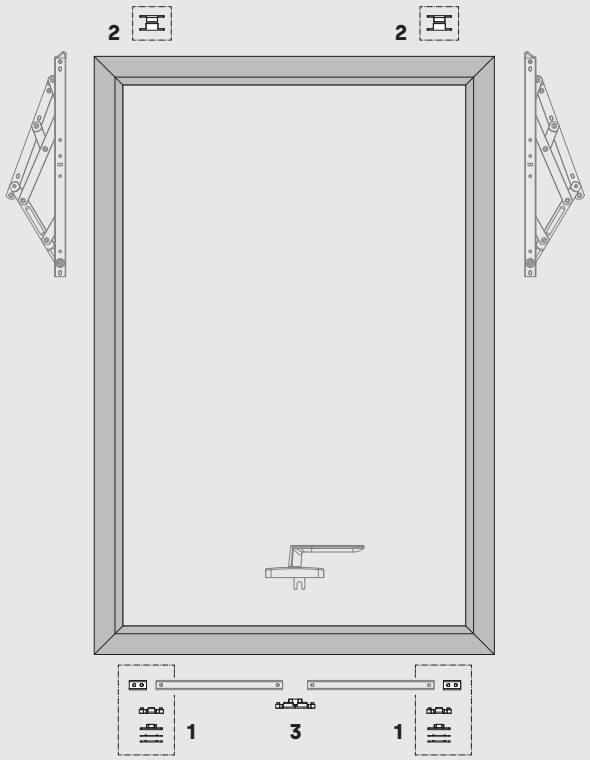


Use aluminium rod to support the sash profile



***EURO GROOVE** FRAME*

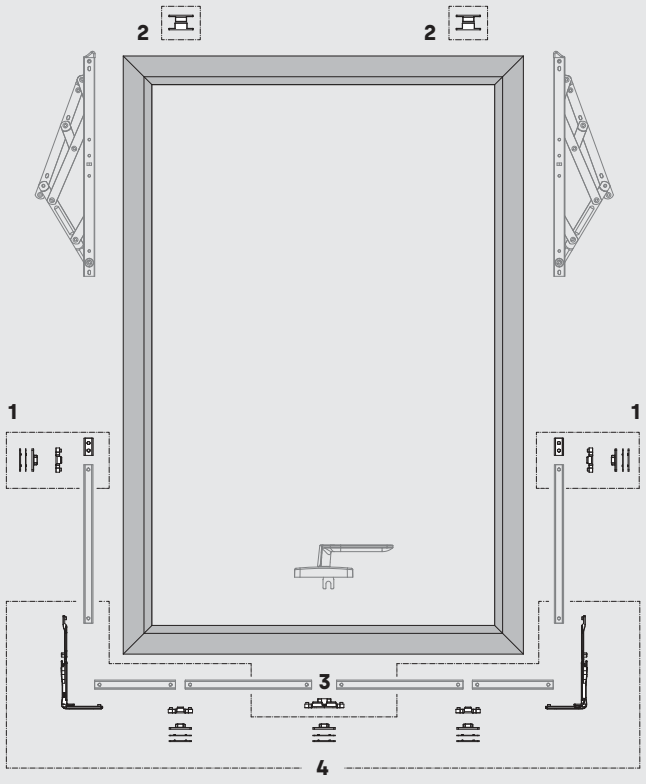




FLAT FRAME
4 POINTS PERIMETER LOCKING

1	183000	**TOP-HUNG LOCKING POINT KIT**	50
2	0518015	**LOCKING POINT (CURTAIN WALL)**	100
3	180100	**TOP-HUNG TRANSMISSION KIT**	500

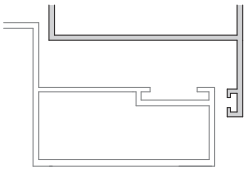
The configuration do not include top-hung friction stays.
Consult compatible friction stay.



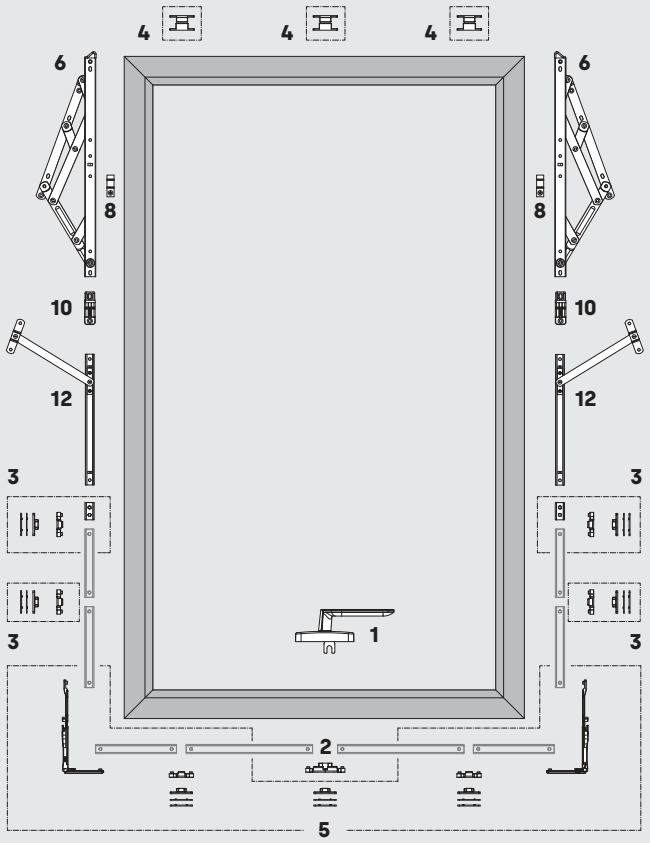
FLAT FRAME
6 POINTS PERIMETER LOCKING

1	183000	**TOP-HUNG LOCKING POINT KIT**	50
2	0518015	**LOCKING POINT (CURTAIN WALL)**	100
3	180100	**TOP-HUNG TRANSMISSION KIT**	50
4	183500	**PERIMETRAL LOCKING KIT**	10

The configuration do not include top-hung friction stays.
Consult compatible friction stay.

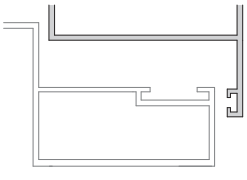


FRAME PROFILE
FLAT (NO GROOVE)

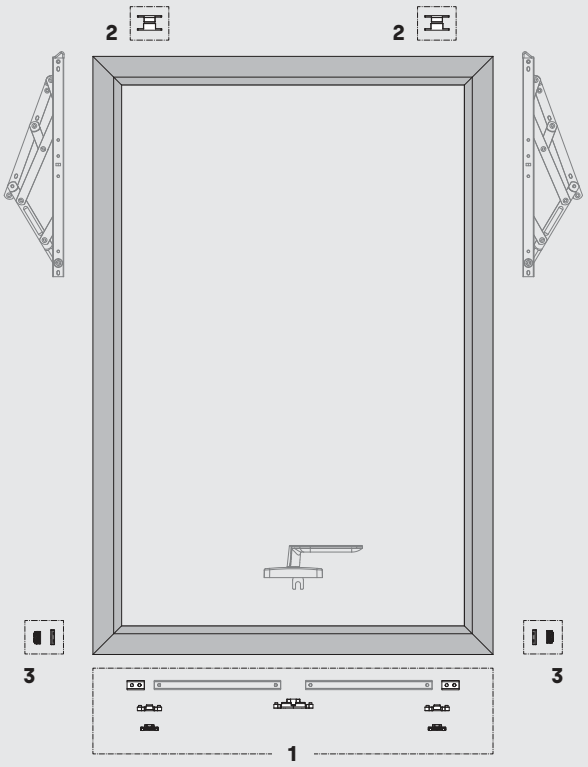


FLAT FRAME
10 POINTS PERIMETER LOCKING

1	**CREMONE** Consult « top-hung opening cremones »		
2	180100	**TOP-HUNG TRANSMISSION KIT**	50
3	183000	**TOP-HUNG LOCKING POINT KIT**	50
4	0518015	**LOCKING POINT (CURTAIN WALL)**	100
5	183500	**PERIMETRAL LOCKING KIT**	10
6	A18011	**CURTAIN WALL FRICTION STAY L = 455**	20
	A18012	**CURTAIN WALL FRICTION STAY L = 705**	
7	18200	**HEIGHT ADJUSTMENT**	100
8	A0518014	**LIMITING PLATE**	200
9	18010	**INOX FRICTION STAY**	10



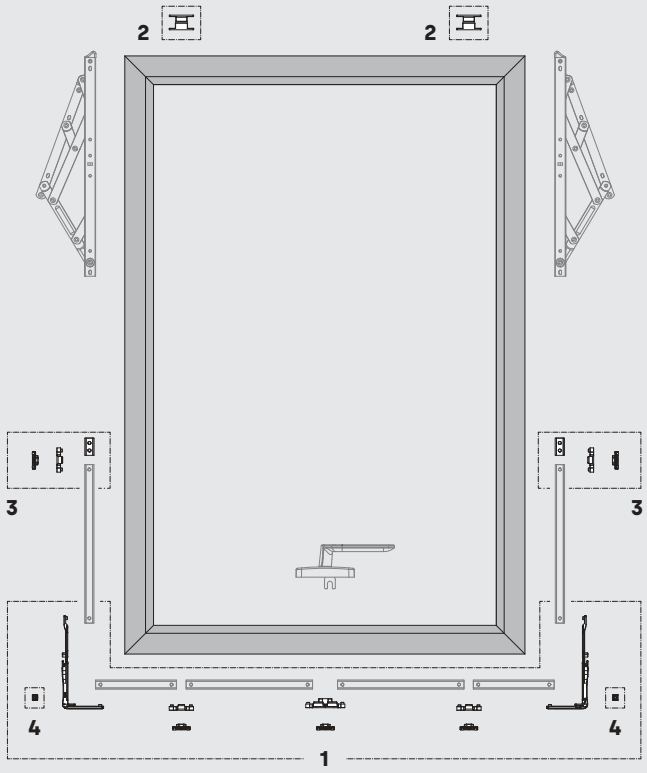
FRAME PROFILE
FLAT (NO GROOVE)



EURO GROOVE FRAME
4 POINTS PERIMETER LOCKING

1	182400	**LOWER LOCKING POINT KIT**	25
2	0518015	**LOCKING POINT (CURTAIN WALL)**	100
3	0710621	**POSITIONING LATCH KIT**	100

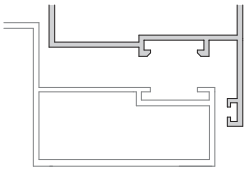
The configuration do not include top-hung friction stays. Consult compatible friction stay.



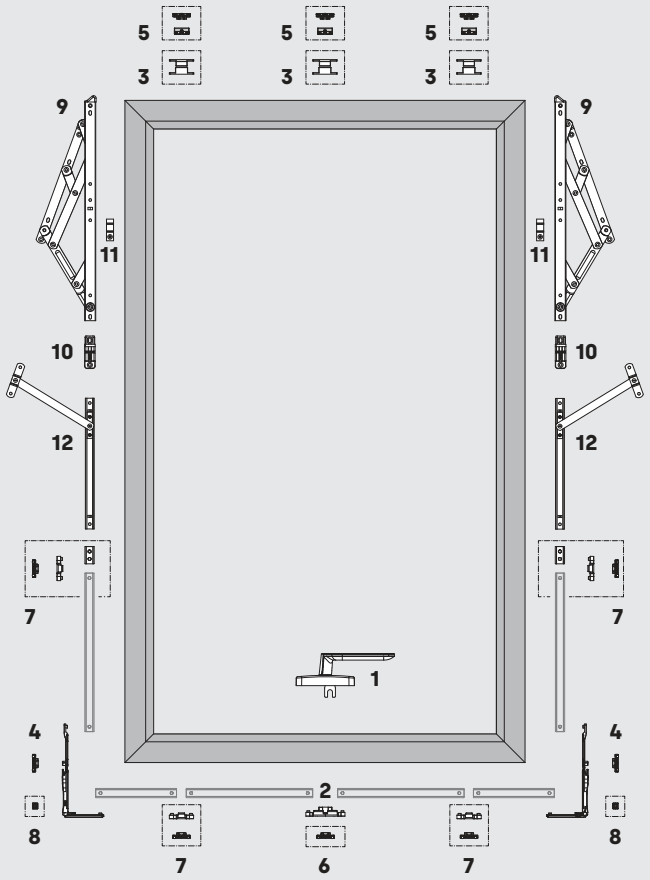
EURO GROOVE FRAME
6 POINTS PERIMETER LOCKING

1	183400	**PERIMETRAL LOCKING KIT**	10
2	0518015	**LOCKING POINT (CURTAIN WALL)**	100
3	113000	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	0517193	**POSITIONING LATCH**	100

The configuration do not include top-hung friction stays. Consult compatible friction stay.

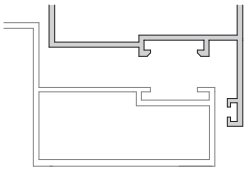


FRAME PROFILE
EURO GROOVE



EURO GROOVE FRAME
10 POINTS PERIMETER LOCKING

1	**CREMONE** Consult « top-hung opening cremones »		
2	180100	**TOP-HUNG TRANSMISSION KIT**	50
3	0518015	**LOCKING POINT (CURTAIN WALL)**	100
4	113420	**HORIZONTAL CORNER DRIVE KIT**	20
5	112500	**CONCEALED LOCKING POINT**	50
6	110200	**ADJUSTABLE LOCKING POINT**	100
7	113000	**SUPPLEMENTARY LOCKING POINT KIT**	50
8	0517193	**POSITIONING LATCH**	100
9	A18011	**CURTAIN WALL FRICTION STAY L = 455**	20
	A18012	**CURTAIN WALL FRICTION STAY L = 705**	
10	18200	**HEIGHT ADJUSTMENT**	100
11	A0518014	**LIMITING PLATE**	200
12	18010	**INOX FRICTION STAY**	10



FRAME PROFILE
EURO GROOVE

12 SLIDING OPENING

254 MINIMALIST handle

256 TITAN flush handle

259 Fixing kits

262 DELTA pull handle

264 Multipoint locks

269 Multipoint locks with key

270 Multipoint locks with hooks

272 Multipoint lock with AM

273 Multipoint with limitation

274 Single-point locks

276 Accessories for multipoint locks

280 TITAN range

284 FENIX range

287 DRACO range

289 Microventilation sliding opening

292 Hooks for recessed handles

294 Keepers for recessed handles

296 LSPLUG adapters

298 HL bogies

304 CLASSIC bogies

minimalist
delta & **gamma**

clic !

Direct installation

Highest durability

RAL colours

Tested robustness

MINIMALIST range

For people who crave a sense of creativity, quality and style in their spaces, STAC introduces the minimalist line of door and window handles to offer a complete range of technical solutions of the highest quality.

» Minimalist

During the design process of the minimalist handles we have done away with all unnecessary elements, developing a hidden mechanism that only exposes the handle.

In this way, we offer technical solutions of great robustness and reliability throughout their service life in accordance with the strictest quality standards.

» Complete range

Our range is compatible with multiple opening systems, with particular emphasis on:

- Windows with side-hung or tilt & turn opening.
- Sliding windows with multipoint lock systems.
- Windows with Euro Groove or Groove 16 profiles.
- Single and double-sash doors with both inward and outward opening.

» Direct installation

The handle is supplied with a positioning and retention mechanism that allows it to be installed directly with a **"click"**, and is just as easy and practical to remove.

» Full range of finishes

There are ever more projects which try to stand out through their used of colour and finishes; the hardware can be painted in the same colour as the aluminium carpentry. Minimalist range is available in both black and white lacquer and the colours that make up the RAL chart.

» Corrosion resistance

The parts that make up the handles have been treated to guarantee resistance to corrosion, exceeding 240 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard.

According to the parameters described in the standard, a grade 4 classification has been obtained, which allows the possibility of using the handles in environments with severe climatic conditions.

Minimalist handles for sliding windows with multipoint lock or systems with 16 groove profile.

- Handles without embellishing escutcheon, with concealed transmission mechanism and guiding system that minimises play.
- They are supplied with floating parts to adapt to different types of profile: with 2 supports (ref. A7711, ref. 7713) or with 4 supports (ref. A7721, ref. 7723).
- In windows with multipoint it is possible to configure the micro-ventilation position.
- Compatible with standard solutions with 7 mm transmission spindle.

DELTA MINIMALIST HANDLE

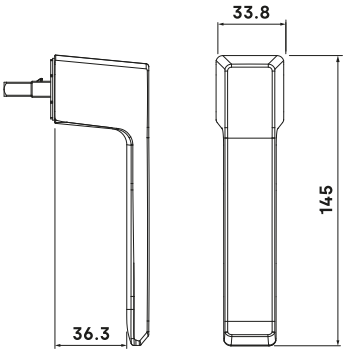
2 supports floating piece

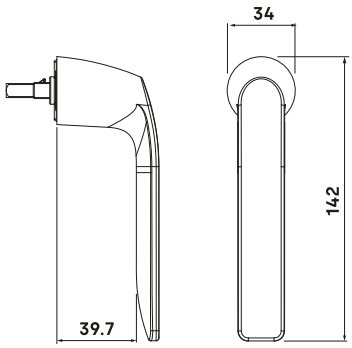
AL ZM AC IN

7

A7711 **

25





GAMMA MINIMALIST HANDLE

2 supports floating piece

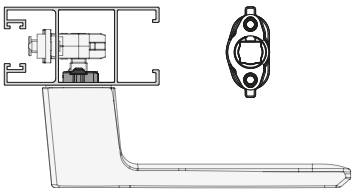
AL ZM AC IN

7

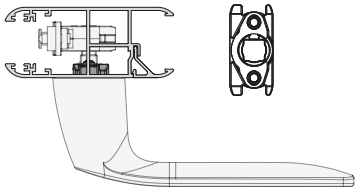
7713 **

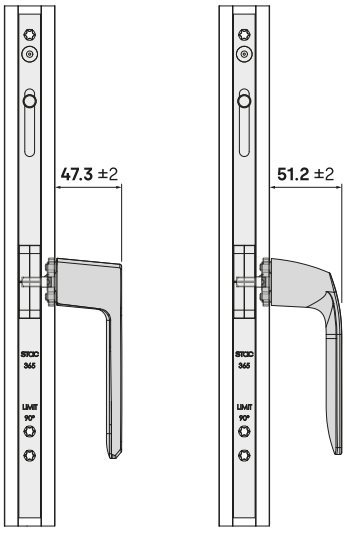
25

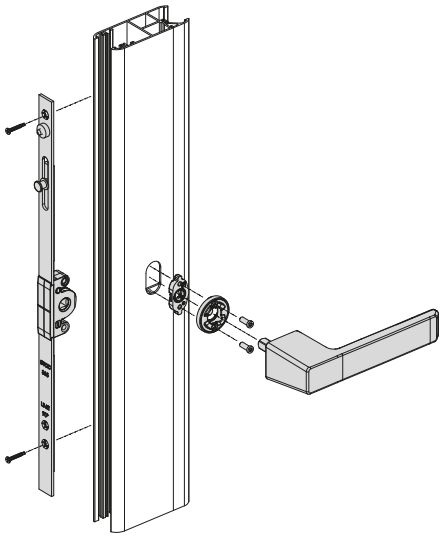
2-SUPPORT *FLOATING PART*



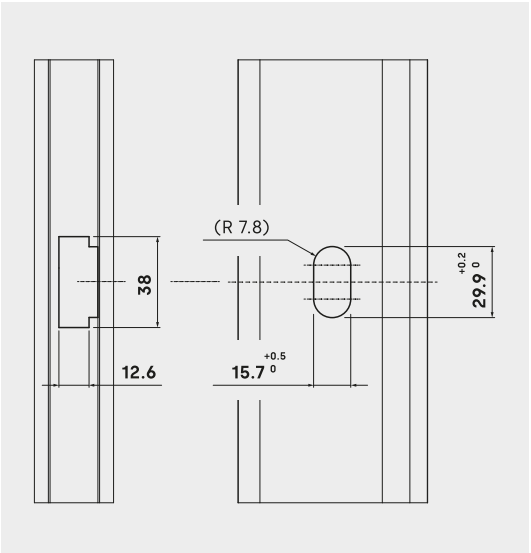
4-SUPPORT *FLOATING PART*







MACHINING PROFILE



TITAN

Flush handles

Crossing sashes

Security system

Direct installation to the profile

Quick and simple assembly

TITAN flush handles

Handles with a design flush with the sash profile that allows modifying the opening system of the sliding window; it makes operation by the user easier, leveraging the unique characteristics of our multipoint locks which allow full, 180° movement of the transmission spindle.

» Aesthetic range

The design has been carefully considered to ensure integration with the other hardware in the TITAN range, keeping their defining characteristics.

- Modern, clean and elegant products.
- Concealed profile fixing system.

» Crossing sashes

The handle for multipoint systems has a design that is flush with the profile of the window leaf and which allows the leaves to pass each other in sliding systems.

A minimum distance of 4 mm between the profiles of the central node is required for the crossing of the sashes.

» Double handle configuration with locking

The TITAN MAX handle allows the installation of a double handle (inside / outside). This configuration, together with the multipoint key locking system, ensures that the window can be securely closed at all times.

The minimum profile section that allows the double handle configuration is 48 millimetres.

» Quick and intuitive installation

As usual in the products we develop, we have given special thought to the professional for whom it is intended. The handles are quick and easy to install.

In this way, the installer can significantly reduce assembly times and better optimise the time spent.

- Machining is easy to carry out.
- Direct installation of the handle base to the profile.
- Plastic covers that conceal the fixing screws.

Handle that allows the crossing of sashes in sliding window systems.

- Available with different spindle lengths: 12 mm (ref. 034430), 15 mm (ref. 034410) and 25 mm (ref. 034420).
- Minimum sash profile width is 32 mm for multipoint with 7.5 mm backset.
- Minimum sash profile width is 28 mm for multipoint with 15 mm backset.
- Compatible with multipoint locks that allow the movement of the transmission spindle from up to 180º.
- The handle maintains the machining dimensions for the recessed handles.

TITAN FLUSH HANDLE

034410 **

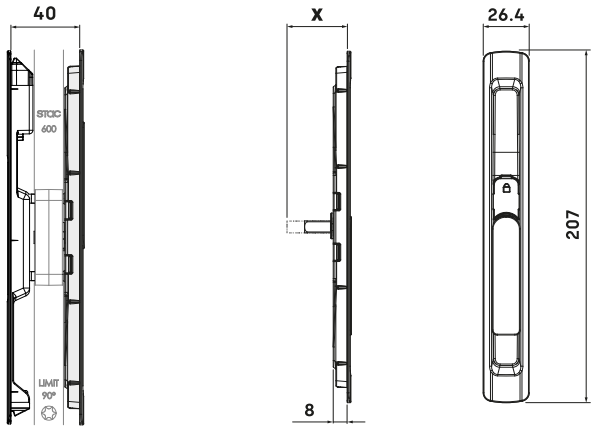
40

ZM IN

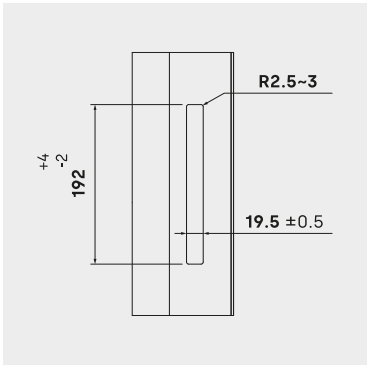
PA

	SPINDLE **DIMENSIONS**	X DIMENSION **LENGTH**	MIN. WIDTH PROFILE **7.5 MULTIPOINT**	MIN. WIDTH PROFILE **15 MULTIPOINT**
034410 **	7 x 15	24.5	34.5	31
034420 **	7 x 25	34.5	34.5	31
034430 **	7 x 12	21.5	34.5	31
034700 **	7 x 20	37.2	-	-

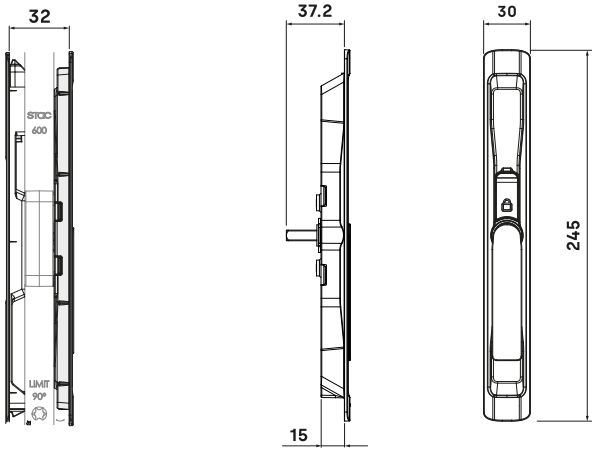
TITAN HANDLE WITH FINGER PULL
MINIMUM SASH THICKNESS
WITH 7.5 MULTIPOINT LOCKING



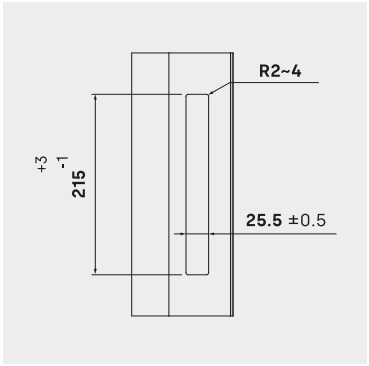
TITAN MACHINING



TITAN HANDLE WITH FINGER PULL
MINIMUM SASH THICKNESS
WITH 15 MULTIPOINT LOCKING



TITAN MAX MACHINING



TITAN MAX FLUSH HANDLE

034700 | **

20

ZM IN PM

Handle for multipoint locks with a flush design that enables the crossing of sashes in sliding window systems.

- Enables the installation of a double interior and exterior handle.
- Fixing kit available for double handle configurations (ref. K0347-01).
- Fixing system to the profile without visible screws. The reference is supplied with quick-fit cover caps to conceal the fixing screws.

The handle is not compatible with microventilation.

TITAN MAX DOUBLE HANDLE KIT

K0347-01

IN

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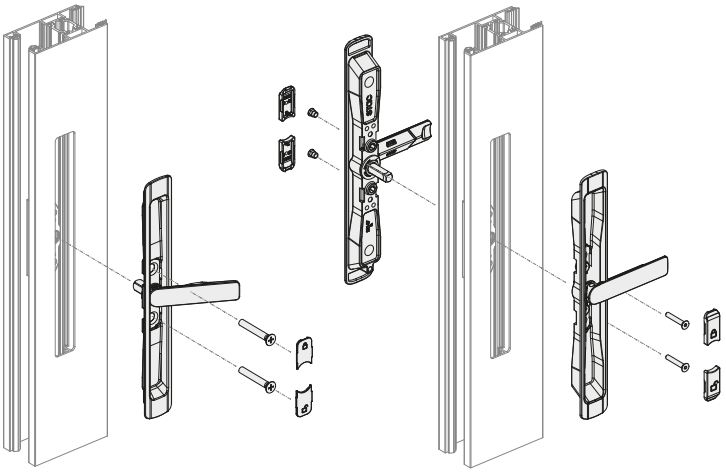
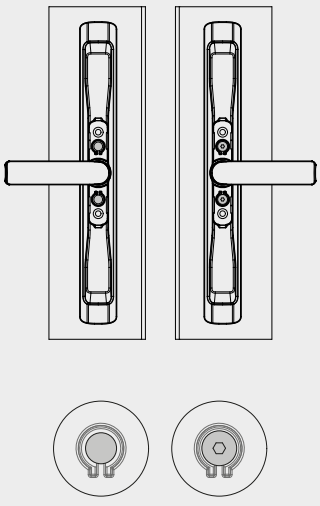
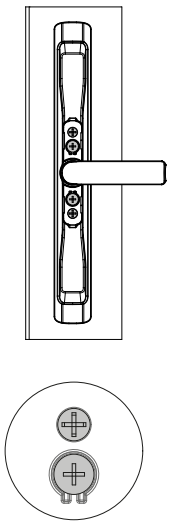


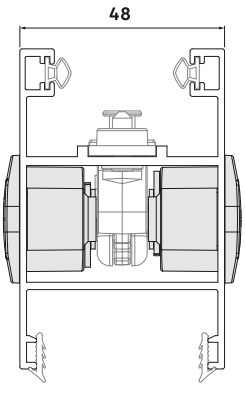


TITAN MAX
DOUBLE CONFIGURATION



TITAN MAX
SINGLE CONFIGURATION

TITAN MAX DOUBLE HANDLE
MINIMUM SECTION



» DOUBLE HANDLE CONFIGURATIONS FIXING KITS

ACCESSORIES	K0771-01	K0771-02	K5771-09	A0507140
POLYAMIDE SHIMS *with fibreglass*	✔	✔	-	-
DIN 965 M4 SCREWS *length (mm)*	L = 80	L = 100	L = 80	L = 65
7 x 7 SPINDLE *length (mm)*	L = 100	L = 130	L = 75	L = 80
M4 CAP NUTS *thread depth (mm)*	P = 13.6	P = 13.6	P = 5.25	P = 13.6

COMPATIBLE HANDLES	K0771-01	K0771-02	K5771-09	A0507140
SIRIUS / **DELTA** *Standard handle*	**J min ~ max** = 26 - 62	**J min ~ max** = 26 - 88	-	-
Fixing screws	**C** = J + 12	**C** = J + 12	-	
Transmission spindle	**C** = J + 32	**C** = J + 32		-
POLARIS *Standard handle*	**J min ~ max** = 30 - 72	**J min ~ max** = 30 - 92	**J min ~ max** = 17 - 55	**J min ~ max** = 30 - 57
Fixing screws	**C** = J + 8	**C** = J + 8	**C** = J + 18	**C** = J + 8
Transmission spindle	**C** = J + 20	**C** = J + 20	**C** = J + 20	**C** = J + 20
SIRIUS PRO / **DELTA PRO** *Standard handle*	**J min ~ max** = 30 - 72	**J min ~ max** = 30 - 92	**J min ~ max** = 17 - 55	**J min ~ max** = 30 - 57
Fixing screws	**C** = J + 4	**C** = J + 4	**C** = J + 14	**C** = J + 4
Transmission spindle	**C** = J + 13	**C** = J + 13	**C** = J + 13	**C** = J + 13
ALTAIR *Standard handle*	**J min ~ max** = 27 - 68	**J min ~ max** = 27 - 88	**J min ~ max** = 26 - 45	**J min ~ max** = 27 - 50
Fixing screws	**C** = J + 11	**C** = J + 11	**C** = J + 21	**C** = J + 11
Transmission spindle	**C** = J + 26	**C** = J + 26	**C** = J + 26	**C** = J + 26
TITAN MAX *Flush handle*	**J min ~ max** = 72 - 115	**J min ~ max** = 72 - 134	**J min ~ max** = 48 - 95	**J min ~ max** = 72 - 99
Fixing screws	**C** = J - 34	**C** = J - 34	**C** = J - 24	**C** = J - 34
Transmission spindle	**C** = J - 22	**C** = J - 22	**C** = J - 22	**C** = J - 22
TITAN FLAP *Pull handle with cam*	**J min ~ max** = 42 - 83	**J min ~ max** = 42 - 103	**J min ~ max** = 25 - 62	**J min ~ max** = 42 - 67
Fixing screws	**C** = J - 4	**C** = J - 4	**C** = J + 6	**C** = J - 4
Transmission spindle	**C** = J - 12	**C** = J - 12	**C** = J + 12	**C** = J - 12
DELTA FLAP *Pull handle with cam*	-	-	**J min ~ max** = 24 - 55	-
Fixing screws	-	-	**C** = J	-
Transmission spindle			**C** = J + 18	

(**J**) Profile width (mm), (**C**) Cutting length (mm)

FIXING KIT FOR DOUBLE HANDLE

K0771-01

100

IN

PA

Fixing kits that allow double handle installations and prevent access from the outside of the window.

• The polyamide shims allow the countersunk head screws to be fitted to SIRIUS or DELTA mechanisms.
• The cap nuts prevent manipulation and disassembly of the hardware and so must be installed on the outside face of the window.

K0771-01

• The fixing kit is supplied with polyamide shims, 7 x 7 x 100 mm transmission spindle, M4 x 80 mm DIN 965 screws and M4 cap nuts.

K0771-02

• The fixing kit is supplied with polyamide shims, 7 x 7 x 130 mm transmission spindle and M4 x 100 mm DIN 965 screws and M4 cap nuts.

Polyamide shims are not required for PRO or POLARIS mechanisms.

FIXING KIT FOR DOUBLE HANDLE

K0771-02

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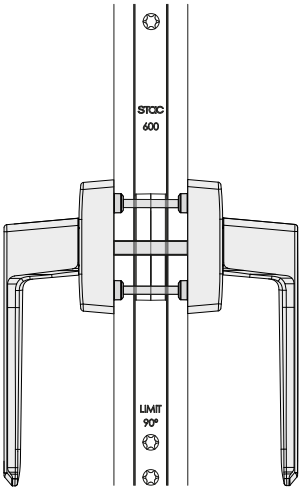
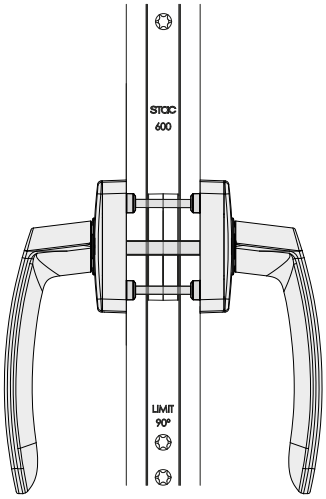
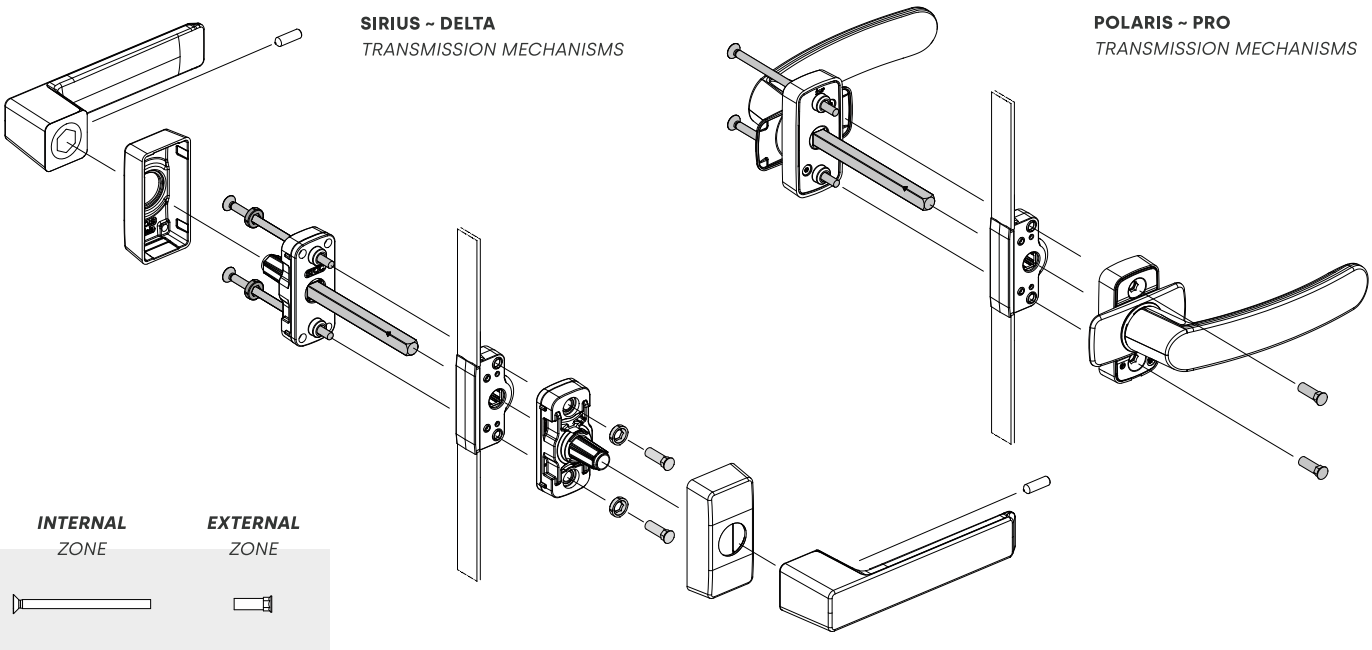




SIRIUS ~ DELTA
TRANSMISSION MECHANISMS

POLARIS ~ PRO
TRANSMISSION MECHANISMS



Fixing kits that allow double handle installations and prevent access from the outside of the window.

• The cap nuts prevent manipulation and disassembly of the hardware and so must be installed on the outside face of the window.

K5771-09

• The fixing kit is supplied with polyamide shims, 7 x 7 x 75 mm transmission spindle, M4 x 80 mm DIN 965 screws and M4 cap nuts.
• Compatible with DELTA FLAP pull handle with cam.

A0507140

• The fixing kit is supplied with polyamide shims, 7 x 7 x 80 mm transmission spindle and M4 x 65 mm DIN 965 screws and M4 cap nuts.

FIXING KIT FOR DOUBLE HANDLE

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K5771-09

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FIXING KIT FOR DOUBLE HANDLE

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A0507140

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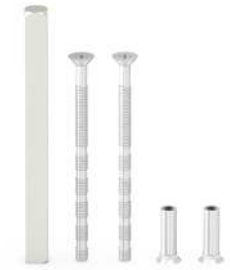
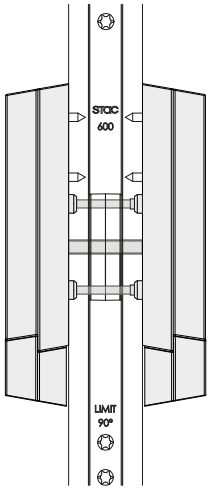
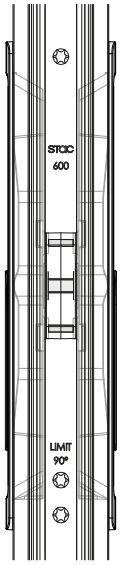
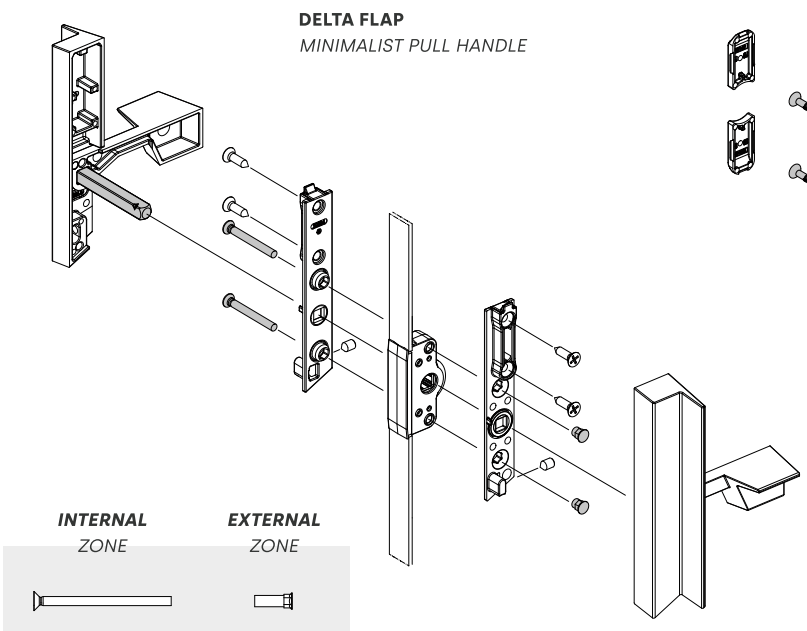


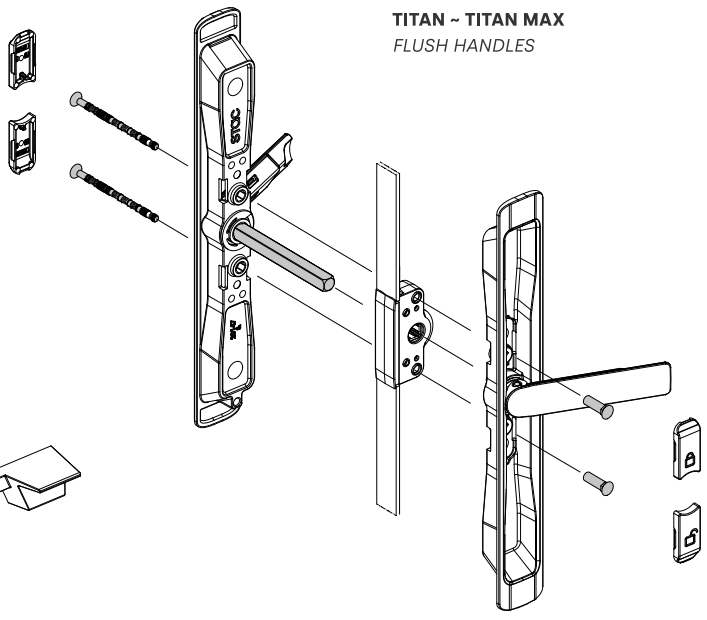




DELTA FLAP
MINIMALIST PULL HANDLE



TITAN ~ TITAN MAX
FLUSH HANDLES



DELTA

MINIMALIST PULL HANDLES

Quick assembly

Minimalist aesthetics

Double configurations

Opening hand

Minimalist DELTA pull handles

Modern architecture seeks to offer simpler spaces, reducing all superfluous elements and leaving only the essential qualities. There is a strong tendency to minimize the visual impact of doors and windows, with regard to both the frames and the accessories.

With this in mind, we have developed handles with reduced dimensions and minimalist aesthetics, with a design in line with the new series of sliding windows on the market.

» Design range

Clean lines never go out of fashion. The design of the pull handles distils the style and character concepts of the DELTA product family.

The restrained dimensions cause less visual impact and allow to adapt both visually and geometrically to any type of sliding system. The base is integrated within the body of the handle allowing the installation base to be hidden, achieving a clean aesthetic without joint lines between the components.

» Now also for wood and PVC systems

We have adapted our minimalist concept to the opening systems for wood and PVC frames, offering both handles for this type of carpentry.

» DELTA FLAP & DELTA FIX pull handles

The FLAP handle, with dynamic cam-operated device, allows single or double configurations in conjunction with the fixed FIX handle.

- Simple installation according to the opening hand.
- Interior ~ exterior double installation in balconies with exterior access and possibility of combining with multipoint lock with key to leave the sliding securely closed.
- Combined installation by means of handle with cam (on the inside) and fixed handle (outside area).

» Useful and functional products

Reduced assembly times quick and easy installation of the handle via a counter-piece and lateral fixation by means of a grub screw.

Total stability of the handle assembly guaranteed by the front fixing of the screws, both to the leaf frame profile and to the system's multipoint lock.

Firm, smooth and fluid operation with maximum positions marked at 0° and 90° of cam rotation.

Lever-action handle that allows the opening of large sliding doors.

- The lever-operated handle significantly reduces the effort required to move the locking points.
- Valid for openings with multi-point locking, both keyed and keyless.
- Allows double handle configurations for internal and external opening.
- Adaptable to any multipoint lock with 7 millimetre spindle.

The lever handle is not compatible with the micro-ventilation position.

DELTA FLAP PULL HANDLE

2001	D I	**

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ZM

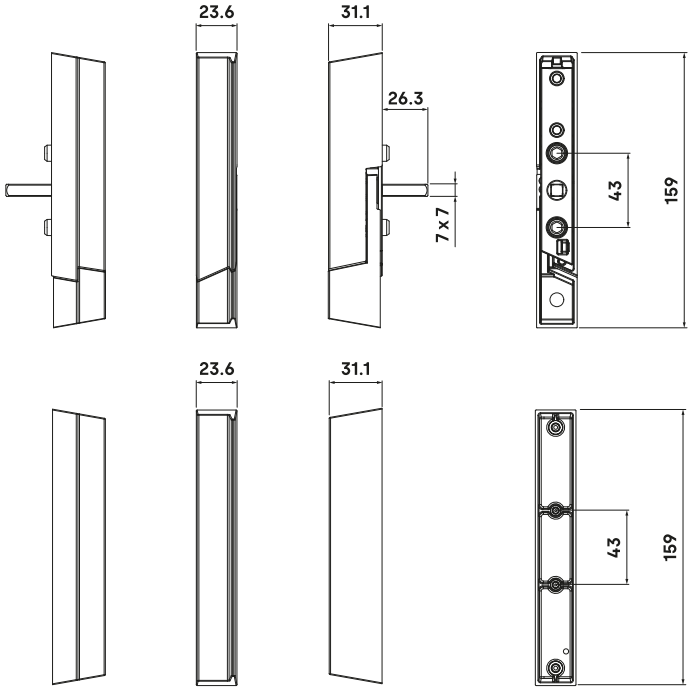
PM

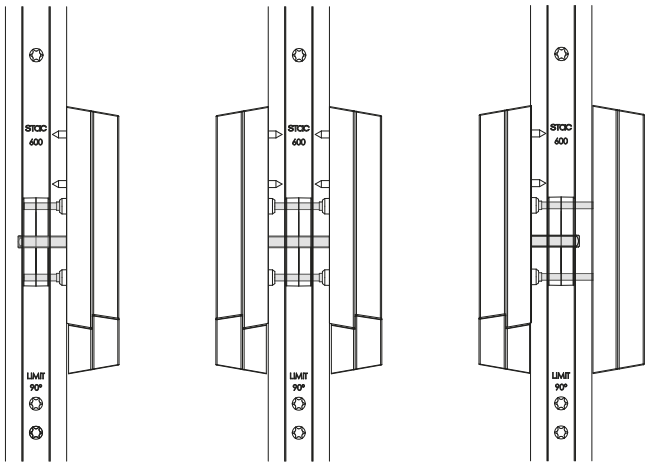
DELTA FIX PULL HANDLE

2002	**

5

ZM





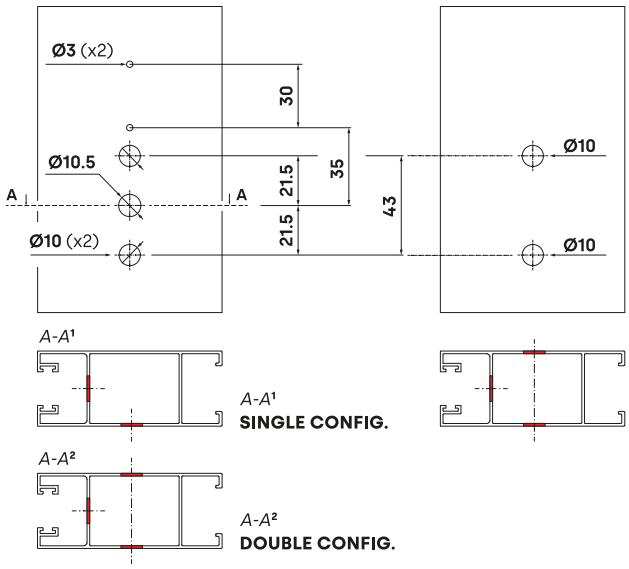
FLAP PULL HANDLE *CONFIGURATION*

FLAP DOUBLE PULL HANDLE *CONFIGURATION*

FLAP PULL HANDLE + FIX *CONFIGURATION*

DELTA FLAP *machining*

DELTA FIX *machining*



MULTIPOINT LOCKS

180°

Handle movement

Security system

Areation position

Quick & simple assembly

Multipoint locks

Multipoint locks are special locks that provide greater security and watertightness in sliding or balcony doors compared to standard locks thanks to the different locking points distributed along the system according to the height of the sash.

By turning the handle, we can simultaneously block the locking or anchoring points. With the sash closed, the multipoint provides greater rigidity to the structure, increasing its resistance to possible unwanted access attempts.

» High-performance technical solutions

- Available with 7.5, 15 and 30 millimetre backset to suit different profile series.
- Different sizes to fit a wide range of sash sizes (from 400 to 2000 millimetres).
- Compatible with all standard or flush handles with 7 mm transmision spindle.
- Exclusive mechanism that discriminates three hardware positions in a 180° handle movement, enabling the airing position by using the 0520052 keeper. *
- Anti-lifting system that prevents the sash from disassembling when it is in the closed position.
- Damping stop that prevents the sash from hitting directly against the metal frame.

* *Not available in the ECO line of multipoint locks.*

» Multipoint locks with key

Systems specially designed for balcony doors with access to the outside and which guarantee a high level of security thanks to the insertion of the cylinder with key.

» Multipoint locks with hooks

The slotted frames allow the locks to be embedded in the window profile so that they do not protrude, maintaining the clean aesthetics of the sliding opening system.

By turning the handle 180° on straight handles and positioning the strikers in the frame in the right way, it is possible to reduce the distance between the cross profiles, significantly increasing the leaf opening.

» Multipoint locks with anti-false manoeuvre system

It prevents movement of the handle when the window sash is open. The system only allows the hardware to be operated when the sash is fully closed., avoiding misuse of the locking mechanism by preventing the locking points of the leaf from colliding with the frame keepers.

» Multipoint lock with 30 backset

The new multipoint lock has a novel internal mechanism which makes the forces of turning and sliding independent, thereby allowing a much smoother operation.

Zinc plated steel or AISI 304 stainless steel multipoint locks with 7.5 mm backset for sliding windows.

- Available in different sizes: 600, 1000 and 1600 mm.
- They are available with aeration opening (keeper ref. 0520052 installation necessary).
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.
- Specific references in AISI 304 stainless steel (ref. P0520).

MULTIPOINT LOCKS 180º B7.5
Zinc plated steel

M0520

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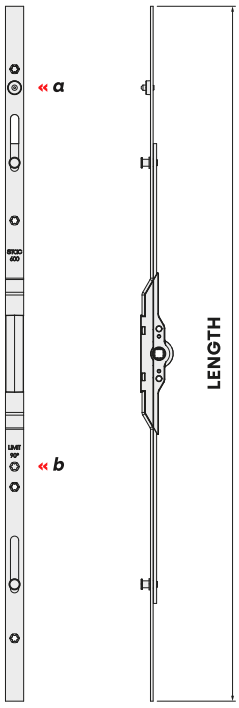
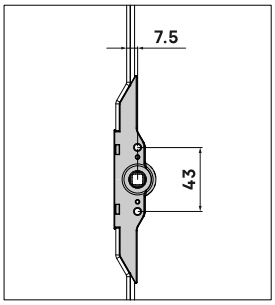
MULTIPOINT LOCKS 180º B7.5
AISI 304 stainless steel

P0520

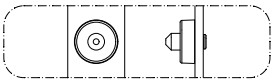
10

ZM AC IN

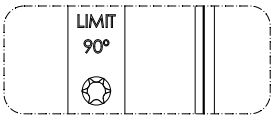




*Shock absorber stop and anti-lifting device (**a**)*

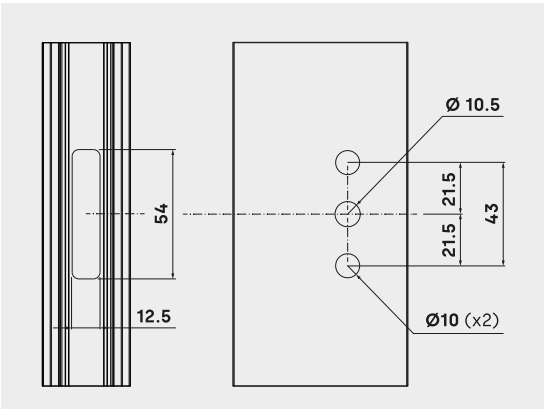


*90º turn limitation system (**b**)*



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05200600	7.5	600	2	4	750	10
M05201000		1000	2	6	1150	10
M05201600		1600	4	6	1750	10
P05200600	7.5	600	2	4	750	10
P05201000		1000	2	6	1150	10
P05201600		1600	4	6	1750	10

PROFILE MACHINING



MULTIPOINT LOCKS 180º B15
Zinc plated steel

M0521

10

ZM AC IN

MULTIPOINT LOCKS 180º B15
AISI 304 stainless steel

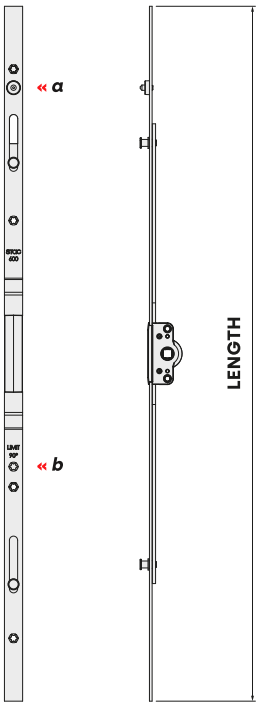
P0521

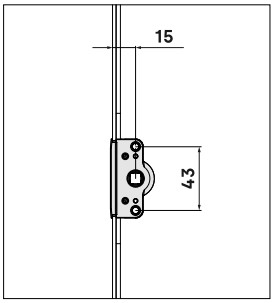
10

ZM AC IN

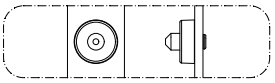
Zinc plated steel or AISI 304 stainless steel multipoint locks with 15 mm backset for sliding windows.

- Available in different sizes: 600, 1000, 1600 and 2000 mm.
- They are available with aeration opening (keeper ref. 0520052 installation necessary).
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.
- Specific references in AISI 304 stainless steel (ref. P0521).

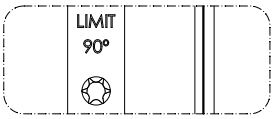




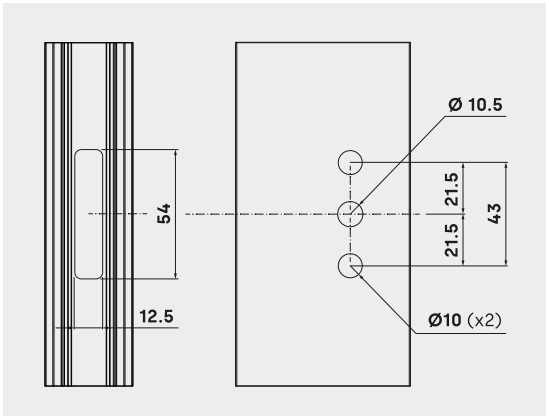
*Shock absorber stop and anti-lifting device (**a**)*



*90º turn limitation system (**b**)*



PROFILE MACHINING



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05210600	15	600	2	4	750	10
M05211000		1000	2	6	1150	10
M05211600		1600	4	6	1750	10
M05212000		2000	7	8	2150	10
P05210600	15	600	2	4	750	10
P05211000		1000	2	6	1150	10
P05211600		1600	4	6	1750	10

Zinc plated steel multipoint locks with 30 mm backset for sliding windows.

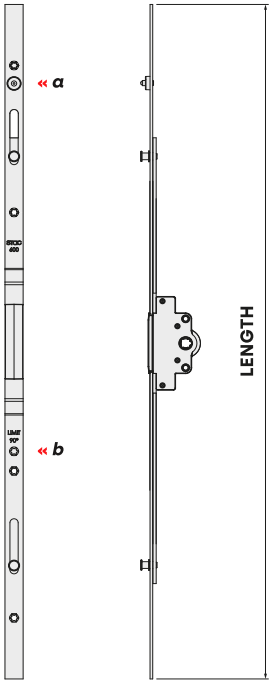
- Available in different sizes: 600, 1000 and 1600 mm.
- They are available with aeration opening (keeper ref. 0520052 installation necessary).
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.

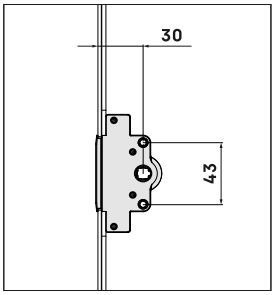
MULTIPOINT LOCKS 180º B30
Zinc plated steel

A0522

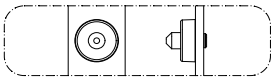
8

ZM AC IN

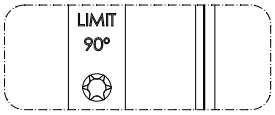


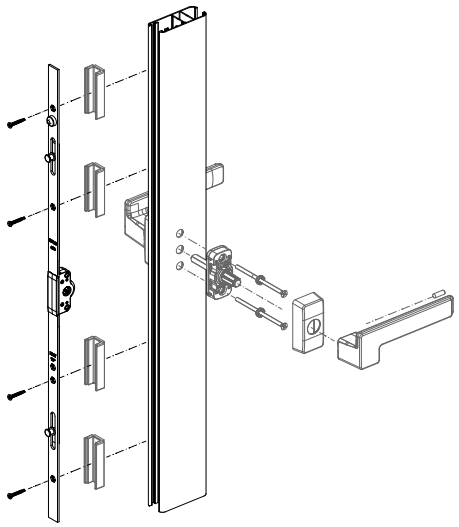


*Shock absorber stop and anti-lifting device (**a**)*



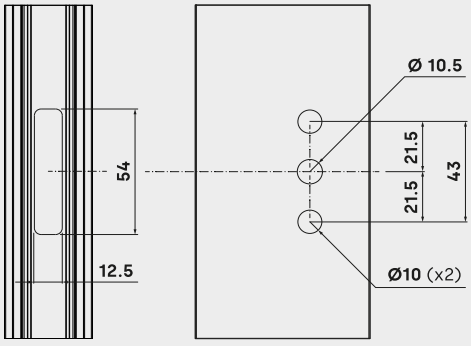
*90º turn limitation system (**b**)*





	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
A05220600	30	600	2	4	750	8
A05221000		1000	2	6	1150	8
A05221600		1600	4	6	1750	8

PROFILE MACHINING



MULTIPOINT LOCKS 90° B7.5
Zinc plated steel

S0520

20

ZM AC IN

MULTIPOINT LOCKS 90° B15
Zinc plated steel

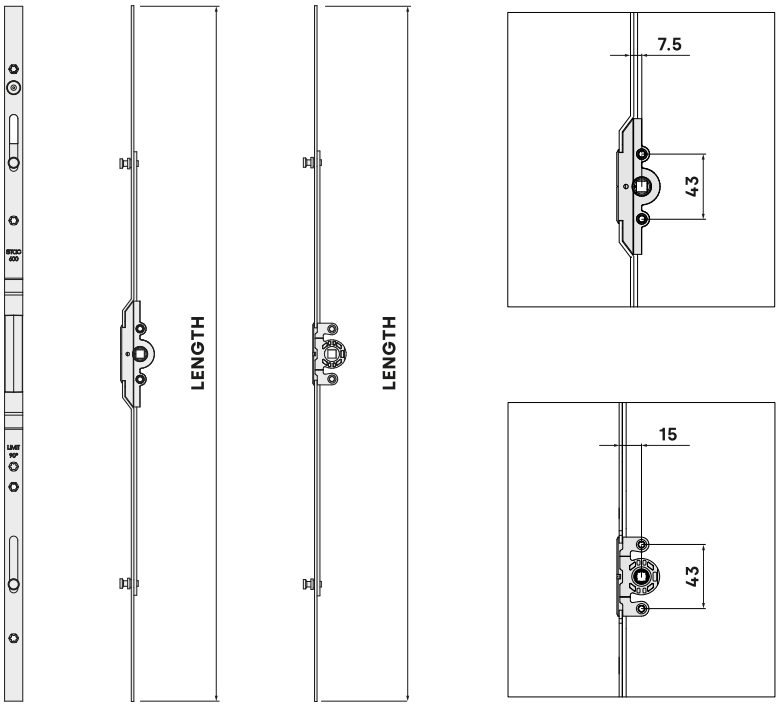
S0521

20

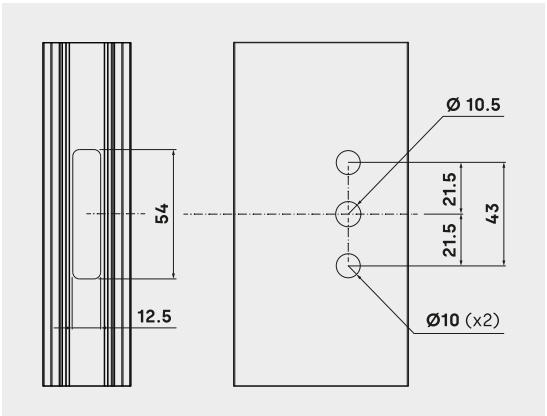
ZM AC IN

Zinc plated steel multipoint locks with 7.5 mm backset for sliding windows.

- Available in different sizes: 400, 600, 1000 and 1600 millimetres.
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.
- 90° opening: the multipoint locks are not compatible with TITAN, TITAN MAX flush handles or with the micro-ventilation position.



PROFILE MACHINING



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
S05200400	7.5	400	2	4	550	20
S05200600		600	2	4	750	20
S05201000		1000	3	7	1150	20
S05201600		1600	3	7	1750	20
S05210400	15	400	2	4	550	20
S05210600		600	2	4	750	20
S05211000		1000	3	7	1150	20
S05211600		1600	3	7	1750	20

Zinc-plated steel or stainless steel multipoint locks with 15 millimetre backset and lock for european cylinder with long cam for sliding windows.

- Multipoint locks designed for balcony windows with access to the street.
- Available in different lengths: 1600 and 1800 millimetres.
- Anti-lifting system that prevents the sash from disassembling when in closed position.
- Damping stop that prevents the leaf from hitting directly against the frame.

Stainless steel multipoint locks (Ref. P0523)

- Stainless steel multipoint locks offering high corrosion resistance for high salinity environments.

BLOCK MULTIPOINT LOCKS 180º B15

M0523

10

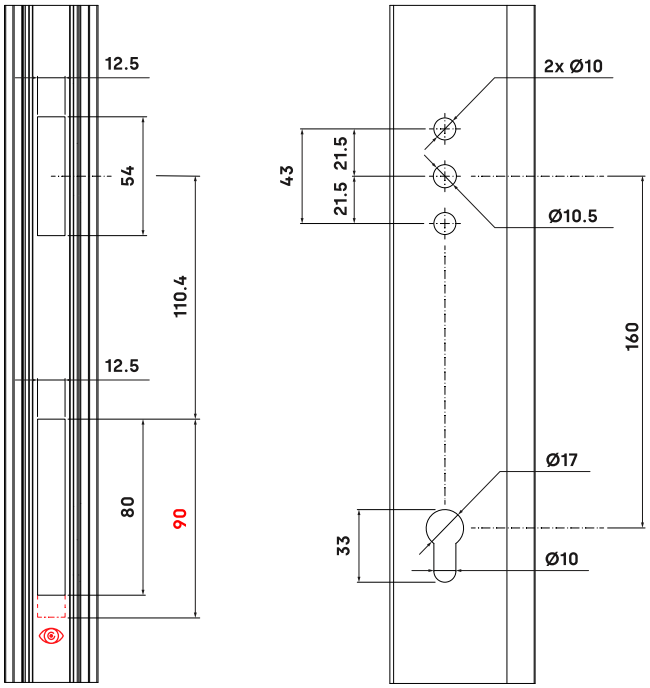
ZM AC IN

BLOCK MULTIPOINT LOCKS 180º B15

P0523

10

ZM AC IN

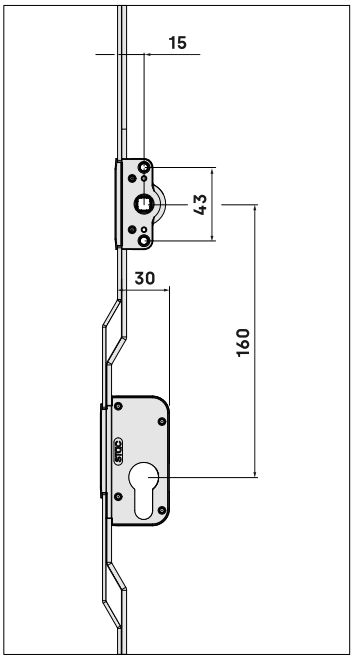


Additional machinning for multipoints with key with discontinued references

	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05231600	15	1600	3	7	1750	10
M05231800		1800	4	9	1950	10
P05231600	15	1600	3	7	1750	10
P05231800		1800	4	9	1950	10
05231600	15	1600	3	10	1750	-
05231800		1800	4	10	1950	-

discontinued

discontinued



MULTIPOINT LOCKS WITH HOOKS 180° B7.5
Zinc-plated steel

M0527

10

ZM AC IN

MULTIPOINT LOCKS WITH HOOKS 180° B15
Zinc-plated steel

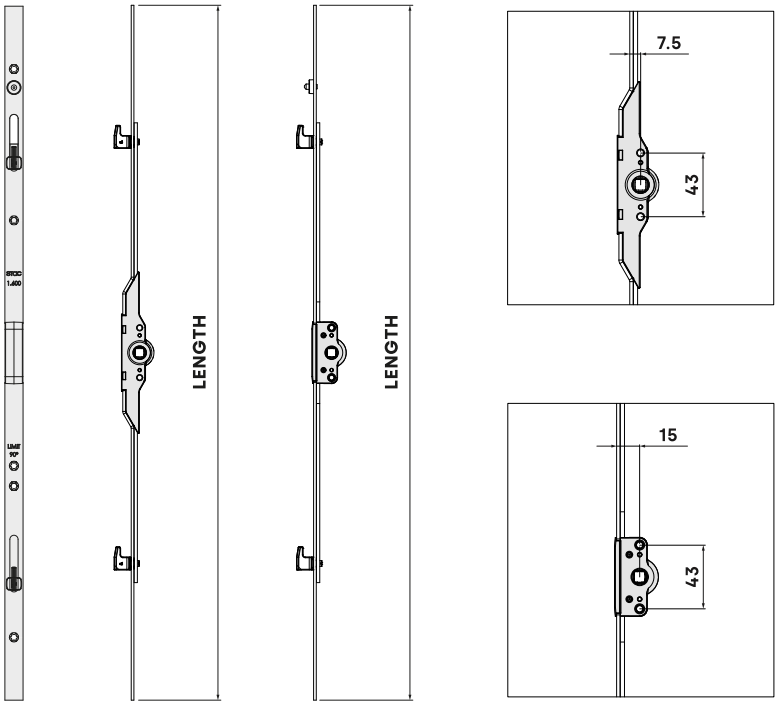
M0528

10

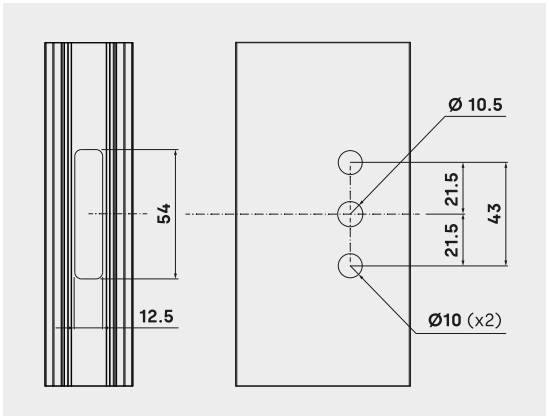
ZM AC IN

Zinc-plated steel multipoint locks with 7.5 millimetre backset and hook-shaped locking points for sliding windows.

- Available in different sizes: 600, 1000 and 1600 mm.
- Suitable for sliding doors with grooved frame profile series.
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.



PROFILE MACHINING



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05270600	7.5	600	2	4	750	10
M05271000		1000	2	6	1150	10
M05271600		1600	4	6	1750	10
M05280600	15	600	2	4	750	10
M05281000		1000	2	6	1150	10
M05281600		1600	4	6	1750	10

Zinc-plated steel multipoint locks with 7.5 millimetre backset and hook-shaped locking points for sliding windows.

- Available in different sizes: 600 and 1600 mm.
- Suitable for sliding doors with grooved frame profile series.
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.

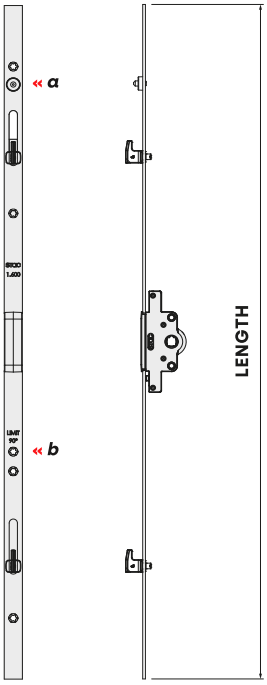
MULTIPOINT LOCKS WITH HOOKS 180º B25

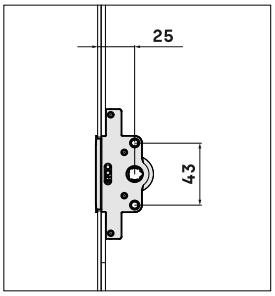
Zinc-plated steel

ZM AC IN

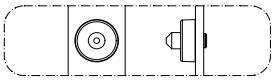
0526

8

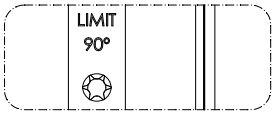




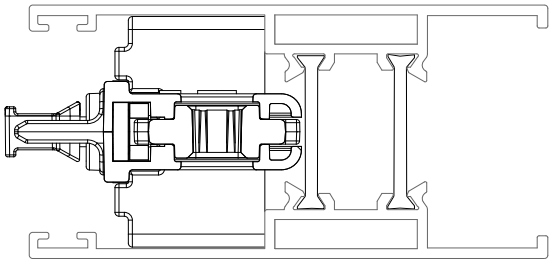
*Shock absorber stop and anti-lifting device (**a**)*



*90º turn limitation system (**b**)*

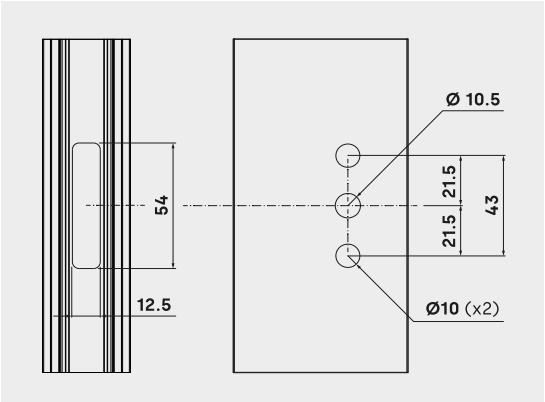


RANURED FRAME PROFILE



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
05260600	25	600	2	4	750	8
P05261600		1600	4	6	1750	8

PROFILE MACHINING



MULTIPOINTS ANTI FALSE MANOEUVRE 180° B15

Zinc-plated steel

M0529

10

ZM AC

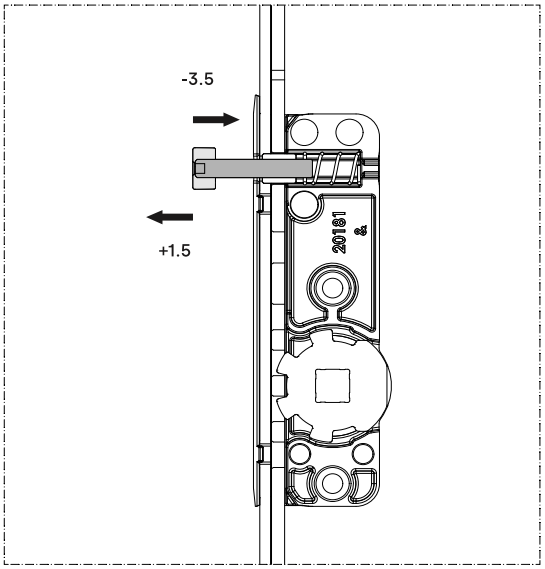
IN

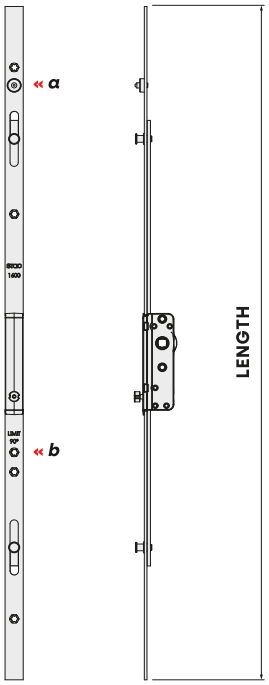
Zinc-plated steel multipoint locks with 15 mm backset and "anti-false manoeuvre" system for sliding windows.

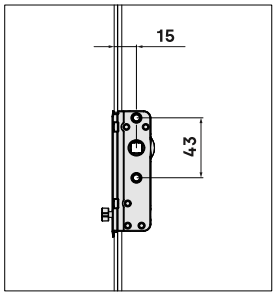
- "Anti-false manoeuvre" system that prevents movement of the handle when the window sash is open. Allows the hardware to be operated only when the sash is fully closed.
- Available in different sizes: 600, 1000 and 1600 mm.
- Device integrated in the gearbox itself, avoiding additional machining on the sash.
- Available plastic shims kit (ref. 0520184) that facilitates the installation in configurations where the multipoint keeper has an adapting shim.
- Equipped with anti-lifting system that prevents the sash from disassembling when in closed position.

Multipoint locks are not compatible with the micro-ventilation position.

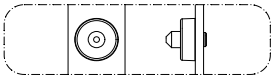
ADJUSTMENT -3.5, +1.5



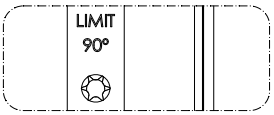




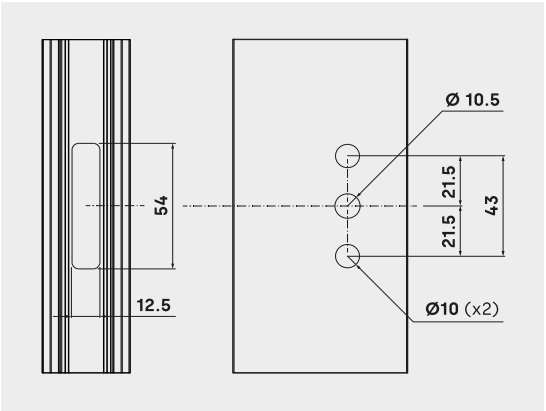
*Shock absorber stop and anti-lifting device (**a**)*



*90° turn limitation system (**b**)*



PROFILE MACHINING



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05290600	15	600	3	4	750	10
M05291000		1000	4	7	1150	10
M05291600		1600	4	7	1750	10

Multipoint lock with 15 mm spindle with limiting device for sliding windows.

- Available in various lengths: 1000 mm (ref. L05311000) or 1600 mm (ref. L05311600).
- Offers an airing position by limiting the opening of the leaf.
- The opening sequencing is performed by turning the handle and the different locking positions provided by the cylinder.
- Equipped with anti-lift system which prevents removal of the sash when in the closed position.
- Damping stopper to prevent the leaf from hitting directly against the metal frame.

MULTIPOINT LOCK WITH LIMITATION B15

ZM AC PA

L05311000

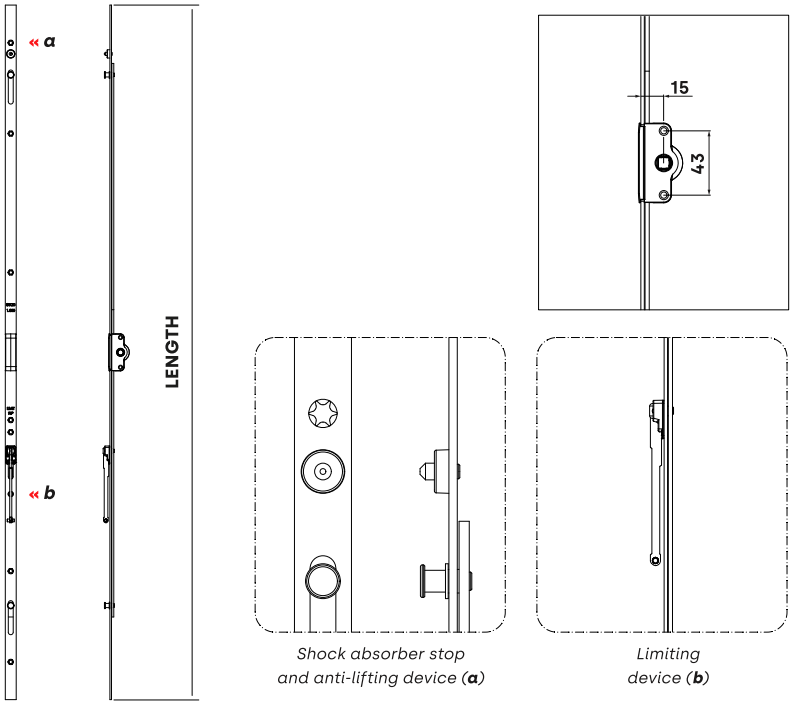
10

MULTIPOINT LOCK WITH LIMITATION B15

ZM AC PA

L05311600

10

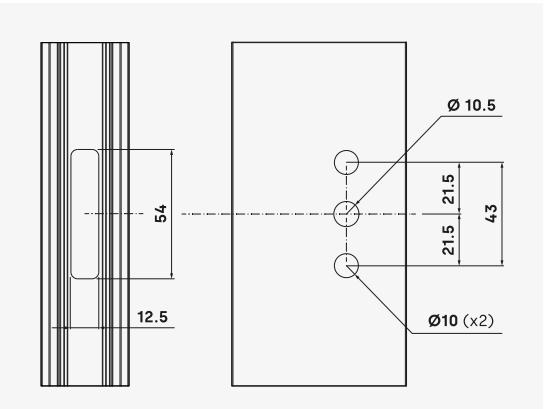


*Shock absorber stop and anti-lifting device (**a**)*

*Limiting device (**b**)*

	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
L05311000	15	1000	2	7	1150	10
L05311600	15	1600	4	9	1750	10

PROFILE MACHINING



SINGLE POINT LOCK B15 90°

Zinc-plated steel

05240230

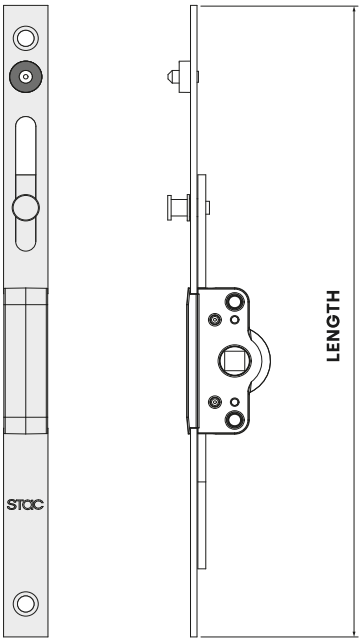
10

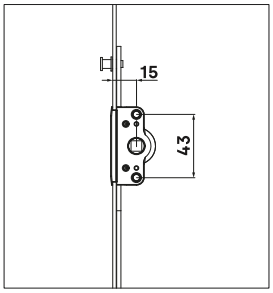
ZM AC IN

Zinc-plated steel single-point lock with 15-millimetre backset for sliding windows.

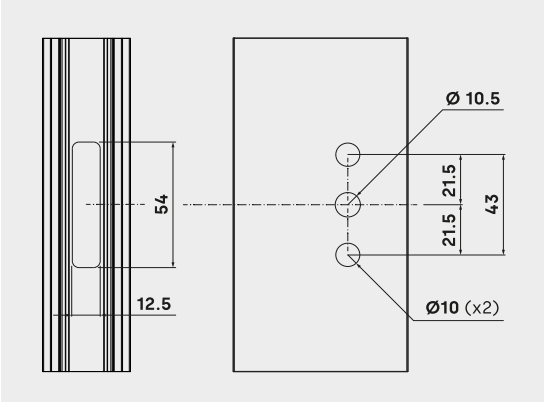
- Available in 230-millimetre length.
- Anti-lifting system that prevents the sash from dismantling when it is in the closed position.
- Damping stop that prevents the sash from hitting the frame directly.

The single-point lock is not compatible with the micro-ventilation position.





PROFILE MACHINING



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
05240230	15	230	1	2	445	10

Zinc-plated steel single-point lock with 7.5 or 15 millimetre backset for sliding windows.

- Available in 7.5 backset with a length of 380 millimetres and 15 backset with a length of 365 millimetres.
- Compatible with the micro-ventilation position (installation of the keeper ref. 0520052 is necessary).
- Anti-lift system that prevents the sash from disassembling when in the closed position.
- Absorber top to prevent the sash from hitting the frame directly.

SINGLE POINT LOCK B7.5 180º

Zinc-plated steel

M05250300

10

ZM AC IN

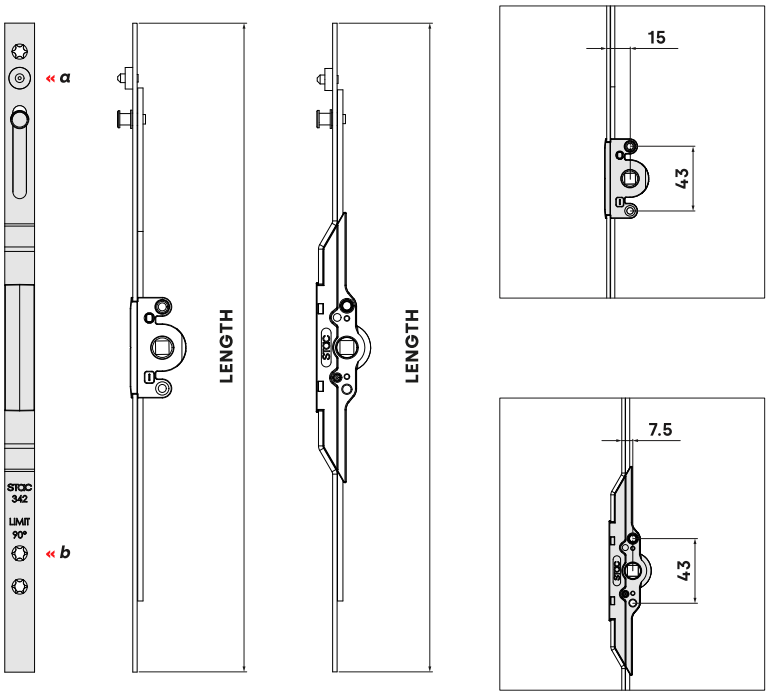
SINGLE POINT LOCK B15 180º

Zinc-plated steel

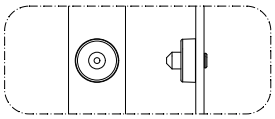
M05240300

10

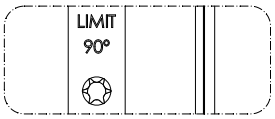
ZM AC IN



*Shock absorber stop and anti-lifting device (**a**)*

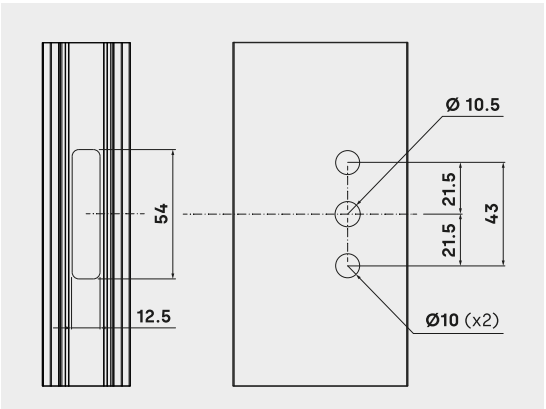


*90º turn limitation system (**b**)*



	BACKSET	LENGTH	LOCKING POINTS	SHIMS	MINIMUM SASH HEIGHT	
M05240300	15	365	1	2	445	10
M05250300	7.5	380	1	2	520	10

PROFILE MACHINING



MULTIPOINT LOCK KEEPER

A0520001

IN

400

Stainless steel keepers for multipoint locks.

- Supplied in AISI 304 stainless steel.
- Fixing to the profile by means of self-tapping screws (not included).
- Plastic shims are available to adapt to different profile series.

Standard keeper (ref. A0520001)

- Valid for sliding window series with standard groove.

EVO keeper (ref. 0520170)

- Conical geometry facilitates the guiding of the multipoint locking point.
- Suitable for open or grooved groove series as well as for series with integrated guide.
- It allows to absorb possible misalignments that may occur between the sash profile, the multipoint lock and the frame profile.

EVO MULTIPOINT KEEPER

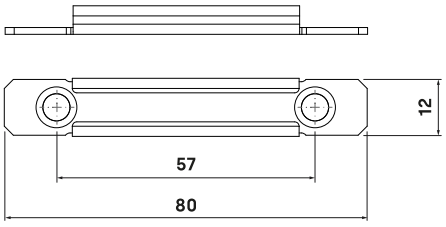
0520170

IN

400

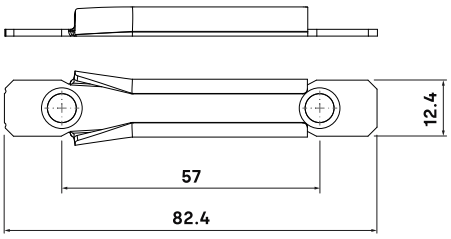
A0520001

MULTIPOINT KEEPER

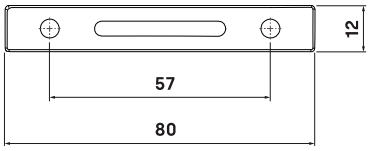


0520170

EVO MULTIPOINT KEEPER

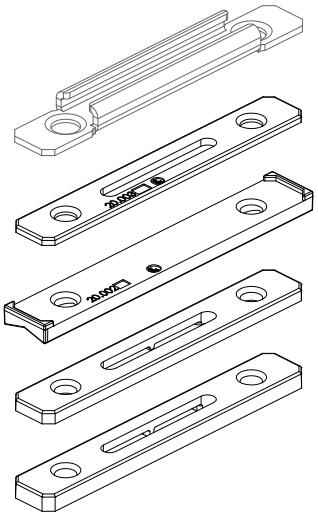


ADAPTATION SHIMS



		HEIGHT	MATERIAL	
0520003		1.7	ABS	400
A0520002		2	ABS	400
0520004		3	ABS	400
0520005		4	ABS	400

References A0520001 / 0520170 do not include adaptation shims



The self-mounting fixings allow the installation of multipoint keepers directly into groove or open groove frame profiles.

- Made of special polymer materials that provide high resistance when fastening, but the sufficient flexibility to allow them to be clipped in place.
- Supplied with DIN 7982 3.9 x 9.5 mm screws.
- The self-mounting fixing allow a certain degree of adjustment.

* The self-mounting fixings are not universal, they depend on the shape of the groove or channel of the profile, so it is necessary to request a prior study from the technical department.

SELF-MOUNTING FIXING

PA

B0520195

100

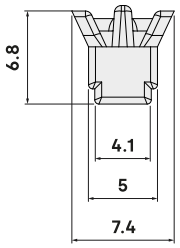
SELF-MOUNTING FIXING

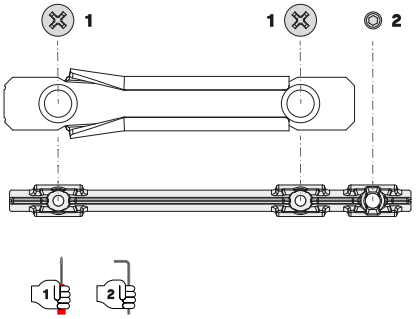
PA

B0520179

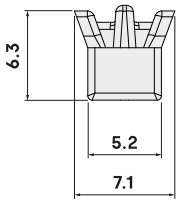
100

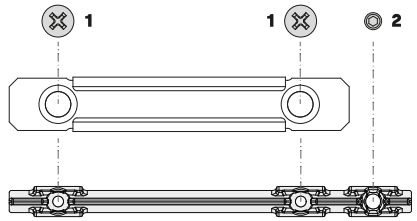
B0520195

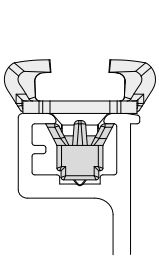


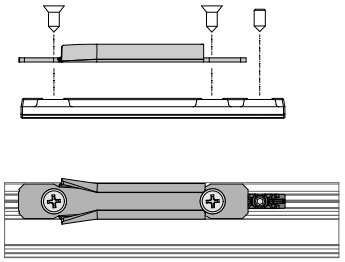


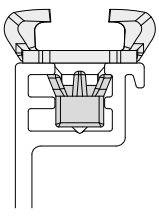
B0520179

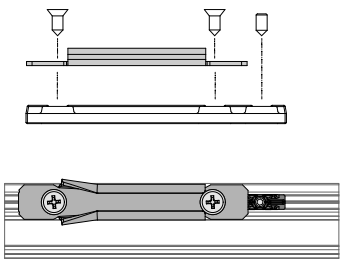












		HEIGHT	MATERIAL	
B0520195		6.8	PA6 POLYAMIDE	100
B0520179		6.3	PA6 POLYAMIDE	100

MICRO-VENTILATION KEEPER

A0520052

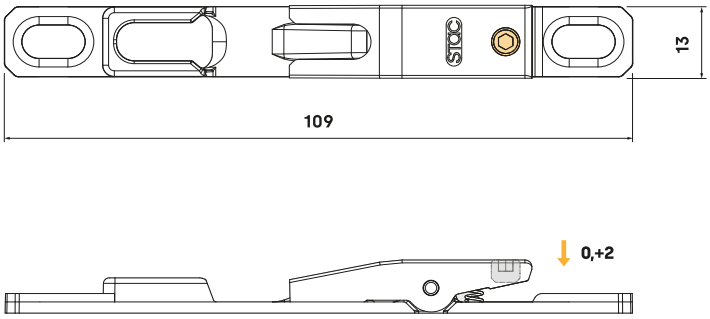
20

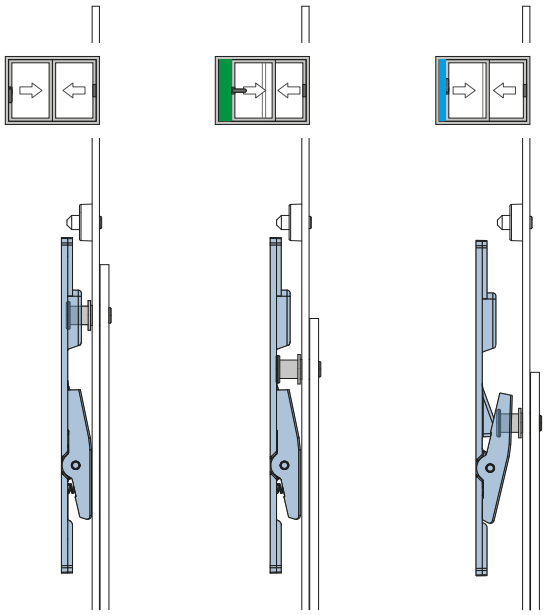
ZM

IN

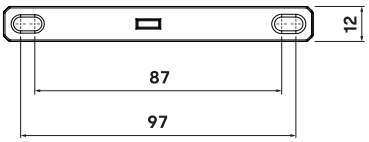
Keeper for multipoint lock that allows the micro-ventilation position.

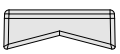
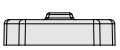
- Class 1 air permeability in opening position (UNE-EN 12207).
- 3 ventilation positions with 4 mm opening with adjustment up to 6 mm.
- Includes plastic shims to adapt to different series of the profile.



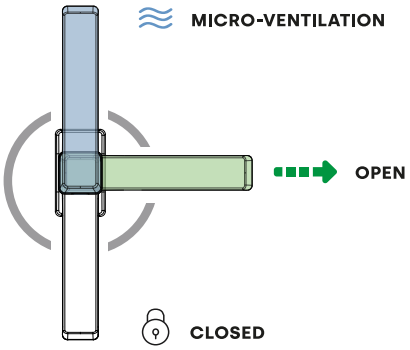


ADAPTATION SHIMS



		HEIGHT	MATERIAL	
A0520052		1.7	ABS	1
		2	ABS	1
		3	ABS	1
		4	ABS	1

Reference A0520052 includes adaptation shims

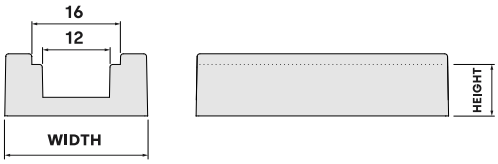


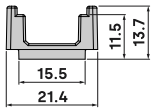
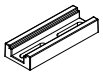
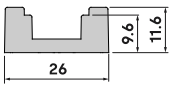
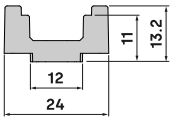
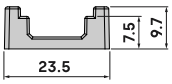
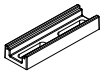
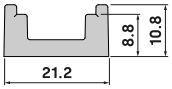
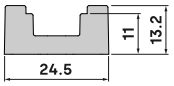
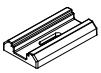
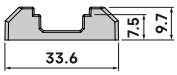
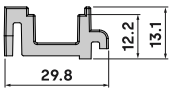
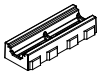
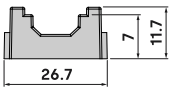
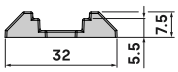
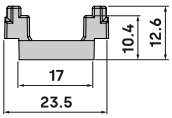
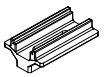
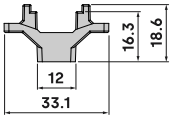
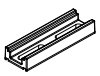
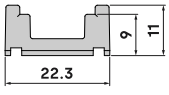
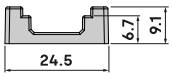
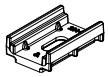
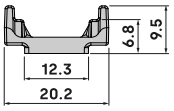
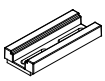
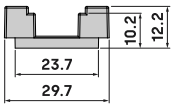
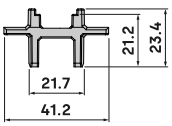
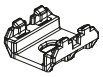
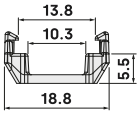
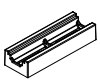
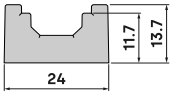
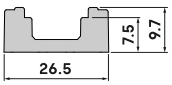
Shims to adapt the multipoint locks to the different series on the market.

• Shims manufactured in ABS to provide rigidity and stability to the fixation.

SHIMS FOR MULTIPOINT LOCK

CALZOS



0599595			600	0520009			400	0520016			400
0599596			1000	0520010			400	0520017			400
0599597			600	0520011			400	0520049			500
0599675			1000	0520012			400	0520050			400
0520006			400	0520013			400	0520146			400
0520007			400	0520014			400	0520198			400
0520008			400	0520015			1000				

TITAN

Flush handles & pull handles

Quick installation

Simple adjustments

No visible screws

RAL colour chart

TITAN range

We have evolved our range of flush handles, flush locks and handles for sliding windows with modern designs but above all by incorporating new solutions.

» Updated aesthetics

Updating the design lines and providing cleanliness have been the aesthetic objectives set when developing the TITAN line, obtaining a modern and elegant product line that does not leave any fixing system in sight.

» Simple installation

Automatic assembly. Both the flush locks and the ffinger pull are equipped with an automatic assembly system that allows for a hole length dispersion of ±3 mm, as well as wider flanges that conceal any machining errors.

Automatic - manual operation. By simply removing a spring, the transition from automatic to manual actuation of the lock has been made as easy as possible.

» Safety systems

Anchoring system. The anchoring system is moved from the bottom to the top. This makes the removal of the lock less intuitive, as the locks are operated by pressing downwards.

Retention system. The retention system with front access prevents the window lock from being removed after installation.

Recessed handle with and whithout key. For customers looking for increased security, a version with a key-operated lock to lock the window opening is available.

» Ergonomics

Product ergonomics were an important factor during the design process, eliminating uncomfortable ridges for users and maximising the pushing area without compromising on design.This all contributes to significantly increasing the intrinsic product quality perceived by the user.

Self-assembly recessed handle that allows the opening of sliding windows.

- Allows the crossing and overlapping of the sashes.
- Retaining system by means of a grub screw to prevent disassembly.
- Hook adjustment by means of direct access from the front of the lock.
- Assembled for automatic operation, it is possible to switch to manual function by removing a spring.

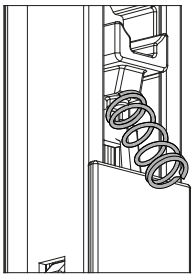
TITÁN BLOCK Recessed handle (ref. 034100)

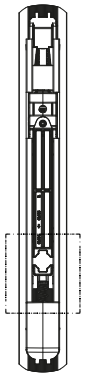
- Recessed handle with locking and unlocking by key.

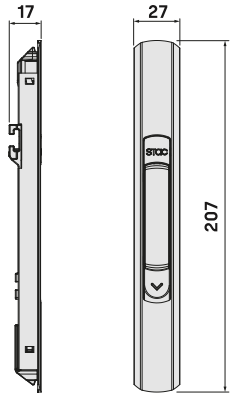
TITÁN finger pull (ref. 0342000)

- Quick fixing to the profile by pressing on the upper part of the finger pull.

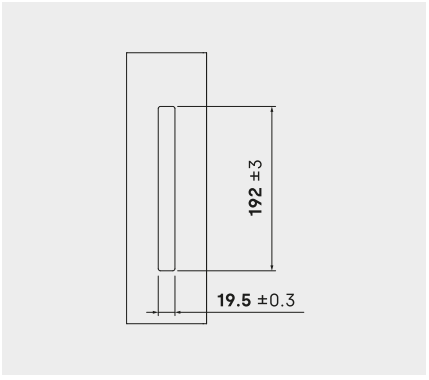
RELEASE THE REAR SPRING TO SWITCH FROM AUTOMATIC TO MANUAL OPERATION

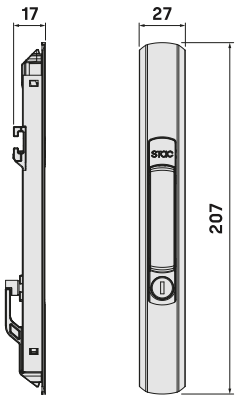






TITAN RECESSED HANDLE MACHINING

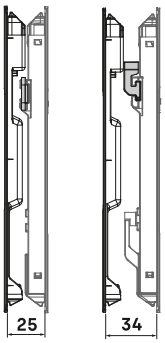


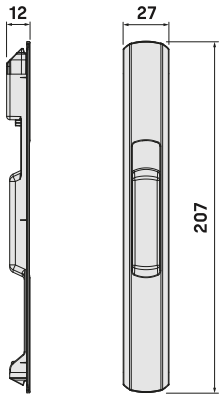


RECESSED HANDLE WITH FINGER PULL
MINIMUM SASH THICKNESS

WITH FRONT HOOK
25 mm

WITH PERIMETER HOOK
34 mm





TITAN RECESSED HANDLE

034000 **

40

TITAN BLOCK RECESSED HANDLE

034100 **

40

TITAN FINGER PULL

0342000 **

50

TITAN BASIC PULL HANDLE

034500 **

10

AL ZM IN PM

Pull handle that allow the opening of sliding windows.

- It is recommended for use as exterior hardware on large sliding windows.

TITAN BASIC pull handle (ref. 034500)

- The pull handle is valid for any type of opening.
- Front installation of the pull handle without any pre-machining of the profile.

TITAN pull handle (ref. 034300)

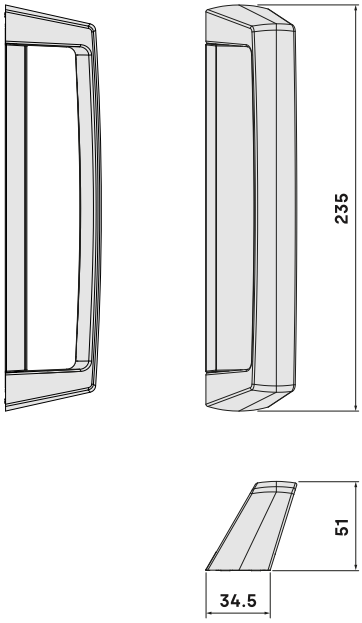
- The pull handle is always used in combination with a recessed handle.
- Fully integrated design that prevents the vision of the fixing screws.
- Fixing system to the profile by means of self-tapping screws.

TITAN PULL HANDLE

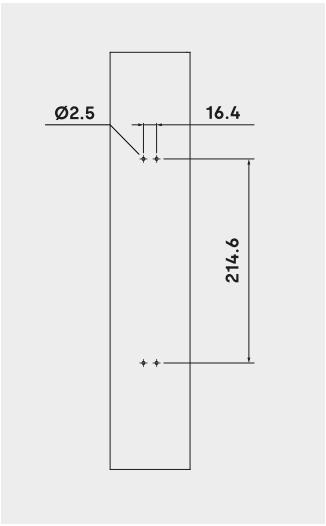
034300 **

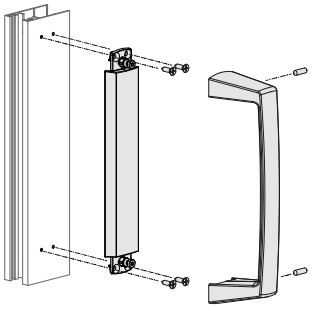
10

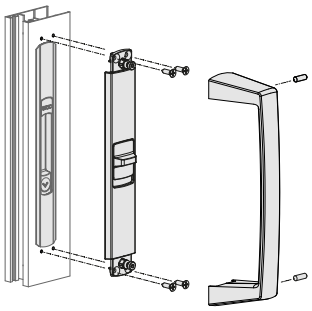
AL ZM IN PA

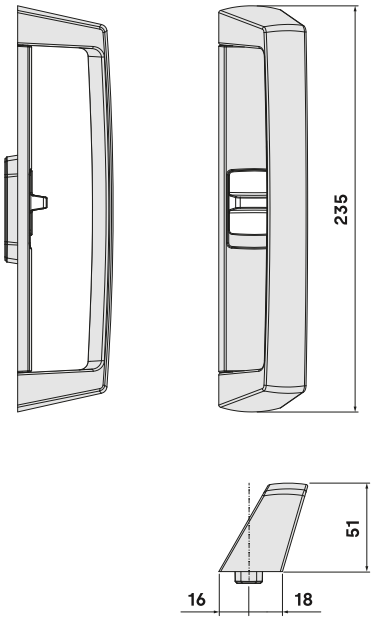


TITAN BASIC MACHINING

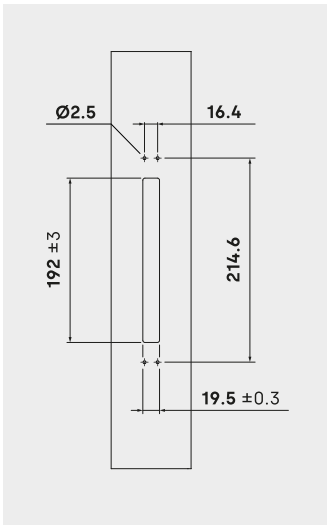








TITAN MACHINING



Pull handle with a cam drive that allows the opening and closing of large sliding windows.

- Compatible with multipoint locks with and without key.
- The cam-operated handle significantly reduces the effort required to move the closing points.
- Allows double handle configuration on the window (internal / external opening).
- Adaptable to any multipoint lock with 7 mm square.

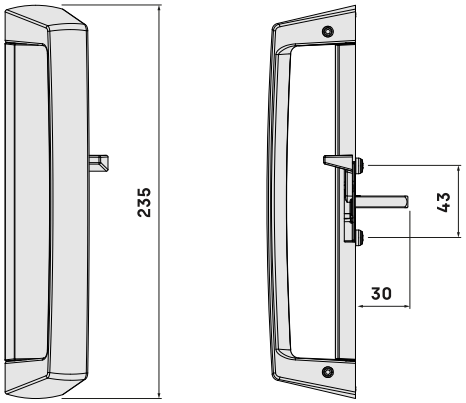
TITAN FLAP pull handle is not compatible with the micro-ventilation position.

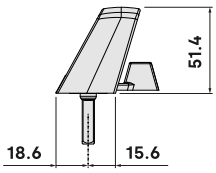
TITAN FLAP PULL HANDLE
Right-hand opening

034800 **

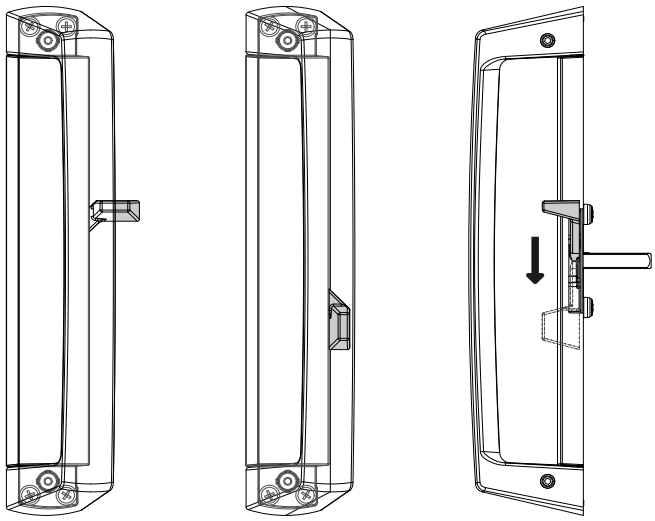
10

IN ZM PM





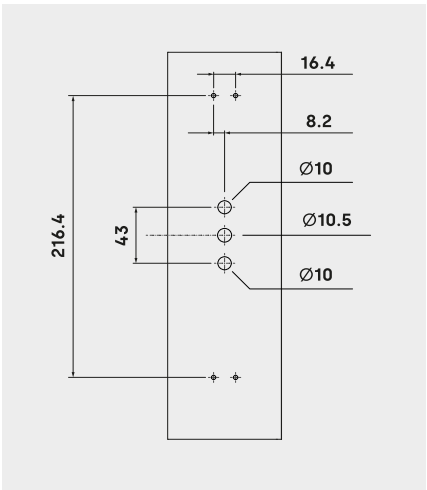
	DESCRIPTION
034800 **	TITAN FLAP R with cam for right-hand openings
034900 **	TITAN FLAP L with cam for left-hand openings



OPENING POSITION

CLOSING/BLOCK POSITION

TITAN FLAP MACHINING



FENIX RECESSED HANDLE
Automatic closing

032000 **

40

AL ZM IN PA

Self-assembly recessed handles and finger pull that allow the opening, the crossing and overlapping of the sliding windows sashes.

FENIX automatic recessed handle (ref. 032000)

- Quick fixing to the profile by pressing on the lower part of the lock with no need to use screws.
- Lock with locking device for manual operation available (ref. 032000C)
- Damping system in the nib that allows it to descend when the sash is closing, with the push button remaining static

FENIX finger pull (ref. 032100)

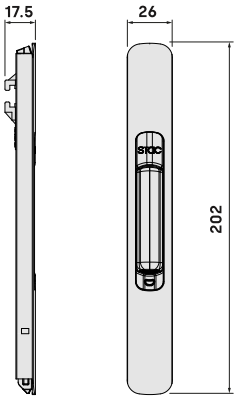
- Quick fixing to the profile by pressing on the upper part.
- Optionally the finger pull can be fixed with screws from the inside for more security.

FENIX RECESSED HANDLE
Manual closing

032000C **

40

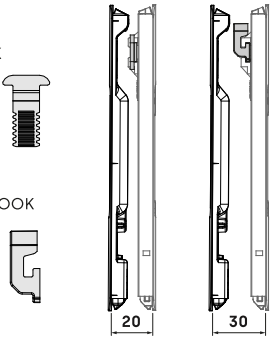
AL ZM IN PM



RECESSED HANDLE WITH FINGER PULL
MINIMUM SASH THICKNESS

WITH FRONT HOOK
20 millimetres

WITH PERIMETER HOOK
30 millimetres

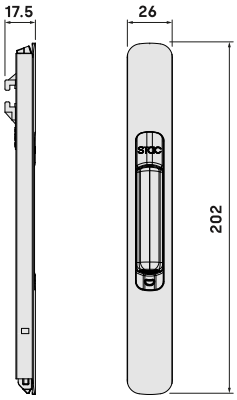


FENIX FINGER PULL

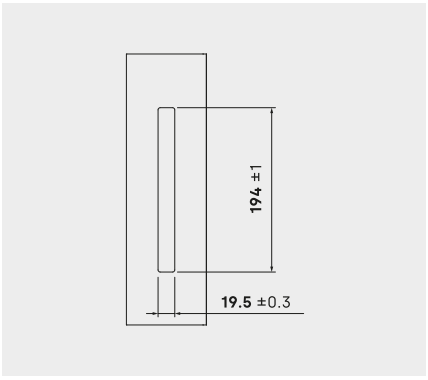
032100 **

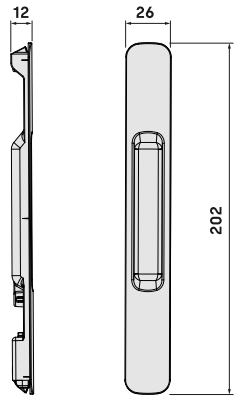
50

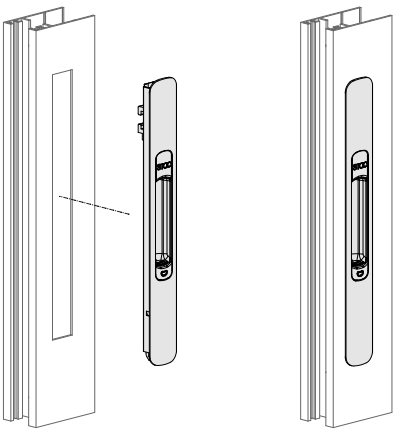
ZM PA



FENIX RECESSED HANDLE MACHINING







Self-assembly recessed handles and finger pull that allow the opening, the crossing and overlapping of the sliding windows sashes.

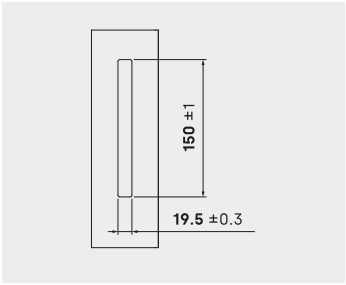
- Quick fixing to the profile by pressing on the lower part of the lock with no need to use screws.
- High tolerance in the machining of the profile thanks to its fixing system.
- By default, the recessed handles work in automatic mode and can be set to manual mode.

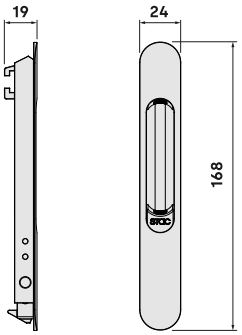
FENIX MINI RECESSED HANDLE

B03300	**

40

FENIX MINI MACHINING



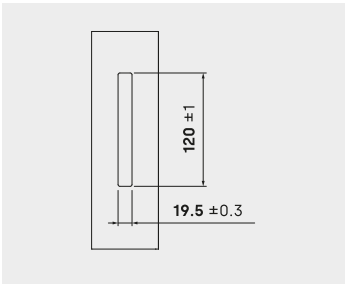


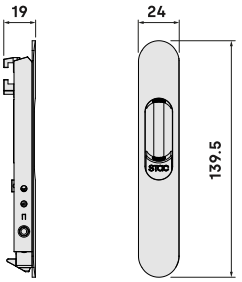
FENIX MICRO RECESSED HANDLE

035000	**

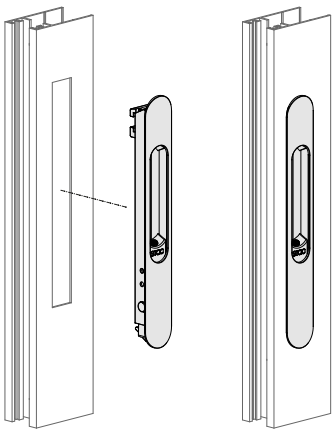
40

FENIX MICRO MACHINING

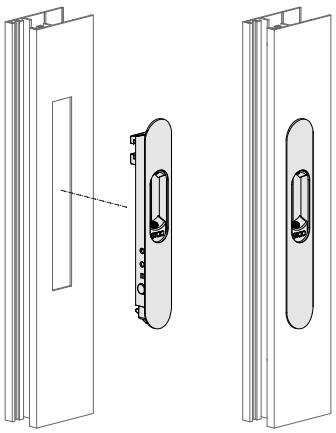




RECESSED HANDLE
FENIX MINI

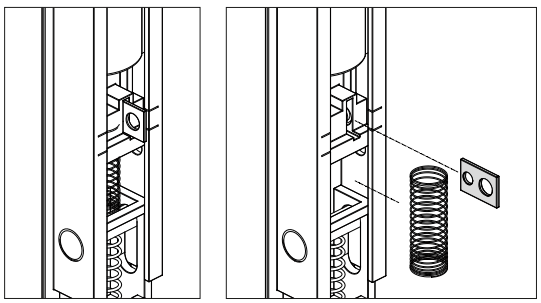


RECESSED HANDLE
FENIX MICRO



CHANGE OF OPERATING MODE
AUTOMATIC TO MANUAL

REMOVE THE PLATE AND THE SPRING FROM THE REAR OF THE LOCK TO SWITCH FROM AUTOMATIC MODE TO MANUAL MODE



FENIX PULL HANDLE

0320 **

50

AL

IN

Optional pull handles that allow the opening of sliding windows.

- The pull handle is always used in combination with a recessed handle.
- The fixing screws are available in different finishes depending on the finish of the pull handle
- The pull handles offer the possibility of fixing by means of riveted nuts (not included).

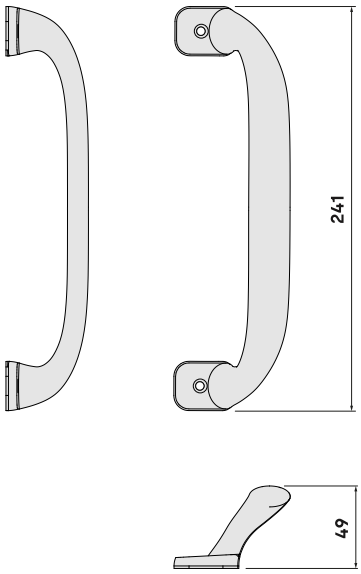
FENIX MINI PULL HANDLE

0503087 **

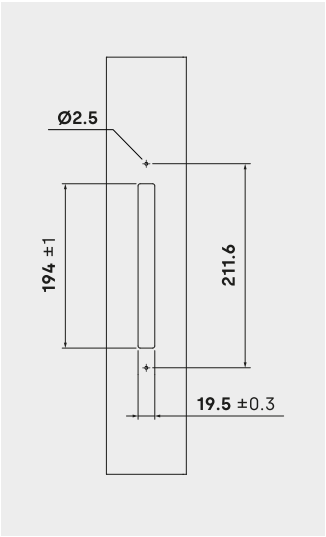
50

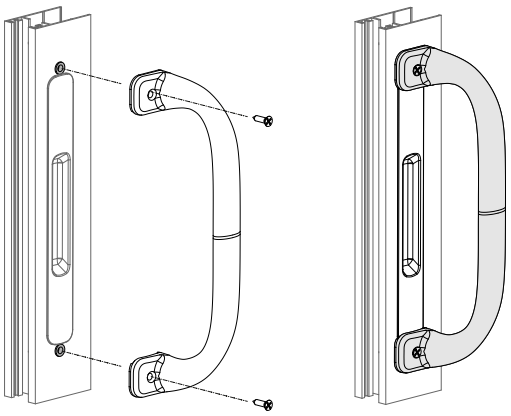
AL

IN

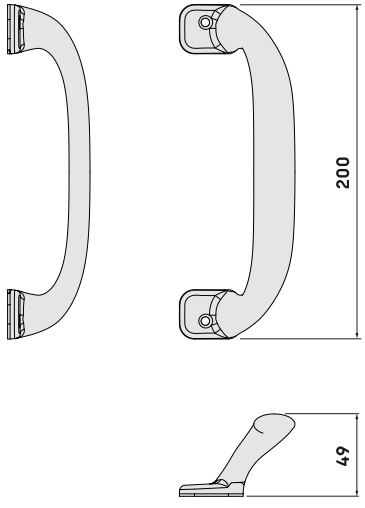


FENIX HANDLE *MACHINING*

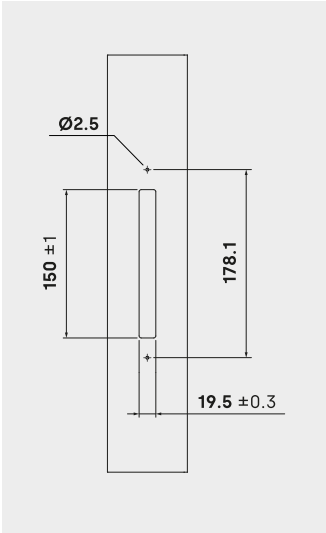




FIXING BY MEANS OF RIVETED NUTS (NOT INCLUDED)



FENIX MINI HANDLE *MACHINING*



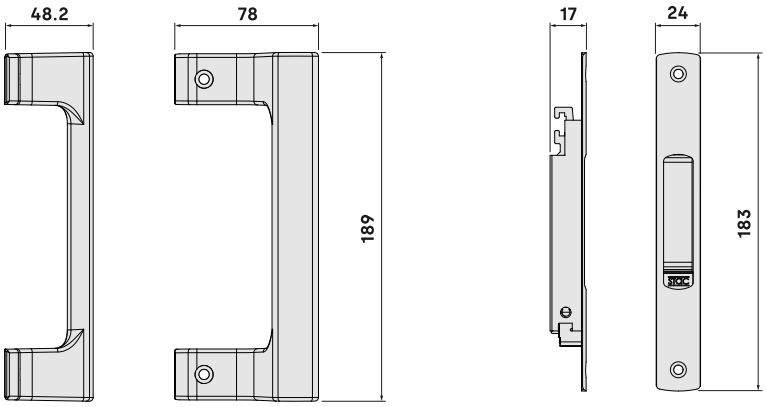
Self-assembly recessed handles that allows the opening, the crossing and overlapping of the sliding windows sashes.

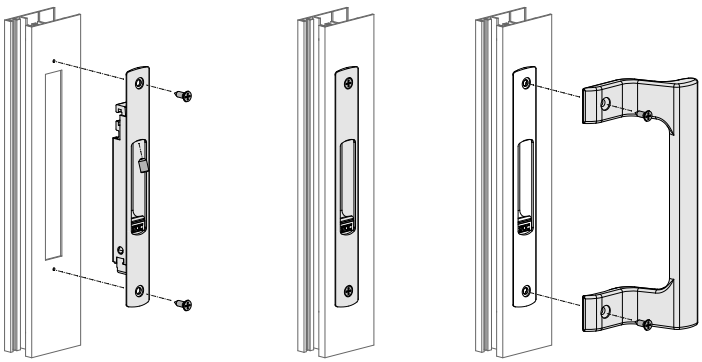
DRACO LINE recessed handle (ref. 030000)

- Fastening to the profile without screws by pressing on the lower part.
- Lock with locking device for manual operation available (ref. 030000C).
- The fixing system allows a high tolerance in the machining of the profiles.
- By default, the recessed handle operates in automatic mode, and can be configured in manual mode.

DRACO pull handle (ref. 0310)

- Optional pull handle that allow the opening of large sliding windows.
- The pull handle is always used in combination with a recessed handle.
- The fixing screws are available in different finishes depending on the finish of the pull handle.





DRACO LINE RECESSED HANDLE
Automatic closing

81 82
AL ZM
IN PP

030000 **

50

DRACO LINE RECESSED HANDLE
Manual closing

81 82
AL ZM
IN PP

03000C **

50

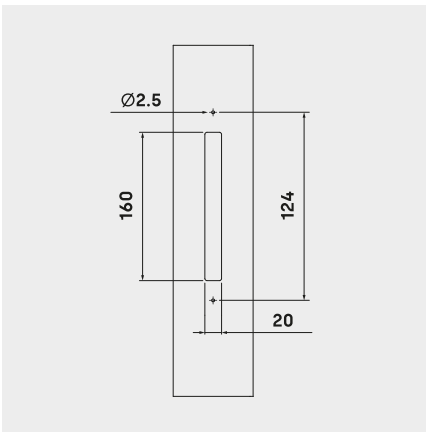
DRACO PULL HANDLE

AL
IN

0310 **

50

DRACO LINE RECESSED HANDLES AND PULL HANDLES MACHINING



DRACO MAX RECESSED HANDLE
Automatic closing

031000 **

50

81 82

AL ZN

IN PP

Self-assembly recessed handles that allow the opening, the crossing and overlapping of the sliding windows sashes.

DRACO MAX automatic recessed handle (ref. 031000)

- Fastening to the profile without screws by pressing on the lower part.
- Lock with locking device for manual operation available (ref. 031000C).
- The fixing system allows a high tolerance in the machining of the profiles.
- By default, the recessed handle operates in automatic mode, and can be configured in manual mode.

DRACO pull handle (ref. 0310)

- Optional pull handle that allow the opening of large sliding windows.
- The pull handle is always used in combination with a recessed handle.
- The fixing screws are available in different finishes depending on the finish of the pull handle.

DRACO MAX RECESSED HANDLE
Manual closing

031000C **

50

81 82

AL ZN

IN PP

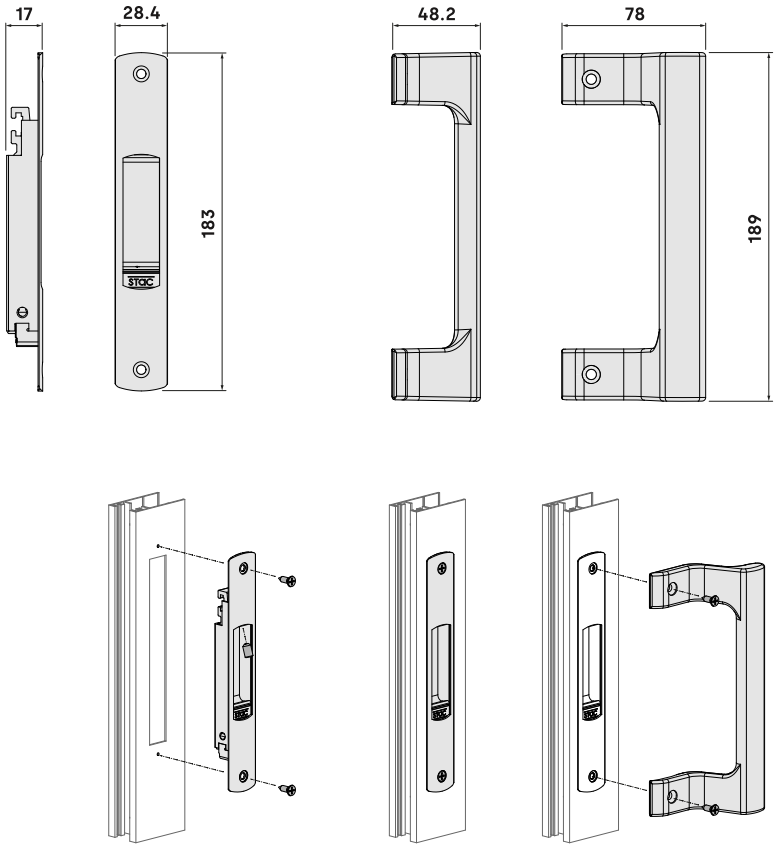
DRACO PULL HANDLE

0310 **

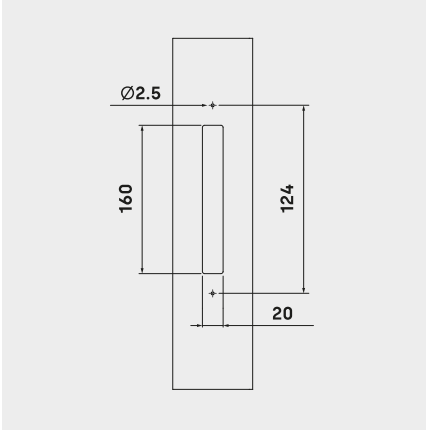
50

AL

IN



DRACO MAX RECESSED HANDLES AND PULL HANDLES MACHINING



Accessory kit that allows an aeration position in perimeter sliding windows.

- Supplied with a keeper (ref. 0403049) and a perimeter nib (ref. 0403050).
- The device is installed on an automatic acting recessed handle.
- The keeper allows two positions: closing position and micro-ventilation.
- The device complies with all the requirements of the Technical Building Code regarding indoor air quality, guaranteeing the required ventilation.
- Air permeability according to the UNE EN 12207: 2017 standard in the micro-opening position (class 1 test).

PERIMETRAL MICROVENTILATION KIT

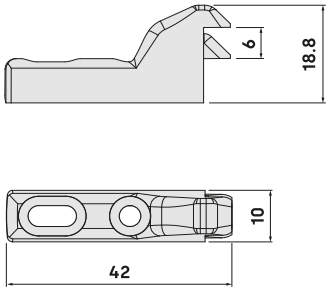
ZM

A05034400

100

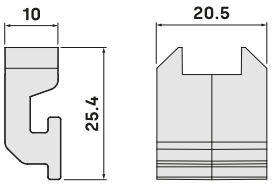
OVERLAPPED KEEPER

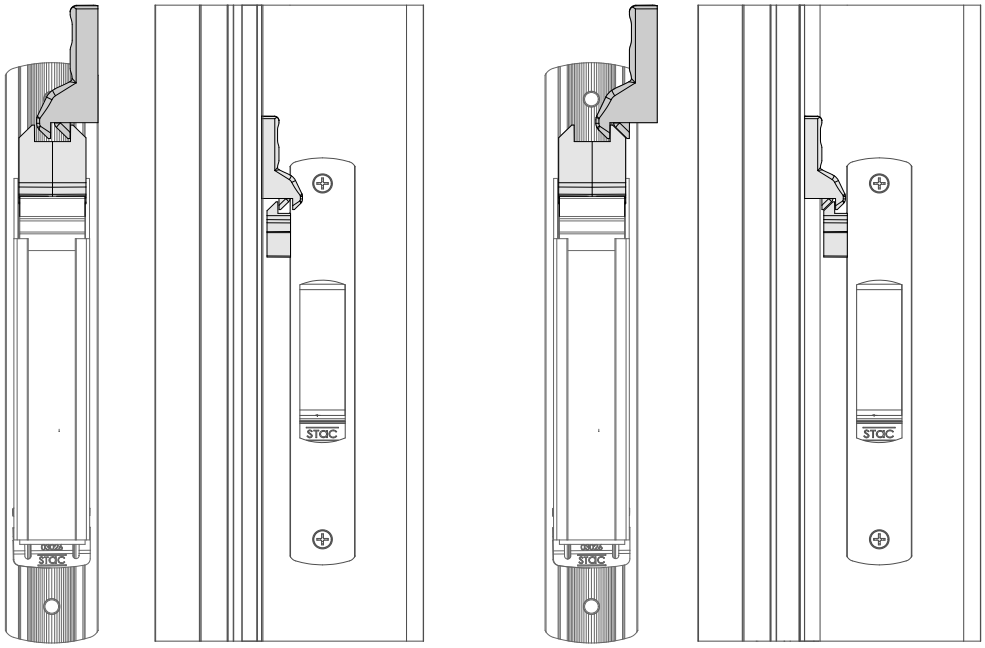
0503004



PERIMETER HOOK

0403050





MICROVENTILATION KIT (S5000)

05035000

100

ZM

IN

Accessory kit that allows an aeration position on sliding windows with straight cut profile.

• The kits can be customized: hooks and keepers can be combined to suit different series.

S5000 KIT (ref. 05035000)

• Valid for 5000 series sliding window.
• Supplied with keeper (ref. 0503009) and frontal hook (ref. 03068).

S5200 KIT (ref. 05035200)

• Valid for 5200 series sliding window.
• Supplied with keeper (ref. 0503010) and frontal hook (ref. 03068).

MICROVENTILATION KIT (S5200)

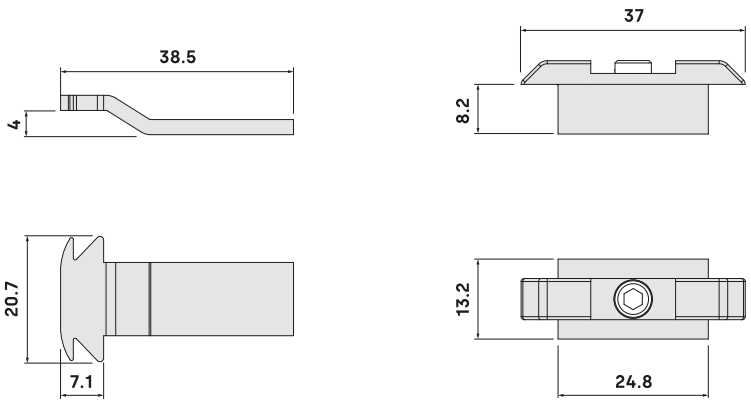
05035200

100

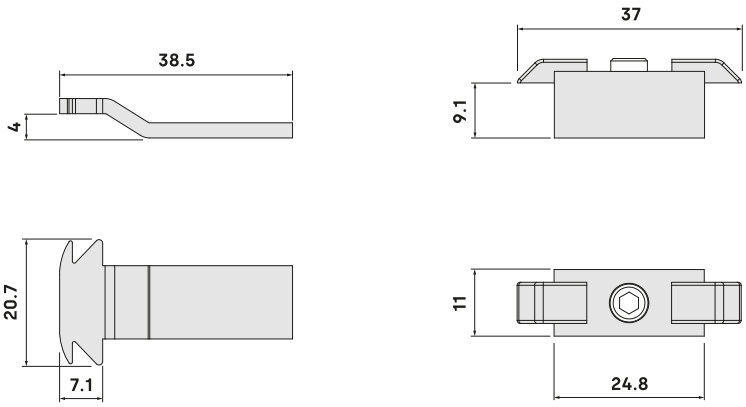
ZM

IN

05035000



05035200



Accessory kit that allows an aeration position on sliding windows with straight cut profile.

• The kits can be customized: hooks and keepers can be combined to suit different series.

S6200 KIT (ref. 05036200)

• Valid for 6200 series sliding window.
• Supplied with keeper (ref. 0503011) and frontal hook (ref. 0503073).

S6500 KIT (ref. 03KT6502)

• Valid for 6500 series sliding window.
• Supplied with keeper (ref. 0503012) and frontal hook (ref. 03065).

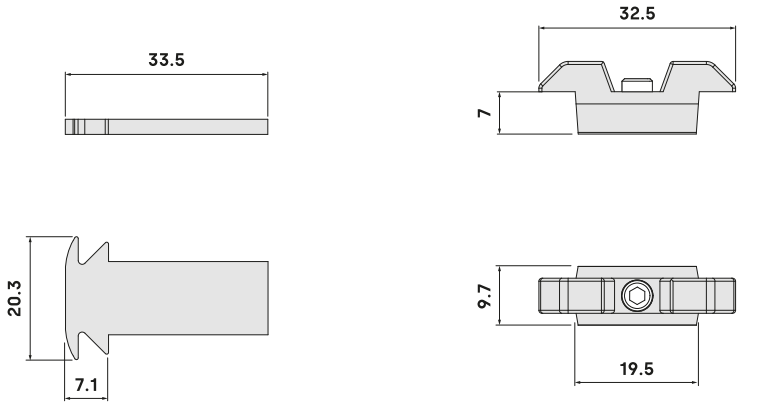
MICRO-VENTILATION KIT (S6200)

ZM
IN

05036200

100

05036200



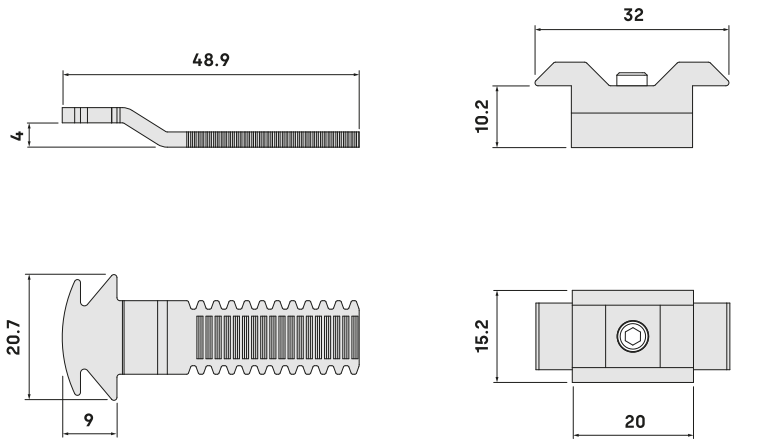
MICRO-VENTILATION KIT (S6500)

ZM
IN

03KT6502

100

03KT6502

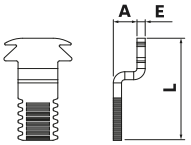


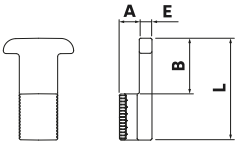
FRONTAL HOOKS

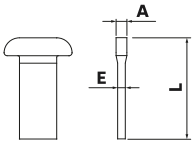
FRONTALES

The hooks in combination with a keeper enable the locking of sliding window systems.

- The special toothing allows metric adjustment of the hook.
- Adaptable to different series, both traditional and perimeter.
- The hooks are suitable for TITAN, FENIX and DRACO mortise locks.







		V	T	F	D
0503073	A - E 2.5 L 33.45	•	•	•	•
0503148	A 7.2 E 2.5 L 32	•	•	•	•
0503162	A 2 E 2.5 L 38	•	•	•	•
0503172	A 11.4 E 2.5 L 44.25	•	•	•	•
03150	A 2 E 2.5 L 38		•	•	•
03151	A - E 2.5 L 34		•	•	•
03152	A 4 E 2.5 L 37.85		•	•	•
03153	A 4 E 2.5 L 45.85		•	•	•
03154	A 1.5 E 2.5 L 32.5		•	•	•
0503170	A 13.9 E 2.5 L 41.25		•	•	•

		V	T	F	D
0503197	A 6.5 E 2.5 L 35.4		•	•	•
0503207	A 1.5 E 2.5 L 27.5		•	•	•
P03153	A 4 E 2.5 L 45.5		•	•	•
0503019	A 3.5 E 2.5 L 34			•	•
0301902	A 3.5 E 2.5 L 40			•	•
03007	A 4 E 2.5 L 37.85			•	•
03031	A 4 E 2.5 L 45.85			•	•
0403048	A 6.9 E 3.5 B 13.5 L 27.5			•	•
0503051	A 2 E 2.5 L 38			•	•
0503069	A - E 2.5 L 34			•	•

		V	T	F	D
03070	A 2 E 2.5 L 27			•	•
0503071	A 5.5 E 2.5 L 46.4			•	•
0503072	A 5.3 E 2.5 L 36			•	•
03078	A 4 E 2.5 L 37			•	•
0503083	A 5.5 E 2.5 L 40			•	•
0503097	A 6.4 E 3.5 B 18 L 33			•	•
0503098	A 5.7 E 3.5 B 32 L 47			•	•
0503129	A 4 E 2.5 L 37.85			•	•

V MICRO-VENTILATION
T TITAN KEEPERS
F FENIX | FENIX MINI KEEPERS
D DRACO MAX | DRACO LINE KEEPERS

The hooks in combination with a keeper enable the locking of sliding window systems.

- The special toothing allows metric adjustment of the hook.
- Adaptable to different series, both traditional and perimeter.
- The hooks are suitable for TITAN, FENIX and DRACO mortise locks.

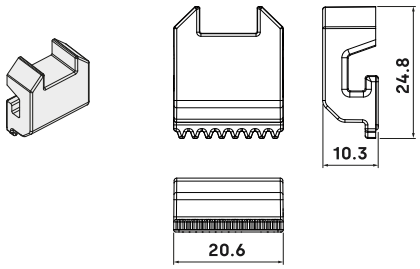
PERIMETRAL HOOKS

PERIMETRALES

A0503050

V T F D

● ● ● ●



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V T F D

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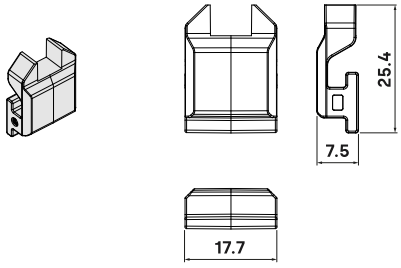

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V T F D

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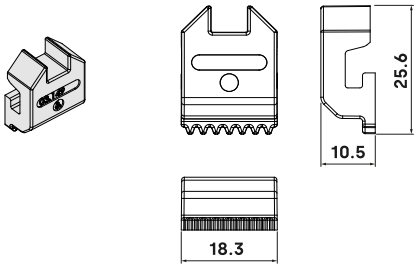

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V T F D

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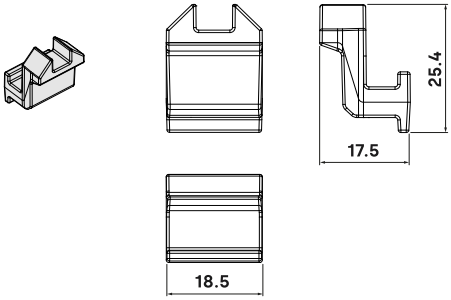


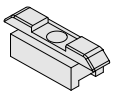
GROOVED KEEPERS

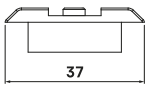
ACANALADOS

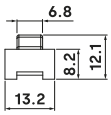
The keeper in combination with a perimetral hook allow the closing of the sliding windows.

- References include screws.
- Adjusted to the frame to regulate with the closing hook.
- The hooks can be adapted to different series of sliding windows, either traditional or perimeter.

0503009







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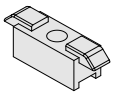

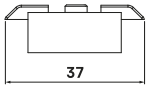

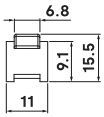

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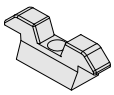

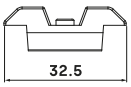

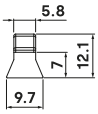

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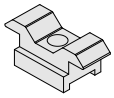

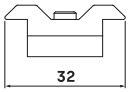

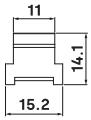

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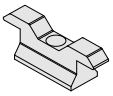

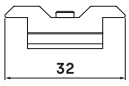

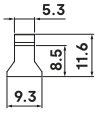

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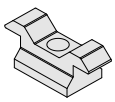

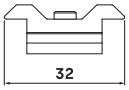

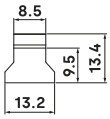

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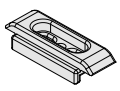

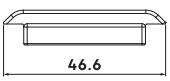

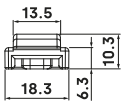

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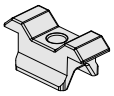

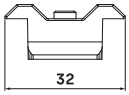

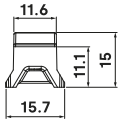

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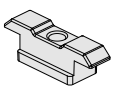

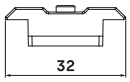

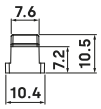

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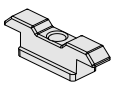

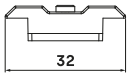

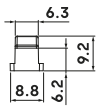

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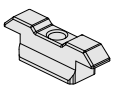

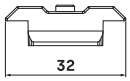

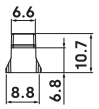

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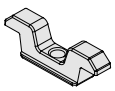

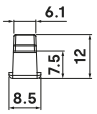

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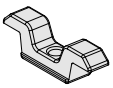

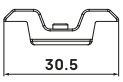

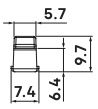

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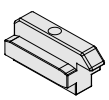
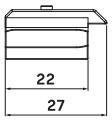
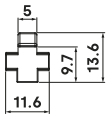
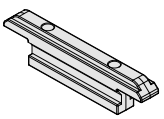
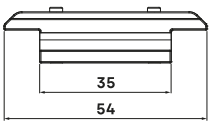


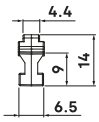




03079C





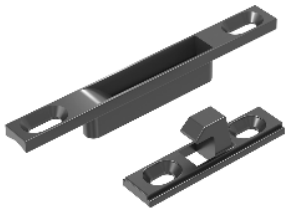


The keeper in combination with a perimetral hook allow the closing of the sliding windows.

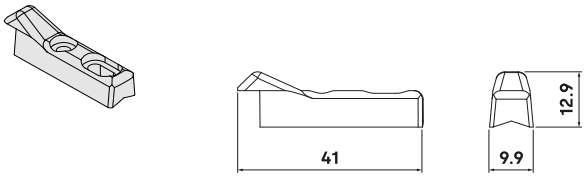
- Adjusted to the frame to regulate with the closing hook.
- The hooks can be adapted to different series of sliding windows, either traditional or perimeter.

OVERLAPPED KEEPERS

SUPERPUESTOS



0503004



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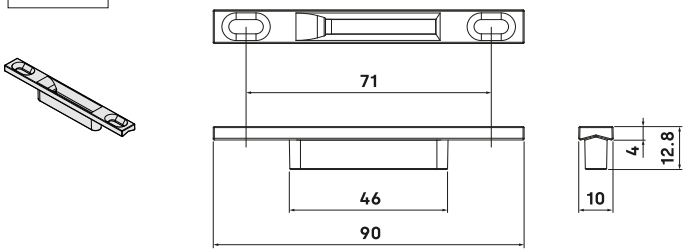

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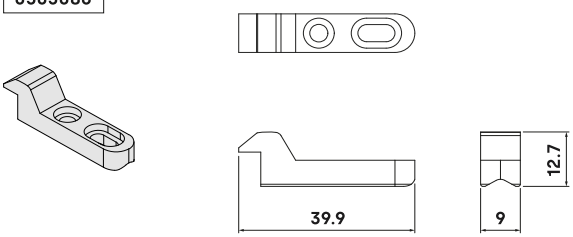
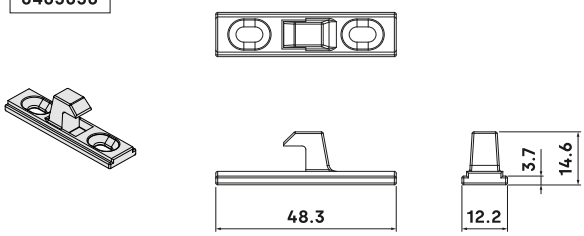
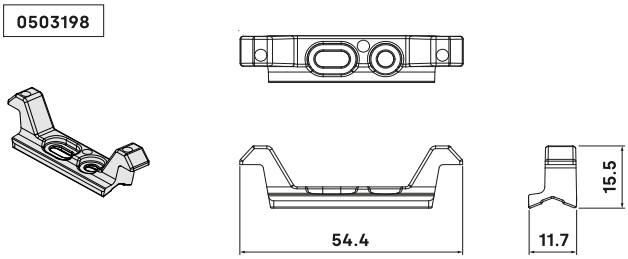


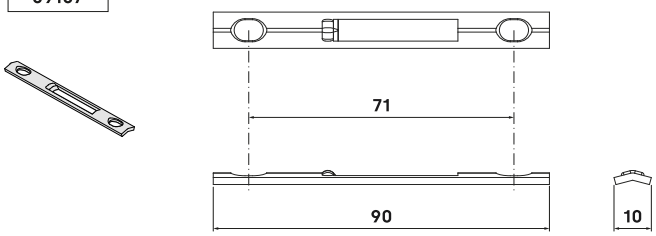
0403030



0503198



09107



LS 200 & 400 systems

Up to 400 kg load

Quick and simple instalation

Corrosion resistance

LSPLUG adapters

The sliding window has undergone a notable evolution in specification and ease of use, primarily focused on meeting today's thermal and acoustic efficiency requirements. This results in profiles with large thermal breaks and considerably thicker glass, this leads to a significant increase in the weight of the slides, especially with large sashes.

As a result, the profiles used in this type of enclosures have to be sufficiently robust, and it is common to use profiles from lifting systems in traditional sliding systems, especially when the weights of the windows exceed certain values.

» Lift & slide systems

The LSPLUG adapters allow the use of our lift & slide rear bogies, adapting the profiles of lift & slide systems as traditional sliding systems.

» High weight-bearing capacity

The LSPLUG adapters are the ideal solution for adapting lift & slide systems to sliding systems with high capacity (two bogies up to 300 - 400 kg / sash).

Allows independent height adjustment in the range from 0 to + 2.5 mm by means of a 5 mm Allen key.

» Corrosion resistance

The zamak parts have been treated to guarantee a resistance to corrosion of more than 480 hours in a saline mist chamber in compliance with the UNE-EN 1670 standard, obtaining a grade 5 classification that grants a high resistance to corrosion and the possibility of use in environments with severe climatic conditions.

The LSPLUG mechanism makes it possible to adapt the profiles of lift & slide systems as traditional sliding systems.

- The adapter is valid for profiles with a 22-millimetre groove.
- Allows height adjustment of 2.5 millimetres, accessible and independent.

LSPLUG LS400 adapter (ref. 237008)

- Maximum load capacity per sash 400 kg with 2 LS400 bogies.

LSPLUG LS200 adapter (ref. 237009)

- Maximum load capacity per sash 300 kg with 2 LS200 bogies.
- The reference 237009 includes centring shims for 22 mm grooves.

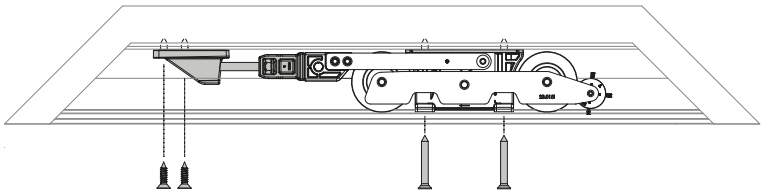
LSPLUG ADAPTER (237008)
LS400 lift & slide system

ZM
AC

237008

50

237008



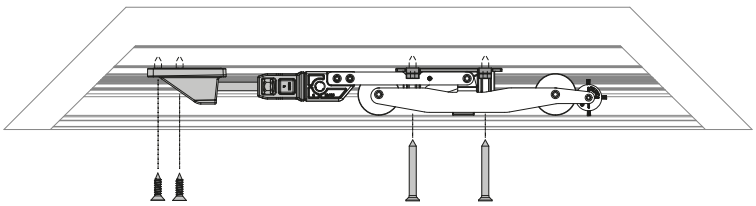
LSPLUG ADAPTER (237009)
LS200 lift & slide system

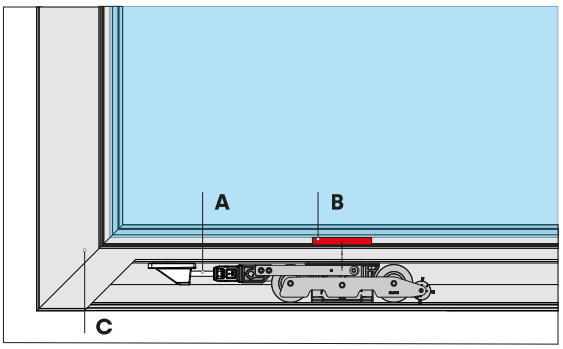
ZM
AC

237009

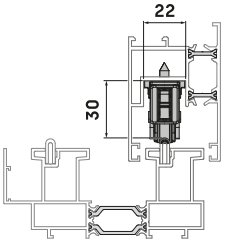
50

237009

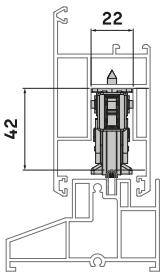




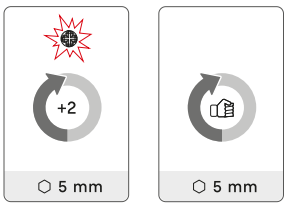
LS200 GROOVE



LS400 GROOVE



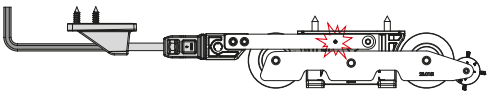
HEIGHT ADJUSTMENT 0, +2.5



A BEFORE FIXING THE DEVICE TO DE SASH PROFILE, MAKE SURE THAT THE SCREW IS IN TENSION.

B FROM 200KG SASH WEIGHT THE GLASS SHIM WILL BE POSITIONED CENTRALLY WITH THE BOGIE.

C LEAVE SPACE ENOUGH TO AVOID INTERFERENCE BETWEEN THE SCREWS AND THE CORNER JOINT.



HL BOGIES

tilting bogies

±2.3 mm adjustment

Up to 200 kg per sash

2 wheels system

Needle roller

HL tilting bogies

Over recent years, the sliding window has undergone a notable evolution in specification and ease of use. This evolution is focused primarily on meeting current demands for thermal and acoustic efficiency. This results in profiles with large thermal breaks and considerably thicker glass. As a result of this, the dimensions of the profiles have increased greatly.

HL tilt bogies stand out for the fact that they can adapt to all types of sliding window system, they are height-adjustable and can bear loads up to 200 kg per sash.

» Tilting system

Mainly, the bogies for sliding windows fail due to wear of the wheel bearings. This is due to the high forces that some wheels bear compared to others, i.e. the weight is not evenly distributed to all wheels.

The design of the bogies has been developed with the premise of balancing the load which each wheel bears via a novel rocker system which increases lifetime expectancy.

» Adaptability

The bogies are available in different sizes to allow distinct wheel widths and thus be usable in any rail system

The runner is manufactured in POM (polyoxymethylene) which guarantees its durability and avoids wear of the wheel and rail.

» Effective adjustment ±2.3 millimetres

If required during the installation of the windows, it is possible to adjust the height of the bogies directly, by use of a 5 mm Allen key.

» Corrosion resistance

The structure of the bogies is made of zamak with a natural zinc treatment. This allows it to withstand up to 480 hours in a saline mist chamber, ensuring a high level of corrosion resistance.

Tilting bogies designed for heavy loads on sliding window.

- Bearings with needle bushing (HK0810 WWB).
- Maximum load per sash 200 kg and height adjustment ±2.3 mm.
- The tilting system allows the load on each wheel to be balanced.

Please refer to the technical table for compatible frame geometry.

96001 HL BOGIE

ZM IN PA PM

96001

20

96002 HL BOGIE

ZM IN PA PM

96002

36

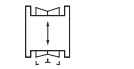
96003 HL BOGIE

ZM IN PA PM

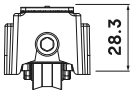
96003

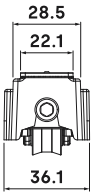
40

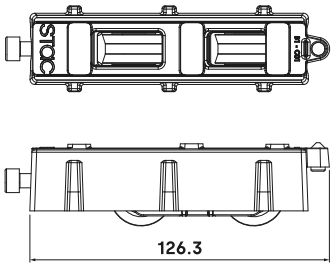
96001



MIN. **28.6**
MAX. **33.2**

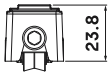


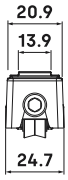


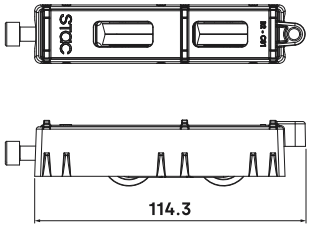


96002

MIN. **24.4**
MAX. **29**

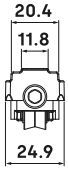


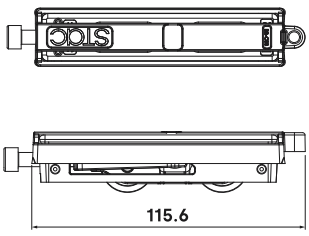




96003

MIN. **20.8**
MAX. **25.4**





96004 HL BOGIE

96004

24

ZM IN PA PM

Tilting bogies designed for heavy loads on sliding window.

- Bearings with needle bushing (HK0810 WWB).
- Maximum load per sash 200 kg and height adjustment ±2.3 mm.
- The tilting system allows the load on each wheel to be balanced.

Please refer to the technical table for compatible frame geometry.

96005 HL BOGIE

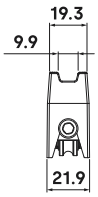
96005

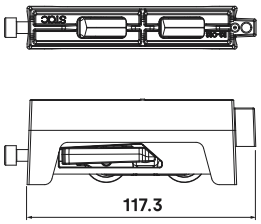
24

ZM IN PA PM

96004

MIN. **38.4**
MAX. **43**





96006 HL BOGIE

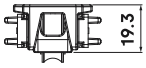
96006

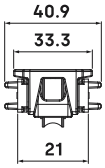
40

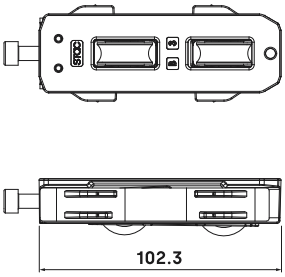
ZM IN PA PM

96005

MIN. **20.3**
MAX. **24.9**

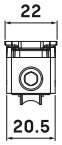


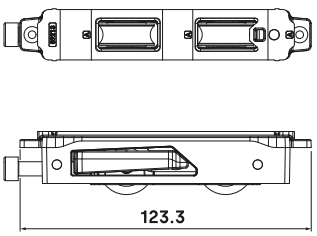




96006

MIN. **22.7**
MAX. **27.3**





Tilting bogies designed for heavy loads on sliding window.

- Bearings with needle bushing (HK0810 WWB).
- Maximum load per sash 200 kg and height adjustment ±2.3 mm.
- The tilting system allows the load on each wheel to be balanced.

Please refer to the technical table for compatible frame geometry.

96007 HL BOGIE

ZM IN
PA PM

96007

40

96008 HL BOGIE

ZM IN
PA PM

96008

50

96009 HL BOGIE

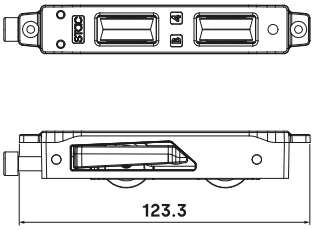
ZM IN
PA PM

96009

50

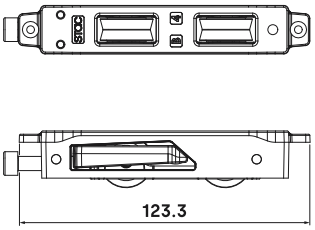
96007

MIN. **17.7**
MAX. **23.3**



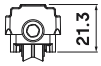
96008

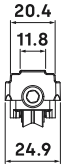
MIN. **19.1**
MAX. **23.7**

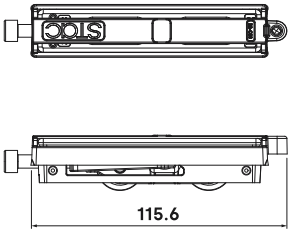


96009

MIN. **20.8**
MAX. **25.4**







96010 HL BOGIE

96010

50

ZM IN
PA PM

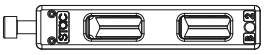
Tilting bogies designed for heavy loads on sliding window.

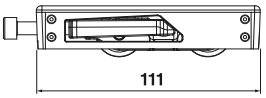
- Bearings with needle bushing (HK0810 WWB).
- Maximum load per sash 200 kg and height adjustment ±2.3 mm.
- The tilting system allows the load on each wheel to be balanced.

Please refer to the technical table for compatible frame geometry.

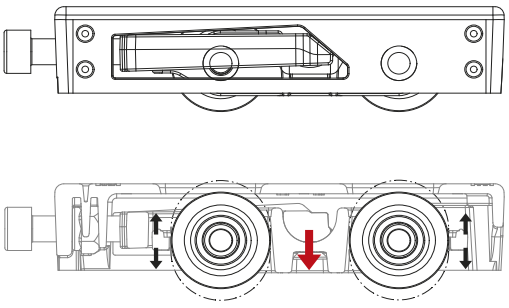
96010

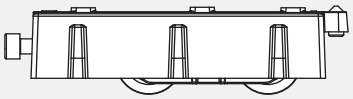
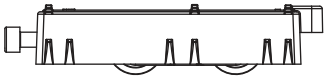
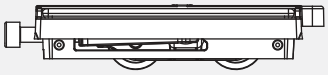
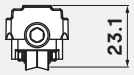
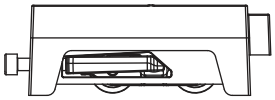
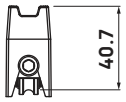
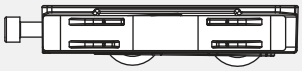
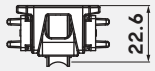
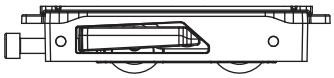
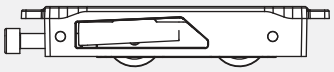
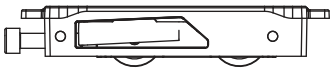
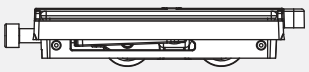
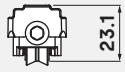
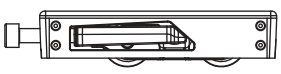
MIN. **19.6**
MAX. **24.2**





» TILTING SYSTEM



	FRAME GEOMETRY	WHEEL MATERIAL	MAX. WEIGHT PER SASH (kg)	ADJUSTMENTS MINIMUM/ MAXIMUM (mm)		
96001	4.3 ~ 143°	PM	200 kg		30.9	min 28.6 max 33.2
96002	150°	PM	200 kg		27.2	min 24.4 max 29
96003	1 ~ 130°	PM	200 kg		 23.1	min 20.8 max 25.4
96004	R2.6	PM	200 kg		 40.7	min 38.4 max 43
96005	R5	PM	200 kg		 22.6	min 20.3 max 24.9
96006	R5	PM	200 kg		25	min 22.7 max 27.3
96007	150°	PM	200 kg		23.6	min 17.7 max 23.3
96008	150°	IN	200 kg		21.4	min 19.1 max 23.7
96009	1 ~ 130°	PM	200 kg		 23.1	min 20.8 max 25.4
96010	2.2 ~ 130°	PM	200 kg		21.9	min 19.6 max 24.2

All references include backset roller bearings (HK0810 WWB).
The adjustment of the tilting bogies is made using a 5 millimetre Allen key.

C99003 BOGIE

C99003

350

PA

The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

C99003 (ref. C99003)

- Maximum load per sash: 30 kg (2 bogies per sash).

C99004 (ref. C99004)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.

C99005 (ref. C99005)

- Maximum load per sash: 30 kg (2 bogies per sash).

C99004 BOGIE

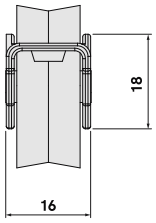
C99004

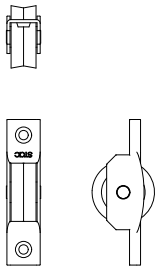
150

ZM

PA

C99003





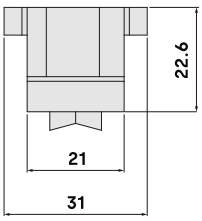
C99005 BOGIE

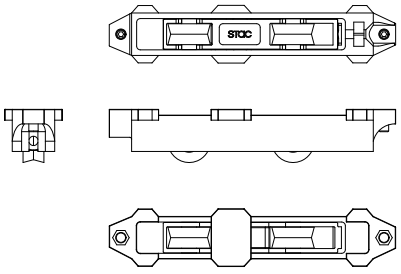
C99005

450

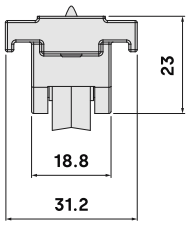
PA

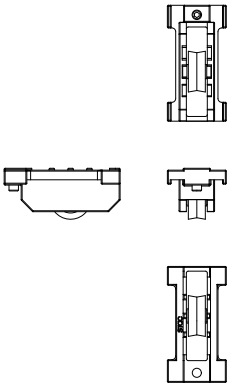
C99004





C99005





The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

C99006 (ref. C99006)

- Maximum load per sash: 100 kg (2 bogies per sash).
- 2-wheeled trolley available: 200 kg maximum load (ref. C99007).

S2000 (ref. 0519028)

- Maximum load per sash: 20 kg (2 bogies per sash).
- Adjustable up to 1.5 millimetres.
- 2-wheeled trolley available: 40 kg maximum load (ref. 0509028).

S2000 (ref. 0519023)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.
- 2-wheeled trolley available: 100 kg maximum load (ref. 0509023).

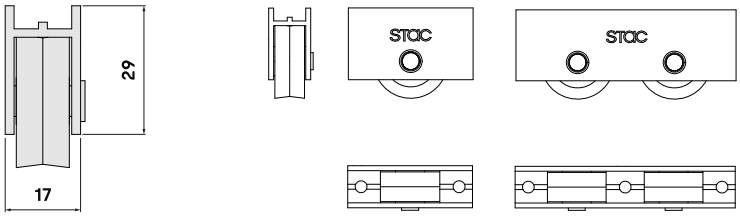
BOGIE C99006

AL
PE

C99006

350

C99006 C99007



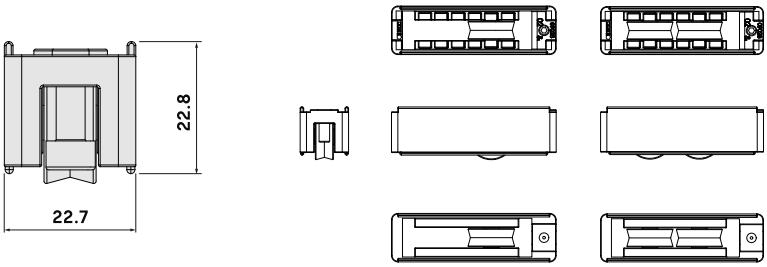
S2000 BOGIE

PE
PA

0519028

250

0519028 0529028



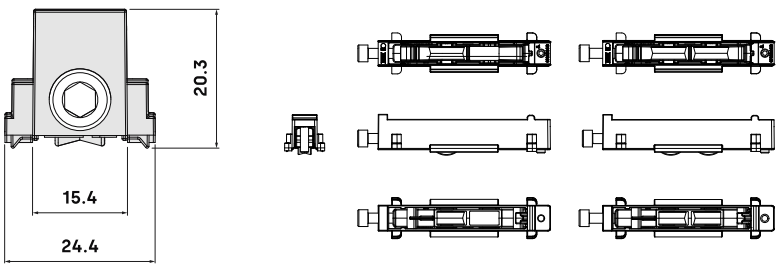
S2000 BOGIE

ZM
PA

0519023

150

0519023 0509023



S4200 BOGIE

A0519021

150

ZM

PA

The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

S4200 (ref. A0519021)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.
- 2-wheeled trolley available: 100 kg maximum load (ref. A0509021).

S60P (ref. 0509039)

- Maximum load per sash: 40 kg (2 bogies per sash).
- Adjustable up to 1 millimetre.

S70P (ref. 0509040)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 1 millimetre.

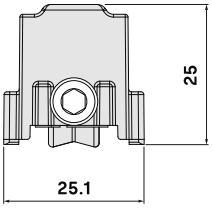
S60P BOGIE

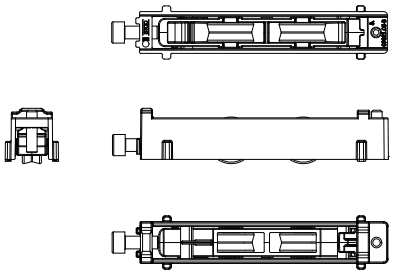
0509039

250

PP

A0519021 A0509021





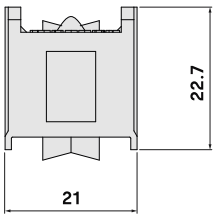
S70P BOGIE

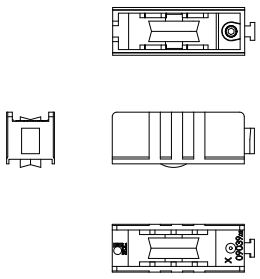
0509040

250

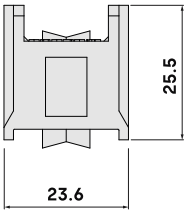
PA

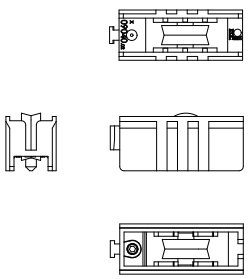
0509039





0509040





The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

S70 S92 (ref. 0509134)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.

S62 (ref. 0509037)

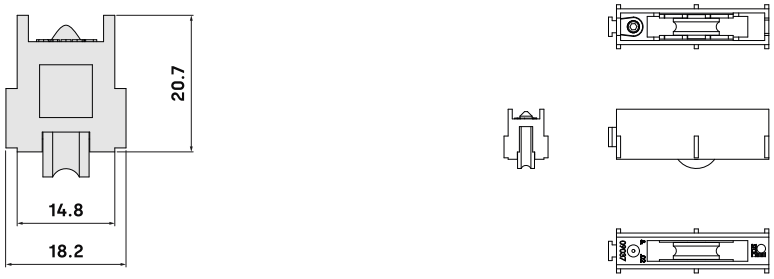
- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 1 millimetre.

S5000 G90 (ref. 0519022)

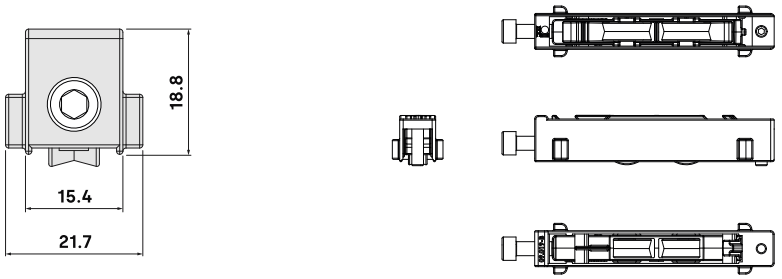
- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.

0509134

0509037



0519022



S70 S92 BOGIE

PE

0509134

250

S62 BOGIE

PA

0509037

100

S5000 G90 BOGIE

ZM

PA

0519022

150

S5000 S2000 HR BOGIE

0509122

150

ZM

PA

S5000 S5200 BOGIE

0509002

250

PE

PA

S6200 BOGIE

0519019

250

PA

The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

S5000 S5200 HR (ref. 0509122)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.
- 2-wheeled trolley available: 100 kg maximum load (ref. 0509022).

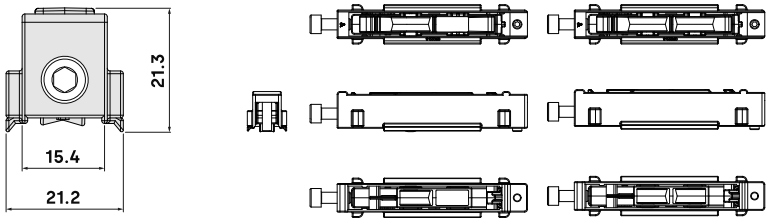
S5000 S5200 (ref. 0509002)

- Maximum load per sash: 20 kg (2 bogies per sash).
- Adjustable up to 1.5 millimetres.
- 2-wheeled trolley available: 40 kg maximum load (ref. 0509001).

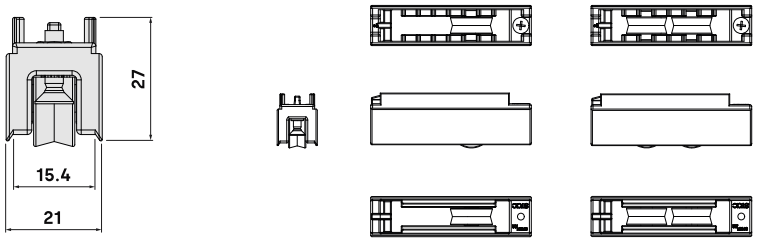
S6200 (ref. 0519019)

- Maximum load per sash: 10 kg (2 bogies per sash).
- Adjustable up to 1.5 millimetres.
- 2-wheeled trolley available: 20 kg maximum load (ref. 0519009).

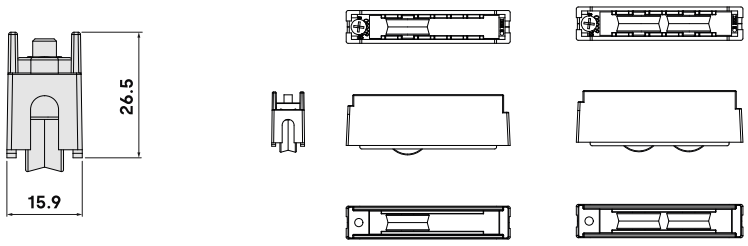
0509122 0509022



0509002 0509001



0519019 0519009



The bogies allow the sash of the sliding window to slide smoothly and without jerks on the frame rail.

S6200 (ref. 0509109)

- Maximum load per sash: 40 kg (2 bogies per sash).
- Adjustable up to 1.5 millimetres.

S6500 (ref. 0509024)

- Maximum load per sash: 50 kg (2 bogies per sash).
- Adjustable up to 2.5 millimetres.

S6500 (ref. 0509025)

- Maximum load per sash: 50 kg (2 bogies per sash).

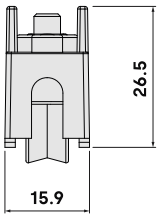
S6200 BOGIE

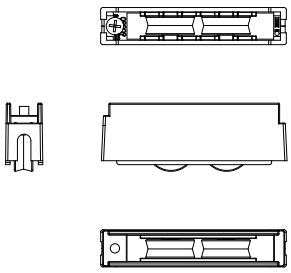
PA

0509109

250

0509109





BOGIE S6500

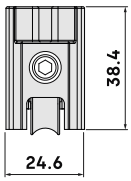
ZM

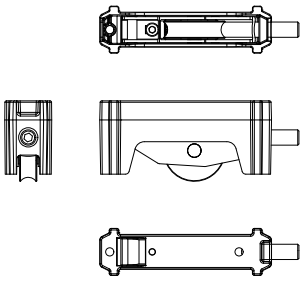
PA

0509024

100

0509024





S6500 FIXED BOGIE

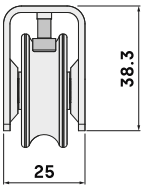
AL

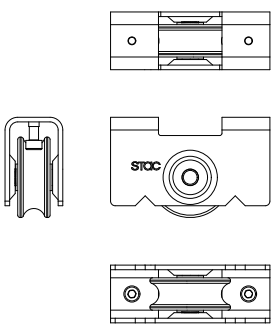
PA

0509025

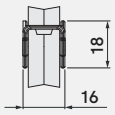
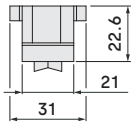
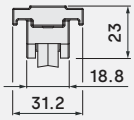
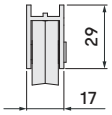
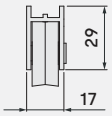
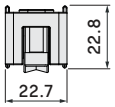
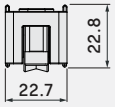
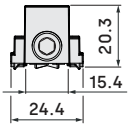
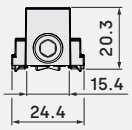
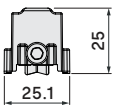
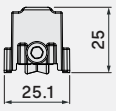
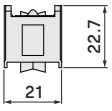
100

0509025

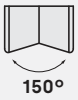
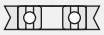
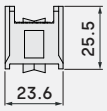
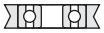
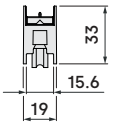
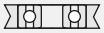
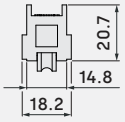
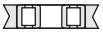
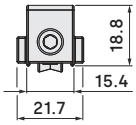
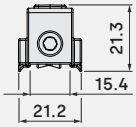
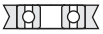
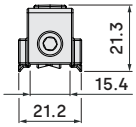
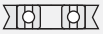
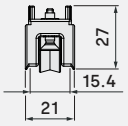
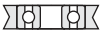
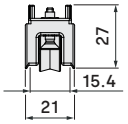
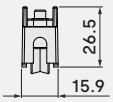
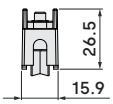
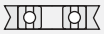
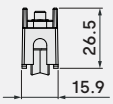
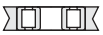
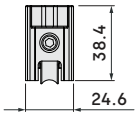
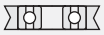
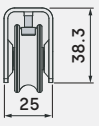




CLASSIC bogies

	FRAME GEOMETRY	WHEELS	MATERIAL	MAX. WEIGHT	BEARING	ADJUSTMENTS	
C99003	160°		PM AC	50 kg		-	
C99004	150°		PM ZM	120 kg		± 2.5 mm	
C99005	160°		PA PA	30 kg		± 1 mm	
C99006	166°		PA AL	100 kg		-	
C99007	166°		PA AL	200 kg		-	
0519028	150°		PA PE	20 kg		± 1.5 mm	
0509028	150°		PA PE	40 kg		± 1.5 mm	
0519023	150°		PA ZM	50 kg		± 2.5 mm	
0509023	150°		PA ZM	100 kg		± 2.5 mm	
A0519021	150°		PA ZM	50 kg		± 2.5 mm	
A0509021	150°		PA ZM	100 kg		± 2.5 mm	
0509039	150°		PA PP	40 kg		± 1 mm	

Ball bearing *Double ball bearing* *Pin bearing* *Simple bearing*

	FRAME GEOMETRY	WHEELS	MATERIAL	MAX. WEIGHT	BEARING	ADJUSTMENTS	
0509040	 150°		PA PA	50 kg		± 1 mm	 25.5 23.6
0509134	R2.2		PA PE	50 kg		± 2.5 mm	 33 15.6 19
0509037	R2.2		PA PE	50 kg		± 1 mm	 20.7 14.8 18.2
0519022	150°		PA ZM	50 kg		± 2.5 mm	 18.8 15.4 21.7
0509122	150°		PA ZM	50 kg		± 2.5 mm	 21.3 15.4 21.2
0509022	150°		PA ZM	100 kg		± 2.5 mm	 21.3 15.4 21.2
0509002	150°		PA PE	20 kg		± 1.5 mm	 27 15.4 21
0509001	150°		PA PE	40 kg		± 1.5 mm	 27 15.4 21
0519019	150°		PA PA	10 kg		± 1.5 mm	 26.5 15.9
0519009	150°		PA PA	20 kg		± 1.5 mm	 26.5 15.9
0509109	150°		PA PA	40 kg		± 1.5 mm	 26.5 15.9
0509024	R5		PM ZM	50 kg		± 2.5 mm	 38.4 24.6
0509025	R5		PA AL	50 kg		-	 38.3 25

13 LIFT & SLIDE OPENING

318 **LS200 lift & slide**

320 **LS400 lift & slide**

322 **Drive gears**

324 **Lift & slide opening accessories**

332 **Locking points for lift & slide opening**

lift & slide

Balanced elevation

Minimum effort

Cleaning mechanism

Wide load range

Lift & slide hardware systems

The lift & slide system is the most airtight and secure option that allows you to cover large areas with the comfort of a sliding window.

» Lift & slide LS200

The best choice on the market for its performance range. The smooth, quiet and effortless sliding of the sashes allows for clear spaces through the various applications of up to 6 sashes with different combinations.

For configurations ≤ 200 kg per sash, the installation of front and rear bogies is required. In addition, for configurations between 200 - 300 kg, the use of two additional bogies is required.

» Lift & slide LS400

Specially designed for heavy and dimensional sashes, allows a totally vertical lifting of the slide. The sashes do not head in the lowering and lifting movements.

The equal distribution of the weight of the sash on each wheel throughout the lifting mechanism, guarantees the correct operation of the bearings without seizure, breakage or noise.

» High performance solutions

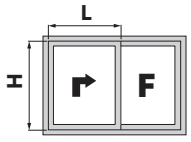
Our hardware solutions enable the configuration of high-performance lifting systems.

- **SOFT-CLOSING**: allows a damped and silent braking of the sash, regardless of the force exerted on it.
- **SOFT-LIFT**: damping of the sash to prevent the handle from recoiling abruptly during the sash lowering movement of the lift sash.
- **COMPENSATING SPRING**: balances the weight by reducing the lifting effort of the sash, minimising the return torque of the handle when lowering.
- **EASY-HANDLING**: accessible cremone fronts that lower the height of the handle by positioning it 440 millimetres from the floor and allow the use of lifts to avoid movements that require high efforts.

	LIFT & SLIDE LS200	LIFT & SLIDE LS400
LIFT SASH		
WEIGHT RANGE PER SASH	200 ~ 300 kg	300 ~ 400 kg
MINIMUM ~ MAXIMUM WIDTH	550 - 3300 mm	550 - 3300 mm
MINIMUM ~ MAXIMUM HEIGHT	950 - 3300 mm	950 - 3800 mm
GROOVE PROFILE		
GROOVE HEIGHT	30 mm	42 mm
GROOVE WIDTH	16, 18 y 22 mm *	22 mm
DRIVE GEAR		
BACKSET	27.5 mm	37.5 mm
C/C DISTANCE	80 mm	80 mm

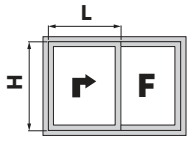
** Groove widths of 18 and 22 millimetres available with adaptation shims.*

LS200		**≤ 200 kg** CONFIGURATION	**201 - 300 kg** CONFIGURATION
233501	FRONT BOGIE	1 unit	1 unit
233502	REAR BOGIE	1 unit	1 unit
A233503	SUPPLEMENTARY BOGIE	-	2 units
233504	FIXED BOGIE	2 units	4 units

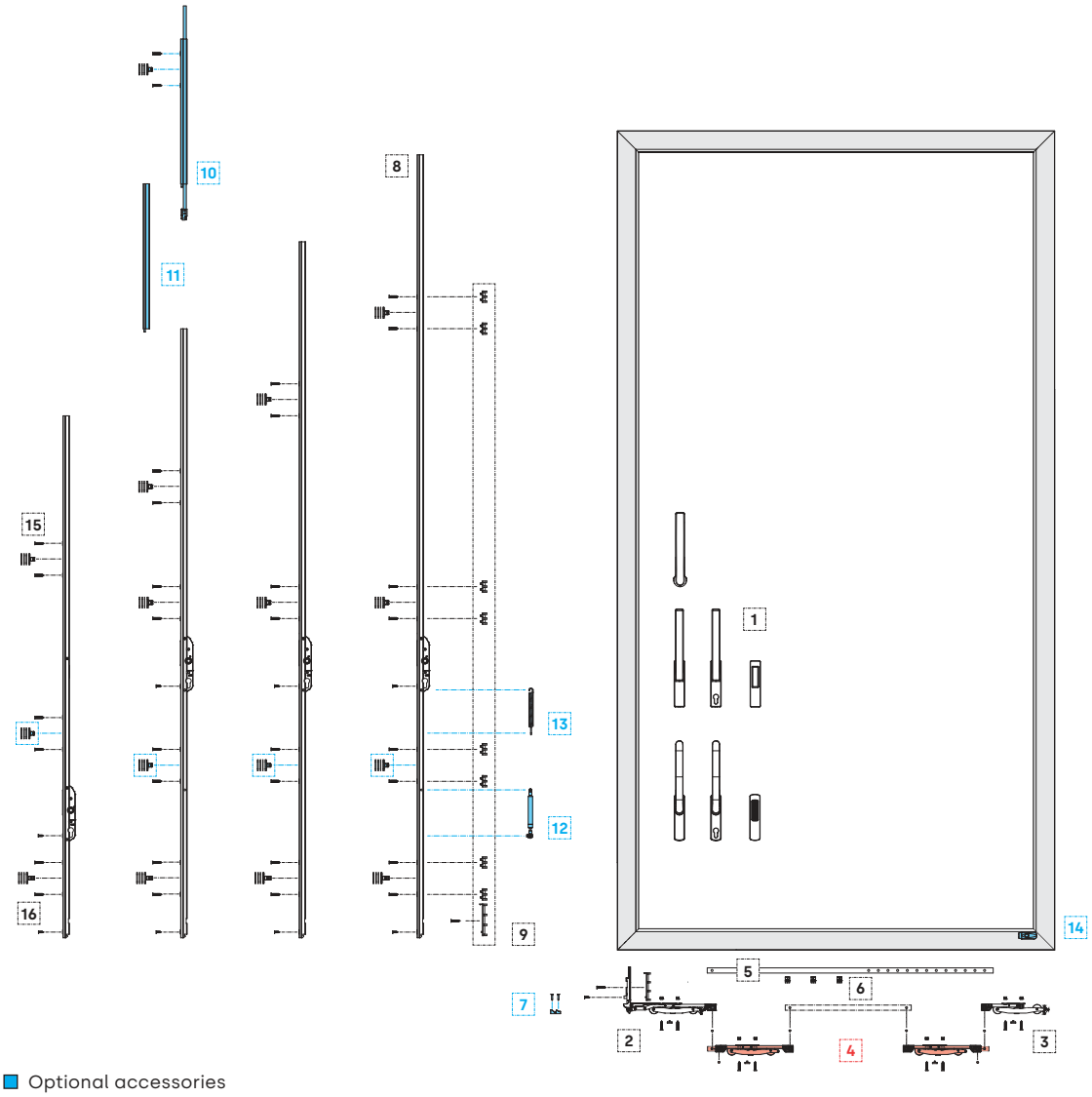


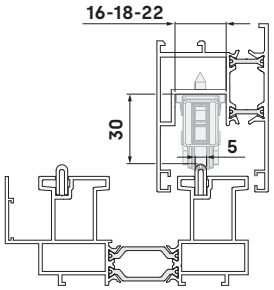
config. ≤ 200 kg H = 950 - 3300 mm, L = 550 - 3300 mm, **config. 201 - 300 kg** H = 950 - 3300 mm, L = 1100 - 3300 mm

LS400		**≤ 300 kg** CONFIGURATION	**301 - 400 kg** CONFIGURATION
233001	FRONT BOGIE	1 unit	1 unit
233002	REAR BOGIE	1 unit	1 unit
233003	SUPPLEMENTARY BOGIE	-	2 units
233004	FIXED BOGIE	2 units	4 units



config. ≤ 300 kg H = 950 - 3300 mm, L = 550 - 3300 mm, **config. 301 - 400 kg** H = 950 - 3300 mm, L = 1100 - 3800 mm





Optional accessories

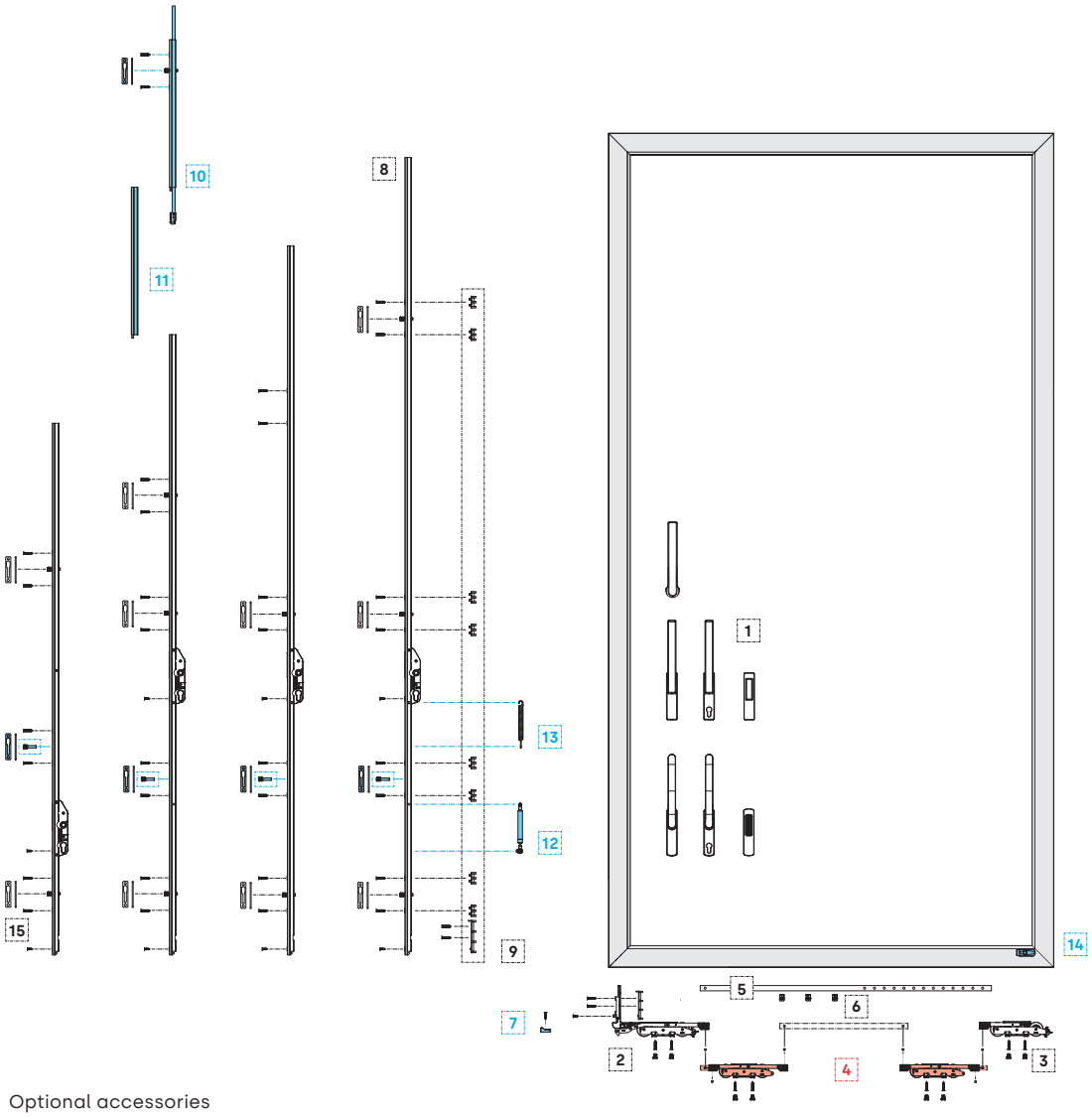
≥ 200 kg / sash configurations

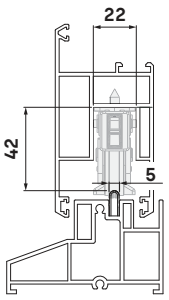
1	**PULL HANDLE | FINGER PULL** *Consult the range of finger and pull handles*	
2	**FRONT BOGIE**	233501
3	**REAR BOGIE**	233502
4	**SUPPLEMENTARY BOGIE**	A233503
-	**FIXED BOGIE**	233504
5	**CONNECTING RODS**	23301 (•)
6	**BOTTOM GUIDES** FOR CONNECTING RODS	237101
7	**HORIZONTAL KEEPER**	234322
8	**DRIVE GEAR** WITH LOCKING POINT	C23450 (•)
	DRIVE GEAR WITH LOCKING POINT (BOLT)	
9	**SHIM KIT (PA)** FOR DRIVE GEAR AND FRONT BOGIE	A2352 (•)
10	**EXTENSION ROD** WITH LOCKING POINT	A234011
	EXTENSION ROD WITH LOCKING POINT (BOLT)	234012
11	**TRIM** FOR EXTENSION ROD	234010

12	**DAMPING CYLINDER**	23700 (•)
13	**COMPENSATING SPRING**	237001
14	**DAMPING STOP**	23402 (•)
-	**FRAME KEEPER**	234042
-	**KIT WITH FRAME KEEPER**	234308
-	**PA SHIM KIT** (1,2,3 mm)	234203
15	**SHORT LOCKING POINT** WITH PA SHIM KIT (1,2,3 mm)	234320
	LONG LOCKING POINT WITH PA SHIM KIT (1,2,3 mm)	A234020
16	**MV SHORT LOCKING POINT** WITH PA SHIM KIT (1,2,3 mm)	234321
	MV LONG LOCKING POINT WITH PA SHIM KIT (1,2,3 mm)	A234021
-	**SHORT LOCKING POINTS KIT** WITH PA SHIM KIT (1,2,3 mm)	234301
-	**LONG LOCKING POINTS KIT** WITH PA SHIM KIT (1,2,3 mm)	234302
-	**SHORT LOCKING POINT KIT** WITH ZM SHIM KIT (9 mm)	234305
-	**SHORT LOCKING POINT KIT** WITH ZM SHIM KIT (15 mm)	234306

-	**SHORT LOCKING POINT** WITH ZM SHIM (9 mm)	234035
-	**MV SHORT LOCKING POINT** WITH ZM SHIM (9 mm)	234036
-	**SHORT LOCKING POINT** WITH ZM SHIM (15 mm)	234037
-	**MV SHORT LOCKING POINT** WITH ZM SHIM (15 mm)	234038
-	**LONG LOCKING POINT** WITH ZM SHIM (9 mm)	234033
-	**MV LONG LOCKING POINT** WITH ZM SHIM (9 mm)	234034
-	**LONG LOCKING POINT** WITH ZM SHIM (15 mm)	234023
-	**MV LONG LOCKING POINT** WITH ZM SHIM (15 mm)	234024
-	**SCREW KIT** 7 x (4.8 x 19 mm) / 12 x (4.8 x 38 mm)	236007
-	**SCREW KIT** 7 x (4.8 x 32 mm) / 4 x (4.8 x 38 mm)	236008

(•) Reference root. See all available options





■ Optional accessories

■ ≥ 300 kg / sash configurations

1	**PULL HANDLE | FINGER PULL** *Consult the range of finger and pull handles*	
2	**FRONT BOGIE**	233001
3	**REAR BOGIE**	233002
4	**SUPPLEMENTARY BOGIE**	233003
-	**FIXED BOGIE**	234004
5	**CONNECTING RODS**	23301 (•)
6	**BOTTOM GUIDES** FOR CONNECTING RODS	237101
7	**HORIZONTAL KEEPER**	234022
8	**DRIVE GEAR** WITH LOCKING POINT	C23400 (•)
	DRIVE GEAR WITH LOCKING POINT (BOLT)	
9	**SHIM KIT (PA)** FOR DRIVE GEAR AND FRONT BOGIE	A2353 (•)
10	**EXTENSION ROD** WITH LOCKING POINT	A234011
	EXTENSION ROD WITH LOCKING POINT (BOLT)	234012
11	**TRIM** FOR EXTENSION ROD	234010

12	**DAMPING CYLINDER**	23700 (•)
13	**COMPENSATING SPRING**	237001
14	**DAMPING STOP**	23402 (•)
15	**FRAME KEEPER**	234042
-	**KIT WITH FRAME KEEPER**	234112
-	**PA SHIM KIT** (1,2,3 mm)	234203
-	**SHORT LOCKING POINT** WITH PA SHIM KIT (1,2,3 mm)	234320
	LONG LOCKING POINT WITH PA SHIM KIT (1,2,3 mm)	A234020
-	**MV SHORT LOCKING POINT** WITH PA SHIM KIT (1,2,3 mm)	234321
	MV LONG LOCKING POINT WITH PA SHIM KIT (1,2,3 mm)	A234021
-	**LONG LOCKING POINTS KIT** WITHOUT SHIM	234101
-	**LONG LOCKING POINTS KIT** WITH PA SHIM KIT (1,2,3 mm)	234108
-	**LONG LOCKING POINT KIT** WITH ZM SHIM KIT (9 mm)	234111
-	**LONG LOCKING POINT KIT** WITH ZM SHIM KIT (15 mm)	234102

-	**LONG LOCKING POINT KIT** WITH ZM SHIM KIT (15 mm)	234109
-	**LONG LOCKING POINT KIT** WITH PA SHIM KIT (1,2,3 mm)	234110
-	**SHORT LOCKING POINT** WITH ZM SHIM (9 mm)	234035
-	**MV SHORT LOCKING POINT** WITH ZM SHIM (9 mm)	234036
-	**SHORT LOCKING POINT** WITH ZM SHIM (15 mm)	234037
-	**MV SHORT LOCKING POINT** WITH ZM SHIM (15 mm)	234038
-	**LONG LOCKING POINT** WITH ZM SHIM (9 mm)	234033
-	**MV LONG LOCKING POINT** WITH ZM SHIM (9 mm)	234034
-	**LONG LOCKING POINT** WITH ZM SHIM (15 mm)	234023
-	**MV LONG LOCKING POINT** WITH ZM SHIM (15 mm)	234024
-	**SCREW KIT** FOR SHIM > 18 mm	236004
-	**SCREW KIT** FOR SHIM ≤ 18 mm	236005
-	**SCREW KIT**	236010

(•) Reference root. See all available options

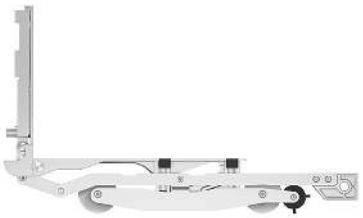
LS200 FRONT BOGIE

233501

1

IN

LS200



LS200 bogies suitable for profiles with 16 mm groove.

- It will support configurations of up to 200 kg per sash.
- Compatible with 18 / 22 mm groove by adaptation shims (included).
- For configurations of up to 250 kg, additional bogies must be added.
- Pozidrive head screws used for fixing to the profile.
- Especially reinforced resin wheels to support heavy loads.
- A rotational cleaning system synchronised by wheel movement.
- 1 and 2 mm shims are available to compensate for the height of the sash.

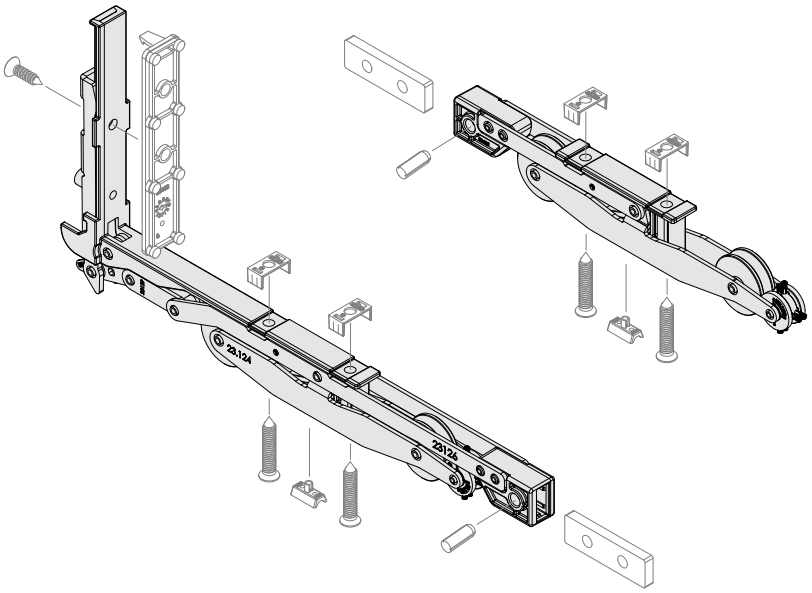
LS200 REAR BOGIE

233502

1

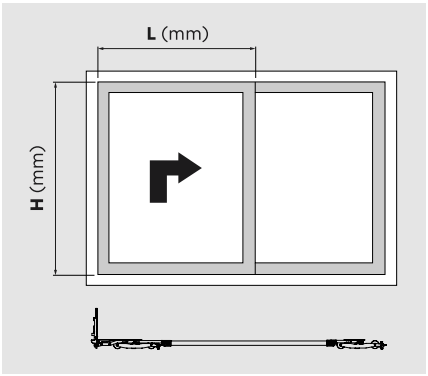
IN

LS200

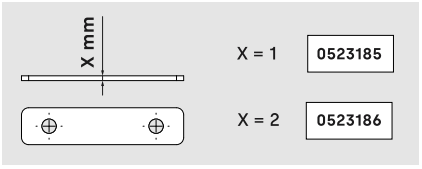


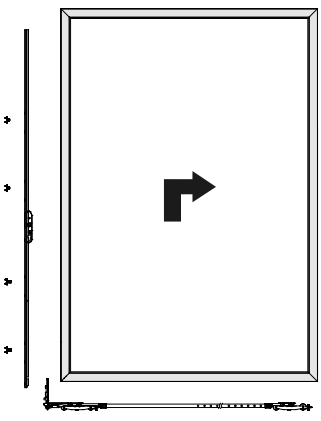
» **< 200 kg** CONFIGURATIONS

	MIN.	MAX.
L	550	3300
H	950	3300



1 - 2 mm SHIMS





LS200 bogies suitable for profiles with 16 mm groove.

- Compatible with 18 / 22 mm groove by adaptation shims (included).
- Pozidrive head screws used for fixing to the profile.
- 1 and 2 mm shims are available to compensate for the height of the sash (ref. 0523185 / ref. 0523186).

Supplementary bogie (ref. A233503)

- For 200 – 300 kg/sash configurations, additional bogies must be added.
- The supplementary bogies are supplied in an individual box with connection rods, screw kit and groove adaptation shims.

Bogie for fixed sash (ref. 233504)

- Allows the fixed sash to be positioned at the same height as the movable sash.
- For configurations with less than 200 kg/sash: 2 units.
- For configurations of up to 200 kg/sash: 4 units.

LS200 SUPPLEMENTARY BOGIE

IN

LS200

A233503

25

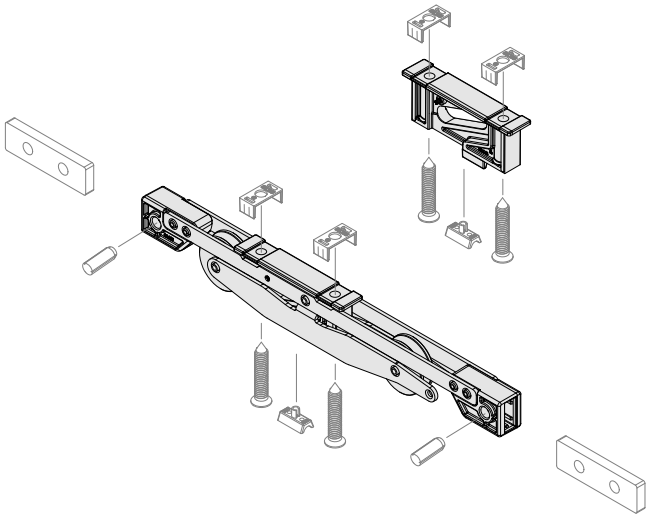
LS200 FIXED BOGIE

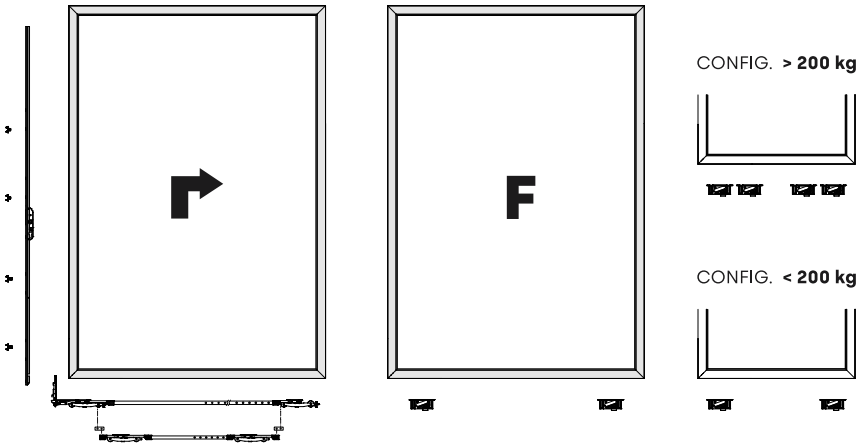
IN

LS200

233504

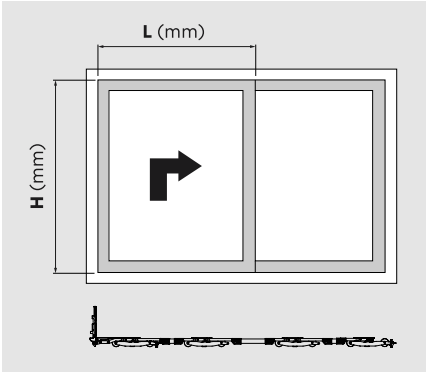
10





» **200 - 300 kg** CONFIGURATIONS

	MIN.	MAX.
L	1100	3300
H	950	3300



LS400 FRONT BOGIE

233001

1

IN

LS400

LS400 bogies suitable for profiles with 22 mm groove.

- Bogies support configurations of up to 300 kg per sash.
- For configurations of up to 400 kg, additional bogies must be added.
- Pozidrive head screws used for fixing to the profile.
- Especially reinforced resin wheels to support heavy loads.
- A rotational cleaning system synchronised by wheel movement.
- 2 mm shims are available to compensate for the height of the sash.

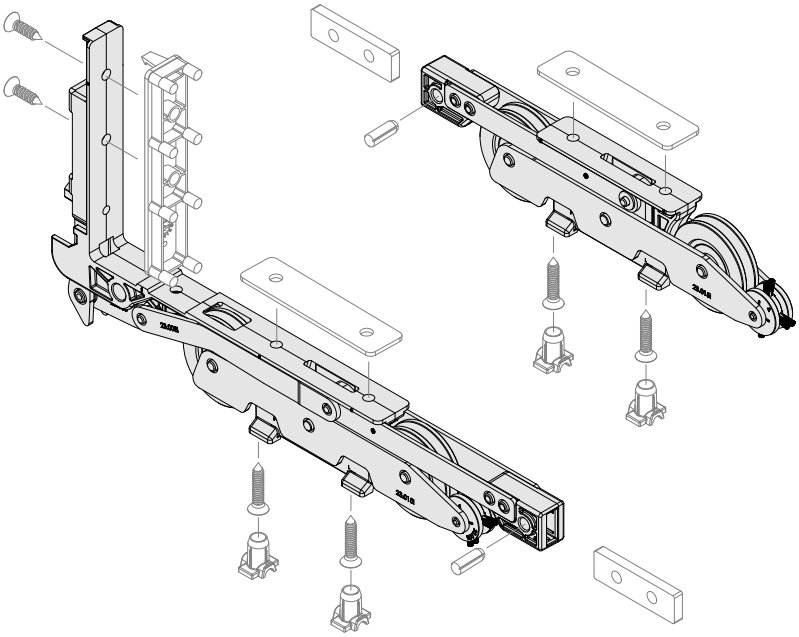
LS400 REAR BOGIE

233002

1

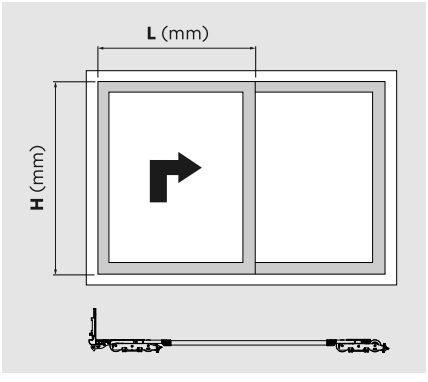
IN

LS400

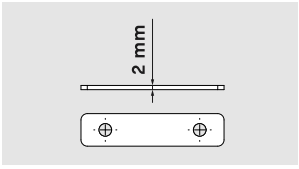


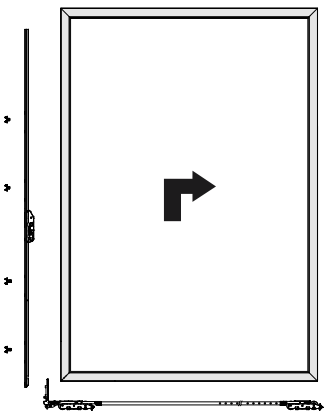
» **< 300 kg** CONFIGURATIONS

	MIN.	MAX.
L	550	3300
H	950	3300



2 mm SHIM 0523092





LS400 bogies suitable for profiles with 22 mm groove.

- Pozidrive head screws used for fixing to the profile.
- 2 mm shims are available to compensate for the height of the sash.

Supplementary bogie (ref. 233003)

- For 300 – 400 kg/sash configurations, 2 additional bogies must be added.
- The supplementary bogies are supplied in an individual box with connection rods and screw kit.

Bogie for fixed sash (ref. 233004)

- Allows the fixed sash to be positioned at the same height as the movable sash.
- For configurations with less than 300 kg/sash: 2 units.
- For configurations of up to 300 kg/sash: 4 units.

LS400 SUPPLEMENTARY BOGIE

IN

LS400

233003

2

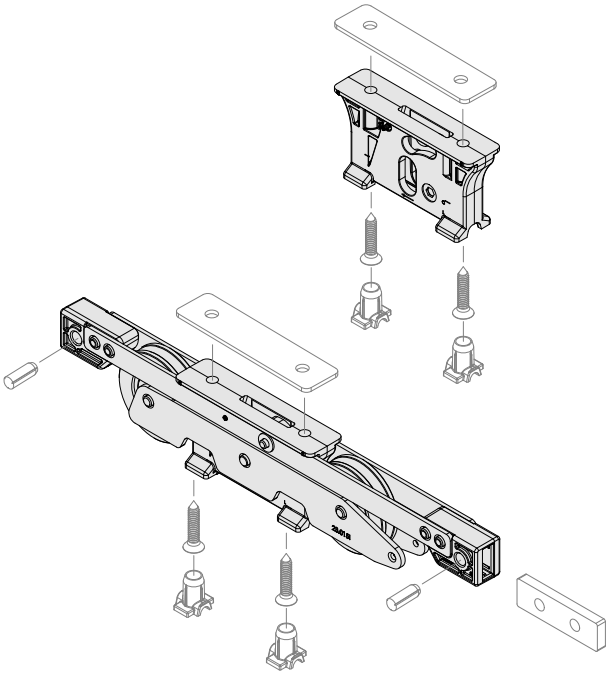
LS400 FIXED BOGIE

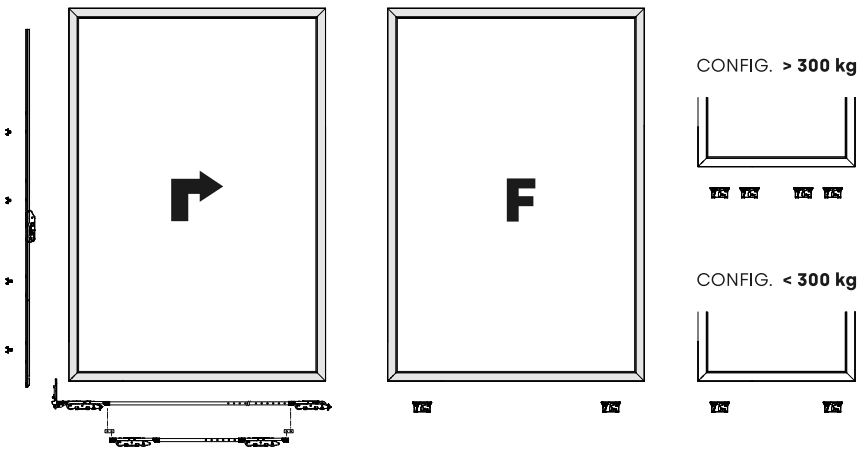
IN

LS400

233004

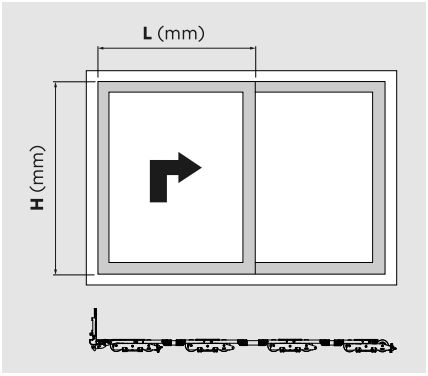
10





» **300 - 400 kg** CONFIGURATIONS

	MIN.	MAX.
L	1100	3800
H	950	3300



A27.5 DRIVE GEAR

C23450

10

AL

ZM

LS200

Drive gear with 27.5 and 37.5 mm backsets.

- 27.5 backset for LS200 lift & slide system.
- 37.5 backset for LS400 lift & slide system.
- The distance between fixing centres of the drive gears is 80 mm.
- The drive gears are delivered with European cylinder pre-installation.
- Bolts: adjustment system that allows an adjustment of -3, 5 millimetres.
- Compatible with the extension rods with locking point (ref. A234011, ref. 234012) for configurations with a sash height greater than 2800 millimetres.

A37.5 DRIVE GEAR

C23400

10

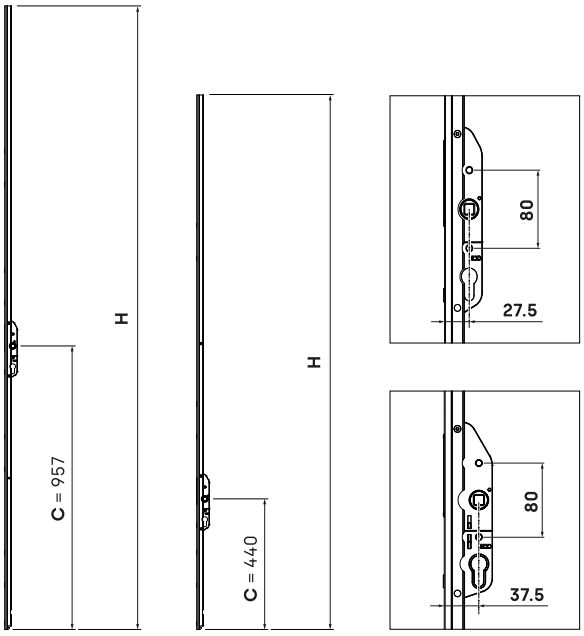
AL

ZM

LS400

DRIVE GEARS **WITH LOCKING POINTS IN THE FRAME PROFILE**

	DRIVE GEAR BACKSET	DRIVE GEAR HEIGHT (H)	SPINDLE HEIGHT (C)
C234501	27.5	2100	957
C234502		2400	957
C234503		2700	957
C234504		1800	440
C234001	37.5	2100	957
C234002		2400	957
C234003		2700	957
C234004		1800	440

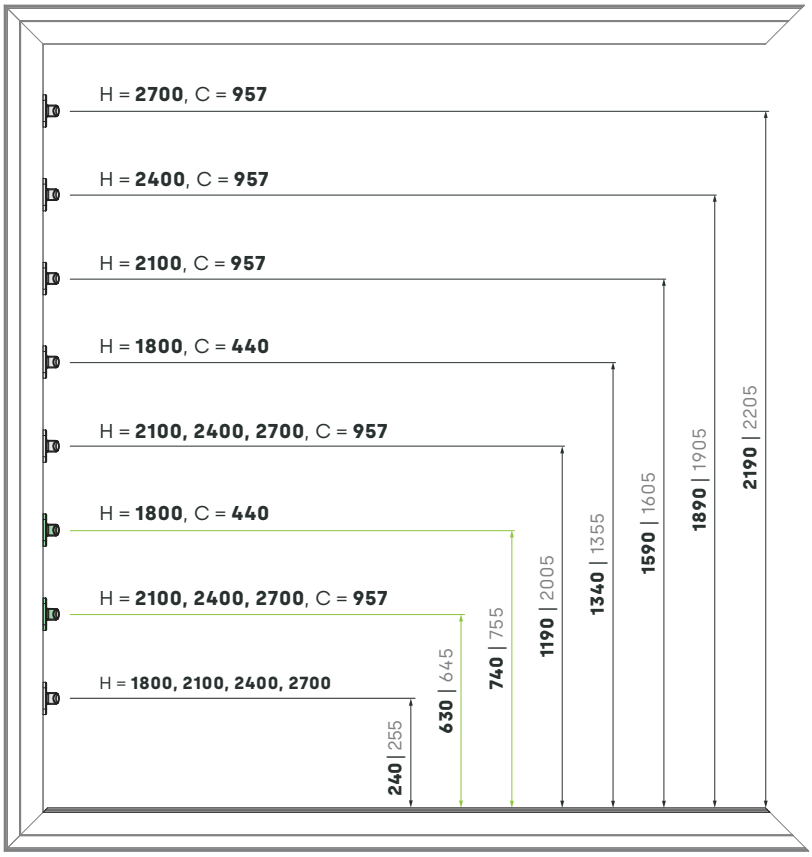


DRIVE GEARS **WITH LOCKING POINTS IN THE SASH PROFILE**

	DRIVE GEAR BACKSET	DRIVE GEAR HEIGHT (H)	SPINDLE HEIGHT (C)
C234505	27.5	1800	440
C234506		2100	957
C234507		2400	957
C234508		2700	957
C234005	37.5	1800	440
C234006		2100	957
C234007		2400	957
C234008		2700	957

» INSTALLATION OF THE LOCKING POINTS ON THE FRAME PROFILE

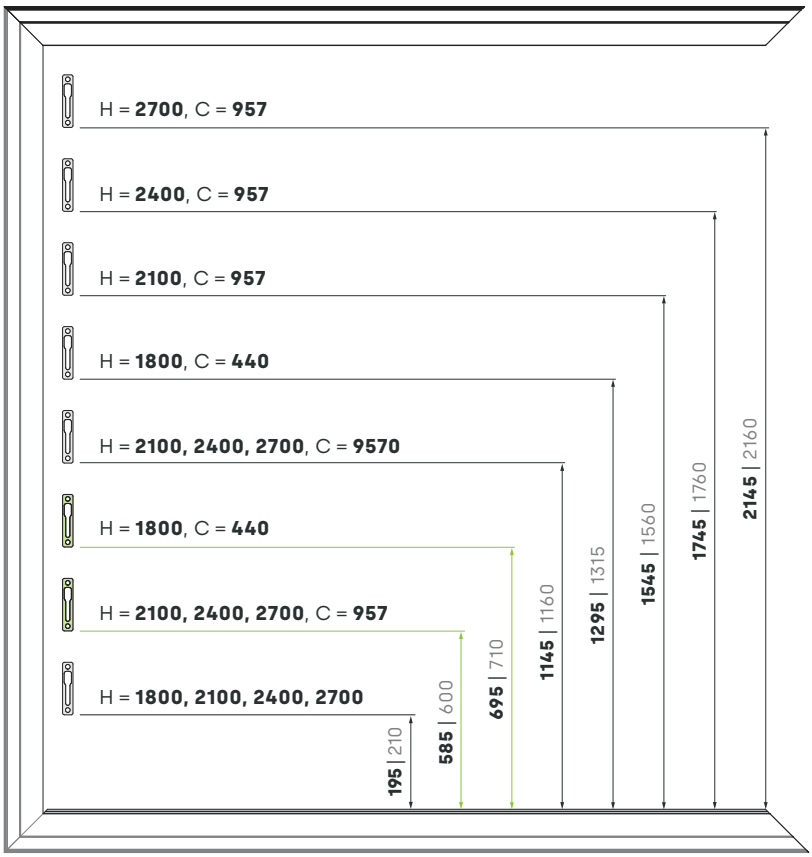
CONFIGURATIONS WITH LOCKING POINTS WITHOUT ADJUSTMENT



	H (mm)	C (mm)	
C234501	2100	957	27.5 mm
C234502	2400	957	
C234503	2700	957	
C234504	1800	440	
C234001	2100	957	37.5 mm
C234002	2400	957	
C234003	2700	957	
C234004	1800	440	

■ LS200 LOCKING POINT HEIGHT
□ LS400 LOCKING POINT HEIGHT
■ OPTIONAL LOCKING POINTS

CONFIGURATIONS WITH LOCKING POINTS WITH ADJUSTMENT



	H (mm)	C (mm)	
C234505	1800	957	27.5 mm
C234506	2100	957	
C234507	2400	957	
C234508	2700	440	
C234005	1800	957	37.5 mm
C234006	2100	957	
C234007	2400	957	
C234008	2700	440	

■ LS200 LOCKING POINT HEIGHT
□ LS400 LOCKING POINT HEIGHT
■ OPTIONAL LOCKING POINTS

SHIMS KIT (LS200)

A2353

1

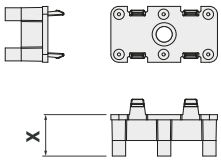
PA

LS200

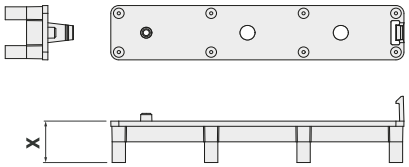
Plastic shims for drive gears and LS200 lift & slide front bogie.

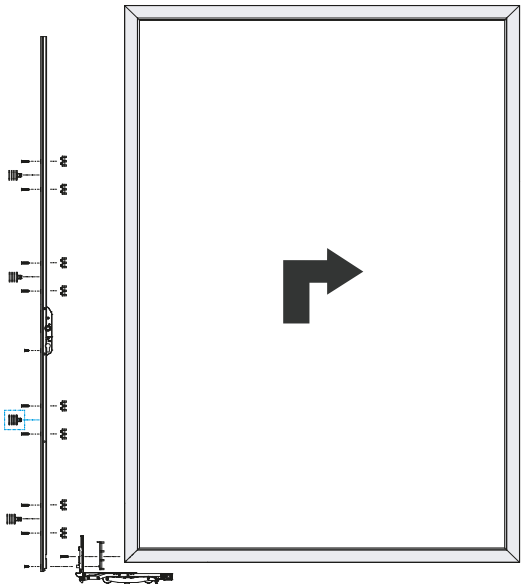
- Made of PA6 polyamide with 10% of fibreglass.
- Available shim width: from 4 to 23 mm.
- Shims allow the thermal break to be preserved.
- Supplied with 8 shims for the drive gear and 1 shim for the front bogie.

SHIM
FOR DRIVE GEAR



SHIM
FOR LS200 FRONT BOGIE





SHIM KITS

	WIDTH (X mm)		WIDTH (X mm)
A235309	4	A235307	14
A235310	5	A235301	16
A235308	6	A235306	17
A235311	7	A235303	17.5
A235312	8	A235302	19
A235313	9	A235304	20
A235314	10	A235305	23
A235315	12		

SHIMS FOR DRIVE GEARS

	WIDTH (X mm)		WIDTH (X mm)
A235409	4	A235407	14
A235410	5	A235401	16
A235408	6	A235406	17
A235411	7	A235403	17.5
A235412	8	A235402	19
A235413	9	A235404	20
A235414	10	A235405	23
A235415	12		

Plastic shims for drive gears and LS400 lift & slide front bogie.

- Made of PA6 polyamide with 10% of fibreglass.
- Available shim width: from 12 to 21 mm.
- Shims allow the thermal break to be preserved.
- Supplied with 8 shims for the drive gear and 1 shim for the front bogie.

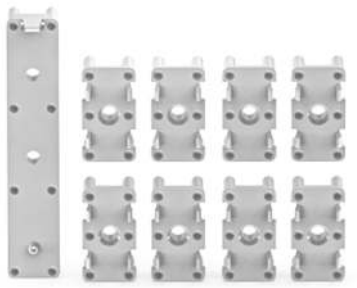
SHIMS KIT (LS400)

PA

LS400

A2352

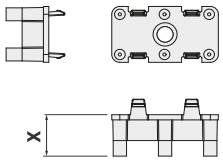
1



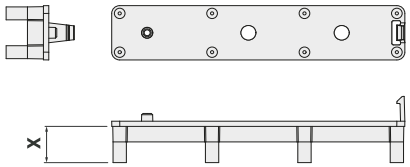
SHIM KITS

	WIDTH (X mm)		WIDTH (X mm)
A235207	12	A235202	17
A235201	14	A235204	18
A235206	15	A235205	21
A235203	15.5		

SHIM
FOR DRIVE GEAR

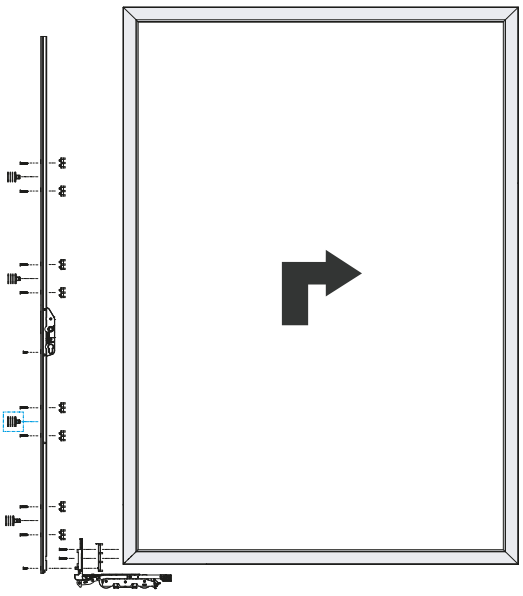


SHIM
FOR LS400 FRONT BOGIE



SHIMS FOR DRIVE GEARS

	WIDTH (X mm)		WIDTH (X mm)
A235007	12	A235002	17
A235001	14	A235004	18
A235006	15	A235005	21
A235003	15.5		



EXTENSION ROD

A234011

10

AL

ZM

LS200

LS400

The extension rods, in combination with the drive gear trim cut to size between the two fittings, provide an additional locking point to any drive gear.

- Additional locking point in configurations with heights above 2800 mm and can reach sash heights of up to 3300 mm.
- The extension rod is fixed by means of a grub screw and fixing to the profile is by means of screws with Pozidrive head.
- Zinc-plated steel parts to guarantee a corrosion resistance of 240 hours in salt spray chamber (UNE-EN 1670).

EXTENSION ROD

234012

10

AL

ZM

LS200

LS400

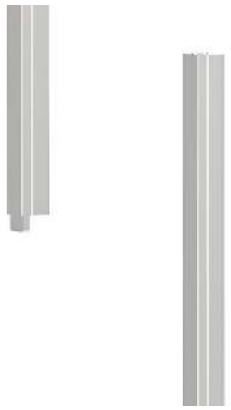
CONFIGURATIONS		X	X min	X max
A234011		940	-	-
A234012		940	-	-
A234011	234010	-	> 940	≤ 1430
234012	234010	-	> 940	≤ 1430
A234011	234010	-	≥ 500	< 940
234012	234010	-	≥ 500	< 940

*(**X**) DISTANCE BETWEEN LOCKING POINTS*

TRIM ROD

234010

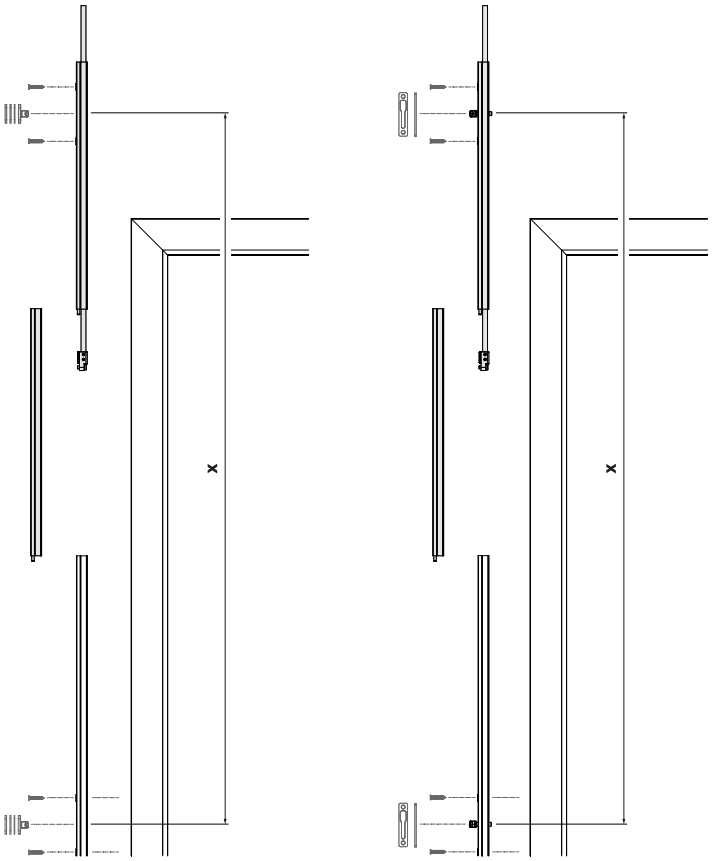
10



AL

LS200

LS400



Steel plates for connecting the bogies to the lift & slide systems.

- The plates are fixed to the bogies by means of steel pins.
- The multiple perforation allows them to be adapted to a wide range of leaf widths.

Centring shims

- They prevent buckling and vibrations of the carriage connecting plate during the transmission of movement.
- 3 shims are supplied in the kits with locking points: ref. 234101, ref. 234103, ref. 234302 and ref. 234303.

CONNECTING RODS

AC

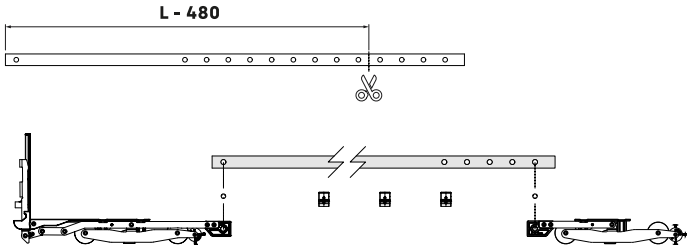
LS200

LS400

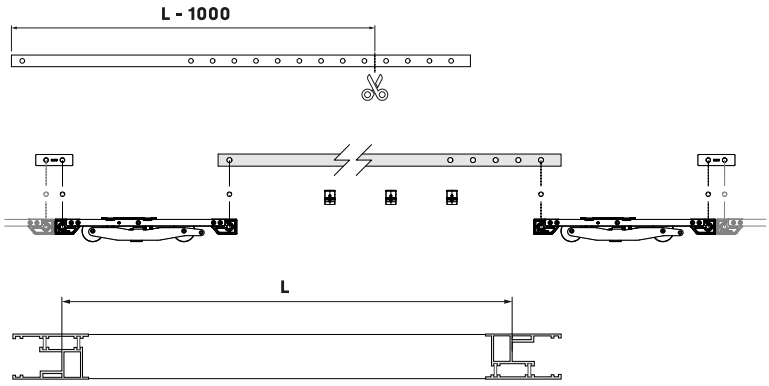
23301

10

CONFIGURATION WITHOUT SUPPLEMENTARY BOGIES



CONFIGURATION WITH SUPPLEMENTARY BOGIES



CENTRING SHIM

LS200

LS400

237101

200

	ROD LENGTH	SASH WIDTH W/ SUPPLEMENTARY BOGIES	SASH WIDTH WO/ SUPPLEMENTARY BOGIES	BOTTOM GUIDE
233010	2760 mm	2301 ~ 2300 mm	2801 ~ 3800 mm	6
233011	1760 mm	1801 ~ 2300 mm	2301 ~ 2800 mm	3
233012	1260 mm	1301 ~ 1800 mm	1301 ~ 2300 mm	2
233013	755 mm	600 ~ 1300 mm	1100 ~ 1600 mm	1

SOFT LIFT DEVICE

23700

1

Damping and compensation systems for drive gears.

SOFT LIFT device

Damping system that allows the handle to be moved to the closing position in a smooth and intuitive way.

- Compatible with 2100, 2400 and 2700 mm drive gears.
- It avoids sudden impacts of the handle and considerably increases the durability of the lifting mechanism.

Compensation spring (ref. 237001)

Allows the weight to be compensated by reducing the sash's lifting force.

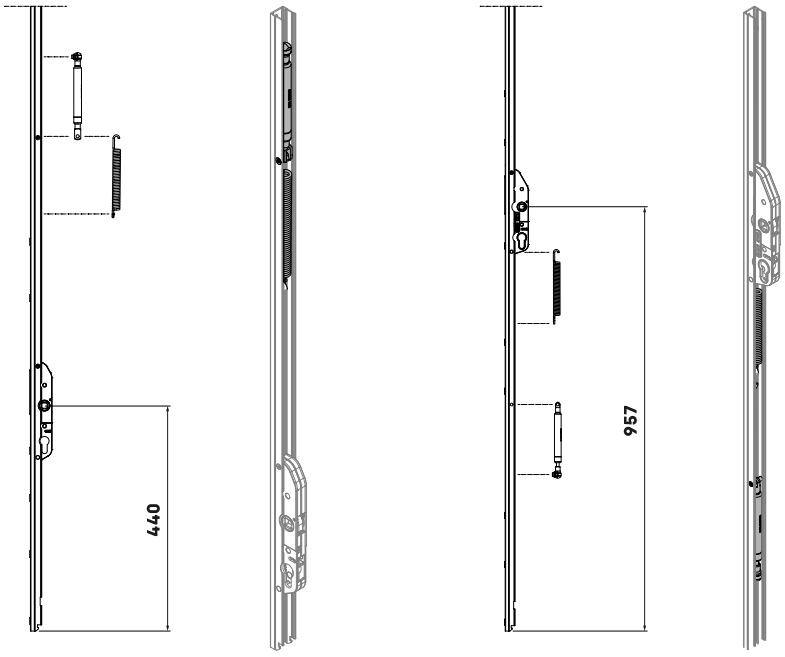
- Minimises the return torque of the door handle when lowering the sash.
- Recommended for sash weights of 200 kg and above.

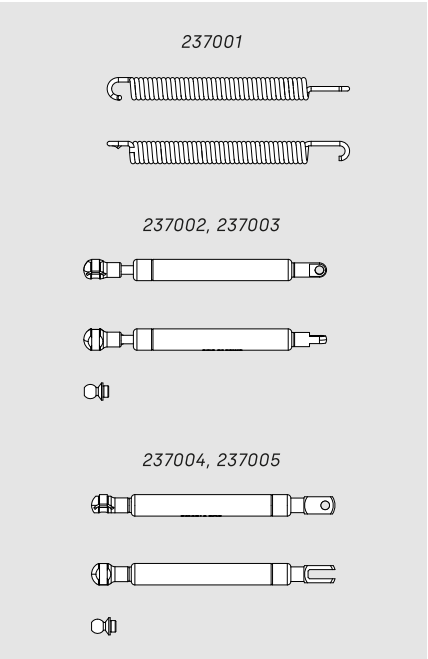
SPRING COMPENSATOR

237001

1

	WEIGHT PER SASH (kg)	RODS (mm)	COMPATIBILITY
237002	150 ~ 250	2100 ~ 2700	LS200
237003	251 ~ 400		LS400
237004	150 ~ 250	1800	LS200
237005	251 ~ 400		LS400





Active security system designed especially for public zones and areas where there are children. The system allows for damping braking of the sash regardless of the force applied to it, preventing possible entrapment.

- Compatible with LS400 lift & slide system.
- Maximum sash weight allowed 300 kg and position adjustment of ± 8 mm.
- It includes adaptation shims for different profile series (aluminium, wood and PVC).

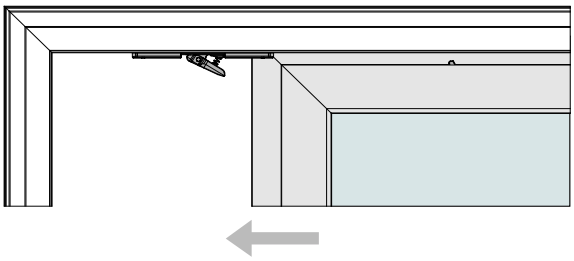
SOFT CLOSING DEVICE

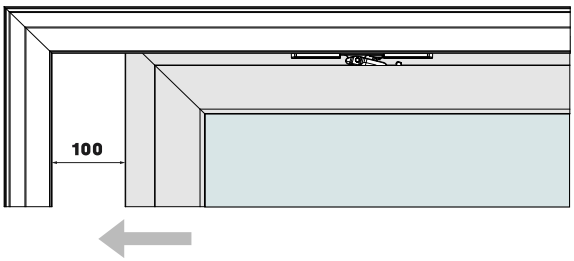
LS400

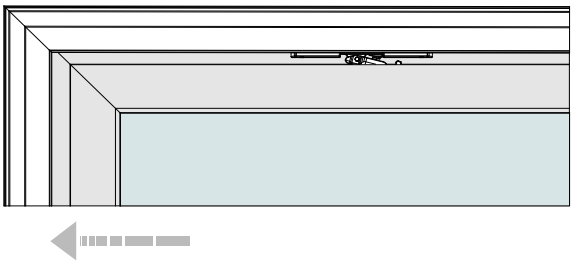
237007

1

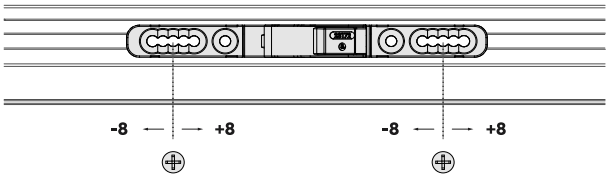
SASH CLOSING
DAMPED AND SMOOTH

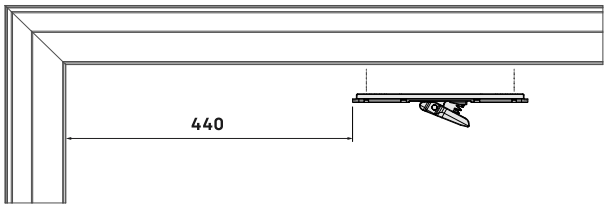


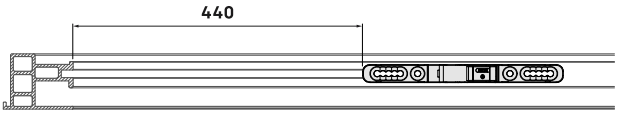




ADJUSTMENT
± 8 mm







KEEPER (234022)

234022

100

ZM

LS200

LS400

Frame keepers for lift & slide systems.

- The keepers are adjustable to ± 3 mm.
- Fixing to the profile by means of Pozidrive screws.
- The parts have been zinc-plated to guarantee a corrosion resistance of 240 hours in a saline mist chamber in accordance with the UNE-EN 1670 standard.

Keeper (ref. 234022)

- Side retention device with adjustment for intermediate sash.
- Supplied with screw DIN 7982 4.8 x 25 mm.

Keeper (ref. 234322)

- Side retention device to allow aeration position.
- Supplied with screw DIN 7982 4.2 x 22 mm.

KEEPER (234322)

Microventilation opening

234322

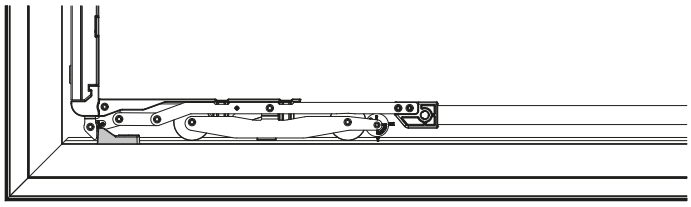
100

ZM

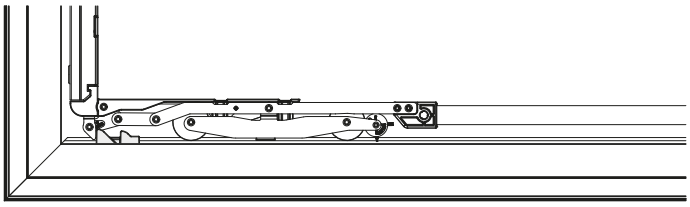
LS200

LS400

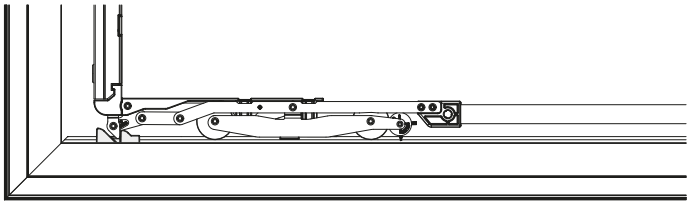
	ADJUSTEMENT	AERATION	COMPATIBILITY
234022	± 3 mm	-	LS200, LS400
234322	± 3 mm	✔	LS200, LS400



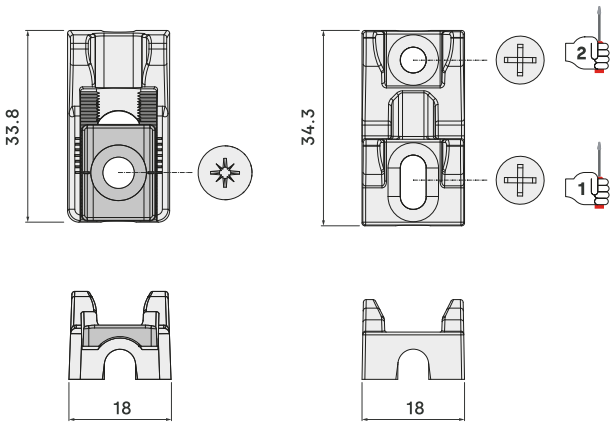
234022 CLOSING POSITION



234322 CLOSING POSITION



234322 MICRO-VENTILATION POSITION



DAMPING STOP

Rubber stop that absorbs sash impact.

- Fixed to the profile using an anti-rotation screw and bolt.
- One unit to be fitted for every two sashes.
- Supplied bagged and packed in individual boxes together with screws.
- Available in black finish (ref. 234026).

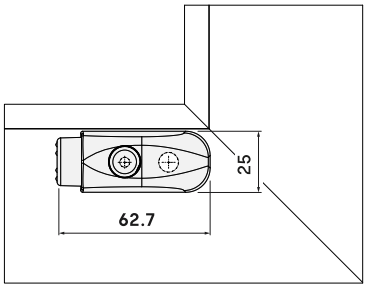
234025

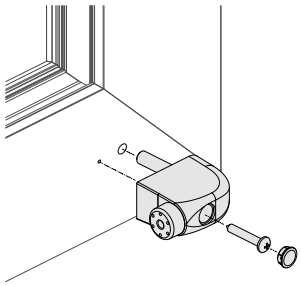
10

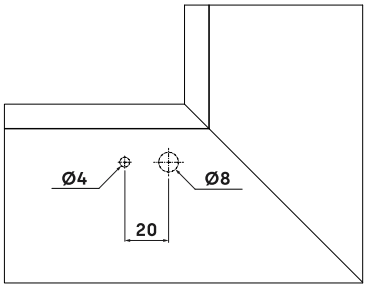
PA

LS200

LS400







SCREWS KITS

2360

LS200

LS400

		SYSTEM	4.8 x 19	4.8 x 32	4.8 x 38	4.8 x 50	4.8 x 55	4.8 x 60	
236004	*	LS400	1 unit	12 units	-	8 units	-	-	100
236005	*	LS400	1 unit	12 units	-	-	-	8 units	250
236007		LS200	7 units	-	12 units	-	-	-	100
236008		LS200	-	7 units	4 units	8 units	-	-	100
236010		LS400	1 unit	12 units	-	-	8 units	-	100

* ***236004*** *for ≤ 18 mm shims,* ***236005*** *for > 18 mm shims.*

SHORT LOCKING POINT WITH SHIM

9 mm zamak shim

234035

10

ZM

IN

LS200

LS400

Short stainless steel locking points for straight cut profile series.

- Supplied with standard locking point or with micro-ventilation and with a 9 or 15 mm zamak shim.
- The kits are suitable for LS200 and LS400 lift & slide systems.
- Fixing to the profile by means of Pozidrive screws.

The short MV locking point enables the micro-ventilation position on the lift & slide system.

SHORT MV LOCKING POINT WITH SHIM

9 mm zamak shim

234036

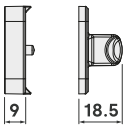
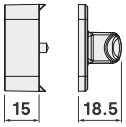
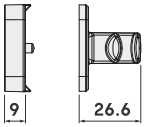
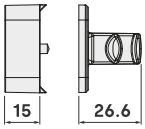
10

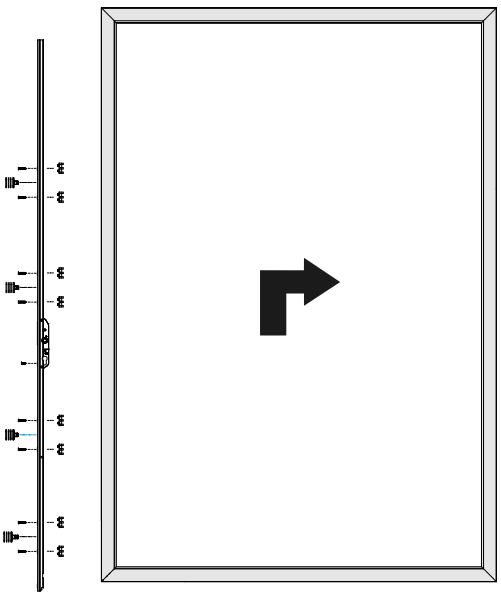
ZM

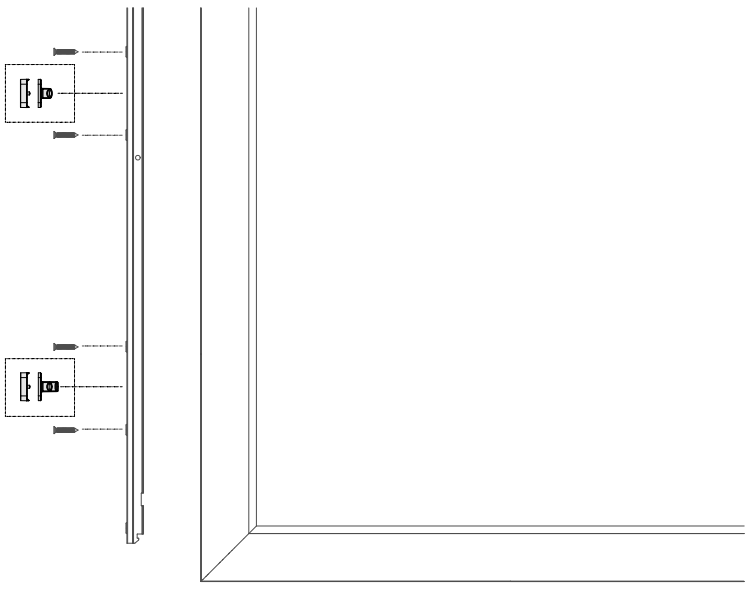
IN

LS200

LS400

		AERATION	SHIM
234035		-	9 mm
234037		-	15 mm
234036		✔	9 mm
234038		✔	15 mm

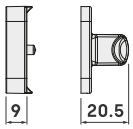
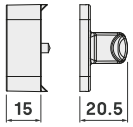
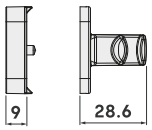
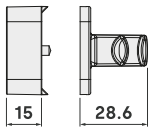




Long stainless steel locking points for straight cut profile series.

• Supplied with standard locking point or with micro-ventilation and with a 9 or 15 mm zamak shim.
• The kits are suitable for LS200 and LS400 lift & slide systems.
• Fixing to the profile by means of Pozidrive screws.

The long MV locking point enables the micro-ventilation position on the lift & slide system.

		AERATION	SHIM
234033		-	9 mm
234023		-	15 mm
234034		✔	9 mm
234024		✔	15 mm

LONG LOCKING POINT WITH SHIM
9 mm zamak shim

ZM
IN
LS200
LS400

234033

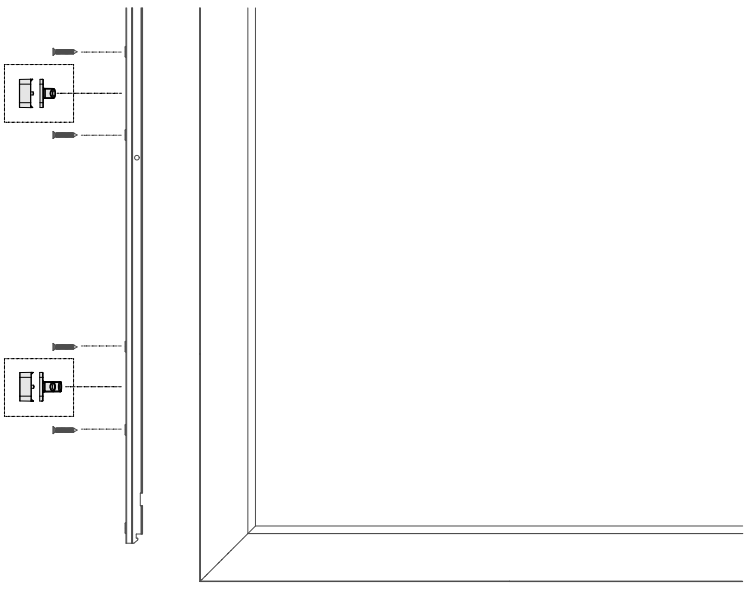
10

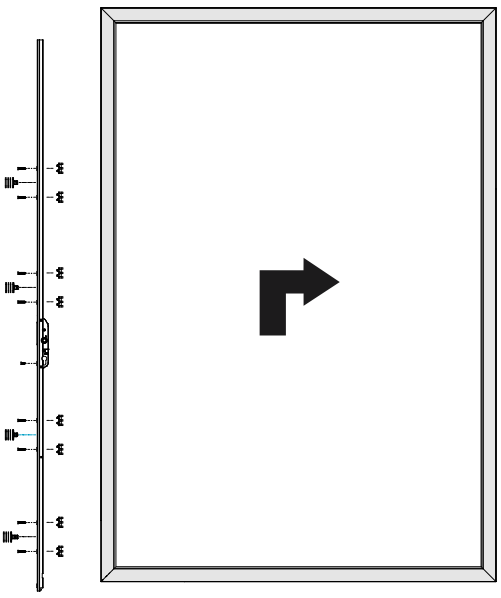
LONG MV LOCKING POINT WITH SHIM
9 mm zamak shim

ZM
IN
LS200
LS400

234034

10





SHORT LOCKING POINTS KIT
15 mm zamak shim

234306

10

IN PA ZM PM

LS200

SHORT LOCKING POINTS KIT
9 mm zamak shim

234305

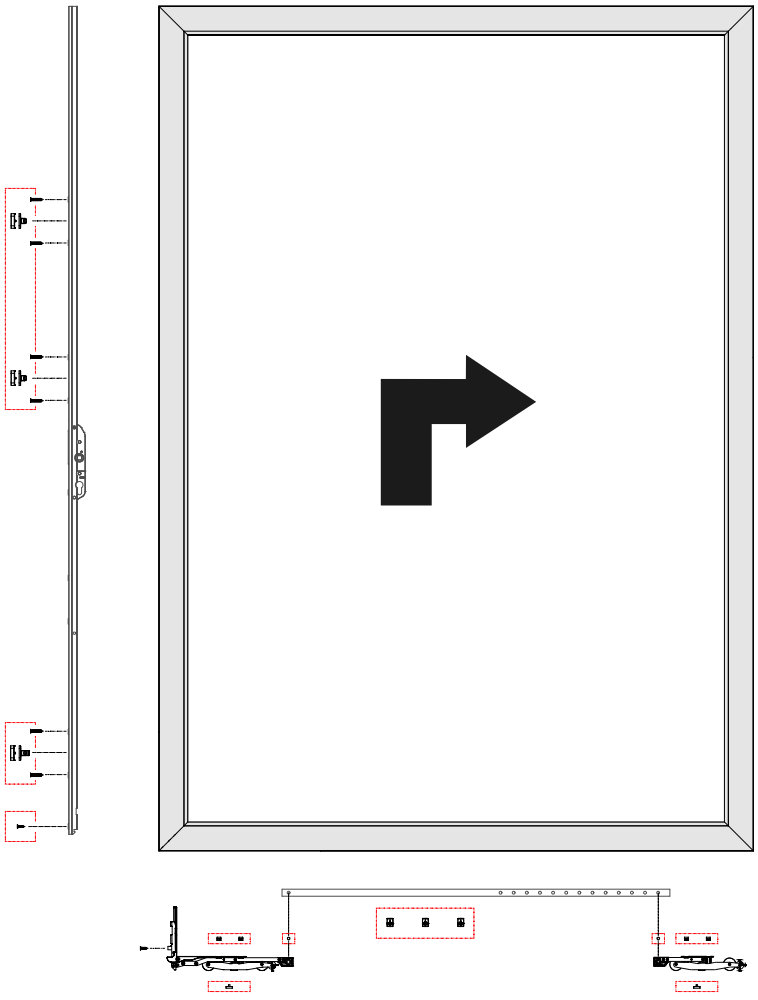
10

IN PA ZM PM

LS200

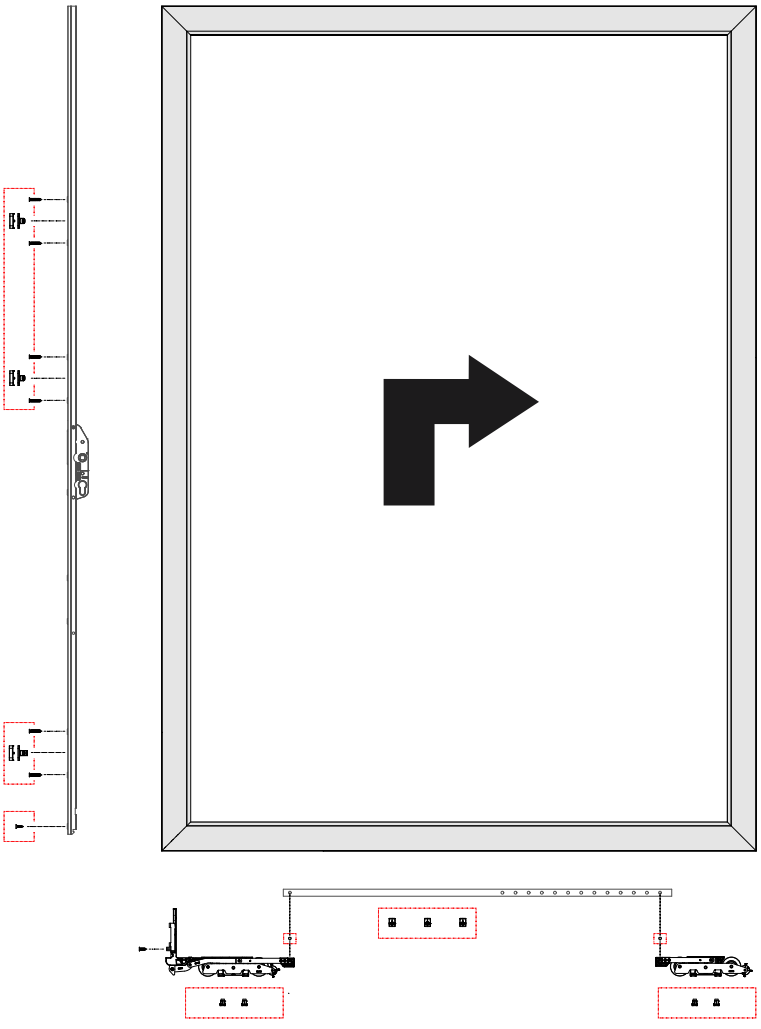
Kits with short stainless steal locking points compatible with LS200 lift & slide.

- Supplied with two short locking points, a locking point for micro ventilation and 9 mm (ref. 234305) or 15 mm(ref. 234306) zamak shims.
- The kits include various accessories for the lift & slide bogies.



Kits with long stainless steel locking points compatible with the LS400 lift & slide system.

- Supplied with two long locking points, one locking point for micro ventilation and 6 mm (ref. 234102), 9 mm (ref. 234109) or 15 mm (ref. 234111) zamak shims.
- The kits include different accessories for the bogies of the lift & slide system.



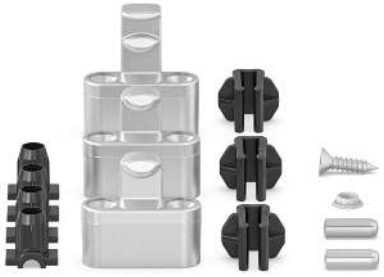
LONG LOCKING POINTS KIT
15 mm zamak shim

ZM IN PA PM

LS400

234102

10



LONG LOCKING POINTS KIT
9 mm zamak shim

IN PA AL PM

LS400

234109

10

LONG LOCKING POINTS KIT
6 mm zamak shim

IN PA ZM PM

LS400

234111

10

FRAME PROFILE KEEPER

234042

126

ZM IN PA PM

LS200 LS400

Stainless steel frame keeper for drive gears with bolts.

- Single sash kits are available specific to each system.
- Supplied with different accessories for the bogies.

FRAME PROFILE KEEPER KIT

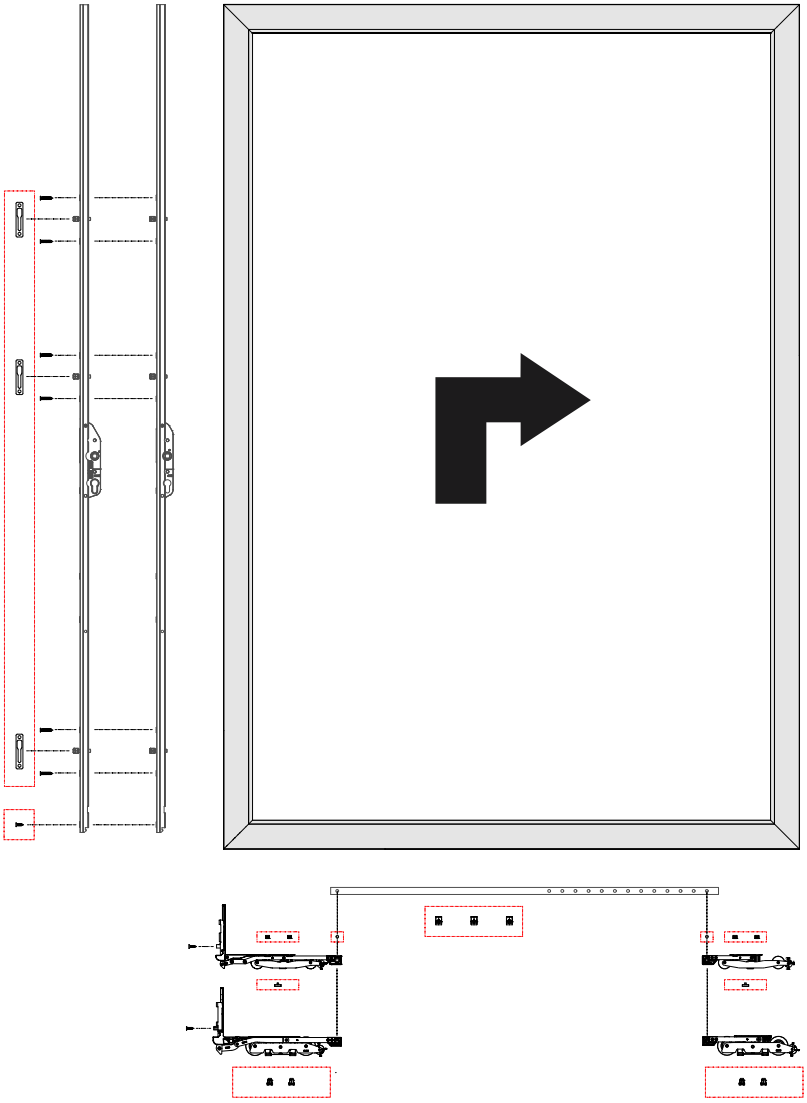
LS200 lift & slide system

234308

40

ZM IN PA PM

LS200



FRAME PROFILE KEEPER KIT

LS400 lift & slide system

234112

80

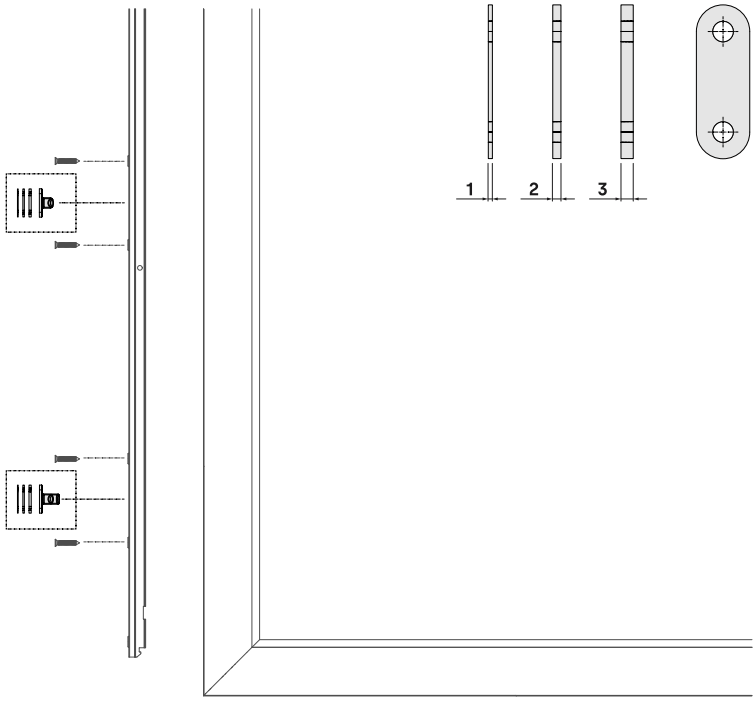
ZM IN PA PM

LS400

1, 2 and 3 mm plastic shims made of PA6 polyamide with 10% glass fibre to adapt the locking points to different types of profiles.

- Valid for kits with locking points LS200 and LS400.
- Micro-ventilation locking points available (ref. 234321, ref. A23402).
- Weight reduction compared to 9 mm and 15 mm zamak shims.

		AERATION	SHIM
234320	18.5	-	234203
A234020	20.5	-	234203
234321	26.6	✔	234203
A234021	28.6	✔	234203



SHIM KIT (PA)

PA
LS200
LS400

234203

50

SHORT LOCKING POINT WITH SHIMS (PA)

Polyamide shims

IN
PA
LS200
LS400

234320

200

LONG LOCKING POINT WITH SHIMS (PA)

Polyamide shims

IN
PA
LS200
LS400

A234020

50

SHORT LOCKING POINTS KIT (PA)

Polyamide shims

234301

40

ZM IN PA PM

LS200

LONG LOCKING POINTS KIT (PA)

Polyamide shims

234302

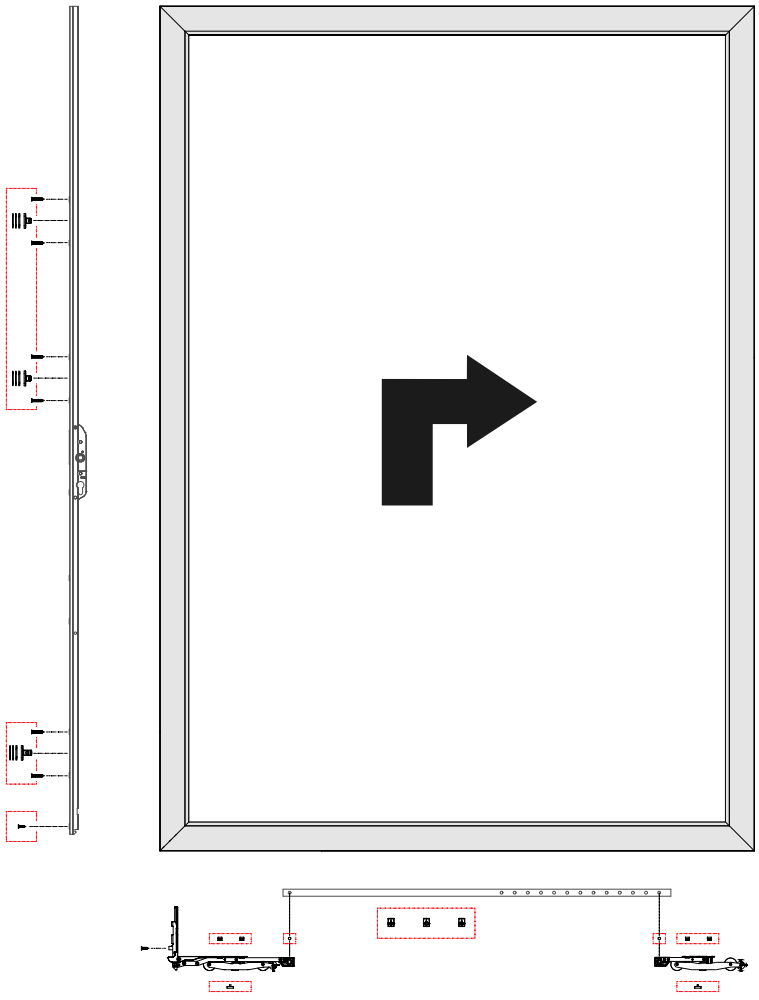
40

ZM IN PA PM

LS200

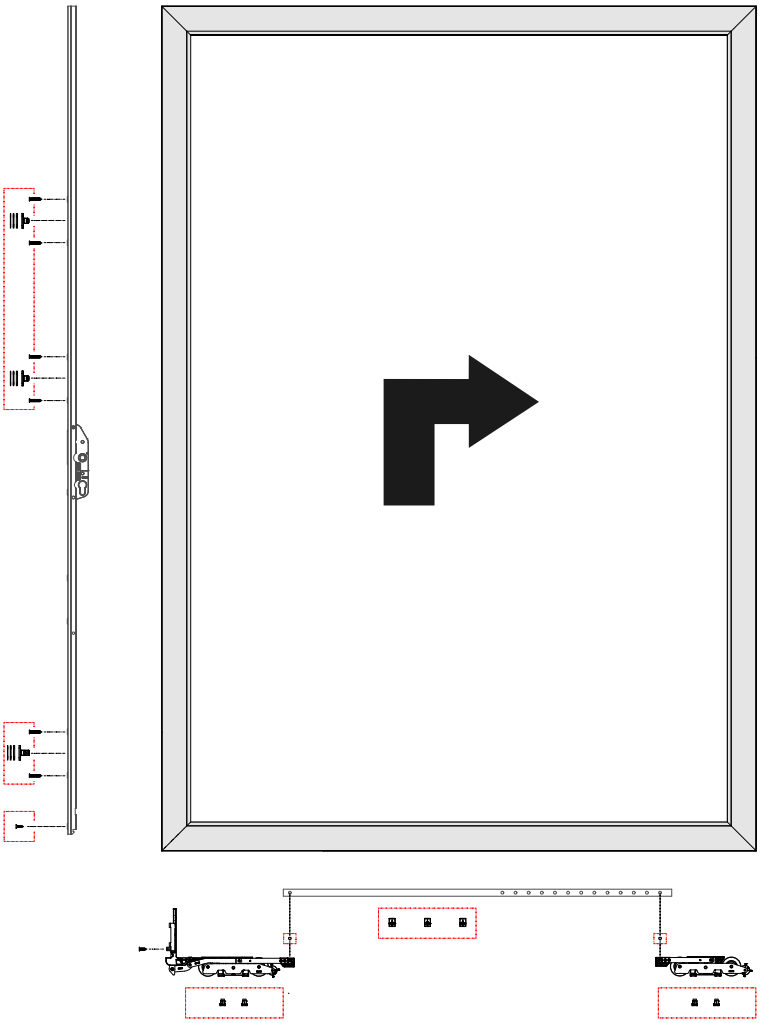
Kits with short stainless steel locking points with polyamide shims compatible with the LS200 lift & slide system.

- Supplied with two short (ref. 234301) or long (ref. 234302) locking points, a locking point for micro-ventilation and polyamide shims of 1, 2 and 3 millimetres.
- The kits include different complements for the bogies of the lift & slide system.



Kits with long stainless steel locking points compatible with the LS400 lift & slide system.

- Supplied with two long locking points and one locking point for micro-ventilation.
- Kit available with 1, 2 and 3 mm polyamide shims (ref. 234108).
- Kit with spike-shaped locking points available (ref. 234110).
- The kits include different complements for the bogies of the lift & slide system.



LONG LOCKING POINTS KIT
Without shims

ZM IN PA PM

LS400

234101

10

LONG LOCKING POINTS KIT (PA)
Polyamide shims

IN PA ZM PM

LS400

234108

10

LONG LOCKING POINTS KIT (PA)
Polyamide shims

IN PA ZM PM

LS400

234110

10

INTERMEDIATE SASH KIT
LS200 lift & slide system

234303

LS200

Kit of elements for intermediate sashes that do not require locking points.

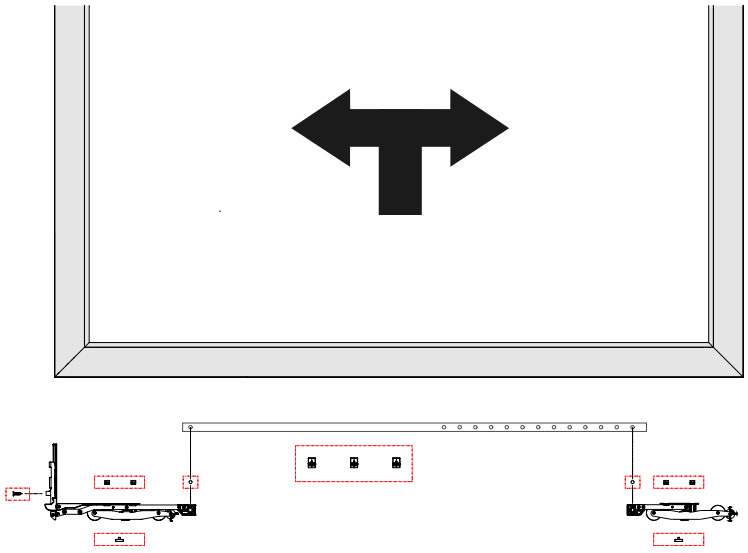
• Specific single sash kits are available for LS200 (ref. 234303) and LS400 (ref. 234103) lift & slide systems.

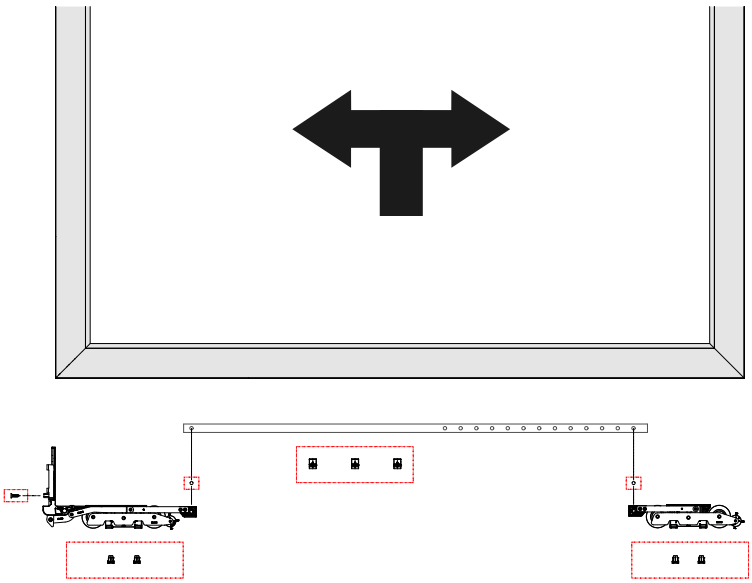
INTERMEDIATE SASH KIT
LS400 lift & slide system

234103

10

LS400





14 PASSIVE SASH

342 PS shootbolt levers

343 Two-way shootbolt levers

346 Shootbolt levers

348 Extendable shootbolt levers

349 Transmission rod levers

350 Passive sash accessories

351 Sash perimeter lockings

352 Door flush bolts

354 Door flush bolts accessories

HP SHOOTBOLT LEVERS KIT

111630

20

Locking kits for Euro Groove passive sash with universal shootbolts.

- Valid for tilt & turn openings with locking to inverter or locking to frame.
- Compatible with standard or eccentric locking points (EVO SECURITY).
- The terminals are made of stainless steel, providing greater strength and surface finish.
- They allow direct clipping of transmission rods for very large door sash dimensions and are the ideal solution for balcony door profiles.

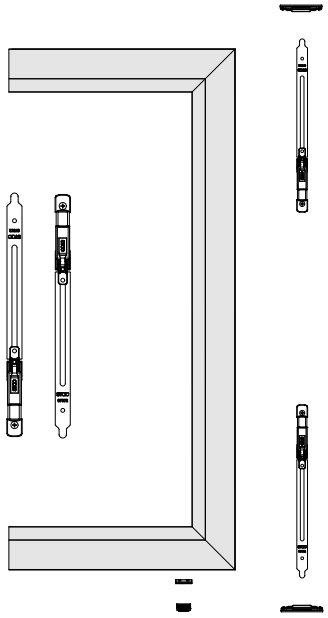
Two-way opening kit (ref. 111631)

- Supplied with two-way lever which allows both terminals to be opened or closed in a single action.
- The terminals are fitted directly onto the rod plate without the need for adaptation parts.

TWO-WAY HP SHOOTBOLT LEVER KIT

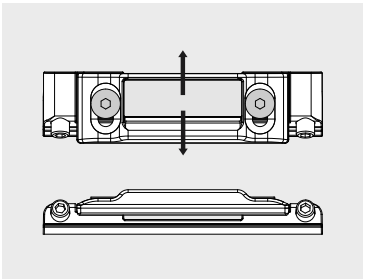
111631

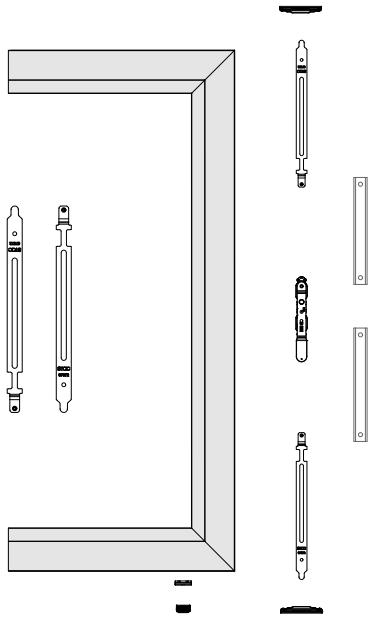
20



ADJUSTMENT ± 2.5

110501





Locking kits for passive sash with Euro Groove with two-way shootbolt lever.

- Supplied with zamak pointers, two-way lever, adjustable keepers and latch kit (sash and frame).
- Two-way lever drives the pointers by means of transmission rods.
- Pointers incorporate locking points tightening adjustment (± 1.5 mm).

Two-way shootbolt lever kit (ref. 111601)

- Valid for tilt & turn windows with locking to inverter.

Two-way shootbolt lever kit (ref. 111602)

- Valid for tilt & turn windows with locking to frame.

TWO-WAY SHOOTBOLT LEVER KIT
Locking to inverter

111601

20

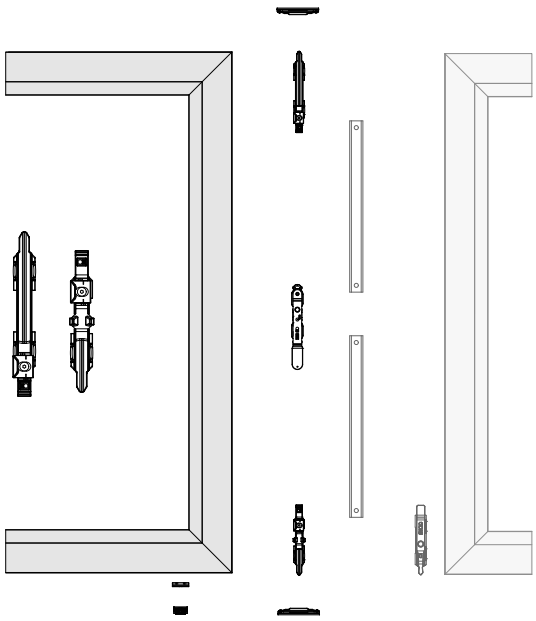
PV ZM IN PA

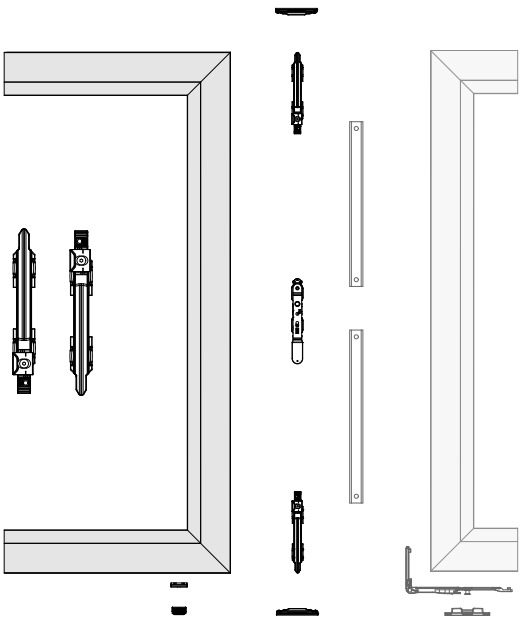
TWO-WAY SHOOTBOLT LEVER KIT
Locking to frame

111602

20

ZM IN PA PV

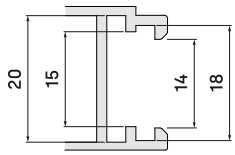




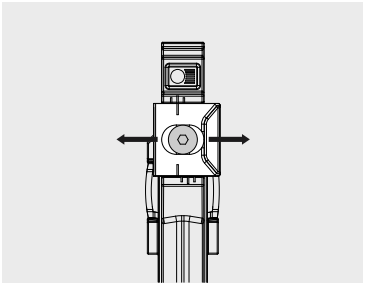
EURO GROOVE

FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

SASH~ FRAME INVERTER PROFILE

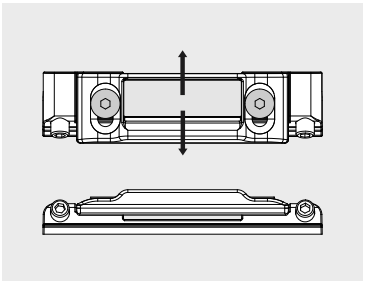


ADJUSTMENT ± 1.5



ADJUSTMENT ± 2.5

110501



TWO-WAY SHOOTBOLT LEVER KIT WITH INOX TERMINALS

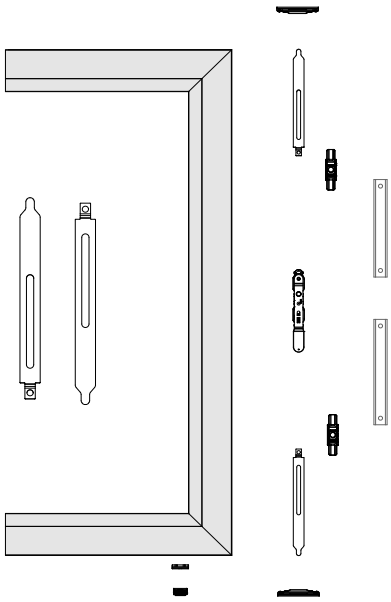
111621

20

ZM IN
PA PV

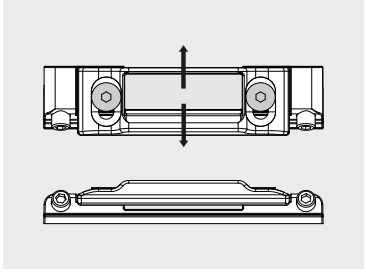
Locking kits for passive sash with Euro Groove with two-way shootbolt lever.

- Valid for tilt & turn windows with locking to inverter or locking to frame.
- Two-way lever drives the pointers by means of transmission rods.
- Supplied with stainless steel pointers, two-way lever, supplementary locking points and latch kit (sash and frame).



ADJUSTMENT ± 2.5

110501



Locking kits for passive sash with Euro Groove with two-way shootbolt lever.

- Valid for side-hung opening windows.
- Supplied with polyamide pointers, two-way lever, adjustable keepers and latch kit (sash and frame).
- Two-way lever drives the pointers by means of transmission rods.

Kit with metallic tip pointers (ref. 071601)

- Supplied with polyamide pointers with zinc plated steel with ±1 mm adjustment.

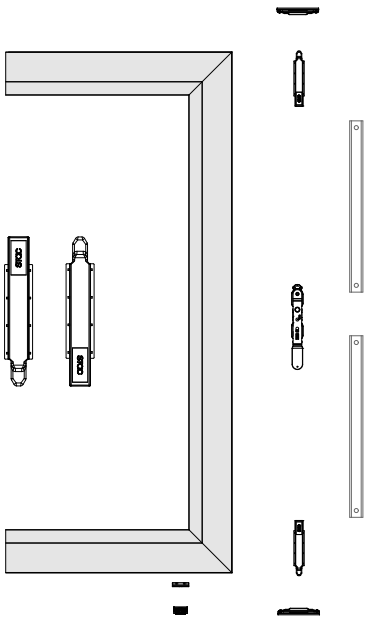
TWO-WAY SHOOTBOLT LEVER KIT

Side-hung opening

071600

20

ZM IN PA PV



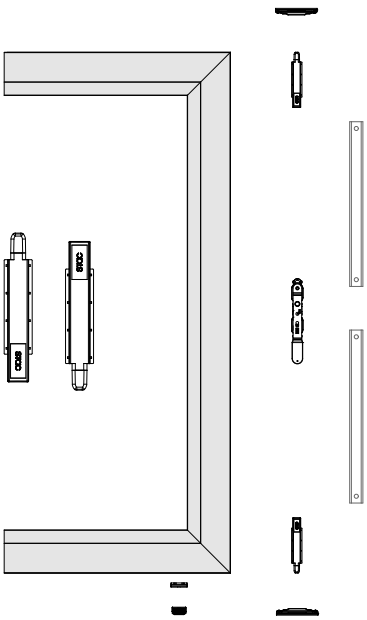
TWO-WAY SHOOTBOLT LEVER KIT WITH ADJUSTMENT

Side-hung opening

071601

20

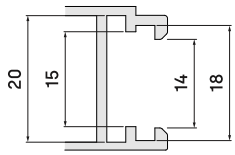
IN PV ZM PA



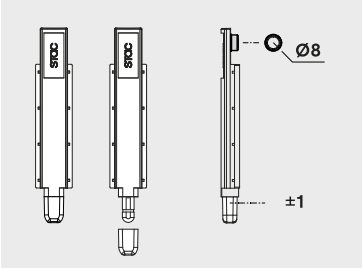
EURO GROOVE

FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

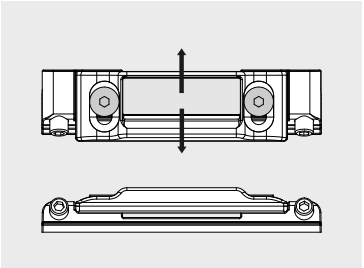
SASH ~ FRAME INVERTER PROFILE



ADJUSTMENT ± 1 — 0507082



ADJUSTMENT ± 2.5 — 110501



SHOOTBOLT LEVER KIT
Locking to inverter

A111600

20

IN ZM PA

Locking kits for Euro Groove passive sash with shootbolt levers.

• Supplied with zamak shootbolt levers, adjustable keepers and latch kit (sash and frame).
• Quick installation by means of eccentric cam allowing pre-positioning of the shootbolt lever.
• The shootbolt levers are equipped with locking points with pressure adjustment (± 1.5 mm).

Shootbolt lever kit (ref. A111600)

• Valid for tilt & turn window with locking to the inverter profile (H min = 665 mm).

Shootbolt lever kit (ref. A113800)

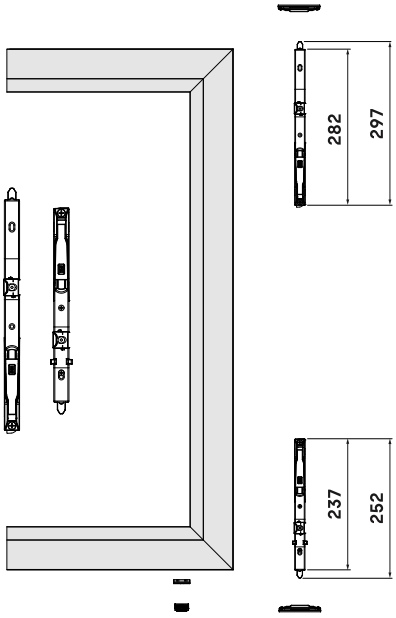
• Valid for tilt & turn window with locking to the frame profile (H min = 665 mm).

SHOOTBOLT LEVER KIT
Locking to frame

A113800

20

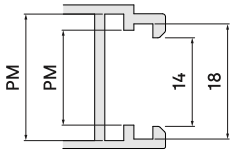
IN ZM PA

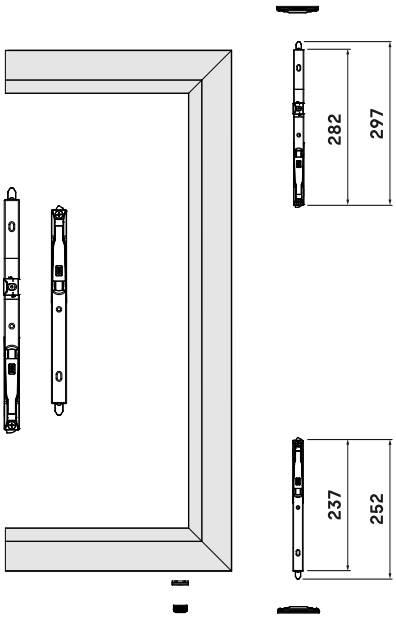


EURO GROOVE

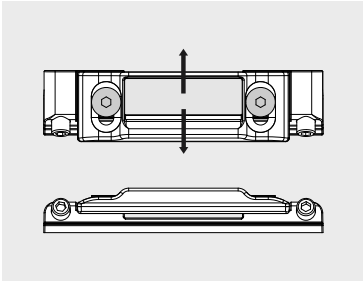
FRAME PROFILE	14 - 18 mm
	15 - 20 mm
SASH PROFILE	14 - 18 mm

SASH ~ FRAME INVERTER PROFILE

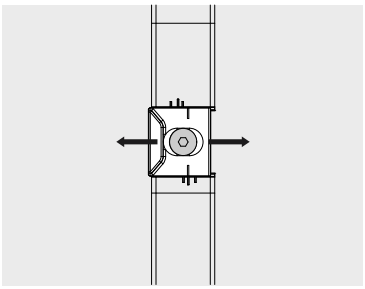




ADJUSTMENT ± 2.5 110501



ADJUSTMENT ± 1.5



Locking kits for Euro Groove passive sash with shootbolt levers.

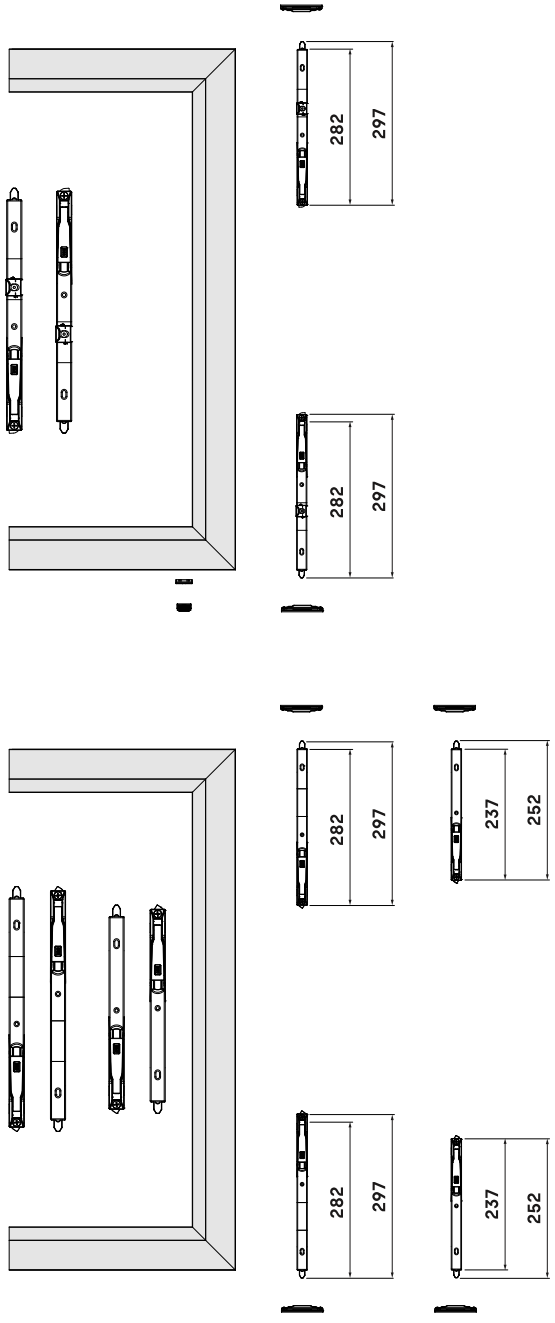
- Supplied with zamak shootbolt levers and keepers with adjustment.
- The adjustable keeper (± 2.5 mm) allows a progressive adjustment by means of anti-slip groove.
- Quick installation by means of eccentric cam allowing pre-positioning of the shootbolt lever.

Kit with shootbolt lever with locking point (ref. A113700)

- Available kit with individual shootbolt lever (ref. A112720).
- The shootbolt levers are equipped with locking points with pressure adjustment (± 1.5 mm).
- Two-way shootbolt lever kit (ref. A113700) is supplied with latch for sash and frame profile.

Kit with shootbolt lever without locking point (ref. A112710)

- Available kit with long shootbolt (ref. A112710) or short shootbolt (ref. A112610).



SHOOTBOLT LEVER KIT WITH LOCKING POINT

Side-hung opening

ce
ce

ZM IN
PA

A113700

20

SHOOTBOLT LEVER KIT WITHOUT LOCKING POINT

Side-hung opening

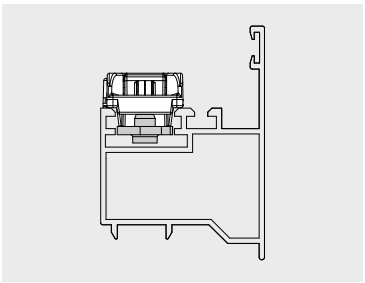
ce
ce

AL IN
PA

A112710

40

ECCENTRIC CAM



EXTENDABLE SHOOTBOLT LEVER KIT
Locking to inverter

111610

20

ZM IN PA

EXTENDABLE SHOOTBOLT LEVER KIT
Locking to frame

111611

20

ZM IN PA

EXTENDABLE SHOOTBOLT LEVER KIT

111620

20

ZM IN PA

Locking kit for the Euro Groove passive sash with extendable shootbolts lever, specially developed for balcony door profiles.

• The shootbolts, which are adjustable in both height and tightness, can be extended using transmission rods or intermediate locking points.
• Safety system with a built-in lever lock, latch kit for sash and frame profile and anti-looseness clip.

Extendable shootbolts kit (ref. 111610)

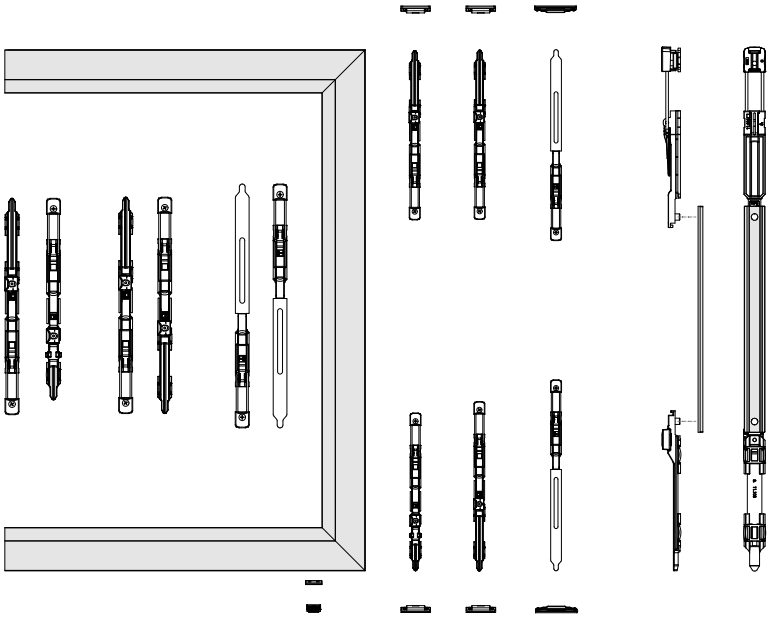
• Valid for tilt & turn opening with locking to inverter (H min = 750 mm).
• Supplied with polyamide keeper (ref. 0507059).

Extendable shootbolts kit (ref. 111611)

• Valid for tilt & turn opening with locking to frame (H min = 750 mm).
• Supplied with polyamide keeper (ref. 0507059).

Extendable shootbolts kit (ref. 111620)

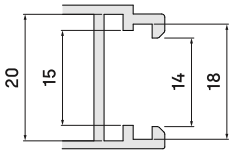
• Valid for tilt & turn opening with locking to frame or locking to inverter (H min = 750 mm).
• Supplied with stainless steel pointers and adjustable keeper (± 2.5 mm).



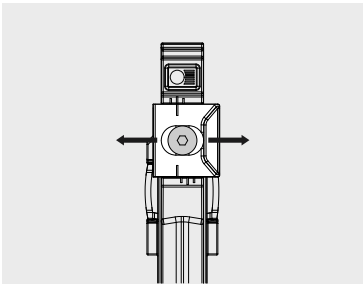
EURO GROOVE

FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

SASH ~ FRAME INVERTER PROFILE

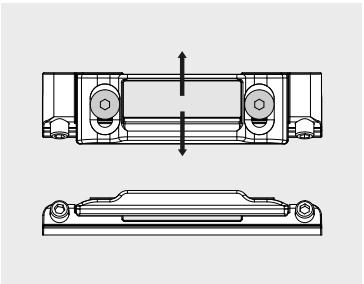


ADJUSTMENT ± 1.5



ADJUSTMENT ± 2.5

110501



Locking kit for Euro Groove passive sash with transmission rod levers.

• Supplied with polyamide rods, aluminium plates, button injected with polyamide, adjustable keepers and latch kit (sash and frame).
• Adjustable keeper (± 2.5 mm) allows a progressive adjustment by means of anti-slip groove.

15 ~ 20 transmission rod levers kit (ref. 112000)

• Valid for 15 ~ 20 mm frame inverters.

14 ~ 18 transmission rod levers kit (ref. 112001)

• Valid for 14 ~ 18 mm frame inverters.

TRANSMISSION ROD LEVER KIT

112000

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TRANSMISSION ROD LEVER KIT

112001

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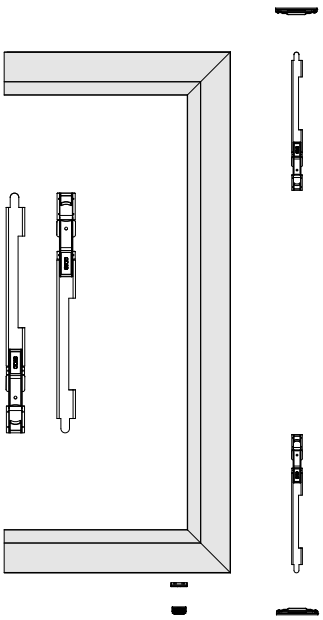
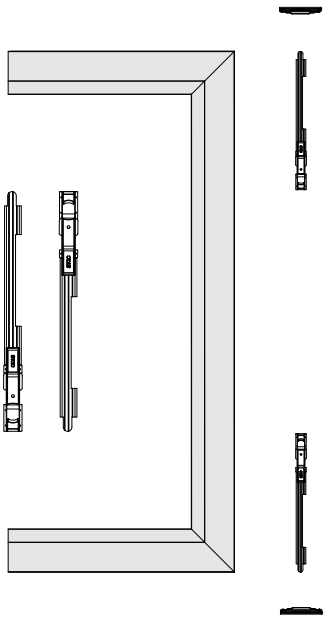
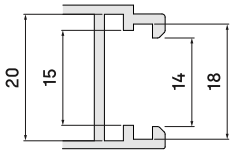




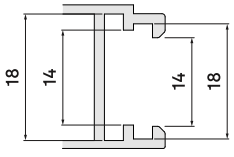
EURO GROOVE	112000
FRAME PROFILE	15 - 20 mm
SASH PROFILE	14 - 18 mm

SASH ~ FRAME INVERTER PROFILE



EURO GROOVE	112001
FRAME PROFILE	14 - 18 mm
SASH PROFILE	14 - 18 mm

SASH ~ FRAME INVERTER PROFILE



RETAINER CLIP KIT

0507047

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Accessories for secondary or passive sash.

Retainer clip (ref. 0507047)

- The mechanism allows the secondary sash to be kept locked when closed.
- It is possible to adjust the mechanism on the lateral groove.
- The retaining clip is deactivated by pulling the sash directly and both the clip and the frame retaining mechanism are removable.

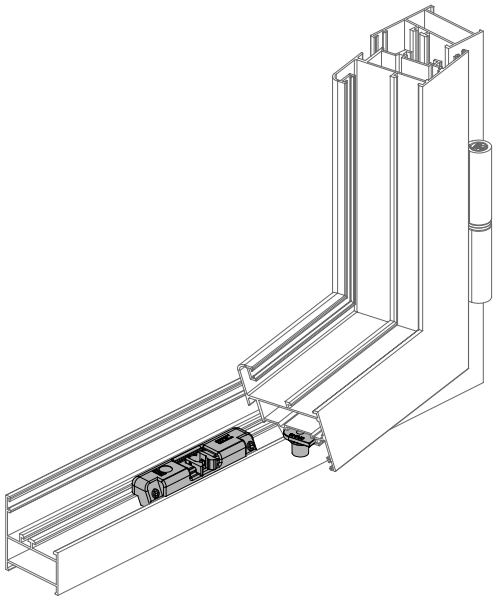
Locking point for the inverter (ref. 110210)

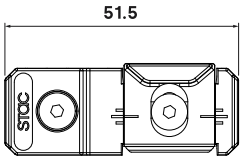
- Accessory for inverter profile to enable a locking point with tightening adjustment.
- Installation of the locking point is recommended for flange thicknesses less than 1.8 mm.

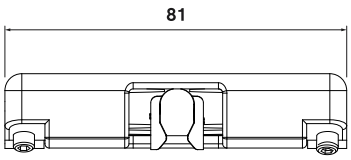
LOCKING POINT FOR INVERTER

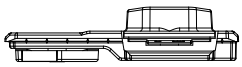
110210

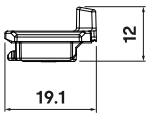
100

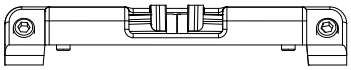


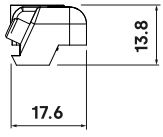


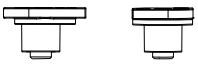












Locking kit with EVO hardware that allows the perimeter closing of the secondary sash.

• Supplied with two-way lever that allows the movement of both return angles simultaneously.
• Easy adaptation to the different profile series.

EVO SOFT perimeter closing (ref. 111604C)

• Locking kit with accessories that integrate encased hardware that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.

EVO SECURITY perimeter closing (ref. 111604S)

• Locking kit with accessories that integrate eccentric hardware that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.

EVO SOFT PERIMETER CLOSING

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EVO SECURITY PERIMETER CLOSING

ce
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IN

111604S

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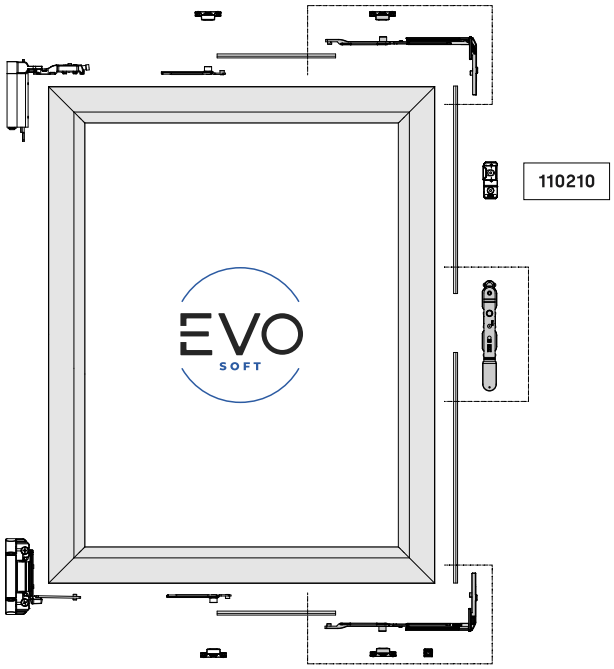

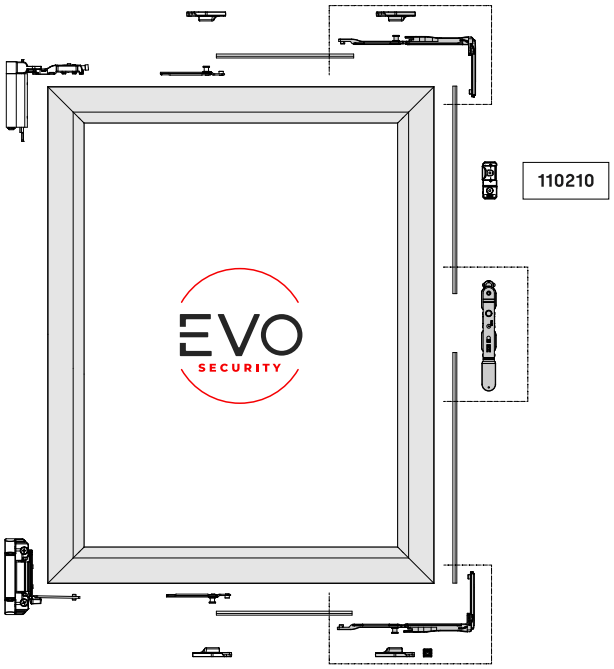

DENEB

Flush bolts for doors

Wide range of profiles groove

High traffic public areas

Available in different lengths

Simple & quick installation

DENEB flush bolts

The DENEB products offer a complete range of solutions that stand out for their versatility, functionality and quality finishes. The flush bolts are designed to fit standard door profiles, both Euro Groove and Centred Groove.

» Public areas

The flush bolt has been designed for buildings or public areas that require greater guarantees of closure and where there is a high frequency of use by users: common areas of shopping centres, health facilities, educational centres, office buildings, airports, passenger stations, etc.

» Adaptability

The flush bolts are designed to fit all standard door profile grooves on the market. The geometry of the flush bolt means it can also be fitted on Euro Groove windows. The above characteristics allow one single product to be used on all market-standard series.

» Resistance

Flush bolts made of aluminium and zamak, lever in nickel-plated zamak and the main components, which support stress and wear, in stainless steel.

» Versatility

Available in different lengths.

- **140 mm flush bolt:** recommended for installation on doors with multipoint locks.
- **140 mm EVO flush bolt**: designed to provide a solution for doors with high foot traffic.
- **240 | 450 mm flush bolts**: designed to provide a solution for doors with high foot traffic. They allow the lever to be brought closer to the user to facilitate operation in all situations.

» Stylish design

In most flush bolt designs the aesthetic aspects are a secondary concern. At STAC we have reinforced this part of the design, giving the DENEB flush bolts a uniform, stylish design with a high-quality finish.

» Smooth operation

The internal mechanism of the flush bolt follows the actuating movement via two springs which provide greater smoothness when the lever is activated.

Lever flush bolts designed to fit all standard grooves.

DENEB FLUSH BOLT

- Valid for doors with Euro Groove or centred groove profiles.
- Available in lacquered finishes: white, black and grey.

DENEB 140 flush bolt (ref. 052401)

- Bolt length: 140 millimetres.
- Recommended for use in doors with multipoint locks.

DENEB EVO 140 flush bolt (ref. 052404)

- Bolt length: 140 millimetres.
- Designed to provide a locking solution for doors with high foot traffic.
- Internal stainless steel reinforcement piece that guarantees greater resistance to breakage from blows.

DENEB 240 / 450 flush bolts (ref. 052402, ref. 052403)

- Bolt length: 240 or 450 millimetres.
- Designed to provide a locking solution for doors with high foot traffic.
- The flush bolts allow the user to get close to the lever in order to facilitate its operation.

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0524	01	**
	02	
	03	
	04	

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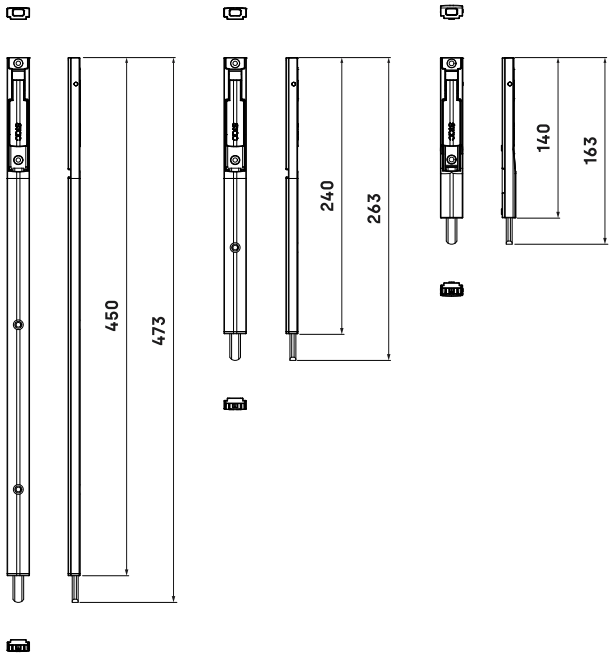
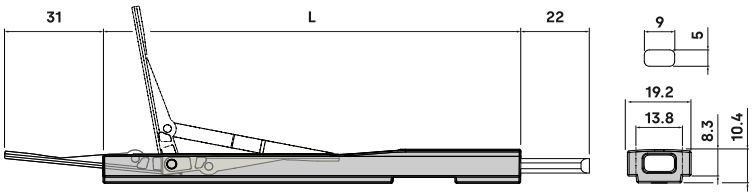
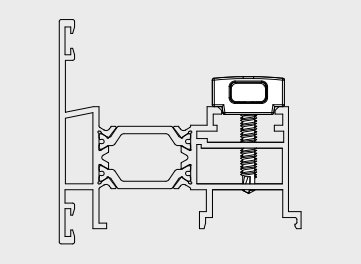


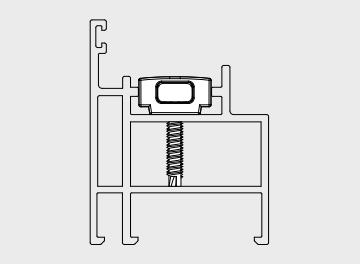


	L (mm)				
052401 **	140	05240101	05240102	05240103	15
052402 **	240	05240201	05240202	05240203	15
052403 **	450	05240301	05240302	05240303	15
052404 **	140	05240101	05240102	05240103	15

EURO GROOVE PROFILE



CENTRED GROOVE PROFILE



UPPER CLOSING POINT

0524020

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ZM

Accessories for DENEB door flush bolts.

Upper closing point (ref. 0524020)

- Self-tapping screw DIN 7504P PH4 4.2 x 25 mm.
- Natural zinc finish.

Dust cover (ref. 05241001)

- Dimensions: 29.8 millimetres high with a diameter of 13.6 millimetres.
- Highly durable materials.
- Quick and easy installation.

DUST COVER

05241001

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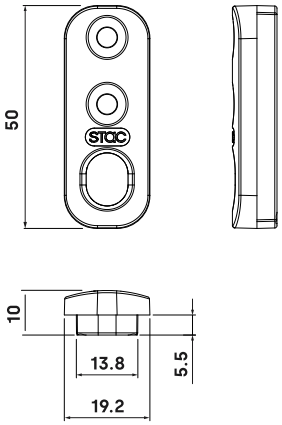
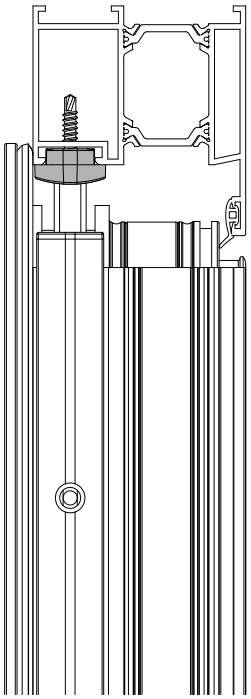
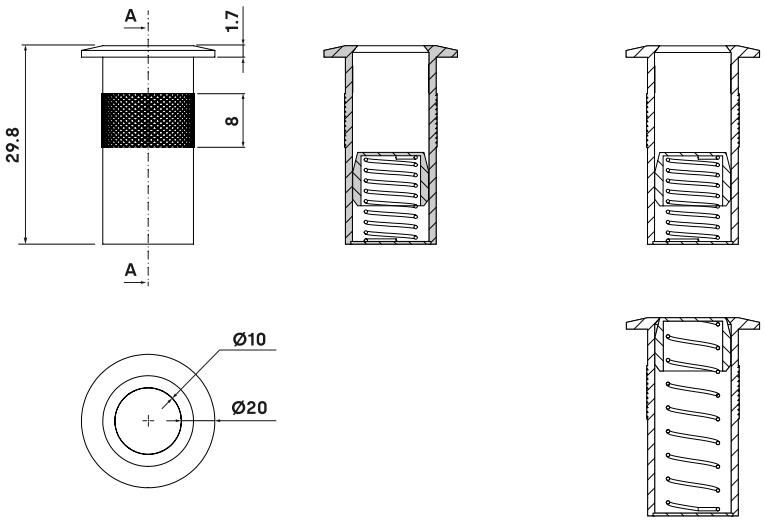






15 G16 HARDWARE

357 G16 hinges

360 G16 side-hung opening

367 G16 PRO tilt & turn

368 G16 tilt & turn

370 G16 tilt & turn friction stays

372 G16 tilt & turn accessories

381 Areation devices

389 Locking points

391 G16 passive sash accessories

394 Transmission rods

G16

Hardware for
16 millimetres groove

Time optimisation

Modular systems

Smooth manoeuvrability

Direct groove clipping

GROOVE 16

The hardware is developed based on the characteristics of PVC systems, while maintaining the particularities that define the hardware for the Euro Groove.

» Frontal assembly

Reduces the assembly times and offers the possibility of assembling the hardware manually, on assembly beds, or can be assembled completely automatically.

» Modularity

The accessories are always the same, independently from the sash size and type of opening which translates into an important reduction of stock references to be held in a warehouse.

» Materials

All the metallic components are manufactured in zamak and stainless steel which have a low galvanic couple with the aluminium of the profile and they have a corrosion resistance of more than 240 hours tested in a saline mist chamber (UNE-EN 1670).

» Smoothness of operation

The locking points that are incorporated in the part components have gyratory bushings that make sash locking much smoother.

» Groove clipping system

The components are incorporated with a groove clipping system that allows us to quickly fit all the hardware so that all the component parts can be screwed in one easy step.

» Profile characteristics

Sash groove. The hardware is designed to adapt to the two types of 16 mm groove used in PVC systems: with flanges and without, so that they have specific rods.

Frame groove. The hardware fits into the standard frame groove (Euro Groove 18 / 14).

Frame / sash face to face. All accessories are suitable for profiles with a 13 mm shaft.

Profile flange and overlap. Sashes with up to 20 mm flange and 8 mm overlap for improved water tightness.

12 mm chamber. Greater adjustment possibilities, cutting error absorption, locking point interference.

The maximum supported weight is 120 kg. Please consult the technical catalogue to obtain all the assembly information and available configurations.

Hinges for side-hung windows with Groove 16 profiles.

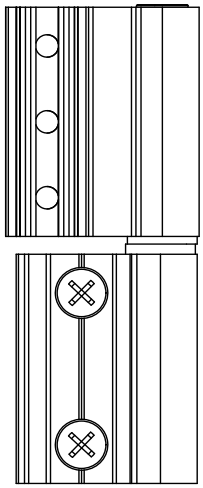
- Maximum load with 2 hinges: 70 kg.
- Hinge with opening hand: right (ref. 1760D) or left-hand opening (ref. 1760I).
- Supplied with an eccentric pin that allows tightening and horizontal adjustment.

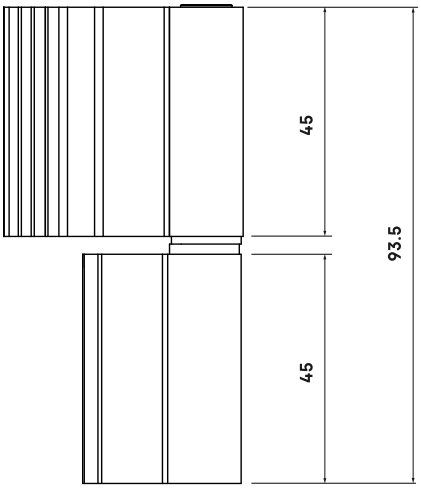
C16 HINGE
Side-hung opening

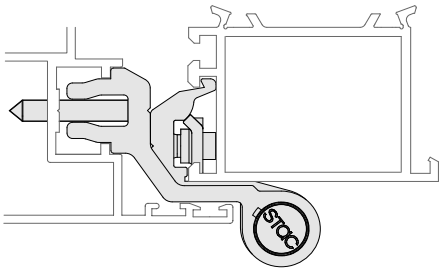
1760	D / I	**

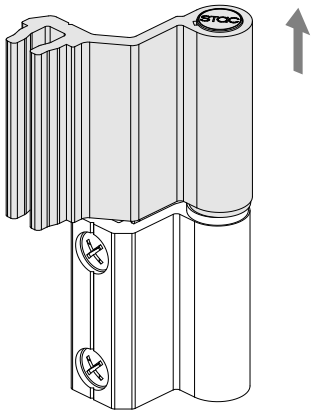
40

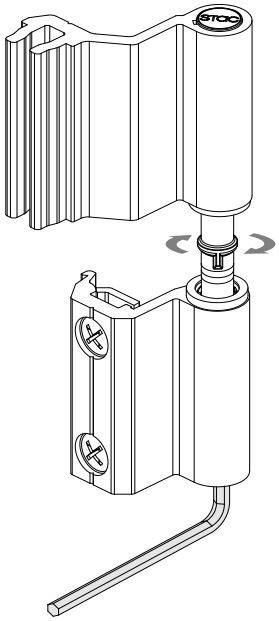
AL IN PA

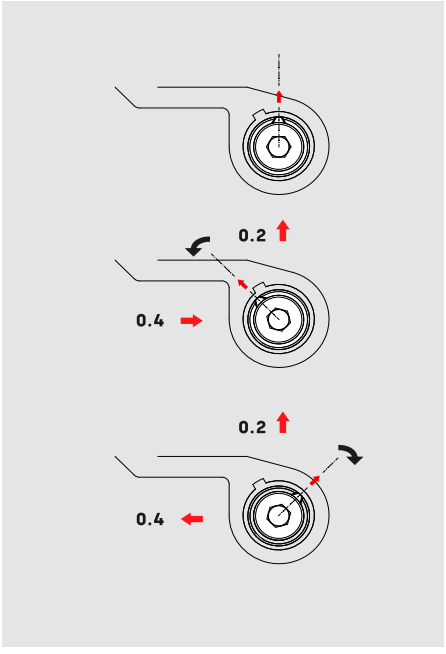












ACRUX C16 HINGE

017	D / I	**

12

C16

AL IN PA PM

Hinge for 16 Groove door profiles with 13 mm shaft.

- Lateral adjustment up to ± 1.2 mm by installing the kit with ref. STC-MC-0011.
- For security reasons, from 120 kg per sash is compulsory to install the reinforcement piece (ref. 0517182).

ACRUX 2-leaf hinge (ref. 017)

- Maximum load: 100 kg with 2 units and 120 kg with 3 units.
- Hinge with opening hand: right (ref. 017D) or left-hand opening (ref. 017I).

ACRUX 3-leaf hinge (ref. P017D)

- Maximum load: 120 kg with 2 units and 150 kg with 3 units.
- Ambidextrous hinge: supplied pre-assembled for right-handed opening but quickly interchangeable for left-handed opening.

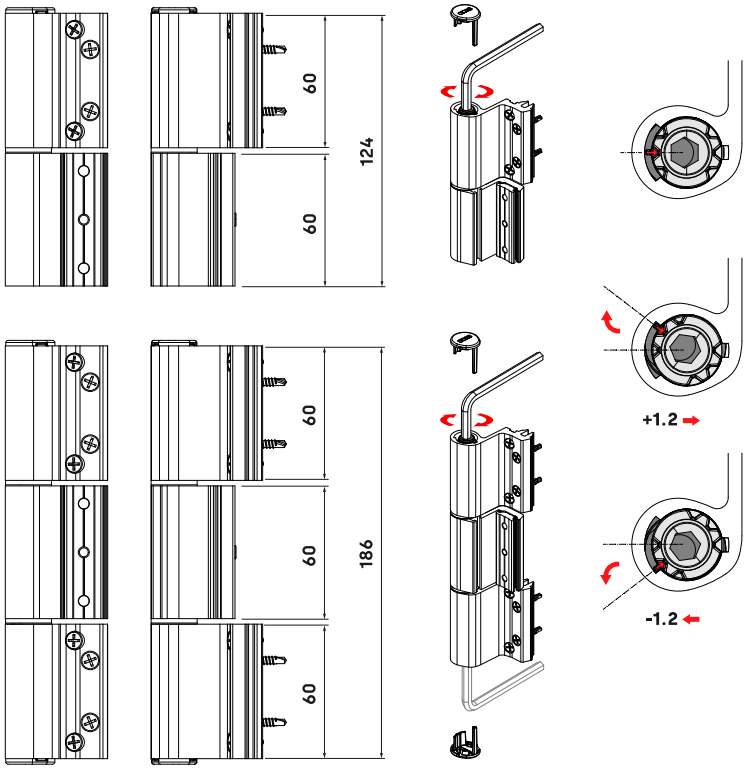
ACRUX C16 HINGE

P017D	**

6

C16

AL IN PA PM

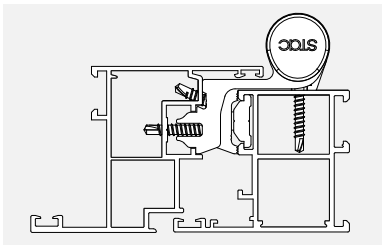


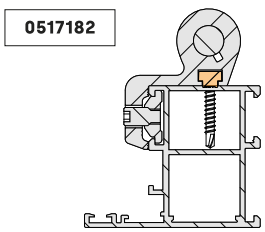
REINFORCEMENT PIECE

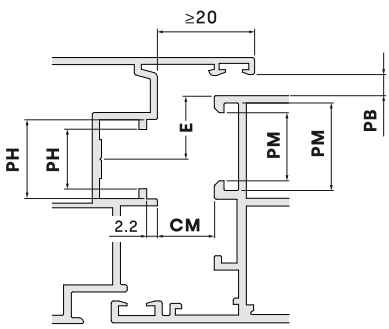
0517182

500

ZM







16 GROOVE (mm)

FRAME PROFILE **PM**	14 - 18
SASH PROFILE **PH**	12.2 - 12.5 16.2 - 16.5
CHAMBER **CM**	12
HINGE PITCH **PB**	MIN. 4.5
SHAFT **E**	13

USAGE RECOMMENDATIONS

Considerations Consult the technical data sheet to access the table of maximum weights supported by each hinge depending on the door dimensions. In any case, for public works, a weight reduction of 20% must be applied to the values shown in the table.

017 D/I **	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	100 kg	120 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	80 kg	100 kg
WITH DOOR CLOSER WITH RESTRICTOR	-	80 kg
WITH DOOR CLOSER W/O RESTRICTOR	-	70 kg

P017D **	2 UNITS	3 UNITS
ONE-FAMILY HOUSE	120 kg	150 kg
MULTI-FAMILY HOUSING AND PUBLIC BUILDINGS	100 kg	130 kg
WITH DOOR CLOSER WITH RESTRICTOR	80 kg	110 kg
WITH DOOR CLOSER W/O RESTRICTOR	70 kg	100 kg

C16 SIDE-HUNG OPENING KIT FOR LOCKING TO INVERTER

With double pointer

17K1HCI

50

C16

ZM

IN

C16 SIDE-HUNG OPENING KIT FOR LOCKING TO FRAME

With double corner drive

17K1HCM

1

C16

ZM

IN

Side-hung opening kits for windows with 16 Groove profiles.

Locking to inverter with double pointer

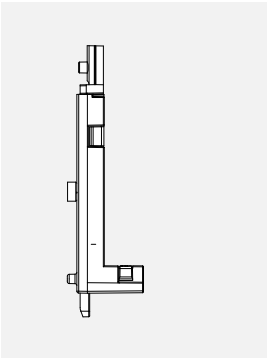
• Single sash configuration (ref. 17K1HCI) or double sash configuration (ref. 17K2HCI).

Locking to frame with double corner drive

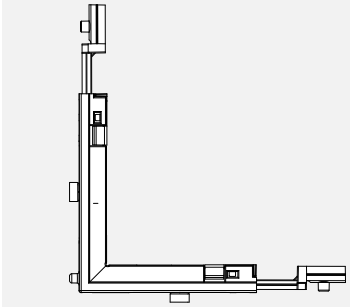
• Single sash configuration (ref. 17K1HCM) or double sash configuration (ref. 17K2HCM).

		LOCKING TO INVERTER	LOCKING TO FRAME
17K1HCI		✔	
17K1HCM			✔
17K2HCI		✔	
17K2HCM			✔

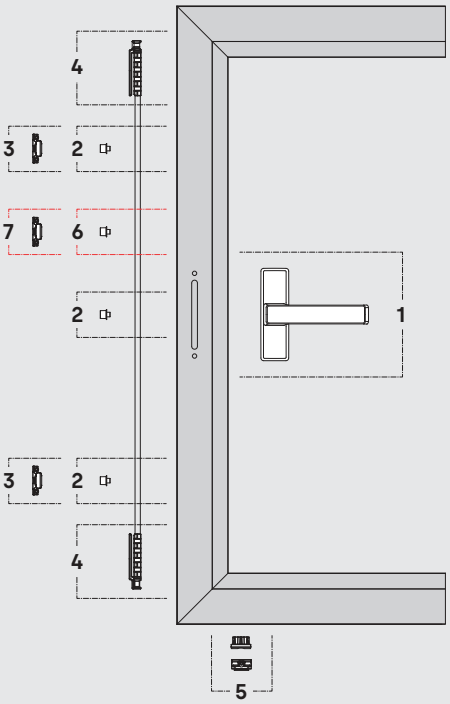
171000 POINTER



173101 DRIVE CORNER

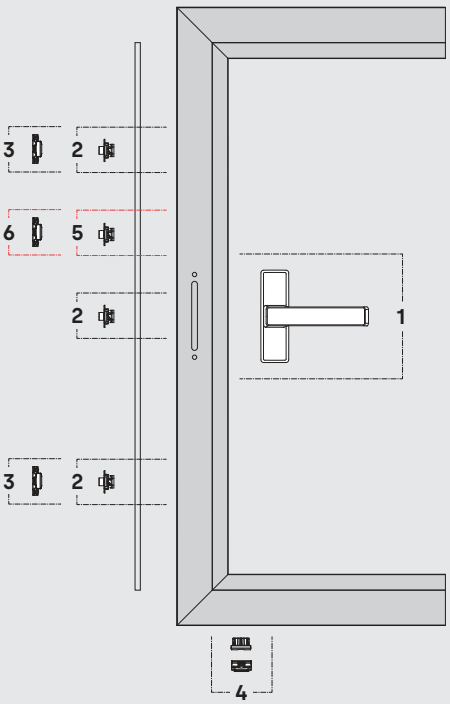


Accessories for sash height over 1200 millimetres



SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH RIVET LOCKING POINTS

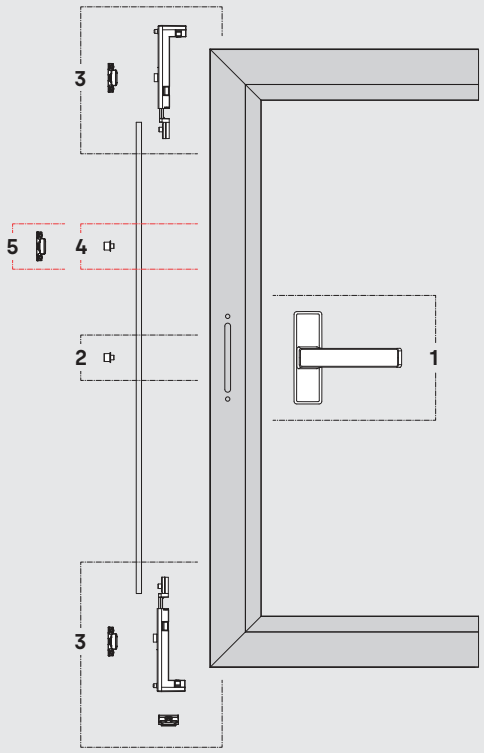
1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
4	**0517083**	**ROD TERMINAL TRIM**	50
5	**0517226**	**POSITIONING LATCH KIT**	50
6	**0517002**	**Ø6 RIVET FOR ROD**	50
7	**170200**	**ADJUSTABLE LOCKING POINT**	100



SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH SLOTTED LOCKING POINTS

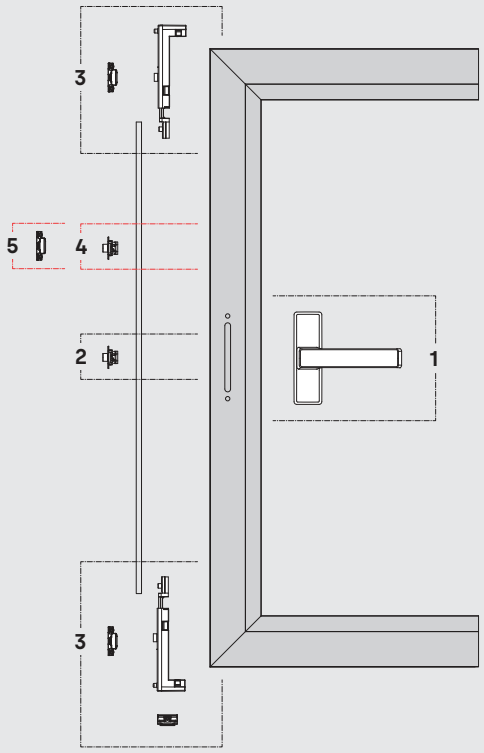
1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
4	**0517226**	**POSITIONING LATCH KIT**	50
5	**178000**	**SLOTTED LOCKING POINT**	100
6	**170200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height over 1200 millimetres



SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS AND RIVET LOCKING POINTS

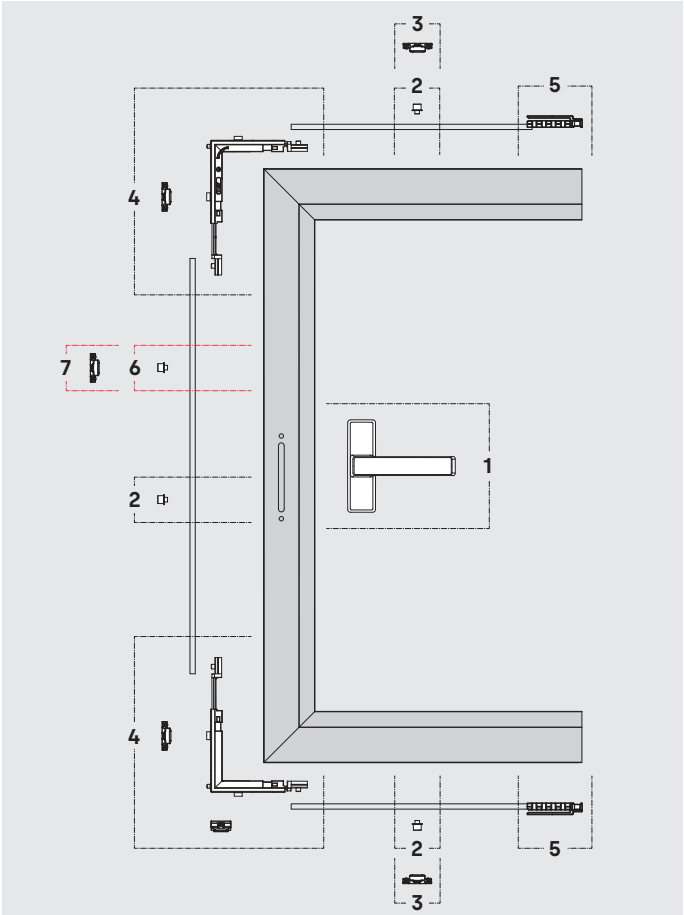
1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
		POINTERS	
3	**17K1HCI**	**ADJUSTABLE LOCKING POINT**	50
		FRAME ANTI-SLIP	
4	**0517002**	**Ø6 RIVET FOR ROD**	50
5	**170200**	**ADJUSTABLE LOCKING POINT**	100

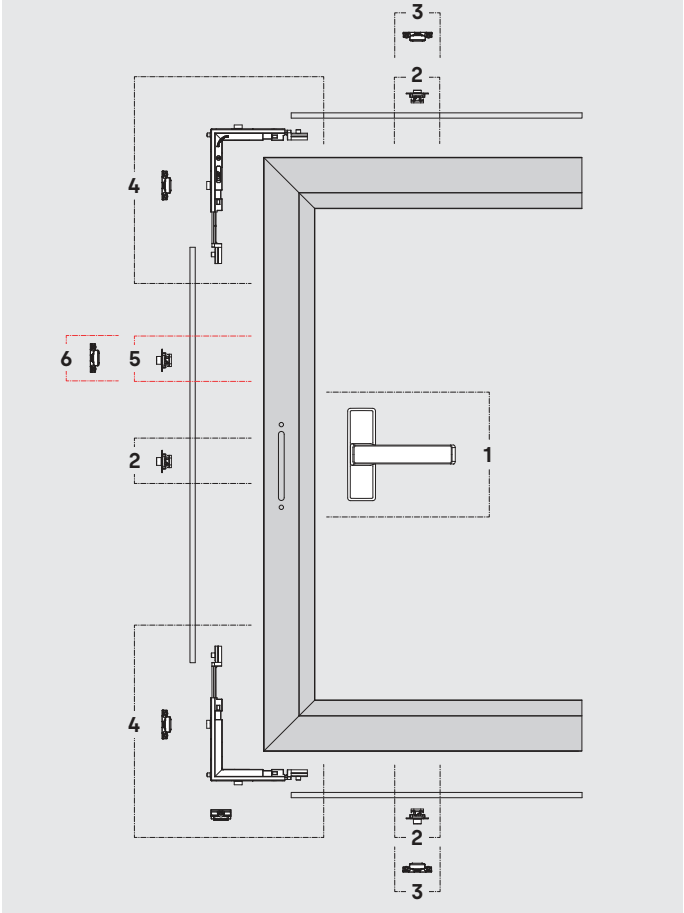


SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS AND SLOTTED LOCKING POINTS

1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
		POINTERS	
3	**17K1HCI**	**ADJUSTABLE LOCKING POINT**	50
		FRAME ANTI-SLIP	
4	**178000**	**SLOTTED LOCKING POINT**	100
5	**170200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height over 1200 millimetres





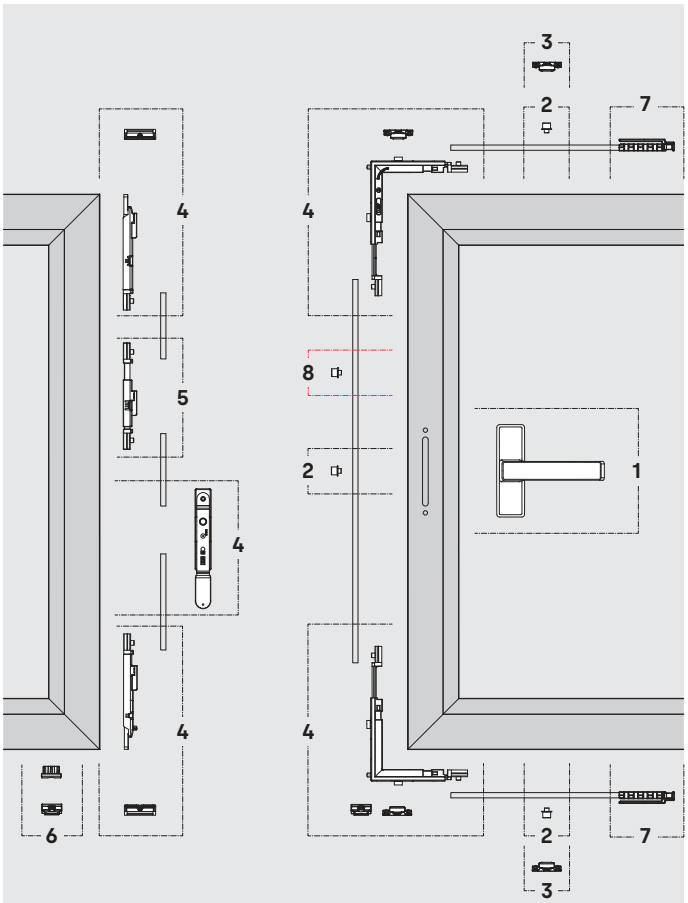
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND RIVET LOCKING POINTS

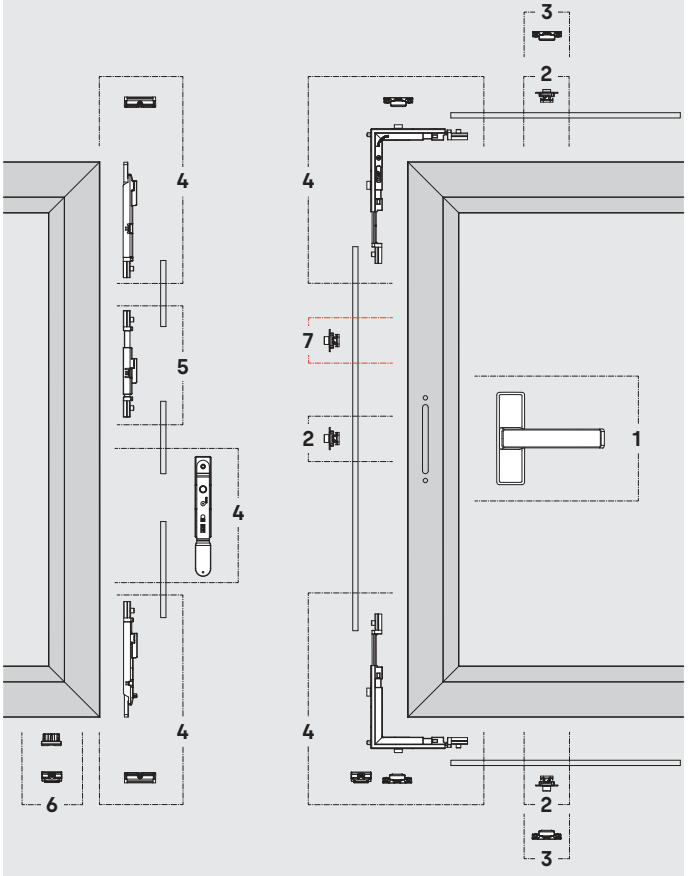
1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
		CORNER DRIVE	
4	**17K1HCM**	**ADJUSTABLE LOCKING POINT**	1
		FRAME ANTI-SLIP	
5	**0517083**	**ROD TERMINAL TRIM**	50
6	**0517002**	**Ø6 RIVET FOR ROD**	50
7	**170200**	**ADJUSTABLE LOCKING POINT**	100

SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH DRIVE CORNER AND SLOTTED LOCKING POINTS

1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
		CORNER DRIVE	
4	**17K1HCM**	**ADJUSTABLE LOCKING POINT**	1
		FRAME ANTI-SLIP	
5	**178000**	**SLOTTED LOCKING POINT**	100
6	**170200**	**ADJUSTABLE LOCKING POINT**	100

Accessories for sash height over 1200 millimetres





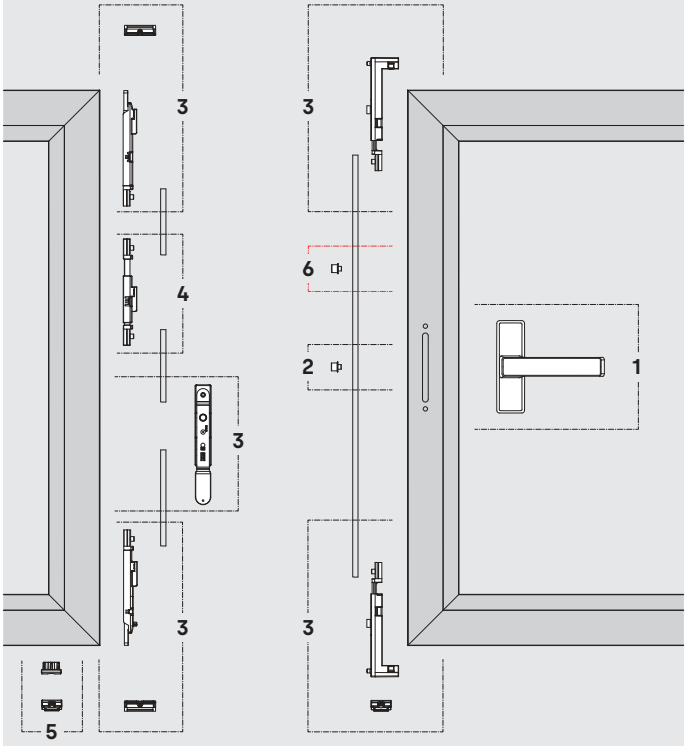
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND RIVET LOCKING POINTS

1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
4	**17K2HCM**	**CORNER DRIVE** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
5	**177000**	**KEEPER FOR INVERTER PROFILE**	50
6	**0517226**	**POSITIONING LATCH KIT**	50
7	**0517083**	**ROD TERMINAL TRIM**	50
8	**0517002**	**Ø6 RIVET FOR ROD**	50

SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND SLOTTED LOCKING POINTS

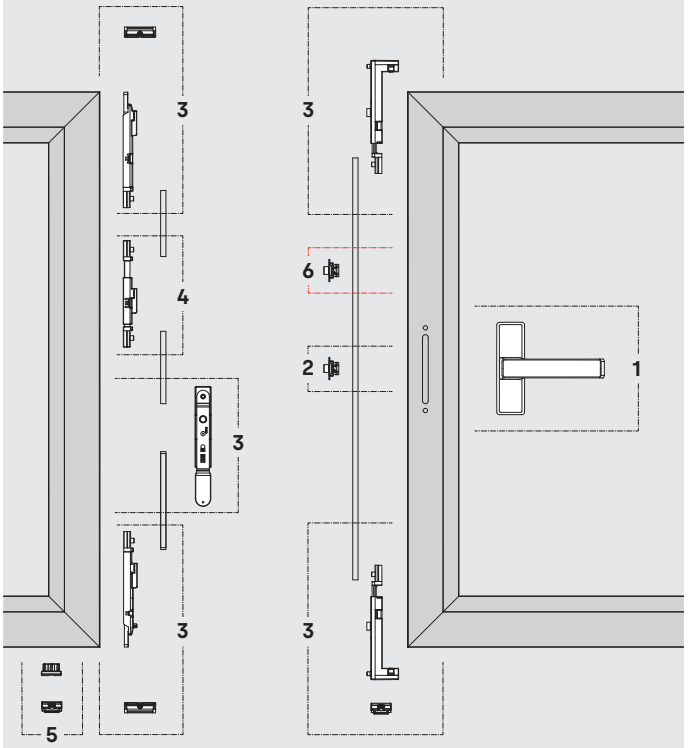
1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
3	**170200**	**ADJUSTABLE LOCKING POINT**	100
4	**17K2HCM**	**CORNER DRIVE** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
5	**177000**	**KEEPER FOR INVERTER PROFILE**	50
6	**0517226**	**POSITIONING LATCH KIT**	50
7	**178000**	**SLOTTED LOCKING POINT**	100

Accessories for sash height over 1200 millimetres



SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS AND RIVET LOCKING POINTS

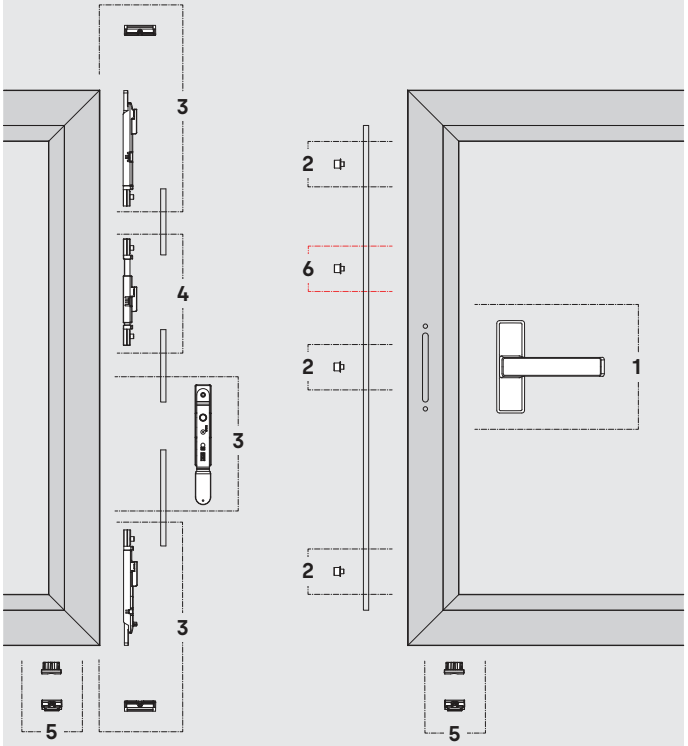
1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
3	**17K2HCI**	**POINTERS** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	1
4	**177000**	**KEEPER FOR INVERTER PROFILE**	50
5	**0517226**	**POSITIONING LATCH KIT**	50
6	**0517002**	**Ø6 RIVET FOR ROD**	50



SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS AND SLOTTED LOCKING POINTS

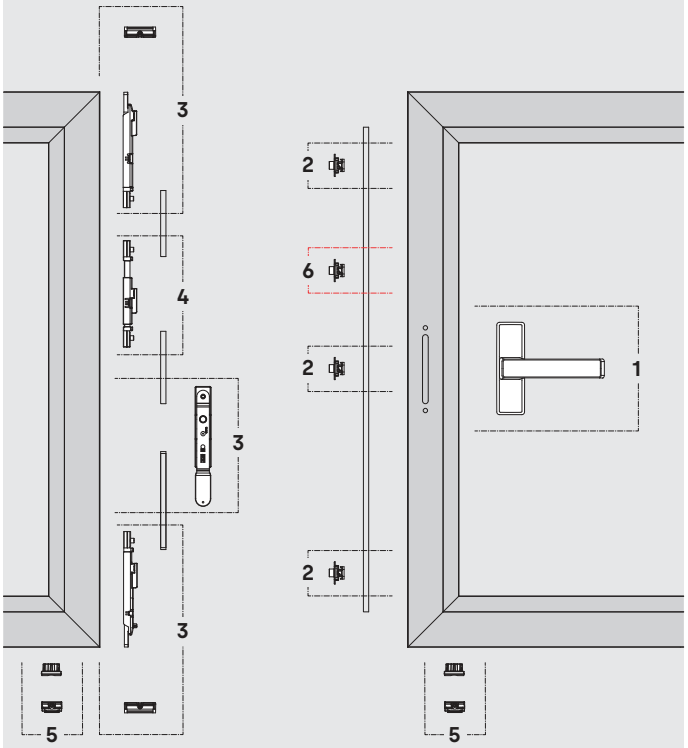
1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
3	**17K2HCI**	**POINTERS** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	1
4	**177000**	**KEEPER FOR INVERTER PROFILE**	50
5	**0517226**	**POSITIONING LATCH KIT**	50
6	**178000**	**SLOTTED LOCKING POINT**	100

Accessories for sash height over 1200 millimetres



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH RIVET LOCKING POINTS

1	**CREMONE**	*Consult cremone section*	
2	**0517002**	**Ø6 RIVET FOR ROD**	50
3	**171600**	**TWO-WAY SHOOTBOLT KIT**	20
4	**177000**	**KEEPER FOR INVERTER PROFILE**	50
5	**0517226**	**POSITIONING LATCH KIT**	50
6	**0517002**	**Ø6 RIVET FOR ROD**	50



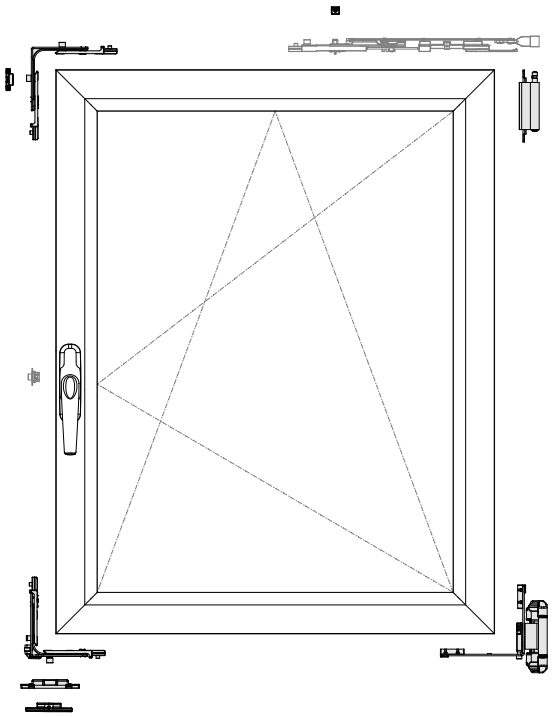
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH SLOTTED LOCKING POINTS

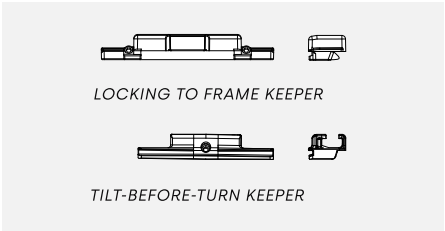
1	**CREMONE**	*Consult cremone section*	
2	**178000**	**SLOTTED LOCKING POINT**	100
3	**171600**	**TWO-WAY SHOOTBOLT KIT**	20
4	**177000**	**KEEPER FOR INVERTER PROFILE**	50
5	**0517226**	**POSITIONING LATCH KIT**	50
6	**178000**	**SLOTTED LOCKING POINT**	100

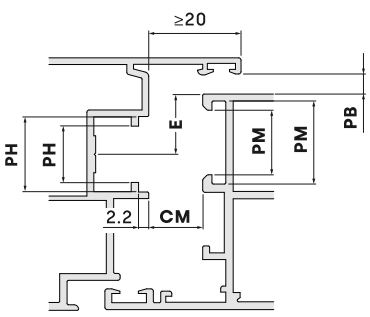
Tilt & turn kit with C16 hinges for locking to frame windows.

- Maximum load per sash up to 120 kg.
- The reference is supplied without cremone (available with LYRA cremone, ref. P1732).
- The kits are valid for windows with 13 mm shaft and 20 mm flange.
- The elements of the sash profile incorporate a clip that allows the pre-positioning of the accessories in the assembly.
- The tilt-before-turn kit (ref. P1723) is suitable for the use of cremones with keys as they provide two locking positions that allow the window to be tilt but not opened.

The kits do not include a tilt & turn friction stay. See compatible friction stays.







16 GROOVE (mm)

FRAME PROFILE **PM**	14 - 18
SASH PROFILE **PH**	12.2 - 12.5 16.2 - 16.5
CHAMBER **CM**	12
HINGE PITCH **PB**	MIN. 4.5
SHAFT **E**	13

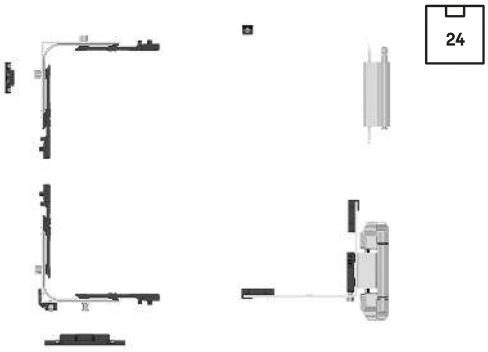
C16 PRO KIT FOR LOCKING TO FRAME
Without cremone

C16

AL ZM IN PA

P1739	D I	**

24

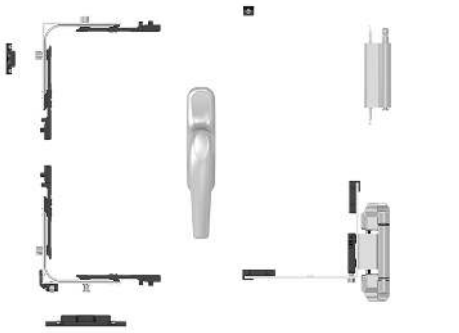


C16 PRO KIT FOR LOCKING TO FRAME
With LYRA cremone

C16

AL ZM IN PE

P1732	D I	**



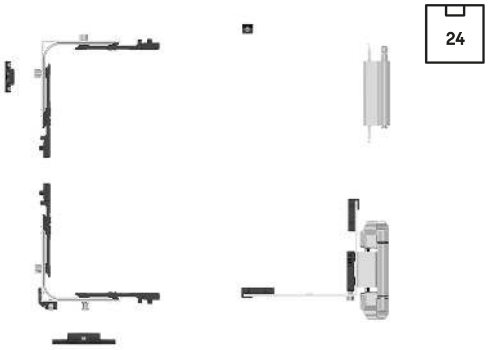
G16 PRO KIT FOR TILT-BEFORE-TURN
Without cremone

C16

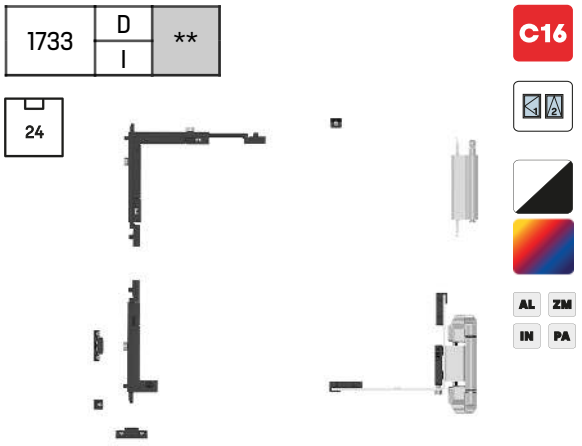
AL ZM IN PA

P1723	D I	**

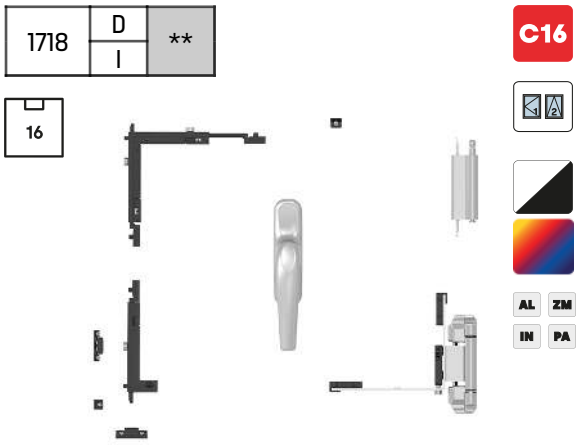
24



G16 KIT FOR LOCKING TO INVERTER
Without cremone



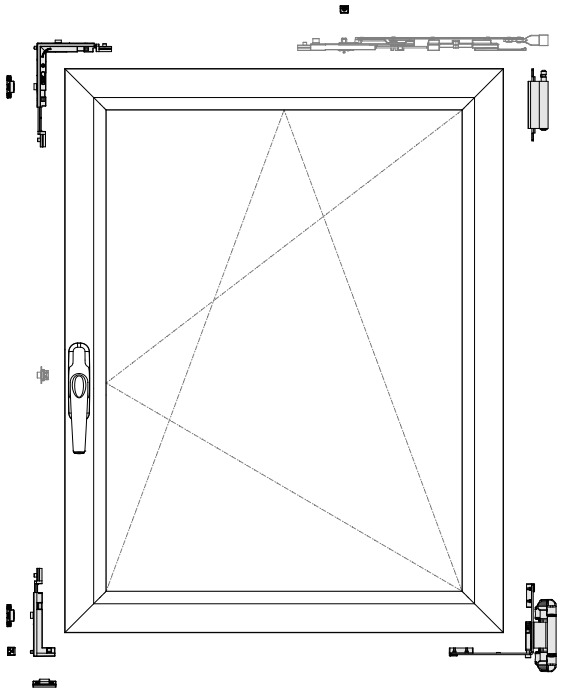
G16 KIT FOR LOCKING TO INVERTER
With LYRA cremone



Tilt & turn kit with C16 hinges for locking to inverter window.

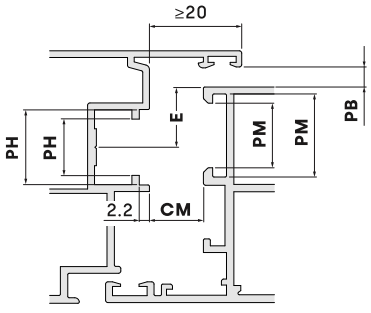
- Maximum load per sash up to 120 kg.
- The reference is supplied without cremone (available with LYRA cremone, ref. 1718).
- The kits are valid for windows with 13 mm shaft and 20 mm flange.
- The elements of the sash profile incorporate a clip that allows the pre-positioning of the accessories in the assembly.

The kit does not include a tilt & turn friction stay. See compatible friction stays.



16 GROOVE (mm)

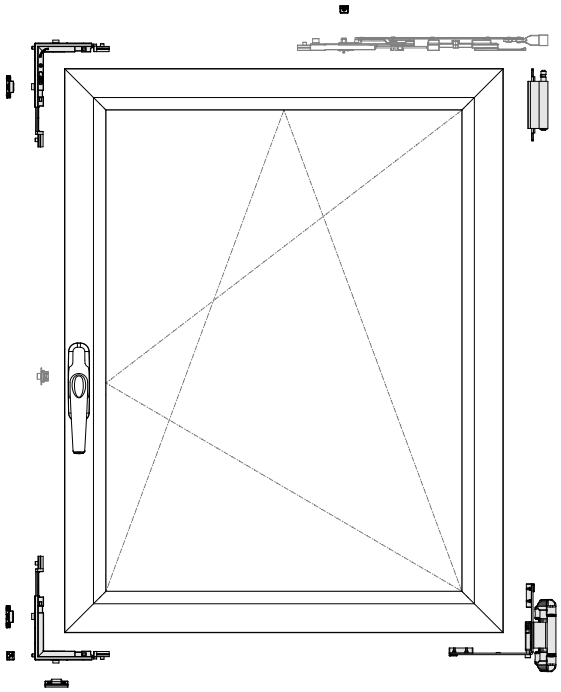
FRAME PROFILE **PM**	14 - 18
SASH PROFILE **PH**	12.2 - 12.5 16.2 - 16.5
CHAMBER **CM**	12
HINGE PITCH **PB**	MIN. 4.5
SHAFT **E**	13



Tilt & turn kit with C16 hinges for locking to frame windows.

- Maximum load per sash up to 120 kg.
- The reference is supplied without cremone (available with LYRA cremone, ref. 1732).
- The kits are valid for windows with 13 mm shaft and 20 mm flange.
- The elements of the sash profile incorporate a clip that allows the pre-positioning of the accessories in the assembly.

The kits do not include tilt & turn friction stays. See compatible friction stays.



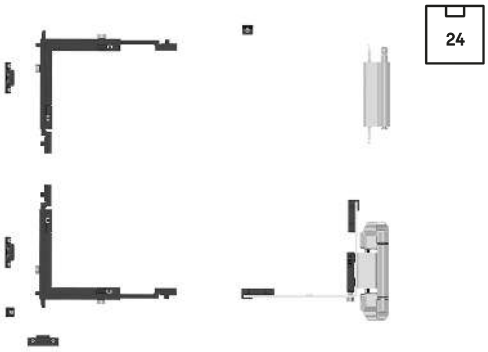
G16 KIT FOR LOCKING TO FRAME
Without cremone

C16

AL ZM IN PA

171010	D / I	**

24



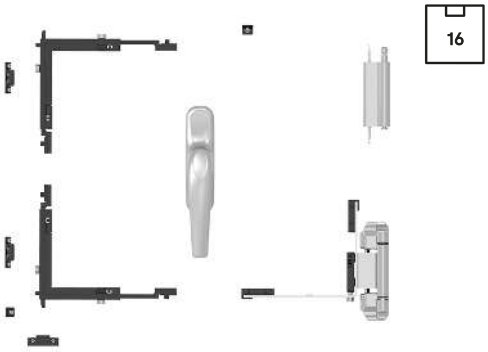
G16 KIT FOR LOCKING TO FRAME
With LYRA cremone

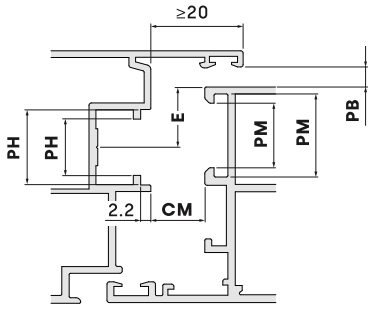
CC

AL ZM IN PA

1732	D / I	**

16





16 GROOVE (mm)

FRAME PROFILE **PM**	14 - 18
SASH PROFILE **PH**	12.2 - 12.5 16.2 - 16.5
CHAMBER **CM**	12
HINGE PITCH **PB**	MIN. 4.5
SHAFT **E**	13

TILT & TURN FRICTION STAY
Tilt & turn

COMPAS MT

10

C16

AL ZM
IN PA

Friction stays for 16 Groove profiles that allow tilt and turn openings in combination with different accessories.

- Specific references for tilt and turn configurations with traditional manoeuvre (locking to the inverter and locking to the frame) or tilt-before-turn.
- They incorporate an adjustment mechanism for adjusting the sash opening in the tilt position and a stop for positioning the friction stay.
- Recessed eccentric that allows adjustment of the closing pressure (± 1 mm).

TILT & TURN FRICTION STAY
Tilt-before-turn

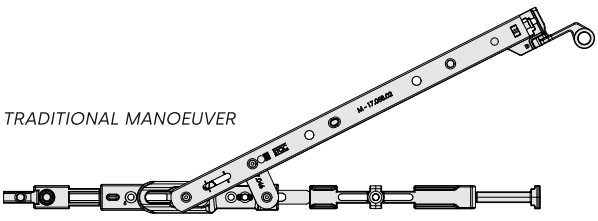
COMPAS ML

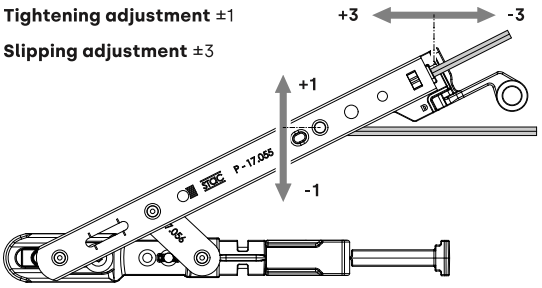
10

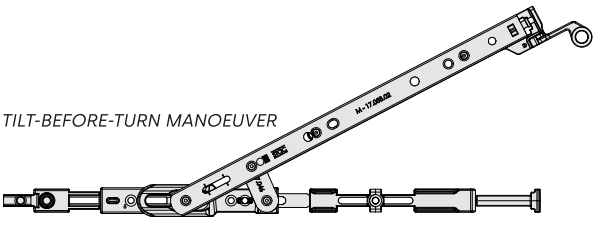
C16

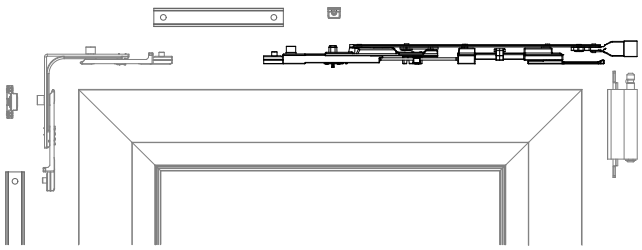
AL ZM
IN PA

TRADITIONAL MANOEUVER		WEIGHT (kg)	SUPPLEMENTARY (130400)
170301 D I **	LARGE FRICTION STAY	120	✔
170302 D I **	MEDIUM FRICTION STAY	120	-
170303 D I **	SMALL FRICTION STAY	120	-
170306 D I **	MINI FRICTION STAY	120	-
TILT-BEFORE-TURN MANOEUVER			
170304 D I **	LARGE FRICTION STAY	120	✔
170305 D I **	MEDIUM FRICTION STAY	120	-
170307 D I **	SMALL FRICTION STAY	120	-

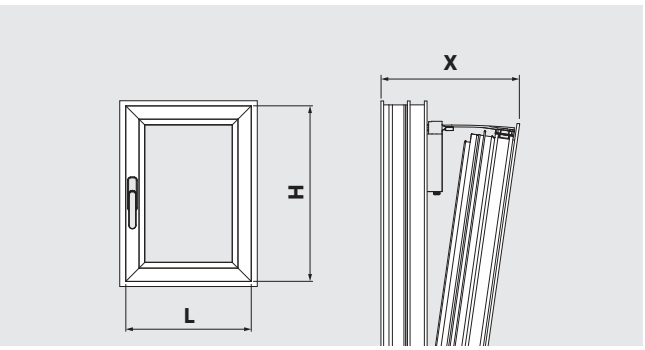








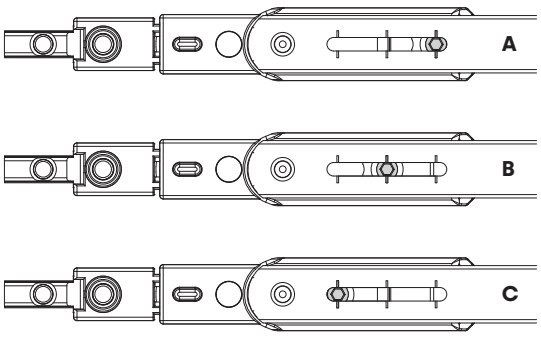
				L min	L max	H min	H max	X max		
G16 HINGES								POSITION **1**	POSITION **2**	POSITION **3**
	170301 D I **	**LARGE** FRICTION STAY		582	1600	440	2400	191	155	101
	170302 D I **	**MEDIUM** FRICTION STAY		498	1200	440	2400	168	112	-
	170303 D I **	**SMALL** FRICTION STAY		405	800	440	2400	130	85	-
	170306 D I **	**MINI** FRICTION STAY		370	445	440	1200	130	85	-
	170304 D I **	**LARGE** FRICTION STAY		582	1600	440	2400	191	155	101
	170305 D I **	**MEDIUM** FRICTION STAY		498	1200	440	2400	168	112	-
	170307 D I **	**SMALL** FRICTION STAY		405	800	440	2400	130	85	-
WITH SUPPLEMENTARY FRICTION STAY										
	170301 D I **	**130400**		980	1600	440	2400	191	155	101
	170304 D I **	**130400**		980	1600	440	2400	191	155	101



L = WINDOW WIDTH

H = WINDOW HEIGHT

X = OPENING IN TILT POSITION



A = POSITION 1, **B** = POSITION 2, **C** = POSITION 3

G16 SUPPLEMENTARY FRICTION STAY

A130400

15

C16

AL ZM IN

The supplementary friction stay acts as a reinforcement to the main friction stay by reducing the forces and vibrations on the friction stay when the window is in the tilt position.

- Recommended for installing on windows with more than 1200 mm sash width.
- Supplied pre-assembled that allowing for quick installation.
- Compatible with bottom-hung window opening configurations with perimeter closing.

G16 supplementary corner drive (ref. 172100)

- Supplementary corner drive kit for tilt & turn opening windows that allows additional locking points between hinges.
- Supplied with adjustable locking point (ref. 170200) and transmission rod terminal trim (ref. 0517083).
- Recommended for installing on windows with more than 1200 mm sash height.

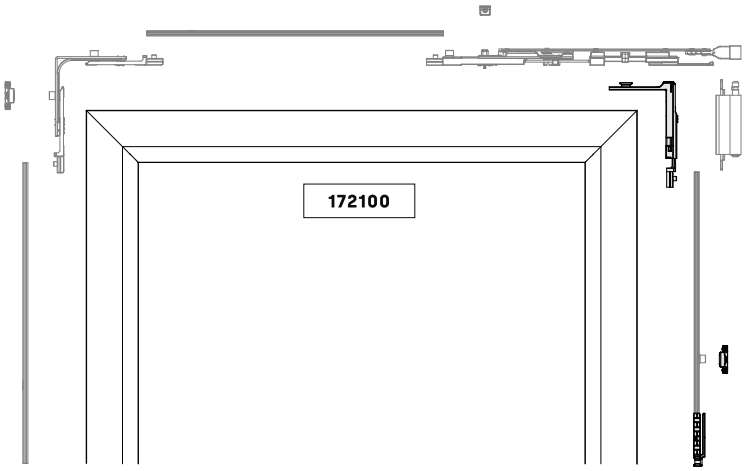
G16 SUPPLEMENTARY CORNER DRIVE

172100

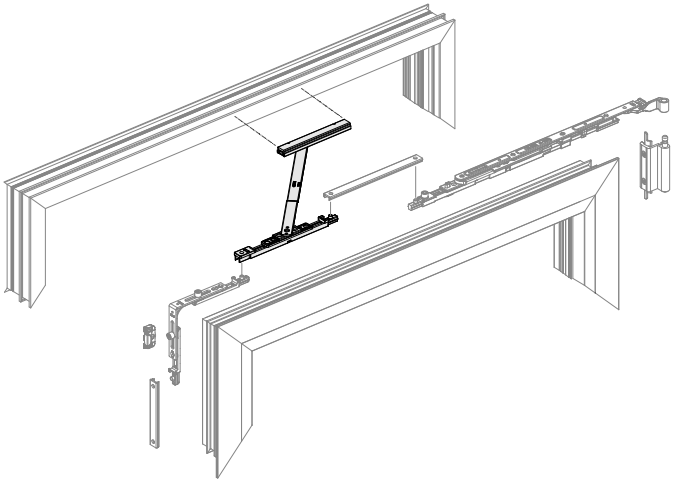
25

C16

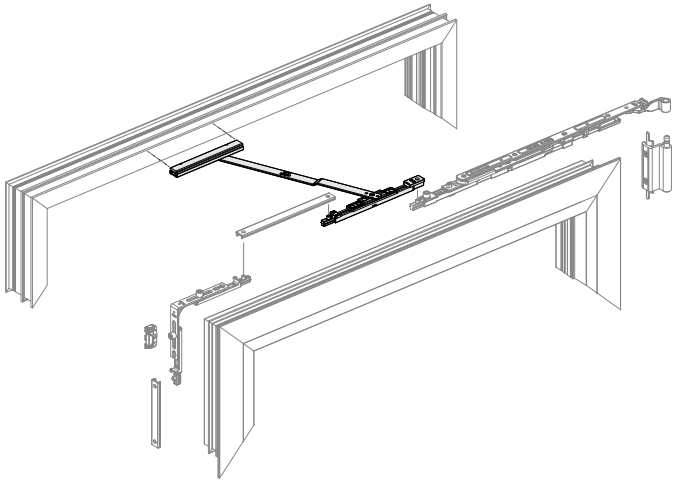
ZM IN PA



A130400 **TRADITIONAL MANOEUVER** CONFIGURATION



A130400 **TILT-BEFORE-TURN MANOEUVER** CONFIGURATION



Friction stay that allows to limit the opening on hinged windows.

- Compatible with side-hung, tilt & turn and bottom-hung openings.
- The guide has an adjustable stop that allows a greater or lesser opening of the sash.

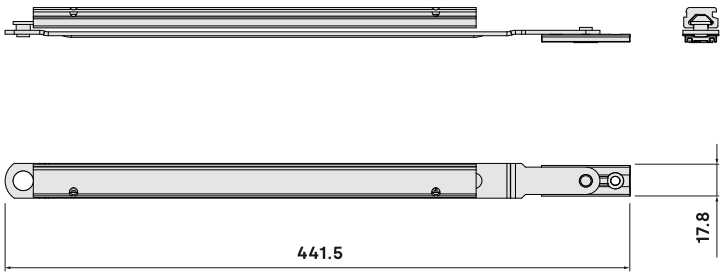
G16 FRICTION STAY WITH RESTRICTION

C16

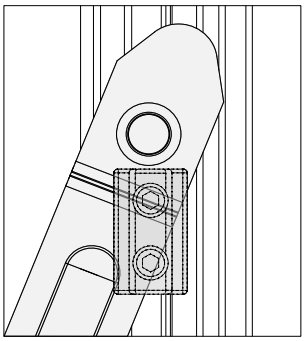
AL ZM IN

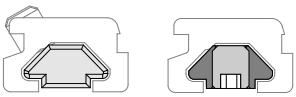
130500

50

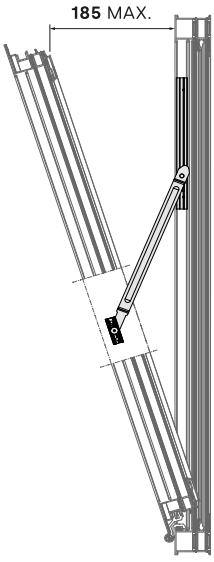


STOP GUIDE

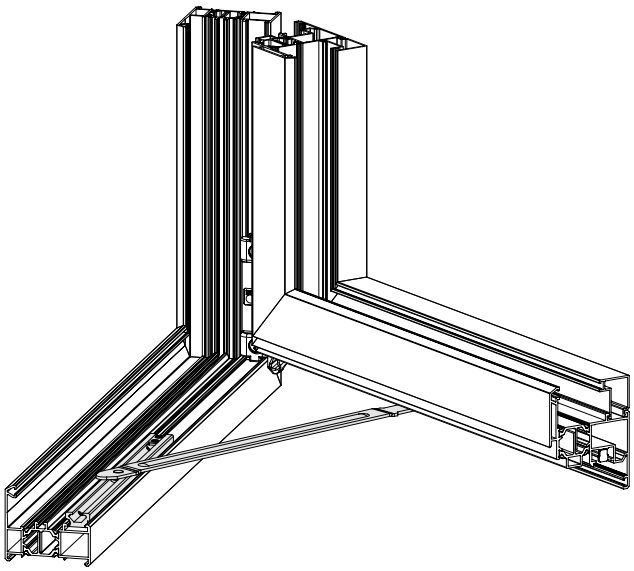




BOTTOM-HUNG OPENING



SIDE-HUNG OPENING



ONE-WAY TRANSMISSION (B15)

0517078

20

C16

AL

ZM

Transmission mechanisms which, by means of a handle, allow to move the rods and the locking points that operate the window.

- Valid for use with 7 mm spindle handles.
- 180° mono-directional movement and 15 mm (ref. 0517078) or 35 mm backset (ref. 0517194).
- They incorporate a stop and an encased locking point to limit the travel of the gearbox depending on the type of window opening: side-hung, bottom-hung, top-hung or outward opening.

ONE-WAY TRANSMISSION (B35)

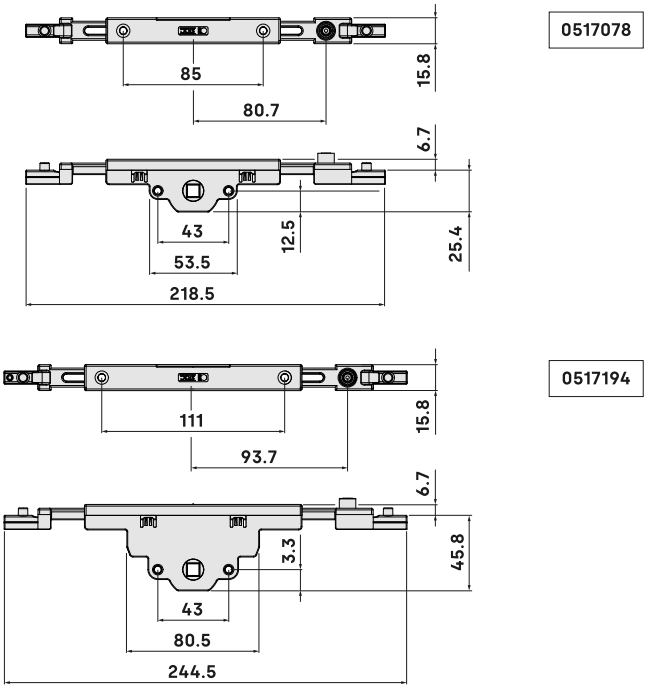
0517194

15

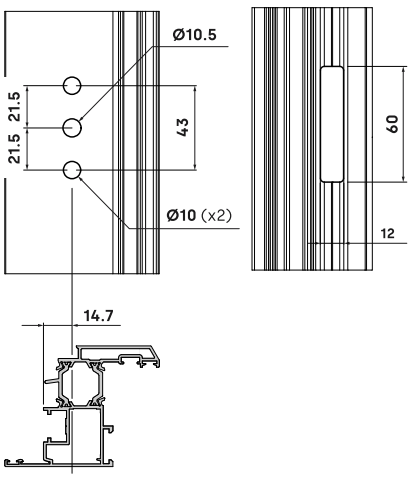
C16

AL

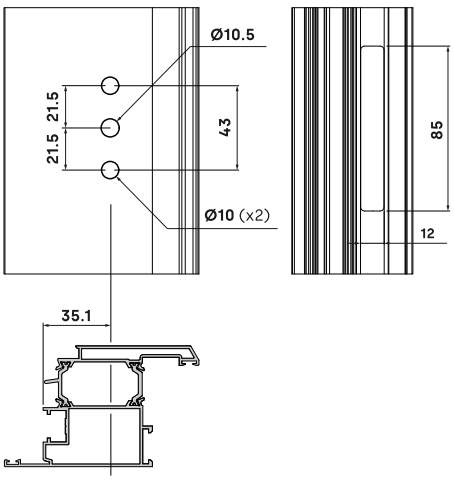
ZM



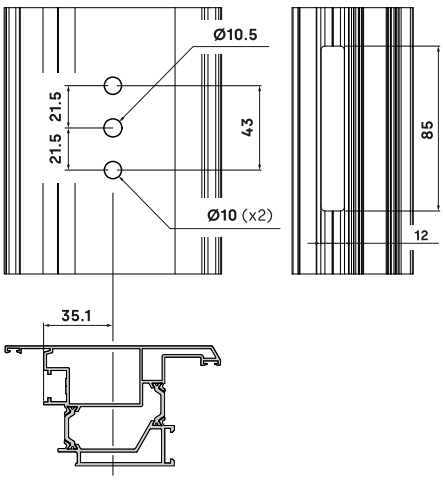
INWARD OPENING
PROFILE

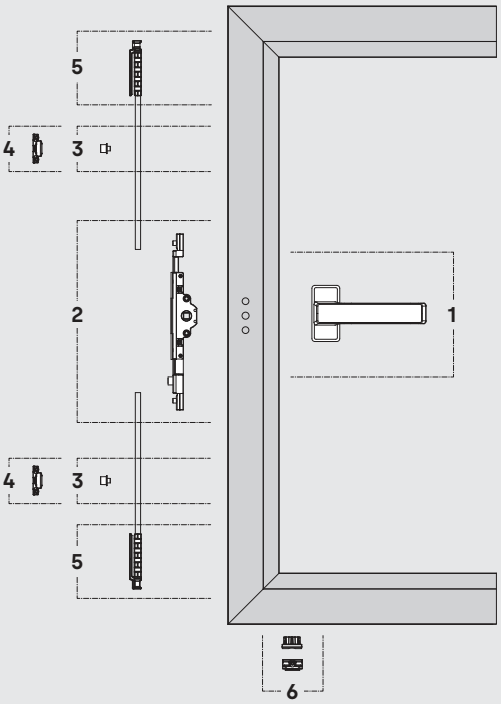


BALCONY DOOR
PROFILE



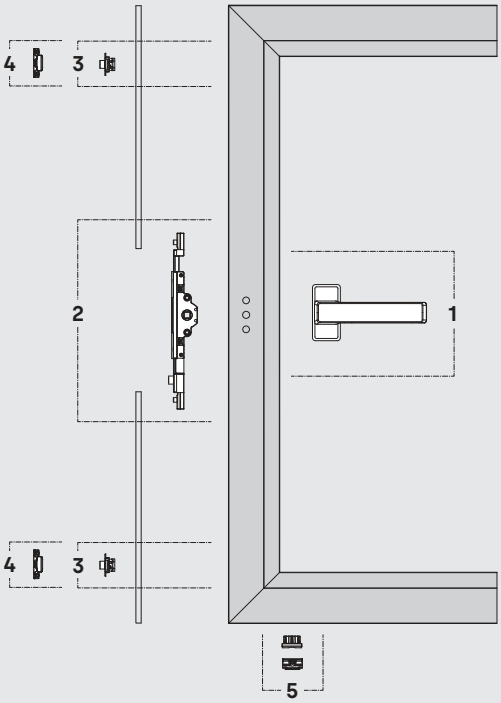
OUTWARD OPENING
PROFILE





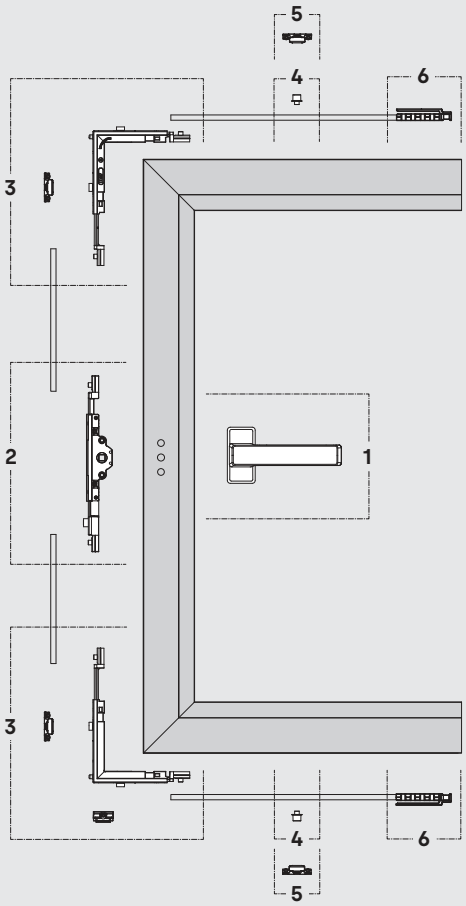
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH RIVETS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**0517002**	**Ø6 RIVET FOR ROD**	50
4	**170200**	**ADJUSTABLE LOCKING POINT**	100
5	**0517083**	**ROD TERMINAL TRIM**	50
6	**0517226**	**POSITIONING LATCH KIT**	50



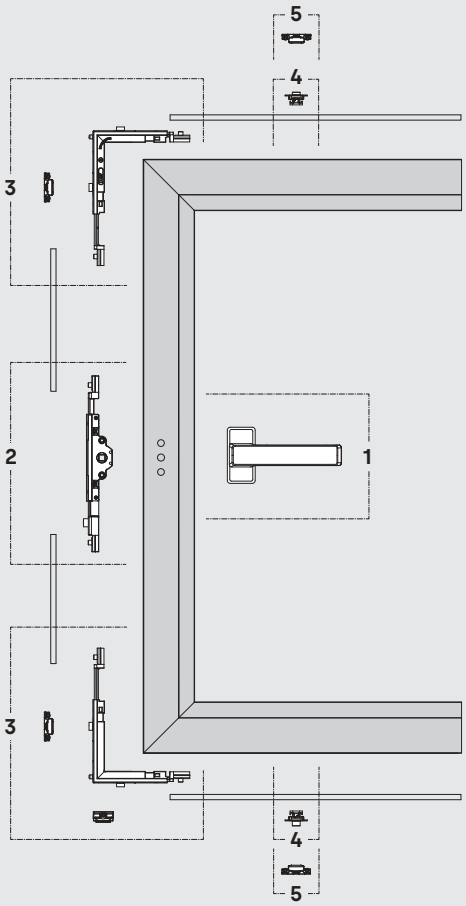
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH SLOTTED LOCKING POINT

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**178000**	**SLOTTED LOCKING POINT**	100
4	**170200**	**ADJUSTABLE LOCKING POINT**	100
5	**0517226**	**POSITIONING LATCH KIT**	50



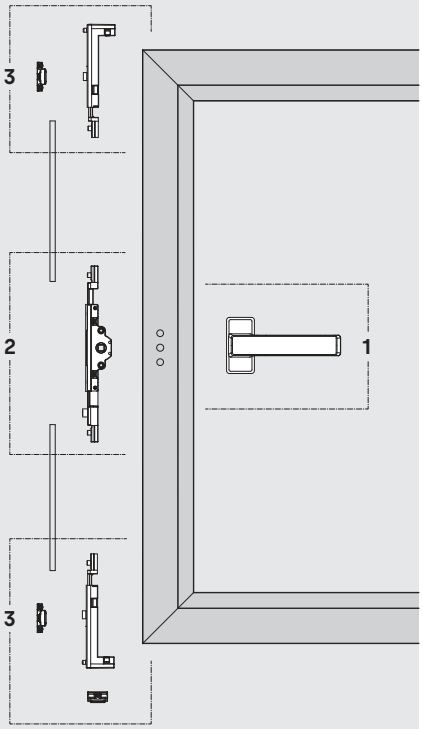
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND RIVETS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
		CORNER DRIVE	
3	**17K1HCM**	**ADJUSTABLE LOCKING POINTS**	1
		FRAME ANTI-SLIP	
4	**0517002**	**Ø6 RIVET FOR ROD**	50
5	**170200**	**ADJUSTABLE LOCKING POINT**	100
6	**0517083**	**ROD TERMINAL TRIM**	50



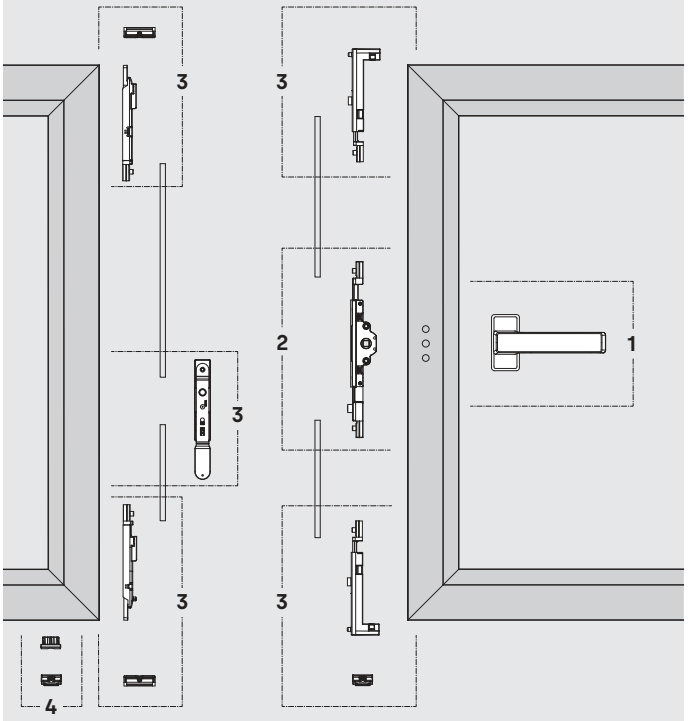
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND SLOTTED LOCKING POINTS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
		CORNER DRIVE	
3	**17K1HCM**	**ADJUSTABLE LOCKING POINTS**	1
		FRAME ANTI-SLIP	
4	**178000**	**SLOTTED LOCKING POINT**	100
5	**170200**	**ADJUSTABLE LOCKING POINT**	100



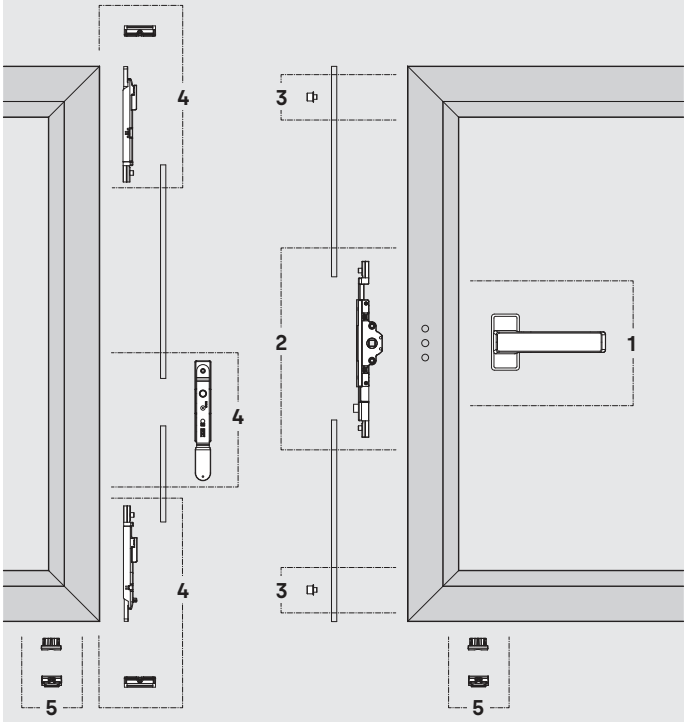
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**17K1HCI**	**POINTERS** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP**	50



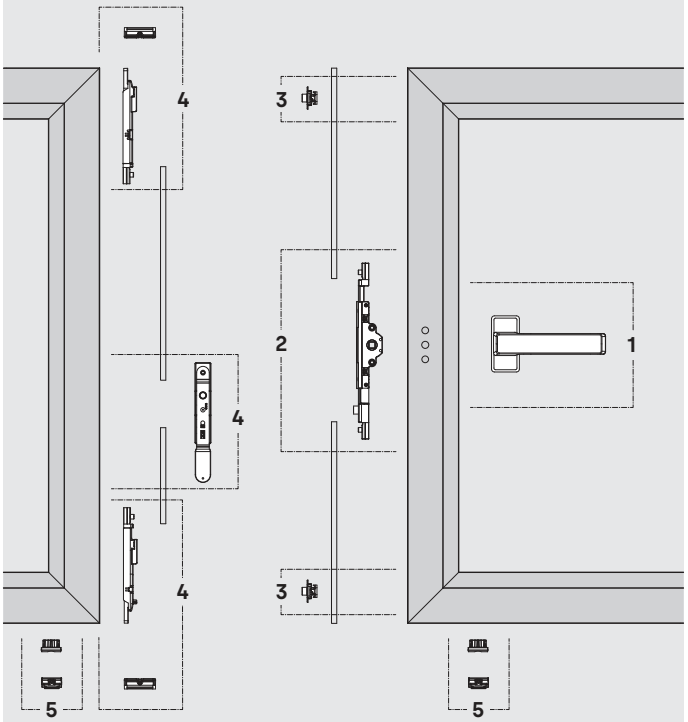
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**17K2HCI**	**POINTERS** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	1
4	**0517226**	**POSITIONING LATCH KIT**	50



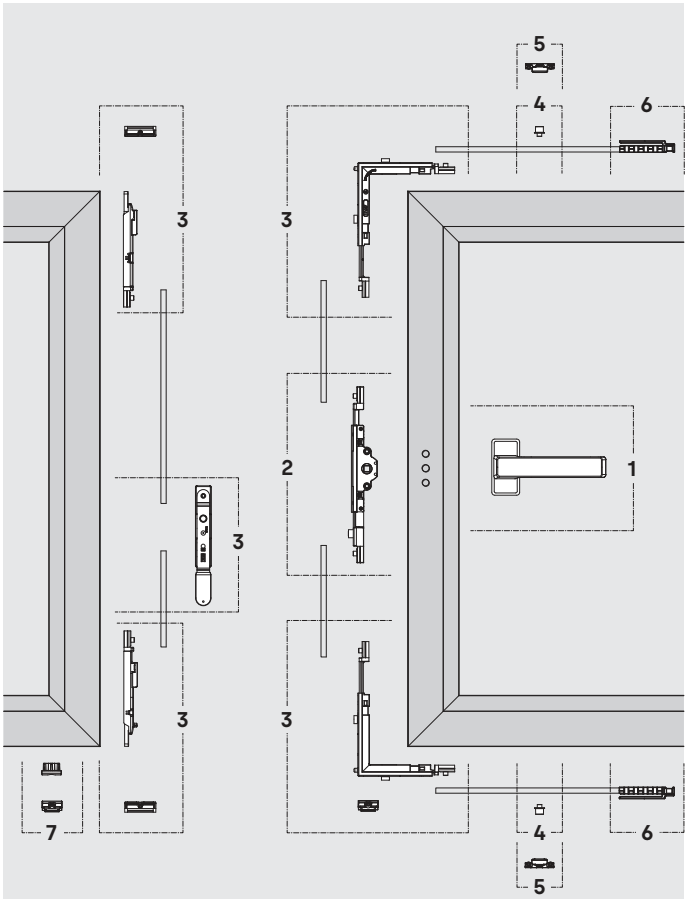
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH RIVETS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**0517002**	**Ø6 RIVET FOR ROD**	50
4	**171600**	**TWO-WAY SHOOTBOLT KIT**	20
5	**0517226**	**POSITIONING LATCH KIT**	50



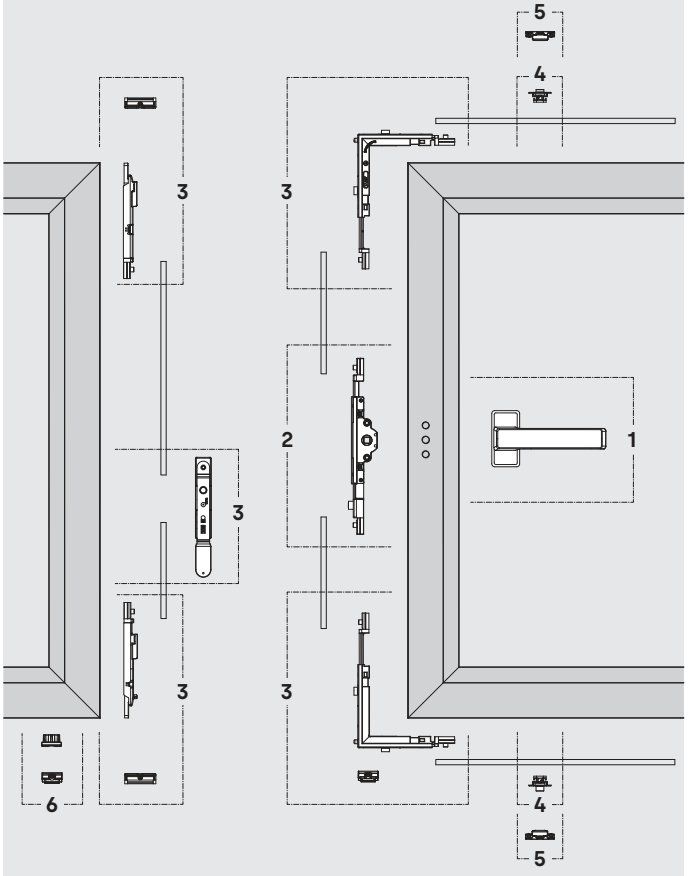
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH SLOTTED LOCKING POINTS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**178000**	**SLOTTED LOCKING POINT**	100
4	**171600**	**TWO-WAY SHOOTBOLT KIT**	20
5	**0517226**	**POSITIONING LATCH KIT**	50



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND RIVETS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**17K2HCM**	**CORNER DRIVE** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
4	**0517002**	**Ø6 RIVET FOR ROD**	50
5	**170200**	**ADJUSTABLE LOCKING POINT**	100
6	**0517083**	**ROD TERMINAL TRIM**	50
7	**0517226**	**POSITIONING LATCH KIT**	50



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND SLOTTED LOCKING POINTS

1	**HANDLE**	*Consult handle section*	
2	**0517078**	**C16 GEARBOX**	20
3	**17K2HCM**	**CORNER DRIVE** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
4	**178000**	**SLOTTED LOCKING POINT**	100
5	**170200**	**ADJUSTABLE LOCKING POINT**	100
6	**0517226**	**POSITIONING LATCH KIT**	50

UPPER PRO CORNER DRIVE

05370900

15

C16

ZM AC IN

Stainless steel angle returns for 16 Groove profile systems.

• Valid for hinge windows with tilt & turn or side-hung opening.

Upper corner drive (ref. 05370900)

• Supplied with two encased locking points that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.
• Compatible with top-hung and bottom-hung opening windows.

Lower corner drive (ref. 05373400)

• The lower stainless steel corner drive has a front encased locking point and a mushroom for tilt & turn opening, which is also used as an anti-lifting device and security locking point.

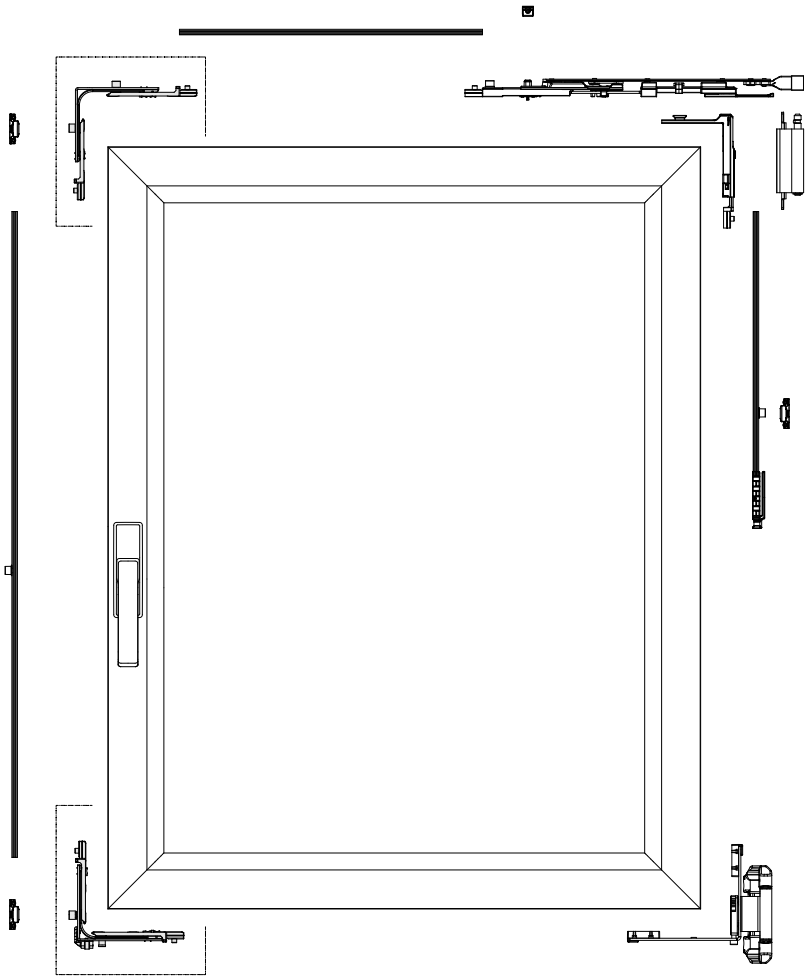
LOWER PRO CORNER DRIVE

05373400

15

C16

ZM AC IN



Friction stay that allows stable ventilation positions to be set for side-hung windows.

- The ventilation position is set by means of a retaining spring.
- Allows adjustment of the window opening in the ventilation position.
- Quick installation thanks to direct pre-mounting on sash and frame.
- Additional fixing by means of self-tapping screws (not included).

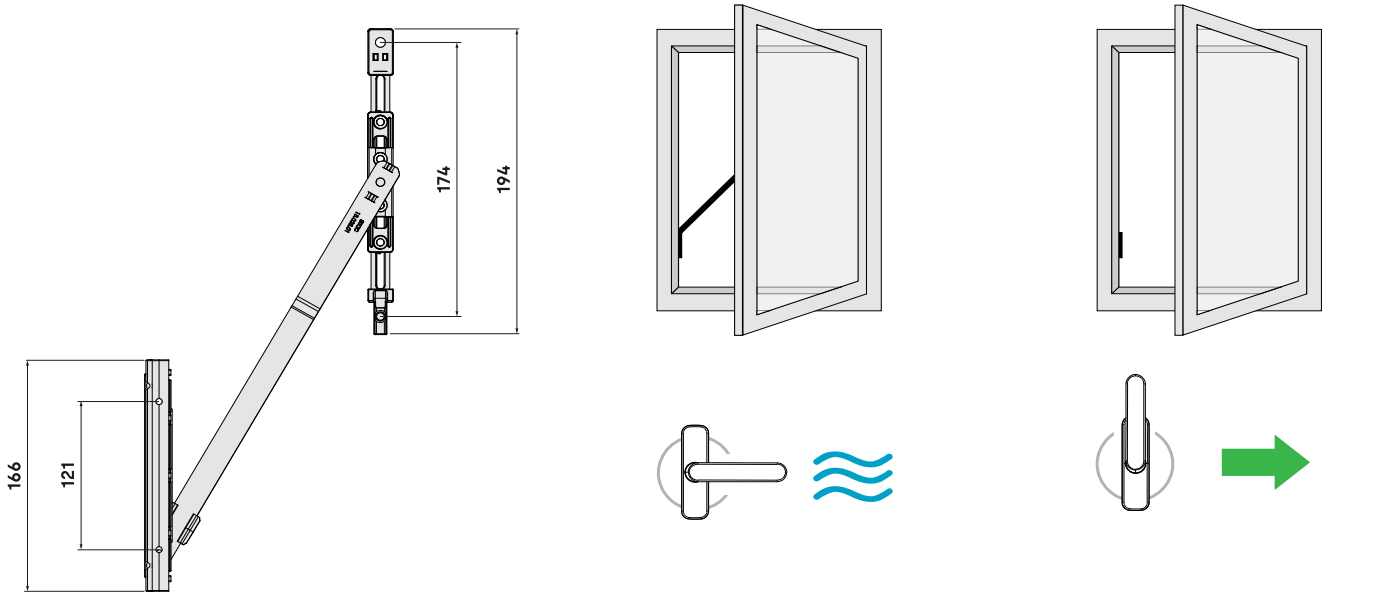
LIMITING FRICTION STAY WITH AERATION

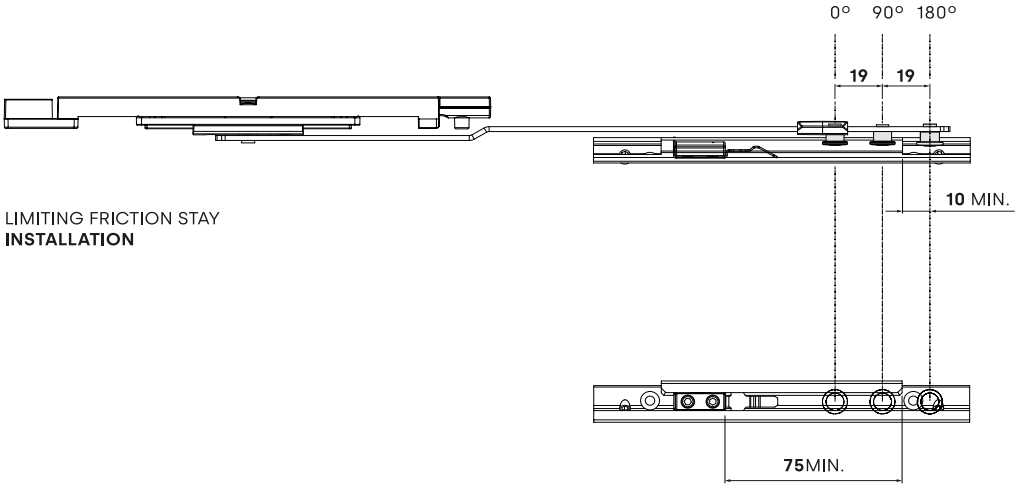
C16

AL ZM IN

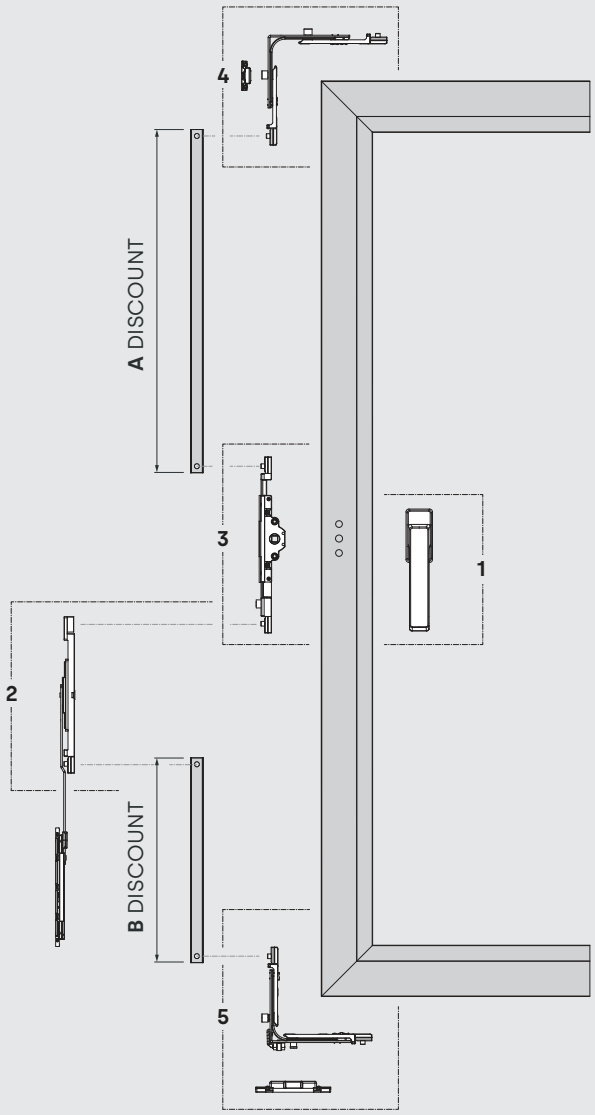
130720

15



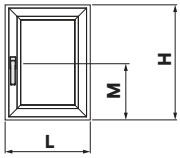


LIMITING FRICTION STAY
INSTALLATION



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE

1	**HANDLE**	*Consult handle section*	
2	**130720**	**LIMITING FRICTION STAY**	15
3	**0517078**	**C16 GEARBOX**	20
4	**05370900**	**UPPER CORNER DRIVE**	15
5	**05373400**	**LOWER CORNER DRIVE**	14



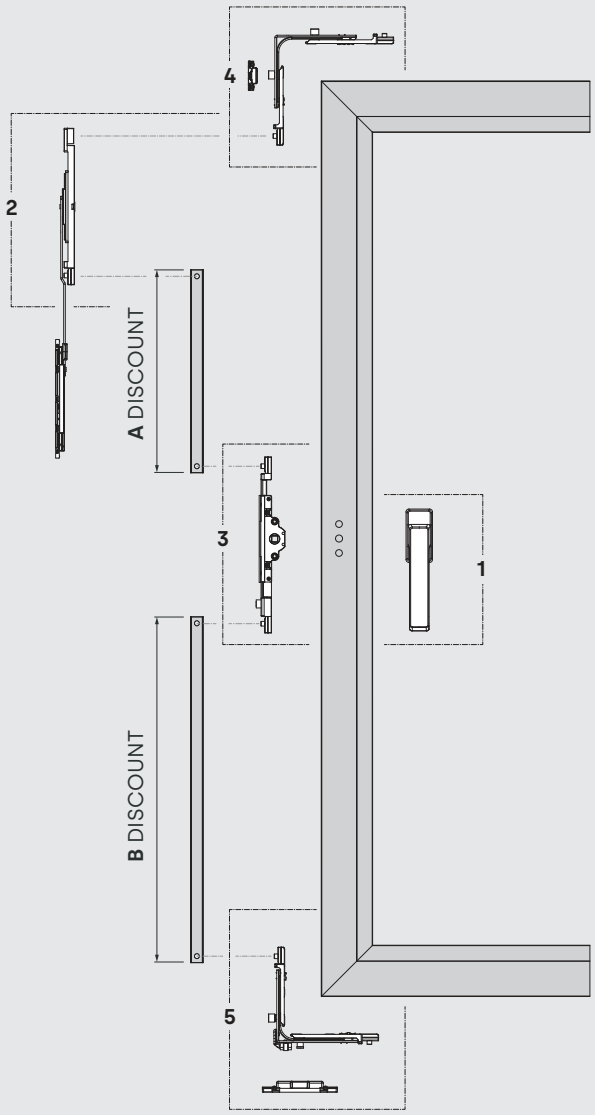
H min. = 900 mm

M min. = 615 mm

DISCOUNTS

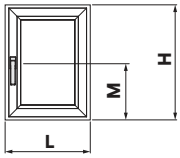
A	H - M - 245
B	M - 404

***For sash heights greater than 1800 millimetres**, we recommend the configuration with the limiting friction stay attached to the bottom of the gearbox.*



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE

1	**HANDLE**	*Consult handle section*	
2	**130720**	**LIMITING FRICTION STAY**	15
3	**0517078**	**C16 GEARBOX**	20
4	**05370900**	**UPPER CORNER DRIVE**	15
5	**05373400**	**LOWER CORNER DRIVE**	14



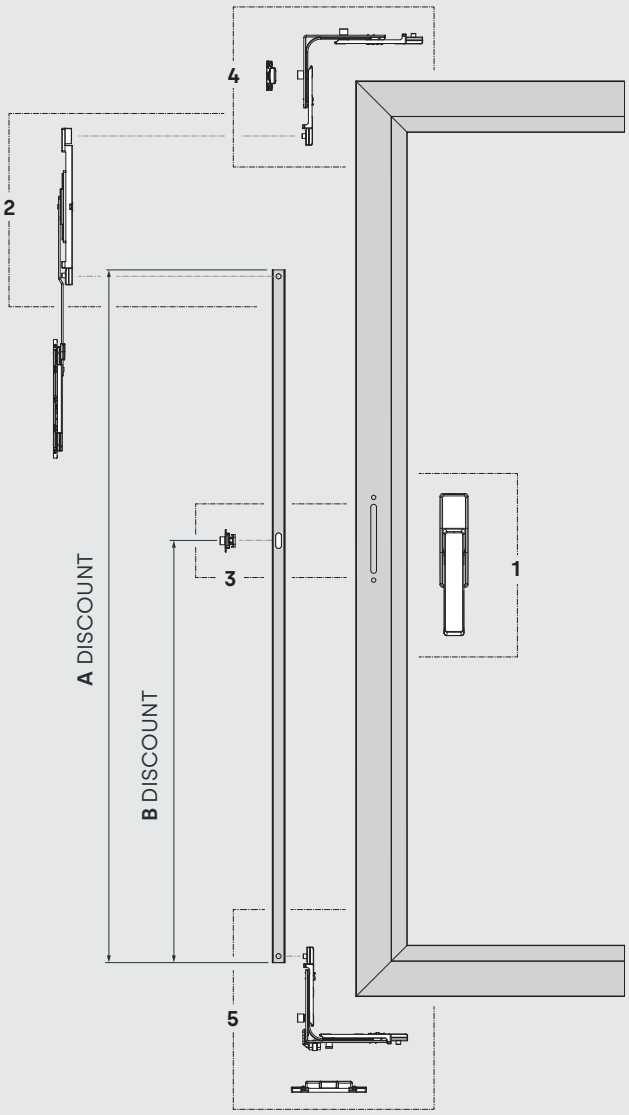
H min ~ H max. = 900 ~ 1800 mm

M max. = 270 mm

DISCOUNTS

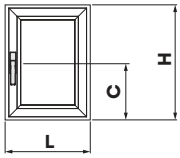
A	H - M - 419
B	M - 230

***For sash heights greater than 1800 millimetres**, we recommend the configuration with the limiting friction stay attached to the bottom of the gearbox.*



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH SLOTTED LOCKING POINT

1	**CREMONE**	*Consult cremone section*	
2	**130720**	**LIMITING FRICTION STAY**	15
3	**178000**	**SLOTTED LOCKING POINT**	100
4	**05370900**	**UPPER CORNER DRIVE**	15
5	**05373400**	**LOWER CORNER DRIVE**	14

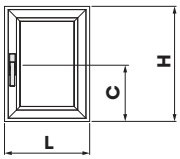


H min. ~ max. = 700 ~ 1800 mm

C max. = 220 mm

DISCOUNTS

A	H - 470
B	C - 148



H min. ~ max. = 700 ~ 1800 mm

C max. = 220 mm

***For sash heights greater than 1800 millimetres**, we recommend the configuration with the limiting friction stay attached to the bottom of the gearbox.*

Accessory kit that allows a ventilated position in hinged windows (tilt & turn or side-hung).

• The design of the keeper allows the sash to be opened between 1 and 6 mm when turning the handle or the cremone.
• The device complies with all the requirements of the Technical Building Code regarding indoor air quality, guaranteeing the required ventilation.

Tests (class 1)

• Air permeability according to the UNE EN 12207: 2017 standard in the micro-opening position.
• Values obtained according to the UNE-EN 1026 standard carried out at notified body 0370 in accordance with report number 09 / 32302792.

DEVICE FOR MICROVENTILATION

C16

ZM

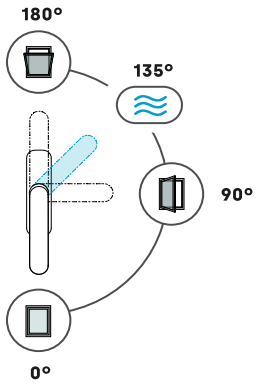
IN

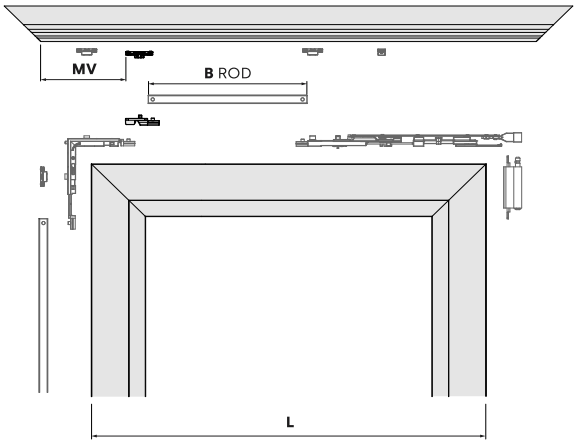
174401

100

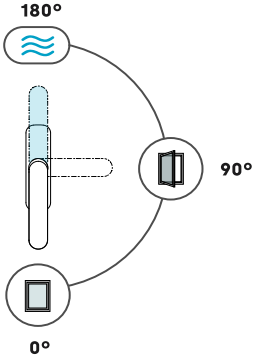
DISCOUNTS **FOR MICROVENTILATION**		**MIN L** (mm)	**MV** (mm)	**B ROD**(mm)
170301 D I **	LARGE FRICTION STAY	665	128	L - 626
170302 D I **	MEDIUM FRICTION STAY	582	128	L - 582
170303 D I **	SMALL FRICTION STAY	488	128	L - 488
170306 D I **	MINI	410	128	L - 410
170301 D I ** 130400	LARGE FRICTION STAY WITH SUPPLEMENTARY	?	?	?

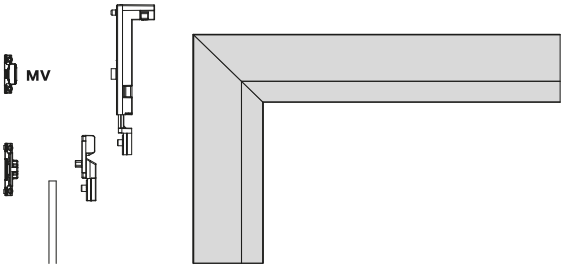
TILT & TURN
OPENING

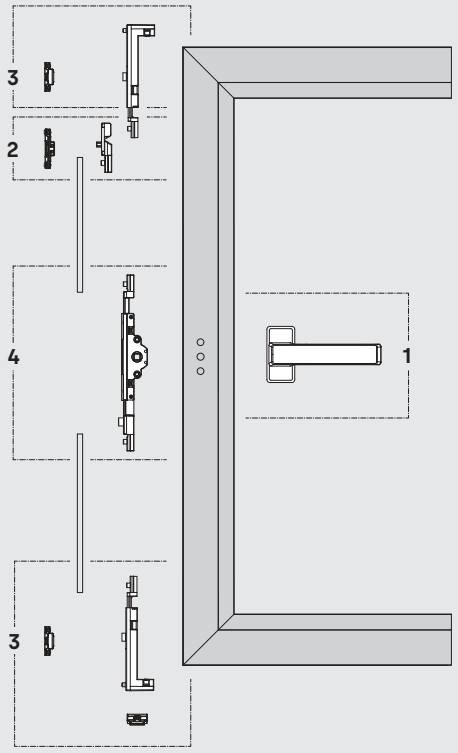




SIDE-HUNG
OPENING

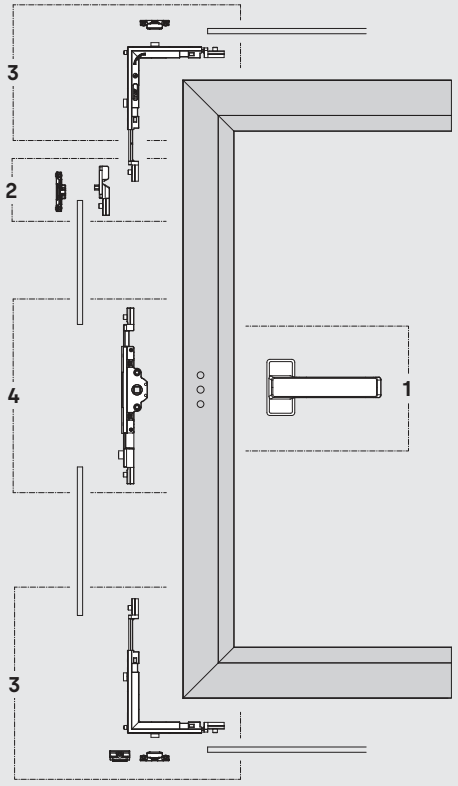






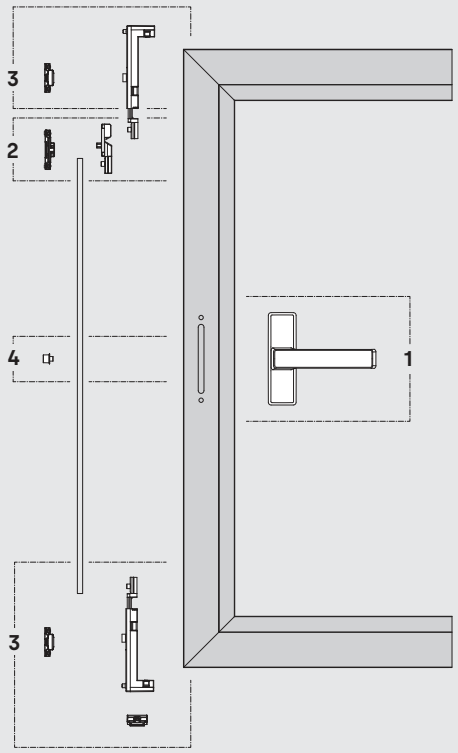
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTERS AND GEARBOX

1	**HANDLE**	*Consult handle section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
		POINTERS	
3	**17K1HCI**	**ADJUSTABLE LOCKING POINTS**	50
		FRAME ANTI-SLIP	
4	**0517078**	**C16 GEARBOX**	20



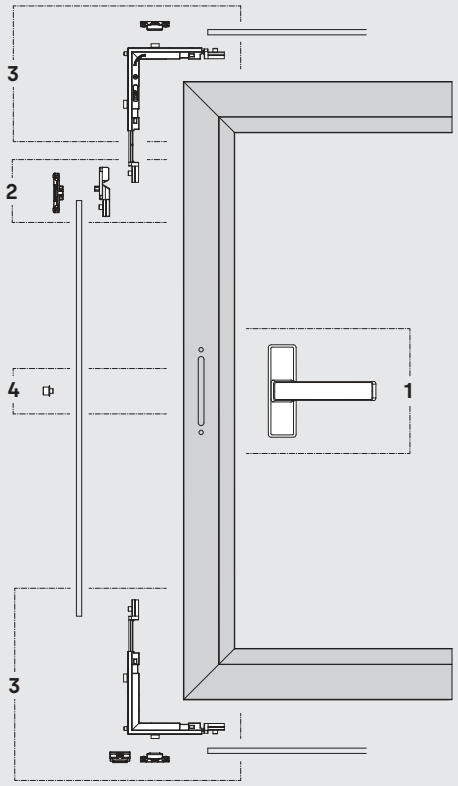
SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND GEARBOX

1	**HANDLE**	*Consult handle section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
		CORNER DRIVE	
3	**17K1HCM**	**ADJUSTABLE LOCKING POINTS**	1
		FRAME ANTI-SLIP	
4	**0517078**	**C16 GEARBOX**	20



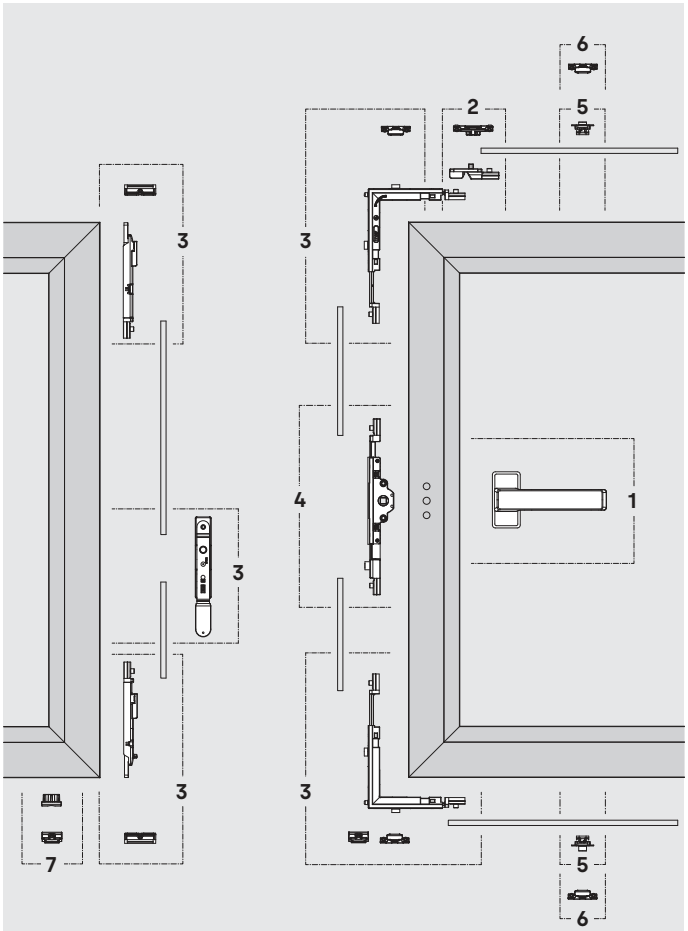
SIDE-HUNG OPENING (LOCKING TO INVERTER)
WITH POINTER AND RIVET

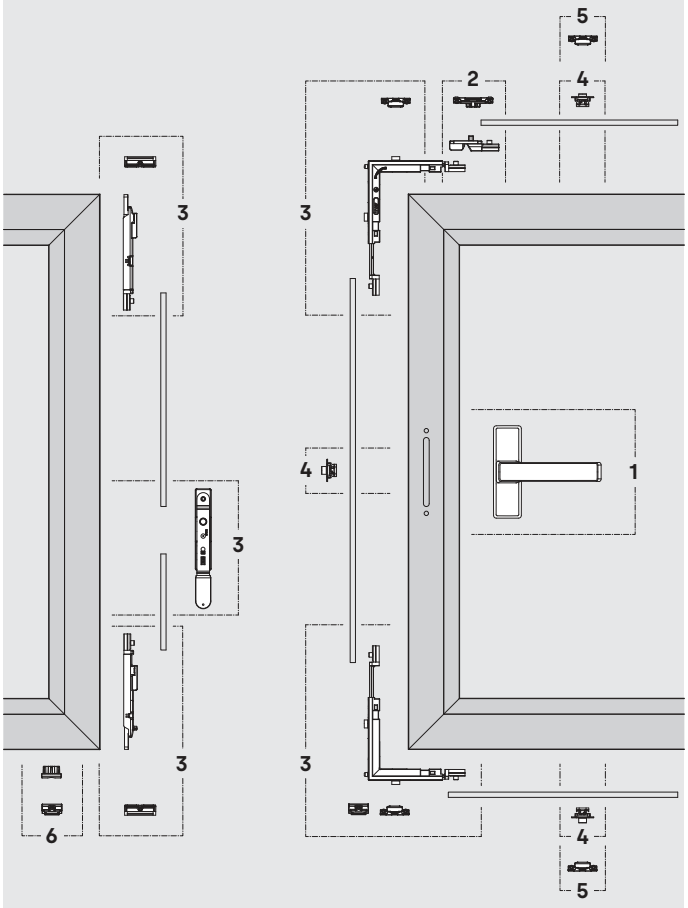
1	**CREMONE**	*Consult cremone section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
		POINTERS	
3	**17K1HCI**	**ADJUSTABLE LOCKING POINTS**	50
		FRAME ANTI-SLIP	
4	**0517002**	**Ø6 RIVET FOR ROD**	100



SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND RIVET

1	**CREMONE**	*Consult cremone section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
		CORNER DRIVE	
3	**17K1HCM**	**ADJUSTABLE LOCKING POINTS**	1
		FRAME ANTI-SLIP	
4	**0517002**	**Ø6 RIVET FOR ROD**	100





SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND GEARBOX

1	**HANDLE**	*Consult handle section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
3	**17K2HCM**	**POINTERS** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
4	**0517078**	**C16 GEARBOX**	20
5	**178000**	**SLOTTED LOCKING POINT**	100
6	**170200**	**ADJUSTABLE LOCKING POINTS**	100
7	**0517226**	**POSITIONING LATCH KIT**	50

SIDE-HUNG OPENING (LOCKING TO FRAME)
WITH CORNER DRIVE AND SLOTTED LOCKING POINT

1	**CREMONE**	*Consult cremone section*	
2	**174401**	**MICRO-VENTILATION DEVICE**	100
3	**17K2HCM**	**CORNER DRIVE** **ADJUSTABLE LOCKING POINTS** **FRAME ANTI-SLIP** **TWO-WAY LEVER KIT**	50
4	**178000**	**SLOTTED LOCKING POINT**	100
5	**170200**	**ADJUSTABLE LOCKING POINTS**	100
6	**0517226**	**POSITIONING LATCH KIT**	50

Keepers for profile systems with 16 Groove.

Keeper for the inverter profile (ref. 177000)

• Keeper for the window inverter profile to provide an additional vertical locking point.
• Installation recommended from 1200 mm sash height.

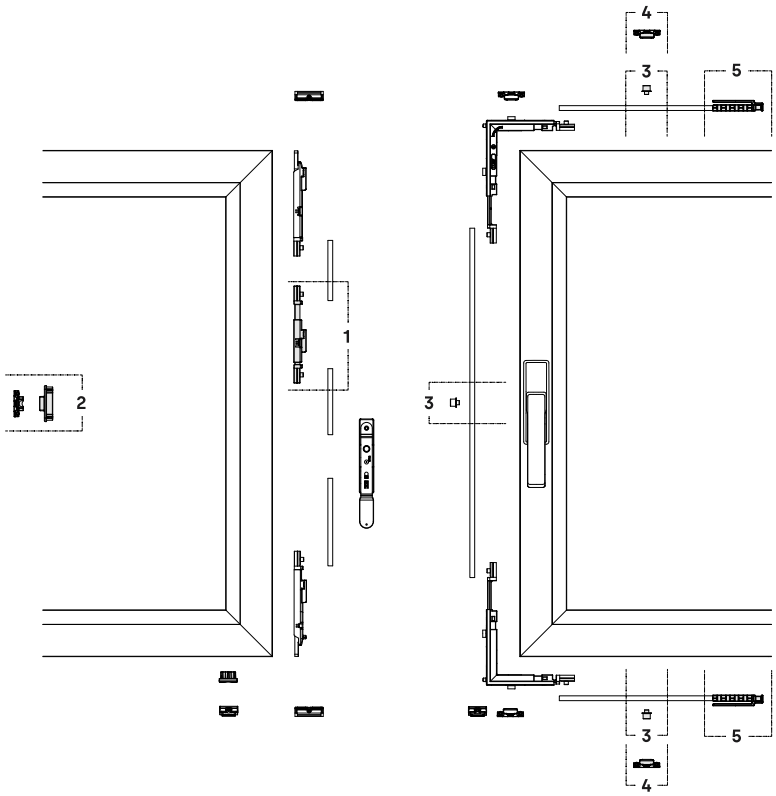
Hidden locking point (ref. 172500)

• Additional locking point for passive sash in tilt & turn opening and for both sashes (active ~ passive) in side-hung opening.
• Installation recommended from 1200 mm sash height.

Transmission rod trim (ref. 0517083)

• The terminal trim is attached to the sash to secure the free ends of the rods in configurations that have a rivet as an additional locking point.

1	177000	KEEPER FOR THE INVERTER PROFILE
2	172500	HIDDEN LOCKING POINT KIT
	178000	Ø8 SLOTTED
3	178000C	Ø8 ENCASED SLOTTED
	0517002	Ø6 RIVET
4	170200	ADJUSTABLE LOCKING POINT
5	0517083	ROD TERMINAL TRIM



KEEPER FOR THE INVERTER PROFILE

C16 ZM IN

177000

50

HIDDEN LOCKING POINT

C16 ZM IN

172500

50

ROD TERMINAL TRIM

C16 ZM

0517083

50

SLOTTED LOCKING POINT

178000

100

C16

ZM

IN

Ø6 RIVET

0517002

50

CE

ZM

ADJUSTABLE LOCKING POINT

170200

100

C16

ZM

IN

Locking points for profile systems with 16 Groove.

Ø8 slotted locking point (ref. 178000)

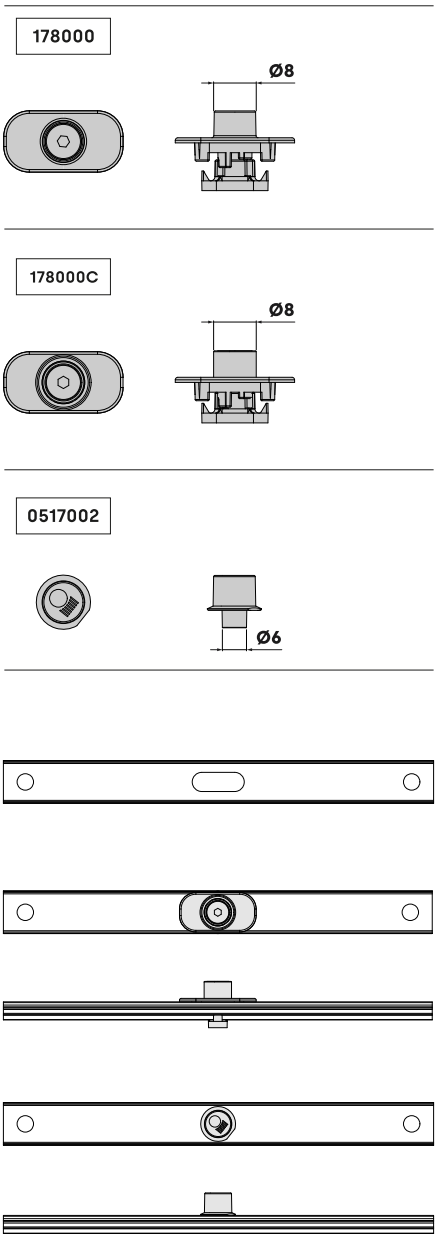
- Allows additional locking points to be obtained manually.
- The cremone lever acts on the locking point, without the need for a cover.
- Available with a slotted encased locking point (ref. 178000C) that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.

Ø6 rivet (ref. 0517002)

- It allows to obtain a locking point in whatever point where the operating rod passes through without having to be machined.
- The cremone lever acts on the locking point, without the need for a cover.
- Compatible with die-cutting machines for 16 Groove: TFS-05 and TFS-06.

Adjustable locking point (ref. 170200)

- Accessory for frame profile that allows the addition of a locking point with tightening adjustment (± 1.5 mm).



C16 hinge kit (ref. 1714)

- Hinges with opening hand: right-hand opening (ref. 1714D) or left-hand opening (ref. 1714I).
- Valid for installation on both the active and passive sash to obtain the same settings as in tilt & turn opening.

Corner drive (ref. 05374700)

- Supplied with an encased locking point that prolongs the durability of the keepers and reduces wear due to friction, while facilitating greater smoothness in the closing manoeuvre of the window.
- Compatible with the two-way lever (ref. 05171668).

C16 HINGE KIT

C16

AL ZM IN PA

1714	D / I	**

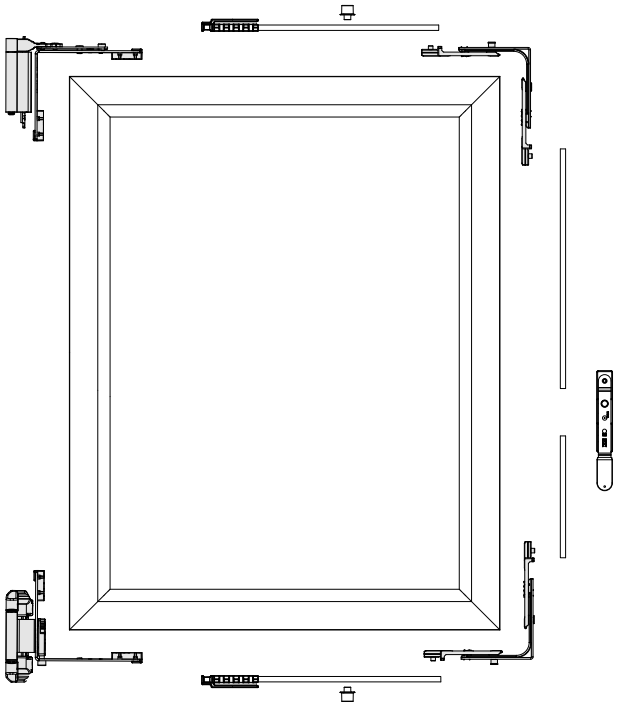
20

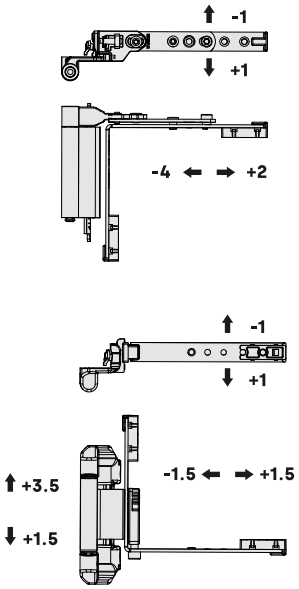
CORNER DRIVE

C16

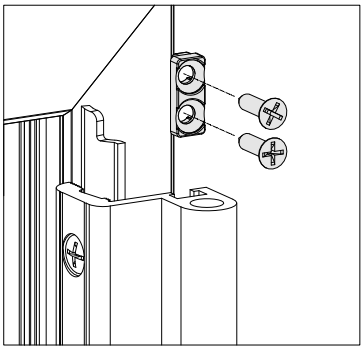
ZM IN AC

05374700

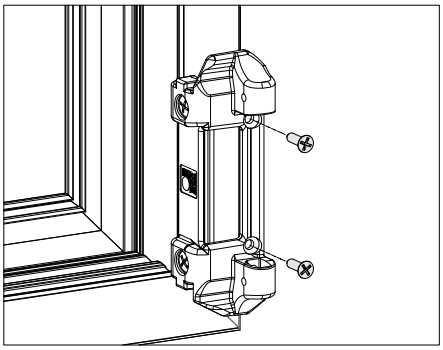




UPPER HINGE
REINFORCEMENT PIECE



FRAME PROFILE
LEAF



For safety reasons, *for tilt & turn openings with sash width and sash height of 1000 and 1800 mm respectively, it is mandatory to screw the upper hinge lever to the frame profile and to install the reinforcement piece in the upper hinge.*

G16 TWO-WAY LEVER

05171668

20

C16

ZM

IN

Accessories that allow the opening or closing of the passive sash in profile systems with 16 Groove.

G16 two-way lever (ref. 05171668)

• The two-way lever allows the opening or closing of the sash by means of the connection with transmission plates.

G16 two-way lever (ref. 171600)

• Supplied with terminals, double keepers and two-way lever.
• The terminals of the kit incorporate locking points with a ± 2.2 mm tightening adjustment and an anti-lifting projection on the lower terminal.

G16 TWO-WAY LEVER KIT

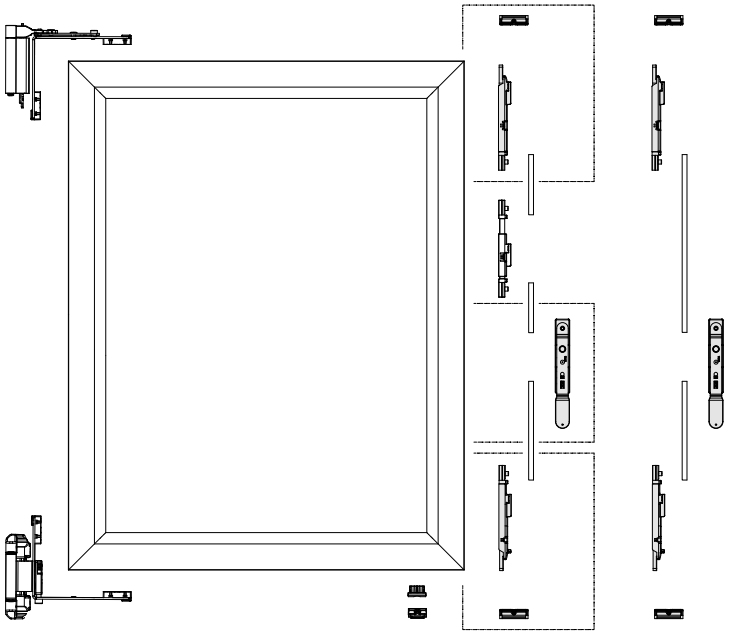
171600

20

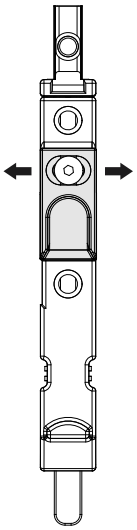
C16

ZM

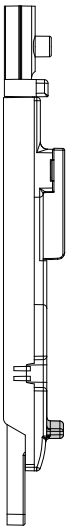
IN



LOCKING POINT ***WITH ADJUSTMENT***



ANTI-LIFTING *PROJECTION*



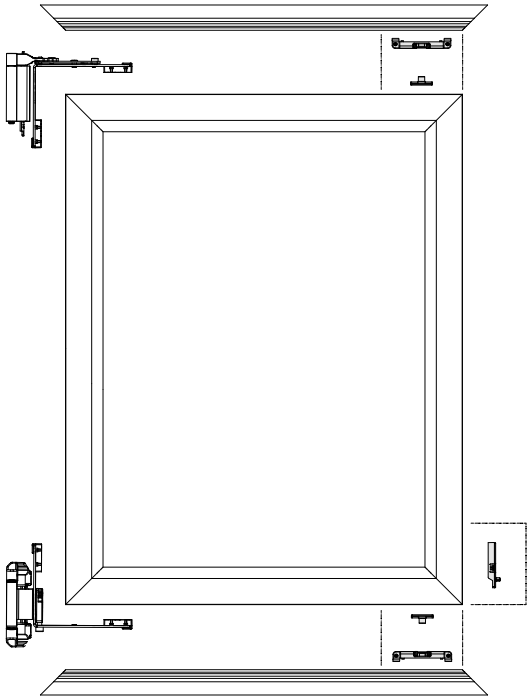
Accessories for secondary or passive sash of systems with 16 Groove profiles.

Kit with retaining clip (ref. 0517047)

- Allows the passive sash to remain locked when it is closed.
- The mechanism can be adjusted on the lateral groove.
- The retaining clip is self-assembling and is deactivated by pulling the sash directly.

Anti-lifting stop (ref. 0517091)

- Prevents disassembly of the sash when it is in the closed position.
- Installation of the anti-lift stop with the retaining clip is required.



RETENTION CLIP KIT

C16

ZM

IN

0517047

50

ANTI-LIFT STOP

C16

ZM

0517091

50

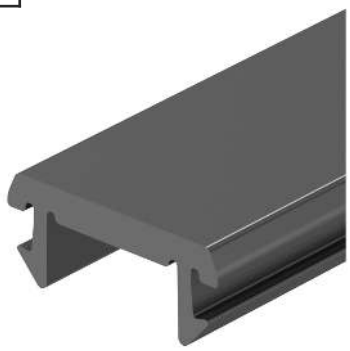
G16 OPERATING ROD

059234

200

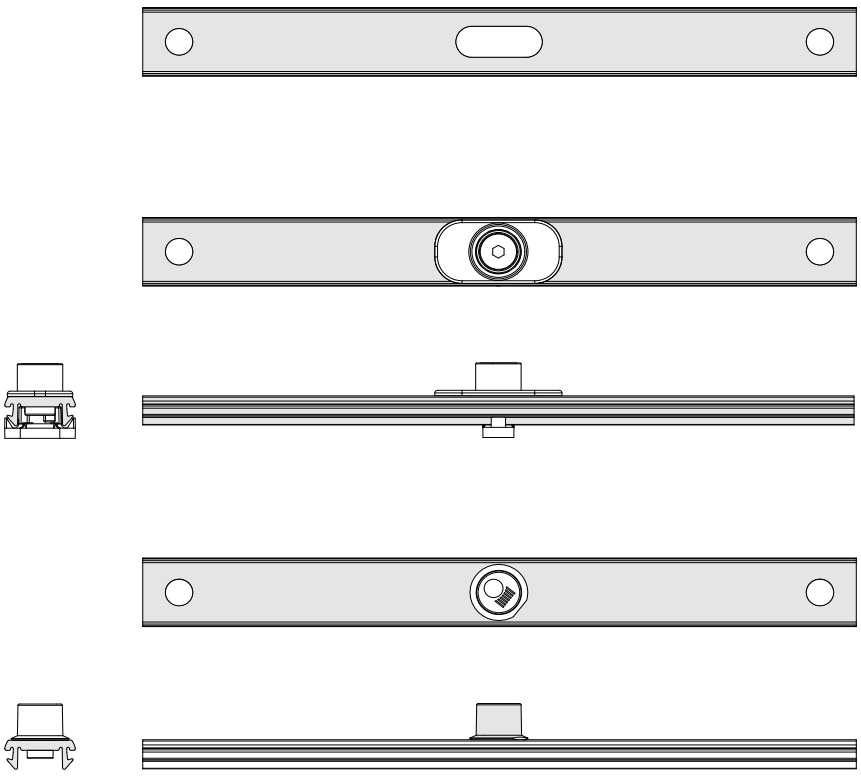
C16

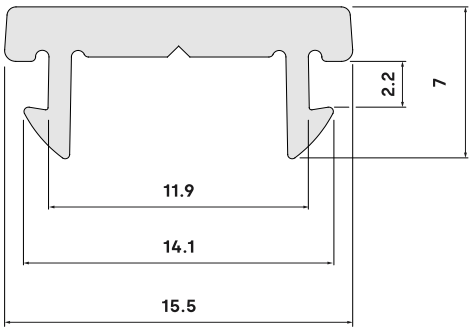
PA

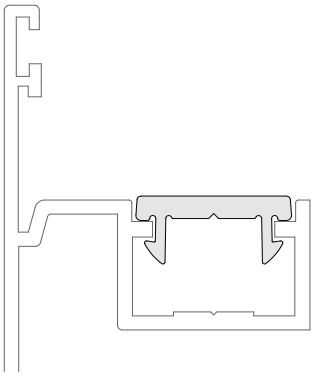


Transmission rod that transfers the movement of the different hardware and accessories that allow the operation of the windows.

- The C16 rod is suitable for systems with 16 Groove.
- Allows the installation of additional locking points.
- Smooth, fluid and jerk-free sliding through the profile groove.
- Cutting and machining to size using TFS G16 die-cutting machines.







16 ACCESSORIES

396 Sequencer devices

398 Locking points

399 Positioning latches

401 Areation devices

406 Limiting friction stays

410 Gearboxes

415 Machining templates

416 Transmission rods

417 Transom joints

418 Plastic injection moulding

TILT-BEFORE-TURN SEQUENCER

110220

10

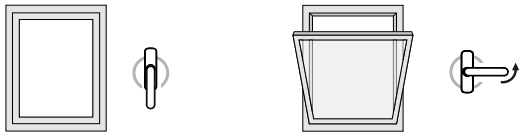
Mechanical device that limits the opening of side-hung and tilt & turn windows using keyless handles.

- The sequencer can be installed in different sash positions.
- The locking or unlocking is done by means of a TORX20 key.
- Allows an additional locking point on larger sashes.
- Compatible with tilt & turn systems with opening limitation.

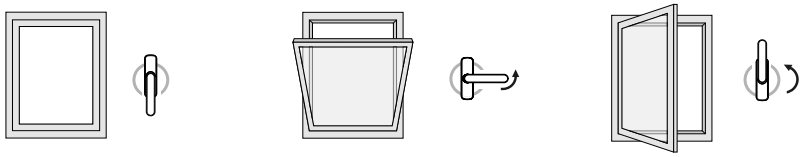
ZM

IN

SEQUENCER WITH LOCK
ALLOWS THE TURNING HANDLE UP TO 90°



SEQUENCER WITHOUT LOCK
ALLOWS THE TURNING HANDLE UP TO 180°



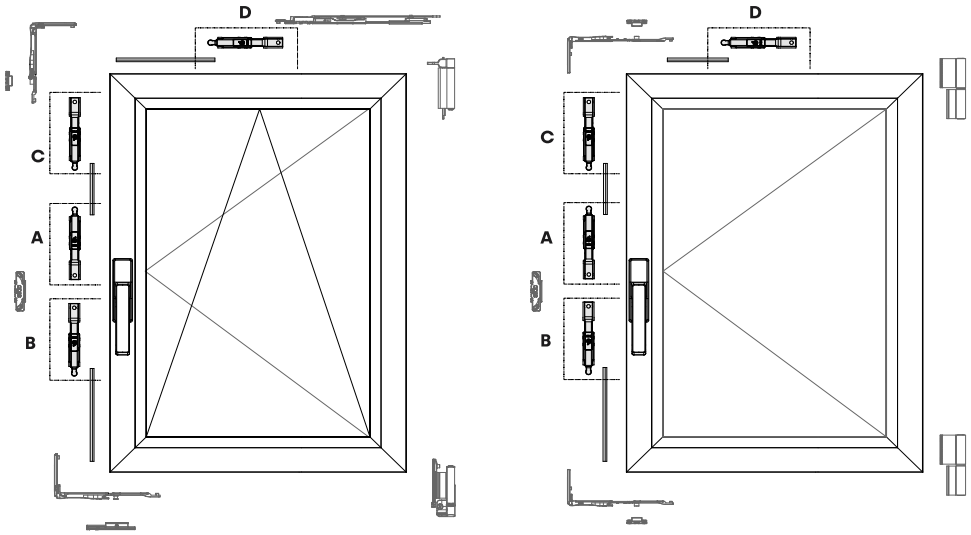
COMPATIBLE POSITIONS

A **UPPER PART**
OF THE CREMONE OR HANDLE

B **LOWER PART**
OF THE CREMONE OR HANDLE

C **COUPLED**
TO THE CORNER DRIVE

D **COUPLED**
TO THE FRICTION STAY



Accessory kit allows a balcony door to fully assume the functions of a door.

• The door sequencer enables a European cylinder locking system with 15 mm cam to be added to tilt & turn systems.
• Allows double handle configurations for indoor and outdoor access.
• The balcony door profile must have a minimum length of 48 mm.
• The use CLX hinges is recommended because of the wide range of adjustment it offers with respect to the load capacity of the hinges.

Tilt & turn door sequencer (CM) (ref. 110230).

• Valid for configurations with lower drive corner (locking to frame).

Tilt & turn door sequencer (CI) (ref. 110231).

• Valid for configurations with lower pointer (locking to inverter).
• Includes hardware connector (ref. 110212) that allows different accessories to be connected without the need for grooving in the profile.

Euro cylinder and cylinder covers are not included in the references.

TILT & TURN SEQUENCER

ce ce ZM PA

110230

5

HARDWARE CONNECTOR

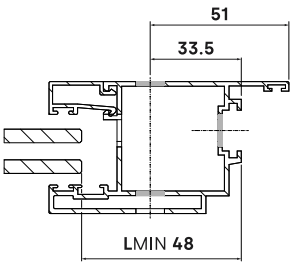
ZM

110212

1

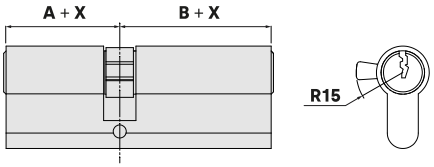
	LOCKING **TO INVERTER**	LOCKING **TO FRAME**
110230	–	✔
110231	✔	–

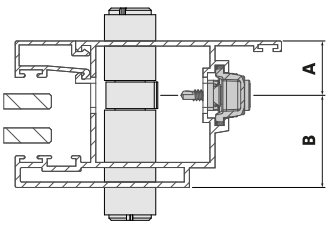
MINIMUM PROFILE LENGTH 48

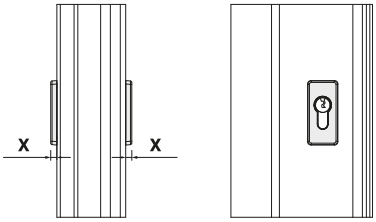


CYLINDER **COVERS**

0508077 **	DELTA	**X** 6.3 mm
0508053 **	SLIM	**X** 6.2 mm
0508029 **	SIRIUS	**X** 8.7 mm
0508028 **	GÉMINIS	**X** 10 mm



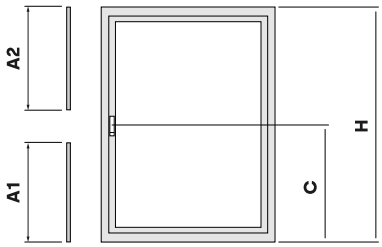




DISCOUNTS OF THE RODS

LOCKING TO FRAME CORNER DRIVE	
A2	H – C – 249
A1	C – 364

LOCKING TO INVERTER POINTER	
A2	H – C – 249
A1	C – 266



ADJUSTABLE LOCKING POINT

110200

100

Locking points for systems with Euro Groove profiles.

Adjustable locking point (ref. 110200)

• Accessory for frame profile that allows the addition of a locking point with tightening adjustment (±1.5 mm).

Supplementary locking point kit (ref. 113000IM)

• Supplied with an adjustable locking point (ref. 110200) and a sliding locking point.

Locking point for inverter profile (ref. 110210)

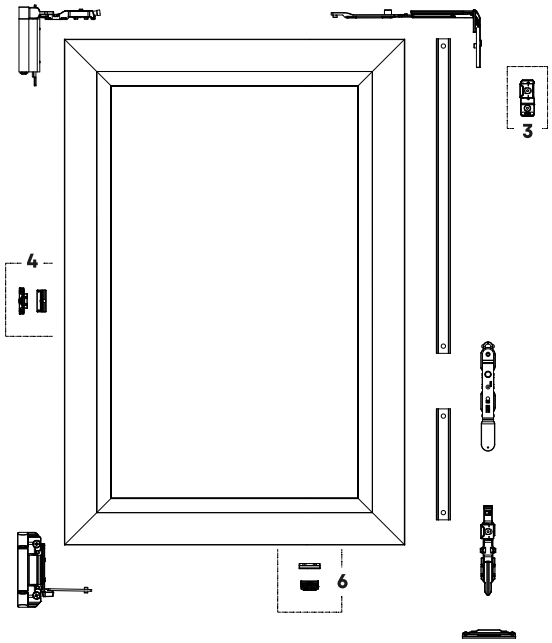
• Accessory for frame profile that allows the addition of a locking point with tightening adjustment (±1.5 mm).
• Recommended for flange thickness less than 1.8 mm.

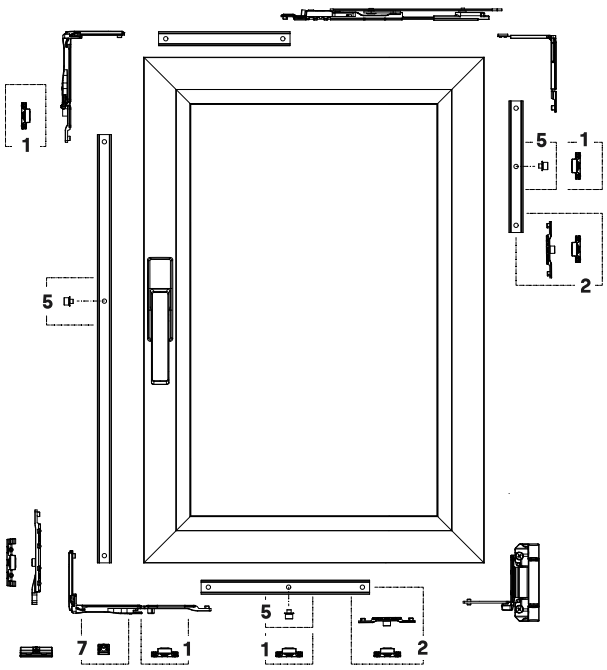
LOCKING POINT FOR INVERTER

110210

100

1	110200	LOCKING POINT WITH ADJUSTMENT
2	113000IM	SUPPLEMENTARY LOCKING POINT KIT
3	110210	LOCKING POINT FOR INVERTER
4	112500	HIDDEN LOCKING POINT
5	0511088	Ø8 RIVET
	0517002	Ø6 RIVET
	0511149	Ø6 EVO RIVET
6	0710621	POSITIONING LATCH KIT
7	0517193	POSITIONING LATCH





Accessories for side-hung, tilt & turn or top-hung windows.

Hidden locking point kit (ref. 112500)

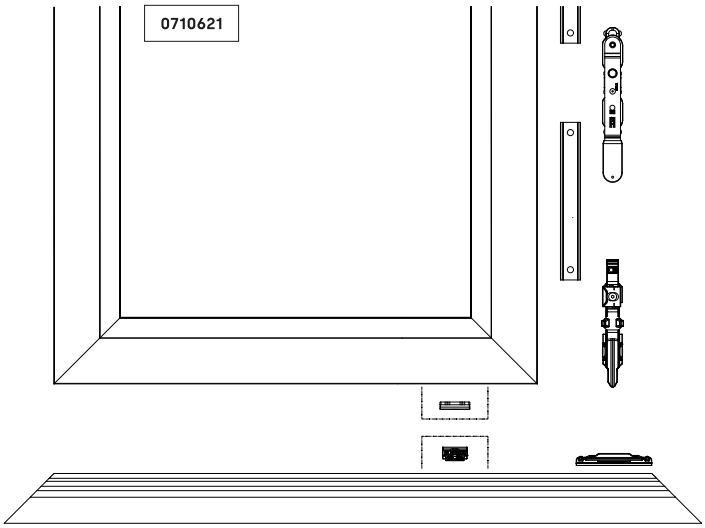
- Supplementary locking point for passive sash in tilt & turn window and for both sashes (active ~ passive) in side-hung opening.
- Recommended for installation from 1200 mm sash height.

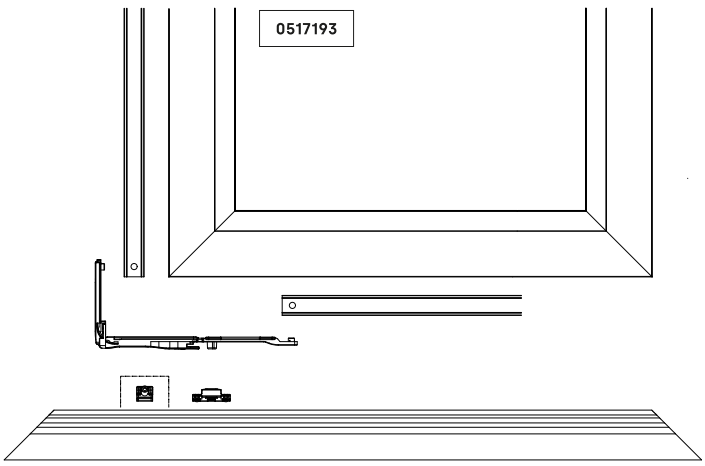
Positioning latch kit (ref. 0710621)

- It makes it easier to position the sash when closing the window.
- Valid for both active and passive sash in tilt & turn window with locking to the inverter in single or double sash configuration.
- Supplied with accessories for sash and frame profile.

Positioning latch (ref. 0517193)

- It makes it easier to position the sash when closing the window.
- Valid for passive sash in top-hung, tilt & turn and side-hung opening.
- Compatible with 16 Groove profiles in locking to frame configurations.





HIDDEN LOCKING POINT KIT

CE
ZM
IN

112500

50

POSITIONING LATCH KIT

CE
CE
ZM
IN

0710621

100

POSITIONING LATCH

CE
C16
CE
ZM
IN

0517193

100

HD KEEPER

110250

1

ZM

IN

Ø6 RIVET

0517002

50

ZM

EVO SOFT Ø6 RIVET

0511149

50

ZM

IN

EVO

Locking points for profile systems with Euro Groove or 16 groove.

HD keeper kit (ref. 110250)

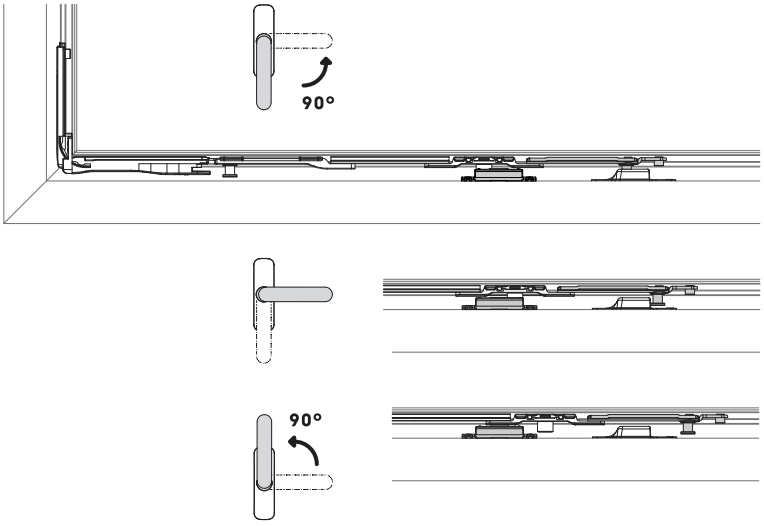
- Closure system for side-hung openings in hinged windows and large format patio doors.
- Supplied with adjustable locking point ± 1 millimetre, EVO SOFT terminal locking point and 8 mm diameter increase washer.

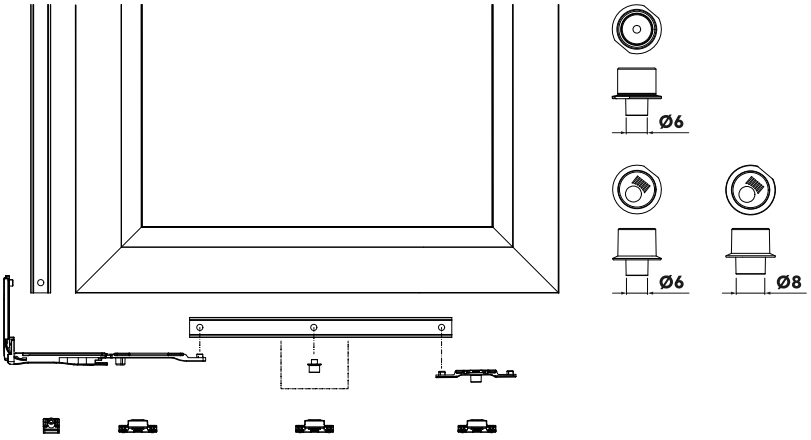
Ø6 rivet (ref. 0517002)

- Available rivet with Ø8 mm with the same characteristics (ref. 05111088)
- They make it possible to obtain a locking point anywhere a transmission rod passes without machining it.
- The blade of the cremone acts directly on the rivet, without the need for a transmission cover.
- Ø6 rivet: compatible with TFS-04, TFS-05 and TFS-06 die-cutting machines.
- Ø8 rivet: compatible with TFS-02 die-cutting machines.

Ø6 EVO SOFT rivet (ref. 0511149)

- EVO SOFT encased locking point for profile series with Euro Groove or 16 groove systems.
- They make it possible to obtain a locking point anywhere a transmission rod passes without machining it.
- The blade of the cremone acts directly on the rivet, without the need for a transmission cover.
- Compatible with die-cutting machines.





A mechanism that allows for a ventilation position.

• The special lock design makes the cremone turn and the sash is positioned in such a way that it remains open between 1 & 6 mm.
• It complies with all the CTE Technical Building Code requirements regarding the quality of interior air and guaranteeing the ventilation required.

Tests (1 class)

• Air permeability according to the UNEEN 12207: 2017 standard in the micro-opening position.
• Values obtained in accordance with the UNE-EN 1026: 2017 standard by notified body 2242 in accordance with report number PR110009.

MICROVENTILATION DEVICE

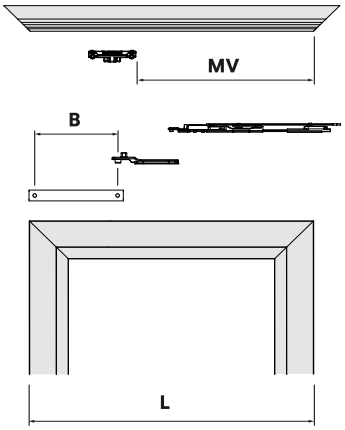
A114401

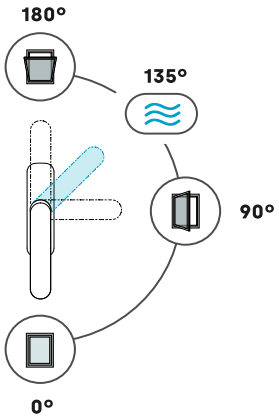
100

ZM

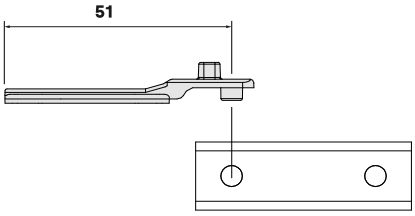
IN

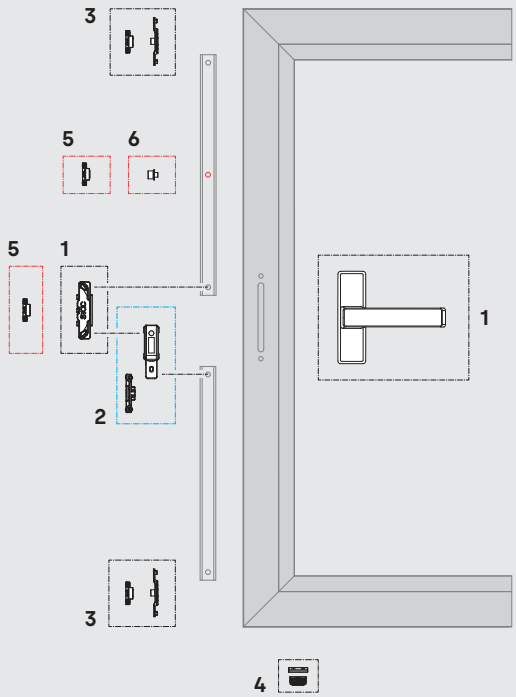
DISCOUNTS **FOR MICROVENTILATION**			**L MIN** (mm)	**MV** (mm)	**B ROD** (mm)
110309		MICRO FRICTION STAY	435	307	L - 360
B110302		SMALL FRICTION STAY	496	371	L - 424
B110301		LARGE FRICTION STAY	618	493	L - 546
T110301		CLX LARGE FRICTION STAY	618	493	L - 646
B110301	A130301	LARGE FRICTION STAY WITH SUPPLEMENTARY	1000	493	L - 646
T110301	A130301	CLX LARGE FRICTION STAY WITH SUPPLEMENTARY	1000	493	L - 646
H1213 R1		CH180 PLUS HINGE WITH SMALL FRICTION STAY	638	490	L - 567
H1213 L1		CH180 PLUS HINGE WITH SMALL FRICTION STAY	638	490	L - 567
H1213 R3		CH180 PLUS HINGE WITH LARGE FRICTION STAY	518	406	L - 483
H1213 L3		CH180 PLUS HINGE WITH LARGE FRICTION STAY	518	406	L - 483
H1213 R1	A130301	CH180 PLUS HINGE WITH SUPPLEMENTARY	1000	490	L - 667
H1213 L1	A130301	CH180 PLUS HINGE WITH SMALL FRICTION STAY	1000	490	L - 667





DISCOUNT **FOR ROD**

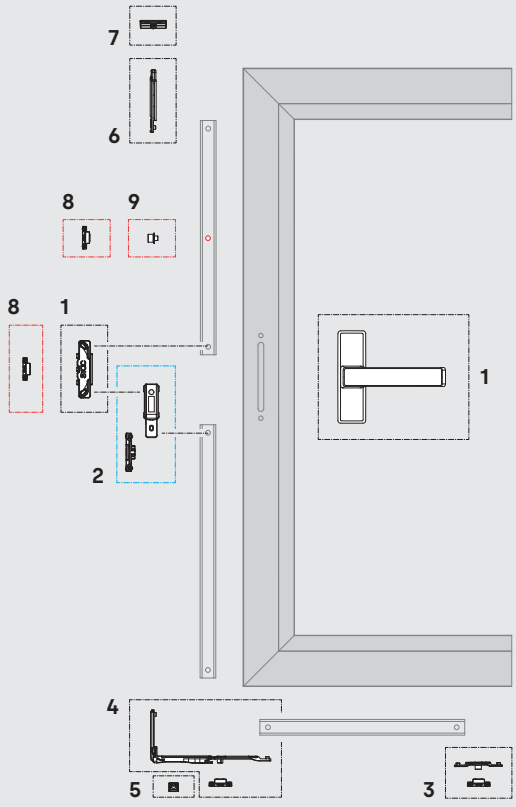




SIDE-HUNG OPENING WITH LOCKING TO INVERTER
WITH SUPPLEMENTARY LOCKING POINTS

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**0710621**	**POSITIONING LATCH KIT**	100
5	**110200**	**ADJUSTABLE LOCKING POINT**	100
6	**0517002**	**Ø6 TRANSMISSION RIVET**	50

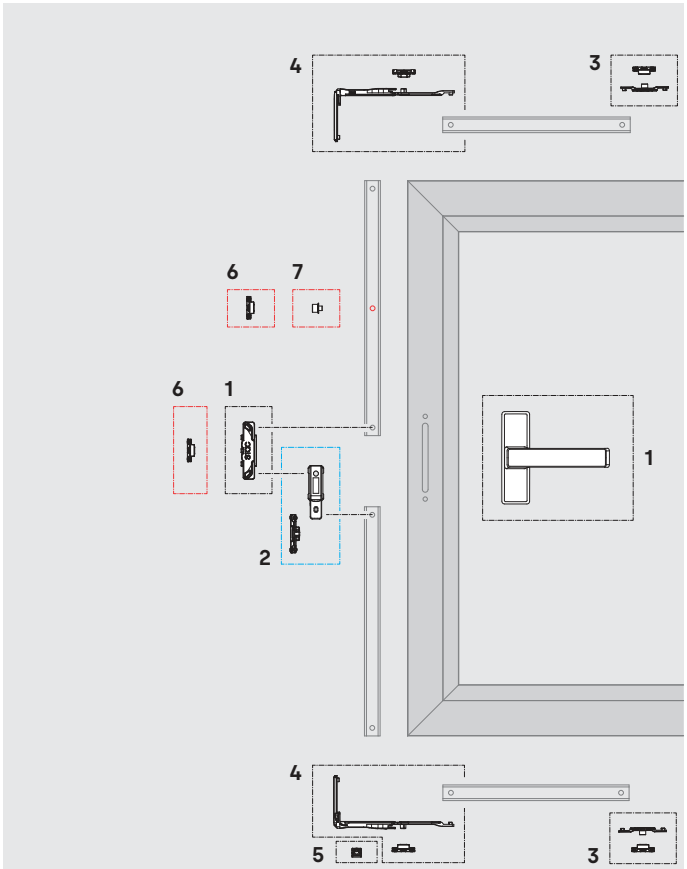
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DRIVE CORNER AND POINTER

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**07085**	**Ø6 SIDE-HUNG OPENING POINTER**	1000
7	**0507105**	**UNIVERSAL KEEPER**	50
8	**110200**	**ADJUSTABLE LOCKING POINT**	100
9	**0517002**	**Ø6 TRANSMISSION RIVET**	50

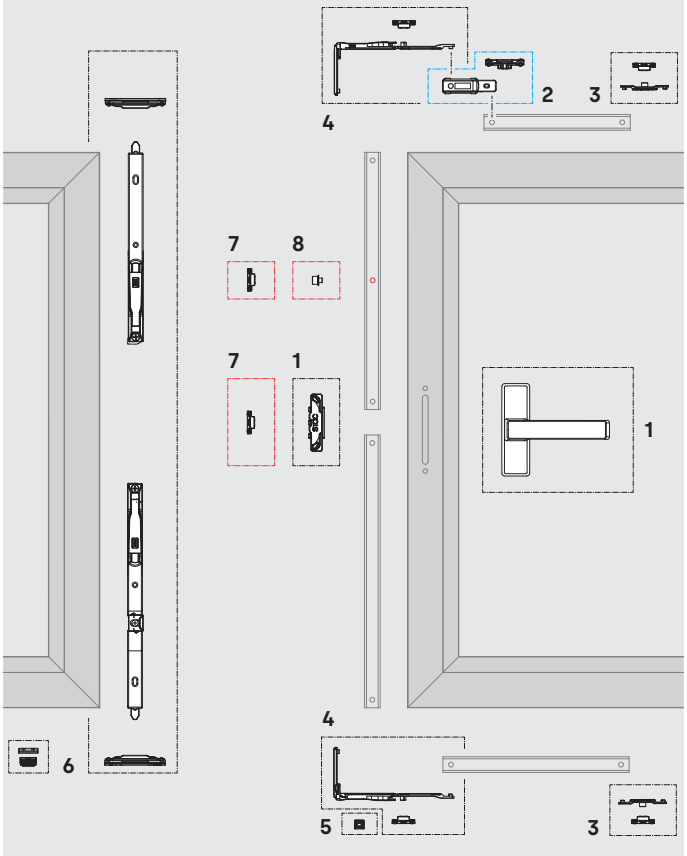
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DOUBLE CORNER DRIVE

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**110200**	**ADJUSTABLE LOCKING POINT**	100
7	**0517002**	**Ø6 TRANSMISSION RIVET**	50

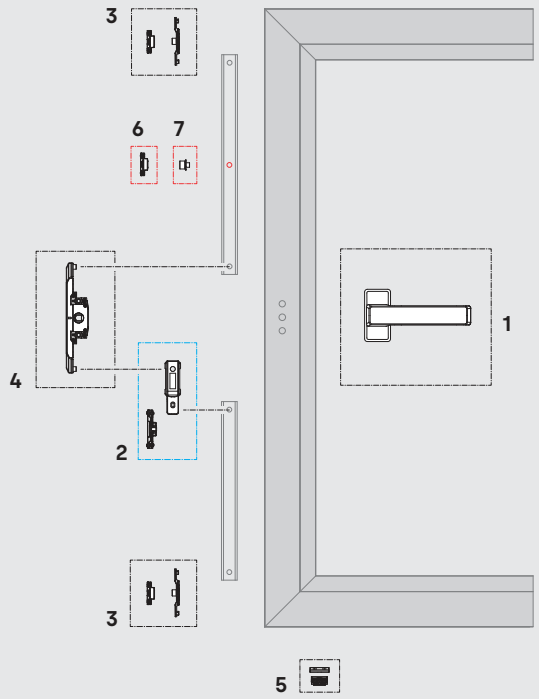
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DOUBLE CORNER DRIVE

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**A113800**	**SHOOTBOLT LEVER KIT**	20
7	**110200**	**ADJUSTABLE LOCKING POINT**	100
8	**0517002**	**Ø6 TRANSMISSION RIVET**	50

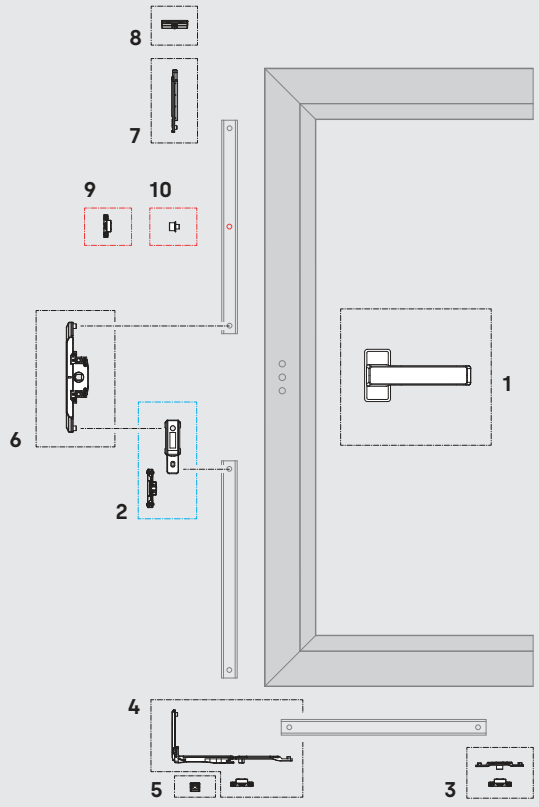
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO INVERTER
WITH SUPPLEMENTARY LOCKING POINTS

1	DELTA, SIRIUS OR LEO CREMONE		
2	A114401	MICRO-VENTILATION KIT	100
3	113000IM	SUPPLEMENTARY LOCKING POINT KIT	50
4	0507063	GEARBOX	50
5	0517193	POSITIONING LATCH	100
6	110200	ADJUSTABLE LOCKING POINT	100
7	0517002	Ø6 TRANSMISSION RIVET	50

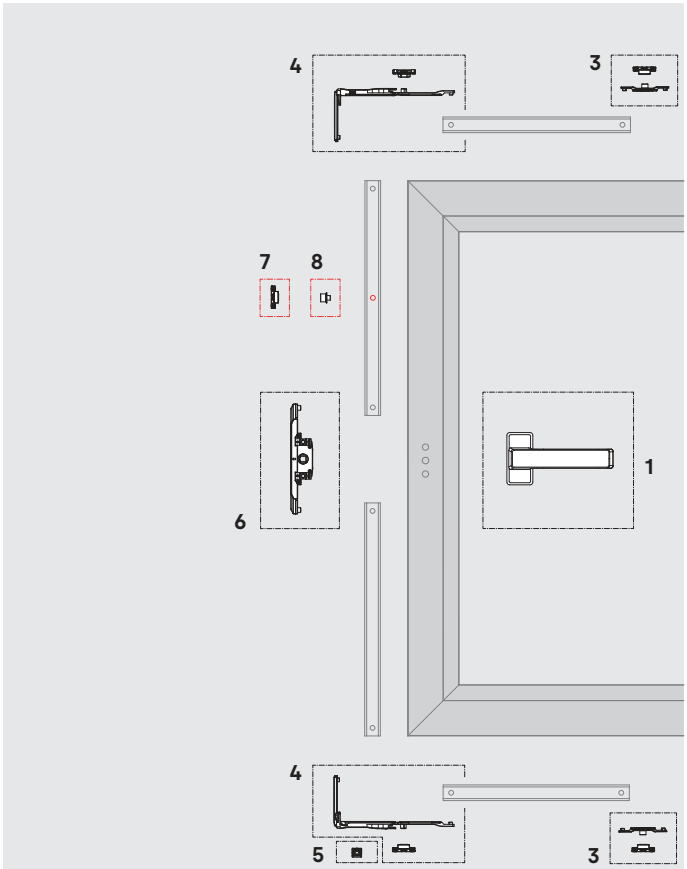
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DRIVE CORNER AND POINTER

1	DELTA, SIRIUS OR LEO CREMONE		
2	A114401	MICRO-VENTILATION KIT	100
3	113000IM	SUPPLEMENTARY LOCKING POINT KIT	50
4	113420	HORIZONTAL CORNER DRIVE KIT	20
5	0517193	POSITIONING LATCH	100
6	0507063	GEARBOX	50
7	07085	Ø6 SIDE-HUNG OPENING POINTER	1000
8	0507105	UNIVERSAL KEEPER	50
9	110200	ADJUSTABLE LOCKING POINT	100
10	0517002	Ø6 TRANSMISSION RIVET	50

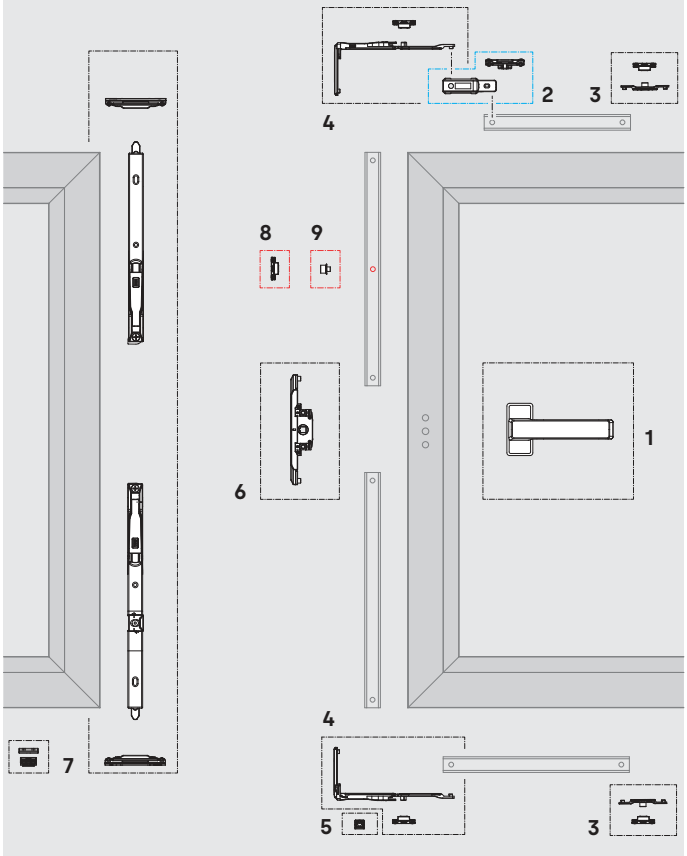
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DOUBLE CORNER DRIVE

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**0507063**	**GEARBOX**	50
7	**110200**	**ADJUSTABLE LOCKING POINT**	100
8	**0517002**	**Ø6 TRANSMISSION RIVET**	50

H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DOUBLE CORNER DRIVE

1	DELTA, SIRIUS OR LEO CREMONE		
2	**A114401**	**MICRO-VENTILATION KIT**	100
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL CORNER DRIVE KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**0507063**	**GEARBOX**	50
7	**A113800**	**SHOOTBOLT LEVER KIT**	20
8	**110200**	**ADJUSTABLE LOCKING POINT**	100
9	**0517002**	**Ø6 TRANSMISSION RIVET**	50

H > 1200 mm

LR LIMITING FRICTION STAY

130110

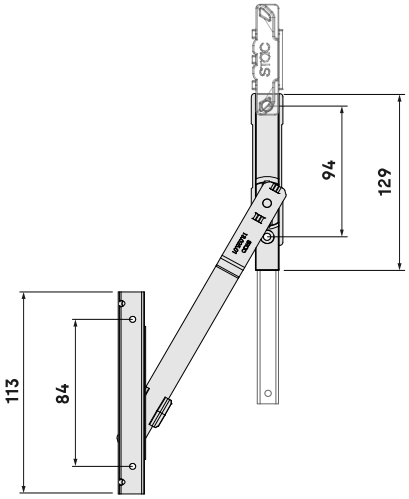
15

ZM

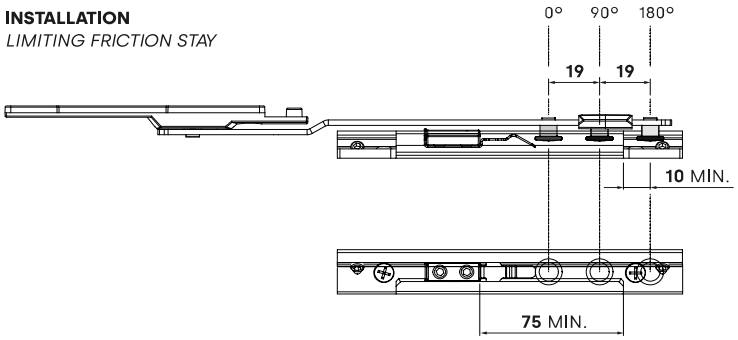
IN

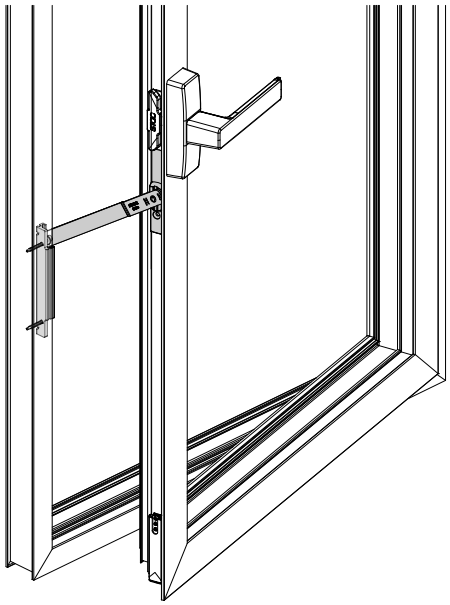
Limiting friction stay for setting stable ventilation positions in side-hung opening windows.

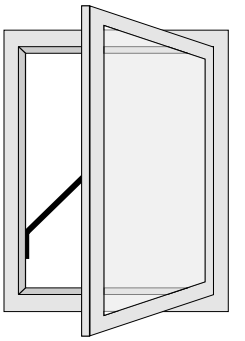
- The ventilation position is set by means of a locking spring.
- Allows adjustment of the window opening in the ventilation position.
- Restrictor friction stay allows an open position with restricted use.
- Quick installation thanks to direct pre-assembly on sash and frame.
- The friction stay allows additional fixing by means of self-tapping screws.

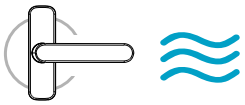


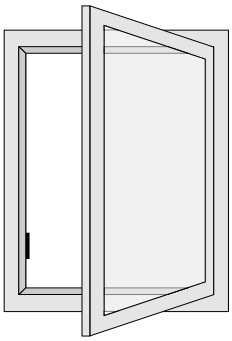
INSTALLATION
LIMITING FRICTION STAY

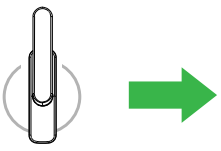


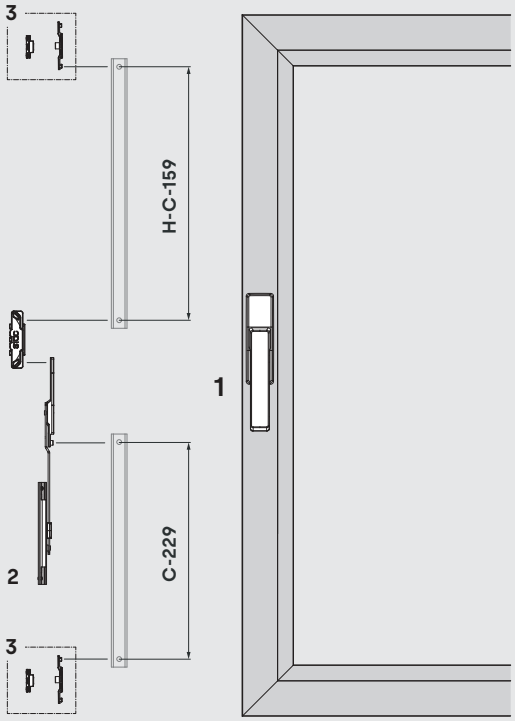






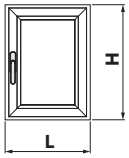






SIDE-HUNG OPENING
LR FRICTION STAY WITH RETENTION

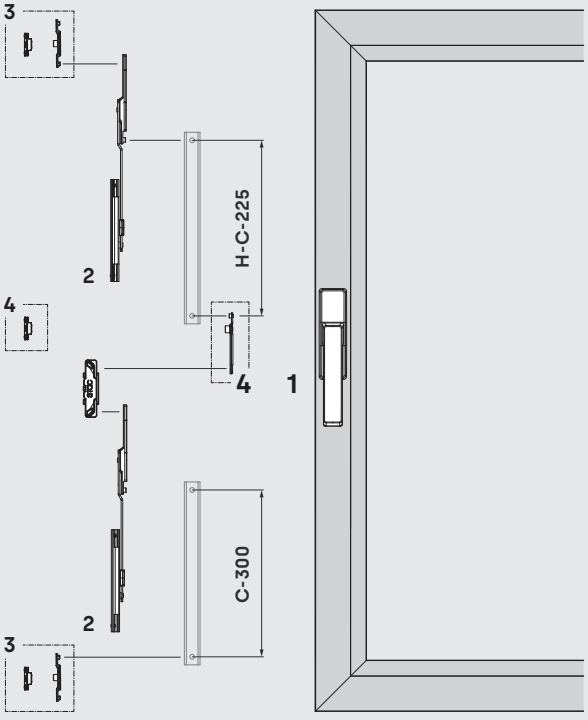
1	**SIDE-HUNG OPENING CREMONE** WITH TRANSMISSION COVER		
2	**130110**	**LR FRICTION STAY WITH LIMITATION**	15
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50



1 SASH CONFIGURATION

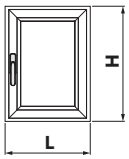
H = 915 ~ 1200 mm

L = 300 ~ 1200 mm



SIDE-HUNG OPENING
LR FRICTION STAY WITH RETENTION

1	**SIDE-HUNG OPENING CREMONE** WITH TRANSMISSION COVER		
2	**130110**	**LR FRICTION STAY WITH LIMITATION**	15
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113004C**	**TERMINAL LOCKING POINT KIT**	20

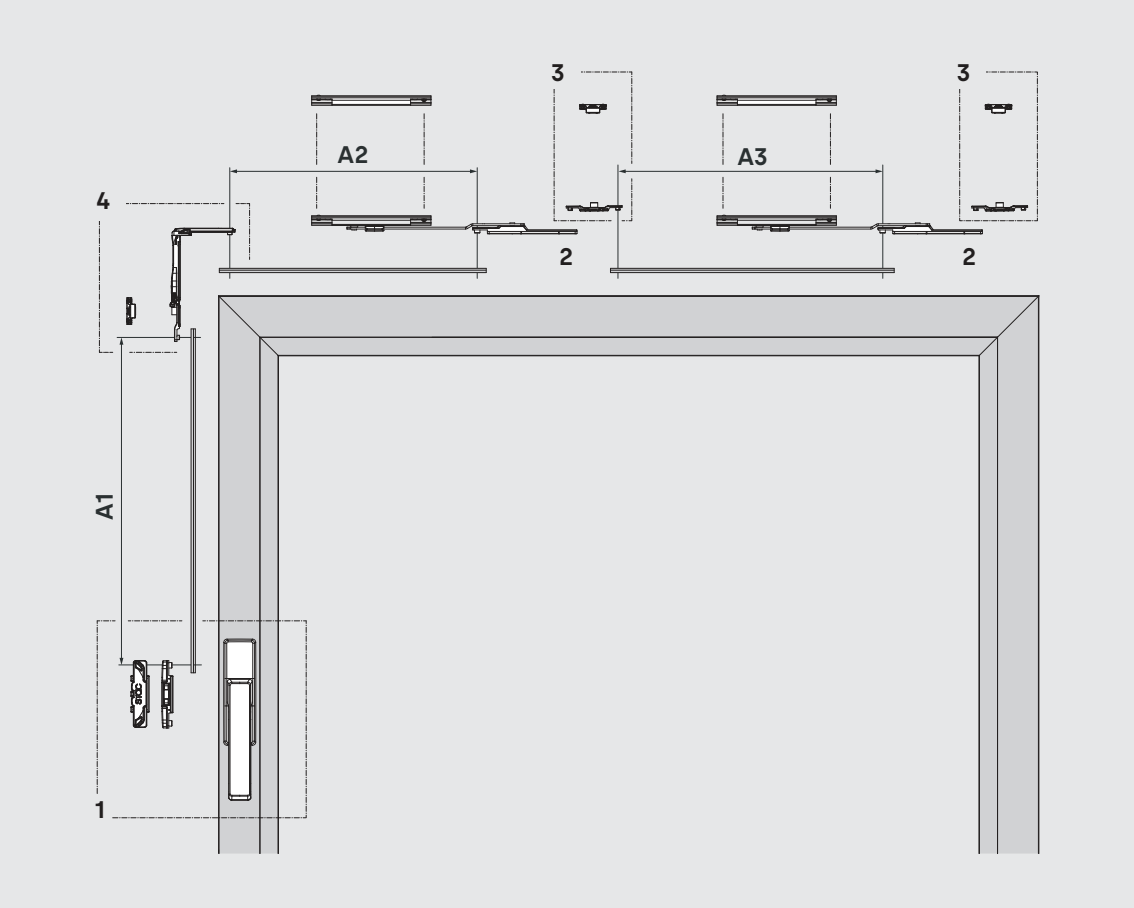


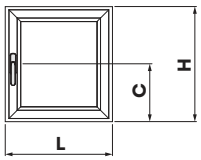
1 SASH CONFIGURATION

H = 1201 ~ 2200 mm

L = 300 ~ 1200 mm

BOTTOM-HUNG OPENING WITH LATERAL TRANSMISSION
LR FRICTION STAY WITH RETENTION





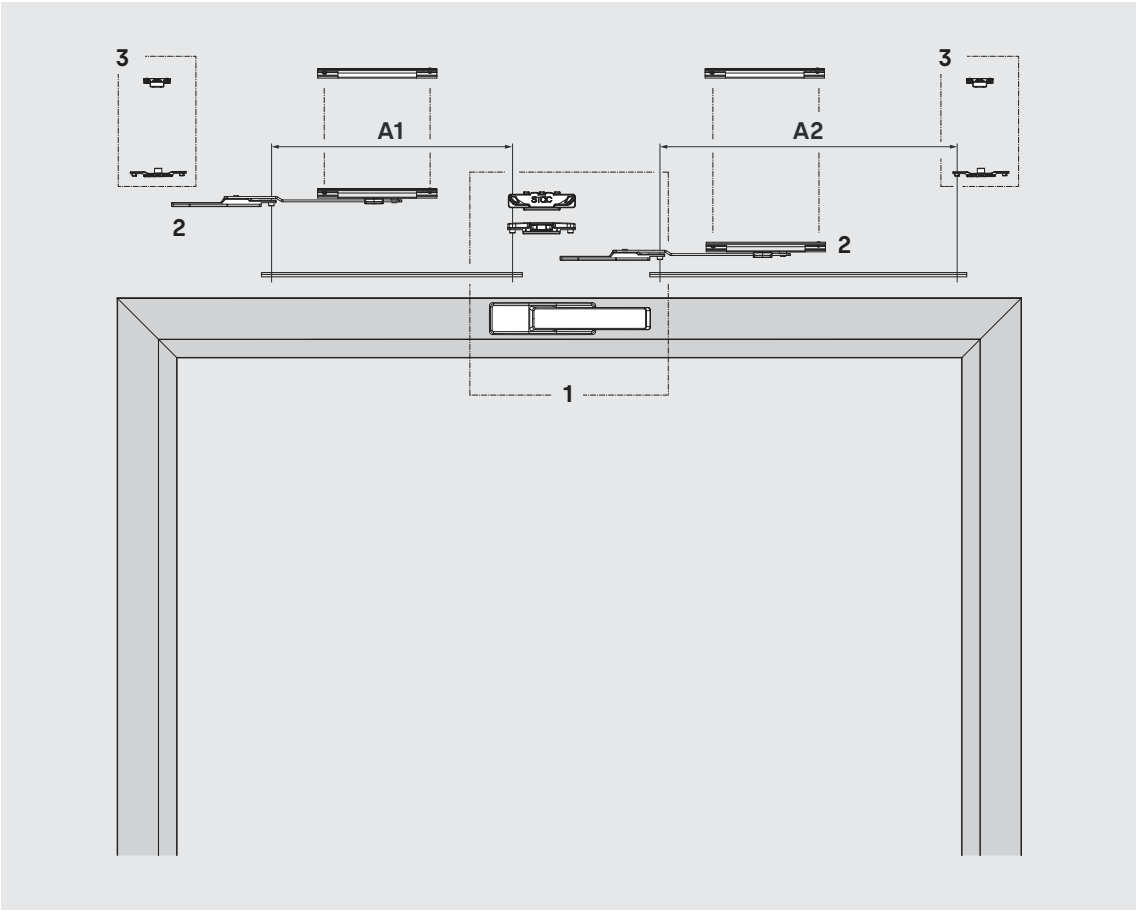
DISCOUNTS	
A1	**H - C - 101**
A2	**C - 220**
A3	**L - 690**

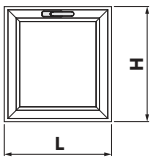
L minimum = 800 mm

1	**TILT & TURN OPENING CREMONE** WITH TRANSMISSION COVER		
2	**130110**	**LR FRICTION STAY WITH LIMITATION**	15
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL DRIVE CORNER KIT**	20

For safety reasons, in public areas the installation of limiters is compulsory from 40 kg per sash. *Consult compatible top-hung friction stay.*

BOTTOM-HUNG OPENING
LR FRICTION STAY WITH RETENTION





DISCOUNTS

A1	**L / 2 - 286**
A2	**L / 2 - 248**

L minimum = 1000 mm

1	**TILT & TURN OPENING CREMONE** WITH TRANSMISSION COVER		
2	**130110**	**LR FRICTION STAY WITH LIMITATION**	15
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50

For safety reasons, in public areas the installation of limiters is compulsory from 40 kg per sash. *Consult compatible top-hung friction stay.*

ONE-WAY TRANSMISSION (B7)

0507063

50

CE

ZM

PA

Transmission mechanism that through a handle allows to move the transmission rods and the locking points that open the window.

- One-way mechanism with a 180º movement and 7 mm backset.
- The gearbox incorporates a stop that limits the rotation of the handle to 90º for side-hung window opening.
- It incorporates adjustable pieces to adapt to any type of groove.
- The gearbox is fixed to the profile with self-tapping screws.

Minimalist gearbox (ref. A770001)

- Designed for use with minimalist handles (ref. A7701, ref. 7703).
- Valid for 8 x 8 mm transmission spindles.

MINIMALIST ONE-WAY TRANSMISSION (B8)

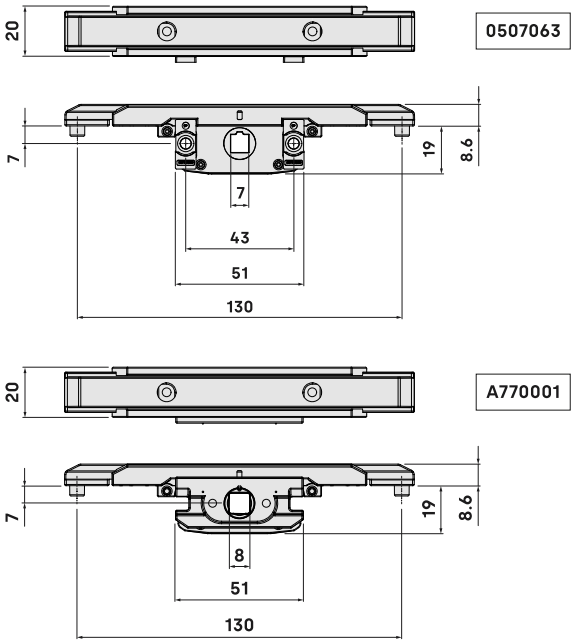
A770001

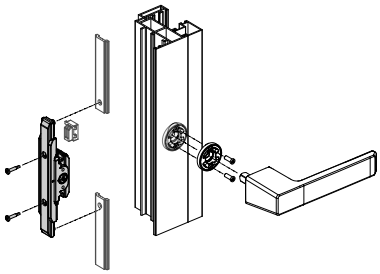
25

CE

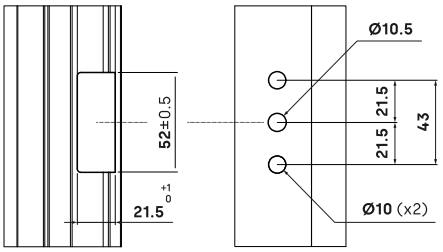
ZM

IN

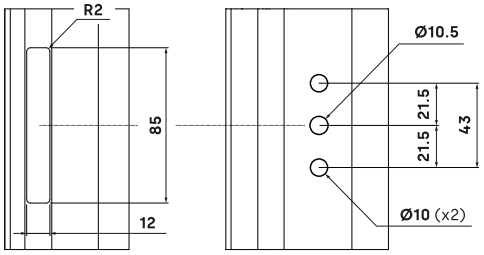


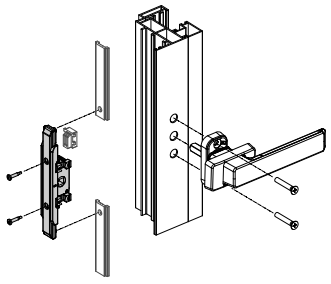


INWARD OPENING *PROFILE*

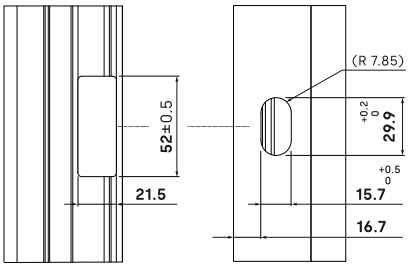


OUTWARD OPENING *PROFILE*

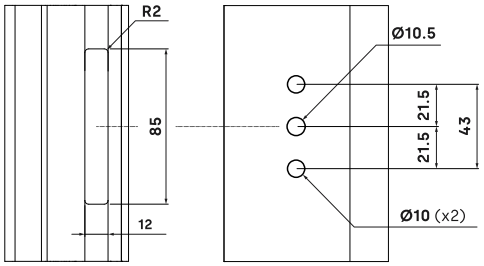




(MINIMALIST HANDLE) INWARD OPENING *PROFILE*



BALCONY DOOR *PROFILE*



Transmission mechanism that through a handle allows to move the transmission rods and the locking points that open the window.

One-way gearbox: 90º movement and 33.5 mm backset (ref. 0507177)

• Designed for installations in inward side-hung openings or outward openings in balcony profiles or door.

One-way gearbox: 180º movement and 33.5 mm backset (ref. 0507178)

• Designed for installations in inward side-hung openings with micro-ventilation or tilt & turn openings in balcony profiles.

Two-way gearbox: 90º movement and 33.5 mm backset. (ref. 0507248)

• Designed for installations in inward side-hung openings or outward openings in balcony profiles or door.

ONE-WAY TRANSMISSION (B33.5)

0507177

25

ONE-WAY TRANSMISSION (B33.5)

0507178

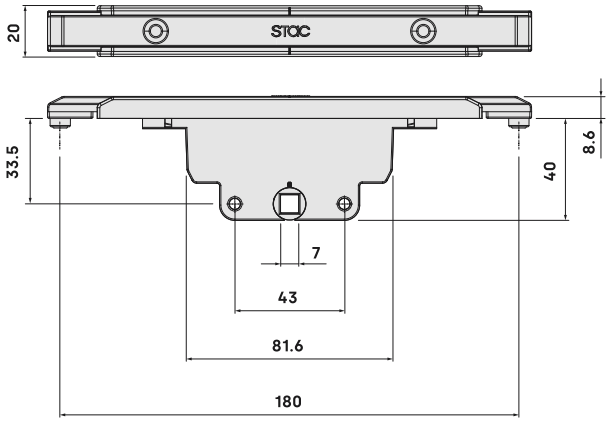
25

TWO-WAY TRANSMISSION (B33.5)

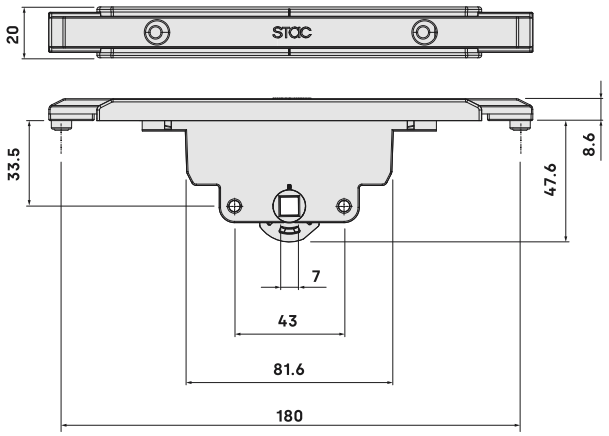
0507248

25

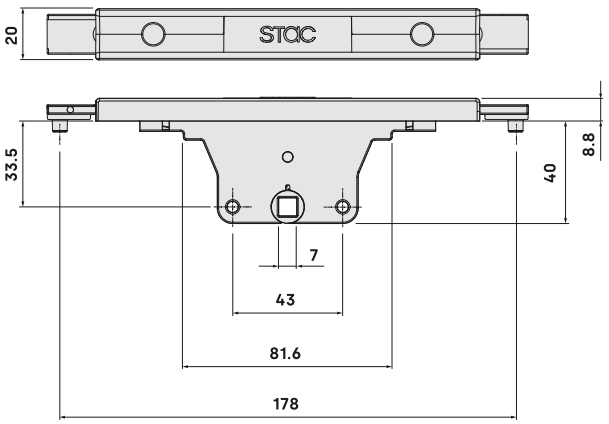
0507177

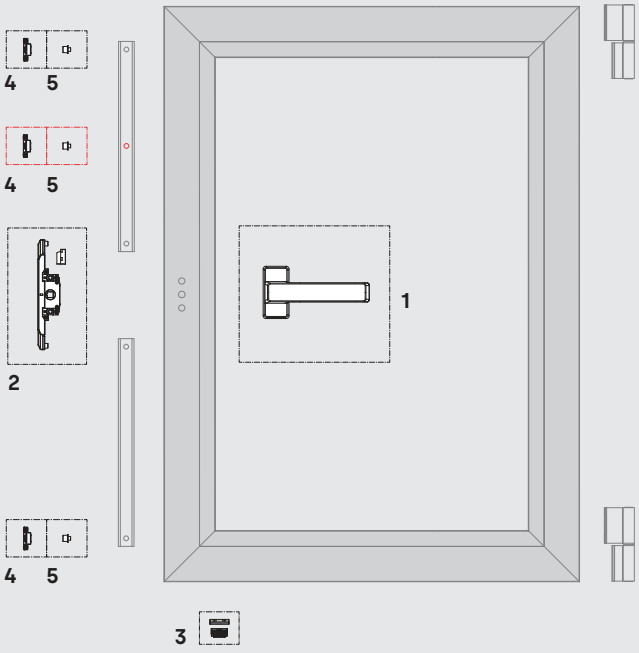


0507178



0507248

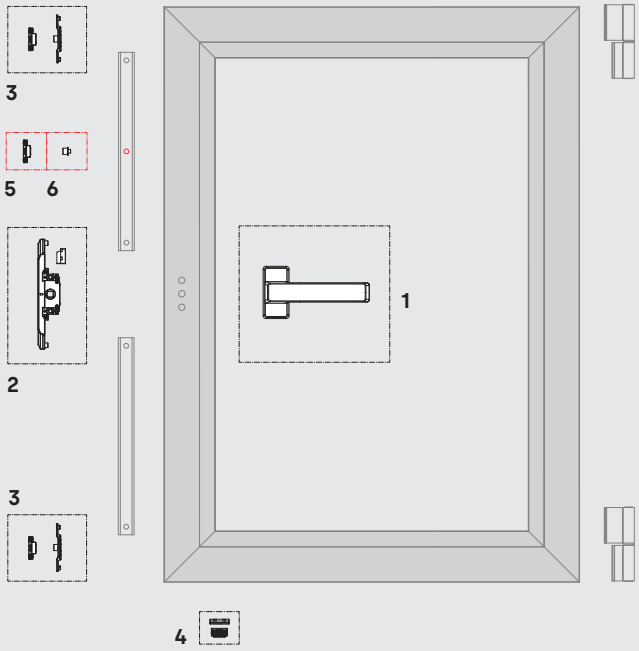




SIDE-HUNG OPENING WITH LOCKING TO INVERTER
WITH TRANSMISSION RIVETS

1	DELTA, SIRIUS OR POLARIS HANDLE		
2	**0507063**	**GEARBOX**	50
3	**0710621**	**POSITIONING LATCH KIT**	100
4	**110200**	**ADJUSTABLE LOCKING POINT**	100
5	**0517002**	**Ø6 TRANSMISSION RIVET**	50

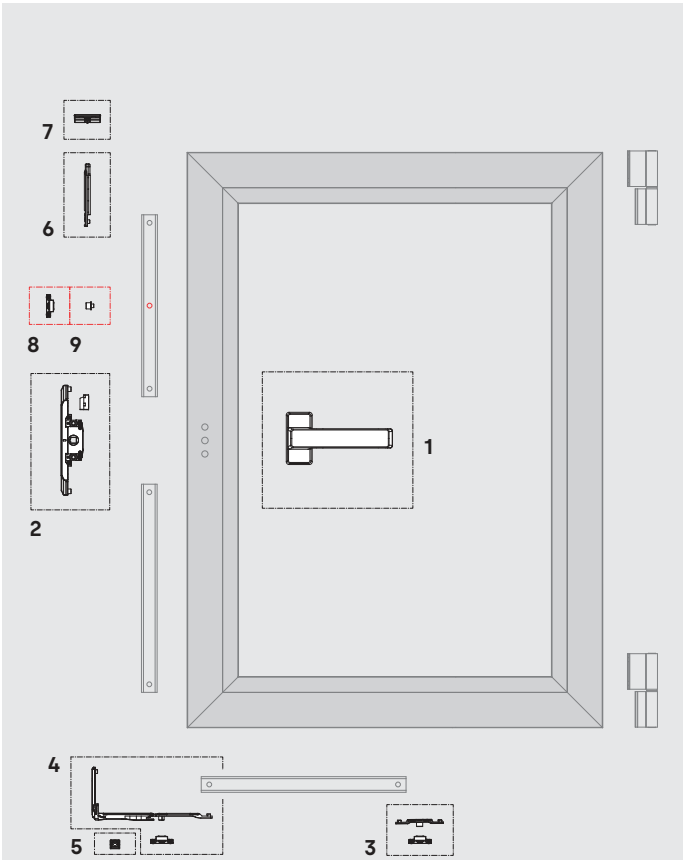
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO INVERTER
WITH SUPPLEMENTARY LOCKING POINTS

1	DELTA, SIRIUS OR POLARIS HANDLE		
2	**0507063**	**GEARBOX**	50
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**0710621**	**POSITIONING LATCH KIT**	100
5	**110200**	**ADJUSTABLE LOCKING POINT**	100
6	**0517002**	**Ø6 TRANSMISSION RIVET**	50

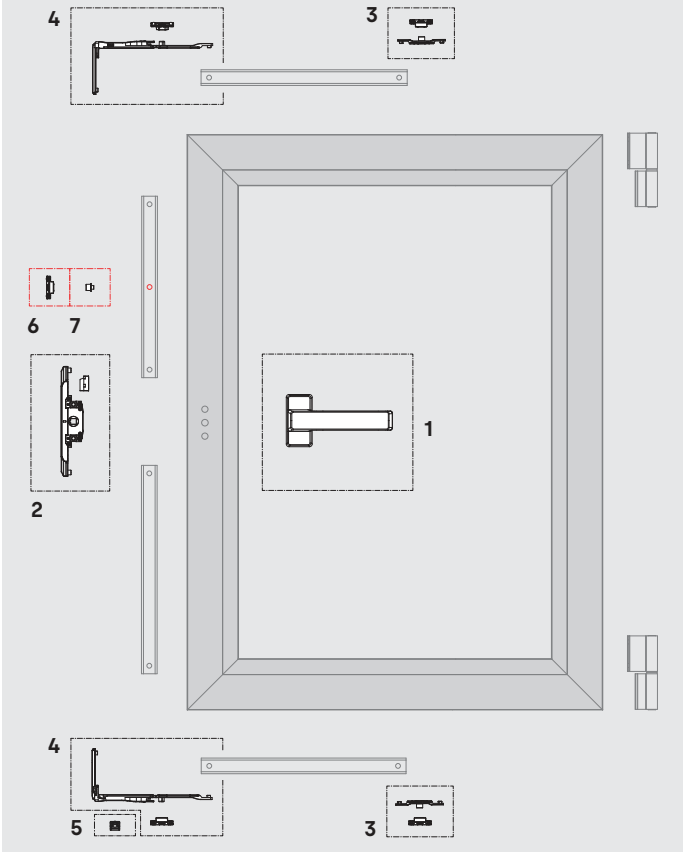
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO FRAME
WITH DRIVE CORNER AND POINTER

1	DELTA, SIRIUS OR POLARIS HANDLE		
2	**0507063**	**GEARBOX**	50
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL DRIVE CORNER KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**07085**	**PUNTERO AP. PRACTICABLE Ø6**	1000
7	**0507105**	**CERRADERO UNIVERSAL**	50
8	**110200**	**ADJUSTABLE LOCKING POINT**	100
9	**0517002**	**Ø6 TRANSMISSION RIVET**	50

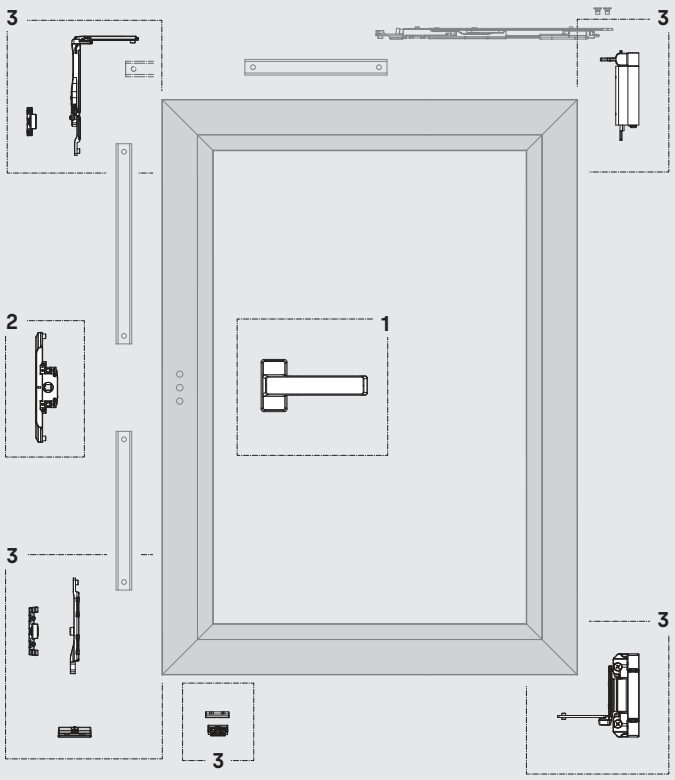
H > 1200 mm



SIDE-HUNG OPENING WITH LOCKING TO INVERTER
WITH DOUBLE DRIVE CORNER

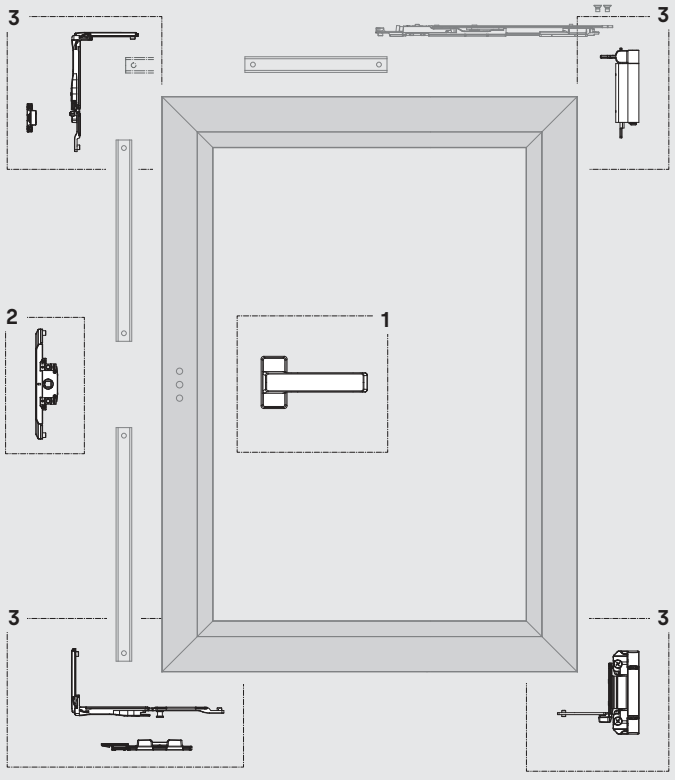
1	DELTA, SIRIUS OR POLARIS HANDLE		
2	**0507063**	**GEARBOX**	50
3	**113000IM**	**SUPPLEMENTARY LOCKING POINT KIT**	50
4	**113420**	**HORIZONTAL DRIVE CORNER KIT**	20
5	**0517193**	**POSITIONING LATCH**	100
6	**110200**	**ADJUSTABLE LOCKING POINT**	100
7	**0517002**	**Ø6 TRANSMISSION RIVET**	50

H > 1200 mm



TILT & TURN OPENING
WITH LOCKING TO INVERTER

1	DELTA, SIRIUS OR POLARIS HANDLE			
2	**0507063**		**GEARBOX**	50
3	**1133**	**	**LOCKING TO INVERTER KIT** WITHOUT CREMONE	24

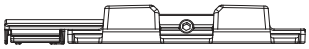


TILT & TURN OPENING
WITH LOCKING TO FRAME or TILT-BEFORE-TURN MANEUVER

1	DELTA, SIRIUS OR POLARIS HANDLE			
2	**0507063**		**GEARBOX**	50
3	**1139**	**	**LOCKING TO FRAME KIT** WITHOUT CREMONE	24
	1123	**	**TILT-BEFORE-TURN MANEUVER** WITHOUT CREMONE	

tilt-before-turn keeper

locking to frame keeper



STAC hardware machining templates for carpentry workshops.

- Supplied in AISI 304 stainless steel.
- The same template can be used for multiple machining sizes.
- By using the templates in the aluminium copying machines, the machining is done directly by means of feeler gauges.

MACHINING TEMPLATES

IN

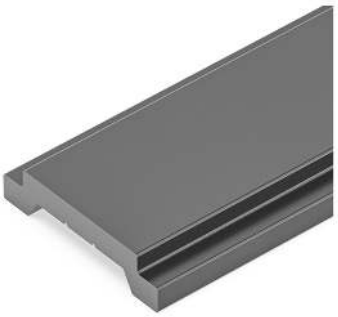
STC-MC

	STC-MC-0013	STC-MC-0014	STC-MC-0015	STC-MC-0016	STC-MC-0017	STC-MC-0018	STC-MC-0030
MINIMALIST HANDLES	●					●	
STANDARD HANDLES	●			●			●
ATRIA PULL HANDLE & FINGER PULL	●	●			●		
DRIVEGEARS	●	●			●		
7 GEARBOXES	●	●				●	
33.5 GEARBOXES	●	●					●
TITAN RECESSED HANDLES	●					●	
FENIX RECESSED HANDLES	●			●			
FENIX MINI RECESSED HANDLES	●		●				
DRACO RECESSED HANDLES	●		●				
HYDRA CREMONE	●	●		●			
MULTIPOINT LOCKS		●		●			
HADES CONCEALED HANDLE		●				●	
TILT & TURN DOOR SEQUENCER							●
DIMENSIONS	*124 x 480 mm*	*124 x 480 mm*	*100 x 220 mm*	*100 x 220 mm*	*100 x 230 mm*	*100 x 220 mm*	*124 x 480 mm*

OPERATING ROD
Polyamide transmission rod

059233

250



PA

TFS 08

Transmission rods that allow the movement of the different hardware and accessories that allow the operation of the windows.

- The transmission rods are valid for Euro Groove profiles.
- Supplied in polyamide (ref. 059233) or in PVC (ref. 059238).
- Smooth, fluid and jerk-free sliding through the profile groove.
- Machining and cut to size through die-cutting machines.
- The transmission rods allow the installation of as many additional locking points as required and at the desired location.

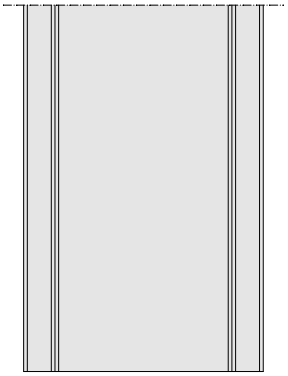
OPERATING ROD
PVC transmission rod

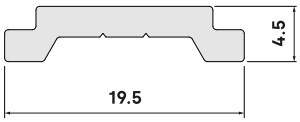
059238

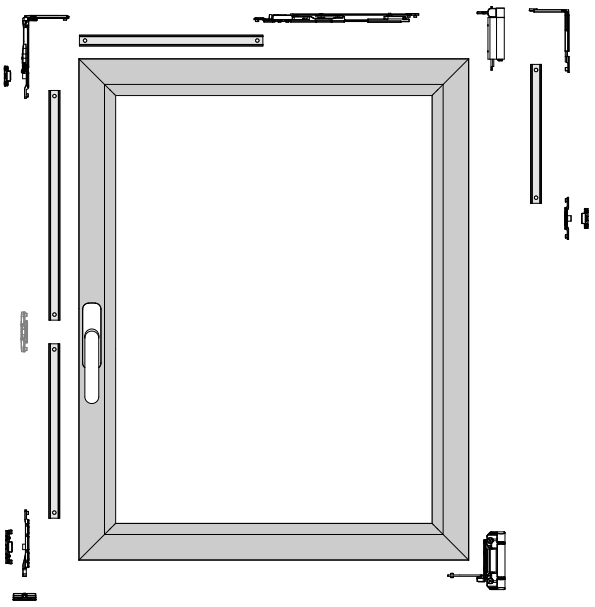
250

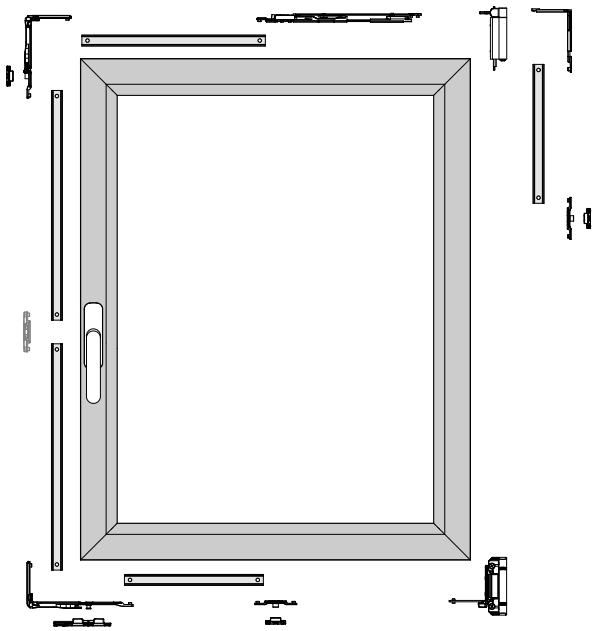
PV

TFS 08









Accessories that allow the assembly of window and door transoms.

- Valid for Euro Groove series.
- Supplied with screws.

TRANSOM CORNER JOINT

CE

AL

IN

04001

250

TRANSOM CORNER JOINT

CE

AL

IN

04005

250

TRANSOM CORNER JOINT

CE

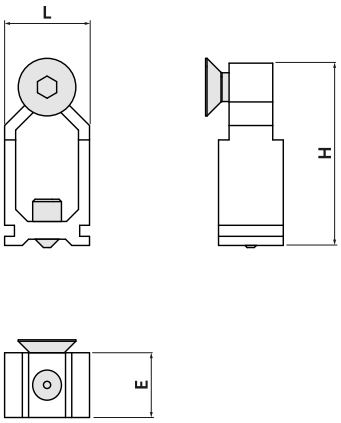
AL

IN

04006

250

	H	L	E	
04001	38	17.7	13.5	*250*
04005	38	17.7	30.4	*250*
04006	38	17.7	19.5	*250*



GLAZING BEAD CLIP

A99009

PA

3500

Plastic accessories that complement other products.

Glazing bead clip

- Allows the window glazing bead to be fixed.

Drainage caps

- References available in different colours.

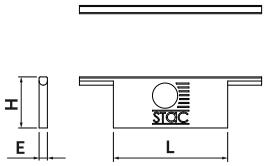
Deflectors

- Refs. 09006 and P99024 are available in different finishes.
- Valid for 30 x 5 mm machining with Euro Groove profile series.

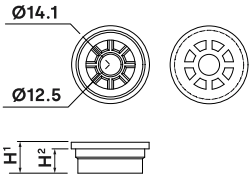
DRAINAGE CAPS

DESAGÜE

PA



	H	L	E	
09007	9.3	20.7	1.5	2000
99025	8.1	13.5	1.5	2000

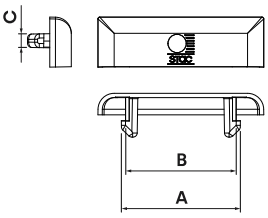


	H¹	H²	Ø	
09027	5.6	4.4	14.1	6000

DEFLECTORS

DEFLECTOR

PA



	A	B	C	
09006	33.2	30.9	4	2000
09016	31.1	27.7	3.5	3000
A09020	33.4	30.9	4	4000
P99024	22.6	20.2	4	2000

17 WINDOW CORNER JOINTS

420 Machined corner joints

421 Pressed corner joints

422 Die cast corner joints

423 Alignment corner joints

424 Flat corner joints

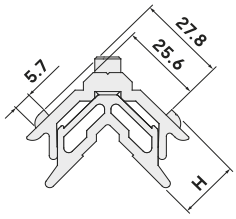
MACHINED CORNER JOINT

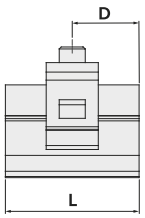
E.MECANIZADAS

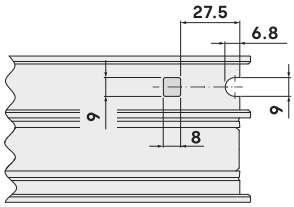
AL

Machined corner joints for sash and frame profile assembly.

• Machined corner joints with grooved and milled pins that guarantee better precision when tightening the miter.
• Positioning of the upper element depending on the system and series. This makes it versatile, as it can be moved to serve a different series. This displacement allows the absorption of possible irregularities in the die-cutting of the profile.







	H	L	D	
A02M111C45	12	10.8	C	300
02M2513C54	12	24	C	250
B02M1914C47	13.7	19	C	400
02M1914D48	13.7	19	13.6	250
B02M2014C50	13.7	19.4	C	350
B02M2214C06	13.7	21.6	C	350
B02M2614C09	13.7	24.9	C	400
02M2614D40	13.7	24.9	10.7	250
02M2814D36	13.7	27.5	15	-
02M2914C32	13.7	28.6	C	250
02M3014D35	13.7	29.8	12.5	250
B02M3714D14	13.7	36.9	15	250
02M4114D21	13.7	41	15	250
02M4214D24	13.7	41.5	15	250
B02M4214C46	13.7	41.5	C	300
02M4016D55	15.8	39.7	15	200

	H	L	D	
02M2516D38	15.8	24.7	16	350
02M1316C53	15.9	12.1	C	250
02M3718D15	17.6	37	15	200
02M3818D18	17.6	37.2	16	250
02M4218D25	17.6	41.5	15	200
02M1120C43	19.2	10.8	C	250
02M2020C49	19.2	19	C	200
02M2620D31	19.2	24.7	16	200
02M2520D39	19.2	25	16	250
02M4220C26	19.2	41.1	C	-
B02M1724C52	23.8	16.3	C	250
02M2626C10	25.2	24.7	C	200
02M3826D19	25.2	36.5	16	125
B02M1730C51	30	16.3	C	250
02M1131C42	30.9	10.8	C	250
02M1531C54	30.9	14.3	C	250

	H	L	D	
02M2231C07	30.9	21.8	C	175
02M2531D37	30.9	24.7	16	150
02M2631C11	30.9	24.7	C	150
02M2632D30	30.9	24.7	16	125
02M3231C13	30.9	31.5	C	150
02M3731D16	30.9	36	15	225
02M4231D27	30.9	41	15	175
02M2235D08	33.2	22	15	-
B02M2636D12	35.2	25.6	11	125
02M3736D17	35.2	36	15	175
02M4236D28	35.2	40.8	15	175
02M0839C48	39.2	7.8	C	250
02M1141C44	39.2	10.8	C	250
B02M3840D20	39.2	37.5	15	150
02M4142C23	41.2	40.5	C	250

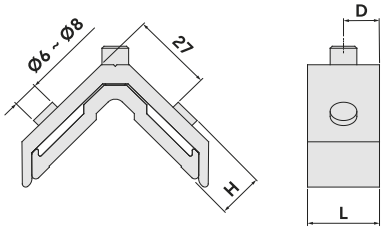
C *CENTRED PIN*

Pressed corner joints for sash and frame profile assembly.

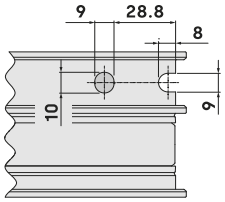
- The upper and lower elements share the same cutting width.
- The round pins ensure greater precision during mitre clamping.

PRESSED CORNER JOINT

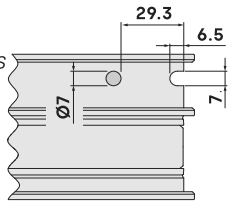
E.ESTAMPADAS



PRESSED CORNER JOINTS
DIAMETER 8



PRESSED CORNER JOINTS
DIAMETER 6



Ø6	H	L	D	
02E2514C39	14.1	25	C	-
02E1614C29	14.2	15.6	C	-
02E2014C33	14.2	19.2	C	250
02E1118C42	17.5	11.5	C	250
02E1218C02	17.5	12.3	C	250
02E1822C40	21.5	17.8	C	-
02E4124D37	23.5	41	C	180
02E1836C41	35.5	17.4	10	-
02E2638C38	37.5	26	C	-

Ø8	H	L	D	
14221401	13.6	21.6	C	250
14251403	13.6	24.7	C	250
14271415	13.6	27	12	-
14291416	13.6	28.4	10.5	200
14301417	13.6	29.8	12.5	250
14311418	13.6	30.4	C	250
14371402	13.6	36.6	15	250
14361427	13.6	37	C	-
14411419	13.6	41	15	-
14421404	13.6	41.5	15	250
14371820	17.3	37	15	250
14371821	17.3	37	16	-
14421822	17.3	41.5	15	150
14412023	19.2	41	C	-
14252624	25.2	24.7	C	200
14372625	25.2	37	16	-

Ø8	H	L	D	
14223105	29.2	21.6	C	150
14253006	29.2	24.7	C	150
14303029	29.2	30.5	10	250
14323026	29.2	31.9	C	200
14363028	29.2	35.9	C	200
14373007	29.2	36.6	15	200
14413008	29.2	41	15	200
14223409	33.2	21.8	15	-
14233610	35.2	25.8	11	100
14363630	35.3	35.9	C	175
14363611	35.3	36.6	15	-
14423612	35.3	41.5	15	150
14384013	39.2	37.5	15	150
14414214	41	40.5	C	-
14454231	41.5	44.6	C	100

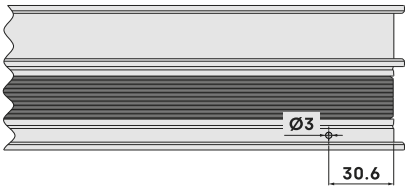
C *CENTRED PIN*

DIE CAST CORNER JOINTS

Corner joints for sash and frame profile alignment.

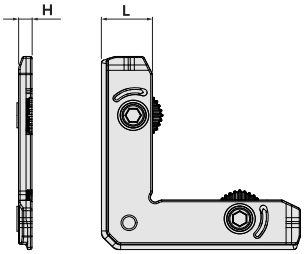
- The corner joints are supplied with tightening mechanism.

E.INYECTADAS

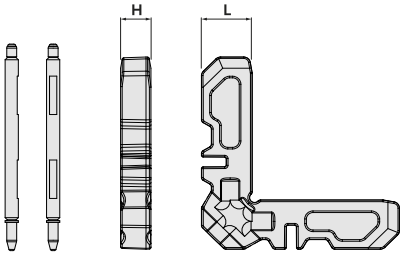


DIE CAST CORNER JOINTS
WITH CAM

	H	L	
0502139	11.3	4.5	*100*
0502141	15.9	4.2	*100*



DIE CAST CORNER JOINTS
WITH CAM



DIE CAST CORNER JOINTS
BULLET TYPE

DIE CAST CORNER JOINTS
BULLET TYPE

	H	L	
0512002	15.4	9.2	*300*
0512004	24.5	3.3	*500*
0512001	30.2	9	*200*
0512005	39.1	4	*350*

A12003	head Ø3 x 8.8	*1000*

Elements for aligning the sash and frame profiles.

• Allows perfect alignment of sash and frame.

ALIGNMENT CORNER JOINTS

ZM

PA

E.ALINEAMIENTO

ALIGNMENT	H	L	E		
99027	6.3	2.5	25	PA	5000
0599089	7.3	4.1	34.1	ZM	350
0599145	8.8	3.8	29.7	ZM	500

ASSEMBLED CORNER JOINTS

AL

E.ENSAMBLAJE

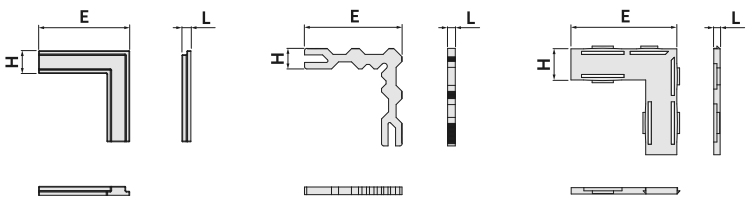
ASSEMBLY	H	L	E		
99137	13.7	5.1	65	AL	500
99127	13.7	22	65	AL	500
99126	13.7	25.2	65	AL	500

OVERLAPPED CORNER JOINTS

PA

E.SOLAPE

OVERLAP	H	L	E		
99004	14.1	25	C	PA	2000
09999055	14.2	15.6	C	PA	800



ALIGNMENT
CORNER JOINTS

ASSEMBLY
CORNER JOINTS

OVERLAP
CORNER JOINTS

FLAT CORNER JOINTS

E.PLANAS

IN

Elements for aligning the sash and frame profiles.

- Allows perfect alignment of sash and frame.

RECESSED CORNER JOINTS

E.EMBUTIDAS

IN

FLAT	H	L	E		
A99018	7.2	1.2	25.3	IN	3000
A99019	12.7	1	38	IN	3000
0599868	7.3	0.8	25.3	IN	3000

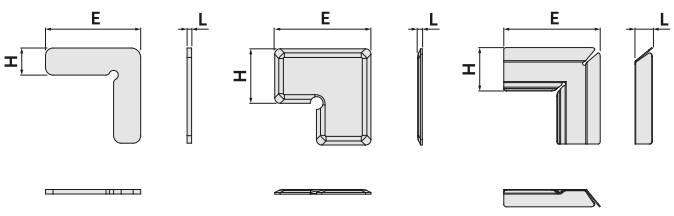
RECESSED	H	L	E		
A99020	14.7	1.1	39.2	IN	3000
A99021	22.7	2.3	45.2	IN	2000
99017	25.6	2.2	45.2	IN	2000

FOLDED CORNER JOINTS

E.PLEGADAS

IN

FOLDER	H	L	E		
09999062	17	5.9	49.7	IN	250
99068	23.8	5	52.9	IN	1250
09999049	26.3	4.1	53.3	IN	2000



FLAT
CORNER JOINTS

RECESSED
CORNER JOINTS

FOLDER
CORNER JOINTS

18 MACHINERY

426 TFS die-cutting machines

431 Hardware workbench

TFS-08 DIE-CUTTING MACHINE

TFS08-CE-N00

1

Machine that allows to carry out the different machining and riveting of the transmission rods for side-hung and tilt & turn windows.

- Machining of transmission rods for Euro Groove systems.
- Operating mode: **manual**.
- It allows to perform the cutting and punching of Ø6 mm.
- Machine compatible with the most common gearbox and transmission cover configurations (handle or cremone).
- Unique cutting system, no arithmetic operations and direct discounts from the window.

TFS-08 SUPPORT

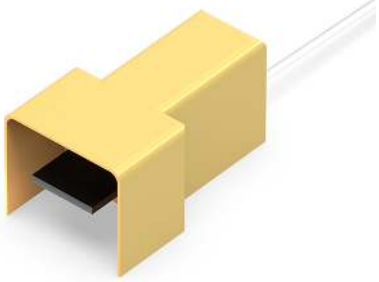
TFS08-SOP-00

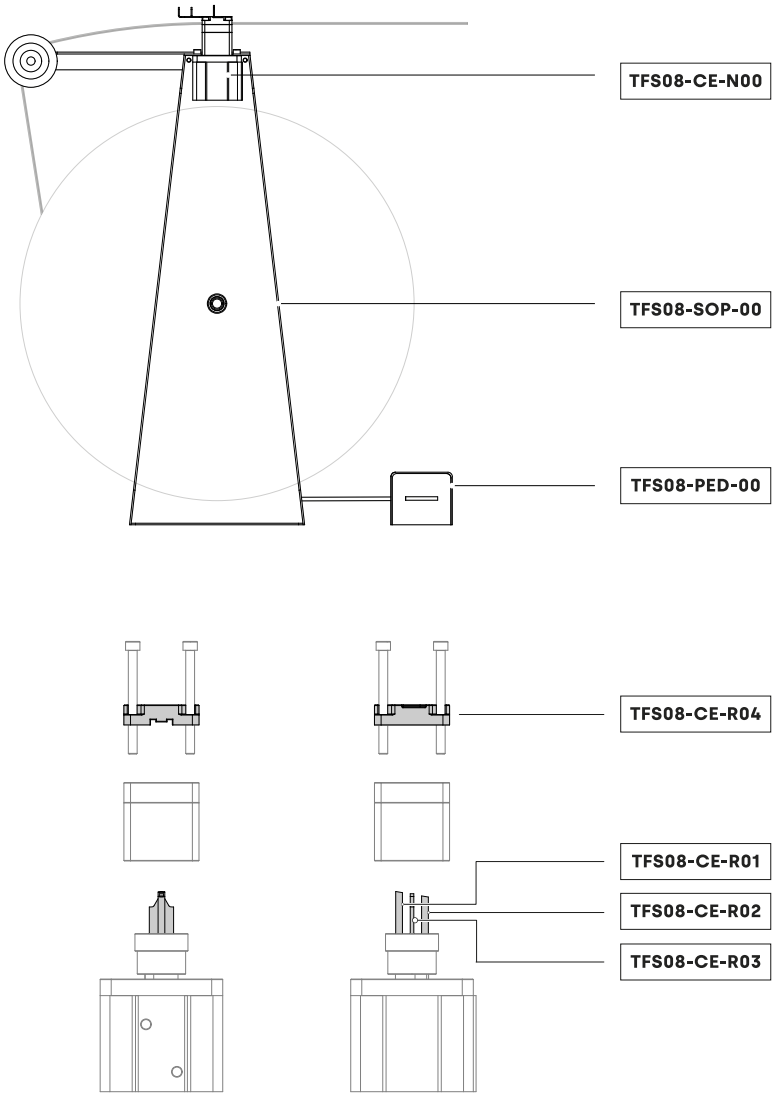
1

TFS-08 PEDAL

TFS08-PED-00

1





Telescopic guide (ref. TFS08-BA-00)

• It allows to obtain directly the measurements on the window to transfer them to the die-cutting machine (not included in the reference).

Discount templates (ref. TFS08-P)

• They allow to obtain directly the discounts to transfer them to the die-cutting machine (not included in the reference).

TFS-08 TELESCOPIC GUIDE

CE

AL AC IN PA

TFS 08

TFS08-BA-00

1

DISCOUNT TEMPLATES

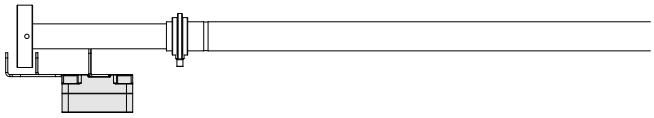
CE

TFS 08

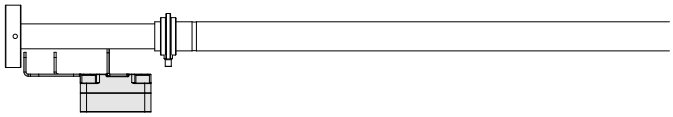
TFS08-P

1

*RODS FOR **STANDARD CONFIGURATION***

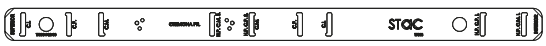


*RODS FOR **CONFIGURATION WITH MICRO-VENTILATION***

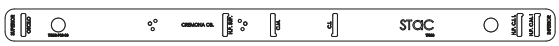


*RODS FOR **CONFIGURATION WITH AN INTERMEDIATE LOCKING POINT***

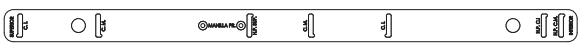
TFS08-P01-00



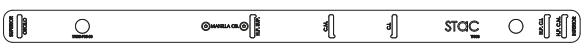
TFS08-P02-00



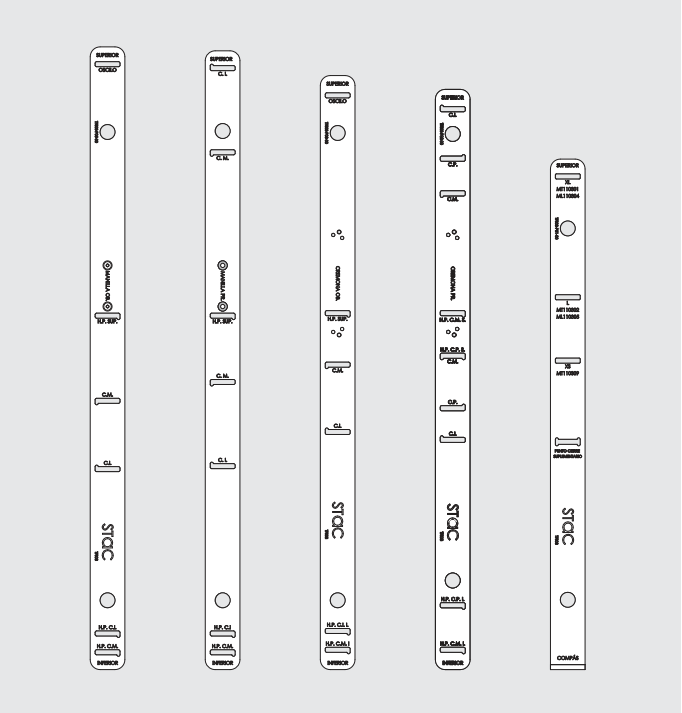
TFS08-P03-00



TFS08-P04-00

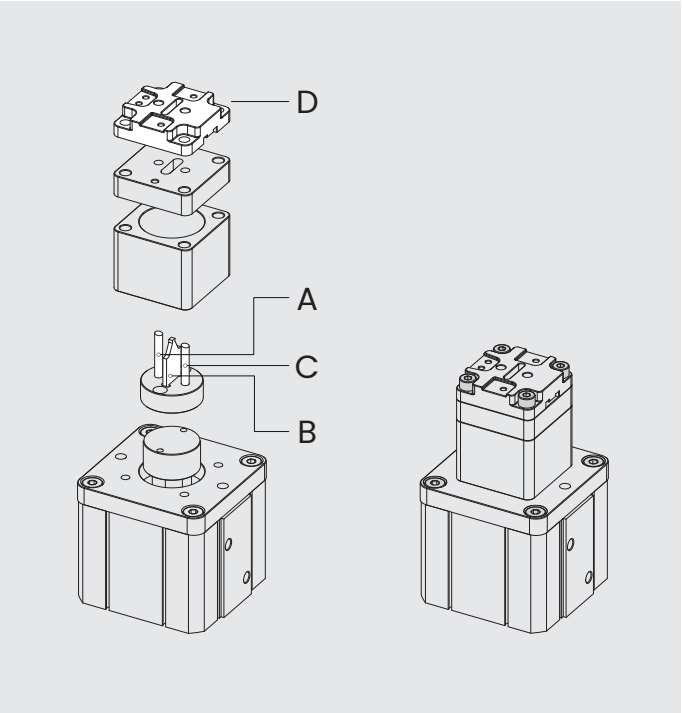


TFS08-P05-00



TFS 08 ***TFS-08 DIE-CUTTING MACHINE***
DISCOUNT TEMPLATES

TFS08-P01-00	DISCOUNT TEMPLATE **TILT & TURN FRICTION STAY**	1
TFS08-P02-00	DISCOUNT TEMPLATE **SIDE-HUNG CREMONE**	1
TFS08-P03-00	DISCOUNT TEMPLATE **TILT & TURN CREMONE**	1
TFS08-P04-00	DISCOUNT TEMPLATE **SIDE-HUNG HANDLE**	1
TFS08-P05-00	DISCOUNT TEMPLATE **TILT & TURN HANDLE**	1



TFS 08 ***TFS-08 DIE-CUTTING MACHINE***
SPARE PARTS

A	**TFS08-CE-R01**	Ø6.1 x 52.5 PUNCH **FOR CYLINDRICAL DRILLING**	1
B	**TFS08-CE-R02**	Ø6.1 x 50 PUNCH **FOR CYLINDRICAL DRILLING**	1
C	**TFS08-CE-R03**	BLADE **FOR CUTTING THE ROD**	1
D	**TFS08-CE-R04**	**CUTTING MATRIX**	1
	TFS08-SOP-00	**MACHINE SUPPORT**	1
	TFS08-PED-00	**PEDAL OPERATION**	1

*In the **operation and maintenance manual for the TFS-08 rod die-cutting machine**, all possible configurations are described in detail.*

Machines that allow to carry out the different machining and riveting of the transmission rods for side-hung and tilt & turn windows.

• **Semi-automatic** operation: the transmission rods are carried out automatically and riveting is done manually.
• The TFS machines facilitate the activity of the technician providing greater efficiency and speed in their work as they make fewer cuts in the rods.
• Space-saving dimensions: 240 x 250 x 250 x 365 mm.

TFS-05 die-cutting machine (ref. 4449)

• Machining of rods for Euro Groove or 16 Groove systems.

Machine and reel holder (ref. 4450)

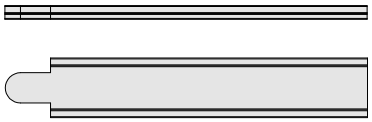
• Support for polyamide transmission rod reel holder.
• Compatible with TFS-04 and TFS-05 machines.

Reel holder (ref. 4446)

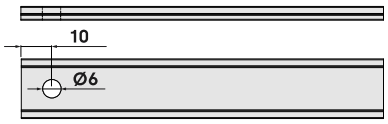
• Support for polyamide transmission rod reel holder.
• Compatible with TFS-01 and TFS-02 machines.

EURO GROOVE

POINTED STAMPING



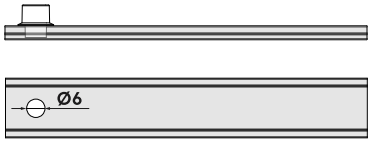
CUT AND PUNCHING



LOAD AND RIVETING (INDEPENDENT STEP)

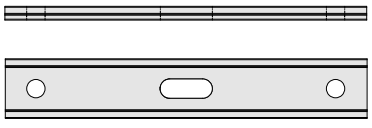
0517002

0511149

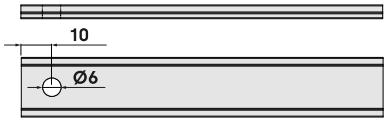


16 GROOVE

PUNCHING AND GROOVED

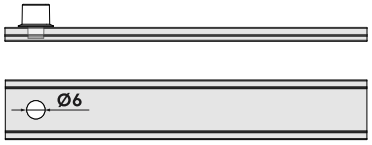


CUT AND PUNCHING



LOAD AND RIVETING (INDEPENDENT STEP)

0517002



TFS-05 DIE-CUTTING MACHINE

CE C16

4449

1

REEL HOLDER

CE C16

TFS 04 TFS 05 TFS 06

4450

1

REEL HOLDER

CE

TFS 01 TFS 02

4446

1

TFS-06 DIE-CUTTING MACHINE

4451

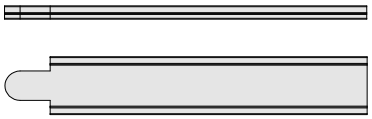
1

Machine that allows to carry out the different machining and riveting of the transmission rods for side-hung and tilt & turn windows.

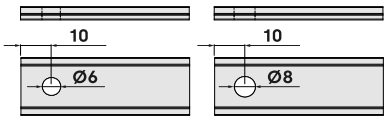
- Machining of rods for Euro Groove or 16 Groove systems.
- **Automatic** operation: the transmission rods and riveting are carried out automatically.
- The machine can be programmed according to the customer's needs.
- It allows to perform the cutting and punching of Ø6 mm as well as the punching and riveting of the plate.
- Highly efficient working system: less assembly time, less cuts, more versatility equal to more productivity.
- The machine is supplied with plates reel holder.

EURO GROOVE

POINTED STAMPING

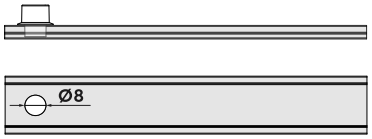


CUT AND PUNCHING



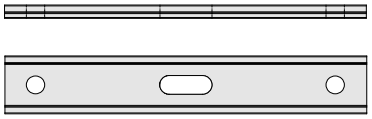
LOAD AND RIVETING

0511088

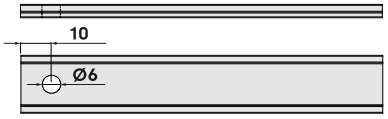


16 GROOVE

PUNCHING AND GROOVED

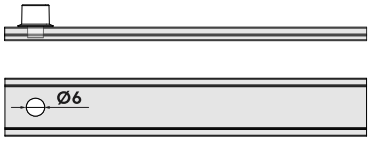


CUT AND PUNCHING



LOAD AND RIVETING

0517002



Hardware bench for machining, screwing, cutting and assembly of the different accessories.

- Compatible with STAC hardware: Euro Groove, 16 Groove and PVS systems.
- Pneumatic and ergonomic tilting bench.
- It has automatic window sash centering devices for the machining of the cremone.
- It has adjustable stops for milling or drilling cremone (fixed or variable height) and transmission cutting (depending on the hardware system).
- Supplied with a screwdriver with automatic screw loading.

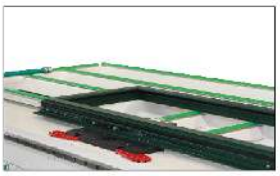
HARDWARE ASSEMBLY CENTRE

CE
C16

BHS-01

1

HARDWARE BENCH CAPACITY	
FRAME MAXIMUM HEIGHT	3000 x 3000 mm
FRAME MINIMUM HEIGHT	300 x 300 mm
MAXIMUM WITH	100 mm
VARIABLE SECTION	25 ~ 100 mm



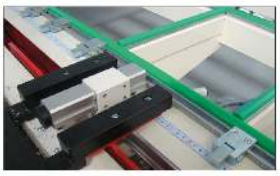
WORK BENCH

LOWER MILLING UNIT
FOR CREMONE

TILTING CYLINDERS

POLYAMIDE ROD
SHEAR

AUTOMATIC
SCREW LOADER



STOP SYSTEM
FOR FIXED NUMBER

SCREWING MODULE

CONTROL PANEL

19 ANODISED HARDWARE

434 Anodised finish

ANODIC

ANODISED HARDWARE

ANODISED HARDWARE

There are more and more projects where, looking for their singularity through colours or finishes, the accessories are lacquered in the same colours as the carpentry.

We understand that architecture is colour, so we wanted to bring new finishes and colours to our products, which are sure to impress.

Thanks to the anodising process, we can artificially increase the thickness of the aluminium's natural layer of oxide, achieving greater levels of corrosion resistance and significantly increasing the resistance and durability of the material. This makes it particularly ideal for projects in severe climatic conditions, such as projects in coastal areas.

In the catalogue there are different products that can be supplied with an anodised finish. These products are indicated in the catalogue with the ANODIC icon, indicating that a product can be ordered in any of the available anodised finishes.

- **Anodised 81**: natural matt
- **Anodised 82**: stainless steel matt
- **Anodised 83**: acier grata

» Tilt & turn kit with hinges

For tilt & turn hardware kits, the anodised finish depends on the selected cremone and hinges.

Anodised references are generated by adding the codes " **81** " or " **82** " as finish.

In the particular case of kits with anodised hinges, it is necessary to indicate the opening hand: « **D** » for right-hand opening or « **I** » for left-hand opening.

More information on each reference can be found in the corresponding product chapter.

The following references are available with an anodised finish: natural, stainless steel or acier.
•For tilt & turn opening kits, the anodised finish depends on the hinges and the cremone.
•In the particular case of kits with anodised hinges, it is necessary to indicate the opening hand: "D" for right-hand opening or "I" for left-hand opening.
•Further information on each reference can be found in the relevant product chapter.

Consult availability of the finish and delivery time.

ANODISED FINISHES

ANODIC

S1132	D / I	**	S
A	B	C	D

DESCRIPTION

A	S1132	PRODUCT REFERENCE ROOT	
B	D	PRODUCT VALID FOR RIGH-HAND OPENING	
	I	PRODUCT VALID FOR LEFT-HAND OPENING	
C	**	AVAILABLE IN DIFFERENT FINISHES	
	R	PRODUCT WITH ADJUSTMENT	
D	S	THE PRODUCT INCLUDES SIRIUS CREMONE	
	C	MANUAL OPERATION	

ANODISED FINISHES

81	NATURAL MATT ANODISED	83	ACIER GRATA ANODISED
82	STAINLESS MATT ANODISED		

ANODISED PRODUCTS

S1132	D	81	S	**CLX EVO-SECURITY KIT** LOCKING TO FRAME WITH CREMONE	S5771	81	**POLARIS HANDLE** 7 x 27 TRANSMISSION SPINDLE
010T260	D	82		**2-LEAF ARIES HINGE** SIDE-HUNG OPENING	052411	83	**TAU DOOR PULL HANDLE L = 450** STRAIGHT AESTHETICS

SIDE-HUNG HINGES		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
ACRUX 2-LEAF Eurogroove 14 ~ 18	**010** D / I ******	●	●	
ACRUX 3-LEAF Eurogroove 14 ~ 18	**P010D** ******	●	●	
ARIES 2-LEAF Center distance: 60 mm	**010T260** ******	●	●	
ARIES 2-LEAF Center distance: 66 mm	**010T266** ******	●	●	
ARIES 2-LEAF DCenter distance: 80 mm	**010T280** ******	●	●	
ARIES 2-LEAF Center distance: 93 mm	**010T293** ******	●	●	
ARIES 3-LEAF Center distance: 60 mm	**010T360** ******	●	●	
ARIES 3-LEAF Center distance: 66 mm	**010T366** ******	●	●	
ARIES 3-LEAF Center distance: 80 mm	**010T380** ******	●	●	
ARIES 3-LEAF Center distance: 93 mm	**010T393** ******	●	●	
NORMA FAST	**025** D / I ******	●	●	
HERCULES	**C0170** ******	●		

HANDLES (TRANSMISSION SPINDLE)		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
POLARIS Transmission spindle 7 x 7 x 27 mm	**S5771** ******	●	●	
POLARIS Transmission spindle 7 x 7 x 64 mm	**S5775** ******	●	●	

CREMONES (TRANSMISSION COVER)		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
SIRIUS Side-hung window (right-hand)	**A0790** **	●	●	
SIRIUS Side-hung window (left-hand)	**A0791** **	●	●	
SIRIUS WITH SHORT LEVER French side-hung window	**A0794** **	●	●	
SIRIUS WITH TRANSMISSION COVER Tilt & turn window	**A1154** **	●	●	
SIRIUS WITHOUT TRANSMISSION COVER Tilt & turn window	**A1157** **	●	●	

PULL HANDLES & FINGER PULL		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
ATRIA PULL HANDLE with keyhole for cam cylinder 15	**2311** **	●	●	
ATRIA PULL HANDLE without keyhole for cam cylinder 15	**2312** **	●	●	
ATRIA PULL HANDLE SET with keyhole for cam cylinder 15	**2313** **	●	●	
ATRIA PULL HANDLE SET without keyhole for cam cylinder 15	**2314** **	●	●	
ATRIA PULL HANDLE + FINGER PULL with keyhole for cam cylinder 15	**2315** **	●	●	
ATRIA PULL HANDLE + FINGER PULL without keyhole for cam cylinder 15	**2316** **	●	●	
ATRIA FINGER PULL with footprint depth: 9.5 mm	**2320** **	●	●	

DOOR HANDLES & CYLINDER COVER PLATES		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
POLARIS DOOR HANDLE SET	**05182** **	●	●	
POLARIS DOOR HANDLE	**05183** **	●	●	
SLIM CYLINDER COVER PLATE Depth: 6.2 mm	**0508053** **	●	●	

DOOR PULL HANDLES		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
TAU STRAIGHT LINE Pull handle length: 450 mm	**052411** **	●		●
TAU OFFSET LINE (LEFT-HAND) Pull handle length: 450 mm	**052412** **	●		●
TAU OFFSET LINE (RIGHT-HAND) Pull handle length: 450 mm	**052413** **	●		●
TAU STRAIGHT LINE Pull handle length: 800 mm	**052451** **	●		●
TAU OFFSET LINE (LEFT-HAND) Pull handle length: 800 mm	**052452** **	●		●
TAU OFFSET LINE (RIGHT-HAND) Pull handle length: 800 mm	**052453** **	●		●
TAU STRAIGHT LINE Pull handle length: 1200 mm	**052461** **	●		●
TAU OFFSET LINE (LEFT-HAND) Pull handle length: 1200 mm	**052462** **	●		●
TAU OFFSET LINE (RIGHT-HAND) Pull handle length: 1200 mm	**052463** **	●		●
DELTA EASY Frontal fixation with diabolos	**0508084** **	●		●
DELTA EASY Lateral fixing with screws	**0508086** **	●		●

TILT & TURN KITS		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
CLX 120 TILT & TURN HINGE KIT Active sash	**T1113** D/I **	●	●	
CLX 120 TILT & TURN HINGE KIT Passive sash	**T1114** D/I **	●	●	
CLX 120 TILT & TURN KIT Locking to inverter without cremone	**T1133** D/I **	●	●	
CLX 120 TILT & TURN KIT Locking to inverter with SIRIUS cremone	**T1118** D/I ** **S**	●	●	
CLX 120 TILT & TURN KIT Locking to frame without cremone	**T1139** D/I **	●	●	
CLX 120 TILT & TURN KIT Tilt-before-turn	**T1123** D/I **	●	●	

RECESSED HANDLES		81	82	83
		NATURAL MATT	*INOX MATT*	*ACIER GRATA*
DRACO MAX Automatic drive	031000 **	●	●	
DRACO MAX Manual drive	031000 ** C	●		
DRACO LINE Automatic drive	030000 **	●		
DRACO LINE Manual drive	030000 ** C	●		

STAC
more than you see

We are near to you

STAC is at your disposal for any queries. For more information visit our website or contact us, we will study a solution tailored to your needs. We have technical support in Spanish, English, French and Italian.

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